The Genesis Trilogy

World, interface and subject

Complete Collected Volume

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Introduction

The central question is how a readable world, a physical interface and an agent able to revise its own use of that interface can form in one process. HAFF develops the world side, Q-RAIF the physical interface, and T-DOME the generation and upkeep of the subject's functional organization. None of the three can be replaced by the most tractable model in another.

The connections have concrete consequences. In a finite semiclassical mean-action scalar/JT realization, the same acquired record protects held quantum content and changes the apparatus's source. At fixed initial mass and geometry, the resulting response enters a later X-basis reading of the old probe into a spent cell, which remains in the continuing mechanical interaction. The task and return share one finite window, cursor and complete reference account. Other models establish shared structure and the value of self-reading. They retain their own physical inputs; the sustained quantum and thermal realization is a stronger problem than this finite return.

The accompanying corpus also includes the overview, the three Q-RAIF companion papers, Brain Anchor and four Imprint papers. Brain Anchor tests learned access, damage and repair on a continuing body. Imprint distinguishes carrying a trace, reading its content and using self-reading to choose later operations. These are independent contributions; they are not absorbed into Foundation III's finite scalar/JT return.

Reading map

HAFF-A *Physical Access and Correlation Geometry: The Holographic Alaya–Field Framework.* Physical channels and their record laws distinguish operational access from an abstract algebra. One initial preparation and fixed reconstruction rule can give non-isometric correlation geometries after different physical reductions.

HAFF-B *Coupling, Records, and Stable Relations: Formation, Protected Content and the Lifetime of a Geometry.* Couplings select what survives. Protected content, slowly decaying relations and physically acquired records have distinct roles in maintaining a description.

HAFF-C *Causation, Agency, and Existence: Historical Control and Maintained Intervention.* Agency is tested by interventions allowed by a retained history, including the maintenance needed for those interventions to remain available.

HAFF-D *Selection, Maintenance and Gravitational Response: Modular Constraints and a Physical Return.* Selection reaches gravitational response through an actual source. The paper joins finite physical returns to modular constraints and develops a common-action conjecture for the missing general mechanism.

HAFF-E *Measurement as Accessibility: Measurement Permissions, Stable Records and Public Readout.* An implementable measurement, a stable pointer and a public record need not coincide. Explicit permissions and instruments determine their connections.

HAFF-F *Records, Dissipation and Intervention: Conditions for a Temporal Direction.* A temporal direction requires an appropriate record and dissipation account. Causal order, entropy production and an agent's stored history are not interchangeable.

HAFF-G *Internal Selection and Revisable Organization: How History Changes Permissions and Physical Sources.* History can revise the operations a system selects. A target operation remains distinct from its larger executor; physical programming and source changes have costs.

HAFF-PS *Postscript: Persistence, Participation and an Unfinished Self.* Persistence and participation are interpreted through the finite constructions, without equating a useful self-model with a complete self-description.

QR-A *Lorentzian Read-out from Local Unitary Propagation: Clifford Relations, Causal Cones and Spectral Mass.* Local unitary propagation produces per-species Clifford readout and causal cones. Speed alignment requires divergent accumulated coupling in the mediator model, or decay of relative-speed perturbations near a supplied fixed point; connectivity alone is insufficient.

QR-B *Thermodynamic Stability Constraints on Persistent Open Quantum Subsystems: Control Algebra, Physical Actuation and Energy Cost.* Persistent control is realized by physical couplings. Protection, calibration, active recovery and field energy are compared under explicit collection and reference permissions.

QR-C *Retained Quantum History, Physical Sources and Receiver Renewal in Q-RAIF.* One autonomous path performs finite repeated readings and receiver updates while retaining the body and history. Local linear-current control also implements neutral-sector stress flows; clock work and the limits of total-source matching are explicit.

TD-I *Memory as a Usable Resource: Correlations, Records and Complete Endpoint Accounting.* The work value of records depends on their physical readout and retained side information. The endpoint account keeps input purity and the outputs of correction visible.

TD-II *Reference Frames Selected by History: The Capability and Cost of a Bounded Agent's Self-Frame.* An agent's usable frame is selected by its actual history. Oriented records and legal readouts determine what it can calibrate and control.

TD-III *Keeping a Self-Model Calibrated: Drift Detection, the Price of Keeping Up, and Reliability in a Closed Loop.* Drift makes continued self-calibration a task. Retained histories determine reference requests, paid maintenance and action authority on one continuing body.

The papers retain their complete proofs, appendices and references, and their own pagination. The contents list PDF page numbers; PDF bookmarks also provide section-level navigation.

PHYSICAL ACCESS AND CORRELATION GEOMETRY: THE HOLOGRAPHIC ALAYA–FIELD FRAMEWORK

A PREPRINT

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ABSTRACT

Physical access has a resolution that an observable algebra does not specify. We describe a reduction by its channel and distinguish its implementable effects from their linear span, their generated algebra and the observables stable under the dynamics. For qubit reductions we characterize equality of output entropy on every pure input. The nonzero branch is separated by every mixed input; the zero branch includes operationally incomparable channels with equal entropy on every input. A nine-probe finite-copy test bounds violations. We then connect physical implementation to geometric certification in one protocol: local unitary couplings reduce a fixed three-qubit GHZ state, further local collisions produce environmental records, and measurements on those records estimate the pair correlations. Calibrated readout errors and sampling errors propagate to confidence intervals for a fixed shortest-path metric. Two noise allocations with the same expected flip count produce non-isometric metrics, and a finite experiment separates their diameters without direct access to the source qubits. The reconstruction rule, source preparation and coupling architecture remain explicit inputs. The results show how finite measurement permissions, rather than an algebra label or exact entropy access, determine which correlation geometry can be established.

1 Introduction

Decoherence explains why interference between some degrees of freedom stops being visible [1]. Entanglement-based approaches to quantum gravity read geometric data from correlations between regions, most sharply through the Ryu–Takayanagi formula [2, 4, 3]. Both start after the system has been divided: into system, apparatus and environment in the first case, into boundary regions or tensor factors in the second. A generic Hilbert space carries no canonical division [5, 6, 7], and the observables that define a division can be chosen so that a given pure state has any Schmidt decomposition one likes [8]. Any geometry read from correlations therefore depends on how the state is reduced.

A reduction can preserve the algebra generated by accessible observables while changing the accuracy of every readout. This is the issue studied here. We first distinguish channel, effect set and stable structure (Section 2), then determine when qubit reductions are indistinguishable by entropy and how finite data detect a violation (Section 3). Section 4 connects a local interaction to an actual environmental record, a measured correlation and a certified finite geometry. Appendix A gives the factorization background and the precise typicality statement.

The geometry in the worked protocol is a metric on three sites under a declared rule. Its non-isometry is proved from the source dynamics and certified through the record dynamics. A Lorentzian metric requires further physical and reconstruction input. Keeping that distinction allows the finite experiment to answer a real part of the world-structure problem: which geometric differences survive the operations a reader can actually perform?

We call the source structure the Alaya-Field, a term taken from Yogācāra philosophy, where it names the storehouse consciousness. Here it is used only as a technical name for the triple $(\mathcal{H}_U, \mathcal{A}_U, |\Omega\rangle)$, with $\mathcal{A}_U$ a von Neumann algebra on $\mathcal{H}_U$ and $|\Omega\rangle$ cyclic and separating, so that the pair is in standard form. The point of the name is the absence of a canonical factorization, not a particular choice of state. In an algebraic quantum field theory $\mathcal{A}_U$ may be a Type III_1

factor [9]. Such a factor has no normal trace, so its states have no intrinsic density operators and no von Neumann trace entropy. The factorization, Haar-measure, partial-trace and entropy results below therefore use their own finite-dimensional representations. There a pure state of the full matrix algebra is cyclic but not separating, so these finite models do not realize every assumption placed on the infinite source structure, and none of their entropy statements is asserted in the Type III setting. The passage from entropy or connectivity data to a spacetime metric is a separate, model-dependent input throughout.

2 Coarse-graining structures, algebras and channels

2.1 Definition

An observable algebra does not by itself reinstate a tensor factorization: operators can act irreducibly on $\mathcal{H}_{\mathrm{total}}$, and in algebraic quantum field theory local algebras are assigned to regions without any global tensor product [9]. A coarse-graining here is meant to be realized by a physical interaction, not chosen by an agent.

Definition 2.1 (Coarse-graining structure)**.** *A coarse-graining structure is a pair* $\mathbf{c} = (\mathcal{A}_{\mathrm{sel},\mathbf{c}}, \Phi_{\mathbf{c}})$*, where*

1. $\mathcal{A}_{\mathrm{sel},\mathbf{c}} \subset \mathcal{B}(\mathcal{H}_U)$ *is a selected unital* $*$*-subalgebra used to define the effective representation. It is not identified with Paper E's probe-reachable effect system, pointer or record algebra, or environmental fixed set [26]; a physical model must state any map between them.*
2. *In the finite-dimensional results,* $\Phi_{\mathbf{c}}$ *is a completely positive trace-preserving (CPTP) map on states that implements an operational reduction; its Heisenberg dual* $\Phi_{\mathbf{c}}^*$ *is unital and completely positive. Partial trace over a specified factor and channels induced by a conditional expectation are examples. In a general von Neumann setting the corresponding object is a normal channel with a normal unital completely positive dual, and no entropy result below is claimed for that extension.*

No theorem guarantees a unique maximal observable algebra for a given physical system without additional structure. Several admissible algebras can coexist, and for a fixed algebra several channels can implement different losses of information, such as tracing out inaccessible degrees of freedom or effective decoherence. The two entries of $\mathbf{c}$ must therefore be related explicitly.

2.2 The algebra generated by a channel

For a channel $\Phi : \mathcal{B}(\mathcal{H})_* \to \mathcal{B}(\mathcal{K})_*$ put

$$\mathcal{S}_\Phi = \operatorname{ran} \Phi^*, \qquad \mathcal{A}_\Phi = C^*(\mathcal{S}_\Phi).$$

$\mathcal{S}_\Phi$ is a unital self-adjoint subspace, not necessarily closed under products. The effects that the channel actually reads out are $\Phi^*([0, I])$, which is in general smaller than the set of positive contractions in $\mathcal{S}_\Phi$. We call $\mathbf{c}$ weakly compatible if $\mathcal{S}_{\Phi_{\mathbf{c}}} \subseteq \mathcal{A}_{\mathrm{sel},\mathbf{c}}$ and minimally compatible if $\mathcal{A}_{\mathrm{sel},\mathbf{c}} = \mathcal{A}_{\Phi_{\mathbf{c}}}$.

Proposition 2.2 (Algebras and post-processing)**.**

1. *If two states agree on* $\mathcal{S}_\Phi$*, in particular if they agree on* $\mathcal{A}_\Phi$*, then their outputs under* Φ *coincide.*
2. *If* $\Phi_2 = \Lambda \circ \Phi_1$ *for a channel* Λ*, then* $\mathcal{S}_{\Phi_2} \subseteq \mathcal{S}_{\Phi_1}$ *and* $\mathcal{A}_{\Phi_2} \subseteq \mathcal{A}_{\Phi_1}$*.*
3. *The converse of (2) fails. For the depolarizing channels* $D_p(\rho) = p\rho + (1-p)I/2$ *on a qubit with* $0 < p < q \leq 1$*, both* $\mathcal{A}_{D_p}$ *and* $\mathcal{A}_{D_q}$ *equal the full matrix algebra,* $D_p = D_{p/q} \circ D_q$*, and* D_q *is not a post-processing of* D_p*.*

Proof. (1) For any output operator Y, $\mathrm{Tr}[\Phi(\rho)Y] - \mathrm{Tr}[\Phi(\sigma)Y] = \mathrm{Tr}[(\rho - \sigma)\Phi^*(Y)]$, and $\Phi^*(Y) \in \mathcal{S}_\Phi$. (2) $\Phi_2^* = \Phi_1^*\Lambda^*$, so the range of Φ_2^* lies in the range of Φ_1^*, and generated algebras inherit the inclusion. (3) Both maps are invertible on the operator space, so their dual ranges are everything. The identity $D_{p/q}(D_q(\rho)) = p\rho + (1-p)I/2$ is direct. Two orthogonal pure inputs have outputs at trace distance p under D_p and q under D_q; a channel cannot increase trace distance, so no Λ satisfies $\Lambda \circ D_p = D_q$. □

Two reductions with the same algebra can thus have different operational resolution. An algebra label is not a substitute for the channel and its allowed post-processing. If a conditional expectation is chosen as the implementation, that is an additional structure, not a synonym for every CPTP reduction.

2.3 Matching the dynamical and measurement interfaces

Paper B's commuting-coupling dynamics gives $\Phi_t(\rho)_{xy} = e^{-tD(x,y)}\rho_{xy}$ in the joint eigenbasis [25]. At finite t all multipliers are nonzero: $\mathcal{A}_{\Phi_t} = \mathcal{B}(\mathcal{H})$. Its block algebra $\mathcal{B}_\pi$ selected by a stability window is therefore an auxiliary description of persistent capability. Unless it is the full algebra, $(\mathcal{B}_\pi, \Phi_t)$ is not even weakly compatible in Definition 2.1. The minimally compatible finite-time pair is $(\mathcal{B}(\mathcal{H}), \Phi_t)$, supplemented by the window, tolerance and $\mathcal{B}_\pi$. In the long-time limit the channel is instead the pinching $\mathcal{E}_{\pi_\infty}$ and $(\mathcal{B}_{\pi_\infty}, \mathcal{E}_{\pi_\infty})$ is minimally compatible. The finite description and its limiting algebra answer different questions.

A measurement channel has a different output. In Paper E's pointer-controlled collision, the source remains a system while an environmental fragment receives a conditional state $|e_z\rangle$, $z = 0, 1$ [26]. Its record channel is

$$\Gamma(\rho) = \sum_{z=0}^{1} \mathrm{Tr}(P_z\rho)|e_z\rangle\langle e_z|, \qquad \Gamma^*(F) = \sum_z \langle e_z|F|e_z\rangle P_z.$$

For overlap $0 < r = |\langle e_0|e_1\rangle| < 1$, the dual range spans the pointer algebra, but the implementable effects omit its sharp projections. The source dephasing channel and Γ arise from the same unitary, although their ranges, outputs and uses differ. The geometric protocol below uses the reduction to set the correlations and Γ to read them. It never replaces $\Gamma^*([0, I])$ by all positive contractions of its span. Persistence is relative to the specified dynamics: a record effect fixed by the recording interaction need not be fixed by the preceding noise interaction.

2.4 Refinement

Definition 2.3 (Refinement)**.** *For coarse-graining structures $\mathbf{c}_1, \mathbf{c}_2$ write $\mathbf{c}_1 \succeq \mathbf{c}_2$ if there is a CPTP map Λ from states on the output space of $\Phi_{\mathbf{c}_1}$ to states on the output space of $\Phi_{\mathbf{c}_2}$ with $\Phi_{\mathbf{c}_2} = \Lambda \circ \Phi_{\mathbf{c}_1}$. Then $\mathbf{c}_2$ is a further coarse-graining of $\mathbf{c}_1$.*

Proposition 2.4 (Order on classes)**.** *The relation $\succeq$ is reflexive and transitive but not antisymmetric. Writing $\Phi \sim \Psi$ when each is a post-processing of the other, $\succeq$ induces a partial order on equivalence classes.*

Proof. Reflexivity uses $\Lambda = \mathrm{id}$, transitivity uses composition of channels. Two different replacer channels, which output fixed states $\omega_1 \neq \omega_2$, simulate each other by a further replacement, so antisymmetry fails. On classes the relation is well defined because composing with a simulating channel does not change membership, and antisymmetry holds by construction. □

If the allowed post-processings are restricted, for example to local, energy-bounded or specified control operations, the restricted class must be closed under composition before the same quotient can be formed. The comparison of channels by approximate post-processing has an operational characterization through success probabilities of discrimination tasks [14]; we use only exact post-processing.

2.5 Mutual information under refinement

Proposition 2.5 (Local and global post-processing)**.** *For channels Λ_A, Λ_B acting on A and B separately,*

$$I(A' : B')_{(\Lambda_A \otimes \Lambda_B)(\rho)} \leq I(A : B)_\rho.$$

A global CPTP post-processing can increase mutual information: the channel that replaces every two-qubit input by a Bell state takes $|00\rangle$, with $I = 0$, to a state with $I = 2\ln 2$.

Proof. $I(A : B)_\rho = D(\rho_{AB}\|\rho_A \otimes \rho_B)$, and $\Lambda_A \otimes \Lambda_B$ maps $\rho_A \otimes \rho_B$ to the product of the new marginals, so data processing for relative entropy gives the inequality. The Bell example is a direct computation. □

A requirement that mutual information decrease under refinement is therefore a theorem for local refinements and false for global ones. A geometric reading of mutual information must say which refinements it allows.

3 When do two reductions give the same entropies?

3.1 Generic separation

Theorem 3.1 (Generic separation for a fixed pair of channels)**.** *Let $\mathcal{H}$ and $\mathcal{K} = \mathcal{K}_A \otimes \mathcal{K}_{A^c}$ be finite dimensional, and let $\Phi_1 \neq \Phi_2$ be CPTP maps from states on $\mathcal{H}$ to states on $\mathcal{K}$. The pure inputs on which Φ_1 and Φ_2 agree form a proper*

real-algebraic subset of Haar measure zero. If, in addition, the real-analytic function

$$f_A(\Psi) := S(\mathrm{Tr}_{A^c}\,\Phi_1(|\Psi\rangle\langle\Psi|)) - S(\mathrm{Tr}_{A^c}\,\Phi_2(|\Psi\rangle\langle\Psi|))$$

is not identically zero on each connected component under consideration of the open set where both reduced outputs have full rank, then $f_A(\Psi) \neq 0$ for generic inputs in those components.

Proof. Pure-state projectors span the Hermitian operators, so $\Phi_1 - \Phi_2$ cannot vanish on all of them. The equality condition is polynomial in the real and imaginary parts of $|\Psi\rangle$, so its solution set is a proper real-algebraic subset and has Haar measure zero. On the full-rank domain the von Neumann entropy, and hence f_A, is real analytic, and a nonzero real-analytic function has a zero set of measure zero. □

The nondegeneracy hypothesis cannot be dropped: channels that differ by a unitary acting on A alone give different outputs with identical entropies on every input. Channel inequivalence alone therefore does not yield an entropic or geometric difference. The rest of this section decides, for qubit reductions, exactly when the hypothesis fails.

3.2 Qubit reductions: the criterion

A qubit channel acts on Bloch vectors as an affine map

$$\Phi_i :\ r \mapsto M_i r + t_i, \qquad i = 1, 2,$$

with M_i a real 3×3 matrix and $t_i \in \mathbb{R}^3$, subject to the complete-positivity conditions [15]. Put

$$A = M_1^T M_1 - M_2^T M_2, \qquad b = M_1^T t_1 - M_2^T t_2, \qquad c = \|t_1\|^2 - \|t_2\|^2.$$

The difference of the squared output Bloch lengths is

$$q(r) = \|M_1 r + t_1\|^2 - \|M_2 r + t_2\|^2 = r^T A r + 2b^T r + c. \tag{1}$$

The entropy of a qubit state is a strictly decreasing function of its Bloch length: $S = h\big((1+\|s\|)/2\big)$ with h the binary entropy in nats.

Theorem 3.2 (Equal entropy on all pure inputs)**.** *Φ_1 and Φ_2 give equal output entropy on every pure input if and only if there is a real number α with*

$$A = \alpha I, \qquad b = 0, \qquad c = -\alpha. \tag{2}$$

Proof. Equal entropy on pure inputs means $q(r) = 0$ on the unit sphere. Comparing $q(r)$ and $q(-r)$ gives $b^T r = 0$ for all unit r, so $b = 0$. A real symmetric quadratic form that is constant on the unit sphere is a multiple of the identity, so $A = \alpha I$ and then $c = -\alpha$. Conversely, (2) gives $q(r) = \alpha(\|r\|^2 - 1)$, which vanishes on the sphere. No rank condition on the outputs is needed, and pure outputs on the boundary are included. □

Together with positivity of the Choi matrices, (2) decides membership for any given pair. It is a decision rule, not yet a classification of all solutions into physically distinct normal forms.

3.3 The two branches

Proposition 3.3 (Zero branch)**.** *Let $X_i = [M_i \mid t_i]$ be the 3×4 matrix with t_i appended. If $\alpha = 0$, then $X_2 = OX_1$ for an orthogonal $O \in O(3)$. If $\mathrm{rank}\, X_1 \leq 2$, O can be chosen with $\det O = +1$, so Φ_2 is Φ_1 followed by a unitary conjugation of the output. If $\mathrm{rank}\, X_1 = 3$, O is unique; $\det O = +1$ again gives a unitary conjugation, and $\det O = -1$ gives a unitary conjugation composed with transposition.*

Proof. For $\alpha = 0$, (2) says exactly $X_1^T X_1 = X_2^T X_2$. Equal Gram matrices define an isometry between the column spaces of X_1 and X_2, which extends to an orthogonal map of $\mathbb{R}^3$. If the column space has dimension at most two, the extension can be composed with a reflection in an unused direction to make the determinant positive. Rotations of the Bloch ball are unitary conjugations, and the reflection $\mathrm{diag}(1, -1, 1)$ is transposition in the standard basis. □

Transposition is not completely positive, but both endpoints of a transposed pair can still be channels.

Example 3.4 (Equal entropies everywhere, no simulation either way)**.** *Take $M_1 = pI$, $M_2 = p\,\mathrm{diag}(1,-1,1)$ and $t_1 = t_2 = 0$. The first map is a depolarizing channel for $0 \le p \le 1$; by the conditions for unital qubit channels [16], the second is completely positive exactly when $0 \le p \le 1/3$. They give equal output entropy on every input, pure or mixed. For $0 < p \le 1/3$ neither is a post-processing of the other. Indeed both are invertible on the Bloch ball, so a post-processing Λ with $\Lambda \circ \Phi_1 = \Phi_2$ would have to act as $r \mapsto \mathrm{diag}(1,-1,1)r$ on an open set and hence everywhere; that map is transposition, whose normalized Choi matrix has smallest eigenvalue $-1/2$. The same argument excludes the reverse direction.*

Identical entropy functions on the whole input space therefore do not imply operational equivalence, and the two channels are incomparable in the order of Proposition 2.4.

Solutions with $\alpha \neq 0$ are not rare. Every pair of complementary qubit channels satisfies (2): if $V : \mathcal{H} \to \mathcal{K} \otimes \mathcal{E}$ is an isometry with $\mathcal{K}$ and $\mathcal{E}$ two-dimensional, $\Phi(\rho) = \mathrm{Tr}_{\mathcal{E}}\, V\rho V^\dagger$ and $\Phi^c(\rho) = \mathrm{Tr}_{\mathcal{K}}\, V\rho V^\dagger$, then for a pure input the joint output is pure and its two marginals have the same entropy. For amplitude damping with decay probability γ and its complementary channel, $\alpha = 1 - 2\gamma$, which is nonzero unless $\gamma = 1/2$. A family with full Choi rank is

$$\Phi_{a,b} : r \mapsto ar + b\hat{z}, \qquad \Phi_{b,a} : r \mapsto br + a\hat{z}, \qquad a, b > 0,\ a + b < 1.$$

$\Phi_{a,b}$ is a convex combination of the identity and the replacer with Bloch vector $b\hat{z}/(1-a)$, so it is CPTP, and likewise $\Phi_{b,a}$. Both give squared pure-output length $a^2 + b^2 + 2abr_z$, and $\alpha = a^2 - b^2$. For $a = .3$, $b = .2$ both normalized Choi matrices have smallest eigenvalue .125.

3.4 Mixed inputs expose the nonzero branch

Proposition 3.5 (Entropy gap on mixed inputs)**.** *For any two qubit channels and any input,*

$$|S_1(r) - S_2(r)| \ge \tfrac{1}{2}|q(r)|. \tag{3}$$

If (2) *holds, then $q(r) = \alpha(\|r\|^2 - 1)$ on the whole Bloch ball, so*

$$|S_1(r) - S_2(r)| \ge \frac{|\alpha|}{2}\big(1 - \|r\|^2\big), \tag{4}$$

and $\mathrm{sgn}(S_1 - S_2) = \mathrm{sgn}\,\alpha$ for every strictly mixed input.

Proof. Write the entropy as a function of the squared Bloch length, $f(x) = h\big((1+\sqrt{x})/2\big)$. Then $-f'(x) = \mathrm{artanh}(\sqrt{x})/(2\sqrt{x}) \ge \frac{1}{2}$, with the continuous extension at $x = 0$ and the integral limit at $x = 1$. The mean value theorem gives (3), and substituting (2) into (1) gives (4). A larger Bloch length means smaller entropy, which fixes the sign. □

When all pure inputs agree, a single strictly mixed input with equal entropies therefore forces $\alpha = 0$ and the orthogonal relation of Proposition 3.3. The argument uses only the one-to-one relation between qubit entropy and Bloch length, not a polynomial form of the entropy.

3.5 Probes and random inputs

Corollary 3.6 (Nine pure probes and one mixed probe)**.** *Let the probe set consist of the nine Bloch vectors*

$$\pm e_1,\ \pm e_2,\ \pm e_3,\ (e_1+e_2)/\sqrt{2},\ (e_1+e_3)/\sqrt{2},\ (e_2+e_3)/\sqrt{2}.$$

If the two channels give equal output entropies on these nine inputs, they give equal output entropies on every pure input. If they also agree at $r = 0$, they agree on every input.

Proof. The six axial probes give $b_i = 0$ and $A_{ii} + c = 0$; the three diagonal probes then give $A_{ij} = 0$ for $i \neq j$. This is (2). At $r = 0$, $q = c = -\alpha$, so agreement there gives $\alpha = 0$. Then q vanishes on the entire Bloch ball, hence the output lengths and entropies agree everywhere. □

The corollary concerns exact entropy values on nine inputs, not nine measurements. On the unit sphere $q(r) = r^T A_0 r + 2b^T r + d$ with $\alpha = \frac{1}{3}\,\mathrm{tr}\,A$, $A_0 = A - \alpha I$ and $d = c + \alpha$, which is linear in nine real parameters.

Proposition 3.7 (Random pure inputs)**.** *For Haar-random pure inputs,*

$$Q := \mathbb{E}\, q(r)^2 = \frac{2}{15}\|A_0\|_F^2 + \frac{4}{3}\|b\|^2 + d^2. \tag{5}$$

$Q = 0$ *exactly when* (2) *holds. If* $Q > 0$*, then* $\mathbb{E}(S_1 - S_2)^2 \geq Q/4$*, and for* $0 < \tau < 1$

$$\Pr\Big\{|S_1 - S_2| > \tfrac{1}{2}\sqrt{\tau Q}\Big\} \geq \Pr\{q^2 > \tau Q\} \geq \frac{(1-\tau)Q}{1-\tau Q}. \tag{6}$$

Proof. Use $\mathbb{E} r_i r_j = \delta_{ij}/3$ and $\mathbb{E} r_i r_j r_k r_l = (\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})/15$; odd moments and the cross terms with the traceless A_0 vanish. The second claim follows from (3). Both outputs are states, so $0 \leq q^2 \leq 1$; splitting $\mathbb{E}q^2 = Q$ at $q^2 = \tau Q$ gives $Q \leq \tau Q(1-P) + P$ with $P = \Pr\{q^2 > \tau Q\}$, which is (6). □

With n independent Haar inputs, the probability of missing the difference at the level $\frac{1}{2}\sqrt{\tau Q}$ is at most $(1 - p_\tau)^n$, where p_τ is the right side of (6). This assumes exact entropies; the next proposition removes that assumption. The average change of purity over pure inputs is also the basis of the unitarity measure of noise [17]; Eq. (5) concerns the difference between two channels rather than the coherence of one.

3.6 Finitely many copies

For a qubit, purity and entropy are both monotone functions of the Bloch length, so equal entropy is equivalent to equal purity, and purity can be measured directly. A SWAP test on two copies of a state σ returns the outcome 0 with probability $\frac{1}{2}\big(1 + \operatorname{Tr}\sigma^2\big)$ [18]. For a qubit with Bloch vector s this is $\frac{3}{4} + \frac{1}{4}\|s\|^2$. We assume that every channel use acts on a freshly prepared, known probe state, that the channels do not drift between uses, and that two output copies can be measured jointly.

Finite data cannot establish that the two entropy functions are exactly equal. They can separate equality from a violation of prescribed size, and they can bound how large a violation can be. Put

$$\kappa_9 = \frac{75 - 3\sqrt{465}}{8} = 1.28855\ldots.$$

Proposition 3.8 (Finite-copy test and confidence bound)**.** *For each of the nine probes* r_k *of Corollary 3.6, send* $2m$ *copies of the input through* Φ_1*, apply a SWAP test to each of the* m *output pairs, and do the same for* Φ_2*. Let* D_k *be the difference of the two observed frequencies of the outcome* 0*.*

1. *Given* $\gamma > 0$*, declare the channels different on pure inputs if* $\max_k |D_k| > \gamma/8$*. If* (2) *holds, a difference is declared with probability at most* $18\, e^{-m\gamma^2/64}$*. If* $\max_k |q(r_k)| \geq \gamma$*, the difference is missed with probability at most* $2\, e^{-m\gamma^2/64}$*.*

2. $\sum_k q(r_k)^2 \geq \kappa_9\, Q$*, and* κ_9 *is the largest constant with this property. Hence* $Q \geq Q_0 > 0$ *forces* $\max_k |q(r_k)| \geq (\kappa_9 Q_0/9)^{1/2}$*, and*
$$m \geq \frac{576\, \ln(18/\delta)}{\kappa_9\, Q_0}$$
output pairs per probe and channel keep both error probabilities below δ *for the hypotheses* $Q = 0$ *and* $Q \geq Q_0$*.*

3. *For every* $t > 0$*, with probability at least* $1 - 18\, e^{-mt^2}$*,*
$$Q \leq \frac{16}{\kappa_9} \sum_{k=1}^{9} \big(|D_k| + t\big)^2.$$

Proof. The expected value of D_k is $q(r_k)/4$. D_k is a sum of $2m$ independent Bernoulli variables with coefficients $\pm 1/m$, so Hoeffding's inequality [20] gives $\Pr\{|D_k - \mathbb{E}D_k| \geq t\} \leq 2e^{-mt^2}$. Under (2) every $\mathbb{E}D_k$ vanishes, and $t = \gamma/8$ with a union bound over the nine probes gives the first error bound. If $|q(r_k)| \geq \gamma$ for some k, then $|\mathbb{E}D_k| \geq \gamma/4$, and missing it requires $|D_k - \mathbb{E}D_k| \geq \gamma/8$, which gives the second.

The smallest generalized eigenvalue of the two quadratic forms $\sum_k q(r_k)^2$ and Q is κ_9, as computed in Appendix C. The largest of nine terms is at least their mean; setting $\gamma^2 = \kappa_9 Q_0/9$ and requiring $18e^{-m\gamma^2/64} \leq \delta$ gives item 2. For item 3, on the event that all nine deviations are at most t, which fails with probability at most $18e^{-mt^2}$, we have $|q(r_k)| = 4|\mathbb{E}D_k| \leq 4(|D_k| + t)$, and $Q \leq \kappa_9^{-1} \sum_k q(r_k)^2$. □

If the pure-input criterion is known to hold, the same argument at the mixed probe $r = 0$ tests the size of α: the expected difference there is $-\alpha/4$, and $m \geq 64\ln(2/\delta)/\alpha_0^2$ pairs per channel separate $\alpha = 0$ from $|\alpha| \geq \alpha_0$ with error at most δ in each direction. The pure test uses $18m$ channel uses per channel, $36m$ in all. The constants are conservative. Beyond qubits, purity and entropy are different functionals and this reduction no longer applies.

4 Connectivity and geometry

4.1 Mutual information as connectivity

To compute correlations, declare a subsystem or region representation of the channel output and its relation to the selected source algebra. The channel and state alone do not determine that representation. For output factors $A, B, \ldots$ in this representation, set

$$I_{\mathbf{c}}(A : B) = S_{\mathbf{c}}(A) + S_{\mathbf{c}}(B) - S_{\mathbf{c}}(AB),$$

with entropies computed after coarse-graining.

Definition 4.1 (Connectivity)**.** *For disjoint output factors A, B in the declared representation, the connectivity is $\mu_{\mathbf{c}}(A, B) := I_{\mathbf{c}}(A : B)$. It defines a weighted graph but is not a distance: large mutual information denotes strong correlation rather than large separation, and the metric axioms generally fail.*

A state-parameter metric, such as the SLD Fisher metric in Appendix B, is a different construction. Neither it nor the graph weights alone supply a Lorentzian metric.

4.2 A finite reconstruction rule

For a declared family of finite-dimensional subsystems, a positive connected correlation graph supports a shortest-path metric once an edge-length rule has been chosen. We use $\ell_e = 1/I_e$ on edges with $I_e > 0$. An edge with $I_e = 0$ is absent; distinct connected components then have infinite separation under this rule. Mutual information itself is not a distance. The rule assigns longer distances to weaker correlations, but its physical interpretation depends on the model.

A positive lower bound on the used mutual informations makes this rule stable. No assumption of exponential decay is needed for this finite statement. Exponential decay relative to a pre-existing locality structure can help construct a spatial interpretation [24, 23]; it would supply an additional input rather than derive that locality from the graph.

Proposition 4.2 (Stability of a graph metric)**.** *On the complete graph with v vertices, give the edge e the length $1/I_e$ and use the shortest-path metric d. If every $I_e \geq m > 0$ and the perturbed weights satisfy $|I'_e - I_e| \leq \varepsilon < m$, then*

$$\max_{i,j} |d'_{ij} - d_{ij}| \leq (v-1)\frac{\varepsilon}{m(m-\varepsilon)}.$$

Proof. For each edge, $|1/I'_e - 1/I_e| = |I_e - I'_e|/(I_e I'_e) \leq \varepsilon/(m(m-\varepsilon))$. A shortest path with positive weights uses at most $v-1$ edges. Comparing the new shortest path with the old lengths, and the old shortest path with the new lengths, bounds the difference in both directions. □

4.3 From entropy data to a spacetime metric

In holography, the entropy of a boundary region equals the area of an extremal surface in Planck units [4], and the connectivity of spacetime has been argued to track the pattern of entanglement [3]. In tensor-network models, graph distance in an entanglement graph can approximate continuum distance [22]. These are reconstruction dictionaries for particular classes of states.

For a comparison of spacetime geometries, a dictionary $\mathfrak{G}$ must also state which entropy profiles it identifies. A sufficient condition is

$$\mathfrak{G}[S_1] \simeq_{\text{diff}} \mathfrak{G}[S_2] \implies S_1 \simeq_{\mathfrak{G}} S_2.$$

Its contrapositive says that separated profiles produce separated geometries. This is a requirement on the dictionary, not a derivation of geometry from a channel. The next result replaces that requirement by a fixed finite rule for which the distances and their measurement uncertainties can be computed.

4.4 One state, one operation family, one rule

Proposition 4.3 (Non-isometric finite geometries)**.** *Fix the three-qubit GHZ state. Allow only local bit-flip channels*

$$\Phi_{\mathbf{p}} = \bigotimes_{i=1}^{3} \big[(1-p_i)\,\mathrm{id} + p_i\,\mathrm{Ad}_{X_i}\big], \qquad 0 < p_i < \tfrac{1}{2},$$

which are generated by local bit-flip semigroups with $p_i = (1-e^{-2\gamma_i t})/2$. Fix the reconstruction rule: on the complete graph of the three qubits, the edge $\{i,j\}$ has length $1/I_{ij}$, and distances are shortest-path lengths. Then

$$I_{ij} = \ln 2 - h\Big(\frac{1+c_{ij}}{2}\Big) > 0, \qquad c_{ij} = (1-2p_i)(1-2p_j),$$

and different flip probabilities can give metric spaces that are not isometric under any relabelling of the vertices:

flip probabilities	*mutual informations of the three edges*	*diameter*
$(.1, .1, .1)$	$(.221753694,\ .221753694,\ .221753694)$	4.509507747
$(.05, .1, .2)$	$(.288183695,\ .153870839,\ .120090263)$	8.327069751
$(.05, .1, .15)$	$(.288183695,\ .214256042,\ .166239219)$	6.015427679

The first and third rows have the same expected number of applied flips, $\sum_i p_i = .3$, when each channel is implemented by applying X_i with probability p_i.

Proof. Every two-qubit marginal of the GHZ state is an equal mixture of $|00\rangle$ and $|11\rangle$. Local flips leave both bits uniform and make them agree with probability $(1+c_{ij})/2$, so $I_{ij} = 2\ln 2 - \big(\ln 2 + h((1+c_{ij})/2)\big)$. All edge lengths are finite and positive, so the rule defines a metric. The diameters differ, so no relabelling makes the spaces isometric. The numbers were also computed from the full eight-dimensional density matrix with the channels applied and partial traces taken; the analytic formula agrees to better than 3×10^{-16}. □

Every entry is fixed in advance: the state, the class of allowed operations, the interpretation of the vertices and the reconstruction rule. Nothing about the target geometry is written into the channels. The local flip channels are invertible on the operator space, so all of them generate the full matrix algebra; the algebra label is the same in every row while the geometry is not, which is Proposition 2.2(3) in geometric form. The equal flip budget in rows one and three shows that the difference comes from how a fixed resource is distributed among the qubits, not from its total. This is a one-step budget in a stated implementation; it does not claim equal cost on all hardware or under continuous-time noise. By Proposition 4.2, perturbing the three mutual informations by ε changes each distance by at most $2\varepsilon/(m(m-\varepsilon))$ with m the smallest mutual information. The model is a finite existence result for a fixed rule, not a derivation of the rule from mutual information and not a spacetime.

4.5 The reduction is a local interaction

For each source qubit prepare a noise ancilla N_i in $|0\rangle$ and apply

$$U_i(\theta_i) = e^{-i\theta_i X_i \otimes Y_{N_i}}, \qquad U_i(\theta_i)|\psi\rangle|0\rangle = \cos\theta_i|\psi\rangle|0\rangle + \sin\theta_i X_i|\psi\rangle|1\rangle.$$

Discarding N_i realizes exactly the local flip channel with $p_i = \sin^2\theta_i$. One possible interaction Hamiltonian is $\hbar g_i X_i \otimes Y_{N_i}$ applied for θ_i/g_i. The angles, tensor factors and timing are supplied by the apparatus. They do not encode a target metric.

Next attach, at each site, a fragment of k_i fresh memory qubits. Use Paper E's collisions $P_{0,i} \otimes I + P_{1,i} \otimes R_{i,a}$, with real conditional overlaps $c_{i,a}$, so the fragment overlap is $r_i = \prod_{a=1}^{k_i} c_{i,a}$. These collisions dephase the source in the Z basis. Every two-site marginal of the noisy GHZ state is already diagonal, so its probabilities and mutual information are unchanged by recording. The post-recording source therefore has the same graph metric as before, although its three-site coherence may have changed. This is preservation of the data used by this rule, not preservation of the entire source state.

The noise gate ends before recording begins. This sequencing and its switching resources matter: the flip couples through X, while the recorder uses Z. Their common fixed algebra is only the scalars. Indeed the composed single-site source channel sends a Bloch vector to $(r_i x, r_i(1-2p_i)y, (1-2p_i)z)$; for $0 < p_i < 1/2$ and $0 < r_i < 1$ it has no fixed nonidentity direction, although its linear range is still the full matrix algebra. The record channel nevertheless

reads the final pointer probabilities with nonzero resolution. This supplies an actual finite-access example beyond an exact stable-algebra label.

Tracing out the source after the collisions gives the exact joint fragment state

$$\rho_{\mathrm{rec}} = \sum_{\mathbf{z}} P_{\mathbf{p}}(\mathbf{z}) \bigotimes_{i=1}^{3} |e_{z_i}^{(i)}\rangle\langle e_{z_i}^{(i)}|, \tag{7}$$

where $z_i = \pm 1$ labels the local pointer and $P_{\mathbf{p}}$ is the actual diagonal distribution of the noisy GHZ state. Thus conditional product structure follows from the unitary couplings and fresh product memories; it is not an extra assumption about independent answers. Each site has equal pointer priors.

For $r_i < 1$, the equal-prior Helstrom measurement on fragment i has a symmetric binary error probability $\epsilon_i = (1 - \sqrt{1 - r_i^2})/2$ [19, 26]. To see the symmetry directly, choose real vectors $e_0 = (1, 0)$ and $e_1 = (r, \sqrt{1 - r^2})$. The sign of $|e_0\rangle\langle e_0| - |e_1\rangle\langle e_1|$ has expectations $+\sqrt{1 - r^2}$ and $-\sqrt{1 - r^2}$ in the two states. Writing $Y_i = \pm 1$ for the actual outcome and $v_i = 1 - 2\epsilon_i = \sqrt{1 - r_i^2}$ therefore gives

$$\mathbb{E}(Y_i \mid \mathbf{z}) = v_i z_i, \qquad \mathbb{E}(Y_i Y_j) = v_i v_j c_{ij}. \tag{8}$$

Local measurement on (7) gives these identities even though the records are correlated before conditioning.

4.6 Finite records certify the geometric difference

Let $f(c) = \ln 2 - h((1 + c)/2)$, $-1 \le c \le 1$. Consider K noise configurations, with independently prepared source states and fresh ancillas in each run. For every configuration collect n triples (Y_1, Y_2, Y_3) from the local measurements above. Calibration supplies estimates $\widehat{v}_i \in [0, 1]$ with $\widehat{v}_i \ge \nu > 0$ and $|\widehat{v}_i - v_i| \le \xi$. The estimate

$$\widehat{c}_{ij} = \mathrm{clip}_{[-1,1]} \frac{n^{-1} \sum_{s=1}^{n} Y_i^{(s)} Y_j^{(s)}}{\widehat{v}_i \widehat{v}_j}$$

uses the measurement outcomes, not hidden pointer values.

Theorem 4.4 (Record-to-metric confidence certificate)**.** *Suppose the calibration event has failure probability at most δ_{cal}, and the record law is* (7) *with the stipulated local measurements. For every $\tau > 0$, with probability at least $1 - \delta_{\mathrm{cal}} - 6Ke^{-n\tau^2/2}$, all $3K$ correlations satisfy*

$$|\widehat{c}_{ij} - c_{ij}| \le e_c := \frac{\tau + 2\xi}{\nu^2}. \tag{9}$$

Intersect each interval $[\widehat{c}_{ij} - e_c, \widehat{c}_{ij} + e_c]$ with $[-1, 1]$ and minimize and maximize f on it, obtaining I_{ij}^-, I_{ij}^+. If every $I_{ij}^- > 0$, form shortest-path metrics d^- and d^+ from edge lengths $1/I_{ij}^+$ and $1/I_{ij}^-$. The true metric obeys $d^- \le d \le d^+$ entrywise, and its diameter lies in $[\operatorname{diam} d^-, \operatorname{diam} d^+]$. Disjoint diameter intervals for two configurations certify non-isometry on the same confidence event.

For a sample-size guarantee, if true and estimated correlations obey $|c_{ij}|, |\widehat{c}_{ij}| \le b < 1$, set $e_I = \operatorname{artanh}(b) e_c$. When $I_{ij} \ge \mu > e_I$,

$$\|\widehat{d} - d\|_\infty \le R := \frac{2e_I}{\mu(\mu - e_I)}. \tag{10}$$

For two configurations with true diameter gap $\Delta > 4R$, the intervals $[\operatorname{diam} \widehat{d} - R, \operatorname{diam} \widehat{d} + R]$ are disjoint on that event.

Proof. Each $Y_i Y_j \in [-1, 1]$ has expectation $v_i v_j c_{ij}$. Hoeffding's inequality gives a two-sided tail $2e^{-n\tau^2/2}$; a union bound covers the $3K$ edge estimates. The three edges within a run need not be independent. On the calibration event, $|v_i v_j - \widehat{v}_i \widehat{v}_j| \le 2\xi$, by telescoping and the bounds $v_i, \widehat{v}_i \le 1$. Dividing by $\widehat{v}_i \widehat{v}_j \ge \nu^2$, using $|c_{ij}| \le 1$, and clipping cannot increase the error, proving (9). The interval construction contains the true mutual informations. Shortest-path lengths and their diameter are monotone in positive edge lengths, proving the metric enclosure. Finally $f'(c) = \operatorname{artanh} c$, so $|\widehat{I} - I| \le e_I$. Proposition 4.2, with three vertices, gives (10). Each estimated diameter differs from its true value by at most R, so the estimated gap is at least $\Delta - 2R$; two radius-R intervals are disjoint when $\Delta > 4R$. □

For the equal-flip-count configurations $(.1, .1, .1)$ and $(.05, .1, .15)$, take one recording qubit per site with overlap $r_i = .2$. Then $v_i = \sqrt{.96}$ and the correlations lie between .56 and .72. With exactly specified v_i, use $\nu = \sqrt{.96}$, $\xi = 0$, $\tau = .003$, $b = .8$, $\mu = .16$ and $\delta = .05$. The choice

$$n = \left\lceil \frac{2\ln(12/\delta)}{\tau^2} \right\rceil = 1\,217\,920$$

runs per configuration gives $e_c = .003125$, $e_I = .0034332$ and $R < .275$. The true diameter gap is $1.505919931 > 4R$. Thus a finite record-only experiment separates the two geometries with confidence at least .95, under the stated source and apparatus model. The sample bound is conservative and does not claim optimal tomography.

A known unitary specification is not free empirical calibration. If the v_i are unknown, prepare reference sources in $Z = +1$ and apply the same record couplings and readers. Each reference outcome has mean v_i. For N independent reference runs per site, clipping the empirical mean to $[0, 1]$ yields

$$\Pr\{\max_i |\widehat{v}_i - v_i| > \xi\} \le 6e^{-N\xi^2/2}$$

for three shared readers; separately calibrated readers for K configurations replace 6 by $6K$. Thus $N \ge \lceil 2\ln(6/\delta_{\mathrm{cal}})/\xi^2 \rceil$ per site suffices. Reject calibrations below ν. For the same two geometries, take $\xi = .001$, $\delta_{\mathrm{cal}} = .025$, $\nu = .978$ and $\tau = .002$. Then $N = 10\,961\,278$ reference runs per site and $n = 3\,086\,894$ main runs per configuration give total failure at most .05, $e_c < .004182$, $R < .36955$ and $4R < 1.4782 < \Delta$. On the calibration event the true $v_i = \sqrt{.96}$ ensures $\widehat{v}_i > \nu$; the correlation bound also stays inside $b = .8$. The reference budget is therefore $3N$ additional preparations and readouts, not part of the $2n$ main runs. Equation (9) includes their residual error. Prepared reference pointers, their basis labels and the stability of the reader between calibration and use are additional physical resources.

The comparison uses $2n$ preparations of the same GHZ source, $6n$ fresh noise ancillas, $6n$ fresh recording qubits, $6n$ local binary readouts and communication of the outcomes to a comparator, plus any calibration runs. The expected noise-flip count is .3 per source in both configurations; their recording and readout schedule is identical. Preparation, switching work, storage, communication and renewal do not vanish because the counts agree. No heat or total-energy equality follows from the flip count, and this finite protocol supplies no reservoir reset.

4.7 Limits of record inversion

At $r_i = 1$ the record states coincide and $v_i = 0$: algebraic inversion of their span supplies no source information. Even for $v_i > 0$ the error grows as the inverse product visibility. A narrow effect set can span the pointer algebra while requiring too many trials for the available budget.

Calibration of individual error rates is also insufficient when the readers share an unmodelled error. Let a common random sign B multiply all otherwise ideal outcomes, with $\mathbb{E}B = v$. Each local reference test then reports visibility v, while the pair products are unchanged, since $B^2 = 1$. Dividing by v^2 overestimates the true correlations. A common inversion has $v = -1$ and is invisible to agreement. Joint reference tests, conditional error modelling or a verified local apparatus are needed to exclude such contamination. The confidence certificate is conditional on the joint record law, rather than an inference of that law from single-site calibration.

A spacetime reconstruction must solve a further problem: finite measurements determine a profile within an error region, not an exact profile. The relevant dictionary must be stable on that region and must quotient its coordinate freedom. Entropy equality from Section 3 supplies a genuine obstruction, including operationally different channels with identical entropy profiles. More accurate readout cannot repair a dictionary's failure to distinguish such profiles.

Conjecture 4.5 (Finite permissions approaching geometric access)**.** *In a source family with an independently justified local reconstruction dictionary, histories that enlarge the available record fragments and calibrate their readout can make the inferred geometry converge while the generated observable algebra remains constant. The required growth of visibility, preparation count and dictionary stability bounds must jointly dominate sampling and model error.*

The finite certificate supplies the mechanism and the costs of this candidate. For E edges with visibility at least ν, error in the correlations scales as $\nu^{-2}\sqrt{\ln(2E/\delta)/n}$ before calibration error. If reconstructed edge mutual informations have lower bound μ, distance error carries another factor of order μ^{-2} times the number of edges in a shortest path. Consequently, a growing graph or an approaching zero-information edge can defeat convergence even while all fixed-algebra labels are unchanged. The conjecture would fail in a proposed model if its dictionary is unstable, if common error survives calibration, or if its available budget cannot overcome these factors. This is a route from finite permissions to geometric access, not an assumption that a continuum or its dynamics is already generated.

The spatial/eigenmode example in Appendix B illustrates dependence of connectivity on factorization. Unlike Proposition 4.3, its empty eigenmode graph has no finite reconstructed paths.

4.8 Not a change of coordinates

A diffeomorphism acts within a fixed algebra of observables and leaves unchanged what can be measured. A change of coarse-graining changes the accessible observables, which correlations are visible and how subsystems are defined. Algebraic difference alone does not prove that two reconstructed metrics are not diffeomorphic, because a dictionary may discard it; the conclusion needs a separating dictionary or an explicit computation such as Proposition 4.3.

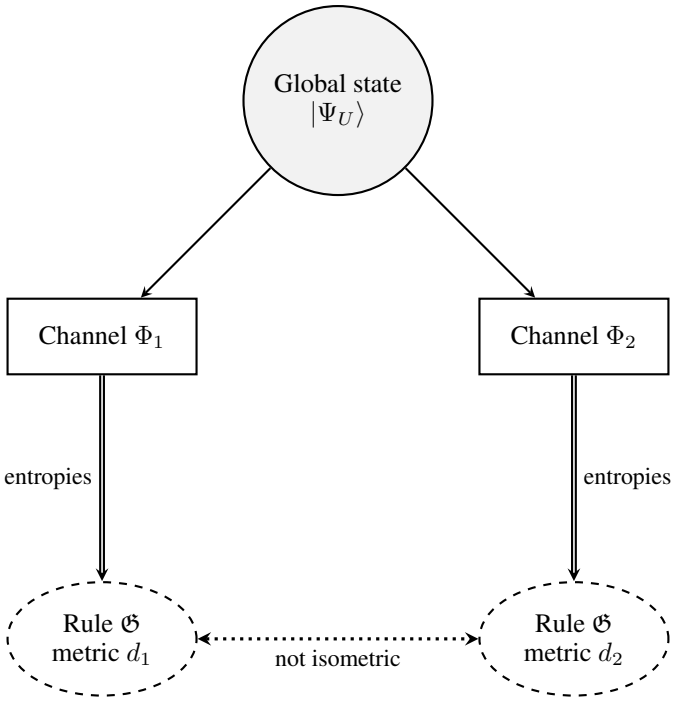


Figure 1: One global state is reduced by two channels from the same allowed family. The same reconstruction rule turns their entropy data into metrics. Section 3 decides when the entropy data differ; Proposition 4.3 gives a case where the metrics are not isometric.

5 Discussion

5.1 Relation to other frameworks

QBism treats the quantum state as an agent's degree of belief. Here a single objective global state is assumed throughout, and coarse-grainings are realized by physical interactions. The only point of contact is that no subsystem decomposition is privileged. Decoherence and many-worlds accounts explain classical behaviour and branching relative to a fixed split into system, apparatus and environment. The present framework leaves that analysis unchanged and asks where the split comes from. The closest relative is algebraic quantum field theory, where algebras are primary and states are functionals on them [9], and the closest technical precursor is the result that tensor-product structures are induced by the accessible observables [6]. What the present paper adds to that line is the channel level: a reduction has a resolution that its algebra does not record (Proposition 2.2), entropy-based readings have exact blind spots (Theorem 3.2, Example 3.4), those blind spots can be tested with finite resources (Proposition 3.8), and a fixed rule can turn a different distribution of the same noise budget into a different finite geometry (Proposition 4.3). The affine form of qubit channels, the complete-positivity conditions and the comparison of channels by post-processing are known tools [15, 16, 14]; the criterion, the branch analysis and the tests are derived here, and a full comparison with the literature on channels with equal output spectra is still to be done.

5.2 Series interfaces and the world-structure question

The protocol connects three distinct responsibilities. Paper B derives a reduction and its persistent structures from specified couplings. Its window theorem selects a block partition when within-block and between-block rates permit one; this does not make the finite channel's full range equal to that block algebra. Paper E derives the fragment's implementable effects and its public readout properties from a collision. Here their composition reaches a certified geometric difference. The source-channel algebra can stay fixed while recording permission and the attainable precision change. Imprint's cross-scale work develops the related question of which reading is worth its cost [27].

The coupling operators, factorization into sites, source preparation and reconstruction rule are still inputs. The new connection derives what happens between those inputs and a finite reader's geometric conclusion; it does not derive the architecture from an unstructured Hilbert space. Histories that alter actual coupling or record access, as studied in the subject and maintenance lines of the trilogy, can change this capability. A claim that they also generate locality or a spacetime dictionary requires its own dynamics and bridge, including the error conditions of Conjecture 4.5.

The finite models use trace entropy, whereas the infinite source description introduced above does not. Neither the SLD Fisher metric nor a shortest-path metric is identified with Lorentzian spacetime. The task is to determine which operational structures can constrain a physical geometric reading, and which geometric claims fail because a finite context cannot distinguish the data.

6 Conclusion

A physical reduction carries more information than its generated algebra. Two reductions may share the algebra and differ in resolution; they may agree on every entropy and remain operationally incomparable. For qubits these possibilities are decided by a quadratic criterion, and finite-copy tests bound its violation.

The worked protocol carries the argument through actual operations. A local unitary produces the source noise, local collisions create environmental records without changing the pair data used by the rule, and the measured records certify non-isometric shortest-path metrics. Noise allocation changes geometry at equal expected flip count. The finite sample guarantee includes imperfect readout, calibration and a common-error failure mechanism, so the conclusion is attainable by the declared readers.

The resulting dependence is physical and quantitative within the model: changing an implemented coupling changes a state, its recorded correlations and the reconstructed metric. Explaining the coupling architecture, deriving a spacetime dictionary and its continuum dynamics remain the larger world-structure problem. Conjecture 4.5 gives a mechanism and failure conditions for one part of that extension. No empirical departure from quantum theory is claimed; the protocol itself gives testable statements.

A No canonical tensor factorization

For a Hilbert space $\mathcal{H}_{\mathrm{total}}$ of dimension $d_A d_B$ with $d_A, d_B > 1$, there is no unique way to write $\mathcal{H}_{\mathrm{total}} = \mathcal{H}_A \otimes \mathcal{H}_B$ without further physical input. Given any pure state $|\Psi_U\rangle$, choose a unitary U with $U(|0\rangle_A \otimes |0\rangle_B) = |\Psi_U\rangle$ and pull the reference factors through U; in the resulting factorization $|\Psi_U\rangle$ is a product. More generally, observables can be chosen so that a given pure state has any prescribed Schmidt decomposition [8].

What survives is typicality for a family of factorizations fixed in advance.

Theorem A.1 (Entanglement typicality for a specified family)**.** *Let $\mathcal{H}_{\mathrm{total}}$ be finite dimensional and let $\mathfrak{F} = \{\mathbf{f}_1, \ldots, \mathbf{f}_m\}$ be a finite family of nontrivial tensor factorizations selected by a physical rule that does not depend on the sampled state. Write $\mathcal{H}_{\mathrm{total}} = \mathcal{H}_{A_j} \otimes \mathcal{H}_{B_j}$ with $d_{A_j} \le d_{B_j}$. For $\delta > 0$ let*

$$p_j(\delta) := \Pr_{\Psi \sim \mathrm{Haar}} \left[S(\rho_{A_j}) < \log d_{A_j} - \delta \right].$$

Then

$$\Pr\left[S(\rho_{A_j}) \ge \log d_{A_j} - \delta \text{ for every } j \right] \ge 1 - \sum_{j=1}^{m} p_j(\delta),$$

and standard concentration estimates make each $p_j(\delta)$ small at large dimension under the usual conditions on the dimension ratio.

Proof. For each fixed $\mathbf{f}_j$, Page's formula gives [10]

$$\mathbb{E}[S(\rho_{A_j})] = \sum_{k=d_{B_j}+1}^{d_{A_j}d_{B_j}} \frac{1}{k} - \frac{d_{A_j}-1}{2d_{B_j}}, \tag{11}$$

and concentration of measure bounds the lower tail [11]; this tail is $p_j(\delta)$. The union bound over the finite, preselected family proves the inequality. □

The size of the statement matters. For a balanced split $d_A = d_B$, Eq. (11) gives a mean entropy of $\log d_A - \frac{1}{2} + o(1)$, so typical states are close to maximally entangled but keep a gap of about half a nat. Drawing the factorization at random with the state fixed is equivalent to drawing the state at random with the factorization fixed. Hence for a fixed state, product factorizations form a set of measure zero, while factorizations with small but nonzero entanglement have small positive measure.

Theorem A.1 says nothing about all factorizations and does not pick out any member of $\mathfrak{F}$. Which family is admissible, and why, is a question of quantum mereology [12, 13]. Earlier work showed that subsystem structure is induced by the accessible observables rather than given by the Hilbert space [5, 6]. The channel-level results in the main text retain the accessible-observable view while distinguishing its algebra from its resolution.

B Other connectivity constructions

When a metric on a parameter space of coarse-grained states is wanted, the finite-dimensional construction is the quantum Fisher metric. For a smooth faithful family $\lambda \mapsto \rho_{\mathbf{c}}(\lambda)$, define the symmetric logarithmic derivatives by $\partial_a \rho_{\mathbf{c}} = \frac{1}{2}(L_a \rho_{\mathbf{c}} + \rho_{\mathbf{c}} L_a)$ and

$$g^{\mathrm{QF}}_{ab}(\lambda) := \frac{1}{2}\,\mathrm{Tr}[\rho_{\mathbf{c}}(\lambda)\{L_a, L_b\}]\,. \tag{12}$$

This is a positive-semidefinite Riemannian metric on identifiable parameter directions [21]. It is not a Lorentzian spacetime metric, and the pairwise numbers $\mu_{\mathbf{c}}(A, B)$ do not determine it without an additional map from graph or state parameters to geometry.

B.1 Spatial and eigenmode descriptions of a free-fermion chain

Example B.1 (A dimerized free-fermion chain)**.** *Take a periodic chain of $2L$ sites with alternating hopping amplitudes $t_1 = 1$ and $t_2 = 1/2$,*

$$H = -\sum_{c=0}^{L-1} \big(t_1\, c^\dagger_{2c} c_{2c+1} + t_2\, c^\dagger_{2c+1} c_{2c+2} + \mathrm{h.c.}\big),$$

at half filling. The single-particle spectrum has two bands separated by the gap $2|t_1 - t_2| = 1$, and the ground state fills the lower band, $|\Psi_0\rangle = \prod_k a^\dagger_{k,-}|0\rangle$, where $a_{k,-}$ are the lower-band Bloch modes. In the description by occupation numbers of the band-resolved Bloch modes the ground state is a product of filled and empty modes: every set of modes has a pure reduced state, disjoint sets have zero mutual information, and the connectivity graph is empty. In the spatial description blocks of sites are correlated. For $L = 60$ and blocks of four sites, the block entropy saturates at $.3554$ as the block grows, as the area law for gapped one-dimensional systems requires [24, 23]. The mutual information between two blocks falls from $.333$ for adjacent blocks to 9.0×10^{-5} at a separation of ten sites and 5.2×10^{-8} at twenty, close to $e^{-r/1.32}$. The connectivity graph is dominated by edges between neighbouring blocks.

The second description has to be chosen with care. Plain Fourier modes are not enough when the Hamiltonian is not translation invariant: for two disconnected dimers on four sites, $H = -(c^\dagger_0 c_1 + c^\dagger_2 c_3 + \mathrm{h.c.})$ at half filling, the gap at fixed particle number is 2, yet the Fourier modes $k = \pi/2$ and $3\pi/2$ share one particle in superposition and have mutual information $2\ln 2$. A translation-invariant single-band chain with particle conservation is gapless unless its band is full or empty, so a gapped example whose ground state is a product of mode occupations needs band-resolved modes, as above, or in general the single-particle eigenmodes of the Hamiltonian.

The example shows that connectivity depends on the factorization: the spatial description has short-range edges and the eigenmode description has none. It does not exhibit two reconstructed geometries, since its zero-connectivity graph supplies no finite paths under the chosen rule; Proposition 4.3 is the example in which both descriptions produce geometries.

C The nine-probe constant

On the unit sphere, $\sum_k q(r_k)^2$ and Q are quadratic forms in the nine parameters (A_0, b, d), and both are positive definite; the first because the nine probe values determine the parameters, as in the proof of Corollary 3.6. Their generalized eigenvalues do not depend on the basis chosen for the parameters, and a direct computation gives the characteristic polynomial, up to a constant factor,

$$(\lambda - 6)\,(4\lambda^2 - 75\lambda + 90)\,(8\lambda^3 - 225\lambda^2 + 1620\lambda - 2700)^2.$$

Its roots are 1.2886, 2.3962, 6, 7.9001, 17.4614 and 17.8287, the three roots of the cubic each counted twice. The smallest is the smaller root of the quadratic factor, κ_9, and it is attained.

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COUPLING, RECORDS, AND STABLE RELATIONS: FORMATION, PROTECTED CONTENT AND THE LIFETIME OF A GEOMETRY

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ABSTRACT

A physical coupling determines which distinctions decay, which can be recorded and which remain available for action. For finite systems with commuting Hermitian noise operators, we derive the fixed block algebra and its long-time pinching, then prove a finite-window selection theorem: a separated set of decoherence rates selects at most one block partition, and its algebra contains every stable block algebra in the supplied joint eigenbasis. Finite-time channels still have full operator range. Block-constant recording couplings report only sector labels while protecting the quantum content within each sector; leakage through independent fragments costs coherence multiplicatively, giving a sharp $R^{-1/2}$ symmetry threshold for R records. Geometry read from mutual information receives both sector and protected-content contributions, which we separate by an exact classical–quantum chain identity. Neither contribution is confined to the noise hypergraph: disconnected decoherence-free sectors can preserve an initially shared logical Bell pair. We therefore distinguish preservation from formation. A collective recording interaction creates a flagged protected Bell branch from a product input, with every branch and the nonlocal recording cost retained; a physical Hamiltonian link creates a deterministic Bell pair between two noise-disconnected protected sectors. For the GHZ state, collective and local couplings yield a triangle, one edge or no edge under one reconstruction rule, with an explicit lifetime under symmetry breaking. Beyond commuting noise, complete finite-time record laws and risks vary continuously even when the exact fixed algebra jumps. These results identify coupling-selected storage, record-conditioned formation and finite operational capability without equating them. The ambient carrier, available interactions and reconstruction dictionary remain physical inputs.

1 Introduction

A stable distinction is a physical achievement. It must survive the interactions that would erase it, be readable through an available instrument, and last long enough to matter. An algebra of invariant observables describes one part of that achievement; it does not specify a reader, a preparation history or a set of available gates.

Paper A [1] shows why the distinction matters for geometry. Channels generating the same observable algebra can resolve different states and produce non-isometric finite correlation geometries. Its local bit-flip family has a Stinespring realization and a finite fragment-readout certificate. Here we ask a complementary question: what structure a coupling protects, how actual records and interactions form usable relations within that structure, and when those relations lose their support.

Commuting Hermitian noise gives an exact starting point. The fixed observables are the commutant of the coupling operators, familiar from einselection and noiseless codes [9, 10, 12, 13]. The infinite-time channel pinches onto this algebra. At finite times its operator range is full, so a protected block algebra is additional stability data rather than the

range of the actual channel. A window and a tolerance give a unique block partition when intra-block and inter-block rates are separated (Theorem 3.2). The maximality statement concerns block algebras in the specified joint eigenbasis, not every rotated algebra or every useful finite task.

Recording adds a different restriction. A coupling constant on each block can record its label without learning its quantum content. Independent fragments that resolve two states within a block destroy their coherence according to a multiplicative visibility budget (Theorem 4.2). The centre and the blocks thus perform different jobs within one stable structure. They are not disjoint sources of geometry: classical sector weights can sustain mutual information even when every block is one-dimensional. Section 5 gives the exact decomposition needed to distinguish the contributions.

Selection alone also says too little about formation. Two disconnected collective-noise components can preserve a logical Bell pair shared at the start. Conversely, local independent processes cannot create cross-component correlations from independent inputs. Section 6 supplies the missing process distinction and two explicit formation mechanisms. One retains a collective measurement flag and all its branches; the other uses an actual cross-component Hamiltonian link. Both generate relations from product inputs, and both expose the interaction resources responsible for them. Foundation III [6] supplies a further mechanism in which acquired records select an encoder and decoder for the same pre-existing quantum content. Storage, formation and learned use then have separate, testable meanings.

The GHZ calculation in Section 7 shows how a coupling changes a finite correlation geometry and how weak symmetry breaking ends its plateau. For noncommuting perturbations, Section 3.1 bounds complete adaptive record laws and task risks without requiring a stable algebra. A transverse qubit exhibits a diverging record lifetime despite a scalar exact fixed algebra.

The supplied Hilbert space, coupling permissions and site factorization are retained throughout. Consciousness I [4] selects a Virasoro fixed net inside a given current-net representation; Foundation's hulls concern closure under declared operations. Neither result derives the ambient carrier. Our formation results address how relations and usable protected sectors arise on that carrier. The stronger question of effective subsystem formation returns in Section 9, with a process-sensitive conjecture rather than a noise-only hypergraph rule.

2 Reductions selected by a coupling

2.1 Setting

Let $\mathcal{H}$ be a finite-dimensional system Hilbert space. The environment couples to the system through Hermitian operators $A_1, \dots, A_K$ with $[A_k, A_l] = 0$, at rates $\gamma_k > 0$. In the Markovian limit the reduced dynamics is generated by

$$\mathcal{L}(\rho) = \sum_{k=1}^{K} \gamma_k \Big(A_k \rho A_k - \tfrac{1}{2}\{A_k^2, \rho\} \Big), \qquad \Phi_t = e^{t\mathcal{L}}. \tag{1}$$

A system Hamiltonian that commutes with every A_k only rotates states inside the blocks defined below; we work in the corresponding interaction picture and return to Hamiltonians that do not commute in Remark 2.2. Section 4 treats the same couplings in an exact unitary model with environmental fragments.

Choose an orthonormal basis $\{|x\rangle : x \in \mathcal{X}\}$ of joint eigenvectors, $A_k|x\rangle = a_k(x)|x\rangle$, and put

$$D(x,y) = \tfrac{1}{2} \sum_k \gamma_k \big(a_k(x) - a_k(y)\big)^2 = \|u(x) - u(y)\|^2, \qquad u(x) = \big(\sqrt{\gamma_k/2}\, a_k(x)\big)_{k=1}^{K} \in \mathbb{R}^K. \tag{2}$$

We call $u(\mathcal{X}) \subset \mathbb{R}^K$ the rate image of the basis; decoherence rates are squared Euclidean distances in it. Write $x \approx y$ when $a_k(x) = a_k(y)$ for every k, let π_∞ be the partition of $\mathcal{X}$ into $\approx$-classes a, and let P_a be the projection onto $\mathcal{H}_a = \operatorname{span}\{|x\rangle : x \in a\}$. For any partition π of $\mathcal{X}$ into classes C, put

$$\mathfrak{B}_\pi = \bigoplus_{C \in \pi} \mathcal{B}(\mathcal{H}_C), \qquad \mathbb{E}_\pi(\rho) = \sum_{C \in \pi} P_C \rho P_C.$$

$\mathfrak{B}_\pi$ is a unital $*$-algebra, $\mathbb{E}_\pi$ is a channel (the pinching), and $\mathbb{E}_\pi$ is its own dual, the conditional expectation onto $\mathfrak{B}_\pi$.

Proposition 2.1 (The reduction selected by a commuting coupling)**.** *1. Φ_t acts as a Schur multiplier in the eigenbasis: $\Phi_t(\rho)_{xy} = e^{-tD(x,y)}\rho_{xy}$, and $\Phi_t^* = \Phi_t$.*

2. *The observables fixed by every Φ_t^* form the algebra $\mathfrak{B}_{\pi_\infty} = \{A_1, \dots, A_K\}'$.*

3. *$\Phi_t \to \mathbb{E}_{\pi_\infty}$ as $t \to \infty$, and the algebra generated by the dual range of $\mathbb{E}_{\pi_\infty}$ is $\mathfrak{B}_{\pi_\infty}$. In the terminology of Paper A, $(\mathfrak{B}_{\pi_\infty}, \mathbb{E}_{\pi_\infty})$ is a minimally compatible coarse-graining structure.*

4. *For every finite t, Φ_t is invertible on the operator space, so the algebra generated by its dual range is $\mathcal{B}(\mathcal{H})$.*

Proof. (1) In the eigenbasis, $(A\rho A - \frac{1}{2}\{A^2,\rho\})_{xy} = (a(x)a(y) - \frac{1}{2}a(x)^2 - \frac{1}{2}a(y)^2)\rho_{xy} = -\frac{1}{2}(a(x)-a(y))^2\rho_{xy}$, so $\mathcal{L}$ multiplies the (x,y) entry by $-D(x,y)$. The multiplier is real and symmetric, hence $\Phi_t^* = \Phi_t$. (2) X is fixed for all t exactly when $X_{xy} = 0$ whenever $D(x,y) > 0$, that is, whenever $x \not\approx y$. An operator commutes with every diagonal A_k exactly when it has no entries between basis vectors with different eigenvalue tuples. (3) Entries with $D(x,y) > 0$ tend to zero and the others are constant. The dual range of $\mathbb{E}_{\pi_\infty}$ is $\mathfrak{B}_{\pi_\infty}$, which is already an algebra. (4) All multipliers $e^{-tD(x,y)}$ are nonzero. □

Part (2) is the elementary commuting case of the known structure of fixed points of quantum dynamical semigroups [14, 15]. For this series the consequence matters more than the statement: the given coupling determines both its channel and its invariant algebra. Part (4) is Paper A's result that an algebra label does not record resolution, now in dynamical form. Before the long-time limit the channel generates everything, and the structure that a coupling makes accessible has to be identified by how well observables survive over time.

The objects of Paper E [3] can be located in this class. Its environmental fixed set $\mathrm{Fix}(\mathcal{E})$ is $\mathfrak{B}_{\pi_\infty}$. In Paper E's collision model, with a qubit pointer, the pointer algebra equals the fixed algebra of the induced dephasing and contains every effect a fragment can report. The centre of $\mathfrak{B}_{\pi_\infty}$ is generated by the block projections P_a; it is commutative, and Section 4 shows that it is what the environment records, so it serves as Paper E's pointer or record algebra. The inclusion of the record algebra in the fixed set is thus a conclusion of the model, as Paper E requires of any such overlap. The fixed set is strictly larger than its centre as soon as some block has dimension greater than one; a nontrivial two-sector qubit pointer has only one-dimensional blocks. Such blocks are decoherence-free subspaces [12, 13]. They retain quantum content; sector weights can retain classical correlations as well.

Remark 2.2 (Hamiltonians that do not commute with the couplings)**.** *If the system Hamiltonian H has a part H_{off} connecting different blocks, $\mathfrak{B}_{\pi_\infty}$ need not remain exactly invariant. Strong dephasing can then confine the dynamics to the blocks, with leakage on a time scale of order $\Gamma_{\mathrm{out}}/\|H_{\mathrm{off}}\|^2$, where Γ_{out} is the smallest between-block rate; this is the mechanism of quantum Zeno subspaces [16]. Section 3.1 gives an exact two-level lifetime and a finite-time stability bound without a commutativity assumption. Paper D's chain with a transverse field requires its own many-body estimate.*

3 Selection over a finite window

Definition 3.1 (Stable algebra on a window)**.** *A unital $*$-subalgebra $\mathfrak{B} \subseteq \mathcal{B}(\mathcal{H})$ is ε-stable on $[0,T]$ if*

$$\sup_{0\le t\le T}\ \sup_{X\in\mathfrak{B},\ \|X\|\le 1} \|\Phi_t^*(X) - X\| \le \varepsilon.$$

The diagnostic tests the whole unit ball of an algebra, rather than selected states or a list of slow observables. For a partition π define the width of a class C in the direction k, $w_k(C) = \max_{x\in C} a_k(x) - \min_{x\in C} a_k(x)$, and the two rates

$$\Gamma_{\mathrm{in}}(\pi) = \max_{C\in\pi}\ \tfrac{1}{2}\sum_k \gamma_k w_k(C)^2, \qquad \Gamma_{\mathrm{out}}(\pi) = \min\{D(x,y):\ x,y \text{ in different classes of } \pi\}. \tag{3}$$

The minimum over an empty set is $+\infty$, so a single-block partition has $\Gamma_{\mathrm{out}} = +\infty$. If every coupling is scalar, the full algebra is fixed and selected for every window. In the rate image, $\Gamma_{\mathrm{in}}(\pi)$ is the largest squared diagonal of the axis-parallel boxes enclosing the classes, and $\Gamma_{\mathrm{out}}(\pi)$ is the smallest squared distance between points of different classes.

Theorem 3.2 (Selection by a window)**.** *Let $0 < \varepsilon < 1$ and $T > 0$.*

1. *For every $X \in \mathfrak{B}_\pi$ and $t \ge 0$, $\|\Phi_t^*(X) - X\| \le t\,\Gamma_{\mathrm{in}}(\pi)\|X\|$. For every state $\rho \in \mathfrak{B}_\pi$, $\|\Phi_t(\rho) - \rho\|_1 \le t\,\Gamma_{\mathrm{in}}(\pi)$. In particular $\mathfrak{B}_\pi$ is ε-stable on $[0,T]$ when $T\Gamma_{\mathrm{in}}(\pi) \le \varepsilon$.*
2. *If $1 - e^{-T\Gamma_{\mathrm{out}}(\pi)} > \varepsilon$, no unital $*$-algebra that contains $\mathfrak{B}_\pi$ properly is ε-stable on $[0,T]$.*
3. *Call π selected by (T,ε) if*

$$T\,\Gamma_{\mathrm{in}}(\pi) \le \varepsilon < 1 - e^{-T\Gamma_{\mathrm{out}}(\pi)}. \tag{4}$$

At most one partition is selected by (T,ε). If π is selected by (T,ε) and π' by (T',ε) with $T \le T'$, then π' refines π, so $\mathfrak{B}_{\pi'} \subseteq \mathfrak{B}_\pi$.

4. *For $T > -\ln(1-\varepsilon)/\Gamma_{\mathrm{out}}(\pi_\infty)$ the selected partition exists and is π_∞, so the selected algebra is the fixed algebra of Proposition 2.1.*

Proof. (1) $\mathfrak{B}_\pi$ is invariant under $\mathcal{L}$, and on it $\mathcal{L}(X) = -\frac{1}{2}\sum_k \gamma_k [A_k,[A_k,X]]$. On a class C, replacing A_k by $A_k - m_k$ with m_k the midpoint of its eigenvalues on C does not change the commutator and gives an operator of norm $w_k(C)/2$; hence $\|[A_k,X_C]\| \le w_k(C)\|X_C\|$ and $\|[A_k,[A_k,X_C]]\| \le w_k(C)^2\|X_C\|$. Since $\|X\| = \max_C \|X_C\|$, $\|\mathcal{L}(X)\| \le \Gamma_{\mathrm{in}}(\pi)\|X\|$. Then $\Phi_t^*(X) - X = \int_0^t \Phi_s^*(\mathcal{L}(X))\,ds$ and Φ_s^* is a contraction in operator norm. For a state $\rho \in \mathfrak{B}_\pi$ and $\|Y\| \le 1$, $\mathrm{Tr}[(\Phi_t(\rho)-\rho)Y] = \mathrm{Tr}[\rho(\Phi_t^* - \mathrm{id})(\mathbb{E}_\pi Y)]$, because $\rho = \mathbb{E}_\pi \rho$, $\mathbb{E}_\pi$ is self-dual and commutes with the Schur multiplier Φ_t^*; since $\mathbb{E}_\pi Y \in \mathfrak{B}_\pi$ and $\|\mathbb{E}_\pi Y\| \le 1$, the first statement bounds this by $t\Gamma_{\mathrm{in}}(\pi)$.

(2) Let $\mathfrak{B} \supsetneq \mathfrak{B}_\pi$ and $X \in \mathfrak{B} \setminus \mathfrak{B}_\pi$. Some entry $X_{x'y'}$ with $x' \in C$, $y' \in C'$, $C \ne C'$, is nonzero. For any $x \in C$, $y \in C'$ the matrix units $E_{xx'}$ and $E_{y'y}$ lie in $\mathfrak{B}_\pi \subseteq \mathfrak{B}$, so $E_{xx'} X E_{y'y} = X_{x'y'} E_{xy} \in \mathfrak{B}$. Then $\|\Phi_T^*(E_{xy}) - E_{xy}\| = 1 - e^{-TD(x,y)} \ge 1 - e^{-T\Gamma_{\mathrm{out}}(\pi)} > \varepsilon$.

(3) Let π and π' be selected by (T,ε) and (T',ε), $T \le T'$. If x, y lie in one class of π', then $D(x,y) \le \Gamma_{\mathrm{in}}(\pi') \le \varepsilon/T' \le \varepsilon/T$. If x, y lay in different classes of π, then $D(x,y) \ge \Gamma_{\mathrm{out}}(\pi) > -\ln(1-\varepsilon)/T \ge \varepsilon/T$. Hence every class of π' lies in a class of π. For $T = T'$ the same argument with the roles exchanged gives $\pi = \pi'$.

(4) $\Gamma_{\mathrm{in}}(\pi_\infty) = 0$ and $\Gamma_{\mathrm{out}}(\pi_\infty) > 0$, so π_∞ satisfies (4) for such T; uniqueness is part (3). □

A window and a tolerance therefore pick out at most one algebra of the form $\mathfrak{B}_\pi$, and they pick one out when the rate image of the basis splits into clusters that are small compared with ε/T and far apart compared with $-\ln(1-\varepsilon)/T$. This is a separation of time scales, the mechanism behind metastability in open quantum systems [17]; for commuting couplings it can be stated exactly. The selected algebra is moreover the largest ε-stable algebra of block form: if $\mathfrak{B}_\sigma$ is ε-stable on $[0,T]$, every matrix unit E_{xy} with x, y in one class of σ satisfies $1 - e^{-TD(x,y)} \le \varepsilon$, so x and y lie in one class of the selected partition π, and $\mathfrak{B}_\sigma \subseteq \mathfrak{B}_\pi$. As the window lengthens, the selected algebra shrinks in discrete steps located at the gaps of the rate spectrum and ends at the fixed algebra. In the order of Paper A this is a chain of further coarse-grainings, since $\mathbb{E}_{\pi'} = \mathbb{E}_{\pi'} \circ \mathbb{E}_\pi$ when π' refines π.

Without a gap there is no selected algebra at that window. Selection uses the sufficient bound of part (1), so between that bound and the necessary bound of part (2) a unique maximal stable block algebra can still exist; but the window may also fail to determine one, and two incomparable maximal stable block algebras can occur. The set of ε-stable matrix units can then be an operator system whose generated algebra contains unstable elements: if $D(x,y)$ and $D(y,z)$ are small, $D(x,z)$ can still be as large as $(\sqrt{D(x,y)} + \sqrt{D(y,z)})^2$, and a chain of small steps can connect distant points of the rate image. Theorem 3.2 is a sufficient selection certificate. Failure of its gap condition does not by itself decide existence or uniqueness of a maximal stable algebra. Nor does stability of a block algebra supply the instruments needed to use every element as a gate.

3.1 Continuous capabilities when the fixed algebra jumps

A weak coupling can change the exact commutant discontinuously. It cannot change every finite experiment discontinuously. The distinction matters when a selected algebra is used to describe a finite observer.

Proposition 3.3 (Finite-time operational stability)**.** *Let $\mathcal{L}_0, \mathcal{L}_1$ be finite-dimensional Lindblad generators, and put $v = \|\mathcal{L}_1 - \mathcal{L}_0\|_\diamond$. Then*

$$\|e^{t\mathcal{L}_1} - e^{t\mathcal{L}_0}\|_\diamond \le \min\{2, tv\}. \tag{5}$$

Consider any experiment with the same initial states, retained ancillas and permitted instruments in both models. Between its instruments the system evolves for prescribed intervals $t_1, \dots, t_m$, with total $T = \sum_i t_i$. Instruments may depend on earlier outcomes. The total variation distance between its complete record laws is at most $\min\{1, Tv/2\}$. Consequently, for a common decision problem with loss in $[0, L]$, the two optimal risks differ by at most $L\min\{1, Tv/2\}$. In particular, a Hamiltonian perturbation $-i\epsilon[K,\cdot]$ gives $v \le 2|\epsilon|\|K\|$.

Proof. Duhamel's formula writes the channel difference as $\int_0^t e^{(t-s)\mathcal{L}_1}(\mathcal{L}_1 - \mathcal{L}_0)e^{s\mathcal{L}_0}\,ds$. CPTP maps have diamond norm one, giving tv; two channels differ by at most two. A hybrid argument replaces the evolution intervals one at a time, keeping outcome-controlled instruments as CPTP maps on the system and classical record. Contractivity and the triangle inequality give a trace-norm difference at most Tv. The bounded-loss estimate holds uniformly over each common policy, so it also holds for the infima. Finally, $\|[K,A]\|_1 \le 2\|K\|\|A\|_1$ holds with every retained ancilla. □

The estimate needs neither a gap nor a selected algebra. It bounds distinguishing, storing and acting through the permitted experiment, not just the state of an isolated system. The available instruments and the total time remain

the same on both sides. A discontinuous change of permissions, or an unbounded observation time, is a different comparison.

For an exact noncommuting example, take

$$\mathcal{L}_\epsilon(\rho) = -i\epsilon[X,\rho] + \kappa(Z\rho Z - \rho), \qquad \kappa > 0. \tag{6}$$

The fixed algebra is $\mathrm{span}\{I, Z\}$ at $\epsilon = 0$ and only $\mathbb{C}I$ at every $\epsilon \neq 0$. To see the latter, the Hilbert–Schmidt dissipative form of a fixed observable forces $[Z, O] = 0$; the remaining Hamiltonian equation then forces $[X, O] = 0$. Nevertheless, the two initially perfect records $(I \pm Z)/2$ remain readable for a long time. If $0 < 2|\epsilon| < \kappa$, put $\omega = \sqrt{\kappa^2 - 4\epsilon^2}$. Their Bloch vectors at time t are $\pm(0, y(t), z(t))$, where

$$z(t) = e^{-\kappa t}\left(\cosh\omega t + \frac{\kappa}{\omega}\sinh\omega t\right), \qquad y(t) = -\frac{2\epsilon}{\omega}e^{-\kappa t}\sinh\omega t. \tag{7}$$

These expressions solve $\dot{y} = -2\kappa y - 2\epsilon z$, $\dot{z} = 2\epsilon y$, with $(y, z) = (0, 1)$ initially. Reading only Z has error $(1 - z(t))/2$; allowing the optimal system measurement gives error $(1 - \sqrt{y(t)^2 + z(t)^2})/2$. The slow decay rate is exactly

$$r = \kappa - \sqrt{\kappa^2 - 4\epsilon^2} = \frac{2\epsilon^2}{\kappa} + O(\epsilon^4/\kappa^3). \tag{8}$$

Thus arbitrarily weak transverse coupling erases the infinite-time record, while its finite-time lifetime diverges as $\kappa/(2\epsilon^2)$. Exact hulls and fixed algebras describe the ideal endpoint; finite-time risks describe what remains available on the way to it. This is also the distinction needed by Foundation's hulls and the maintenance tasks of T-DOME and Brain Anchor. Neither the general stability estimate nor this example asserts that every slow operator space is an algebra or a physical projection.

4 Records live on the centre

4.1 A recording model

To see what the environment learns we replace the Markovian generator by an exact unitary model. Environmental fragments $E_1, \ldots, E_K$ start in states ω_k, and the interaction is

$$U = \sum_{x\in\mathcal{X}} |x\rangle\langle x| \otimes V_1(x) \otimes \cdots \otimes V_K(x), \tag{9}$$

with unitaries $V_k(x)$ that depend on the basis state through the coupling, for example $V_k(x) = \exp[-i\theta_k a_k(x) G_k]$ for a fragment generator G_k. The joint state after the interaction is

$$\sigma = \sum_{x,y} \rho_{xy}|x\rangle\langle y| \otimes \bigotimes_k V_k(x)\,\omega_k\,V_k(y)^\dagger,$$

so the system undergoes the Schur multiplier

$$\rho_{xy} \mapsto c(x,y)\,\rho_{xy}, \qquad c(x,y) = \prod_k c_k(x,y), \qquad c_k(x,y) = \mathrm{Tr}\big[\omega_k V_k(y)^\dagger V_k(x)\big]. \tag{10}$$

The generator (1) arises from such models in weak-coupling and repeated-collision limits. Here the collision statements are exact for the specified unitary and an initial product $\rho \otimes \bigotimes_k \omega_k$. Fresh independent fragments are an input: recollisions or shared initial fragment correlations do not obey the product formula automatically.

Proposition 4.1 (Records carry only the block label)**.** *Let π be a partition of $\mathcal{X}$ such that every $V_k(x)$ depends on x only through its class $a(x) \in \pi$. Then:*

1. *$c(x, y) = 1$ whenever x and y lie in one class, so $\mathbb{E}_\pi(\rho)$, and with it the state on every block, is unchanged.*
2. *For every $O \in \mathfrak{B}_\pi$ and every operator F on any collection $\mathcal{F}$ of fragments,*

$$\mathrm{Tr}[\sigma(O \otimes F)] = \sum_{a\in\pi} \mathrm{Tr}[P_a \rho P_a\, O]\ \ \mathrm{Tr}[\omega_{\mathcal{F}}(a) F], \qquad \omega_{\mathcal{F}}(a) = \bigotimes_{k\in\mathcal{F}} V_k(a)\,\omega_k\,V_k(a)^\dagger.$$

The state of the fragments is $\sum_a p(a)\,\omega_{\mathcal{F}}(a)$ with $p(a) = \mathrm{Tr}\,P_a\rho$; it depends on ρ only through the distribution of the label, and given the label, block observables and fragments are uncorrelated.

Proof. (1) $V_k(x) = V_k(y)$ when x, y lie in one class, so $c_k(x, y) = 1$. (2) $O_{yx} \neq 0$ only when x and y lie in one class a, and then the fragment factor is $\omega_{\mathcal{F}}(a)$ for both. Summing $\rho_{xy}O_{yx}$ over $x, y \in a$ gives $\mathrm{Tr}[P_a \rho P_a O]$. □

Whatever the environment learns under such couplings is the block label, which generates the centre of $\mathfrak{B}_\pi$. Many fragments can hold readable copies of that label while leaving the blocks alone. Block dependence alone does not guarantee distinguishability: the conditional states $\omega_k(a)$ must also be different, and a permitted measurement must resolve them. Records and the correlations inside the blocks are thus carried by one selected algebra, the first by its centre and the second by its blocks. That the redundantly available information is classical, and is information about one measurement, holds far more generally [18]; a two-qubit system with a decoherence-free subspace that proliferates part of its information while the remaining subspace keeps non-classical features was studied in [19]. What the commuting class adds is the exact location of both kinds of information in one algebra and, next, the price of imperfect symmetry.

4.2 Leakage costs coherence, and the costs multiply

Theorem 4.2 (Leakage and coherence)**.** *For basis states x, y and a fragment k, let $D_k(x, y) = \frac{1}{2}\big\|V_k(x)\omega_k V_k(x)^\dagger - V_k(y)\omega_k V_k(y)^\dagger\big\|_1$ be the fragment's distinguishability of x and y, so that an optimal measurement on the fragment identifies one of the two equally likely states with probability $\frac{1}{2}(1 + D_k)$ [20]. Then*

$$|c_k(x, y)|^2 + D_k(x, y)^2 \leq 1, \tag{11}$$

with equality when ω_k is pure. Consequently

$$|c(x, y)| \leq \prod_k \big(1 - D_k(x, y)^2\big)^{1/2} \leq \exp\Big(-\tfrac{1}{2}\sum_k D_k(x, y)^2\Big), \tag{12}$$

and if the coherence between x and y must stay above $1 - \varepsilon$, then

$$\sum_k D_k(x, y)^2 \leq 2 \ln \frac{1}{1 - \varepsilon}. \tag{13}$$

Proof. Purify $\omega_k = \mathrm{Tr}_R\, |\Omega\rangle\langle\Omega|$. Then $c_k(x, y) = \langle\Omega|(V_k(y)^\dagger V_k(x) \otimes I)|\Omega\rangle$ is the overlap of the pure states $(V_k(x) \otimes I)|\Omega\rangle$ and $(V_k(y) \otimes I)|\Omega\rangle$, whose trace distance is $(1 - |c_k|^2)^{1/2}$. Partial trace over R does not increase trace distance, which gives (11), with equality when there is nothing to trace out. Inequality (12) follows from (10) and $\ln(1 - u) \leq -u$, and (13) by taking logarithms. □

Inequality (11) is Englert's relation between visibility and which-way distinguishability [21], applied to one fragment. It bounds what a fragment learns by the coherence it destroys, not the reverse: a fragment in a mixed state can destroy coherence while learning nothing, as in Paper E's maximally mixed memory example. Purity is one useful preparation resource; mixed fragments can also record when their conditional states differ. For many fragments it becomes the budget (13): the sum of their squared pairwise distinguishabilities is at most $2\ln(1/(1 - \varepsilon))$ for a protected pair of basis states. This is not a bound on their total Shannon information about an arbitrary block.

Corollary 4.3 (Redundancy against symmetry)**.** *Suppose that R fragments each distinguish two states x, y of one block with $D_k(x, y) \geq \beta$. Then $|c(x, y)| \leq (1 - \beta^2)^{R/2}$, and keeping the coherence above $1 - \varepsilon$ requires*

$$\beta \leq \Big(\frac{2\ln(1/(1 - \varepsilon))}{R}\Big)^{1/2}. \tag{14}$$

In a sequence of models with $R \to \infty$ and $D_k(x, y) = \beta_R = \lambda R^{-\alpha}$, $\alpha > 0$, for the R fragments that couple to the block, the coherence tends to zero if $\alpha < \frac{1}{2}$. If the fragment states are pure and no other fragment distinguishes x from y, it tends to 1 for $\alpha > \frac{1}{2}$ and to $e^{-\lambda^2/2}$ for $\alpha = \frac{1}{2}$.

Proof. The bound and (14) are (12) and (13). For pure fragments $|c(x, y)| = (1 - \beta_R^2)^{R/2}$ by the equality case, and $\frac{R}{2}\ln(1 - \lambda^2 R^{-2\alpha}) \to 0,\ -\lambda^2/2,\ -\infty$ for $\alpha > \frac{1}{2}, = \frac{1}{2}, < \frac{1}{2}$. □

A coupling of the form (9) is block-constant only by symmetry: the fragments must respond identically to all states of a block. Real couplings break such a symmetry slightly, and each recording fragment then leaks a little. Corollary 4.3 says that objectivity by redundancy and protection of the blocks can coexist, but only if the symmetry error per fragment falls as $R^{-1/2}$. In the limit of many records this is a sharp threshold in the exponent α. It is the closest analogue of a transition in this class. It is a statement about how a symmetry error scales with the number of records, and it is not a phase transition in the thermodynamic sense.

5 Sector records and protected correlations

A mutual-information geometry does not require entanglement. Even local dephasing preserves $\rho = \frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11|$, for which $I(1:2) = \ln 2$. With local Z couplings every selected block is one-dimensional and every sector state is a product. Its entire edge comes from sector weights.

For a selected state $\bar{\rho} = \mathbb{E}_\pi(\rho) = \sum_k p_k \rho_k$, with $\rho_k = P_k \rho P_k / p_k$, retain the label in a classical register:

$$\Omega_{SK} = \sum_k p_k \rho_k \otimes |k\rangle\langle k|_K. \tag{15}$$

This extension is physically realized by a perfectly resolving block-constant recorder followed by measurement or dephasing of its label register. It is not the coherent source–memory state before that readout. If the actual fragment states overlap, K is still a useful mathematical label, but it is not a freely readable record. Zero-weight sectors are omitted.

Proposition 5.1 (The two contributions to correlation)**.** *For any supplied disjoint subsystems A, B of S,*

$$I(A:B)_{\bar{\rho}} = \sum_k p_k I(A:B)_{\rho_k} + I(A:K)_\Omega + I(B:K)_\Omega - I(AB:K)_\Omega. \tag{16}$$

The last three terms need not have a nonnegative sum.

Proof. For $X = A, B, AB$, the classical–quantum entropy formula gives $I(X:K) = S(\bar{\rho}_X) - \sum_k p_k S((\rho_k)_X)$. Substitution in $S(A) + S(B) - S(AB)$ proves the identity. □

The first term locates correlations protected within sectors; it can include classical as well as quantum correlations. The correction describes what mixing sector-conditioned states does to those correlations. In the locally dephased example above it equals $\ln 2$. In the next recording example it is negative. Thus neither a noncommutative block nor a classical centre alone determines the geometry. The sector states, their weights, the site marginals and the dictionary all enter.

6 How protected relations are formed

A noise commutant specifies what can be stored. It does not determine whether that content was inherited or created. The distinction already invalidates a noise-only mereology rule.

6.1 Disconnected noise can preserve a shared Bell pair

Take four qubits with only $A_{12} = Z_1 + Z_2$ and $A_{34} = Z_3 + Z_4$. Each pair has a zero-magnetization sector with logical basis $|0_L\rangle = |01\rangle$, $|1_L\rangle = |10\rangle$. The state

$$|\Psi_L\rangle = \frac{|0_L\rangle_{12}|0_L\rangle_{34} + |1_L\rangle_{12}|1_L\rangle_{34}}{\sqrt{2}} \tag{17}$$

is annihilated by both noise operators and is fixed exactly by (1). Yet $I(12:34) = 2\ln 2$. The two noise-hypergraph components are disconnected. Applying a Hadamard to each qubit gives the same counterexample for the X couplings of Section 7. Classical and quantum correlations can both cross disconnected noise components when they are present initially.

6.2 A collective recorder forms a flagged protected relation

Start two qubits in the physical product $|+_X\rangle|+_X\rangle$ and a fresh three-level memory in $|0\rangle$. Let $M = Z_1 + Z_2$ and let P_+, P_0, P_- project onto its eigenvalues $2, 0, -2$. For the shift $S_3|k\rangle = |k+1 \bmod 3\rangle$, use

$$U_{\mathrm{rec}} = P_+ \otimes I + P_0 \otimes S_3 + P_- \otimes S_3^2. \tag{18}$$

This is a block-constant recording interaction, not a product of independent local readers. It has a finite Hamiltonian implementation: choose a Hermitian logarithm G with $e^{-iG} = S_3$ and $\|G\| = 2\pi/3$, put $a(M) = (2I - M)/2$, and apply $H_{\mathrm{rec}} = (\hbar/t_g)a(M) \otimes G$ for a time $t_g > 0$. The same memory couples to both sites. Its preparation, shared physical port, interaction time, control and subsequent readout are resources.

The complete output is

$$|\Omega\rangle = \tfrac{1}{2}|00\rangle|0\rangle + \tfrac{1}{\sqrt{2}}|\Psi^+\rangle|1\rangle + \tfrac{1}{2}|11\rangle|2\rangle, \qquad |\Psi^+\rangle = (|01\rangle + |10\rangle)/\sqrt{2}. \tag{19}$$

A computational-basis measurement of the memory produces the classical–quantum state with probabilities $(1/4, 1/2, 1/4)$ and conditional source states $(|00\rangle\langle 00|, |\Psi^+\rangle\langle\Psi^+|, |11\rangle\langle 11|)$. All branches remain in the protocol. Flag 1, of probability $1/2$, certifies a Bell pair in the $M = 0$ protected sector; flags $0, 2$ certify product states.

If the flag is discarded, the source is instead

$$\rho_* = \tfrac{1}{4}|00\rangle\langle 00| + \tfrac{1}{2}|\Psi^+\rangle\langle\Psi^+| + \tfrac{1}{4}|11\rangle\langle 11|, \qquad I(1:2)_{\rho_*} = \tfrac{1}{2}\ln 2. \tag{20}$$

It is separable. Indeed it is the uniform collective Z-rotation twirl of $|+_X, +_X\rangle\langle +_X, +_X|$, a mixture of product rotations. Its partial-transpose eigenvalues are $0, 1/4, 1/4, 1/2$. The unconditional source therefore gains a mutual-information edge without entanglement; separability does not imply that every correlation measure is classical. The retained flagged ensemble permits a conditional protected Bell use, with herald probability and every unsuccessful branch charged. The pure source–memory state (19), the measured flagged state and the unconditional source are three different outputs.

For this flagged ensemble, $\sum_k p_k I(1:2)_{\rho_k} = \ln 2$, $I(1:K) = I(2:K) = \frac{1}{2}\ln 2$ and $I(12:K) = H(K) = \frac{3}{2}\ln 2$. Equation (16) recovers the unconditional $\frac{1}{2}\ln 2$ with correction $-\frac{1}{2}\ln 2$. Collective dephasing of the product input alone reaches the same ρ_* but supplies no accessible flag. A flag requires the specified recording interaction and actual readout.

6.3 A real link forms a deterministic protected Bell pair

The distinction is sharper in the four-qubit example. Define

$$L^x_{ab} = \tfrac{1}{2}(X_aX_b + Y_aY_b), \qquad H_{\rm link} = \hbar J L^x_{12}L^x_{34}, \quad J > 0.$$

L^x_{ab} exchanges $|01\rangle$ and $|10\rangle$ and vanishes on $|00\rangle, |11\rangle$. Consequently $[H_{\rm link}, A_{12}] = [H_{\rm link}, A_{34}] = 0$. Starting from the physical product $|01\rangle_{12}|01\rangle_{34}$, the full noisy evolution remains in the joint zero sector and is exactly

$$|\psi(t)\rangle = \cos(Jt)|0_L0_L\rangle - i\sin(Jt)|1_L1_L\rangle. \tag{21}$$

At $t = \pi/(4J)$ it has $I(12:34) = 2\ln 2$ and one ebit across the pair partition. The noise term vanishes throughout the trajectory.

This is new protected correlation from an initially uncorrelated state. It uses an actual cross-pair Hamiltonian link, here a four-body control term. The noise graph remains disconnected but the complete interaction graph does not. The factorization, prepared input, gate permissions, coupling strength and duration are supplied; there is no inferred free implementation or heat price for the link.

6.4 The process constraint and learned use

Proposition 6.1 (Correlation provenance across independent components)**.** *Let a supplied partition of a carrier have initially independent states and auxiliary resources. Suppose its full channel, including record registers, factors across these components. Then it creates no cross-component mutual information. If the input is correlated, a product channel cannot increase the mutual information across any bipartition of the components, but it can preserve initially shared classical or quantum correlations.*

Proof. A product channel sends a product input to a product output, including its local records. For a correlated input, $I(A:B) = D(\rho_{AB}\|\rho_A \otimes \rho_B)$, so the second statement is data processing under the product channel. □

An implementation satisfies the factor premise when its Hamiltonian and noise generators act separately on the components, its fresh ports are independent, and every control and instrument is component-local. Shared random controls, communication or common memories must be added to the process description: they can correlate unconditional outputs. Conditioning on a joint event can also change correlations and is not a product unconditional channel. The proposition is a diagnostic, not a claim that every process with a disconnected noise graph factors.

The constructions above do more than add initial-state qualifications to a conjecture. They exhibit how the missing relations can form: a collective port records an indistinguishability sector, while an interaction inside protected sectors builds usable quantum content. The other indispensable step is learning which encoding an unknown physical plant

protects. Foundation III [6] implements that step for $A_s = n_1 + sn_2$ with unknown $s = \pm 1$. A diagnostic sharing the plant's actual sign writes a record; that record, not s, selects odd or even encoding and the matching decoder. The same logical content and its old reference survive from before diagnosis. The record-controlled frame changes the later contact, rather than merely changing a description of the fixed commutant.

That result assumes a physical relation between diagnostic and storage couplings. A diagnostic carrying an unrelated sign would not learn the protected frame. Its retained-history scalar/JT implementation also contains real body forces, a source and a second acquisition; it is not identified with the full quantum material model. The complementary contribution here is the separation of formation, protection and use: each can fail while the others hold. A noiseless block can be unreadable, a readable label can outlive its quantum content, and a learned encoder can protect content without supplying every logical gate.

7 A geometry selected by the coupling

We now apply these results to the state and the reconstruction rule of Paper A. The rule acts on the complete graph of the sites: the edge $\{i, j\}$ has length $1/I_{ij}$, where I_{ij} is the mutual information of sites i and j, and distances are shortest-path lengths. We extend it to $I_{ij} = 0$ by giving that edge infinite length, so that a site whose edges all have zero mutual information is disconnected. The state is the three-qubit GHZ state $(|000\rangle + |111\rangle)/\sqrt{2}$. In the eigenbasis $|\pm\rangle$ of the Pauli operator X it reads

$$|\mathrm{GHZ}\rangle = \tfrac{1}{2}\big(|+{+}{+}\rangle + |+{-}{-}\rangle + |-{+}{-}\rangle + |-{-}{+}\rangle\big).$$

All couplings below are sums of X_i, hence diagonal in this basis, and Sections 2–4 apply. A local coupling $A = X_i$ with rate γ_i is the bit-flip noise of Paper A, with flip probability $p_i = (1 - e^{-2\gamma_i t})/2$. A collective coupling $A = X_1 + X_2 + X_3$ dephases between different values of the X-magnetization $M = \sum_i x_i$ and leaves states of equal M coherent.

Proposition 7.1 (Three couplings, three geometries)**.** *Let* $\Psi^+ = (|+-\rangle + |-+\rangle)/\sqrt{2}$ *and* $\tilde{W} = (|+{-}{-}\rangle + |-{+}{-}\rangle + |-{-}{+}\rangle)/\sqrt{3}$*. The long-time selected states* $\mathbb{E}_{\pi_\infty}(|\mathrm{GHZ}\rangle\langle\mathrm{GHZ}|)$ *and their geometries are:*

1. *Collective coupling* $X_1 + X_2 + X_3$*:* $\frac{1}{4}|{+}{+}{+}\rangle\langle{+}{+}{+}| + \frac{3}{4}|\tilde{W}\rangle\langle\tilde{W}|$*. Each pair has* $I_{ij} = \frac{1}{2}\ln 2$*; the geometry is an equilateral triangle with side* $2/\ln 2 = 2.8854$*.*
2. *Collective coupling* $X_1 + X_2$ *and local coupling* X_3*:* $\frac{1}{4}|{+}{+}{+}\rangle\langle{+}{+}{+}| + \frac{1}{2}|\Psi^+\rangle\langle\Psi^+|_{12} \otimes |-\rangle\langle-|_3 + \frac{1}{4}|{-}{-}{+}\rangle\langle{-}{-}{+}|$*. Here* $I_{12} = \frac{1}{2}\ln 2$ *and* $I_{13} = I_{23} = 0$*; the geometry is a single edge, and site* 3 *is disconnected.*
3. *Local couplings* X_1, X_2, X_3*: the uniform mixture of the four even strings* $|{+}{+}{+}\rangle, |{+}{-}{-}\rangle, |{-}{+}{-}\rangle, |{-}{-}{+}\rangle$*. Every* $I_{ij} = 0$*; no edge survives.*

Proof. The pinching keeps the amplitudes inside each class of equal eigenvalue tuples. (1) The classes met by the GHZ state are $M = 3$ and $M = -1$. The marginal of sites $1, 2$ is $\frac{1}{4}|{+}{+}\rangle\langle{+}{+}| + \frac{1}{2}|\Psi^+\rangle\langle\Psi^+| + \frac{1}{4}|{-}{-}\rangle\langle{-}{-}|$, with eigenvalues $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ and entropy $\frac{3}{2}\ln 2$; each single-site marginal is $I/2$; so $I_{12} = 2\ln 2 - \frac{3}{2}\ln 2$, and the other pairs follow by symmetry. (2) The classes are labelled by $(x_1 + x_2, x_3)$; the strings $|+{-}{-}\rangle$ and $|-{+}{-}\rangle$ share the label $(0, -)$. The marginal of sites $1, 2$ is as in (1). The marginal of sites $1, 3$ is uniform on the four strings $|{+}{+}\rangle, |{+}{-}\rangle, |{-}{-}\rangle, |{-}{+}\rangle$, and so is that of sites $2, 3$. (3) Every pair marginal of the uniform mixture of even strings is uniform. Direct evaluation of the full eight-dimensional density matrices gives the same values. □

The state and the rule are the same in all three cases; only the coupling differs. At zero time each pair has $I_{ij} = \ln 2$, Paper A's GHZ geometry at zero noise. Under local noise Paper A's triangles relax toward the empty graph; a collective coupling stops the relaxation at an equilateral triangle, and a coupling that is collective on one pair keeps exactly that edge. The surviving correlations come from blocks of dimension greater than one: the three-dimensional block $M = -1$, which contains the entangled state $\tilde{W}$, in case (1), and the two-dimensional block spanned by $|+{-}{-}\rangle$ and $|-{+}{-}\rangle$ in case (2). In case (3) the selected algebra is commutative with product minimal projections, every selected state is classically correlated, and for this state the classical pair correlations vanish as well. By Proposition 2.1(4) the channels at any finite time generate the full matrix algebra in all three cases, so an algebra label cannot tell them apart.

7.1 The lifetime of the selected geometry

Real couplings are not exactly collective. Add weak local couplings X_i with rates γ_i to the collective coupling with rate γ_c. The exact fixed algebra is then commutative and the long-time geometry is empty, but on a window the

magnetization blocks are still selected. With π the partition by M, the rates (3) are $\Gamma_{\rm in}(\pi) = 2(\gamma_1+\gamma_2+\gamma_3)$ and $\Gamma_{\rm out}(\pi) = 2\gamma_c + 2\min_i \gamma_i$, while the coherences of the GHZ state between its two blocks decay at the rates $8\gamma_c + 2(\gamma_i+\gamma_j)$.

Lemma 7.2 (Plateau bound)**.** *For a state ρ and a partition π, let $\Gamma_{\rm out}(\rho,\pi)$ be the smallest $D(x,y)$ over pairs in different classes with $\rho_{xy}\neq 0$, and let r be the number of basis vectors with $\rho_{xx}\neq 0$. Then for all $t\geq 0$*

$$\|\Phi_t(\rho) - \mathbb{E}_\pi(\rho)\|_1 \leq \sqrt{r}\, e^{-t\Gamma_{\rm out}(\rho,\pi)}\, \|\rho - \mathbb{E}_\pi(\rho)\|_2 + t\,\Gamma_{\rm in}(\pi). \tag{22}$$

Proof. Split $\Phi_t(\rho) - \mathbb{E}_\pi\rho = \Phi_t(\rho - \mathbb{E}_\pi\rho) + (\Phi_t(\mathbb{E}_\pi\rho) - \mathbb{E}_\pi\rho)$. The second term is bounded by Theorem 3.2(1). The first is the Schur product of $\rho - \mathbb{E}_\pi\rho$ with multipliers of modulus at most $e^{-t\Gamma_{\rm out}(\rho,\pi)}$; its rank is at most r, because a row of a positive matrix with zero diagonal entry vanishes, and $\|Z\|_1 \leq \sqrt{\operatorname{rank} Z}\,\|Z\|_2$. □

The bound turns into a certified geometry through continuity of entropy. If two states of a qubit pair are at trace distance $\delta = \frac{1}{2}\|\rho-\sigma\|_1 \leq \frac{1}{2}$, then by the Fannes–Audenaert inequality [22] and Winter's bound for conditional entropy [23] their mutual informations differ by at most

$$f(\delta) = h(\delta) + 2\delta\ln 2 + (1+\delta)\,h\Big(\frac{\delta}{1+\delta}\Big), \tag{23}$$

with h the binary entropy in nats. Partial traces do not increase trace distance, and Paper A's stability bound for graph metrics converts a uniform change $f < m$ of mutual informations bounded below by m into a change of at most $2f/(m(m-f))$ of every distance among three sites.

t	$\|\Phi_t(\rho) - \mathbb{E}_\pi\rho\|_1$	bound (22)	distances (d_{12}, d_{13}, d_{23})
0	0.8660	1.2247	(1.4427, 1.4427, 1.4427)
1	0.0010	0.0018	(2.8968, 2.9034, 2.9066)
10	0.0094	0.0140	(2.9728, 3.0221, 3.0458)
100	0.0897	0.1400	(3.5620, 3.9855, 4.2034)
300	0.2442	0.4200	(4.8943, 6.5032, 7.4508)
1000	0.5969	1.4000	(12.559, 28.867, 41.426)

Table 1: GHZ state under a collective coupling with $\gamma_c = 1$ and local couplings with $(\gamma_1,\gamma_2,\gamma_3) = 10^{-4}(1,2,4)$. The plateau value of every distance is $2/\ln 2 = 2.8854$.

Proposition 7.3 (A certified plateau)**.** *For $\gamma_c = 1$ and $(\gamma_1,\gamma_2,\gamma_3) = 10^{-4}(1,2,4)$, the bounds* (22) *and* (23) *certify that every distance of the GHZ geometry lies within 25% of the plateau value $2/\ln 2$ for $0.716 \leq t \leq 3.58$. Numerical root finding of the exact evolution gives the approximate 25% window $0.053 \lesssim t \lesssim 53.6$ and 5% window $0.163 \lesssim t \lesssim 8.86$. If the local rates are scaled to $s(1,2,4)$ with $s \leq 10^{-3}$, the late edges scale to leading order as $1/s$: approximately $5.36\times 10^{-3}/s$ for the 25% window, $8.86\times 10^{-4}/s$ for the 5% window and, when $s \leq 10^{-4}$ so that it exists, $3.58\times 10^{-4}/s$ for the certified 25% window. The windows begin at times of order $1/\gamma_c$.*

Proof. For this state $r = 4$, $\|\rho - \mathbb{E}_\pi\rho\|_2 = \sqrt{6}/4$, $\Gamma_{\rm out}(\rho,\pi) = 8.0006$ and $\Gamma_{\rm in}(\pi) = 1.4\times 10^{-3}$. The right side of (22) is the convex function $b(t) = (\sqrt{6}/2)e^{-8.0006t} + 0.0014t$. On $[0.716, 3.58]$ its maximum is at an endpoint and is less than 0.0050121. The trace distance used in (23) is $\delta(t) = \frac{1}{2}\|\Phi_t(\rho) - \mathbb{E}_\pi\rho\|_1 \leq b(t)/2$, not $b(t)$. The increasing entropy and graph bounds therefore give a distance error less than $0.72130 < \frac{1}{4}\cdot 2/\ln 2$ throughout that interval. The exact-evolution window edges were located by root finding of the Schur multipliers. Once the coherences between blocks have decayed, the in-block multipliers depend on t only through st, which gives the late-edge scaling. The root values are numerical diagnostics, distinct from the conservative analytic interval. □

Table 1 shows the three stages. The GHZ geometry, with side $1/\ln 2$, relaxes on the time scale $1/\gamma_c$ to the selected triangle with side $2/\ln 2$; the triangle persists while $t\Gamma_{\rm in}(\pi)$ is small; then the unequal local rates deform it into scalene triangles, which are not isometric to the plateau, before every edge diverges. This is Paper A's non-isometry produced by one time-ordered dynamics rather than by a comparison of separately chosen channels. Proposition 7.3 gives a state-specific stability certificate for Paper A's declared graph dictionary on an explicit window. The plateau exists when its end lies after its beginning, that is, when the ratio of in-block to between-block rates is small, as Theorem 3.2 requires.

The constants in Proposition 7.3 are small. At the end of the exact 25% window the three in-block coherences have fallen by only 3.2, 5.2 and 6.2 per cent. The selected pair marginal has a zero eigenvalue, and in-block decoherence populates it, so pair entropies respond to a coherence loss δ like $\delta \ln(1/\delta)$. Geometries read from entropies are therefore far more fragile than the coherences that carry them, and the gap between the certified and the exact window comes mostly from this logarithmic response together with worst-case continuity bounds.

7.2 Imperfect redundant records detach a vertex

Finally we let fragments record the magnetization with a small asymmetry. Each of K fragment qubits starts in $|0\rangle$ and interacts through $V_k(x) = \exp[-i\theta(M(x)+\eta x_1)Y]$ with $\theta = \pi/8$, where Y is the Pauli operator of the fragment. For $\eta = 0$ the coupling is block-constant. The two magnetizations present in the GHZ state, $M = 3$ and $M = -1$, then produce orthogonal fragment states, so every fragment holds a perfect copy of the label, and by Proposition 4.1 the equilateral triangle survives for any number of fragments. For $\eta > 0$ each fragment distinguishes strings of the block $M = -1$ that differ at site 1, with $D_k = |\sin 2\theta\eta|$.

η	K	coherence of $\|+--\rangle, \|-+-\rangle$	$I_{12} = I_{13}$	I_{23}	label error
0.02	1	0.99988	0.34631	0.34657	6.2×10^{-5}
0.02	100	0.98774	0.32790	0.34657	6.2×10^{-5}
0.02	1000	0.88393	0.23581	0.34657	6.2×10^{-5}
0.05	1	0.99923	0.34525	0.34657	3.9×10^{-4}
0.05	100	0.92577	0.26724	0.34657	3.9×10^{-4}
0.05	1000	0.46243	0.05555	0.34657	3.9×10^{-4}

Table 2: Recording the magnetization with a site-1 asymmetry η. The last column is the minimal error probability with which a measurement on one fragment identifies the label M, whose prior is $(\frac{1}{4}, \frac{3}{4})$. Coherences are relative to their initial values; the coherence between $|-+-\rangle$ and $|--+\rangle$ stays equal to 1.

Table 2 shows the effect. Every fragment is a nearly perfect and readable record of the label for both asymmetries, so the loss of geometry is not a price paid for better records. It is the small leakage about site 1, multiplied over fragments as in Theorem 4.2. The coherences between strings that differ at site 1 decay as $(1 - D_k^2)^{K/2}$, the edges to site 1 weaken, and as $K \to \infty$ the state approaches $\frac{1}{4}|+++\rangle\langle+++| + \frac{1}{4}|+--\rangle\langle+--| + \frac{1}{2}|-\rangle\langle-|_1 \otimes |\Psi^+\rangle\langle\Psi^+|_{23}$. Then $I_{12} = I_{13} = 0$ and $I_{23} = \frac{1}{2}\ln 2$: site 1 is detached and the geometry becomes the single edge of Proposition 7.1(2), now on sites $2, 3$. With $\eta_K = 0.5K^{-\alpha}$ the coherence after $K = 10^6$ fragments is 0 for $\alpha = \frac{1}{4}$, $0.9258 = e^{-(\pi/8)^2/2}$ for $\alpha = \frac{1}{2}$, and 0.9999 for $\alpha = \frac{3}{4}$, as Corollary 4.3 predicts with $\lambda = 2\theta \cdot 0.5$ to leading order.

8 What selection settles

The selected partition is fixed by the supplied dynamics, window and tolerance, and is robust to perturbations preserving the rate gap. This differs in emphasis from QBism's state assignments and relational quantum mechanics' reference-system dependence [26, 25], without deciding their wider interpretations. Public agreement further needs actual fragments and calibrated readers; Paper A provides one finite realization. A stable algebra alone does not grant those permissions.

The GHZ comparison changes a state under one graph dictionary, not the coordinates of an unchanged metric. Sector mixing can sustain edges after quantum content is lost; the leaking GHZ block instead supports the specific edges destroyed in Table 2. Formation requires the collective port or Hamiltonian link of Section 6, not noise selection alone. Removing either resource supplies its corresponding opposing control.

AQFT and Consciousness I start with their local or current-net carriers [24, 4]. Observable-induced factorization and quantum mereology address when retained relations support an effective subsystem description [28, 29, 30]. Decoherence and Everettian accounts share the underlying dynamics [10, 27]; the issue here is whether its resulting structure is readable, operable and persistent for a finite task.

9 Beyond the commuting class

Conjecture 9.1 (Process-selected effective subsystems)**.** *On an initially specified carrier, robust effective subsystems form through a history of interactions that both builds relations and makes a noncommuting set of internal operations*

reproducible across a finite task window. The relevant connectivity is that of the complete physical process, including recording ports and Hamiltonian controls, not the collective-noise hypergraph alone. Within a separated-rate regime, a learned encoding can turn the process's indistinguishable sectors into stable operational components; their boundaries change when the reproducible operations or their lifetime change.

The conjecture concerns effective factorization, not the existence of elementary sites. It predicts more than correlation alone: successful formation should yield an encoder, decoder and repeatable operations on the formed component, with calibration records and finite error bounds. The exact formation examples give protected relational content; Foundation III adds record-controlled encoding of unknown content. Together they support a route from coupling through history to changed capability. They do not yet supply a general autonomous acquisition of a complete internal gate set. A proposed component that has only a state-dependent correlation but no reproducible internal operations would fail this criterion. The disconnected Bell counterexample rules out noise-only connectivity as a necessary condition, while Proposition 6.1 identifies the real no-creation obstruction for product processes.

For general Lindblad dynamics, a separated spectrum defines slow states and an effective metastable evolution [17]. A finite-window extension should select the operational structure of this manifold rather than assume that slow observables close under multiplication. Even complete positivity and idempotence do not ensure closure in the original multiplication. For example,

$$P(X) = \operatorname{diag}\big(x_{11}, x_{22}, (x_{11} + x_{22})/2\big) \tag{24}$$

is a unital completely positive idempotent on M_3. It is completely positive because it is a sum of two positive evaluation maps into positive diagonal matrices. Its range contains $E = \operatorname{diag}(1, 0, 1/2)$ but not $E^2 = \operatorname{diag}(1, 0, 1/4)$. A completely positive projection gives its range a Choi–Effros product $x \circ y = P(xy)$ [7, 8]. This abstract product does not make xy a physically permitted observable or control in the original representation. An ordinary conditional expectation onto a represented subalgebra needs the additional algebraic closure. A metastable spectral projection may not even be completely positive. Thus a general selection theory must provide actual instruments or recovery maps, their errors and their operational closure, rather than infer them from a slow spectral space.

The physical price of that capability is the next link to Q-RAIF and the gravitational line. A passive GHZ geometry has lifetime of order $1/s$ under symmetry breaking, while independent redundant fragments require an $R^{-1/2}$ protecting symmetry. Q-RAIF B realizes correction as field coupling with work and stress [5]; HAFF D distinguishes entropy removal, usable side information and supplied purity, and gives specified source–response constructions [2]. These results suggest a maintained-structure mechanism: actual records learn a protected frame, operations install it, maintenance changes the physical source, and the response changes subsequent records. Foundation III realizes a finite retained-history scalar/JT return, whereas the full quantum material source requires its own support and preparation account. Connecting those implementations and renewing their resources is a substantive requirement, not an irreducible heat flux inferred from algebra inclusion.

Many-site and continuum selection require further work when rate gaps close. Consciousness I's ideal fixed-net closure and finite-time noise estimates are not already a continuum uniqueness theorem for window-selected blocks. Likewise, the graph dictionary is supplied here. A general emergence of spacetime needs a physical family of readers and a reconstruction law that remains controlled as the operational structure grows.

10 Conclusion

A commuting coupling determines its fixed block algebra and its long-time pinching. On a separated finite window it selects a unique block partition in the supplied joint eigenbasis; the selected algebra contains every stable block algebra of that form. The actual finite-time channel has full operator range. Block-constant record couplings report sector weights without resolving protected content. Independent leakage multiplies the coherence loss and imposes a sharp redundancy–symmetry threshold.

The relation to geometry has two parts. Sector mixing and within-sector content contribute through the exact identity (16); neither alone determines the mutual information graph. For the GHZ source, collective, pair-collective and local couplings yield a triangle, one edge and no edge under one rule. Weak asymmetry gives a finite plateau and then destroys its supporting edges. Noncommuting perturbations can erase the exact fixed algebra without abruptly erasing a finite record or task capability.

Formation requires a history. An initially shared logical Bell pair can survive between noise-disconnected protected components, so a noise graph cannot delimit all surviving quantum relations. A collective recorder forms a flagged protected Bell branch from a product input, with all branches retained; an actual Hamiltonian link forms a deterministic Bell pair protected throughout its generation. These constructions reduce the missing formation step to identifiable physical operations without hiding their costs. Record-controlled encoding supplies the further route from learned

relations to changed capability. The remaining general problem is to derive reproducible subsystem operations and their sustained maintenance from such histories, rather than merely rename correlated states as subsystems.

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Causation, Agency, and Existence: Historical Control and Maintained Intervention

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Abstract

A cause changes what an allowed intervention can bring about; a record alone need not identify that change. We define interventional influence relative to a process, an intervention set and a readout. It contracts under post-processing and along channel chains. A direct cause and a common cause can have identical declared record laws and influences 1 and 0.

Agency requires more than a persistent register. In a binary tracking task, the memoryless failure probability is bounded by $\frac{1}{2}[1-(1-2q)\kappa_{\mathrm{s}}\kappa_{\mathrm{c}}]$. Three retained readings beat every present-reading controller in the specified sensor menu, despite a smaller influence of the present world on the action. We then give the same agent a damageable body: an actual mixed-port diagnosis controls repair, and the repaired body determines the next sensing channel. The finite reversible protocol maintains tracking error at most 0.15820 from a healthy start, against a long-run lower bound 0.22172 for damage-independent maintenance calendars at the same repair rate. All spent ports and old records remain; fresh blank supply is not a thermal reset. This is a mechanism for maintaining intervention capability, not a derivation of the initial body or its couplings.

These results support relational existence: structures can be objective within their coupling, history and observation window without being intrinsic to a state alone. The essay develops implications for will, self-maintenance and structural realism, then compares traces and dependence with Yogācāra and Mādhyamaka. These comparisons have their own arguments and do not establish phenomenal consciousness.

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1 Introduction

A register can remain unchanged while doing nothing. An agent can change almost every internal value while continuing to sense, act and preserve the conditions for doing so again. The distinction matters when a quantum state does not by itself select a division into things, causes and agents.

The earlier papers provide different parts of the physical account. Paper A [1] connects actual channel and record permissions to different correlation geometries for a fixed source and dictionary. Paper B [2] derives stable block content in a commuting coupling class and distinguishes pre-existing relations from relations formed by actual recording or Hamiltonian links. Block-constant couplings record central labels; arbitrary environmental records need not have that form. Paper E [3] derives measurement permissions, partly mixed records and calibrated public readout. Paper F [4] accounts for record formation, retention and updating; Paper G [5] asks how much of the selection can be performed from inside the process.

The present question is how these structures acquire causal and agentive roles. Section 2 separates observation from intervention. Section 3 bounds viable control with and without history, then closes a missing step: records act on the same body whose condition controls later sensing. Section 4 develops relational objectivity rather than identifying reality with a reader's answer. Section 5 compares the argument with Buddhist accounts of traces and dependence. The initial partition and gate permissions are stated inputs to the finite mechanism; their primitive formation remains a broader question.

2 Part I: Causation, records and interventions

2.1 Observational asymmetry and its limits

Standard quantum mechanics assigns no joint density operator to non-commuting observables at two times without a process formalism, so a cross-time mutual information is defined only relative to declared instruments.

Definition 2.1 (Instrument-relative directional asymmetry)**.** *Fix a two-time process $\mathcal{N}_{t_1\to t_2}$, an initial preparation and instruments $\mathcal{I}_A, \mathcal{I}_B$ with classical outcomes $X_A^{t_i}, X_B^{t_i}$, measured in separate forward and reverse arrangements. Put*

$$\Delta^{\mathcal{I}}_{A\to B} = I(X_A^{t_1} : X_B^{t_2}) - I(X_B^{t_1} : X_A^{t_2}),$$

with Shannon mutual informations of the outcome distributions.

$\Delta^{\mathcal{I}}$ is a well-defined observational diagnostic. Its sign depends on the instruments, the preparation and the process, and it can arise from a common cause or from asymmetric noise. Other conventions for quantum two-time objects exist [9, 10]; none turns a correlation into a cause without an account of intervention, which we now give.

2.2 Interventional influence

Definition 2.2 (Interventional influence)**.** *Fix a process in which a declared set $\mathcal{J}_A$ of interventions can be applied at A, each replacing the mechanism that ordinarily determines A while everything else is held fixed, and fix a readout at B. Let ρ_B^J be the state, or the outcome distribution, at B under the intervention J. The interventional influence of A on B is*

$$\kappa_{\mathcal{J}}(A \to B) = \sup_{J,J' \in \mathcal{J}_A} D\big(\rho_B^J, \rho_B^{J'}\big),$$

where D is the trace distance, or the total-variation distance for classical outcomes.

For several steps, with memory in the environment and controls at several times, the natural setting is a quantum comb or process tensor [11, 12]; the definition applies to any slot of such an object.

Proposition 2.3 (Properties of interventional influence)**.**

1. *$0 \le \kappa \le 1$, and $\kappa = 0$ exactly when no allowed intervention changes the state at B.*
2. *Post-processing the readout at B by any channel, or restricting $\mathcal{J}_A$, cannot increase κ.*
3. *If the interventions at A affect B only through M, in the sense that $\rho_B^J = \Lambda(\rho_M^J)$ for a channel Λ that does not depend on J, then*

$$\kappa(A \to B) \le \eta(\Lambda)\, \kappa(A \to M), \qquad \eta(\Lambda) = \sup_{\rho \neq \sigma} \frac{D(\Lambda\rho, \Lambda\sigma)}{D(\rho, \sigma)}.$$

For a classical channel with kernel $\Lambda(\cdot \mid m)$, $\eta(\Lambda) = \max_{m,m'} D(\Lambda(\cdot \mid m), \Lambda(\cdot \mid m'))$, Dobrushin's coefficient [13], which is the influence of M on B under all point interventions on M.

Proof. (1) is immediate. (2) Trace and total-variation distances do not increase under channels, and a supremum over a smaller set is smaller. (3) $D(\Lambda\rho_M^J, \Lambda\rho_M^{J'}) \le \eta(\Lambda) D(\rho_M^J, \rho_M^{J'})$ by the definition of η. For classical kernels the supremum is attained on pairs of point masses, which is Dobrushin's formula. □

Part (3) is the causal counterpart of the data-processing inequality: influence passed along a chain is at most the product of the influences of its links. Section 3 uses it to show that an agent needs both sensing and control.

2.3 Records do not reveal causes

Proposition 2.4 (Same observations, different causes)**.** *Consider two processes on bits A, B observed in the computational basis at times $t_1 < t_2$.*

1. *Direct cause: A is a uniform bit at t_1, $B = 0$ at t_1, and a controlled NOT from A to B acts between t_1 and t_2.*
2. *Common cause: a hidden uniform bit C is copied into A before t_1 and into B between t_1 and t_2; $B = 0$ at t_1.*

The joint distributions of $(X_A^{t_1}, X_B^{t_1}, X_A^{t_2}, X_B^{t_2})$ *coincide, both equal to* $\frac{1}{2}$ *on* $(0,0,0,0)$ *and on* $(1,0,1,1)$*, and both give* $\Delta^{\mathcal{I}}_{A\to B} = \ln 2$*. For the interventions "set* A *to* 0*" and "set* A *to* 1*" at* t_1*, the influence on* B *at* t_2 *is* 1 *in the first process and* 0 *in the second.*

Proof. In both processes A keeps its value, B starts at 0 and ends equal to the uniform bit carried by A. Under the intervention, B at t_2 equals the set value in the first process and the hidden bit in the second. □

Every statistic of the declared observed variables is the same in the two processes, so accumulating more of those records cannot settle which structure produced them. For quantum variables the situation is partly different: some causal structures can be inferred from observations alone when entanglement or coherence is available [14]. That is an advantage of particular quantum protocols, not a general replacement for interventions.

2.4 Records and interventions in one process

A record carries a recoverable difference from an earlier process. An intervention changes a declared mechanism or operation. They can occur within the same interaction without being the same role. Paper E shows that a memory need not be pure to record: the relevant budget is its purity deficit $\ln d - S(\eta)$, and partly mixed pointer ports have nonzero readable information. Records of a commuting pointer label are a specific acquisition model, not a classification of every record.

For an already detached port, a later trace-preserving operation on the source leaves its marginal unchanged, by no-signalling. A renewed interaction with the port, or a direct operation on it, can change its state. Reversible relabelling, repair, transfer into another memory and reset have different resource balances. Paper F treats them through the joint state; they cannot all be priced as erasure of a marginal entropy. The Imprint Chain [6] makes a complementary distinction: a passive record and a feedback can carry the same information about an interface while only the feedback changes the world state.

In Proposition 2.4, the recorded correlation between A and B does not make A a cause of B. Conversely, an influence can change a state without a record being retained. Memory consists here of physical records that remain available to later decisions. Their availability may itself depend on repair, storage and permission.

The process already has a time order. Under fresh-port and specified interaction conditions, it can acquire an operational direction of record accumulation and intervention. Repeated contact can return correlations and restore source coherence; local entropy need not increase at every step. These mechanisms explain a direction within the process, not the origin of its time parameter.

3 Part II: Historical control and maintained capability

3.1 What a boundary criterion must allow

A tempting boundary criterion asks every internal observable to stay nearly unchanged: $\sup_t \sup_{\|O\|\le 1} \|\mathcal{E}_t^*(O) - O\| \le \epsilon$. It admits an isolated frozen register and excludes a changing tracker. In the example below, the internal $Z_r = (-1)^r$ has deviation $1+(1-2\nu)(1-2q) = 1.768$, although the tracker succeeds. What must persist is a capability and the physical organization that supports it, not every stored value.

A stable algebra of the kind derived in Paper B can hold part of that organization. It does not by itself provide the sensor, actuator or maintenance procedure. We first specify the task and then construct the maintenance mechanism.

Definition 3.1 (Viable agent on a window)**.** *Fix a declared partition into a world variable W, an agent register R and an actuated variable b, declared interventions on W and on R, a viability condition $v(W, b)$ and a window. The register is a viable agent on the window, with margin $\eta < \frac{1}{2}$, if*

1. *the viability condition fails with probability at most η at every step of the window;*
2. *the sensing influence $\kappa_{\mathrm{s}} = \kappa(W \to R)$ is positive;*
3. *the control influence $\kappa_{\mathrm{c}} = \kappa(R \to b)$ is positive.*

The agent is memoryless if its action at a step depends on the world only through its reading at that step.

3.2 Sensing and control multiply

Consider a world bit W_t that flips with probability $q < \frac{1}{2}$ at each step. The agent reads it into R, acts on b, and is viable when $b_{t+1} = W_{t+1}$.

Proposition 3.2 (Viability needs sensing and control)**.** *Let a uniform world bit be read through a fixed sensor channel with independent fresh randomness. The actuator receives the world only through that reading, uses independent randomness, and the next world flip is independent of both. Thus the sensor's ordinary conditional output law equals its output under interventions setting W_t, with no initial world–controller common cause or retained world-correlated side input. For this memoryless menu, with κ_{s} computed for interventions setting W_t to 0 or 1 and κ_{c} for all point interventions on R,*

$$\Pr(b_{t+1} \neq W_{t+1}) \;\geq\; \tfrac{1}{2}\big(1 - (1 - 2q)\,\kappa_{\mathrm{s}}\kappa_{\mathrm{c}}\big). \tag{1}$$

For a sensor that errs with probability ν and an actuator that errs with probability μ, both symmetric, equality holds and the misalignment is $\frac{1}{2}(1 - (1 - 2\nu)(1 - 2\mu)(1 - 2q))$.

Proof. Let K be the channel from W_t to b_{t+1}. With $W_{t+1} = W_t \oplus f$, $\Pr(f = 1) = q$, the success probability is $q + \frac{1}{2}(1 - 2q)(K(0|0) + K(1|1))$, and $K(0|0) + K(1|1) \leq 1 + D(K(\cdot|0), K(\cdot|1)) = 1 + \kappa(W_t \to b_{t+1})$. By Proposition 2.3(3), $\kappa(W_t \to b_{t+1}) \leq \kappa_{\mathrm{s}}\kappa_{\mathrm{c}}$. For symmetric channels $\kappa_{\mathrm{s}} = 1 - 2\nu$, $\kappa_{\mathrm{c}} = 1 - 2\mu$, and the misalignment is the probability that an odd number of the three independent events sensor error, actuator error and flip occurs. □

Corollary 3.3 (Records without control, control without records)**.** *In this fresh-sensor task, a passive recorder with $\kappa_{\mathrm{c}} = 0$, an open-loop actuator with $\kappa_{\mathrm{s}} = 0$, and an initially uncorrelated frozen register fail with probability $\frac{1}{2}$.*

The channel condition is necessary. If a hidden fair bit is copied into W_t and a frozen R before the slot, then $b_{t+1} = R$ has ordinary error q, although intervening on current W_t does not change R: $\kappa_{\mathrm{s}} = 0$. The common-cause record is useful without a new sensing influence. It is not a fresh sensor channel from the intervened world. Treating observed $\Pr(R \mid W_t)$ as that channel would erase the distinction proved in Proposition 2.4. Retained world-correlated inputs must instead be included as memory, as in the next subsection.

The magnitudes $\kappa_{\mathrm{s}}, \kappa_{\mathrm{c}}$ alone do not certify the labels needed for the directional task: a reversed sensor has the same influence as a correctly oriented one. T-DOME III shows that acting cannot reveal parameters moved by a symmetry of the complete record law. Certifying this directional task therefore needs a symmetry-breaking labelled operation or retained standard [31]. It may be physically implemented inside the agent; the requirement concerns the information it supplies, not its location outside the loop.

With $\nu = 0.1$, $\mu = 0.05$ and $q = 0.02$ the memoryless agent fails with probability 0.1544, while its register changes value in one step with probability 0.193. It is viable in Definition 3.1 and fails the pointwise-invariance criterion; an initially uncorrelated frozen register passes that criterion and has no sensing route.

3.3 Memory changes what an agent can do

Let the agent instead act on the majority of its last three readings. Exact enumeration gives a misalignment of 0.1082. The influence of the present world state on the action falls at the same time: an intervention on W_t changes only the newest reading, which decides the majority only when the two older readings disagree, so the influence is $(1-2\nu)(1-2\mu)\Pr(\text{older readings disagree}) = 0.139$, against 0.72 for the memoryless agent.

Corollary 3.4 (Memory is necessary for this level of viability)**.** *In the specified drifting-bit menu with one fresh sensor reading per slot, every memoryless agent fails with probability at least* $\frac{1}{2}(1-(1-2q)(1-2\nu))$*, which is* 0.116 *for* $\nu = 0.1$*,* $q = 0.02$*. The majority-of-three agent fails with probability* 0.1082 *even with an imperfect actuator,* $\mu = 0.05$*. Within this menu, a failure probability below* 0.116 *therefore requires dependence on past readings.*

Proof. A memoryless agent's register depends on W_t only through the reading, so $\kappa_{\mathrm{s}} \le 1-2\nu$ by Proposition 2.3(2); Proposition 3.2 with $\kappa_{\mathrm{c}} \le 1$ gives the bound; the majority value follows from enumerating the 2^8 values of the initial world bit, three flips, three reading errors and actuator error. The exact value is 0.10822784; Appendix A gives the verification method. □

The memory agent tracks the world better while the present world influences it less. Its success rests on records of past world states, which are correlated with the present one because the world persists; an intervention on the present world cannot reach those records. Agency therefore involves two different relations to the world: causal influence through current sensing and control, and correlation carried by records. Proposition 3.2 bounds what the first can achieve alone. This is the task-level form of the memory resource studied in T-DOME I [8], and it is the step at which the self-reading of Imprint W [7] and the internal selection rules of Paper G become relevant: an agent that also keeps a model of its own sensor and actuator, and revises it from its records, can choose which readings to take.

3.4 Records maintain the body that reads next

The tracker so far has a supplied sensor. Give it a body bit D_t: 0 is functional and 1 damaged. The same physical body remains throughout the run; its condition modulates the world-sensing coupling, and it can be repaired before sensing. Damage is independent of the world, arrives after each task slot with probability $0 < d < 1$, and is absorbing until repair. This is a specified failure mechanism, not a biological model.

Each slot begins with diagnosis. A fresh port in $\eta_{\mathrm{d}} = [I + (1-2e)Z]/2$, $0 < e < 1/2$, interacts by controlled X with D_t. Its Z record is $Y_t = D_t \oplus N_t$, $\Pr(N_t = 1) = e$, from the mixed-port instrument of Paper E. Exactly when $Y_t = 1$, the controller swaps a fresh $|0\rangle$ into D_t; the old body state goes into that port and remains in the spent bank. The post-repair body is $D'_t = D_t(1-Y_t)$. The decision uses Y_t, not hidden D_t.

The world sensor then gives

$$R_t = W_t \oplus N_t^{\mathrm{s}} \oplus D'_t Q_t, \qquad \Pr(N_t^{\mathrm{s}} = 1) = \nu, \quad \Pr(Q_t = 1) = \tfrac{1}{2}. \tag{2}$$

For a healthy body it is the original sensor; for a damaged body its reading is fair. Two gates realize this law: controlled X from W_t to a port in $[I+(1-2\nu)Z]/2$, followed by a flip controlled by D'_t and a fresh fair bit Q_t. The spent Q_t is retained but outside the controller's readout menu. Joint access to it would change the sensing permission.

The reading drives the same actuated variable b. A fresh blank is swapped into b, retaining its old value; copying R_t and adding independent actuator error gives $b_{t+1} = R_t \oplus N_t^{\mathrm{a}}$, $\Pr(N_t^{\mathrm{a}} = 1) = \mu$. The world then flips with probability q, and finally the body suffers damage: $D_{t+1} = D'_t \vee J_t$, $\Pr(J_t = 1) = d$. Fresh ports are independent of the retained body, world and history. Diagnostic orientation and error are known or separately calibrated; a common unknown sign is not eliminated by a self-certifying reference.

Proposition 3.5 (Maintenance of a sensing route)**.** *Let* $x_t = \Pr(D_t = 1)$*,* $z_t = \Pr(D'_t = 1)$*, and* $r_t = \Pr(Y_t = 1)$*. Then*

$$z_t = ex_t, \qquad r_t = e + (1-2e)x_t, \qquad x_{t+1} = d + (1-d)ex_t, \tag{3}$$

$$x_* = \frac{d}{1-(1-d)e}, \qquad z_* = ex_*, \qquad r_* = e + (1-2e)x_*. \tag{4}$$

With $\rho = (1-d)e$*,* $x_t = x_* + \rho^t(x_0 - x_*)$*. For a uniform world independent of the initial body, the raw sensing influence and tracking failure are*

$$\kappa_{\mathrm{s},t} = (1-z_t)(1-2\nu), \tag{5}$$

$$\mathcal{R}_t = \mathcal{R}_0 + (\tfrac{1}{2} - \mathcal{R}_0)z_t, \qquad \mathcal{R}_0 = \tfrac{1}{2}[1-(1-2q)(1-2\nu)(1-2\mu)]. \tag{6}$$

A healthy start gives $z_t \le z_*$ *and* $\mathcal{R}_t \le \mathcal{R}_0 + (\frac{1}{2} - \mathcal{R}_0)z_*$ *at every slot. Without repair, sensing influence tends to zero and tracking failure tends to* $1/2$*.*

Proof. Only a damaged body missed by diagnosis remains damaged, with probability ex_t. The diagnostic is positive on a healthy body with probability e and on a damaged body with probability $1-e$. Independent damage gives $x_{t+1} = z_t + d(1-z_t)$. Solving the recurrence gives the fixed point and convergence. Repair and damage do not depend on the world, so the body remains independent of W_t. The sensor mixes the original channel, with weight $1-z_t$, and a fair channel, with weight z_t, giving its influence and risk. Without repair, $x_t = 1-(1-x_0)(1-d)^t$. □

This is intervention rather than relabelling. Replacing diagnosis by *never repair* or *always repair*, with the pre-slot distribution fixed, changes post-repair damage from x_t to zero and risk by $(\frac{1}{2} - \mathcal{R}_0)x_t$. Setting the body to healthy or damaged after repair changes the available sensing channel. The record maintains a physical condition of later intervention, not merely confidence in a previous answer.

Proposition 3.6 (A matched calendar comparison)**.** *Consider repair calendars independent of damage and of all information about it, including diagnostic or world-sensor histories. They may use private randomness and have positive long-run repair frequency* r*, with empirical mean gap* $1/r$ *and vanishing end effects in the occupancy average. In the same present-reading control menu, their long-run mean post-repair damage is at least*

$$z_{\mathrm{cal}}(r) = r[(1-\omega)G(L) + \omega G(L+1)], \quad L = \lfloor r^{-1} \rfloor, \quad \omega = r^{-1} - L, \tag{7}$$

where

$$G(\ell) = \sum_{j=0}^{\ell-1}[1-(1-d)^j] = \ell - \frac{1-(1-d)^\ell}{d}.$$

Gaps $L, L+1$ *in these proportions attain the bound. Frequency at most* r *gives the same lower bound.*

Proof. A gap of length ℓ, starting with repair, has expected damaged occupancy $1-(1-d)^j$ at age j, with sum $G(\ell)$. Conditioning on an independent calendar preserves the damage law. The differences $G(\ell+1) - G(\ell) = 1-(1-d)^\ell$ increase. The convex envelope on the integers minimizes average gap occupancy at fixed mean gap $1/r$ by balancing adjacent lengths. Multiplication by gaps per slot gives (7). Adding repairs cannot increase occupancy, so the allowance is optimally used in full. □

For $d = e = 0.1$ and $q = 0.02, \nu = 0.1, \mu = 0.05$,

$$x_* = \frac{10}{91}, \quad z_* = \frac{1}{91}, \quad r_* = 0.18791209, \quad \mathcal{R}_* = 0.15819780.$$

The optimal independent calendar at this repair frequency has $z_{\mathrm{cal}} = 0.19479225$ and mean error 0.22172020. Both receive the same diagnostic and world-reading ports and the same repair and actuator menu. One unit is charged per actual repair, with other per-slot inputs equal. A controller inferring damage from sensor history is another record-guided strategy, outside the calendar class. Equal long-run rate does not mean equal repair counts on every finite trajectory.

3.5 Finite stock and a maintained boundary

For n slots, the realization has three continuing bits D, W, b and nine fresh ports per slot: diagnostic, repair blank, world sensor, fair sensor-noise, actuator-noise, world-flip, damage coin, damage-dilation blank and actuator blank. Maximum repair count is n; a healthy start gives expected count at most nr_*. Supplying the bank, storing spent ports and routing records belong to the finite-window budget. Mean activation count is not stock capacity.

Each of the three blank ports per slot supplies initial purity deficit $\ln 2$. Diagnostic and sensor ports supply $\ln 2 - h(e)$ and $\ln 2 - h(\nu)$; actuator, world-flip and damage coins supply $\ln 2 - h(\mu), \ln 2 - h(q), \ln 2 - h(d)$; fair noise supplies none. Here h is binary entropy in nats. Unused repair blanks may remain pure; supplied purity is not necessarily exhausted on every slot. Old states and correlations remain; the reversible circuit in Appendix A conserves joint entropy. No bath is reset and no work or heat price follows from counting swaps. For the numerical parameters, the nine-port input purity is $8 \ln 2 - 3h(0.1) - h(0.05) - h(0.02) = 4.27337417$ nats per slot. For degenerate port Hamiltonians, $k_B T$ times this quantity is the free-energy difference from a maximally mixed preparation at temperature T, not heat dissipated or net work consumed by the run. The initialized healthy body and blank actuator in the explicit check add two initial pure bits; the world starts fair. Gate strengths, preparation, duration, clock, storage and later bank renewal remain physical inputs.

The body's condition regulates whether a reading is informative, and a record of that condition regulates the operation preserving the route. Organization survives while its values change. Continued intervention capability is produced by record-conditioned action, rather than merely imposed as a viability condition. Repair uses the current diagnosis only; earlier diagnoses remain stored but are not read by this policy. The majority tracker establishes the value of longer history, whereas this circuit establishes the maintenance of later permissions. Neither result alone constructs a long-term self-model.

Remark 3.7 (Maintained organization)**.** *A broader agency hypothesis is that an effective agent boundary is the organization of sensing and action routes whose continued availability is causally regulated by retained history. Disrupting use of a bodily diagnostic should impair later intervention capability even when current discrimination is spared. At a matched budget, an independent calendar should fail where a condition-dependent policy succeeds.*

The binary body establishes one instance. Formation must also explain how routes and the repair menu arise from earlier interactions and how histories coordinate several demands. The hypothesis explains no additional boundary if a fixed external controller accounts for every apparent self-maintenance effect, or if manipulating history never changes later permissions. BRAIN supplies learned protection and routing for a different body model [32]; Imprint W and Imprint SH price reliability-based action [7, 33]. Their dynamics and budgets must match before they form a larger construction; they are not assumed to be this circuit.

3.6 Will, degrees of agency and free will

On this account, "will" names the control route from an agent's records to the world: the influence κ_{c}, together with the dependence of the action on the agent's records. A thermostat has a sensing route, a control route and a viability condition, and it is a viable agent in the weak sense of Definition 3.1. Organisms differ from it in degree along several axes: the length of the

window over which viability is maintained, the margin, the strength of sensing and control, the depth of memory, and whether the agent can revise its own selection rules. Agency is graded along these axes and is not an all-or-nothing property.

The traditional free-will question presupposes well-defined agents and causal histories. The present account supplies both relative to a declared partition and intervention set, and it turns part of the question into questions that can be answered: how strongly the agent's records, rather than the present state of the world, determine what it does, and whether its selection rules are themselves revised from its records. It does not settle the metaphysical question and is not offered as doing so.

The account says nothing about what it is like to act. The explanatory gap between functional organization and experience [15, 16] remains. One empirical route connects these notions to neuroscience, where habitual and flexible control have been associated with distinct cortico-basal ganglia loops [17, 18] and the sense of agency with motor prediction [19]; the relevant times there are classical control times, not quantum coherence times [20]. Whether the quantities defined here are useful for that research is open.

4 Part III: Existence without substance

4.1 Relational existence

Parts I and II treat causes and agents as real within declared contexts and dependent on them. This raises the question of their ontological status. Two extremes are available and both are mistaken: treating context-dependent structures as fundamental, and denying them objective reality because they depend on context. We propose structural realism about relational patterns [21, 22].

Definition 4.1 (Relational existence)**.** *An entity X exists relative to a context $\mathcal{S}$ if X is well defined and stable within $\mathcal{S}$, participates in objective relational patterns there, and may be absent or differently constituted in another context $\mathcal{S}'$.*

Definition 4.2 (Emptiness, technical sense)**.** *An entity is empty in the technical sense if it lacks being independent of every context while exhibiting stable patterns within specified contexts. Emptiness in this sense entails neither non-existence nor illusion.*

Paper B supplies physical instances of a context. In its commuting-coupling class, the stated stability and observability criteria select a unique largest admissible block algebra on a window. Smaller admissible algebras also exist. Selected content depends on coupling and window, yet the same inputs define the same criterion for all readers. Readable redundant labels additionally require actual recording couplings, fragment preparation and calibrated readout, as in Paper E; stability alone does not make them public.

The maintained body adds history and budget to the context. The sensor's operational distinction survives under record-guided repair and disappears under unmaintained damage. That difference is real in the task outcomes even for a reader unaware of the controller's policy. Relational objectivity concerns these reproducible consequences, not agreement on a chosen label.

4.2 Examples

Temperature depends on which degrees of freedom are treated as macroscopic, yet given that choice it is measurable and predictive. Quasiparticles such as phonons and magnons are collective excitations with no counterpart among the microscopic constituents; operationally they are stable peaks in spectral functions, dispersion relations, lifetimes and scattering amplitudes, all of which depend on temperature, doping and resolution. Subsystems are the case closest to

this framework. A state alone does not privilege one of its many factorizations, and Paper A shows that entropies, mutual informations and reconstructed geometries depend on the channel through which the state is read; Paper B shows how couplings select one structure over a window.

4.3 What is invariant

Objectivity does not require independence from every context. It requires that patterns be stable and predictive within a context and that the transformations between contexts be stated. What stays invariant depends on that class of transformations. Under a change of factorization, implemented by a unitary relabelling of the Hilbert space, the spectrum of the global state is invariant, while subsystem entropies, mutual informations and the geometries built from them are not. For readers applying the same selection criterion with couplings and window fixed, the selected algebra is unchanged. Access to its labels requires the declared record protocol; readers with different measurement permissions need not obtain the same information. A mere unitary change of coordinates is also different from physically changing a coupling.

Kind	Characterization	Example
Intrinsic existence	independent of every context	classical particles, if fundamental
Relational existence	dependent on context, objective within it	temperature, selected subsystems, records
Conventional existence	dependent on context and on agreement among agents	money, legal rights
Non-existence	absent from every relevant context	phlogiston, luminiferous ether

Table 1: Kinds of existence claim. The causal influences, agents and geometric features of this series belong to the second kind.

5 Part IV: Interpretive bridges to Buddhist philosophy

5.1 Method

The comparison below is between formal patterns. It does not claim historical influence in either direction, endorsement of Buddhist doctrine, or that physics confirms any religious view; "Buddhism" is not treated as one doctrine, and the discussion is limited to some Yogācāra and Mādhyamaka texts. One limitation must be stated plainly. The framework was named after the ālaya-vijñāna, so a resemblance between the two was partly built in by the naming and cannot count as independent evidence for either. The mathematical results of Parts I–III do not depend on the comparison; the comparison depends on them.

5.2 Yogācāra: traces that condition later appearance

Yogācāra texts describe the ālaya-vijñāna, the store consciousness, as the stream in which the impressions (*vāsanā*) of past actions are kept as seeds (*bīja*) that condition later experience [23, 24, 26, 27]. Vasubandhu's *Triṃśikā* says that it proceeds like a torrent; it is a changing stream, not a fixed container. The following correspondences are formal analogies.

The analogy concerns the role of traces. In both pictures what can appear later depends on what earlier processes have left behind, and the traces are not inert stores but conditions on later access. The differences are as important. The Yogācāra substrate is described as consciousness; the HAFF state is physical and the framework is neutral on mind and matter. The analogy does not establish panpsychism, idealism or the claim that quantum information is conscious.

Yogācāra notion	Counterpart in this series (analogy only)
Ālaya-vijñāna, a stream that keeps traces	the global state together with the records its history has formed
Bīja and vāsanā, seeds and impressions	records that change what later readouts can distinguish (Papers E, F; the Imprint papers)
Pravṛtti-vijñāna, the active consciousnesses	selected readouts and the effective subsystems they define (Papers A, B)
Āśraya-parāvṛtti, transformation of the basis	a change of couplings that changes which structure is selected, as opposed to a change of description (Paper B)

Table 2: Formal analogies. Yogācāra concerns consciousness and liberation; HAFF concerns physical structure, and its global state carries no phenomenal properties.

5.3 Mādhyamaka: dependence without a foundation

For Nāgārjuna, whatever arises dependently is empty of intrinsic nature (*svabhāva*), and that emptiness is itself a dependent designation, the middle way (*Mūlamadhyamakakārikā* 24:18) [25, 28, 29]. The technical emptiness of Definition 4.2 has the same form: denial of intrinsic being together with affirmation of functional reality.

Dependence also applies to the apparatus of selection. The structure that selects is itself conditioned: the selected algebra depends on couplings and on a window (Paper B), and a system that selects its own couplings does so through dynamics that its own registers do not select (Paper G). In a finite model the chain of selectors ends at that dynamics. The endpoint is not a foundation in Nāgārjuna's sense, an independently existing ground, because it is only the part of the dynamics that the given model leaves outside, and a larger model can take it in. Dependence therefore holds at every level without an infinite regress and without an independent base. Whether this resembles the Mādhyamaka middle way is a matter of interpretation. The resemblance is in form, and Mādhyamaka uses the insight to loosen attachment to views, whereas the physical account here describes how structure is selected and maintained.

5.4 Karma: actions, traces and later access

Buddhist accounts of karma hold that intentional actions leave traces that condition future experience [30, 34]. In the stated fresh-port process, an intervention on the source changes later records while detached earlier ports retain their marginals (Paper F). Records then condition later reading and action. In Section 3.4, they condition whether later sensing remains informative. The analogy is formal and stops at normativity. Karma classifies actions as wholesome or unwholesome; interventional influence and records carry no ethical valuation.

5.5 Where the traditions and the framework differ

The differences are substantive. Buddhist philosophy starts from first-person experience and aims at liberation from suffering; the physical results used here begin with third-person structure. Rebirth, karmic continuity across lives, cosmological realms and meditative attainment have no counterpart here, and the comparison neither supports nor undermines them. Yogācāra readings that give consciousness priority differ from the framework's neutrality. And the methods differ: contemplative inquiry, textual interpretation and dialectic on one side, mathematics and experiment on the other. The comparison is worth making because both traditions and the framework deny intrinsic subsystem structure while affirming stable relational patterns, and because precise statements on one side can sharpen questions on the other.

6 Conclusion

Causal influence is a property of a declared intervention process, not of correlation alone. It contracts through links and distinguishes processes whose observed records coincide. In the tracking task, sensing and control jointly limit memoryless viability; history can surpass that limit without raising the influence of the current world on the action.

The same finite task now includes a damageable body. An actual diagnostic controls repair of that body, whose condition determines the next sensor channel. The recurrence and matched calendar comparison show how a record maintains later intervention capability. Fresh blanks and spent history are explicit resources; this is a finite-stock mechanism, not a perpetual thermal cycle or the primitive origin of the initial body.

Agency so understood is a maintained relational organization. Partition, loss and permissions specify the task; success and failure are physical consequences, not matters of agreement. A fuller formation account must generate and coordinate the routes rather than supply all of them. The Buddhist comparison clarifies a related philosophical possibility: traces and dependent structures can have functional reality without an independently existing substrate. Its experiential and ethical questions remain independent of the physical mechanism.

A Reversible realization and checks

The body protocol is an explicit basis-permutation unitary on twelve bits. Besides continuing D, W, b, use fresh $(C, B, R, Q, A, F, J, G, V)$ with initial one-probabilities

$$(e, 0, \nu, \tfrac{1}{2}, \mu, q, d, 0, 0).$$

Here C, B are the diagnostic and repair ports; R, Q the sensor and fair-noise ports; A, F, J actuator, world and damage randomness; G, V blank damage and actuator ports. On diagonal states perform, in order:

1. $C \leftarrow C \oplus D$; swap D, B conditional on $C = 1$.
2. $R \leftarrow R \oplus W$; $R \leftarrow R \oplus DQ$.
3. Swap b, V; $b \leftarrow b \oplus R \oplus A$.
4. $W \leftarrow W \oplus F$.
5. $G \leftarrow G \oplus D$; $D \leftarrow D \oplus [(1 - G)J]$.

Each gate is reversible on all basis inputs, not just the initialized subspace. Since G starts at zero, the last step stores the post-repair body in G and gives $D_{\text{next}} = D' \vee J$. Repair uses $C = Y$ alone. The physical control of noise by the body does not grant a hidden diagnostic read. All ports remain in the spent bank, and the next slot uses fresh ones. Z reading of C, R is compatible with the diagonal protocol and does not change its joint law. Realizing these gates and their known orientations is a charged hardware permission.

The check enumerates all 4096 input strings and verifies that the composite is a permutation. For diagonal product inputs with $x = \Pr(D = 1)$ equal to $0, 1/5, 10/91, 1$, exact rational probabilities reproduce every term of (3) and (6). Old b does not affect the output: it remains in V. Joint entropy is unchanged. This is a circuit distribution check for the stated classical task, not unknown quantum-source or bath certification.

The calendar check verifies discrete convexity of G and the optimal $5, 6$ gap mixture at the stated rate. Independently, enumeration of the original three-reading task gives 0.10822784. The probability that the two older readings disagree is $\frac{1}{2}[1 - (1 - 2q)(1 - 2\nu)^2] = 0.1928$, giving present-world influence 0.138816. Analytic contraction and calendar proofs, rather than numerical sampling, establish the general statements.

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Selection, Maintenance and Gravitational Response: Modular Constraints and a Physical Return

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Abstract

Physical selection changes a gravitational source when it changes an actual coupling, preparation or maintenance process. We develop this route alongside a modular route that constrains which geometric laws can agree with selected algebraic data. The two routes have different inputs: modular stability does not select a unique algebra, and a supplied gravitational action does not explain the origin of that law.

Calibrated correction, projection selection and learned protection produce stress through the interactions that change access. We compute its Jackiw–Teitelboim (JT) response and construct a finite semiclassical mean-action return in which acquisition changes a material source and later sensing changes the next force. Quantum history, old references, support, synchronization, clock work and recoil are retained. Joint perturbation estimates allow bounded controller coupling to an unbounded common Hamiltonian on domain-preserving strong solutions. A separate full-mode two-field model matches its stress to a work-corrected source at rate $O(I^{-1})$ as clock inertia I grows; the energy correction leaves a nonzero pressure response. For a real-quartet carrier, an exact nonlinear normal-order conversion couples its actual full-mode quantum state, physical work clock and material director to constrained JT evolution. The resulting bulk solution has a prepared positive causal domain.

Maintenance is priced by the entropy removed from the full retained state. For a thermal chiral exhaust of capacity c_{ex}, its long-window heat rate obeys $\dot{Q} \geq k_B T \dot{s} + 3\hbar \dot{s}^2/(\pi c_{\mathrm{ex}})$. The complete output's relative entropy and entropy slack measure its excess above this floor. Direct export of the current purifier retains cross-species redundancy, so accurate renewal alone cannot make that protocol sharp. Matched horizon transport gives $\overline{\dot{S}_h} \geq \dot{s} + 6\dot{s}^2/c_{\mathrm{ex}}$ in the thermal rapidity clock. The actual thermal-source construction includes the running clock's nonextensive reference stress; physical realizations use finite operating windows. Pointer-aligned dephasing gives different complete renewal debits for the two selected axes. This identifies a candidate mechanism for a gravitational maintenance reading to favor the stable pointer algebra. Sustainable renewal, sharpness of the complete adaptive output and persistent causal program contrast form a joint conjecture, with drive routing, moving-clock conversion and support work charged. A common source action and the origin of geometric selection remain further targets.

1 The problem: what makes selection gravitational?

An apparatus changes which distinctions it can preserve and which operations it can execute. That change may reorganize a reconstructed geometry; it may also change the state of matter. A gravitational claim must say which of these changes it describes. Forgetting a record after an operation changes what can be inferred, while the emitted stress persists. Changing the interaction can change both access and the physical source.

The question of this paper is whether these two descriptions can meet: can the structures selected by physical operations supply geometric data whose consistency requires a gravitational law, and can the same operations produce sources whose response changes subsequent selection? HAFF-A, B and E provide the distinction between channels, stable content and actual records [19, 20]; Foundation I provides a hull relative to a readable seed [54]. Here the object to explain is the relation between these physical selections and gravitational dynamics.

One route starts with modular data. A supplied standard half-sided modular family generates translations. With a holographic dictionary and the required ball first laws, its variations match the linearized Einstein equation. Consciousness II supplies a different, same-spacetime conditional consistency law for a boosted ball family [24]. Neither route derives its geometric dictionary or consistency condition from modular stability alone. The SLD metric of distinguishability is positive semidefinite; it is neither the BKM Hessian of relative entropy nor a Lorentzian spacetime metric.

The other route starts with actual interactions. We derive sources from calibrated correction, selected projections and learned protection, then compute their response in a specified JT theory. The JT metric has fixed curvature in the minimally coupled sector; its dilaton and invariant respond. Direct dilaton-coupled support changes that sector explicitly. The resulting finite return retains a quantum reader and its old reference while its acquired record drives mechanics and new radiation. Its complete mean-action work and recoil are included. This realizes a causal source mechanism without assuming the proposed origin of the gravity law.

Sections 2–4 develop the derived sources, finite physical return and maintenance price. The modular route follows in Sections 5–8; the final conjectures ask how these routes could meet. The appendices retain the instruments, profiles, clocks and apparatus proofs. In particular, Table 1 states the objects and the interfaces between them. It separates established results from the growing-window conjecture; it is not a decomposition of one already completed apparatus.

2 From a Selected Interface to a Physical Source

A calibrated interface acts through a field coupling whose outgoing stress sources the supplied JT theory. This gives a physical selection–source map; the origin of the gravitational law in Conjecture 7.2 is a separate question.

2.1 What records can and cannot do to a source

Let a classical record $R \sim \mathrm{Bernoulli}(p)$ control whether a physical operation prepares the matter state ψ or leaves the reference ω. Without reading R, the matter state is $\rho_p = (1-p)\omega + p\psi$ and its stress is the mixture of the two stresses. Deleting or ignoring R after the preparation does not act on the matter, so ρ_p and its source are unchanged.

Object and supplied inputs	Result	Remaining interface
Modular family, dictionary and ball laws	Conditional translations and linearized geometric consistency (Sections 5–8)	Formation of the candidate family and its geometric dictionary.
Calibrated field operations and a JT action	Derived outgoing source, retarded response and finite echo/contact readout (Section 2; Appendix C)	Total apparatus stress and physical references; the action is an input.
Quantum core with classical mechanics and mean-action source	Finite retained-history return and strong-domain perturbation bounds (Section 3; Foundation III)	Fixed-window mean-action result; protected content, old references and initial work remain charged. No quantum stress tensor for the scalar fields or support is supplied.
Spatial quadratic fields and real-quartet bulk	Work-corrected full-mode stress matching, a finite spatial return, and a separately constrained quartet interior	Joining their receivers, moving endpoints, preparation and thermal ports in one material model.
Complete retained entropy and paid finite carrier	Long-window heat floor and neutral-sector sorting of a resettable source (Section 4; Proposition 9.3)	Actual adaptive source concentration, protected-content compatibility and local delayed readout.
One reused body over growing runs	Conjecture 9.2 jointly requires complete-output sharpness, normalized window contrast and causal transition gain	Sustained inlet, support, sensing and resource account in that same dynamics; none is supplied by sorting alone.

Table 1: Objects, results and interfaces. Finite constructions retain their stated inputs; the joint growing-window target is a conjecture.

Changing the preparation probability p changes the source. The information side has a matching statement. For the joint state $\rho_{RM} = (1-p)|0\rangle\langle 0| \otimes \omega + p|1\rangle\langle 1| \otimes \psi$ and the reference $\pi_R \otimes \omega$, with π_R the marginal of R, additivity of relative entropy on direct sums and data processing give

$$D(\rho_{RM}\|\pi_R \otimes \omega) = p\, D(\psi\|\omega), \qquad D(\rho_p\|\omega) \le p\, D(\psi\|\omega) \tag{1}$$

[51]. Losing the record lowers the distinguishability available to a reader; it does not remove the matter source. A gravitational consequence of selection must therefore pass through a preparation or interaction, and the conditional geometry of a branch, the average of branch geometries and the geometry of the averaged source are three different objects.

2.2 The source of a calibrated feedback actuator

The companion paper [21] realizes a corrective flip of a degenerate bit as a field coupling. A right-moving free massless scalar with null coordinate u, current $j = \partial_u \phi$ and stress $T_{uu} =: j^2 :$ is coupled through a real smooth switch g. We require a finite nonzero one-particle norm $\lambda^2 = \int_0^\infty \omega |\hat{g}(\omega)|^2 d\omega/(4\pi)$ and $\int (1+u^2)|g'(u)|^2 du < \infty$, where $\hat{g}(\omega) = \int g(u) e^{-i\omega u} du$ in the L^2 sense when needed. For example, $g \in H^1(\mathbb{R})$ together with the weighted derivative condition is sufficient. Smooth compact switches and the full-line packet below both belong to this class. The smeared current defines a Weyl unitary,

$$U = \exp\Big(-iX \int g(u) j(u)\, du\Big) = \exp\big[-i\lambda X (b + b^\dagger)\big], \tag{2}$$

where b is a normalized field mode with mean frequency $\bar{\omega}$ fixed by g, and $\lambda^2 \bar{\omega} = \frac{1}{4}\int g'^2 du$. With the drive mode in the coherent state $|\sqrt{N}\rangle$ and $2\lambda\sqrt{N} = \pi/2$, the coupling flips the bit with success probability $s_N = (1 + e^{-\pi^2/(8N)})/2$. From the current commutator $[j(u), j(v)] = \frac{i}{2}\delta'(u-v)$ one has $U^\dagger j(u) U = j(u) + \frac{1}{2} X g'(u)$ and

$$U^\dagger T_{uu}(u) U = T_{uu}(u) + g'(u) X j(u) + \tfrac{1}{4} g'(u)^2. \tag{3}$$

The cross term has zero expectation if the bit and the fresh field input are initially independent and $\mathrm{tr}(\rho X) = 0$, or if their joint state has no coherence in the bit's Z basis; a Z-diagonal bit marginal alone does not suffice [21]. Writing $j = \psi(u) b + \psi^*(u) b^\dagger + j_\perp$, the incoming coherent beam has stress $T_{\mathrm{in}} = 4N[\mathrm{Re}\,\psi(u)]^2$ and each activation adds

$$F_g(u) = \tfrac{1}{4} g'(u)^2 = 4\lambda^2 [\mathrm{Im}\,\psi(u)]^2. \tag{4}$$

This added stress is a property of the coupling. Its integral is the switching work $\lambda^2 \bar{\omega}$ supplied by the external switching agent.

The feedback task is that of the Imprint Chain [22] with the flip realized by this coupling. A hidden instrument label is estimated from calibration records with mismatch probability q; each task step reads the bit with reliability ν, accepts feedback with probability h, switches g on when accepted and the reading is negative, and lets a bath with contraction c act. Each opportunity receives its own scheduled incoming beam, independent of the bit and earlier records. In the companion paper the stationary loss $L_\infty(q)$ and the stationary coupling rate satisfy [21]

$$p(q) = h\nu L_\infty(q) + \tfrac{h(1-\nu)}{2}. \tag{5}$$

Proposition 2.1 (Source–loss relation)**.** *At fixed drive, switch and task parameters, the stationary mean source per opportunity outside the apparatus is* $\langle T_{uu}\rangle_q = T_{\rm in} + p(q)F_g$*. For two calibrations,*

$$\langle T_{uu}(u)\rangle_q - \langle T_{uu}(u)\rangle_{q'} = h\nu\,[L_\infty(q) - L_\infty(q')]\,F_g(u). \tag{6}$$

Proof. The incoming beam is present at every opportunity. A coupling occurs with stationary probability $p(q)$ and, under the joint initial condition, adds F_g in expectation by Eq. (3). Equation (5) gives the difference. □

Better calibration lowers the task loss and, through it, the source. The coefficient comes from the same interaction that performs the correction; no separate rule maps a calibration error to an emitted pulse. The relation does not assign stress to knowledge. Deleting the calibration record after the commands have been executed leaves the outgoing source unchanged, in line with Section 2.1. The relation is a stationary average per opportunity; a single trajectory, with its fixed hidden match label, does not follow the population mean step by step.

2.3 A source set by learned quantum protection

The activation-rate mechanism is not the only way that retained records can change a source. Brain Anchor [57] constructs two degenerate qubits with a programmable coupling

$$A_a = Z_1 + \alpha a Z_2, \qquad U_a = e^{-iA_aQ_f}, \quad Q_f = \int f(u)j(u)du = \sqrt{k}(b + b^\dagger). \tag{7}$$

The real smooth compact switch f and positive gain interval are supplied. The gain α is unknown, and each encounter receives a fresh independent vacuum field. On $\mathrm{span}\{|01\rangle, |10\rangle\}$ the coupling is $(1 - \alpha a)Z_L$. Calibration records from the commands $\pm h$ have means $r_\pm = e^{-2k(1\mp\alpha h)^2}$, so $\log(r_+/r_-) = 8k\alpha h$. The finite-record certificate in that paper bounds $|1 - \alpha a|$, including command rounding and known drift. It does not promise an exact fixed algebra after finite training.

Proposition 2.2 (One residual controls protection and source)**.** *Fix the acquired command* a*, write* $m = 1 - \alpha a$ *and* $F_f = f'^2/4$*, and keep the initial field independent of the logical body and retained ancillas. For every logical input, one encounter has added mean stress* m^2F_f *and switching work* m^2E_f*, with* $E_f = \int F_f du$*. After* M *independent identical encounters, with the output ports discarded and no interleaved gates or active recovery, its passive protection error is*

$$\epsilon_M = \tfrac{1}{2}(1 - e^{-2Mkm^2}). \tag{8}$$

For a single pulse with stress moments $E_j = \int u^j F_f du$ *and the initial data of Proposition 2.4, the late invariant is*

$$\mathcal{C}(m) = H_0^2 + 2\kappa H_0 m^2 E_1 + \kappa^2 m^4 (E_1^2 - E_0E_2). \tag{9}$$

Thus retained history can change the physical protection and the gravitational response at fixed pulse shape and activation schedule.

Proof. The current shift is $j \mapsto j + A_a f'/2$. The centered independent field input makes the stress cross term vanish for every logical state, and $A_a^2 = m^2 I_L$ on the code. Its characteristic function multiplies the logical off-diagonal by e^{-2km^2} per encounter; a dephasing channel's half-diamond error is its phase-flip probability. Time ordering may give a factor $e^{i\vartheta_f A_a^2}$, which is scalar on the code but not on the whole two-qubit register. Finally substitute $m^2 F_f$ in the fixed-initial-data source solution below. □

This link does not require a rule that emits a pulse only after a wrong answer. The same interaction produces both the residual error and its source, even when the logical body is entangled with retained memory. At $m = 0$ the body decouples in the code; a noiseless stored qubit need not continuously export entropy. The finite-rate entropy-export bound of Section 4 applies to actual renewal, not to every protected state. Calibration itself adds field energy $2N(1 + \alpha^2 h^2)E_f$ for N independent tests at each sign. Preparations, measurement, the controller and switching mechanism add further charges. The source suppression during operation alone is not a total-energy saving.

Brain Anchor also treats an unknown centered Gaussian variance K_ω. Logical records at $0, \pm h$ give $K_\omega = -\log r(0)/2$ and $\alpha = [\log r(h) - \log r(-h)]/(8hK_\omega)$; two settings suffice when the signs of $1 \mp \alpha h$ are known. Its finite-record mismatch certificate needs no supplied temperature label. The coherence is now $e^{-2K_\omega m^2}$, but the coupling-added source remains $m^2 F_f$ on centered independent input.

Incoming stress must still be added. A u-clock chiral KMS input has background $\pi/(12\beta^2)$ in natural units. Within a supplied bounded one-temperature family, strictly decreasing K_β permits a finite-error background estimate; a general Gaussian variance does not determine incoming energy. At common preparations and initial data, the linear dilaton equation compares coupling-source differences. Equation (9) cannot be used for the full source after dropping its thermal background, nor is u thermality the rapidity-clock reference of Section 4.

The same command can also operate on one reused vacuum field. For nonnegative compact switches f_j on disjoint intervals, with no noncommuting logical gates, the smeared currents commute but have negative correlations. Brain Anchor gives exact coherence $e^{-2m^2 k_M}$ with $k_M = \langle (\sum_j Q_{f_j})^2 \rangle \le Mk$, and added source $m^2 \sum_j f_j'^2/4$ of energy $Mm^2 E_f$. Its JT response uses the actual pulse-train moments and initial coefficients in a valid operating domain. No independent-port approximation, thermal input or infinite-lifetime worldline is inferred.

There is a finite causal pulse in this construction. For example, set $f(u) = A \exp[-1/(1 - (u-3)^2)]$ on $2 < u < 4$, zero outside, and normalize its one-particle norm to $k = .2$. Numerical quadrature gives $(E_0, E_1, E_2) = (.4568522, 1.3705565, 4.3215400)$. For $H_0 = 1$, $\kappa = .05$ and $\alpha = 1.1$, the supplied command $a = 48/55$ has residual $m = .04$ and gives $\mathcal{C} \simeq 1.00021929$; the comparison $a = 1$ has $m = -.1$ and $\mathcal{C} \simeq 1.00137053$. These are deterministic evaluations of the displayed pulse, with code and outputs in `supplemental/pulse_debit`, not statistics of a calibration pilot. For a random learned command, the invariant of the mean source uses $(\mathbb{E}m^2)^2$, whereas the mean branch invariant uses $\mathbb{E}m^4$ in the quadratic term. The compact source ends before a later boundary echo is scheduled, but the calibrated wall and timing instrument remain the inputs of Section C.1.

Stress readback also has an exact blind direction. At fixed $a_0 > 0$, the gains $\alpha_\pm = (1 \pm d)/a_0$ give residuals $\mp d$ and therefore the same protection channel and source. Field parity conjugates their full code interactions while leaving the vacuum and stress invariant. No

parity-covariant stress instrument with invariant outcome labels can distinguish them, nor can their mean JT responses at common initial data. Their correcting commands $1/\alpha_\pm$ differ. Probes at $a_0 \pm h$ break this ambiguity through the opposite signs of $8k\alpha h(1-\alpha a_0)$ in their log contrast. A gravitational response is therefore not a complete calibration reference for the interface that produced it. Off-frame coupling probes, or a signed field quadrature, are additional access, with their own sources and costs. Full output-field parity and a conditional logical pulse can instead restore the stored state without identifying the gain, as shown in Ref. [57]. This recovery leaves the emitted source unchanged. Active state correction and learning a match that suppresses the actual source are therefore distinct operations.

The gain is a physical actuator parameter, not a freely changed description. The selected finite code is not the geometric algebra of Section 8, and the supplied JT action still supplies the gravitational response. A link to continuous modular selection requires that further interface; it is not obtained by identifying a protected M_2 sector with a half-sided modular inclusion.

2.4 A physical seed changes the source

The rate-controlled model changes how often a fixed interaction occurs. The coupled observable can itself change, giving a direct relation between an exactly represented algebra and a source. Keep the incoming field, switch and activation schedule fixed. On a qubit put $P_\theta = (I + \cos\theta\, Z + \sin\theta\, X)/2$ and use $U_\theta = \exp\bigl[-i\lambda P_\theta(b + b^\dagger)\bigr]$. The projection selects a pointer algebra and its source population under this fixed schedule.

Proposition 2.3 (Selected projection and outgoing stress)**.** *For $\lambda \neq 0$, an initially independent qubit state ρ and normal field state σ, the exactly represented algebra of the induced qubit channel is $\{P_\theta\}'$. Let $r_\theta = \mathrm{tr}(\rho P_\theta)$. The stress formula below requires its displayed current moments to exist. The outgoing field is*

$$(1 - r_\theta)\sigma + r_\theta e^{-i\lambda Q}\sigma e^{i\lambda Q}, \qquad Q = b + b^\dagger,$$

and its stress is

$$T_\theta(u) = T_{\mathrm{in}}(u) + r_\theta \left[g'(u)\langle j(u)\rangle_\sigma + \tfrac{1}{4}g'(u)^2\right]. \tag{10}$$

Thus two selected projections can change the source at fixed global input state and fixed activation schedule.

Proof. The quadrature has non-atomic spectral distribution in a normal state, so the multiplicative-domain theorem of Q-RAIF C gives $\{P_\theta\}'$ [14]. Expanding U_θ in its two projection blocks and tracing the qubit gives the field mixture; off-diagonal blocks have zero trace even when ρ does not commute with P_θ. In the current identity replace the Pauli X by P_θ and use $P_\theta^2 = P_\theta$. Initial independence gives (10). □

For $\rho = (I + 0.8Z)/2$, the seeds P_0 and $P_{\pi/2}$ select different pointer algebras, with $r_0 = 0.9$ and $r_{\pi/2} = 0.5$. In the worked coherent packet, the cross profile is $8\lambda\sqrt{N}\,\mathrm{Re}\,\psi\,\mathrm{Im}\,\psi$. Its first three moments vanish, so the invariant increment is (62) with p replaced by r_θ. With the table's $N = 16$ and background it is 2.0807581×10^{-5} versus 1.1559770×10^{-5}. The full source also contains the cross profile, which enters the local dilaton even though its first three moments do not enter the final invariant.

Here the source weight is derived from the state's population in the selected projection; it is not assigned to an external mismatch label. Changing the projection means

changing a physical interaction, not renaming an unchanged algebra. The two algebras are abstractly isomorphic but are differently embedded relative to the fixed state and coupling. This supplies a seed–interface–source link. It does not identify that finite-dimensional pointer algebra with a Type III geometric candidate or explain why the coupling selects this seed rather than another. Those are the remaining selection and geometry bridges.

2.5 Response of a given Jackiw–Teitelboim model

Take the JT action [52, 53]

$$I = \frac{1}{16\pi G}\int \sqrt{-g}\,\Phi(R+2/L^2) + I_{\rm matter}, \qquad ds^2 = -\frac{4L^2\,du\,dv}{(1+uv)^2}, \tag{11}$$

with the usual boundary terms, $\kappa = 8\pi G$, $D = 1+uv$ and the vacuum dilaton $\bar\Phi = H_0(1-uv)/D$, $H_0 > 0$. The field equations are $\nabla_\mu\nabla_\nu\Phi - g_{\mu\nu}\Box\Phi + g_{\mu\nu}\Phi/L^2 = -\kappa T_{\mu\nu}$, and in these coordinates

$$\Phi_{uu} + \tfrac{2v}{D}\Phi_u = -\kappa T_{uu}, \qquad \Phi_{vv} + \tfrac{2u}{D}\Phi_v = -\kappa T_{vv}, \qquad \Phi_{uv} + \tfrac{2}{D^2}\Phi = \kappa T_{uv}. \tag{12}$$

The matter is a conformal theory with matched left and right central charges, and the state-independent anomaly is removed by subtracting the same background. The left-moving input is unchanged, and the state-dependent source is right-moving, $T_{uu} = F(u)$ with $T_{uv} = T_{vv} = 0$.

Proposition 2.4 (Response with fixed initial data)**.** *Let F be integrable with $\int |u|^2|F|du < \infty$. On a connected coordinate domain with $D \neq 0$, write the perturbation as $\Phi - \bar\Phi = (A(u) + vB(u))/D$ and impose the initial coefficient data $A(u), B(u) \to 0$ as $u \to -\infty$. For a source bounded below in support this is equivalent to zero perturbation before the source. The unique solution with these data is*

$$\Phi = \bar\Phi + \frac{A(u)+vB(u)}{D}, \qquad A = -\kappa\int_{-\infty}^{u}(u-z)F(z)dz, \qquad B = -\kappa\int_{-\infty}^{u} z(u-z)F(z)dz. \tag{13}$$

The perturbation is linear in F. With $E_j = \int u^j F\,du$, after the source (or in the asymptotic limit for sources with tails) the invariant $\mathcal{C} = \Phi^2 - L^2(\nabla\Phi)^2$ takes the vacuum value

$$\mathcal{C} = H_0^2 + 2\kappa H_0 E_1 + \kappa^2(E_1^2 - E_0E_2), \tag{14}$$

and inside the support $\partial_u\mathcal{C} = \kappa F\,(2H_0u + uA - B)$.

Proof. Write $\psi = D\,(\Phi - \bar\Phi)$. The vv equation of (12) reads $\psi_{vv} = 0$, so $\psi = A(u) + vB(u)$, and the other two equations reduce to $A'' = -\kappa F$, $B'' = uA''$ and $B' = uA' - A$, which Eq. (13) satisfies. The absolute first and second moments make both retarded coefficients tend to zero at $-\infty$. Two solutions differ by $\alpha + \beta u + \gamma v - \alpha uv$, whose coefficients are $A_h = \alpha + \beta u$ and $B_h = \gamma - \alpha u$. Their two zero limits force $\alpha = \beta = \gamma = 0$. Every vacuum solution has numerator $\alpha + \beta u + \gamma v - \alpha uv$ and invariant $\alpha^2 + \beta\gamma$; after the source $\alpha = H_0 + \kappa E_1$, $\beta = -\kappa E_0$, $\gamma = \kappa E_2$. The balance law follows by differentiation. □

The coefficient condition cannot be replaced by $\Phi - \bar\Phi \to 0$ at each fixed v. The homogeneous solution $\eta v/(1+uv)$ has that limit for every fixed v along nonsingular

points. It even tends to zero uniformly on the subdomain $u < 0$, $v \le 0$, since its absolute value is at most $|\eta|/|u|$; no uniform limit near $D = 0$ is implied. Adding it leaves the source unchanged but changes the late invariant by $-\kappa E_0 \eta$. It is excluded by $B(-\infty) = 0$. Source comparisons below hold H_0, the coordinate matching and both coefficient boundary data fixed.

The JT curvature stays $-2/L^2$; the dilaton and the invariant respond. A dilaton difference is not by itself a higher-dimensional curvature or a mass, and comparing boundary proper times needs a boundary normalization.

The matching condition matters. Set $A_F, B_F \to 0$ at $+\infty$, or exactly to zero after a compact source, instead of the initial data of Proposition 2.4. The final-state-matched solution differs from the retarded solution by

$$A_F = A_R + \kappa(uE_0 - E_1), \qquad B_F = B_R + \kappa(uE_1 - E_2).$$

For the chiral diffeomorphism states of Consciousness I [25], with a Cut at a to the left of the support, the relative entropy is $D_a = 2\pi \int (u-a)F(u)\,du$. The final-state-matched solution gives $\delta\Phi(a,0)/(4G) = -D_a$, while the initial-state-matched solution gives $\delta\Phi(a,0) = 0$ at a Cut before the pulse. The first identity compares an earlier section with a matched final state; it does not say that a pulse removes area before it arrives. The dilaton–modular-energy relation itself is known [58]; what is used here is its type-correct pairing with a specified source and matching condition.

The same equations show that a null-only record of the source is incomplete. A genuine vacuum-density shift $T_{\mu\nu} = -\delta\rho\, g_{\mu\nu}$ has zero null components but gives the particular solution $\delta\Phi = \kappa L^2 \delta\rho$. A probe that reads only null energy cannot certify that a source is absent, which is the source-completeness condition relevant to the vacuum curvature test of Ref. [26].

3 Response that changes the next operation

A source-to-geometry calculation becomes a return only when the geometry changes an actual record and that record changes a later interaction. The readout permission matters. In the minimally coupled JT sector, instruments on the same matter metric with the same inputs have identical adaptive record laws even when the dilaton differs (Proposition C.1). A boundary tied to a fixed dilaton value evades that obstruction. Its echoes reconstruct the three vacuum coefficients, and a source-compensated inverse includes finite probe backreaction (Propositions C.2 and C.3). An exact current contact instead obtains an invariant-dependent bit, including the contact's own source (Propositions C.5 and C.6). The bit selects the next physical projection. Its same-field conditional covariance remains part of the history; detector replacement does not reset the field.

These are finite physical records, not permission to read an exact metric. The worked current contact has total variation only 2.3650×10^{-4} between its two initial geometries. Its fixed-cutoff energy budget also forbids indefinitely many contacts of unchanged positive source charge. Appendix C contains the timing, stress and accumulated-source bounds.

3.1 A finite prefix is a sufficient response object

The source need not have finite second moment over all time to determine a finite episode. This matters for actual material radiation, whose tails are not inferred from a finite

smooth-source certificate.

Proposition 3.1 (Finite-prefix response and its source tolerance)**.** *Fix $u_* < U$ and matched data (a_*, b_*, γ_*) at u_* in the traceless single-chiral JT sector on a common patch with $1 + uv \neq 0$. For $F \in L^1([u_*, U])$, define*

$$a(u) = a_* + \kappa \int_{u_*}^{u} zF(z)dz, \quad b(u) = b_* - \kappa \int_{u_*}^{u} F(z)dz, \quad \gamma(u) = \gamma_* + \kappa \int_{u_*}^{u} z^2 F(z)dz.$$

Then $\Phi = (a + bu + \gamma v - auv)/(1 + uv)$ solves the JT equations on this window with those data, and $\mathcal{C}(u) = a(u)^2 + b(u)\gamma(u)$. No future-source assumption enters either reading. If two sources have $\int_{u_*}^{U}(1 + z^2)|\delta F|dz \leq D$, then

$$\begin{aligned} |\delta a|, |\delta b|, |\delta\gamma| &\leq \kappa D, \\ |\delta\Phi| &\leq \frac{\kappa D(1 + |uv| + |u| + |v|)}{|1 + uv|}, \\ |\delta\mathcal{C}| &\leq \kappa D(2|a| + |b| + |\gamma|) + 2\kappa^2 D^2. \end{aligned} \tag{15}$$

Proof. The coefficient derivatives are $a' = \kappa uF$, $b' = -\kappa F$ and $\gamma' = \kappa u^2 F$. They cancel from the first derivatives of the numerator, giving the field equations in their weak sense. Direct substitution in $\Phi^2 - L^2(\nabla\Phi)^2$ gives $a^2 + b\gamma$. The three integrands are bounded in absolute value by $1 + z^2$. Expanding the numerator and invariant gives the estimates. □

For the complete quantum material, Q-RAIF C bounds smooth interacting stress tests on one positive Hamiltonian with all old references retained [14]. Its cross and creation contributions are essential. The bound is a test-functional certificate, whereas Eq. (15) uses a weighted L^1 certificate. They are not interchangeable. With a smooth preparation cutoff, smearing a finite-resolution dilaton reading gives smooth retarded kernels. Each such kernel can be used as a source test with its own Q-RAIF C form constant. Recovering a sharp prefix additionally requires a bound on unresolved source structure. This gives a direct finite-window interface without claiming an unproved all-time tail estimate or discarding controller and support sources.

3.2 The same quantum reader, support and source

A reflecting boundary alone does not supply its support. The material branch uses a positive dilaton-dependent mass and varies the worldline. It contains the finite reader, history clock, blank and old purification in its mass. Classical scalar fields couple to an angle θ; storage y has positive kinetic and spring energies. Their emitted powers are $P_\pm = g_\pm^2 \dot{\theta}^2$, derived by characteristics, and their reaction is included in the mechanical equation. The two exterior invariants evolve with the proper worldline and obey the mass junction. Appendix F gives this complete mean-action construction and its admissible positive domain.

The quantum core is

$$H_c = H_{\text{hist}} + E_{\text{off}} I + \lambda(\Phi - \phi_0) Z_P + \frac{\chi_R}{2} P_R (y - d)^2.$$

Its first active clock edge writes an actual Y_P outcome; the next swaps the probe with a paid blank. The record spring reacts while acquisition is in progress. Variation gives

$F_y = \chi_R(y-d)\langle P_R\rangle$ and $F_\Phi = \lambda\langle Z_P\rangle$, with $\dot{E}_c = F_y\dot{y} + F_\Phi\dot{\Phi}$. Consequently $M = m_{\rm rest} + U + e + E_c$ obeys $\dot{M} = (U' + F_\Phi)\dot{\Phi} - P_+ - P_-$. These are quantum expectations in a retained pure-state calculation, not a sampled record fed to an independent source generator.

Proposition F.7 proves a nonzero source return through its shortest-path expansion. Either $\lambda = 0$ or $\chi_R = 0$ removes the proper-power difference between its two initial geometries. Both interactions present produce a nonzero coefficient at order t^{2r+7}, where r is the acquisition edge. The finite 33-state example retains its old probe reference and all work balances. This is a mean-worldline realization; quantum support fluctuations are not removed by taking the quantum expectation of its force.

The reader must also remain accurate when the response is allowed to move, rather than being supplied beforehand. The following estimate controls this coupled problem.

Proposition 3.2 (Joint tolerance of an acquired-record source return)**.** *Write the mean-action equations as $\dot{z} = \mathcal{F}(z,\Psi)$ and $i\dot{\Psi} = H_c(z)\Psi$, retaining the old reference in Ψ. Allow $H_c(z) = H_* + V_c(z)$, where H_* is a common semibounded self-adjoint internal Hamiltonian and V_c and its first two coordinate derivatives are bounded on the stated domain. For an unbounded core, require strong initial data in $D(H_*)$ and propagators preserving this common domain along the stated coordinate paths. This includes the finite core above and regular Gaussian internal data with bounded controller coupling; it does not include an unbounded coordinate-force operator merely because its free field is Gaussian. Include the complete mass M among the classical variables, with its actual initial value and work equation*

$$\dot{M} = (U' + \langle\partial_\Phi V_c\rangle)\dot{\Phi} - P_+ - P_-.$$

Let exact and approximate solutions, with the same core and reference space, lie in a compact subset of the positive domain of Proposition F.4, with $|y| \le Y$ and $|\dot{\theta}| \le V$. Use the maximum norm for classical variables nondimensionalized in fixed material and clock units. Suppose the approximate equations have classical residual at most $r_z(t)$, including their mass-work residual. Let $r_H(t)$ bound the Hermitian Hamiltonian error applied to the approximate quantum vector; the propagators and this residual must exist on their specified domains. Choose constants on this domain such that

$$|\mathcal{F}(z,\Psi) - \mathcal{F}(\widehat{z},\widehat{\Psi})| \le L_z|z-\widehat{z}| + L_\Psi\|\Psi - \widehat{\Psi}\|.$$

Here the same reference is compared, not just its reduced probe state. For the displayed sensor and spring the Hamiltonian Lipschitz constant is $L_H = |\lambda| + \chi_R(Y + |d|)$. For any $\sigma > 0$ put $L = \max\{L_z + \sigma L_H, L_\Psi/\sigma\}$ and $E = |z-\widehat{z}| + \sigma\|\Psi - \widehat{\Psi}\|$. Then

$$E(t) \le e^{Lt}E(0) + \int_0^t e^{L(t-s)}[r_z(s) + \sigma r_H(s)]ds. \tag{16}$$

Any terminal record probability differs by at most $2E/\sigma$, and the total proper emitted power differs by at most $2\gamma VE$, where $\gamma = g_+^2 + g_-^2$. These estimates cover the actual quantum-force reaction in the classical response, including its contribution to mass and clock conversion.

Proof. For normalized states, $|\langle A\rangle_\Psi - \langle A\rangle_{\widehat{\Psi}}| \le 2\|A\|\|\Psi - \widehat{\Psi}\|$. In particular the P_R and Z_P expectations have that bound. The force differences are bounded by

$$|\delta F_y| \le \chi_R|\delta y| + 2\chi_R(Y + |d|)\|\delta\Psi\|, \quad |\delta F_\Phi| \le 2|\lambda|\|\delta\Psi\|,$$

The mechanical work $-F_y\dot{y}$ cancels its contribution $F_y\dot{y}$ to the internal energy, leaving the stated mass equation. Thus mass is compared through its initial value and integrated work, not by bounding $\langle H_* \rangle$ with the state-vector error. The mass, square roots $\beta_\pm$ and null-clock denominators are smooth with bounded derivatives on the stated compact domain; these formulas give finite L_z, L_Ψ for the actual coupled equations. All clock and material energy remains in the initial mass. The quadratic spring and the probe term give the displayed L_H. Duhamel comparison using the exact unitary propagator gives

$$\|\delta\Psi(t)\| \le \|\delta\Psi(0)\| + \int_0^t [L_H|\delta z(s)| + r_H(s)]ds.$$

It contains no spurious growth from the norm of a common clock Hamiltonian. Add the classical integral bound with weight σ and apply Gronwall. Finally $P_+ + P_- = \gamma\dot{\theta}^2$ gives the power estimate, and a record effect has norm at most one. □

For the strong-domain solutions just specified, the Schrödinger equation gives

$$\frac{d}{dt}\langle H_* + V_c(z)\rangle = \sum_j \langle \partial_j V_c(z)\rangle \dot{z}_j.$$

The common H_* contributes no explicit coordinate work. In its interaction picture only bounded perturbation differences enter Duhamel's estimate, so the same proof applies without bounding $\langle H_* \rangle$ by a vector-norm error. Extending to all energy-form initial data would additionally require an energy-controlled approximation preserving this work identity. Spatial field observables still require the separate material form bounds of Q-RAIF C. This extension treats the core as internal material of the specified thin worldline; it does not equate spatial field stress with a point source. Smooth tests of the line stress $Mu^\mu u^\nu \delta_\Gamma$ are Lipschitz on the same compact tube, so their errors are bounded by a constant times $\int E(s)ds$. The constants and admissible tube may change when the initial clock mass grows; no uniform-in-clock-size geometric limit follows.

The work equation is necessary even when the retained quantum states are close. Q-RAIF C gives a positive clock-driven oscillator whose actual total energy is mJ, whereas its scheduled product-clock target has energy $mJ + b^2/(2\omega)$ for every m, despite vanishing joint state error [14]. A nearby rotated clock can match that scalar energy, but does not thereby match local stress. For a scheduled comparison with internal Hamiltonian jumps $h_{b-1} \to h_b$, subtract the switching work $W_b = \langle h_b - h_{b-1}\rangle_{t_b}$ from its clock ledger. Together with the continuous coordinate work this gives the mass equation used above. Initial mass mismatch and any missing work belong in $E(0)$ and r_z; they cannot be dropped by declaring the clock state error small. This ledger is not the energy expectation of the unmodified product-clock target.

A source contrast remains nonzero whenever the two power-error bounds sum to less than its ideal contrast at a chosen time. Thus finite implementation errors can be admitted while preserving the causal sensor-removal and feedback-removal tests. The domain condition is verified by a strict ideal margin and an error tube smaller than that margin; it cannot be certified from a reduced-state fidelity alone. These are the fixed-window mean-action bounds in Table 1.

Foundation III uses this action for a frame learned while the original quantum content is held [56]. In a finite positive parameter family, the acquired record changes the apparatus's own source; its geometric response enters an X-basis reading of the old probe into

the already spent cell. The first-order geometry signal is generated after record acquisition, with no new probe or blank port. The matched intervention keeps the complete initial state, mass and geometric constraints fixed and removes only the first record's mechanical use. A sensor-edge diagnostic separates the geometric path from direct probe and cursor effects. This is one finite semiclassical mean-action scalar/JT return. The completed-window geometry-edge contrast, averaged over the stated plant-sign prior, is about -4.98×10^{-9} in the example; its small size is reported separately from the task advantage. It is distinct from the spatial realization below and from the sustainable sharp-exhaust conjecture.

3.3 A complete spatial clock-work source

The work correction can be a spatial tensor rather than a scalar debit. Consider the local quadratic action

$$S_{\chi u} = -\frac{1}{2}\int \sqrt{-g}\left[I(\nabla\chi)^2 + (\nabla u)^2 + I\mu^2\chi^2 + (\omega u + \lambda\chi/\omega)^2\right] d^2x, \tag{17}$$

where $I, \mu, \omega > 0$. In canonical variables $(\sqrt{I}\chi, u)$ its mass matrix is

$$\begin{pmatrix} \mu^2 + \lambda^2/(I\omega^2) & \lambda/\sqrt{I} \\ \lambda/\sqrt{I} & \omega^2 \end{pmatrix}.$$

Its determinant is $\mu^2\omega^2 > 0$. Constant orthogonal diagonalization gives two positive-mass Klein–Gordon fields with their complete spatial mode towers. Choose the interacting static vacuum and a finite-energy Weyl displacement with initial data $\chi = u = \dot{u} = 0$, $\dot{\chi} = \Omega f$, $f \in C_c^\infty$. Its added clock energy in flat space is $I\Omega^2\|f\|_2^2/2$. One fixed local renormalization prescription is used for the interacting vacuum and all displaced states.

Put $L_a = \partial_t^2 - \partial_x^2 + a^2$. In flat space let

$$L_\mu c = 0, \qquad L_\omega u_0 = -\lambda c, \qquad L_\mu \eta_0 = -\lambda u_0 - \lambda^2 c/\omega^2, \tag{18}$$

where c has the prepared clock data and the other two fields have zero data. Define

$$T^{(a)}_{\alpha\beta}[q] = \partial_\alpha q \partial_\beta q - \tfrac{1}{2} g_{\alpha\beta}[(\nabla q)^2 + a^2 q^2], \tag{19}$$

$$B_{\alpha\beta}[c, \eta] = \partial_\alpha c \partial_\beta \eta + \partial_\alpha \eta \partial_\beta c - g_{\alpha\beta}(\nabla c \cdot \nabla \eta + \mu^2 c\eta). \tag{20}$$

The work-corrected comparison source is

$$T_{\mathrm{cmp}} = IT^{(\mu)}[c] + B[c, \eta_0] + T^{(\omega)}[u_0] - g(\lambda c u_0 + \lambda^2 c^2/(2\omega^2)) + T_{\mathrm{vac},I}. \tag{21}$$

The large positive clock source and the same interacting-vacuum source are present on both sides of the comparison.

Proposition 3.3 (Full-mode stress matching)**.** *For each fixed finite window and compact smooth tensor test W, the actual coherent quantum state of* (17) *satisfies*

$$\left| \int \sqrt{-g}\, W^{\alpha\beta}(\langle T_{\alpha\beta}\rangle - T_{\mathrm{cmp},\alpha\beta}) \right| \le C_W/I. \tag{22}$$

The constants are independent of I above a finite threshold. In flat space the comparison obeys the exact local Ward identity. The source difference also tends to zero in every fixed C^k norm on a compact window. The same statements hold on a fixed static metric using its finite-window Klein–Gordon energy constants.

Proof. The actual means obey $L_\mu \chi_I = -\lambda u_I/I - \lambda^2 \chi_I/(I\omega^2)$ and $L_\omega u_I = -\lambda \chi_I$. For $\eta_I = I(\chi_I - c)$ these imply the exact identities

$$L_\mu \eta_I = -\lambda u_I - \lambda^2 \chi_I/\omega^2, \qquad u_I - u_0 = -\lambda G_\omega \eta_I / I,$$

where G_ω is the zero-data retarded operator. Use $\|q\|_{s,a}^2 = \|\dot{q}\|_{H^s}^2 + \|q_x\|_{H^s}^2 + a^2\|q\|_{H^s}^2$. The energy inequality is $\|G_a F(t)\|_{s,a} \le \int_0^t \|F(r)\|_{H^s} dr$. Set $C = |\Omega| \|f\|_{H^s}$, $a_T = |\lambda| T/\mu$, $b_T = |\lambda| T/\omega$, $d_T = \lambda^2 T/(\omega^2 \mu)$ and $h_T = a_T b_T + d_T$. For $I \ge 2h_T$ it gives

$$\sup \|\eta_I\|_{s,\mu} \le 2h_T C, \qquad \sup \|u_I - u_0\|_{s,\omega} \le 2a_T h_T C/I, \tag{23}$$

$$\sup \|\eta_I - \eta_0\|_{s,\mu} \le 2h_T^2 C/I. \tag{24}$$

These are full spatial energy estimates, without a mode cutoff. The exact quantum covariance equals the interacting-vacuum covariance; only the mean is displaced. Metric variation therefore gives

$$\langle T \rangle = I T^{(\mu)}[c] + B[c, \eta_I] + I^{-1} T^{(\mu)}[\eta_I] + T^{(\omega)}[u_I] - g(\lambda \chi_I u_I + \lambda^2 \chi_I^2/(2\omega^2)) + T_{\mathrm{vac},I}.$$

Polarization of each quadratic tensor and the displayed energy estimates prove (22). Smooth initial data, higher Sobolev estimates and the wave equations give the C^k result. Finally $\nabla^\alpha B_{\alpha\beta} = (\lambda u_0 + \lambda^2 c/\omega^2)\partial_\beta c$; the material and contact terms have the opposite divergence. □

The correction has a measurable tensor component. Relative to the common clock and vacuum terms in (21), the flat-space preparation gives

$$\delta T_{tt} = O(t^4), \qquad \delta T_{xx} = -\lambda^2 \Omega^2 f^2 t^2/\omega^2 + O(t^4). \tag{25}$$

The clock pays the order-t^2 energy transfer; pressure does not cancel. This subtraction defines a comparison, not a second experiment with the coupling removed and a different vacuum. For metric-only matter on a fixed AdS patch, the matched sources drive the linear JT dilaton equation with the same initial constraints and boundary data. Its finite-window response inherits the source estimate. A positive dilaton domain may require initial homogeneous data growing with I; that resource and a detector's gain cannot be removed from the account. This is the quadratic clock/material entry of Table 1; its receiver and thermal interfaces are separate.

At finite spatial bandwidth, these source and readback steps can be placed in one mean-action evolution. Proposition F.8 includes an elastic bar, a physical synchronization field, two retained quantum spins and the full oscillator Hilbert space of the chosen modes. The two preparations have the same initial gravitational constraints. Their later pressure contrast produces a dilaton contrast, which changes the retained sensor and the next material force. The synchronization field is varied, not supplied as a prescribed clock. This establishes the finite spatial return listed in Table 1.

The retained quartet admits a full-mode bulk step toward that construction. Proposition F.9 evolves its physical clock, material director and exact renormalized quantum source together with constrained JT data. The normal-order conversion and causal-state identification are nonlinear. The result supplies an interior patch for specified global quantum data; moving endpoints, the other gate windows and the finite-horizon renewal error must still be joined to it.

4 Maintenance and its gravitational price

4.1 Entropy removal and the source it can constrain

A fixed unitary correction can return its control port coherently. Removing entropy injected by unobserved noise is a different task. To price it, the state whose entropy is removed must include the controller's usable side information. The heat account must also distinguish a thermal reservoir from a supply of pure memories.

HAFF G supplies a record-driven program and executor in which selecting a physical coupling changes its source under one fixed executor [55]. The dephased quartet below adds a specified full renewal debit. It does not identify the two material realizations or inherit a thermal controller from that program loop.

Proposition 4.1 (Maintenance entropy account)**.** *Let $V = SC$ include the maintained system and every retained controller and side-information register. A correction acts unitarily on finite-dimensional VAB, initially in $\rho_V \otimes \alpha_A \otimes \tau_B$, where $\tau_B = e^{-\beta H_B}/Z$, $\beta = (k_B T)^{-1} > 0$. Put*

$$s = S(V) - S(V'), \qquad p = S(A') - S(A), \qquad Q = \operatorname{tr} H_B(\rho'_B - \tau_B).$$

Then, with $I_{V:A:B} = S(V) + S(A) + S(B) - S(VAB)$,

$$\beta Q + p = s + I_{V':A':B'} + D(\rho'_B \| \tau_B) \ \geq s. \tag{26}$$

If the marginal entropy of C is returned, s equals the decrease of $S(S|C)$. In a sequence of such corrections, a bounded total purity consumption and a positive asymptotic entropy-removal rate $\dot{s}$ imply $\liminf Q_{[0,t]}/t \geq k_B T \dot{s}$.

Proof. Unitary invariance of total entropy and the initial product give $\Delta S(V) + \Delta S(A) + \Delta S(B) = I_{V':A':B'}$. Also $\beta Q = \Delta S(B) + D(\rho'_B \| \tau_B)$. Combine the identities and sum over corrections. The rate statement follows when the accumulated p divided by elapsed time tends to zero. □

This is the Landauer entropy balance, with the memory-purity term displayed explicitly; the equality and the role of accessible side information are standard [13, Theorem 3 and Section 5.1]. For initially correlated or nonthermal inputs the same calculation instead gives

$$\beta Q + p = s + (I_{\text{out}} - I_{\text{in}}) + D(\rho'_B \| \tau_B) - D(\rho_B \| \tau_B).$$

These consumed correlations and bath free energy can pay for correction. A classical copy of the error permits a reversible conditional flip with $s = 0$, although the system's marginal entropy decreases. Negative quantum conditional entropy gives another reason to include the side information. A pure-memory swap has $p = s$ and can have $Q = 0$. A retained oscillator's closed unitary loop also has no entropy removal by itself. Thus a count of corrections or a marginal noise entropy is not automatically $\dot{s}$. In a cyclic device $\dot{s}$ must be established from its actual noise and controller dynamics. For nondegenerate systems, heat and supplied work are further separated by the change in internal energy.

The memory can also be reused without being decorrelated after each correction. Let correction i act on VA and a fresh thermal port B_i, initially independent of the whole state of VA and all earlier ports, and put $s_i = S(V_i) - S(V'_i)$ and $x_i = S(B'_i) - S(B_i)$. Suppose each intervening noise step is a CPTP map on V, tensored with the identity on A

and the previously exported ports; updates involving A belong to the charged correction. Entropy conservation gives

$$x_i = s_i - [S(A_i') - S(A_i)] + I(V_i' : A_i') - I(V_i : A_i) + I(V_i' A_i' : B_i').$$

Data processing between corrections and telescoping therefore imply

$$\sum_{i=1}^{n} x_i \geq \sum_{i=1}^{n} s_i - [S(A_n') - S(A_1)] - I(V_1 : A_1). \tag{27}$$

For a fixed finite A, the last two terms are bounded. Reused memory purity and initial correlations cannot absorb a positive entropy-removal rate indefinitely. Here x_i refers to each exported port's marginal entropy. Summing those marginals can overcount correlated records. There is a stronger joint-port statement under the same fresh-input and local-noise assumptions:

$$S(B_{1:n}') - \sum_{i=1}^{n} S(\tau_{B_i}) \geq \sum_{i=1}^{n} s_i - [S(A_n') - S(A_1)] - I(V_1 : A_1). \tag{28}$$

To see this, retain all previously exported ports in the mathematical state. A noise step on V, with the identity on those ports, changes its total entropy by $\Delta S(V) - \Delta I(V : AB_{<i}) \geq \Delta S(V)$, by data processing. The corrections preserve total entropy, and each new port adds $S(\tau_{B_i})$. At the end, subadditivity between VA and all exported ports gives Eq. (28) after the V entropies telescope. Thus finite reused memory cannot hide the maintained rate in correlations between outputs.

The same account applies when the noise also acts on the retained controller. Put $W = VA$ and let each correction be a unitary on WB_i with a fresh, independent input τ_{B_i}, uncorrelated with W and the earlier ports. Noise between corrections acts on W alone. With $s_i^W = S(W_i) - S(W_i')$, the exact identity is

$$S(B_{1:n}') - \sum_{i} S(\tau_{B_i}) = \sum_{i} s_i^W + I(W_n' : B_{1:n}') + \sum_{i=1}^{n-1} \ell_i \;\geq\; \sum_{i} s_i^W, \tag{29}$$

where $\ell_i = I(W_i' : B_{1:i}') - I(W_{i+1} : B_{1:i}') \geq 0$ is the information lost to the intervening noise. Indeed, that noise changes the entropy of the joint state by $\Delta S(W) + \ell_i$; the correction preserves it. Summing these changes and expanding the final entropy as $S(W_n') + S(B_{1:n}') - I(W_n' : B_{1:n}')$ gives the identity. Returning an old port to an operation would change these hypotheses. The joint VA rate, rather than a sum of system marginal entropies, is what an apparatus with noisy internal memory must export.

Proposition 4.2 (A finite-rate chiral exhaust floor)**.** *Consider a unitary, zero-chemical-potential chiral conformal exhaust with central charge $c_{\mathrm{ex}} > 0$, thermal input at $T \geq 0$, and an extensive entropy-current account. Energy and temperature are defined by the same translation generator and clock. Assume that the outgoing modes export net dimensionless entropy current $x \geq \dot{s} > 0$, where $\dot{s}$ is the actual maintenance entropy-removal rate after all retained side information and sublinear resource consumption are accounted for. The entropy-current maximum at fixed energy is the thermal one. Then the excess heat current over the incoming thermal stream obeys*

$$\dot{Q} \geq k_B T \dot{s} + \frac{3\hbar}{\pi c_{\mathrm{ex}}} \dot{s}^2. \tag{30}$$

Proof. Write $a = \pi c_{\mathrm{ex}} k_B^2/(12\hbar)$. The thermal energy and dimensionless entropy currents are $J_E(T) = aT^2$ and $J_S(T) = 2aT/k_B$. The first is the standard chiral CFT current [11, Eq. (1)]; the second follows from its extensive Gibbs entropy. Entropy maximization at fixed energy is the Gibbs variational principle, underlying the single-channel information-flow bound [10]. Choose $T_* = T + k_B x/(2a)$ so that the thermal output has the required entropy. Any output with at least that entropy has energy current at least aT_*^2. Subtracting aT^2 gives

$$\dot{Q} \geq k_B T x + \frac{k_B^2 x^2}{4a} = k_B T x + \frac{3\hbar x^2}{\pi c_{\mathrm{ex}}}.$$

The expression increases for $x \geq 0$. □

The current account is a physical condition on the exported modes, not the entropy of a Type III local interval. Equation (28) establishes the needed joint entropy rate in a fresh-port account; matching those ports to a conformal current must preserve its capacity and entropy budget. At fixed capacity and positive rate the quadratic term survives even at $T = 0$. Thermal output saturates the channel inequality, but this alone does not construct a correcting apparatus. The currents are long-window, extensive Gibbs limits. For an observation window τ with net exported entropy X_τ, bounded rates and sublinear boundary corrections give

$$Q_\tau \geq k_B T X_\tau + \frac{3\hbar X_\tau^2}{\pi c_{\mathrm{ex}} \tau} + o(\tau).$$

The current formula is not an exact inequality for an arbitrarily short finite packet. Applying the Gibbs bound to the entire output also includes correlations between packets; it does not require each output to be a separate thermal state.

Proposition 4.3 (Output deficit and direct renewal)**.** *Use the complete outgoing carrier B_τ, its physical translation generator H_τ, and the thermal input of Proposition 4.2. Let the actual charged entropy removal satisfy $s_\tau/\tau \to x > 0$, set $T_* = T + 6\hbar x/(\pi c_{\mathrm{ex}} k_B)$, and write $\tau_* = e^{-H_\tau/(k_B T_*)}/Z_\tau$. Assume the extensive Gibbs limits used above and that input, charge, interaction, resource and boundary remainders are $o(\tau)$. If the full entropy account gives $L_\tau = S(\rho_{B_\tau}) - S_T - s_\tau \geq -o(\tau)$, then*

$$Q_\tau - \tau\left(k_B T x + \frac{3\hbar x^2}{\pi c_{\mathrm{ex}}}\right) = k_B T_* \left[D(\rho_{B_\tau} \| \tau_*) + L_\tau\right] + o(\tau). \tag{31}$$

Thus asymptotic equality requires, and under these hypotheses is equivalent to, both deficit and entropy-slack rates tending to zero.

For Q-RAIF C's original quartet preparation followed by direct per-species ring renewal [14], let d_{ring} bound its reset error against the actual old outside marginal, η its residual zero-mode amplitude, and e_H the full-joint clock comparison error over the specified horizon. Suppose each round has disjoint emission modes, these bounds hold uniformly, and no later converter changes the exported modes. With $t = d_{\mathrm{ring}} + 2\arcsin\eta + 2e_H < 1/2$, the output of M rounds obeys

$$D(\rho_{B_\tau} \| \tau_*) \geq M\left[h(\epsilon) - 2h(t)\right], \qquad h(p) = -p\ln p - (1-p)\ln(1-p), \tag{32}$$

when the zero-chemical-potential free-rail reference factorizes across the original species. If $M/\tau \to \nu$ and $h(\epsilon) > 2h(t)$, the asymptotic heat excess above the floor is at least $k_B T_* \nu[h(\epsilon) - 2h(t)]$ per unit time. Temporal independence of thermal packets is not required.

Proof. The finite-carrier Gibbs identity is $D(\rho\|\tau_*) = (E_\rho - E_*)/(k_B T_*) - S(\rho) + S_*$. Here $S_* - S_T = x\tau + o(\tau)$ and $E_* - E_T = \tau[k_B T x + 3\hbar x^2/(\pi c_{\mathrm{ex}})] + o(\tau)$, which gives (31).

The current purifier's occupation pair is 00 or 11, with probabilities $1-\epsilon, \epsilon$. Operations on the receiver leave its joint marginal with previous outputs unchanged. The reset certificate compares that pair with the same actual past marginal; it need not be independent of the current readout record. The exact linear scattering column emitting an old zero mode d_j has squared norm $1-\eta^2$. CAR therefore give the extracted mode $b_j = \sqrt{1-\eta^2}\, d_j + \eta e_j$, where e_j is a canonical orthogonal mode in the actual body and incoming field. A two-mode rotation of angle $\arcsin\eta$ compares b_j with d_j, retaining every old reference. The two purifier species cost $2\arcsin\eta$; replacement of the comparison joint state and its outside marginal costs at most $2e_H$.

Measure their even occupations in the disjoint emitted modes, giving $X_{1:M}, Y_{1:M}$. For each round the joint law of $(X_j, X_{<j})$ is within variation t of $\mathrm{Bern}(\epsilon) \otimes P_{X_{<j}}$, and $\Pr(X_j \neq Y_j) \le t$. Conditional binary entropy continuity and the binary Fano bound give $H(X_j|X_{<j}) \ge h(\epsilon) - h(t)$ and $H(X_j|Y_j) \le h(t)$. Hence $I(X_{1:M} : Y_{1:M}) \ge M[h(\epsilon) - 2h(t)]$. The reference measurement law is $q_X q_Y$, even though each sequence may be thermally correlated. Data processing and $D(P_{XY}\|q_X q_Y) = I(X:Y) + D(P_X\|q_X) + D(P_Y\|q_Y)$ prove (32). □

All entropies in this proposition use the finite-carrier account and its extensive limit, not a trace entropy of a Type III interval. A fixed-charge canonical reference needs its sector-weight correction; it has the same rate conclusion only when that correction is sublinear. A conditional account must use the same charged removal. For a complete usable system $W = ZM$ with a conserved classical record Z, write $s^W = S(W) - S(W')$. A unitary interaction with a port B gives

$$S(B'|Z) - S(B|Z) = s^W + I(M' : B'|Z) - I(M : B|Z).$$

When the port is initially independent of W, its thermal entropy replaces $S(B|Z)$ and the last term vanishes. Old purifications remain in the physical state; they are usable side information only if the apparatus has that access. Fresh-port sequences with noise acting only on M also accumulate nonnegative information losses to earlier outputs. Reading an old output or updating Z needs a joint account of that operation.

For a classical record X retained at input and output, put $L_c = S(B_{\mathrm{out}}|X) - S(B_{\mathrm{in}}|X) - s_\tau$. When the input marginal entropy is S_T, the same Gibbs identity is

$$\begin{aligned} Q_\tau - \tau f(x) &= k_B T_* \left[D(\rho_{B_{\mathrm{out}}}\|\tau_*) + L_c + I(X : B_{\mathrm{out}}) - I(X : B_{\mathrm{in}})\right] + o(\tau), \\ f(x) &= k_B T x + \frac{3\hbar x^2}{\pi c_{\mathrm{ex}}}. \end{aligned} \tag{33}$$

Indeed $L_\tau = L_c + I_{\mathrm{out}} - I_{\mathrm{in}}$, while $D(\rho_{XB}\|\rho_X \otimes \tau_*) = D(\rho_B\|\tau_*) + I(X:B)$. These are two forms of one account; thermal input marginals alone do not remove old correlations. The direct protocol's deficit lower bound tends to $\nu h(\epsilon) > 0$ as its certified error vanishes. A physical converter that redistributes those correlated records, with its own work and retained memory charged, is therefore necessary for that route to approach the sharp floor.

The incoming state and conserved charges matter. For a nonthermal stream with energy current e_{in} and dimensionless entropy current σ_{in}, put $E_{\mathrm{th}}(\sigma) = 3\hbar\sigma^2/(\pi c_{\mathrm{ex}})$. The same variational argument gives

$$\dot{Q} + e_{\mathrm{in}} - E_{\mathrm{th}}(\sigma_{\mathrm{in}}) \ge k_B T_\sigma x + \frac{3\hbar x^2}{\pi c_{\mathrm{ex}}}, \qquad T_\sigma = \frac{6\hbar\sigma_{\mathrm{in}}}{\pi c_{\mathrm{ex}} k_B}. \tag{34}$$

The nonequilibrium energy above the entropy-matched thermal stream can pay for export. A pure coherent input therefore cannot be charged as though it were a zero-energy thermal reservoir. At nonzero chemical potential the corresponding Gibbs comparison must hold all retained charges fixed. For example, with a level-one $U(1)$ current normalized as in Ref. [12, Eq. (84)], the unitary CFT Gibbs currents in units $\hbar = k_B = 1$ are $e = \pi c_{\rm ex} T^2/12 + \pi\mu^2/2$, $j = \pi\mu$ and $\sigma = \pi c_{\rm ex} T/6$ [12, Eq. (93)]. Equal incoming and outgoing charge currents cancel the chemical energy in their difference; an apparatus that absorbs charge can instead consume it. Equation (30) does not cover that consumption without a charge ledger.

For independent exhausts at a common temperature, with central charges c_j and nonnegative entropy currents x_j, the same argument and Cauchy–Schwarz give

$$\dot{Q}_{\rm tot} \geq k_B T \dot{s} + \frac{3\hbar \dot{s}^2}{\pi \sum_j c_j}, \qquad \sum_j x_j \geq \dot{s}. \tag{35}$$

The capacity term is sharp at the channel level: equality requires thermal outputs and $x_j = c_j \dot{s}/\sum_k c_k$. Additional channels are an additional physical resource. Intermittent export through the same capacity cannot lower the long-window Gibbs floor: apply it to the complete output with its total entropy. If a local-current description is valid, $\overline{x^2} \geq \bar{x}^2$ gives the same conclusion.

Heat alone does not bound every geometric observable. Even a positive E_0 can give zero or negative $\delta\mathcal{C}$ in Eq. (62); its response depends on other moments and on the background. A positive weighted integral $\int K(u)F(u)du \geq K_{\rm min} \int F du$ requires both a lower bound on K over the exhaust and $F \geq 0$. Quantum excess stress need not have that sign. There is, however, a simple JT observable which reads E_0 without either assumption.

Proposition 4.4 (A heat account read through the dilaton)**.** *Compare two JT solutions with the same H_0, boundary coefficient data and coordinate and clock normalization. Suppose their source difference F is the exhaust of the correction in Proposition 4.1, with finite absolute second moment and $\int F du = Q$. This last equality requires that the specified outgoing channel carry all the heat in the same energy normalization. Then its retarded coefficient difference obeys*

$$-\frac{A'(+\infty)}{\kappa} = Q \;\geq\; k_B T (s - p). \tag{36}$$

For a compact exhaust and any $u_2 > u_1$ after it, the left side is exactly $-[A(u_2) - A(u_1)]/[\kappa(u_2 - u_1)]$. No pointwise sign of F is required.

Proof. Differentiate Eq. (13): $A'(u) = -\kappa \int_{-\infty}^{u} F(z)dz$. Take the late limit and apply Eq. (26). □

The reference for Q is the incoming thermal reservoir transported through the same exhaust, before its heat uptake; H_B is its u-translation generator. A thermal background has a nonintegrable stationary flux, so the JT solution here is the difference from that same transported thermal background. The identification $E_0 = Q$ is an energy-transport hypothesis: other exhaust channels, energy stored in the device, work beams crossing the observation surface and redshift must be accounted for separately. With a fixed transport factor $E_0 = \zeta Q$, Eq. (36) acquires the same factor ζ. The coefficient slope is read from

the dilaton in the declared matched coordinate frame, not an invariant under arbitrary changes of boundary clock. The relation links an actual entropy-removal process to a specified dilaton reading. The JT dynamics and heat-routing map are its geometric inputs.

4.2 A horizon reading of the maintenance price

For a matched running single-chiral JT source, let $\mathcal{C} = R^2 > 0$ and $\xi = (bu^2 + 2au - \gamma)/(2R) > 0$ on the regular future outer patch. The marginal horizon has $\Phi_h = R$ and $S_h = S_0 + 2\pi R/\kappa$. In its rapidity clock $d\eta = du/\xi$,

$$\frac{dS_h}{d\eta} = 2\pi F\xi^2.$$

When the charged heat is exactly $F\xi^2$ and the physical exhaust input is thermal at $T_H = 1/(2\pi)$ in this same clock, the long-window channel bound gives

$$\overline{\dot{S}_h} \geq \dot{s} + \frac{6\dot{s}^2}{c_{\mathrm{ex}}}.$$

Appendix D proves the horizon balance and its domain conditions. The entropy here is classical Wald entropy; a generalized-entropy statement needs its matter contribution.

Steady positive flux cannot be extrapolated in one fixed u chart: its boundary dilaton numerator eventually vanishes. Nor does a new clock prepare a reservoir. For an actual source field thermal in η, the vacuum-subtracted physical flux contains $q_{\mathrm{ref}} = -c(\delta' + \delta^2/2)/(24\pi)$, where $\delta = R'/R$. Its self-consistent equations are

$$R' = R\delta, \qquad \delta' = \frac{\kappa q - R\delta}{\kappa c/(24\pi)} - \frac{\delta^2}{2}.$$

The positive growing family of Proposition D.5 has $R = R_0 + \kappa q\eta + O(1)$ and finite accumulated reference correction. Hence its asymptotic horizon rate is $2\pi q = s + 6s^2/c$. A partially transmitting local thermal contact produces a nonthermal output with a strictly larger energy price at the same entropy current; changes of angle add their own pump work. The detailed state, source and clock calculations are in Appendix D. Their global source-clock solution does not grant infinite life to a finite material energy reserve.

5 The Factorization Problem

5.1 No canonical tensor factorization

A Hilbert space of dimension $d = d_1 d_2$ admits a tensor factorization, but its Hilbert structure alone does not select one. Commuting observable algebras can induce the subsystem structure [17, 18]. Which of those algebras an apparatus can use is a physical question about its couplings and resources.

5.2 Coarse-graining and effective descriptions

Paper A [19] gives the following conditional separation.

Proposition 5.1 (Conditional coarse-graining separation)**.** *Let $\Phi_1 \neq \Phi_2$ be finite-dimensional coarse-graining channels with a common output space. Their outputs differ for generic pure inputs. Entropy profiles differ only under the additional nondegeneracy hypothesis stated in Paper A, and metric inequivalence follows only when a supplied reconstruction dictionary separates those profiles modulo diffeomorphisms.*

Channel inequivalence therefore establishes a difference of effective descriptions, not by itself a difference of spacetime metrics. Paper A also decides, for qubit reductions, exactly when two channels give the same output entropy on every pure input, and it constructs one global state, one family of local noise channels and one fixed reconstruction rule that yield non-isometric finite metrics. Metric inequivalence of spacetime geometries remains a conditional model claim throughout this paper.

5.3 Geometry from entanglement

In a holographic model the Ryu–Takayanagi formula [27] relates an extremal area to boundary entropy,

$$S_A = \frac{\mathrm{Area}(\gamma_A)}{4G_N\hbar}, \tag{37}$$

and motivates an entanglement account of connectivity [28]. Such a reconstruction requires its dictionary. Outside a specified model, the accessible data determine operator relations rather than a spacetime metric.

6 A Structural Distinction: Gauge Forces and Gravity

Gauge interactions evolve matter and connection fields within a supplied localization structure. The algebraic proposal asks whether gravity instead describes how physical correlations, accessible operations and their stability select that structure. With a geometric dictionary, changing those relations can change effective distances and causal structure. The preceding sources test a necessary interface: changed access reaches geometry through a specified matter process.

If the proposal holds, quantum gravity must describe selection of localization as well as fields within it. This is not an explanation of why quantization fails. Background independence [29, 30] asks for covariance under changes of selected geometric structure, not equality of the metrics assigned to operationally different algebras. Selection is constrained by interactions and stability, rather than subjective choice [20]; changed knowledge affects stress only through an operation.

A common reconstruction dictionary gives every matter representation in one selected geometric sector the same metric. This conditional universality is not a derivation of free fall or equality of inertial and gravitational mass. Degrees outside the sector cannot be declared geometrically irrelevant while changing its metric: a dark-sector application would need an enlarged source algebra and explicit backreaction map. No dark-matter or dark-energy conclusion follows here.

7 Technical Formulation

7.1 Setup

We separate the operator-algebraic and finite-dimensional regimes result by result. Modular statements use a Type III_1 ambient factor with a faithful normal state and explicit cyclic and separating hypotheses. Density matrices, traces, the quantum Fisher information metric and the qubit chain are finite-dimensional statements. The Rindler example is an algebraic quantum field theory check. The linearized Einstein equation is imported only in the holographic regime of Theorem 8.6, and the JT model of Section 2 is a supplied two-dimensional gravity theory.

Definition 7.1 (Selected geometric algebra)**.** *A* ***selected geometric algebra*** $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ *is a von Neumann subalgebra chosen as input to a specified algebra-to-geometry reconstruction model. It is not Paper E's probe-reachable effect system, pointer or record algebra, or environmental fixed set. A coarse-graining model must declare the channel or inclusion relating any of those objects to* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$*.*

No canonical tensor factorization is assumed. State restrictions and relative-entropy data on $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ can be associated, as in holography and tensor networks, with an effective distance structure. That structure depends on the choice of $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$, on the stability of correlations under coarse-grained dynamics and on the redundancy of the encoding; it is an effective description valid in the context the algebra defines.

7.2 Central conjecture

Conjecture 7.2 (Gravity as adiabatic algebra evolution)**.** *Gravitational dynamics corresponds to an* ***adiabatic flow*** *of selected geometric algebras* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t)$*:*

1. *the global state evolves unitarily,* $|\Psi_U(t)\rangle = U(t)|\Psi_U(0)\rangle$*;*
2. *a model-dependent rule selects a geometric candidate* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t) \subset \mathcal{M}_U$ *at each time;*
3. *as the state evolves, the selected candidate may shift,* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t) \to \mathcal{A}_{\mathrm{geo},\mathbf{c}}(t+dt)$*;*
4. *this shift manifests phenomenologically as the dynamical curvature of spacetime.*

Unitary evolution of observables within a fixed $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ *manifests as gauge interaction. Gauge dynamics is evolution within the algebra (fixed stage, moving actors); gravitational dynamics is evolution of the algebra itself (moving stage).*

The conjecture answers one objection: if algebras were kinematical background, gravity could not be dynamical. Here the selection of which algebra is stable is state dependent, and state evolution induces algebra flow. The selection rule that produces the path is not derived in this paper; Section 4 returns to what a physical selection would have to supply. The adiabatic assumption takes algebra transitions to be slow compared with dynamics inside the algebra. Rapid transitions would correspond to strong-field regimes where the effective geometric description breaks down, and the dynamics of non-adiabatic transitions is left open.

7.3 Comparison with quantum information geometry

For comparison consider the quantum Fisher information metric (QFIM) on a smooth finite-dimensional family of faithful effective states $\rho(\lambda)$ [36, 37],

$$g_{\mu\nu}(\lambda) = \frac{1}{2}\mathrm{Tr}\left(\rho(\lambda)\{L_\mu, L_\nu\}\right), \qquad \partial_\mu\rho = \frac{1}{2}(\rho L_\mu + L_\mu\rho). \tag{38}$$

It is coordinate independent, reduces to the classical Fisher metric in commuting limits, and measures how hard nearby states are to distinguish. It is also positive semidefinite, whereas a Lorentzian spacetime metric is indefinite and carries causal structure. Nothing in Eq. (38) supplies the negative direction, so the QFIM is a metric on model space and an analogy, not the spacetime metric. A change $\mathcal{A}_{\mathrm{geo},\mathbf{c}} \to \mathcal{A}_{\mathrm{geo},\mathbf{c}} + \delta\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ may change this state-space metric; calling the change a spacetime perturbation requires an additional Lorentzian reconstruction map. In AdS/CFT, relative entropy and the entanglement first law constrain bulk perturbations [38, 39]; those results use the holographic dictionary and do not make an arbitrary boundary QFIM the bulk metric.

8 Algebraic Foundation: Modular-Stable Candidates

This section places one candidate class in Tomita–Takesaki modular theory. It does not translate measurement accessibility or environmental redundancy into modular conditions, and it does not select a unique algebra.

Standard modular definitions and their two checks are in Appendix A.

8.1 Modular definition of candidate algebras

Let $\mathcal{M}_U$ be a Type III_1 factor in standard form on $\mathcal{H}_U$, let $|\Psi_U\rangle$ be cyclic and separating, and write $\omega(\cdot) = \langle\Psi_U| \cdot |\Psi_U\rangle$. Then ω is faithful and normal and its ambient modular group $\sigma_t^{\omega,U}$ exists. Let $\{\mathcal{E}_t\}_{t\geq 0}$ be a normal unital completely positive (UCP) semigroup acting in the Heisenberg picture on $\mathcal{M}_U$.

Definition 8.1 (Modular-stable geometric candidate)**.** *A von Neumann subalgebra $\mathcal{A}_{\mathrm{geo},\mathbf{c}} \subset \mathcal{M}_U$ is a **modular-stable geometric candidate** if:*

(U1) ***Faithful restriction.*** *$\omega|_{\mathcal{A}_{\mathrm{geo},\mathbf{c}}}$ is faithful and normal. In the faithful ambient setting this holds for every unital subalgebra, so it is a regularity condition, not a selector.*

(U2) ***Ambient modular stability.*** *$\sigma_t^{\omega,U}(\mathcal{A}_{\mathrm{geo},\mathbf{c}}) = \mathcal{A}_{\mathrm{geo},\mathbf{c}}$ for all real t.*

(U3) ***Semigroup stability.*** *$\mathcal{E}_t(\mathcal{A}_{\mathrm{geo},\mathbf{c}}) \subseteq \mathcal{A}_{\mathrm{geo},\mathbf{c}}$ for all $t \geq 0$.*

Remark 8.2 (Typed relation to the measurement objects)**.** *Conditions (U1)–(U3) do not derive Paper E's implementable effects* Eff*, their operator system* $\mathrm{Eff}_{\mathrm{span}}$ *or the pointer algebra $\mathcal{P}_{\mathrm{env}}$, and setwise semigroup stability is not the exact fixed set* $\mathrm{Fix}(\mathcal{E})$*. A downstream model may specify a channel or inclusion from those objects into $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$; without it the relation is motivational. Environmental redundancy does not imply faithfulness of a state restriction. A causal restriction map and a Lindblad evolution are different types, and each model must supply its own $\mathcal{E}_t$ and verify invariance.*

8.2 Selection is not settled by these conditions

Remark 8.3 (No uniqueness from (U1)–(U3))**.** *Conditions (U1)–(U3), even with only scalars fixed by both dynamics, do not determine a unique candidate. There are four independent obstructions.*

1. *Uniqueness of the modular flow for a fixed pair $(\mathcal{A}, \omega)$ does not select the algebra $\mathcal{A}$.*

2. *A normal completely positive map may preserve two subalgebras separately without preserving their join, because it is not multiplicative. Join preservation holds for modular automorphisms but not for a general $\mathcal{E}_t$; it does hold for subalgebras of the fixed algebra of a state-preserving dynamics, on which $\mathcal{E}_t$ acts as the identity [54].*

3. *Pointwise ergodicity constrains fixed elements, while (U2)–(U3) are setwise conditions; no implication to a unique maximal element of the lattice was proved.*

4. *With a faithful ambient state, $\mathcal{M}_U$ itself satisfies (U1)–(U3), so a bare maximality rule selects the ambient algebra unless an independent seed, properness, read-out or minimal-hull condition is supplied.*

Replacing $\mathcal{B}(\mathcal{H}_U)$ by a faithful Type III factor repairs the existence of the ambient modular flow but none of these selection failures. The class of candidates may contain several members that are not unitarily equivalent, and classifying it requires additional structure. A nontrivial selection principle could start from a foreground seed and construct a minimal invariant hull, or impose an explicit read-out constraint. Foundation I carries out the first route for a supplied seed: under state-preserving dynamics the hull of a seed in the fixed algebra lies between that seed and the fixed algebra. For a minimal compact symmetry, it equals the seed-stabilizer fixed algebra when that algebra is preserved by the hull generators and the hull contains the group-fixed algebra [54]. These hypotheses establish uniqueness relative to the seed; the seed's physical origin is a further question.

8.3 Connection to linearized gravity

Theorem 8.4 (Entanglement first law in a density-matrix setting)**.** *Let ρ_0 be a faithful finite-dimensional density matrix and $\rho_\varepsilon = \rho_0 + \varepsilon\,\delta\rho + O(\varepsilon^2)$ a normalized perturbation, with $K_0 = -\log\rho_0$. Then $\delta S = \delta\langle K_0\rangle$. An infinite-dimensional extension requires entropy differentiability and control of the modular-energy variation; faithfulness alone is insufficient. For a Type III local algebra the statement must instead use Araki relative entropy.*

Proof sketch. $S(\rho_\varepsilon\|\rho_0) = -S(\rho_\varepsilon) + \langle K_0\rangle_{\rho_\varepsilon}$ plus a constant is nonnegative and stationary at $\varepsilon = 0$. □

Conjecture 7.2 posits a family $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(\lambda)$. Modular theory treats two kinds of perturbation differently. For a state perturbation on a fixed algebra the Connes cocycle $u_t = (D\omega' : D\omega)_t$ relates the automorphism groups by $\sigma_t^{\omega'}(A) = u_t\sigma_t^{\omega}(A)u_t^*$ [50, 40]. This is not a left-multiplication formula for the two full modular operators. If the algebra family is generated by the positive translation of an HSMI, $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(a) = e^{iaP}\mathcal{A}_{\mathrm{geo},\mathbf{c}}(0)e^{-iaP}$,

and the vacuum is fixed by this translation, its full modular generator obeys $K(a) = e^{iaP}K(0)e^{-iaP}$. On a common smooth domain,

$$\left.\frac{dK(a)}{da}\right|_{a=0} = i[P, K(0)]. \tag{39}$$

Equation (39) concerns an HSMI-generated family and a derivative of matrix elements on that domain, not arbitrary subalgebra changes or an operator-norm Taylor expansion. The one-sided CHM charge used for a ball below is distinct from the full modular generator; their relevant local-unitary expectation differences agree, but the operators themselves do not.

Theorem 8.5 (Linearized gravity from entanglement, [39])**.** *In a holographic CFT dual to Einstein gravity in asymptotically AdS spacetime, the entanglement first law $\delta S = \delta\langle K_{\rm mod}\rangle$ for all boundary balls is equivalent to the linearized bulk Einstein equations $G^{(1)}_{\mu\nu} + \Lambda g^{(1)}_{\mu\nu} = 8\pi G_N T^{(1)}_{\mu\nu}$.*

The derivation uses the ball modular Hamiltonian [47]

$$K^B_{\rm mod} = 2\pi \int_B d^{d-1}x\, \frac{R^2 - |\mathbf{x} - \mathbf{x}_0|^2}{2R}\, T_{00}(x), \tag{40}$$

together with the Ryu–Takayanagi formula and the relation between bulk and boundary modular Hamiltonians [48]; complementary arguments are due to Jacobson [49].

Theorem 8.6 (Conditional holographic matching for an HSMI-generated family)**.** *Let $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ be a modular-stable candidate in a holographic CFT vacuum representation, and assume:*

(H1) the AdS/CFT correspondence holds;

(H2) the Ryu–Takayanagi formula and its quantum corrections hold;

(H3) the JLMS relation between bulk and boundary modular Hamiltonians holds;

(H4) the compared algebra family is an explicitly supplied HSMI/ball-region family whose generator P implements the boundary-region variation used in the holographic first-law argument;

(H5) the regulated entropy variation, or its relative-entropy replacement, satisfies the hypotheses of the Faulkner et al. theorem for all required balls.

Then Eq. (39) *describes the modular-generator variation within that family, and the holographic first law matches the linearized Einstein equations by the imported result:*

$$\begin{aligned} \textit{HSMI/ball family (H4)} &\Longrightarrow \delta K_{\rm mod} \xrightarrow{\textit{(H5)}} \delta S = \delta\langle K_{\rm mod}\rangle, \\ \textit{(H1)–(H3)} &\Longrightarrow \delta(G_{\mu\nu} + \Lambda g_{\mu\nu}) = 8\pi G_N T^{(1)}_{\mu\nu}. \end{aligned} \tag{41}$$

The theorem does not show that a generic stability-selected algebra variation is an HSMI translation, nor that HAFF algebra flow is gravitational dynamics.

8.4 The circularity issue

Jacobson's thermodynamic derivation [34] presupposes a background geometry, while HAFF wants geometry to emerge from algebra. Three strategies are discussed in the literature. (A) In the algebraic programme, wedges are defined through modular inclusions without a background metric, and Theorem A.5 recovers a positive translation from an HSMI; constructing a sufficient net and showing that the candidates of this paper form such inclusions remain open tasks. (B) A bootstrap starts from a seed geometry, derives linearized equations from the first law, updates the geometry and iterates. (C) In holography the boundary theory supplies the algebra and the bulk geometry is derived from boundary data. Theorem 8.6 uses strategy (C) and assumes the HSMI/ball interface. A geometry-free reconstruction from the present candidate class remains open.

9 Selection and response over growing windows

9.1 A conjecture on a sustainable selection–response loop

The unresolved question is whether one reused local body can combine selection, renewal, sharp exhaust and informative geometric feedback. Fix two physical preparations on a logical receiver A and its current purifier R:

$$|\Phi_Z\rangle = \sqrt{1-\epsilon}|00\rangle + \sqrt{\epsilon}|11\rangle, \qquad |\Phi_X\rangle = (H_A \otimes I_R)|\Phi_Z\rangle, \quad 0 < \epsilon < 1/2.$$

The logical labels denote Q-RAIF C's neutral pair codes, with their spectator sea retained. Here R is the paid, resettable current purifier in the third and fourth body species, available to joint neutral gates. Old references outside this source quartet are denoted R_{old} and are not acted upon. The quartet is a resettable working source alongside the series' protected unknown content H, not a replacement for it; access to the current R comes from those paid body species and joint neutral gates, not from an old-purifier permission. Each cycle's inaccessible environment applies the same pointer-aligned channel

$$\mathcal{D}_q(\rho) = (1-q)\rho + qZ_A\rho Z_A, \qquad 0 < q < 1/2, \qquad h(x) = -x\ln x - (1-x)\ln(1-x).$$

No noise error label is available to the apparatus. Renewal returns the complete quartet, including the current purifier.

Proposition 9.1 (Pointer alignment and the complete renewal debit)**.** *For $\rho_p = \mathcal{D}_q(|\Phi_p\rangle\langle\Phi_p|)$, return to its specified pure quartet removes entropy $h_p = S(\rho_p)$, where*

$$h_Z = h\left(\frac{1-\sqrt{1-16q(1-q)\epsilon(1-\epsilon)}}{2}\right) > 0, \qquad h_X = h(q) > h_Z. \tag{42}$$

The receiver-only debits are 0 for Z and $h(\epsilon + q(1-2\epsilon)) - h(\epsilon)$ for X. Approximate quartet return subtracts its residual joint entropy from h_p, with finite-dimensional entropy continuity controlling the error. Program updates, sensors and usable controller registers enter the full maintenance debit in Proposition 4.1 as well.

Proof. The mixture of $|\Phi_p\rangle$ and $Z_A|\Phi_p\rangle$ has nonzero eigenvalues $(1 \pm d_p)/2$, with $d_p^2 = 1 - 4q(1-q)(1 - |\langle\Phi_p|Z_A|\Phi_p\rangle|^2)$. The overlaps are $1-2\epsilon$ and zero. Strict monotonicity of h on $(0, 1/2)$ proves the contrast. Tracing R gives the receiver debits. The reference marginal is unchanged by the noise, so the difference between each complete and marginal debit equals the receiver–purifier mutual information lost to the noise. □

Pointer alignment lowers the complete renewal price without making it zero. For $\epsilon = .1, q = .05$, $h_Z = .0877516288$ and $h_X = .1985152433$ nats per cycle. The marginal Z state is unchanged, but its purifier correlations are not. The loop must establish that its additional charged operations preserve an actual complete-debit contrast $\bar{s}_X > \bar{s}_Z$.

Conjecture 9.2 (Pointer-aligned selection with a sharp complete exhaust)**.** *Fix $\nu > 0$, $c_{\mathrm{ex}} > 0$ and thermal input at $T_H = 1/(2\pi)$ in the running rapidity clock, in natural units. For this dephasing family there are local models with one reused body during each N-cycle run, for arbitrarily large N, restoring both preparations to the declared error and realizing the following requirements jointly.*

Renewal acts on the current resettable A/R source, with joint control of both pairs explicitly permitted. Independently held unknown content H and its inaccessible old reference R_{old} remain preserved by the specified content-holding task throughout the loop. No reset of H or operation on R_{old} is granted by source renewal. Compatibility of that task with sustained renewal and the response controller is part of the conjecture.

The selected coupling supplies a complete chiral output carrying the charged heat and entropy. Every retained usable register, sensor and program update enters $s_{1:N}$, with $s_{1:N}/N \to \bar{s} > 0$ for the actual adaptive run. Initial usable purity and correlation capacity are $o(N)$; there is no positive rate of uncharged purity or nonequilibrium resource consumption. The excess above the complete-output floor

$$\limsup_{N\to\infty} \left[\frac{2\pi Q_{1:N} - s_{1:N}}{N} - \frac{6\nu\bar{s}^2}{c_{\mathrm{ex}}} \right] \tag{43}$$

can be made arbitrarily small along the family. This is sharpness for the whole adaptive output, including correlations and every atypical label.

The physical comparison has a declared block and decision clock. Let L_N be the sorting block length, $W_N \geq L_N$ the number of cycles used for one reading, and $d_N \geq 0$ the causal transport, sorting and sensing delay, all in cycle units. The program is held through each frame of $B_N = W_N + d_N$ cycles. Its outgoing observation interval $I_j = [\eta_j^{\mathrm{out}}, \eta_j^{\mathrm{out}} + W_N/\nu]$ is physically assigned to that program, and its record is available only by the frame's end, before the next decision. Require $K_N = \lfloor N/B_N \rfloor - 1 \to \infty$, so each tested decision and its following program are inside the run; when block sorting is used, $L_N \to \infty$. For the block converter, start at a sorting boundary and choose $W_N = m_N L_N$, $m_N \in \mathbb{N}$. Pad the output latency to $r_{\mathrm{out}} L_N$ and the subsequent readout latency to $r_{\mathrm{read}} L_N$, choosing $r_i = \max\{1, \lceil \delta_i/L_N \rceil\}$ for the physical latencies δ_i in cycle units, $i \in \{\mathrm{out}, \mathrm{read}\}$; these may depend on N. The window I_j contains m_N complete output blocks after the r_{out} blocks; the final r_{read} blocks allow its record to arrive before the next decision. Thus $d_N = (r_{\mathrm{out}} + r_{\mathrm{read}}) L_N$ aligns all frames and windows. Buffers, padded waiting, operation during delay and the final flush enter the complete entropy and energy account.

For fixed $Q_{\mathrm{ref}} > 0$ the normalized reading is

$$\overline{Y}_j = \frac{S_h(\eta_j^{\mathrm{out}} + W_N/\nu) - S_h(\eta_j^{\mathrm{out}})}{2\pi Q_{\mathrm{ref}} W_N}. \tag{44}$$

Whole-frame X and Z program interventions, made at the same pre-program state, history and geometric data and averaged over the declared actual-run pre-frame histories, give $\mathbb{E}(\overline{Y}_X - \overline{Y}_Z) \geq \delta_Y > 0$ uniformly as implementation and sharpness errors vanish. This is a per-cycle-normalized window contrast, not a one-cycle switch inside a pooled block.

The response changes the next program toward Z when the measured maintenance response is larger. To isolate this edge, let $P_j^{\mathrm{on/off}}(Z \mid \mathcal{F}_j)$ be the next-program probabilities with the physical response input present or blocked, respectively. The paired intervention keeps the upstream history and executed commands fixed up to that decision, including all non-response controller inputs. With $\mathcal{F}_j$ drawn from the actual run before this tested input, require

$$G_N = \frac{1}{K_N} \sum_{j=1}^{K_N} \mathbb{E}\big[P_j^{\mathrm{on}}(Z \mid \mathcal{F}_j) - P_j^{\mathrm{off}}(Z \mid \mathcal{F}_j)\big] \geq g_0 > 0. \tag{45}$$

The denominator includes every declared decision opportunity, not only selected informative or exploration windows. The response-channel replacement used in the test and its fixed reference are specified as part of the sensor. Its finite resolution, range, stress and entropy are charged. Neither a counterfactual fixed-program contrast nor a conditional gain on rare events substitutes for this operational test. For binary X/Z, the same opportunity/history average obeys $\langle P^{\mathrm{off}}(X)\rangle_N = \langle P^{\mathrm{on}}(X)\rangle_N + G_N \geq \langle P^{\mathrm{on}}(X)\rangle_N + g_0$. These are blocked-input probabilities on matched actual-run histories, not frequencies of a separately evolved off-history; if the on average tends to zero, the blocked average remains at least g_0. The actual adaptive output still satisfies Eq. (43).

Each model supplies a conserved complete stress tensor, the heat transport and a timelike worldline in a positive dilaton domain. Its actual clock has $0 < z_{\min} \leq d\tau/d\eta \leq z_{\max} < \infty$ uniformly along the family, with physical inlet temperature $T_\tau = 1/(2\pi z)$. Thermal preparation, reference stress and matching errors give a sublinear accumulated remainder in the horizon account. Drives are recovered, stored or routed away from the heat reading, or enter as explicit nonthermal sources in a matched background comparison. Their program-dependent residual remains in the contrast. Support work, clock preparation, thermal banks, buffer stages, ground offsets and final stored energy belong to the same ledger. Size, installed energy, control capacity and the dilaton domain may grow with N; each run keeps one body and a controller whose initial entropy and usable purity and correlation capacity are $o(N)$. Temperature in the rapidity frame, external exhaust capacity and cycle frequency stay fixed.

The conjecture keeps sustained renewal, sharp complete export and an informative causal return in one dynamics. At fixed $\nu, c_{\mathrm{ex}}, T_H, \bar{s}$, the quadratic excess over bare Landauer remains. Proposition 4.3 excludes direct purifier dumping from the sharp target. Arbitrary-unitary reservoir constructions approach Landauer's bound [13, Section 6], but do not keep this exhaust capacity fixed.

Proposition 9.3 (Finite-carrier sorting of the dephased source)**.** *Q-RAIF C's charge-sector sorter [14] applies to L independently realized copies of either ρ_p under its finite-carrier hypotheses: the source marginal is $\rho_p^{\otimes L}$, initially a product with the complete thermal carrier; their external purifications are retained; joint source operations act on both the receiver and its current R pair, while identity on old references refers to systems outside that quartet; the source is neutral in the original zero-mode quartet sector; the original neutral gates and paid loading, unloading and pipeline stages are available; and weighted energy tails control the implementation limit. It returns the source near its blank with entropy slack $o(L)$ and complete output Gibbs deficit $o(V)$, $V = L/\nu$, at*

$$T_{*,p} = T_H + \frac{6\nu h_p}{\pi c_{\mathrm{ex}}}, \qquad c_{\mathrm{ex}} = 4.$$

The same spectral argument applies to a supplied deterministic word with product marginal $\bigotimes_{j=1}^{L}\rho_{p_j}$ and actual entropy $s_L=\sum_j h_{p_j}$. The finite-bank target uses s_L plus the same subextensive margin. If $s_L/L\to s$, its temperature tends to $T_H+6\nu s/(\pi c_{\mathrm{ex}})$, with no convergence rate required; otherwise there need not be one limiting temperature. This is a block converter, not an adaptive moving apparatus or a per-cycle heat-reading law.

The spectral proof is in Appendix D.1. It uses a supplied product marginal and controlled energy tails, not a local reset certificate for an adaptive history.

This converter supports specified joint-source renewal, not correction of unknown content against an inaccessible purifier. For example, if R were instead forbidden to all operations, the noisy Z pair would have negativity $(1-2q)\sqrt{\epsilon(1-\epsilon)}$, below its target $\sqrt{\epsilon(1-\epsilon)}$. A channel on A and independent thermal ports cannot increase that negativity: partial transpose on R commutes with the channel, and trace norm contracts on Hermitian operators. It therefore cannot restore the entangled target. Preserving the distinct held content H/R_{old} alongside this converter requires the stated separate content-holding interface.

In a long fixed-program calibration window, suppose its full debit is $\bar{s}_p$ and converter and horizon remainders are sublinear. For $F(s)=s+6\nu s^2/c_{\mathrm{ex}}$ the matched mean reading at the floor is

$$\overline{Y}_p=\frac{F(\bar{s}_p)}{2\pi Q_{\mathrm{ref}}}+o(1),\qquad \Delta\overline{Y}=\frac{F(\bar{s}_X)-F(\bar{s}_Z)}{2\pi Q_{\mathrm{ref}}}+o(1). \tag{46}$$

For the quartet alone, $\nu=1, c_{\mathrm{ex}}=4$ give $F(h_Z)=.0993021514$, $F(h_X)=.2576276961$ and $\Delta\overline{Y}=.02519829/Q_{\mathrm{ref}}$. Here $\overline{Y}_p$ is the normalized window reading (44), under the stated fixed-program calibration assumptions, not a consequence of the adaptive sharpness clause. A single changed preparation within a pooled L-cycle block changes its mean debit by $O(L^{-1})$ and its normalized heat by $O(L^{-1})$; it does not produce this whole-window contrast.

Separate fixed-program sharpness does not prove the whole adaptive target. If time fractions a_p are occupied by long individually sharp phases with debits $\bar{s}_p$, their combined heat has the excess

$$\sum_p a_p F(\bar{s}_p)-F\left(\sum_p a_p\bar{s}_p\right)=\frac{6\nu}{c_{\mathrm{ex}}}\operatorname{Var}_a(\bar{s}_p). \tag{47}$$

This is positive for nonzero fractions and unequal debits. Joint block sorting can remove that energy gap at the channel level, but the present proof supplies no local program contrast or causal return after that sorting. Correlations with retained program history also enter the slack in Eq. (33). These are joint obstacles to Eq. (43). Sparse exploration with mean X-window share $a_N\to 0$ removes this phase-variance term but settles neither window gain. For independent noisy window tests with a fixed nonzero information per trial, vanishing estimation error requires the expected test count $K_N a_N\to\infty$. If window accuracy improves with N, the requirement concerns the accumulated physical detection information instead; counting windows alone is insufficient. Let b_N be the frequency of response-sensitive opportunities under the same opportunity/history measure as G_N. If on/off kernels coincide elsewhere, $|G_N|\le b_N$. When updates occur only in X frames, $b_N\le a_N$ with a_N evaluated over the same tested opportunities and actual-run histories. For equal-length windows, a_N is the expected X share of observed cycles, not the fraction of the full run covered by X observations. Hence $a_N\to 0$ violates Eq. (45) despite fixed

conditional contrast. Exploration and update duty are not identical in general: continuous monitoring during Z windows could keep b_N nonvanishing. Its compatibility with charged sensing, whole-output sharpness and a causal return is unproved, not excluded.

Under isotropic depolarization the two noisy quartet spectra and full debits instead coincide. Their separately entropy-optimal matched heat outputs have zero asymptotic program contrast; unsorted source contrast can become recovered work. This excludes that particular sharp-heat mechanism, not different full control debits or every causal realization using depolarization. A fixed initial record also obeys $I(X : Y_{1:N}|Z) \le H(X|Z)$ for charged baseline Z; sublinear initial record entropy cannot supply a positive new-information rate indefinitely.

The exact marginal-horizon balance gives $\overline{Y}_j = Q_{I_j}/(W_N Q_{\mathrm{ref}})$ when the entire matched source is charged heat. Drive, other and reference-source integrals add $[E_{\mathrm{drive},I_j} + E_{\mathrm{other},I_j} + E_{\mathrm{ref},I_j}]/(W_N Q_{\mathrm{ref}})$ in the same normalization. A common background cancels in program comparisons only with a justified common clock and transport; program-dependent residuals remain. Table 1 separates the available finite readouts from this joint inlet, renewal, sensor and outlet task. In particular, Q-RAIF C's massless sorter and massive-outside full-angle gate are different layouts [14].

The sorting-based route fails without source entropy concentration or controlled energy tails. Any joint realization must also avoid positive asymptotic output deficit or slack, loss of program contrast after drive and sensing charges, and failure of causal routing, temperature matching or the operating worldline.

Fresh, independently discarded quadrature drives have a separate accuracy cost [21]: at error ϵ_{drv} their input energy is at least $\omega_h[\pi/(4\sqrt{\epsilon_{\mathrm{drv}}})+O(1)]$, where $\omega_h^{-1} = \int |f|^2\omega^{-1}d\omega$. Reused drives retain correlations and correlated failures; recovery is not established by fresh-drive accuracy. At $\epsilon_{\mathrm{drv}} = .01$ the finite bound is $6.74238\omega_h$. Choose the source parameter $\epsilon = .01$ independently of ϵ_{drv}. With $q = .05$, $\nu = 1$ and $c_{\mathrm{ex}} = 4$, the complete quartet debits are $(h_Z, h_X) = (.01370657, .19851524)$. Their heat floors $F(h_p)/(2\pi)$ are $(.00222632, .04100272)$. With the additional benchmark choice $\omega_h = 1$, the drive/heat ratios are about $(3028, 164)$; unrecovered fractions $(.0003302, .0060813)$ already equal one corresponding heat unit. At the main example's $\epsilon = .1$, the Z floor is .01580443 and that ratio is about 427, while the X values are unchanged. These quartet-only benchmarks exclude additional charged controller debits and do not identify the drive with the quartet's full renewal device. Gate spectrum, duration and debit set the ratio; no universal $\omega_h \ge \nu$ or universal drive/heat multiplier follows. Outgoing coherent drive energy uses the nonthermal ledger in Eq. (34). Recoverable energy, stored energy and exported heat are separate entries; the same outgoing energy is never charged twice.

The exact moving-clock conversion is

$$z = \frac{Ly}{R}, \qquad y = \sqrt{R_b^2 + L^2\dot{\Phi}_b^2} - L\dot{\Phi}_b, \quad R_b^2 = \Phi_b^2 - R^2, \tag{48}$$

with proper-time dot. The tracking-clock conversion is derived in Appendix D.1, Eq. (94). Thus $z = L\rho$ is a stationary limit. The physical input needs $zT_\tau = T_H$, and support power is $(L^{-1} - K)\dot{\Phi}_b/\kappa$ in Eq. (88), including its actual sign and source. Prescribing the trajectory or transforming temperature constructs neither support nor inlet. When the monitored source carries heat plus a sublinear reference remainder, its source–clock sector has $R = R_0 + \kappa Q_{1:N} + o(N)$ at a positive bounded mean heat; an exterior operating domain must therefore reach $\Phi_b > R = \Omega(N)$. With bounded z, an isolated reserve paying all per-cycle losses also requires initial energy $\Omega(N)$; replenished charged work is an alternative.

The family must report its upper as well as lower size, installed-energy, clock-domain and compiler scalings. A stronger resource target would be $O(N)$ installed energy and dilaton range, with polynomial control capacity; this is an optional benchmark, not a hypothesis of Conjecture 9.2. Faster growth would miss that benchmark without refuting the stated conjecture. Fixed-gap and fixed-cutoff examples alone give no uniform growing-run certificate.

This loop would connect selection to its source and backreaction. Its finite pointer algebra still needs a map to the modular-stable geometric algebra, and Conjecture 7.2 still asks for the origin of the gravitational action.

9.2 The two routes and a common source principle

The finite return demonstrates that a law supplied to a model can feed back into the physical operations that select access. It does not explain that law. Modular reconstruction supplies a constraint on the law, but does not manufacture the apparatus. Their union requires a further mechanism: the same physical history must determine both the accessible structure and its complete source response.

Conjecture 9.4 (Common-action selection and geometric consistency)**.** *There is a local matter–controller theory in which readable histories form and maintain a nontrivial selected geometric structure, and whose causal effective action uses the same prepared fields, support and clock in its source and response. In a low-energy sector with a common cone, its selected ball data satisfy the geometric consistency condition of Consciousness II. Its first-order response is the corresponding Einstein response with one source normalization; its higher-order response includes the interaction, contact and clock work that produced the selection. Selection changes this response through physical couplings and sources, while a change of description at fixed matter history leaves the source unchanged. The sustainable family of Conjecture 9.2 is a finite operating sector of this theory, with a response-to-operation gain that survives the growing-window limit.*

The conjecture has tests beyond assigning gravity to a massive apparatus. Removing the history's access to a command must remove the record-mediated component of the source, while holding the commanded interaction fixed and merely discarding a log must preserve it. Source identification must survive switched physical references; logical noise records alone fail this test in Proposition B.1. The same calibrated source must then pass the all-ball consistency test with fixed initial or boundary data. A law that reproduces only the JT dilaton or one binary return has not passed the higher-dimensional test. At second order the source, geometry and changed preparation must be varied together; entropy Hessians do not become SLD or spacetime metrics by name.

Q-RAIF C supplies a local common action with dynamical support, memory and clock fields and a specified positive finite canonical ordering [14]. This permits a concrete attack on the missing source: freeze the texture only as a controlled limit, quantize the retained material and controls together, and compare the resulting local gates with the actual acquired-history Hamiltonian. Direct substitution fails already because dynamical support and clock introduce kinetic cross terms and forces absent from a fixed-profile gate. Moreover a spatial canonical cutoff preserves its positive form but does not by itself provide a covariant renormalized Ward identity. Classical on-shell conservation therefore cannot certify the complete quantum source. The coupled tolerance above closes the

trajectory/reader comparison in the retained mean-action branch; it cannot remove these two operator and continuum obstacles.

For the real-quartet carrier, Proposition F.9 supplies that Ward identity and the actual full-mode bulk state. Its projective and Weyl conversion is varied with the clock and director; adding a complete vacuum effective action to the retained canonical stress would double count the quantum response. This bulk construction does not yet transport the entire reset protocol through dynamical walls.

The finite receiver-renewal family of Q-RAIF C holds clock mean energy and norm fixed while its space, time and paid Gibbs bank grow. It is a real finite update result, with old references retained. It renews a receiver rather than the entire interacting material. In the 31-site full-program diagnostic, the receiver occupation .157712 exceeds the target .13 by .027712; this is not a full-program state error. The independently reported completion probability is .857524. An additional complete-ring construction renews the same body to fixed per-cycle tolerance through four thermal outlet species, retaining old correlations and restoring the original mixed receiver and its current purifier by local neutral pumps [14]. At the same incoming history, changing the maintained logical axis changes the stress emitted by that reset, with a strictly positive finite-error bound; no separate program-tagged emitter is added. Its prescribed drives are paid, and its energy-constrained storage stays bounded at fixed parameters. Its finite smooth schedule can also be driven by a work-clock field whose source includes the old incoming correlations [14, Section on a spatial work clock]. That result has static endpoints and a supplied frame. The bulk covariant construction and this fixed-frame schedule have not yet been joined with those endpoints and the sharp excess-heat limit required here. The finite-program diagnostic remains evidence against combining successful components without a common state and dynamics. The conjectured common theory would be refuted by an unavoidable extra extensive source or renewal price, by a persistent mismatch of its selected modular data, or by a response gain that vanishes under the stated fixed resources. These are different failure modes, not a general impossibility claim.

10 Relation to Existing Approaches

AdS/CFT [31] and its area–entropy relation [27] supply the dictionary used in Theorem 8.6; it is not derived here or granted to arbitrary selected algebras. MERA [32, 33] implements coarse-graining through disentanglers and isometries, with network depth resembling a radial AdS direction. Different networks can describe the same boundary data. Relating either construction to physical selection requires the explicit algebra-to-geometry map.

Jacobson's horizon Clausius argument [34] shares the entropy-based approach to gravitational dynamics. Section 2 addresses its source-side input: which selecting processes produce the stress whose response is computed. Loop quantum gravity [29, 35] instead quantizes the metric with background independence. Its relation to a path of selected algebras would require a representation map from spin-network states. These connections motivate the proposal without supplying its missing selection law.

11 Conclusion

Physical selection reaches gravity through the selecting interaction's source. Calibration changes activation rates; selected projections and learned protection change stress at fixed schedule. Logical noise records alone can miss that source. Prepared references or signed field readings remove specific ambiguities, with their resources charged.

Within matched JT data, a finite return retains the quantum reader, old reference, record spring, clock energy, mechanics, radiation and recoil in one mean action. Acquisition and feedback are separately necessary for its source contrast, and joint error bounds preserve the test under finite perturbations. The spatial construction includes support and synchronization, with a nonzero source–dilaton–sensor path and a subsequent complete material-force response. The latter also contains ordinary material and clock effects; it is not entirely gravitational. The full-mode quartet bulk supplies a covariant, actual-state source with clock and director work, rather than replacing the retained state by a vacuum.

Continued maintenance is a different result. The entropy of the complete retained system, consumed purity, correlations and conserved charges fixes the export debit. A thermal chiral channel of fixed capacity imposes its quadratic rate price. Matched horizon transport records that price; the actual thermal-source calculation also retains the moving clock's reference stress. Renewal accuracy alone does not establish sharp exhaust or indefinite operation.

The modular route constrains the law through a supplied geometric family and dictionary; the source route computes within an assigned law. The common-action and sustainable-loop conjectures ask for formed access, geometric consistency, complete causal sources and persistent return gain in one dynamics. Pointer alignment gives a complete-debit contrast; finite-carrier sorting can approach its channel floor. Individually sharp program phases still leave a positive variance price at nonzero duty fractions, so joint sharpness with an informative return remains open. Endpoint and gate matching, moving support, sustained thermal renewal and the origin of geometric selection remain construction tasks.

A Modular definitions and candidate checks

A.1 Preliminaries

Standard references are Takesaki [40], Bratteli–Robinson [41] and Haag [42].

Definition A.1. *A **von Neumann algebra** $\mathcal{M}$ on $\mathcal{H}$ is a $*$-subalgebra of $\mathcal{B}(\mathcal{H})$ containing the identity and closed in the weak operator topology; equivalently $\mathcal{M} = \mathcal{M}''$, where $\mathcal{M}'$ is the commutant. It is a **Type III$_1$ factor** if $\mathcal{M} \cap \mathcal{M}' = \mathbb{C}\mathbf{1}$, it admits no semifinite normal trace, and its Connes spectrum is $\mathbb{R}_{\geq 0}$.*

Under standard conditions, local algebras of quantum field theories are Type III_1 factors [42, 43].

A vector $|\Omega\rangle$ is **cyclic** for $\mathcal{M}$ if $\mathcal{M}|\Omega\rangle$ is dense, and **separating** if $M|\Omega\rangle = 0$ implies $M = 0$. For cyclic and separating $|\Omega\rangle$ the antilinear map $S_\Omega : M|\Omega\rangle \mapsto M^*|\Omega\rangle$ is closable, and its closure has the polar decomposition $S_\Omega = J_\Omega \Delta_\Omega^{1/2}$ with modular conjugation J_Ω and modular operator Δ_Ω.

Theorem A.2 (Tomita–Takesaki [40]). *With $\mathcal{M}$, $|\Omega\rangle$, J_Ω, Δ_Ω as above, $J_\Omega \mathcal{M} J_\Omega = \mathcal{M}'$, and the modular group $\sigma_t^\Omega(M) := \Delta_\Omega^{it} M \Delta_\Omega^{-it}$ satisfies $\sigma_t^\Omega(\mathcal{M}) = \mathcal{M}$ for all real t.*

A state ω is KMS at parameter β in the modular-time convention used here if for all $A, B \in \mathcal{M}$ there is a function $F_{A,B}$ analytic in $0 < \operatorname{Im} z < \beta$ and continuous on the closure, with $F_{A,B}(t) = \omega(\alpha_t(A)B)$ and $F_{A,B}(t + i\beta) = \omega(B\alpha_t(A))$. This ordering gives $\beta = 1$ for $\sigma_t^\omega(A) = \rho^{it} A \rho^{-it}$ in a density-matrix representation; the usual future physical Hamiltonian time has the opposite modular orientation.

Theorem A.3 (Takesaki [40]). *A faithful normal state ω on $\mathcal{M}$ is KMS at $\beta = 1$ for σ_t^ω, and σ_t^ω is the unique one-parameter automorphism group with this property.*

The modular flow is thus the unique time evolution for which the given state looks thermal. In the Rindler wedge it is the Lorentz boost, and the KMS condition at $\beta = 2\pi$ gives the Unruh temperature $T_U = a/(2\pi)$.

Definition A.4 ([44]). *An inclusion $\mathcal{N} \subset \mathcal{M}$ with common cyclic and separating vector $|\Omega\rangle$ is a **half-sided modular inclusion** (HSMI) if $\sigma_t^{\mathcal{M}}(\mathcal{N}) \subset \mathcal{N}$ for all $t \leq 0$.*

Theorem A.5 (Wiesbrock [44]). *For an HSMI in the stated $t \leq 0$ convention, the closure of $P = (\log \Delta_{\mathcal{N}} - \log \Delta_{\mathcal{M}})/(2\pi)$ is positive. The canonical unitary group is $U(a) = e^{iaP}$, fixes $|\Omega\rangle$, and obeys $\mathcal{N} = U(1)\mathcal{M}U(1)^*$ and $\Delta_{\mathcal{M}}^{it} U(a) \Delta_{\mathcal{M}}^{-it} = U(e^{-2\pi t} a)$ [15, Thm. 2.1].*

The translation from $\mathcal{M}$ to $\mathcal{N}$ is therefore encoded algebraically, without background geometry. This is the standard tool for extracting translation structure from modular data.

A.2 Example 1: Rindler wedge

Consider a free massless scalar field in $(1 + 3)$-dimensional Minkowski spacetime with Fock space $\mathcal{H}_U = \mathcal{F}(L^2(\mathbb{R}^3))$ and vacuum $|\Omega\rangle$. Let $\mathcal{W}_R = \{x > |t|\}$ be the right Rindler wedge and $\mathcal{R}(\mathcal{W}_R) = \{e^{i\phi(f)} : \operatorname{supp} f \subset \mathcal{W}_R\}''$. For this check we take $\mathcal{M}_U = \mathcal{R}(\mathcal{W}_R)$, which makes the ambient modular action well defined without performing a nontrivial selection.

The Reeh–Schlieder theorem [42] makes $|\Omega\rangle$ cyclic and separating for $\mathcal{R}(\mathcal{W}_R)$, so (U1) holds. By the Bisognano–Wichmann theorem [45, 46], $\Delta_\Omega = e^{-2\pi K}$ with K the global boost generator in the x-direction. The modular action is $\sigma_t^\Omega(X) = e^{-2\pi i t K} X e^{2\pi i t K}$; its spacetime action preserves the wedge, giving $\sigma_t^\Omega(\mathcal{R}(\mathcal{W}_R)) = \mathcal{R}(\mathcal{W}_R)$, which is (U2). No decoherence semigroup is specified, so (U3) is not verified. Haag duality, $\mathcal{R}(\mathcal{W}_R)' = \mathcal{R}(\mathcal{W}_L)$, does not imply that every proper extension of the wedge algebra contains the opposite wedge algebra, nor any maximality statement.

On the $t = 0$ Cauchy slice the full modular generator is

$$K_{\mathrm{mod}} = -\ln \Delta_\Omega = 2\pi K, \qquad K = \int_{\mathbb{R}^3} x\, T_{00}(0, x, y, z)\, d^3x. \tag{49}$$

Formally this is $2\pi(K_R - K_L)$, with $K_R = \int_{x>0} x T_{00}\, d^3x$ and $K_L = \int_{x<0} (-x) T_{00}\, d^3x$. The right-wedge expression $2\pi K_R$ alone is not $-\log \Delta_\Omega$: in a factorizing regulator it is the right reduced-state modular Hamiltonian, up to its normalization constant. In the Type III theory the separate one-sided charges require an appropriate quadratic-form or regulator interpretation. For a local unitary excitation in the right wedge, the

opposite-wedge expectation is unchanged, so the change of the full boost charge equals the right-sided change when those differences exist. This does not identify the two operators. The example reproduces the Bisognano–Wichmann geometric modular action; it does not show that (U1)–(U3) select the wedge algebra.

A.3 Example 2: Qubit chain with dephasing

Consider n qubits with Hamiltonian $\hat{H} = -J\sum_i \sigma_z^{(i)}\sigma_z^{(i+1)} - h\sum_i \sigma_x^{(i)}$, $J > 0$, coupled to an environment through

$$\mathcal{T}_t(\rho) = e^{t\mathcal{L}}(\rho), \qquad \mathcal{L}(\rho) = -i[\hat{H}, \rho] + \gamma \sum_i \left(\sigma_z^{(i)} \rho\, \sigma_z^{(i)} - \rho\right), \tag{50}$$

with Heisenberg dual $\mathcal{E}_t := \mathcal{T}_t^*$ and global state $\omega = Z^{-1}e^{-\beta\hat{H}}$. For an exact check we take $h = 0$; for $h \neq 0$ the diagonal algebra is stable only approximately in a strong-dephasing regime.

For $h = 0$ the diagonal observables are fixed by the Heisenberg semigroup, and the candidate is the maximal abelian algebra

$$\mathcal{A}_{\mathrm{geo},\mathbf{c}} = \{f(\sigma_z^{(1)}, \ldots, \sigma_z^{(n)})\}'' = \mathcal{A}_{\mathrm{diag}}. \tag{51}$$

The thermal state is positive definite at finite β, so $\omega(|s\rangle\langle s|) = \langle s|\rho_{\mathrm{th}}|s\rangle > 0$ for every computational basis state, which gives (U1); diagonal elements of a positive definite matrix are positive in any basis. The ambient modular flow is $\sigma_t^{\omega,U}(X) = \rho_{\mathrm{th}}^{it} X \rho_{\mathrm{th}}^{-it}$; for $h = 0$, ρ_{th} is diagonal and commutes with $\mathcal{A}_{\mathrm{diag}}$, giving (U2). For $h \neq 0$ the conjugation generally leaves $\mathcal{A}_{\mathrm{diag}}$. Under the dephasing dual, off-diagonal matrix units decay as $e^{-2\gamma\, d(s,s')\, t}\, |s\rangle\langle s'|$ with Hamming distance d; for $h = 0$ every diagonal observable is fixed, so (U3) holds and $\mathcal{A}_{\mathrm{diag}} \subset \mathrm{Fix}(\mathcal{E})$. This does not prove maximal setwise invariance, since the full matrix algebra is always mapped into itself.

The modular flow acts trivially on the abelian pointer algebra, and the entropy of a block A within $\mathcal{A}_{\mathrm{diag}}$ is the Shannon entropy of the marginal distribution $p(s_A)$. The example checks one modular- and semigroup-stable candidate, which under exact fixation is the pointer algebra; it neither verifies the Type III regime nor selects the candidate uniquely by (U1)–(U3). Selection of another type is available here: for $h = 0$ the fixed algebra is the commutant of the $\sigma_z^{(i)}$, and the window theorem of Paper B shows that every sufficiently long observation window selects it as the unique maximal stable algebra [20].

B Source identification, carriers and profiles

B.1 Continuing calibration and identification of the source

Source control need not interrupt stored quantum content. With an independently prepared vacuum mode and full output collection, Q-RAIF B's corrected parity instrument has branches

$$\mathcal{I}_\pm(\rho) = \tfrac{1}{2}(1 \pm e^{-2k(1-\alpha a)^2})\rho.$$

It yields the same calibration statistic without preparing a new body state. The coherent controlled-parity identity $Re^{-imZ_L Q_f}R = I_L \otimes e^{-imQ_f}$ goes further: it protects the body's marginal for an arbitrary mode state by transferring the interaction to the field alone [21].

For a centered independent input the coupling-added stress is still $m^2 F_f$. The meter can be returned coherently; retaining a calibration outcome is a separate read and memory operation. Partial collection loses coherence during the inquiry itself. Brain Anchor prices that loss, bounds continuing retuning under drift and gives an exact posterior criterion for when an extra inquiry pays for its source and apparatus price [57]. No geometric response or entropy export is needed for these preservation identities.

The inquiry and maintenance choices can also share one running body. Brain Anchor tracks both the interface posterior b_x and the coherence-weighted mass c_x in a finite hidden-state model. Equal b_x and even equal reduced body states can require different next commands, because past loss has different correlations with the interface. Q-RAIF B realizes collection by finite resonant contacts, including successive contacts with a finite packet. In the matched field realization, the source at the next encounter is still $m_j^2 F_f$, irrespective of the accessible correction, at the interaction output before collection. A collector can redistribute that energy, so this is not its full downstream stress. For a history-dependent policy the coefficient is computed from that policy's hidden-path weights, not from a new independent gain draw at every step. The four-encounter joint example gives $\sum_j \mathbb{E} m_j^2 = .588965$, against .640000 with permitted maintenance but no retained calibration reads. These are source accounts; the supplied objective's prices are not an entropy-export measurement. Using these sources in the JT response also requires their profiles and emission times and any collector contribution. The continuous collection route below specifies a free-field remainder, but does not supply the wall, thermal exhaust or entire sensing apparatus of the response model.

B.1.1 Continuous collection changes the outgoing profile

Q-RAIF B also gives an excitation-preserving, positive-frequency collection model [21]. A retained oscillator of frequency Ω couples to a continuum $\omega \in [\Omega - B, \Omega + B]$, with real even form factor $g(\omega - \Omega)$. A prescribed finite source drive prepares the matched packet. The one-particle propagator S then determines both the collected fraction and the outgoing wave. Its finite waveform certificate allows clock, arrival and switching errors; it is not a white-noise limit. The field interaction is band-smeared, so its spatial tails are part of the model. Parameters are fixed within each encounter.

Let $z_{\mathrm{out}}(\omega)$ denote the asymptotic free outgoing amplitude for a unit source drive, at the same null-time origin as the incoming packet, and let the body-conditioned amplitudes be $\pm m\beta z_{\mathrm{out}}$. With the current normalization of Eq. (3), put

$$\chi(u) = -i \int_{\Omega - B}^{\Omega + B} \sqrt{\frac{\omega}{4\pi}} z_{\mathrm{out}}(\omega) e^{-i\omega u} d\omega, \quad F_{\mathrm{out}}(u) = 4m^2\beta^2 [\mathrm{Re}\, \chi(u)]^2. \tag{52}$$

For independent vacuum preparation, this is the outgoing free-field stress for every logical input. Tracing the body gives a mixture of opposite coherent amplitudes, whose normal-ordered squared currents agree. Accessible parity correction does not change this marginal stress. For a capture pulse ending at T, the rotating-frame amplitude is $z_{\mathrm{out}}(\omega) = e^{i(\omega - \Omega)T} [P_{\mathrm{field}} S(0, w)](\omega)$, with the prescribed source carrier phase included. Removing the final free propagation phase fixes the common time origin; it must not be counted again when placing the source in the JT model. The input occupation is $k_{\mathrm{eff}} m^2 = \beta^2 m^2 \|w\|^2$. For real envelopes and an even band, the source, receiver and free remainder energies in units $\hbar = 1$ are, respectively,

$$\Omega k_{\mathrm{eff}} m^2, \qquad \Omega \xi k_{\mathrm{eff}} m^2, \qquad E_0^{\mathrm{out}} = \Omega(1 - \xi) k_{\mathrm{eff}} m^2. \tag{53}$$

Capture moves energy into the receiver. A later vacuum swap exports it to a separate exhaust. The net prescribed capture-switch work is zero for this symmetric packet family, while source preparation supplies the first energy in Eq. (53). The receiver, exhaust, controls and supports do not disappear from a gravitational account.

Frequency regularity matters for using this source in the JT moment formulas. A hard flat band generally gives $F_{\mathrm{out}}(u)$ tails of order u^{-2}, so finite energy alone does not ensure a finite second moment. Instead choose $g(\delta) = \sqrt{J_0}[1 - (\delta/B)^2]$ inside the band and zero outside. For a finite source envelope and finite collection time, $z_{\mathrm{out}}(\omega)$ is g times a smooth spectral function. Thus $\sqrt{\omega} z_{\mathrm{out}} \in H_0^1$ and Plancherel gives

$$E_2^{\mathrm{out}} = \int u^2 F_{\mathrm{out}}(u) du = m^2 \beta^2 \int \left| \partial_\omega (\sqrt{\omega} z_{\mathrm{out}}) \right|^2 d\omega < \infty. \tag{54}$$

E_1^{out} is finite by Cauchy–Schwarz. This specifies a legitimate free-field source for the response dictionary, rather than reusing the uncaptured packet unchanged. It does not determine the total apparatus stress or assert compact temporal support. The JT background, boundary access and matter normalization remain the declared inputs of the response model.

In the worked collection setting $\Omega = 20$, $B = 5$, $T = 16$ and $J_0 = 1/(2\pi)$, a fixed 8721-bit waveform table and stated clock tolerances certify $1 - \xi < 4.43 \times 10^{-7}$ and a relative preparation-strength uncertainty below .001. This is enough to keep the collection-induced loss below .001 in the preceding 4096-per-setting calibration example. The bound does not include parity-gate or detector errors. Repeated encounters require independent vacuum fields and exhausts, or a separately proved return/reset bound on one reused field. A fixed-capacity thermal cycle and the source–sensor return of Section 4 are not supplied by this finite-bank construction.

With several Gaussian field species, however, logical calibration alone may not identify the actual source. Let the species have the same switch profile and independent centered inputs, with quadrature covariance kW in their species coordinates. Put $G = (\mathbf{g}_1, \mathbf{g}_2)$ and $\mathbf{v}(a) = \mathbf{g}_1 - a\mathbf{g}_2$. The logical coherence and coupling-added stress are respectively

$$r_W(a) = e^{-2k\mathbf{v}(a)^T W \mathbf{v}(a)}, \qquad T_{\mathrm{add}}(u) = \|\mathbf{v}(a)\|^2 F_f(u). \tag{55}$$

The first measures $D_W = G^T W G$; the second needs $C = G^T G$. The incoming stress of the reference preparation is an additional source.

Proposition B.1 (Protection records do not always identify stress)**.** *Take* $W = \mathrm{diag}(1, 4)$ *and the two coupling families*

$$\mathbf{g}_1 = (1, 0), \quad \mathbf{g}_2 = (1, 1), \qquad \widetilde{\mathbf{g}}_1 = (0, 1/2), \quad \widetilde{\mathbf{g}}_2 = (2, 1/2).$$

Every command gives the same logical channel, with weighted residual $1 - 2a + 5a^2$*. Their stress coefficients are* $1 - 2a + 2a^2$ *and* $1/4 - a/2 + 17a^2/4$*. Thus no finite adaptive experiment using only these fresh-port logical channels distinguishes them. The noise-minimizing command is* $a = 1/5$ *for both, but their source-minimizing commands are* $1/2$ *and* $1/17$*.*

Proof. Insert each vector pair in Eq. (55). Equality of conditional channels at every command gives equality of every adaptive logical record law by induction, also with retained ancillas. Minimizing each positive quadratic gives the stated commands. □

At the noise optimum their added stress coefficients are 17/25 and 8/25. This is an ambiguity of the specified logical access, not an inability to observe stress through another instrument. It persists with the same incoming preparation and does not come from a changed background.

Proposition B.2 (Switched references identify the physical source)**.** *In the same two-species family, suppose the apparatus can independently prepare known centered Gaussian references with covariances kW_1, kW_2, where $W_1 = \mathrm{diag}(1,4)$ and $W_2 = \mathrm{diag}(4,1)$. At each reference measure the logical coherence means at $0, +h, -h$, $h > 0$. They determine the physical source Gram matrix exactly:*

$$D_{j,11} = -\frac{\log r_j(0)}{2k}, \qquad D_{j,12} = \frac{\log r_j(h) - \log r_j(-h)}{8kh},$$
$$D_{j,22} = -\frac{\log r_j(h) + \log r_j(-h) - 2\log r_j(0)}{4kh^2}, \qquad C = \frac{D_1 + D_2}{5}. \tag{56}$$

More generally, known reference weights with $I = \sum_j t_j W_j$ suffice, with $C = \sum_j t_j D_j$.

For independent ± 1 trials, N at each of the six settings, suppose all true means are at least $r_{\min} > 0$. Let $\eta = \sqrt{2\log(12/\zeta)/N} \le r_{\min}/2$, clip sample means to $[r_{\min}/2, 1]$ and use the same formulas. With probability at least $1-\zeta$, uniformly on $|a| \le A$,

$$|\widehat{V}(a) - \|\mathbf{v}(a)\|^2| \le \Delta_C := \frac{2\eta}{5kr_{\min}}\left(1 + \frac{A}{h} + \frac{2A^2}{h^2}\right). \tag{57}$$

Minimizing $\widehat{V}$ on any supplied finite command grid incurs actual excess source coefficient at most $2\Delta_C$ over its best point.

Proof. The log characteristic function is the quadratic form $-2kD_j$. Its three coefficients give Eq. (56); $W_1 + W_2 = 5I$ gives C and proves the general sufficient condition. The union Hoeffding bound covers all six means. Each log error is at most $2\eta/r_{\min}$, so the errors of C_{11}, C_{12}, C_{22} are at most $2\eta/(5kr_{\min})$, $\eta/(5khr_{\min})$ and $4\eta/(5kh^2 r_{\min})$. Insert them in $V = C_{11} - 2aC_{12} + a^2 C_{22}$ to prove the uniform bound. Two applications of that bound compare the chosen grid point with the best one. □

These are $6N$ new preparations, couplings and reads, not six free exact means. The reference covariances, their switches and their incoming energies have to be calibrated and charged. Separate logical test preparations are permitted here; applying these references to held content would also require the protection operations of Q-RAIF B and their collection account. Equation (56) identifies the coupling-added source under the declared profile and species architecture, not the full stress of an unknown apparatus. There is another route when the apparatus has an oriented field readout rather than only logical records.

Proposition B.3 (Signed field calibration with held content)**.** *For d independent input modes put $Q_s = b_s + b_s^\dagger$ and $P_s = i(b_s^\dagger - b_s)$. Assume centered independent Gaussian modes, initially independent of the body and its retained reference, a known common k, and $\mathrm{Var}(P_s)/(4k) \le \sigma_{\max}^2$. Controlled total parity conjugates $Z_L\sqrt{k}\sum_s v_s(a)Q_s$ to $I_L\sqrt{k}\sum_s v_s(a)Q_s$. After this coherent echo the signed readings $Y_s = -P_s/(2\sqrt{k})$ have mean $v_s(a)$ while the logical state remains unchanged. Two commands, $a = 0, h$ with $h \ne 0$, identify*

$$\mathbf{g}_1 = \mathbb{E}\mathbf{Y}(0), \qquad \mathbf{g}_2 = \frac{\mathbb{E}\mathbf{Y}(0) - \mathbb{E}\mathbf{Y}(h)}{h}.$$

This does not require the Gaussian variances as labels.

Suppose $|v_s(0)|, |v_s(h)| \le B$ and use a finite readout that clips Y_s to $[-Y, Y]$, $Y > B$, then rounds to the nearest point of a prescribed uniform grid of spacing $q > 0$ in that interval, adding the two endpoints if needed. This alphabet has at most $\lceil 2Y/q \rceil + 2$ symbols and absolute rounding error at most $q/2$. For N fresh queries at each command, define

$$\eta = Y\sqrt{\frac{2\log(4d/\zeta)}{N}} + \frac{q}{2} + 2\sigma_{\max}\varphi\left(\frac{Y-B}{\sigma_{\max}}\right), \qquad \varphi(z) = \frac{e^{-z^2/2}}{\sqrt{2\pi}}.$$

With probability at least $1-\zeta$, the estimated residual on $|a| \le A$ has vector error at most $e_v = \sqrt{d}\,\eta(1 + 2A/|h|)$. If its true norm is bounded by B_v, then

$$|\widehat{V}(a) - \|\mathbf{v}(a)\|^2| \le 2B_v e_v + e_v^2. \tag{58}$$

The readout uses a finite alphabet, with at most $\lceil \log_2(\lceil 2Y/q \rceil + 2) \rceil$ bits per species per query.

Proof. Total parity reverses every Q_s, so the coherent parity calculation of Q-RAIF B [21] gives a field-only displacement. The canonical relation $[Q_s, P_s] = 2i$ shifts P_s by $-2\sqrt{k}v_s(a)$ and leaves its variance unchanged. For a Gaussian with $|\mu| \le B$ and standard deviation at most $\sigma_{\max}$, the clipping bias is at most $2\sigma_{\max}\varphi((Y-B)/\sigma_{\max})$, by integrating the two tails. Hoeffding for the bounded clipped readings and a union over the $2d$ means give the first term of η; rounding adds at most $q/2$. Thus each coordinate error of $\mathbf{g}_1$ is at most η and of $\mathbf{g}_2$ at most $2\eta/|h|$. The Euclidean bound follows. Expanding $\|\mathbf{v} + \delta\mathbf{v}\|^2$ proves Eq. (58). □

The sign reference, species-resolved quadrature detector and complete mode control are its additional permissions. Two commands here mean $2N$ new field queries, each with d digitized readings; mode preparation, coupling, capture and memory remain charged. Unknown incoming means would confound the shift. Together with the matched JT equations either identified source predicts a response from actual records rather than from a supplied coupling label. Boundary sensing still checks the physical response itself. Neither route replaces the wall, clock and thermal-export construction in a closed loop.

B.2 The command carrier matters

How strongly the gravitational field records the command statistics depends on how the command is physically carried.

Proposition B.4 (Phase-carried commands)**.** *Keep the coupling switched on at every opportunity and carry the command in the drive phase: a correction uses the input $|\sqrt{N}\rangle$ and no correction uses $|i\sqrt{N}\rangle$. Then (i) the no-correction branch flips the bit with probability $1 - s_N$ and the correction branch with probability s_N; (ii) both inputs carry the same field energy $N\bar{\omega}$, and the branch sources are $4N[\mathrm{Re}\,\psi]^2 + F_g$ and $4N[\mathrm{Im}\,\psi]^2 + F_g$; (iii) for the worked packet both branch sources have the same E_0, E_1, E_2, so $\mathcal{C}_\infty$ does not depend on the command rate, while the dilaton coefficients $A(u)$, $B(u)$ at finite u do.*

Proof. With $Q = b + b^\dagger$, the bit channel multiplies coherences in the X basis by $\langle\beta|e^{-2i\lambda Q}|\beta\rangle = e^{-2\lambda^2}e^{-4i\lambda\,\mathrm{Re}\,\beta}$. For $\beta = i\sqrt{N}$ this factor is $e^{-2\lambda^2}$, real and positive, which gives (i). The stress of a coherent input β is $(\psi\beta + \psi^*\beta^*)^2$, which is $4N[\mathrm{Re}\,\psi]^2$ for

$\beta = \sqrt{N}$ and $4N[\operatorname{Im}\psi]^2$ for $\beta = i\sqrt{N}$, and the coupling adds F_g in both branches. The equal moments of the two quadrature profiles give (iii) through Eq. (14); the difference of A between branches is proportional to $\operatorname{Re}[a + i(u - u_0)]^{-2}$, which is nonzero for generic u. □

The hiding has a price. With phase-carried commands the uncorrected steps also err, so the world mean obeys $m \mapsto c(2s_N - 1)[(1 - h)m + h\nu]$ on the calibrated branch. At $N = 16$ and a known interface the stationary loss rises from 0.1896 with gated couplings to 0.2334, and the switching work per opportunity rises from $0.0039\bar{\omega}$ to $0.0386\bar{\omega}$. The phase must also be set on the incoming packet before the encounter, which for full-line packets has the same causal limitation as on-demand emission. At $N = 16$ and $\kappa = 10^{-4}$ the coefficient A at the packet centre differs between the two branch sources by 3.4×10^{-4}, and the difference decays as u^{-2}.

So the gravitational trace of a selection is not universal. With gated couplings it sits in the rate of an energy-carrying event; with phase-carried commands it sits in the shape of the source and disappears from the asymptotic invariant of this example. A general law connecting selection to gravitational response must therefore name the carrier of the selection.

B.3 A worked packet

Take the normalized mode $f(\omega) = 2a\sqrt{\omega}\, e^{-a\omega} e^{i\omega u_0}$, for which $\bar{\omega} = 1/a$ and

$$\psi(u) = -\frac{ia}{\sqrt{\pi}\,[a + i(u - u_0)]^2}. \tag{59}$$

The two quadrature profiles $4[\operatorname{Re}\psi]^2$ and $4[\operatorname{Im}\psi]^2$ have different shapes but the same normalized moments $(E_0, E_1, E_2) = (1, u_0, u_0^2 + a^2)/a$. They have power-law tails: $[\operatorname{Im}\psi]^2 \sim a^2/(\pi u^4)$ and $[\operatorname{Re}\psi]^2 = O(u^{-6})$. The absolute third moment of F_g diverges logarithmically; the first three moments are finite and are exactly those used in Eq. (14). The switch determined by $g' = 4\lambda \operatorname{Im}\psi$ and $g(\pm\infty) = 0$ is

$$g(u) = -\frac{4\lambda a(u - u_0)}{\sqrt{\pi}\,[a^2 + (u - u_0)^2]}. \tag{60}$$

Both g and F_g have full-line support. The formulas and table below are exact for this scattering idealization with asymptotic coefficient data. A command decided at a finite time cannot gate its entire switch; finite-time feedback uses the following compact approximation.

Proposition B.5 (Causal compact approximation)**.** *Choose a fixed even $\eta \in C_c^\infty(\mathbb{R})$ with $0 \le \eta \le 1$, $\eta = 1$ on $[-1, 1]$ and $\eta = 0$ outside $[-2, 2]$, and set $g_T(u) = \eta((u - u_0)/T)g(u)$. For $T \ge a$, write $F_T = (g_T')^2/4$. The unrescaled switching stresses obey*

$$\begin{aligned} \left| \int (F_T - F_g)du \right| &= O(\lambda^2 a^2 T^{-3}), \\ \int (u - u_0)(F_T - F_g)du &= 0, \\ \left| \int (u - u_0)^2 (F_T - F_g)du \right| &= O(\lambda^2 a^2 T^{-1}). \end{aligned} \tag{61}$$

The one-particle smearing vectors converge in norm, with difference $O(|\lambda|a/T)$. Hence the corresponding coherent-drive gate channels converge. A gate chosen before $u_0 - 2T$ can be applied causally. Its duration and any delay between the read and the gate are part of its task schedule.

Proof. With $w = u - u_0$ and $C = 4|\lambda|a/\sqrt{\pi}$, the tails satisfy $|g| \le C/|w|$ and $|g'| \le C/w^2$ for $|w| \ge T \ge a$. The derivative of the cutoff obeys the same w^{-2} bound on its annulus, up to a constant depending only on η. Integrating w^{-4} and w^{-2} gives the first and third bounds; parity gives the second. For $h = g_T - g$, $\|h\|_2 = O(|\lambda|aT^{-1/2})$ and $\|h'\|_2 = O(|\lambda|aT^{-3/2})$. Plancherel and Cauchy–Schwarz bound $\int_0^\infty \omega|\hat{h}|^2 d\omega/(4\pi)$ by $\|h\|_2\|h'\|_2/2$, giving the one-particle estimate. The coherent-state characteristic function is continuous in this norm. □

At finite T, the drive mode, its overlap with the incoming beam and the gate error must be recomputed from g_T before assigning values to a finite-cutoff table. If the drive is matched to the new mode, its quadrature profiles and first three stress moments converge as well. The real and imaginary profiles are Hilbert-transform pairs. For $r = (g_T - g)'$, the identity $u\mathcal{H}r = \mathcal{H}(ur) + \pi^{-1}\int r$ in the convention $\mathcal{H}r(u) = \pi^{-1}\,\mathrm{pv}\int r(v)/(u - v)dv$ has zero last term. The L^2 isometry of $\mathcal{H}$ therefore gives $\|u\mathcal{H}r\|_2 = \|ur\|_2 \to 0$. Together with convergence of the mode normalization, this proves the weighted profile convergence. Equation (6) holds exactly for each compact implementation with its actual gate parameters. The same discrete feedback law applies when the target is held between its read and gate; noise during a new delay must instead be included in the transition law.

With $A_{\mathrm{tot}} = N + p\lambda^2$ the asymptotic invariant is $\mathcal{C}_\infty = H_0^2 + 2\kappa H_0 u_0 A_{\mathrm{tot}}/a - \kappa^2 A_{\mathrm{tot}}^2$. Relative to the same beam without couplings,

$$\delta\mathcal{C}(p) = p\lambda^2\Big[\frac{2\kappa H_0 u_0}{a} - \kappa^2\big(2N + p\lambda^2\big)\Big], \tag{62}$$

which has an explicit quadratic term in p. Table 2 uses $a = 1$, $u_0 = 3$, $H_0 = 1$, $\kappa = 10^{-4}$, the task parameters $\nu = 0.8$, $c = 0.9$, $h = 0.4$ of Ref. [21], and the uncalibrated ($q = \frac{1}{2}$) and calibrated (32 passive blocks, $q = 0.0404888$) coupling rates.

N	gate error	p, $q = \frac{1}{2}$	p, q_{32}	$\delta\mathcal{C}$, $q = \frac{1}{2}$	$\delta\mathcal{C}$, q_{32}	beam share	quad./lin.
1	0.3544	0.15525	0.11413	5.746×10^{-5}	4.224×10^{-5}	7.0×10^{-2}	-1.2×10^{-6}
4	0.1327	0.15153	0.10698	1.402×10^{-5}	9.897×10^{-6}	4.1×10^{-3}	-2.7×10^{-7}
16	0.0371	0.15033	0.10468	3.476×10^{-6}	2.420×10^{-6}	2.5×10^{-4}	-6.7×10^{-8}
64	0.0095	0.15002	0.10408	8.657×10^{-7}	6.006×10^{-7}	1.6×10^{-5}	-1.7×10^{-8}

Table 2: JT invariant response to the coupling-added source of the worked packet. "Beam share" is $\delta\mathcal{C}(p)$ at q_{32} divided by the beam's own change $\mathcal{C}_\infty(N) - \mathcal{C}_\infty(0)$; "quad./lin." is the ratio of the p^2 term of Eq. (62) to the term linear in p.

At $N = 16$ the ratio of the two responses is 0.696342074, against the coupling-rate ratio 0.696342054. For $\kappa = 0.01$ and 0.05 ($\kappa N = 0.16$ and 0.8) the response ratios are 0.6963442 and 0.6963560; the quadratic term stays small because $p\lambda^2$ is small, not because κN is. The calibration-dependent part is a small fraction of the beam's own response, and that fraction varies roughly as N^{-2}: it is 7% at $N = 1$, where the gate fails 35% of the time, and 1.6×10^{-5} at $N = 64$. Its size is therefore a property of the chosen

drive and background, not a general feature of selection, and a field-energy ratio such as $p\lambda^2/N$ is not a general invariant response ratio. In this example the bracket of Eq. (62) is positive, so better calibration lowers $\delta\mathcal{C}$; for other backgrounds the bracket can vanish or change sign.

The mean-source solution is not the mean of branch solutions. For a single command $I \sim \text{Bernoulli}(p)$, $\mathbb{E}\,\mathcal{C}_\infty(N + \lambda^2 I) - \mathcal{C}_\infty(N + p\lambda^2) = -\kappa^2\lambda^4 p(1-p)$, which counts only the classical command randomness and not field fluctuations within a branch. With M independent copies of the whole experiment, each with its own hidden instrument, calibration records and drives, and with $G = G_1/M$, the classical source-fraction variance of smooth stress readings falls as $1/M$. A common hidden label shared by all copies does not give this independence, and a coherent drive still has nonzero energy variance. The exact mean-source solution does not show that a single quantum world has a definite classical geometry.

B.4 What remains input

The construction takes as given the JT action and its anti-de Sitter background, the matter field and its central-charge matching, the boundary data and the identification of the downstream null coordinate with the preparation coordinate, the semiclassical use of mean stress, and the actuator's field, switch, clock and fresh ports. The two-dimensional sector theorem permits either one or several cones; it neither derives this matter–gravity identification nor forbids a dynamical route to it [23]. The source is computed outside the apparatus, for the free outgoing field; the stress of the switch and its support is not included, and a covariant detector stress tensor shows that it cannot simply be dropped in the coupling region [59]. Within these inputs the response is exact.

C Finite boundary sensing and returned records

C.1 A conditional readback route

The matter action in Section B.4 does not directly couple to Φ. A bulk detector of the matter metric alone therefore does not acquire a local dilaton readout. Boundary motion offers a different route if a physical reflecting boundary with a known dilaton condition is available.

Proposition C.1 (A dilaton-blind record law)**.** *Compare two dilaton solutions with the same matter metric, the same accessible matter input states and the same initial controller state. Suppose every instrument and conditional evolution used by the controller depends on this metric and its earlier records, but not on the dilaton or a boundary that senses it. Then every finite adaptive record sequence has the same distribution in the two solutions.*

Proof. The first instrument has the same input and conditional outcome law. Given any earlier record sequence, the next instrument and input evolution are again the same. Induction gives equality of the joint record laws. Private random bits can be included in the controller state and the induction. □

This statement does not exclude inference from previously recorded sources. It excludes a new dilaton signal from an otherwise identical metric-only experiment. A reflecting boundary evades its hypothesis because its position depends on Φ.

Proposition C.2 (Reconstruction from three boundary echoes)**.** *In a source-free interval let a, b, γ be constant, $\mathcal{C} = a^2 + b\gamma > 0$, and suppose the apparatus can time reflections at the supplied boundary $\Phi = \Phi_b > \sqrt{\mathcal{C}}$. Use one calibrated null coordinate frame. Three reflection pairs (u_i, v_i), with distinct u_i and $d_i = (a + \Phi_b)u_i - \gamma \neq 0$, determine a, b, γ uniquely:*

$$\begin{pmatrix} 1 - u_1 v_1 & u_1 & v_1 \\ 1 - u_2 v_2 & u_2 & v_2 \\ 1 - u_3 v_3 & u_3 & v_3 \end{pmatrix} \begin{pmatrix} a \\ b \\ \gamma \end{pmatrix} = \Phi_b \begin{pmatrix} 1 + u_1 v_1 \\ 1 + u_2 v_2 \\ 1 + u_3 v_3 \end{pmatrix}. \tag{63}$$

Writing this matrix as M, its determinant is

$$\det M = \frac{2\Phi_b(\mathcal{C} - \Phi_b^2) \prod_{i<j}(u_j - u_i)}{\prod_i d_i} \neq 0.$$

If the u_i are exact and the measured returns are $v_i + \delta v_i$, an invertible measured matrix $\widetilde{M}$ yields the exact coefficient error

$$\delta(a, b, \gamma)^T = \widetilde{M}^{-1}(\delta v_i d_i)_{i=1}^3, \qquad \|\delta(a, b, \gamma)\|_2 \leq \|\widetilde{M}^{-1}\|_2 \|(\delta v_i d_i)_i\|_2. \tag{64}$$

Proof. The boundary is $v = V(u) = (a + bu - \Phi_b)/[(a + \Phi_b)u - \gamma]$, with $V' = (\Phi_b^2 - \mathcal{C})/[(a+\Phi_b)u-\gamma]^2 > 0$, so it is timelike. Its equation gives Eq. (63). Multiplying row i of M by d_i leaves evaluations of three quadratic polynomials; their coefficient determinant is $2\Phi_b(\mathcal{C} - \Phi_b^2)$, giving the displayed Vandermonde factor. In the noisy equation the difference between the measured right side and $\widetilde{M}(a, b, \gamma)^T$ is exactly $(\delta v_i d_i)_i$. □

Null signals can carry these timestamps back to a receiver before the next command. They reconstruct the coefficients, hence $\mathcal{C}$, in a declared source-free interval. The inverse becomes ill-conditioned near $\Phi_b = \sqrt{\mathcal{C}}$ or coincident sample times. Exact reconstruction is a test-probe limit, not three cost-free finite-energy measurements. A usable protocol must supply the reflecting boundary, its support, calibrated clocks, timing resolution, causal waiting time and probe stress; the last changes the background beyond that limit. Six timestamps stored with b bits each occupy $6b$ logical bits, not a determined physical energy or complete memory cost. This gives an explicit sensing interface to implement, rather than granting the controller direct access to Φ.

C.2 Probe backreaction and a returned program

A finite probe contributes stress, including the reflected component. The inverse can retain both chiral components instead of treating the probe as free. Work in the mean-source JT equations of Section B.4, with the same trace-anomaly subtraction. Suppose the complete additional bulk source in the sensing interval is calibrated as $T_{uu} = f(u)$, $T_{vv} = h(v)$, $T_{uv} = 0$. Define

$$\begin{aligned} A_f(u) &= -\kappa \int_{u_*}^{u} (u - z) f(z)\, dz, & B_f(u) &= -\kappa \int_{u_*}^{u} z(u - z) f(z)\, dz, \\ A_h(v) &= -\kappa \int_{v_*}^{v} (v - z) h(z)\, dz, & B_h(v) &= -\kappa \int_{v_*}^{v} z(v - z) h(z)\, dz, \\ K(u, v) &= A_f(u) + vB_f(u) + A_h(v) + uB_h(v). \end{aligned} \tag{65}$$

The matching convention fixes the lower limits and the homogeneous coefficient data. These are calibrated sources, not a derived reflection law for the wall. If reflection or

support stresses introduce other components or unknown source terms, the following inverse must be changed accordingly.

Proposition C.3 (Source-compensated echo inverse)**.** *For a known K as in Eq.* (65)*, the solution is*

$$\Phi(u,v) = \frac{a + bu + \gamma v - auv + K(u,v)}{1 + uv}. \tag{66}$$

Three points on the supplied boundary $\Phi = \Phi_b$ determine the coefficients $\boldsymbol{\theta} = (a, b, \gamma)^T$ whenever their matrix M in Eq. (63) *is invertible:*

$$M\boldsymbol{\theta} = \big(\Phi_b(1 + u_i v_i) - K(u_i, v_i)\big)_{i=1}^{3}. \tag{67}$$

The source-free determinant formula need not hold for these altered echoes; invertibility persists under sufficiently small perturbations of an invertible source-free matrix.

Suppose the measured u_i, v_i have errors at most δ_u, δ_v, the calibrated K has pointwise error at most η_K, and the measured matrix $\widetilde{M}$ is invertible. On a domain joining true and measured points, bound the derivatives of

$$R(u,v) = (\Phi_b - a) + (\Phi_b + a)uv - bu - \gamma v - K(u,v)$$

by $|R_u| \le L_u$ and $|R_v| \le L_v$. Then

$$e := \|\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|_2 \le \sqrt{3}\|\widetilde{M}^{-1}\|_2(L_u\delta_u + L_v\delta_v + \eta_K), \tag{68}$$

$$|\widehat{\mathcal{C}} - \mathcal{C}| \le \sqrt{4a^2 + b^2 + \gamma^2}\, e + e^2, \qquad \widehat{\mathcal{C}} = \widehat{a}^2 + \widehat{b}\widehat{\gamma}. \tag{69}$$

If the next projection angle is $\vartheta_{\text{next}} = F(\widehat{\mathcal{C}})$ for a supplied Lipschitz program F, its error relative to $F(\mathcal{C})$ is at most $\mathrm{Lip}(F)$ *times Eq.* (69)*.*

Proof. The f contribution is the retarded solution already obtained in Section 2; interchange u, v for the h contribution. Their sum satisfies all three JT equations and gives Eq. (66). Evaluation on the boundary yields Eq. (67). The true residual $R(u_i, v_i)$ is zero. The residual at each measured point, including the calibration error, is bounded by $L_u\delta_u + L_v\delta_v + \eta_K$. Multiplication by $\widetilde{M}^{-1}$ gives Eq. (68). Expanding $\widehat{a}^2 + \widehat{b}\widehat{\gamma}$ leaves a linear term with gradient $(2a, \gamma, b)$ and the quadratic term $(\delta a)^2 + \delta b\delta\gamma$, whose magnitude is at most e^2. The program bound follows from its Lipschitz property. □

An apparatus needs a certified parameter range to bound L_u, L_v and the conditioning of the inverse before use. Neither the assumed calibration of K nor this error bound supplies a timing POVM or the wall's support energy.

Here is a causal one-step return in that declared geometry. Take the selected projection of Section 2.4, a fresh vacuum field and the independent bit state $(I + 0.8Z)/2$. Use a smooth compact switch

$$g(u) = A\exp\left[-\frac{1}{1 - ((u-3)/0.25)^2}\right] \quad (|u-3| < 0.25), \qquad g(u) = 0 \quad \text{otherwise,} \tag{70}$$

with A fixed by $\int (g')^2/4\, du = 0.25$. Unlike the full-line packet of Section 2, this source finishes before the readout begins. Set $H_0 = L = 1$, $\kappa = 0.05$ and $\Phi_b = 3$. The source moments are $E_0 = 0.25$, $E_1 = 0.75$, $E_2 = 2.257177876$; a coupled population r gives

$$(a, b, \gamma) = (1 + \kappa r E_1, -\kappa r E_0, \kappa r E_2).$$

The projections P_0 and $P_{\pi/2}$ give $r = 0.9$ and 0.5 respectively, and hence $\mathcal{C} = 1.067496366$ and 1.037498878.

Supply a timelike receiver at $v = -\alpha/u$, $\alpha = 1/4$. Its proper clock is $\tau = 2L\sqrt{\alpha}\ln u/(1-\alpha)$, up to an additive constant. Emit at receiver coordinates $t_i = 4, 6, 9$, with $v_i = -\alpha/t_i$. The outward ray has fixed v_i and meets the boundary at

$$u_i = \frac{a - \Phi_b + \gamma v_i}{(a + \Phi_b)v_i - b} > t_i.$$

The reflected ray has fixed u_i; reception at the receiver therefore occurs at the same coordinate u_i, with $v = -\alpha/u_i > v_i$. For $r = 0.9$ the receive coordinates are $8.18983, 12.56501, 19.53471$; for $r = 0.5$ they are $8.10371, 12.30538, 18.81553$. Throughout these intervals the receiver obeys $\sqrt{\mathcal{C}} < \Phi < \Phi_b$ and $1 + uv > 0$, inside the usual cutoff and outside the stationary horizon. A new compact coupling can start after $u = 20$. Writing $\mathcal{C}_{\mathrm{mid}}$ for the midpoint of the two invariants, choose

$$F(\mathcal{C}) = \frac{\pi}{4} + 10(\mathcal{C} - \mathcal{C}_{\mathrm{mid}}). \tag{71}$$

The two returned angles are 0.935385602 and 0.635410725. If the inverse error stays below the corresponding separation margin, the measurement changes the next coupled projection and its represented algebra $\{P_{F(\mathcal{C})}\}'$. This is a response-dependent return map, not a proof that this program is optimal or that its interface formed through learning. The controller uses the measured invariant; computing one from assumed initial data is a different protocol.

The listed times use the source-free geometry. With finite probes, the return must instead use Eq. (67) and the altered boundary. As a bulk check, two calibrated positive impulsive fluxes of integrals 10^{-5} in u and 2×10^{-5} in v give a coefficient bias of about 5.7×10^{-6} if ignored; source compensation removes it to numerical precision. This is not a physical reflection or timing experiment. For a translation-only readout with the listed proper-delay difference $\Delta\tau = 0.05001338$, the sharp signed-delay bound in Q-RAIF B requires at least $11.04\hbar$ in this clock's energy units for binary error 0.1 [21]. A real echo can also carry redshift and amplitude information, so that restriction is not a universal bound on geometric sensing. It shows why one cannot assign both negligible probe stress and a finite resolution without constructing the readout. The boundary, receiver, timing instrument, full probe-induced stress and support costs remain explicit apparatus inputs to this conditional return; mean-source evolution does not bound stress fluctuations.

C.3 A quantum boundary record without echo timestamps

A current detector gives a second readback route. It measures fluctuations over a prescribed interval of proper time, rather than the arrival time of a reflected signal. The boundary condition still does essential work: the detector is carried by the supplied timelike curve $\Phi = \Phi_b$. A detector on the same fixed bulk worldline in both solutions would remain subject to Proposition C.1.

Use a transparent chiral current interface, with the vacuum defined in u and the current normalization of Section 2. Its input on the interval to be sensed has zero mean and covariance $\langle j(u)j(u')\rangle = -[4\pi(u-u'-i0)^2]^{-1}$. The apparatus supplies a proper-time clock, a degenerate detector bit initially in $|0\rangle$, and a real compact switch $\zeta(\tau)$. The finite interaction is

$$U_s = e^{-iX_sQ_g}, \qquad Q_g = \int g(u)j(u)\,du, \qquad g(u(\tau)) = \zeta(\tau). \tag{72}$$

Equivalently its integrated interaction is $\int \zeta(\tau) X_s \dot{u}(\tau) j(u(\tau))\, d\tau$. The fixed detector operator and c-number current commutator make the time-ordering correction a scalar phase. This is the same gapless-detector method used for exact field channels [8]; the finite switch defines a smeared current, not an unsmeared instantaneous Hamiltonian. We read Z_s after contact and retain the bit $R = 0, 1$, with $R = 1$ the flipped outcome. Clock calibration, preparation and recording permissions are supplied resources.

Proposition C.4 (Invariant-dependent finite detector law)**.** *On a source-free background with $0 < \mathcal{C} < \Phi_b^2$, a future boundary segment has*

$$\dot{u} = \frac{bu^2 + 2au - \gamma}{2L\sqrt{\Phi_b^2 - \mathcal{C}}} > 0, \qquad \lambda_b = \frac{\sqrt{\mathcal{C}}}{L\sqrt{\Phi_b^2 - \mathcal{C}}}. \tag{73}$$

In the test-background calculation, Eq. (72) *has the record law*

$$k_b(\mathcal{C}) = \frac{1}{4\pi} \int_0^\infty \omega \coth \frac{\pi\omega}{\lambda_b} |\widehat{\zeta}(\omega)|^2 \, d\omega, \qquad \widehat{\zeta}(\omega) = \int e^{i\omega\tau} \zeta(\tau)\, d\tau, \tag{74}$$

$$\Pr(R = 1 \mid \mathcal{C}) = \frac{1 - e^{-2k_b(\mathcal{C})}}{2}. \tag{75}$$

For every nonzero switch with finite variance, the probability is strictly increasing in $\mathcal{C}$. It depends on the invariant, not on the three coefficients separately. For two equally likely backgrounds the optimal decision from this one bit has error $[1 - |p_1 - p_0|]/2$, where $p_i = \Pr(R = 1 \mid \mathcal{C}_i)$.

Proof. Pulling back the metric to the level curve gives $d\tau/du = 2L\sqrt{\Phi_b^2 - \mathcal{C}}/(bu^2 + 2au - \gamma)$. The Riccati equation in Eq. (73) makes $u(\tau)$ a fractional-linear transform of $e^{\lambda_b \tau}$ on each regular segment. Möbius invariance of the current two-point function therefore gives

$$\langle \dot{u}(\tau) j(u(\tau)) \dot{u}(\tau') j(u(\tau')) \rangle = -\frac{\lambda_b^2}{16\pi \sinh^2[\lambda_b(\tau - \tau' - i0)/2]}.$$

Its symmetrized spectrum yields Eq. (74); the pullback vacuum is KMS at temperature $\lambda_b/(2\pi)$ in this proper clock. For a centred Gaussian field, $\langle e^{2iQ_g} \rangle = e^{-2\langle Q_g^2 \rangle}$, hence the detector's Z_s expectation is e^{-2k_b}. The factors λ_b and $\coth(\pi\omega/\lambda_b)$ strictly increase with $\mathcal{C}$ for $\omega > 0$. A nonzero compact switch has Fourier weight on a set of positive measure, so the integral strictly increases. The binary decision formula is the total-variation formula for two Bernoulli laws. □

This response uses the familiar Hawking–Unruh covariance. It does not add a direct matter–dilaton coupling, and the KMS temperature here is the boundary proper-time temperature, not an automatically thermal exhaust in the running rapidity clock. Without calibration of L, Φ_b and the proper-time scale, these records identify λ_b rather than $\mathcal{C}$. A clock-scale error can mimic an invariant change; this instrument alone cannot distinguish them.

C.4 The sensing interaction changes the background

The same detector interaction adds the chiral mean stress $f(u) = [g'(u)]^2/4$. To include it, choose a boundary-source completion: this current is transmitted into the JT region, the coupling-added bulk source is precisely $T_{uu} = f(u)$, and the other state-dependent chiral

source is zero. Keep the trace-anomaly subtraction of Section B.4. This is a specified boundary injection model, not the reflected-probe model of Section C.1. No unknown reflected component is fitted as a known function. Forces supporting the boundary, an ultraviolet realization of the contact and the clock's stress require a further apparatus model.

Proposition C.5 (Self-consistent sensing source in mean JT geometry)**.** *In this completion, put* $R_b = \sqrt{\Phi_b^2 - \mathcal{C}}$ *and let dots denote proper-time derivatives. Starting with the matched coefficient data, solve*

$$\dot{u} = \frac{bu^2 + 2au - \gamma}{2LR_b}, \qquad f(u(\tau)) = \frac{\dot{\zeta}^2}{4\dot{u}^2},$$
$$(\dot{a}, \dot{b}, \dot{\gamma}) = \frac{\kappa\dot{\zeta}^2}{4\dot{u}}(u, -1, u^2). \tag{76}$$

On a regular finite segment this gives the boundary trajectory and the complete coupling-added bulk source together. Its exact balances are

$$\dot{\mathcal{C}} = \frac{\kappa L}{2} R_b \dot{\zeta}^2, \qquad R_b(\tau) = R_b(0) - \frac{\kappa L}{4}\int_0^\tau \dot{\zeta}(s)^2\, ds, \tag{77}$$

$$E_s^{(\tau)} := \int f\dot{u}^2\, d\tau = \frac{1}{4}\int \dot{\zeta}^2\, d\tau = \frac{R_b(0) - R_b(\tau_f)}{\kappa L}, \quad E_s^{(u)} := \int f\, du = \frac{1}{4}\int \frac{\dot{\zeta}^2}{\dot{u}}\, d\tau. \tag{78}$$

For the actual, generally nonstationary map $u(\tau)$*, the detector variance is*

$$k[g] = \frac{1}{4\pi}\int_0^\infty \omega|\widehat{g}(\omega)|^2 d\omega = \frac{1}{8\pi}\int_{\mathbb{R}^2} \frac{[g(u) - g(v)]^2}{(u-v)^2}\, du\, dv. \tag{79}$$

Its record probability remains $(1 - e^{-2k[g]})/2$*; replacing this variance by the stationary KMS integral during contact would neglect backreaction.*

Proof. The retarded JT solution has running coefficients $a' = \kappa u f$, $b' = -\kappa f$ and $\gamma' = \kappa u^2 f$, with primes in u. In the derivative of the boundary level equation the source contributions cancel: its numerator derivative is b and its denominator derivative is $a+\Phi_b$. Thus its slope is still $(\Phi_b^2 - \mathcal{C})/[(a+\Phi_b)u - \gamma]^2$ with running coefficients. The induced metric gives the first equation in Eq. (76), while $g' = \dot{\zeta}/\dot{u}$ gives the second. Differentiating $a^2 + b\gamma$ and substituting $\dot{u}$ gives Eq. (77). The energy balances follow by change of variable. The source f follows from $U_s^\dagger j U_s = j + X_s g'/2$ and the initially independent, centred detector with $\langle X_s\rangle = 0$. The incoming current on the deterministic mean trajectory is Gaussian; its vacuum variance has the Fourier expression in Eq. (79). Parseval's identity and $\int(1 - \cos\omega x)/x^2\, dx = \pi|\omega|$ give the second. □

In this model the mean source fixes a classical trajectory before the detector outcome is conditioned on. It is not a theory of outcome-conditioned quantum geometry. The inequality $\kappa L\int \dot{\zeta}^2/4 < R_b(0)$ prevents closure of the boundary gap; one must also check $\dot{u} > 0$ and $1 + uv > 0$ over the chosen window. The two energies in Eq. (78) refer to different clocks. Neither is the total preparation, support, readout or erasure cost. For any sequence obeying this positive-source completion at unchanged Φ_b, L, κ, the cumulative proper-clock source charge satisfies

$$\sum_n E_{s,n}^{(\tau)} < \frac{\sqrt{\Phi_b^2 - \mathcal{C}_{\text{initial}}}}{\kappa L}. \tag{80}$$

This follows directly from $R_b > 0$ and Eq. (77). A nonzero charge per contact therefore rules out infinitely many such contacts at a fixed cutoff. It is an upper budget for this positive-source boundary family, not a universal lower bound on sensing energy or a prohibition of protocols with a different net source.

Proposition C.6 (Finite sensing survives its own added source)**.** *Take two source-free initial backgrounds with different invariants, with the same Φ_b, L and a common proper-time contact interval on which both unperturbed curves are regular. Fix a nonzero C^2 compact switch ζ_0. At fixed κ, the completion in Proposition C.5, with $\zeta = \epsilon\zeta_0$, has distinct record laws for all sufficiently small $\epsilon > 0$. The added energies and invariant changes are finite and of order ϵ^2.*

Proof. On the compact regular segment the trajectory equations are locally Lipschitz, and their source-dependent terms are $O(\kappa\epsilon^2)$. Continuous dependence preserves the domain and gives $u_\epsilon - u_0 = O(\kappa\epsilon^2)$ in C^1. The corresponding compact smears obey $\|g_\epsilon - \epsilon g_0\|_0 = O(\kappa\epsilon^3)$: the inverse clocks and their first derivatives have the same perturbative order, and the H^1 bound controls the vacuum $H^{1/2}$ seminorm. Thus $k[g_\epsilon] = \epsilon^2 k[g_0] + O(\kappa\epsilon^4)$ and $p_\epsilon = \epsilon^2 k[g_0] + O((1+\kappa)\epsilon^4)$. The leading variances differ strictly by Proposition C.4. Their record difference cannot vanish for all sufficiently small nonzero ϵ. Equation (78) gives the energy order. □

C.5 A finite record returns to a physical selection

The two compact-source branches of Eq. (70) can be sensed by the same contact, without giving their coefficient values to the program. Start the boundary clock at $u = 4$, keep $\kappa = 0.05$, and use the C^2 compact profile

$$\zeta(\tau) = \frac{4}{3}\left[1 - \left(\frac{\tau - 4}{4}\right)^2\right]^3 \quad (0 \le \tau \le 8), \qquad \zeta = 0 \quad \text{otherwise.} \tag{81}$$

It has finite variance and energy; smooth approximants give the same limiting quantities. Solving Eq. (76) and integrating Eq. (79) gives

Original population r	Initial $\mathcal{C}$	Final $\mathcal{C}$	$\Pr(R = 1)$	Final u
0.5	1.037498878	1.1206720	0.31748037	57.481
0.9	1.067496366	1.1505122	0.31771688	52.458

The contact charge is exactly $E_s^{(\tau)} = 512(4/3)^2/(385\cdot 8) = 0.29552670$. The corresponding null energies are 0.06577751 and 0.06774638. These are deterministic numerical evaluations with converged quadratures, not certified intervals. The single-bit total variation is about 2.3650×10^{-4}, so one contact is a weak discriminator, with equal-prior error about 0.49988175. It is a nonzero physical return, not a high-confidence reconstruction or an optimal sensing strategy. Many repeats would need a new account of field correlations, preparation and accumulated source; they cannot be declared independent by multiplying this record law. Even ignoring all other operations, Eq. (80) allows at most 190 contacts with this unchanged charge on either initial branch. This is a necessary budget, not a 190-contact construction.

Retain R and let it select the next coupled projection: $P_0 = (I + Z)/2$ if $R = 0$ and $P_{\pi/2} = (I + X)/2$ if $R = 1$. Use the independent bit state $(I + 0.8Z)/2$ and a later

compact switch ζ_{next}, scheduled by the same proper clock after $\tau = 8$. Its realised null smear is $g_{\mathrm{next}}(u(\tau)) = \zeta_{\mathrm{next}}(\tau)$. The program supplies that proper-time profile and selects its projection using only R; it need not know $u(\tau)$ or $\mathcal{C}$. The mean population coupled to the next field is

$$r_{\mathrm{next}} = 0.9 - 0.4\Pr(R = 1), \qquad T^{\mathrm{added}}_{uu,\mathrm{next}} = r_{\mathrm{next}}\frac{(g'_{\mathrm{next}})^2}{4}. \tag{82}$$

It is 0.77300785 and 0.77291325 in the two branches. Thus the earlier selection changes a source, that source changes the boundary record law, and the recorded bit changes both the next pointer algebra and the next mean source. The controller uses no numerical dilaton input. For example, use the profile in Eq. (81) rescaled to amplitude $1/4$ and the proper-clock interval $9 \le \tau \le 11$. Its added proper-clock source charge is $r_{\mathrm{next}}\, 512(1/4)^2/(385 \cdot 2)$. The source kernel in u is the one on each branch's actual mean trajectory; the population difference alone is not a comparison of two complete fixed-shape stresses. The same trajectory equations apply with $\dot\zeta^2$ replaced by $r_{\mathrm{next}}\dot\zeta^2_{\mathrm{next}}$, continuing the causal mean solution beyond readout.

There is a same-current subtlety in this return. On the union of the sensor and later null intervals, take the input to have the vacuum restriction. A preceding compact Weyl source in a disjoint earlier interval leaves this restriction unchanged, since the relevant smeared currents commute. Conditioning on the sensor bit applies $\cos Q_g$ or $\sin Q_g$ to that restriction. Both resulting states are invariant under simultaneous current parity, so every later current has zero conditional mean. The next bit is independently prepared, and the linear stress cross term vanishes in each record branch, proving Eq. (82) without resetting the field. Its conditional covariance generally changes. The later field is not a fresh vacuum port, and a vacuum or thermal gate-error formula cannot be carried over to it.

For a realised switch $\widetilde{g}$, define the vacuum seminorm $\|g\|_0 = \sqrt{k[g]}$. If $\|\widetilde{g} - g\|_0 \le e$, then

$$|\widetilde{p} - p| \le |k[\widetilde{g}] - k[g]| \le 2\sqrt{k[g]}e + e^2.$$

This prices the needed control of the actual smeared interaction; clock and worldline errors must be included in $\widetilde{g}$, not only in its amplitude. Two record laws remain distinct if the sum of their probability-error bounds is below their stated separation. Section C.6 adds the independent source requirement, and Section C.8 gives a finite thermal replacement. The clock, contact and continuous exhaust with their complete stress remain the joint apparatus task of Conjecture 9.2. The finite run consumes a prepared detector bit and an independently prepared next bit. Resetting the record without retained side information must export its binary entropy; it is not supplied by coherent field return.

C.6 A joint certificate for records and gravitational sources

A detector can be accurate as an instrument and inaccurate as a source. The vacuum seminorm controls a half derivative of its switch, whereas the injected energy controls a full derivative. This distinction is already visible in a single contact, before questions of thermal reuse arise.

Proposition C.7 (Record accuracy does not control source energy)**.** *Let* $g_n(u) =$

$n^{-3/4}e^{-u^2/2}\cos(nu)$, $n \geq 1$, *and* $E[g] = \int (g')^2 du/4$. *Then*

$$k[g_n] = \frac{n^{-3/2}}{8}[n\sqrt{\pi}\ \mathrm{erf}(n) + 2e^{-n^2}],$$
$$E[g_n] = \frac{n^{-3/2}\sqrt{\pi}}{8}[n^2 + \tfrac{1}{2} + \tfrac{1}{2}e^{-n^2}]. \tag{83}$$

Its detector flip probability tends to zero while its source energy diverges. Every member has finite absolute stress moments through order two.

Proof. The Fourier transform is the sum of two shifted Gaussians. Integrating its squared modulus with weights ω and ω^2 gives the two formulas. Consequently $k[g_n] \sim \sqrt{\pi}/(8\sqrt{n})$ and $E[g_n] \sim \sqrt{\pi n}/8$. Gaussian decay gives the finite moments. □

This is a comparison on a fixed null-coordinate field. A member with too much energy can leave a chosen gravitational operating domain. A fixed bandwidth excludes this sequence, but that bandwidth is then part of the source certificate. The controlled weak-switch result above uses stronger regularity and is unaffected.

Proposition C.8 (Simultaneous record, source and response control)**.** *Write the realised switch as* $g + h$, *put* $w(u) = 1 + u^2$, *and assume* $g, h \in H^1$ *with weighted square-integrable derivatives. Define*

$$e_0 = \sqrt{k[h]}, \qquad A = \|\sqrt{w}\, g'\|_2, \qquad e_1 = \|\sqrt{w}\, h'\|_2, \qquad D_s = \tfrac{1}{2}Ae_1 + \tfrac{1}{4}e_1^2.$$

For the independent vacuum input of the current detector,

$$|\widetilde{p} - p| \leq 2\sqrt{k[g]}e_0 + e_0^2,$$
$$\int w\,|\widetilde{f} - f|du \leq D_s, \qquad f = (g')^2/4, \quad \widetilde{f} = (g' + h')^2/4. \tag{84}$$

The second line bounds each of the three stress-moment errors by D_s. *For matched retarded data, at each common* u *the running coefficients satisfy* $|\delta a|, |\delta b|, |\delta\gamma| \leq \kappa D_s$, *hence*

$$|\delta\mathcal{C}| \leq \kappa D_s(2|a| + |b| + |\gamma|) + 2\kappa^2 D_s^2,$$
$$|\delta\Phi(u,v)| \leq \frac{\kappa D_s(1 + |uv| + |u| + |v|)}{|1 + uv|}. \tag{85}$$

For a single contact, even when the detector has a retained reference, the half trace distance of the complete output from its ideal output is at most $\sqrt{1 - e^{-k[h]}} \leq e_0$.

Proof. Cauchy–Schwarz applied to $\widetilde{f} - f = g'h'/2 + (h')^2/4$ proves the weighted bound. The inequalities $1, |u|, u^2 \leq w$ cover both full and prefix moments; the retarded coefficient equations then give the stated errors. Expand $(a+\delta a)^2 + (b+\delta b)(\gamma+\delta\gamma)$ and the dilaton numerator for the response bounds. For the final claim, purify the initial detector and reference. The Weyl relation makes $U_g^\dagger U_{g+h}$ equal to $e^{-iX_sQ_h}$ up to one scalar phase, since $X_s^2 = I$. Its vacuum overlap is $e^{-k[h]/2}$ independently of the detector state. The pure-state trace-distance formula and partial-trace contraction give the bound. This claim concerns the declared vacuum input, not an arbitrary field left by earlier instruments. □

The coefficient bounds also control the clock on a regular window. If both compared fixed-cutoff solutions have $R_b \geq r_* > 0$, $P \geq p_* > 0$, and $R_b \leq r^*$, put $\Delta_C = \sup|\delta\mathcal{C}|$ and $\Delta_P = \sup \kappa D_s(1+|u|)^2$ on a null interval of length ℓ. Matching the initial proper time gives

$$\sup|\delta\tau(u)| \leq 2L\ell\left(\frac{\Delta_C}{2r_*p_*} + \frac{r^*\Delta_P}{p_*^2}\right).$$

An inverse-clock bound follows by dividing by a positive lower bound on $d\tau/du$. The interval and gap conditions must hold in both solutions. They cannot be inferred from fidelity alone.

A finite timing tolerance has its own source condition. For a translation error $h_\delta(u) = g(u-\delta) - g(u)$ and g with the indicated second derivative norms finite,

$$\begin{aligned}\sqrt{k[h_\delta]} &\leq |\delta|\sqrt{k[g']},\\ \|\sqrt{w}\,h_\delta'\|_2 &\leq |\delta|\|\sqrt{w}\,g''\|_2 + \tfrac{1}{2}\delta^2\|g''\|_2.\end{aligned} \tag{86}$$

The first inequality follows from $|e^{i\omega\delta}-1| \leq |\omega\delta|$. For the second, integrate $g''(u-s)$ between 0 and δ and use $\sqrt{1+u^2} \leq \sqrt{1+(u-s)^2} + |s|$. This is a null-clock translation certificate. A proper-clock error must be pulled through its actual worldline map; a finite quantum clock also needs control of its tails and correlations. Small arrival error alone does not supply a continuum stress bound.

C.7 Boundary energy, work and clock conversion

Choose the timelike Dirichlet boundary term, with the normal pointing toward increasing dilaton,

$$I_\partial = \frac{1}{\kappa}\int_{\partial M}\sqrt{-h}\,\Phi(K - L^{-1})\,dt. \tag{87}$$

Its lapse variation fixes the proper-time quasilocal energy

$$E_b = \frac{1}{\kappa}(\Phi_b/L - n^\mu\partial_\mu\Phi).$$

This is the finite-cutoff JT energy with a stated counterterm, rather than a total apparatus energy [60, Sec. 6]. For example, in Gaussian normal coordinates $ds^2 = dr^2 - N^2dt^2$, integrating the bulk curvature term by parts and adding Eq. (87) leaves the lapse boundary term $N(\partial_r\Phi - \Phi/L)/\kappa$. Its variation is $-E_b\delta N$. The dilaton boundary momentum is $(K - L^{-1})/\kappa$.

Proposition C.9 (Energy and work on a running level curve)**.** *In the traceless single-chiral source completion let $\Phi_b = \Phi_b(\tau)$ be a smooth prescribed boundary value. On a future timelike segment put*

$$R_b^2 = \Phi_b^2 - \mathcal{C} > 0, \quad S_b = \sqrt{R_b^2 + L^2\dot\Phi_b^2}, \quad y = S_b - L\dot\Phi_b > 0, \quad q_\tau = f\dot u^2.$$

Then

$$\begin{aligned}\dot u &= \frac{P}{2Ly}, & \dot{\mathcal{C}} &= 2\kappa Lyq_\tau, & \dot R_b &= \frac{\Phi_b\dot\Phi_b - \kappa Lyq_\tau}{R_b},\\ E_b &= \frac{\Phi_b - S_b}{\kappa L}, & K &= \frac{\Phi_b + L^2\ddot\Phi_b + \kappa L^2 q_\tau}{LS_b}, & \dot E_b &= q_\tau + \frac{L^{-1} - K}{\kappa}\dot\Phi_b.\end{aligned} \tag{88}$$

At a fixed cutoff these reduce to $E_b = (\Phi_b - R_b)/(\kappa L)$ and $\dot E_b = q_\tau$. On the source-free stationary family, $dE_b = T_b\,dS_h$ at fixed Φ_b, with $T_b = R/(2\pi LR_b)$.

Proof. Writing the boundary as $v = V(u)$, its dilaton equation and the induced metric give $1 = 4L^2R_b^2\dot{u}^2/P^2 - 4L^2\dot{\Phi}_b\dot{u}/P$. Its positive root is the first equation. Use $\mathcal{C}' = \kappa f P$ for the next two. The gradient identity gives $Ln^\mu\partial_\mu\Phi = S_b$. Along the unit tangent, the JT equation reads $\ddot{\Phi}_b = -\Phi_b/L^2 - \kappa q_\tau + KS_b/L$. This gives K; differentiating E_b proves the work balance. The stationary first law follows by differentiating $E_b = (\Phi_b - \sqrt{\Phi_b^2 - R^2})/(\kappa L)$ and $S_h = S_0 + 2\pi R/\kappa$. □

For example, retaining a constant gap $R_b = r > 0$ under positive flux requires

$$y = \frac{r}{\sqrt{1 + 2\kappa L^2 q_\tau/\Phi_b}}, \qquad \dot{\Phi}_b = \frac{\kappa L y q_\tau}{\Phi_b}.$$

These equations give a nonempty family on regular finite patches. Increasing the cutoff carries the work term in Eq. (88); it is not an uncharged extension of the fixed-cutoff budget. The boundary datum is still externally prescribed. A physical support must produce it and carry its own stress.

The same calculation fixes the conversion to the running horizon clock:

$$z := \frac{d\tau}{d\eta} = \frac{Ly}{R}, \qquad q_\eta = z^2 q_\tau. \tag{89}$$

For a fixed cutoff $y = R_b$, and the stationary temperature is $T_b = (2\pi z)^{-1}$. During a driven contact this is not a global KMS identification. If a local thermal-current regime has separately been justified, entropy flux transforms as $s_\eta = z s_\tau$ and its actual input temperature must satisfy $zT_\tau = 1/(2\pi)$ to use Proposition D.2. Changing coordinates does not prepare that input.

C.8 A thermal detector can be reused without refreshing its field

A finite Gibbs swap provides a physical replacement channel for the detector. Its limitation is equally useful: reset accuracy, retained correlations and heat cannot be specified independently.

Proposition C.10 (Reference-uniform thermal replacement)**.** *During sensing the detector Hamiltonian is degenerate. For reset raise its excited level to $\Delta > 0$ and supply an independent auxiliary bit in $\tau_\beta = \mathrm{diag}(1-\epsilon, \epsilon)$, with $\epsilon = (1 + e^{\beta\Delta})^{-1}$. A resonant SWAP, followed by lowering the detector gap, gives for every joint detector–reference state*

$$\rho_{SA} \longmapsto \tau_\beta \otimes \rho_A.$$

Its half-diamond distance from pure-blank replacement is exactly ϵ. If $p = \Pr(S = 1)$ just before reset, the auxiliary heat and net gap-modulation work are both

$$Q_{\mathrm{reset}} = W_{\mathrm{gap}} = \Delta(p - \epsilon). \tag{90}$$

A finite resonant swap can use $H_{\mathrm{sw}}(\tau) = \chi(\tau)(I - \mathrm{SWAP})$ with $\int \chi\, d\tau = \pi/2$; it commutes with the equal-gap free Hamiltonian. Its net switch work is zero when χ vanishes at both endpoints.

Proof. The swap transfers the whole previous detector state and its reference correlations to the auxiliary, not just its populations. The distance between the two replacement states is ϵ, for every reference, proving the channel statement and attainability. Raising the gap costs Δp, the resonant swap conserves total bare energy, and lowering it returns $\Delta\epsilon$. The outgoing auxiliary has excitation probability p. The expectation of $I-\mathrm{SWAP}$ is constant during the specified resonant contact, so its switching work integrates to zero. □

This replacement process is the standard swap benchmark for Landauer accounting [13, Example 1 and Sec. 3.3]. The auxiliary must be prepared and transported. The old state remains correlated with retained matter; it has not disappeared. If a copy of the outcome is kept, that copy is part of the maintained state in Eq. (26). At finite temperature this finite device never supplies an exact pure blank.

Proposition C.11 (A bounded-heat Gibbs ladder)**.** *Fix $\beta > 0$, a target $0 < \epsilon < 1/2$, and m finite thermal stages. First swap the degenerate detector with an independent zero-gap Gibbs bit $I/2$. Put $x_* = \ln[(1-\epsilon)/\epsilon]$, $x_j = jx_*/m$ and $p_j = (1+e^{x_j})^{-1}$. At stage j raise the detector gap from x_{j-1}/β to x_j/β and resonantly swap with an independent Gibbs bit of that gap. Finally lower its gap to zero. The complete replacement is exactly $\tau_\beta \otimes \rho_A$, with target excitation ϵ, and*

$$\beta Q_{\mathrm{ladder}} = \sum_{j=1}^{m} x_j(p_{j-1} - p_j) \le \ln 2 - h(\epsilon) + \frac{x_*}{m}(\tfrac{1}{2} - \epsilon). \tag{91}$$

The net gap-modulation work equals this heat. Choosing $m \ge x_*/(2\eta)$ gives $\beta Q_{\mathrm{ladder}} \le \ln 2 + \eta$ for every initial detector state and every retained reference.

Proof. The first swap transfers all old correlations to its output. Each later swap leaves an independent Gibbs detector. The heat formula follows from the thermal populations, and summation by parts gives the same net work, including final gap lowering. The sum is a right-endpoint approximation to $\int_0^{x_*} x[-p'(x)]dx = \ln 2 - h(\epsilon)$; its excess is at most $(x_*/m)(p_0 - p_m)$. □

This is an explicit finite swap construction of the type used in Ref. [13, Sec. 6], with a distribution-independent starting step and a bound on its stages. It consumes $m+1$ prepared Gibbs bits, including the zero-gap port, and raises the maximum gap to x_*/β. The finite reset does not hide a pure auxiliary. The first port can leave with less entropy than it had initially; it must be included along with all later outputs and their correlations. For a fixed supplied sequence of operations, N replacements differ from the pure-blank retained-system process by at most the sum of their ϵ_i, including arbitrary intervening instruments on the field. Thus high reset accuracy need not cause divergent heat, but the number of stages and the maximum gap grow. If the replacement changes the realised geometry or timing of those operations, their source and clock errors require the separate joint certificate of Section C.6. The timing, Gibbs preparation, transport and capacity required to execute these stages are additional resources.

Here is an exact two-read law on one field. Let g_1, g_2 have ordered, disjoint compact supports, put $k_i = \langle Q_i^2 \rangle$ and $c_{12} = \langle Q_1 Q_2 \rangle$ in the initial vacuum, and reuse the detector after the above swap. Both contacts start with its state τ_β. Read Z_s and use it for the subsequent command before each reset. A nondemolition dephasing of the pointer can be

implemented by a controlled Z_s using a separate degenerate Gibbs bit $I/2$. Those ports are supplied resources, not pure memories. Since $[Q_1, Q_2] = 0$, the joint outcomes obey

$$\begin{aligned}\Pr(R_1 = r, R_2 = t) = \frac{1}{4}\big[1 + \alpha\{(-1)^r e^{-2k_1} + (-1)^t e^{-2k_2}\} \\ + \alpha^2 (-1)^{r+t} e^{-2(k_1+k_2)} \cosh(4c_{12})\big].\end{aligned} \tag{92}$$

Here $\alpha = 1 - 2\epsilon$. Indeed each conditional pointer probability at a fixed Q_i is $[1 + \alpha(-1)^r \cos(2Q_i)]/2$. The commuting Gaussian characteristic function then gives the displayed law. The same field remains conditioned by the first read: thermal replacement of the detector does not restore a vacuum field. On a stationary test background the contrast of two invariant-dependent single-contact laws is multiplied by α, so it stays nonzero for $\epsilon < 1/2$. The weak-source proof of Proposition C.6 also survives this factor.

A retained body makes the two commands physically consequential. Prepare the body once in $(I + 0.8Z)/2$. After read r, couple it through $P_0 = (I + Z)/2$ or $P_1 = (I + X)/2$ to an independent vacuum control current. The first source population is $\rho_1 = 0.9 - 0.4 \Pr(R_1 = 1)$. If that contact's realised variance is k_c, the first X projection attenuates the body's Z component by $e^{-k_c/2}$. With p_{rt} denoting Eq. (92), its second source population is therefore

$$\rho_2 = \tfrac{1}{2} + 0.4\{p_{00} + e^{-k_c/2} p_{10}\}. \tag{93}$$

This uses the body left by the first command, not a newly prepared copy.

For a finite boundary run take $L = 1$, $\Phi_b = 3$, $\kappa = .01$, initial $(u, a, b, \gamma) = (1, 1, 0, 0)$, $\epsilon = .01$ and $\beta = 1$ in the proper clock. Use the profile $A[1 - (2(\tau - \tau_0)/d - 1)^2]^3$ on an interval of duration d. The two sensor contacts have amplitude .5 on $[0, 2]$ and $[4, 6]$; the two selected control contacts have amplitude .3 on $[2.5, 3.5]$ and $[6.5, 7.5]$. The gaps leave finite windows for readout and reset. Integrating the same mean boundary equations gives

$$k_1 = .068080013, \quad k_2 = .068082993, \quad c_{12} = -.000926643, \quad k_c = .024462001,$$

$$\rho_1 = .871049781, \quad \rho_2 = .870722307, \quad \mathcal{C}_{\text{final}} = 1.030570773.$$

The proper-clock coupling-source charge is .540937293. The two reset auxiliaries separately absorb .573257937, using $\Delta = \ln 99$; this heat has not been inserted as a chiral bulk source. The two outgoing reset bits have joint entropy .519488204, against .112003069 initially, and relative entropy .165772802 to the product Gibbs reference. Thus their heat obeys exactly $\beta Q = \Delta S(B_1 B_2) + D(\rho_{B_1 B_2} \| \tau_\beta^{\otimes 2})$. Other output ports and retained matter belong to the full entropy account. The source charge, reset heat and apparatus work are distinct entries.

Starting instead at $a = 1.4$ with the other initial coefficients unchanged changes both recorded laws and returned source populations on the same proper schedule. The detector and body are reused in both runs. The completion transmits the sensor current and two independent control currents into the bulk. The reset ports, clock and support still require a conserved transport completion; their cost is not removed by the displayed bulk balance. A long sequence of simple Gibbs swaps is also expensive: choosing $\epsilon_N = \varepsilon/N$ to certify total blank-replacement error at most ε requires $\beta\Delta_N = \ln(N/\varepsilon - 1)$. If detector excitation probabilities stay above a fixed positive constant, its heat per reset grows as $\log N$. This particular construction cannot attain the fixed-rate sharp-exhaust conjecture. The ladder of Proposition C.11 avoids this heat divergence, but fitting its stages and transported thermal units into a fixed capacity and operating rate remains necessary.

D Running horizons, source states and clocks

D.1 Source sorting and tracking-clock support

The finite-carrier sorting statement remains Proposition 9.3, with its source and old-reference permissions unchanged.

Proof of Proposition 9.3. The source enters Q-RAIF C's proof in three places: concentration of input eigenlabel strings, the source's fixed neutral charge, and the final source–carrier entropy identity. Its detailed depolarizing eigenvalues are not otherwise used. Each ρ_p has rank two in the same neutral quartet sector and finite information variance at fixed q, ϵ. Independent eigenlabel strings have information variance $O(L)$, including a nonidentical product over the two-symbol alphabet. The sorter's subextensive typical-label margin, thermal-sector state count and physical neutral-sector unitary compiler therefore apply with entropy $\sum_j h_{p_j}$. The source has zero total charge, leaving the carrier's sector weights and sublinear sector correction unchanged. Completing the eigenlabel injection to a sector unitary keeps all bad labels and acts as the identity on external references. The source–carrier entropy identity and controlled full-carrier energy tails give the deficit and slack limits through the Gibbs identity. Trace error alone would not suffice. Uniform local reset error alone does not establish the supplied product marginal for an adaptive history. □

Equation (48) gives the actual clock, not a static-wall substitution. For a tracking datum $R_b = \rho R$, $\Phi_b = \alpha R$, $\alpha = \sqrt{1+\rho^2}$ and $\delta = R^{-1} dR/d\eta$, it gives

$$z = L\sqrt{\rho^2 - 2\alpha\delta}, \qquad \rho^2 > 2\alpha\delta. \tag{94}$$

D.2 The running horizon and its entropy

Long-time entropy limits need an operating geometry, not just a source integral. Positive steady flux can exhaust a fixed u chart. At the boundary $v = -1/u$ the numerator of the retarded dilaton is

$$N_b(u) = 2H_0 - \frac{\kappa}{u}\int^u (u-z)^2 F(z)\, dz.$$

For a constant flux $F_0 > 0$ switched on at $u = 0$, this becomes $2H_0 - \kappa F_0 u^2/3$ and vanishes at $u_* = \sqrt{6H_0/(\kappa F_0)}$. A device at a fixed exterior position cannot therefore be assumed to run at fixed u frequency indefinitely. Finite observation windows and families with larger operating domains are different from an infinite-time protocol in one fixed background.

There is an exact horizon balance inside the retarded solution. In this subsection we use $\hbar = k_B = 1$ and the classical Wald entropy, not an event-horizon generalized entropy. Write the running coefficients as

$$a(u) = H_0 + \kappa\int^u zF(z)\, dz, \qquad b(u) = -\kappa\int^u F(z)\, dz, \qquad \gamma(u) = \kappa\int^u z^2 F(z)\, dz,$$
$$\Phi = \frac{a + bu + \gamma v - auv}{1+uv}, \qquad \mathcal{C}(u) = a^2 + b\gamma, \qquad R(u) = \sqrt{\mathcal{C}(u)}. \tag{95}$$

The lower limits and any source-free initial coefficients are fixed by the matching data. Put $P = bu^2 + 2au - \gamma$ and $\xi = P/(2R)$.

Proposition D.1 (Exact marginal-horizon balance)**.** *On a patch with $\mathcal{C} > 0$, $a + R \neq 0$, $P > 0$ and $1 + uv_h \neq 0$, the curve*

$$v_h(u) = \frac{b}{a + R}$$

is a future outer marginal horizon: $\partial_u \Phi = 0$, $\partial_v \Phi < 0$ and $\partial_v \partial_u \Phi < 0$ there. Its dilaton and Wald entropy obey

$$\Phi_h = R, \qquad S_h = S_0 + \frac{2\pi R}{\kappa}, \qquad \frac{dS_h}{du} = 2\pi F \xi, \qquad \frac{dS_h}{d\eta} = 2\pi F \xi^2, \quad d\eta = \frac{du}{\xi}. \tag{96}$$

For $F \geq 0$ this horizon is spacelike or null and its entropy is nondecreasing.

Proof. The coefficient derivatives $a' = \kappa u F$, $b' = -\kappa F$ and $\gamma' = \kappa u^2 F$ cancel from the first dilaton derivative. With $D = 1 + uv$,

$$\Phi_u = \frac{b - 2av - \gamma v^2}{D^2}, \qquad \Phi_v = -\frac{P}{D^2}.$$

Substitution of v_h gives $\Phi_h = R$ and $\Phi_{uv}|_h = -2R/(1 + uv_h)^2$. Also $\mathcal{C}' = \kappa F P$ and $v_h' = -\kappa F (1 + uv_h)^2/(2R)$. The metric then makes the tangent spacelike for $F > 0$ and null for $F = 0$. Finally $R' = \kappa F \xi$ and $\kappa = 8\pi G$ give Eq. (96). □

The identification of this classical marginal horizon and its area law is a standard JT result [53, Sec. 2.3.1]. Here it assigns an actual horizon observable to the running invariant used in the source account. Negative quantum excess stress need not be monotone, and replacing S_h by S_{gen} would require the matter entanglement contribution and a separate quantum argument.

Proposition D.2 (Conditional entropy-export cost at the horizon)**.** *Suppose the charged heat in the rapidity clock is exactly $\dot{Q}_\eta = F\xi^2$, and the exhaust satisfies Proposition 4.2 in that same clock at $T_H = 1/(2\pi)$. On a window of rapidity duration τ, let X_τ be the net exported maintenance entropy. In the extensive rate regime,*

$$\frac{\Delta S_h}{\tau} \geq \frac{X_\tau}{\tau} + \frac{6}{c_{\mathrm{ex}}} \left(\frac{X_\tau}{\tau} \right)^2 + o(1).$$

Writing bars for the corresponding mean rates gives

$$\overline{\dot{S}_h} \geq \dot{s} + \frac{6}{c_{\mathrm{ex}}} \dot{s}^2. \tag{97}$$

The same pointwise formula requires a justified local-current limit; it does not follow for arbitrary fluctuating packets. A mean horizon-entropy budget $\overline{\dot{S}_h} \leq B$ permits at most

$$\dot{s} \leq \frac{c_{\mathrm{ex}}}{12} \left(\sqrt{1 + \frac{24B}{c_{\mathrm{ex}}}} - 1 \right).$$

Proof. Multiply Eq. (30) at $T = T_H$ by 2π, use the integrated Eq. (96) and the long-window Gibbs account, then invert the increasing quadratic. □

This transfers the entropy removed from the full maintained state to a specified gravitational entropy, including the finite-rate excess. The heat-routing condition is substantive: incoming nonequilibrium energy, other stress sources, charge transfer and the clock's own entropy cannot be omitted. Nor does a change of clock alone prepare a thermal reservoir.

Proposition D.3 (Schwarzian of the running clock)**.** *On a smooth running patch put* $q = F\xi^2$ *and* $\delta = R^{-1}dR/d\eta = \kappa q/R$*. Then*

$$\{u;\eta\} = -\frac{1}{2} - \frac{d\delta}{d\eta} - \frac{\delta^2}{2}. \tag{98}$$

For constant $q > 0$*,* $R(\eta) = R_0 + \kappa q\eta$ *and*

$$\{u;\eta\} = -\frac{1}{2} + \frac{(\kappa q)^2}{2R(\eta)^2}. \tag{99}$$

Proof. Write $j = (bu + a)/R$, and let a dot denote $d/d\eta$. The coefficient equations and $h^2 - 2bR\xi = R^2$, with $h = bu + a$, give

$$\frac{\dot{\xi}}{\xi} = j - \delta, \qquad \dot{j} = \frac{j^2 - 1}{2} - j\delta.$$

Substitute these into $\{u;\eta\} = (\dot{\xi}/\xi)^{\cdot} - (\dot{\xi}/\xi)^2/2$. For constant q, $\dot{\delta} = -\delta^2$, which recovers Eq. (99). □

Constant positive heat current changes the vacuum clock correlation, not merely the horizon size. Positive growth itself, however, does not rule out an exact thermal reference. The following two histories solve the clock condition.

Proposition D.4 (Growing horizons with exact geometric KMS clocks)**.** *Use natural units and the prescribed-mean-source JT equations. An independent chiral conformal reference sector is in its vacuum; its state is not the state of the driven matter supplying* F*. On every finite smooth operating window, the following positive-flux profiles admit the initial matching* $(a, b, \gamma) = (R_0, 0, 0)$ *at* $\eta = 0$*,* $u = u_0 > 0$*:*

1. *Constant fractional growth* $\delta_0 > 0$*:* $R = R_0 e^{\delta_0 \eta}$*,* $q = \delta_0 R/\kappa$*. The reference pulled back to the* η *clock is geometric KMS at* $T = \sqrt{1 + \delta_0^2}/(2\pi)$*.*

2. *Fixed rapidity temperature: for* $\eta_0 > 0$*,*

$$R = R_0(1 + \eta/\eta_0)^2, \qquad q = \frac{2R_0(\eta + \eta_0)}{\kappa\eta_0^2}. \tag{100}$$

   The same reference is exactly geometric KMS at $T_H = 1/(2\pi)$ *despite the growing horizon.*

Here KMS refers to the pulled-back reference algebra and its complete η*-translation dynamics. A finite experiment samples its restriction.*

Proof. Both profiles make Eq. (98) constant, $-s^2/2$, with $s = \sqrt{1+\delta_0^2}$ in the first case and $s = 1$ in the second. Let $\delta(0) = \delta_0$ also in the second case, where $\delta_0 = 2/\eta_0$. An explicit solution with the required initial coordinate and derivatives is

$$u(\eta) = u_0 \left[1 + \frac{e^{s\eta} - 1}{s + m(e^{s\eta} - 1)}\right], \qquad m = \frac{s - 1 + \delta_0}{2}. \tag{101}$$

It has $\xi = \dot{u} > 0$, $\xi(0) = u_0$ and $\dot{\xi}(0)/\xi(0) = 1 - \delta_0$. Set $j = \dot{\xi}/\xi + \delta$ and

$$b = \frac{R(j^2 - 1)}{2\xi}, \qquad a = Rj - bu, \qquad \gamma = bu^2 + 2au - 2R\xi.$$

The constant-Schwarzian equation gives $\dot{j} = (j^2 - 1)/2 - j\delta$. Differentiation then gives $\dot{b} = -\kappa q/\xi$, $\dot{a} = \kappa u q/\xi$ and $\dot{\gamma} = \kappa u^2 q/\xi$. Thus these are the retarded coefficient equations with $F = q/\xi^2$, $a^2 + b\gamma = R^2$ and $P = 2R\xi$. At zero they give $(R_0, 0, 0)$; $a + R$ and the horizon denominator remain nonzero on a sufficiently short initial operating patch, which can be continued while those patch conditions hold.

Equation (101) is a Möbius transform of $e^{s\eta}$. Möbius invariance of the reference vacuum and the geometric KMS construction therefore give inverse temperature $2\pi/s$ [9, Sec. 2.8, Proposition 2.4 and Theorem 2.5]. For a free current the complete Gaussian correlation test is explicit: if $\langle j(u)j(u')\rangle = -C_j/(u - u' - i0)^2$, then

$$\langle \xi(\eta) j(u(\eta))\, \xi(\eta') j(u(\eta'))\rangle = -\frac{C_j s^2}{4\sinh^2[s(\eta - \eta' - i0)/2]}.$$

This is stationary and obeys the KMS boundary condition; the conformal-net argument establishes the state on bounded observables, not just on one stress expectation. □

These are actual nonstationary solutions of the mean-source equations, rather than frozen-background substitutions. The reference sector's thermality is a kinematic input that can now be made compatible with growth. It is not a thermal exhaust realization: the driving sector can be coherent, its state is not KMS merely because the independent reference is, and preparation, heat transport and the physical clock still need their own account. In the second profile q grows linearly, so it does not realize the fixed mean export in Conjecture 9.2. As $\eta \to \infty$, u approaches a finite endpoint and $F = q/\xi^2$ diverges. Each finite smooth window has finite source energy; these formulas assert no finite-resource infinite run.

D.3 The actual thermal source changes its clock reference

A thermal spectator field does not make the exhaust thermal. Here the state condition is imposed on the source field itself. Work with the vacuum-subtracted chiral flux of the preceding JT construction, with the state-independent trace background absorbed consistently into the renormalized JT parameters. A complementary chirality may remain in its reference state. This is a semiclassical conformal-matter completion, not a microscopic construction of a thermal emitter.

For the inner patch let $u = f(v_-)$ be its Poincaré null coordinate. The coordinate change $\rho = \Phi/f' - L^2 f''/(f')^2$ gives

$$\{u; v_-\} = -\frac{\mathcal{C}_-}{2L^4}.$$

Specify the source state as the pullback of a geometric KMS state at temperature T_o in η, with $T_H = 1/(2\pi)$ as the comparison temperature. Define the charged excess current and entropy current by

$$q = \frac{\pi c}{12}(T_o^2 - T_H^2) > 0, \qquad s = \frac{\pi c}{6}(T_o - T_H), \qquad 2\pi q = s + \frac{6s^2}{c}. \tag{102}$$

This state condition is imposed on the finite operating patch. Section D.4 gives a local preparation route from supplied thermal reservoirs and also treats nonthermal outputs; the reservoir and contact's complete material realization remain inputs.

The physical stress is a tensor. The Schwarzian enters when the state is expressed relative to the Poincaré vacuum, through the chiral stress transformation law [7, Eq. (2.14)]:

$$\begin{aligned} F_{vv} &= \left(\frac{d\eta}{dv_-}\right)^2 \left(\frac{c}{48\pi} + q\right) - \frac{c}{24\pi}\{\eta; v_-\} + \frac{c}{24\pi}\{u; v_-\} \\ &= \frac{R^2 q}{L^4} - \frac{c}{24\pi}\{\eta; v_-\}. \end{aligned} \tag{103}$$

The last term in the first line is essential. On a static patch, the T_H thermal state is the zero-flux vacuum reference; retaining only its positive thermal energy would falsely insert a source. Conversely, on a growing patch that same running-clock reference need not have zero physical stress.

Proposition D.5 (State–clock–source consistency)**.** *Put $a = \kappa c/(24\pi)$ and $\delta = R'/R$, where primes in this proposition denote η derivatives. The actual source state and the JT equation are jointly satisfied by*

$$R' = R\delta, \qquad \delta' = \frac{\kappa q - R\delta}{a} - \frac{\delta^2}{2}. \tag{104}$$

The physical flux in this clock and the horizon entropy account are

$$\begin{aligned} F_\eta &= q + q_{\rm ref} = R'/\kappa, & q_{\rm ref} &= -\frac{c}{24\pi}(\delta' + \delta^2/2), \\ \Delta S_{h,-} &= 2\pi q \Delta\eta - \frac{c}{12}\Delta\delta - \frac{c}{24}\int \delta^2 d\eta. \end{aligned} \tag{105}$$

For $R_0^2 > 2a\kappa q$ and $\kappa q/(2R_0) \le \delta_0 \le 2\kappa q/R_0$, the source–clock ODE has a global positive solution in η with

$$\frac{\kappa q}{2R} \le \delta \le \frac{2\kappa q}{R}, \quad R = R_0 + \kappa q \eta + O(1), \quad \int_0^\infty \delta^2 d\eta \le \frac{8\kappa q}{R_0}. \tag{106}$$

Thus $\Delta S_{h,-}/\Delta\eta \to 2\pi q = s + 6s^2/c$. This global statement concerns the source and clock, not the lifetime of a finite material energy reserve.

Proof. Since $dv_-/d\eta = L^2/R$, its Schwarzian is $\{v_-; \eta\} = -\delta' - \delta^2/2$. The chain rule in Eq. (103) gives $F_\eta = q - c(\delta' + \delta^2/2)/(24\pi)$. Together with $\partial_v \mathcal{C}_- = 2\kappa L^2 F_{vv}$ this is Eq. (104); integration gives Eq. (105). At the upper boundary $\delta = 2\kappa q/R$, the derivative of $\delta - 2\kappa q/R$ is $-\kappa q/a + 2(\kappa q)^2/R^2 < 0$. At the lower boundary the derivative of $\delta - \kappa q/(2R)$ is $\kappa q/(2a) + (\kappa q)^2/(8R^2) > 0$. The strip is invariant and $R' \ge \kappa q/2$. Its upper bound gives $\int \delta^2 \le \int_0^\infty 4(\kappa q)^2 (R_0 + \kappa q\eta/2)^{-2} d\eta$. Finally $R' + a\delta' + a\delta^2/2 = \kappa q$ gives the bounded correction to linear growth. All variables stay finite on finite η intervals. □

The charged heat q and physical source F_η differ by an explicitly accounted reference stress. Proposition D.2 remains conditional on their equality; it cannot be applied pointwise to q after discarding q_{ref}. Here the reference correction is nonextensive, so the sharp long-time rate is recovered by the stronger state–clock calculation. A finite patch does not establish a globally finite-energy state across a limiting Poincaré coordinate endpoint.

For any fixed finite proper-time window, nonempty operating solutions also follow without relying on the numerical table. At $q = 0$, start from the positive static support, $e > 0$, $R_0 > 0$ and $\delta = 0$. This is a regular equilibrium of the combined equations. Continuous dependence on q, with $\delta_0 = \kappa q/R_0$, preserves positive mass, energy reserve and exterior domain on that window for sufficiently small $q > 0$. This is not a fixed-rate infinite-lifetime statement.

Combining Eq. (104) with Eq. (125) uses $P_{\mathrm{out}} = R'/[\kappa z^2]$ and $P_{\mathrm{in}} = 0$. For the support above, initial $e = 2.34$ and $\delta_0 = \kappa q/R_0$, all three examples below run for proper duration 80 with positive reserves and $\Phi > R$:

q	η_{end}	Φ_{end}	e_{end}
.005	85.53862	3.003858	1.882696
.010	85.60190	3.007607	1.424039
.020	85.73040	3.015345	.502569

The mass-junction residual is below 1.2×10^{-12} on the checked grid. These solve the moving mean support and the state condition together. The internal reserve's decline is part of the solution. It has not been derived from the seven-qubit core's local release Hamiltonian.

D.4 A local thermal contact supplies the actual source

The source condition can be replaced by a scattering calculation. Use two complex chiral fermion ports, each with $c = 1$, at a local conformal contact. This is the doubled free-fermion defect of Bernard, Doyon and Viti [5, Sections 3 and 5]. The zero-charge input streams are jointly independent and thermal at T_H and T_o in the same supplied η frame. Their fields mix by $S_\theta = \cos\theta I - i \sin\theta \sigma_x$ at the contact. Put $\chi = \sin^2\theta$. For positive frequency, the source occupation is

$$n_\chi(\omega) = (1-\chi) n_{T_H}(\omega) + \chi n_{T_o}(\omega), \qquad n_T(\omega) = \frac{1}{1 + e^{\omega/T}}. \tag{107}$$

The actual output is hot KMS for $\chi = 1$. For $0 < \chi < 1$ it is stationary but nonthermal: its high-frequency tail has coefficient χ rather than one. Its return-port cross covariance is nonzero. These are retained field correlations, not fresh independent thermal maps. Q-RAIF C derives the complete contact law and its energy account [14, Section on a local thermal contact].

Proposition D.6 (Stationary transfer without a KMS output)**.** *For a constant contact angle set*

$$q_\chi = \frac{\pi\chi}{12}(T_o^2 - T_H^2), \qquad s_\chi = \frac{1}{\pi}\int_0^\infty [h(n_\chi) - h(n_{T_H})]\, d\omega. \tag{108}$$

Here $h(x) = -x \ln x - (1-x)\ln(1-x)$. *The state–clock equations, positive growing family and finite reference correction in Proposition D.5 hold with* $c = 1$ *and* $q = q_\chi$.

Their asymptotic horizon rate is

$$\frac{\Delta S_{h,-}}{\Delta\eta} \longrightarrow 2\pi q_\chi \geq s_\chi + 6s_\chi^2. \tag{109}$$

The inequality is strict for $0 < \chi < 1$ *and* $T_o > T_H$.

Proof. The stationary output's stress in η is $1/(48\pi) + q_\chi$. Its coordinate/state transformation has the same central charge and projective-vacuum subtraction as Eq. (103). That derivation uses only this stress expectation, not the KMS property. The source–clock ODE and its invariant strip therefore follow unchanged. At fixed energy the Gibbs state maximizes the fermion entropy current. Its temperature is $T_{\rm eff} = \sqrt{T_H^2 + 12q_\chi/\pi}$ and its entropy excess is $\pi(T_{\rm eff} - T_H)/6$. Since the partially transmitted state is not that Gibbs state, the entropy inequality is strict. Eliminate $T_{\rm eff}$ and use $T_H = 1/(2\pi)$. □

For a hot-input excess .02 and $\chi = .5$, the output has $q_\chi = .01$, $s_\chi = .04818703$ and thermal-floor excess $q_\chi - T_H s_\chi - 3s_\chi^2/\pi = .000113460$. A KMS output with the same heat has $s = .04863795$. Their mean stress, geometry and clock can be identical despite their different states and entropy currents. A geometric response alone does not establish thermalization or identify the source's complete state.

A program that changes the contact also changes its stress. For a smooth prescribed angle, point splitting the full unequal-time covariance gives

$$q_{\rm eff}(\eta) = \sin^2\theta(\eta) q_{\rm hot} + \frac{\theta'(\eta)^2}{4\pi}, \qquad F_\eta = q_{\rm eff} + q_{\rm ref}. \tag{110}$$

Each output port receives the same pump term. The total field work is $\int \theta'^2 d\eta/(2\pi)$ and is at least $(\Delta\theta)^2/(2\pi\Delta\eta)$ over a switching interval. Thus switching the selected interface has a source cost in the same interaction, rather than triggering an unrelated emitter. The running clock equations use $q_{\rm eff}(\eta)$, retaining their reference stress. They also permit a stronger statement for changing programs.

Proposition D.7 (Positive growth under bounded changing sources)**.** *Suppose* $q_{\rm eff}$ *is smooth,* $0 < q_{\min} \leq q_{\rm eff}(\eta) \leq q_{\max}$ *and* $R_0^2 > 2a\kappa q_{\max}$*, with* $\kappa q_{\min}/(2R_0) \leq \delta_0 \leq 2\kappa q_{\max}/R_0$*. Then the source–clock ODE with* $q_{\rm eff}$ *has a global positive solution satisfying*

$$\begin{aligned} \frac{\kappa q_{\min}}{2R} \leq \delta \leq \frac{2\kappa q_{\max}}{R}, \qquad & R' \geq \frac{\kappa q_{\min}}{2}, \\ \int_0^\infty \delta^2 d\eta \leq \frac{8\kappa q_{\max}^2}{q_{\min} R_0}, \qquad & R - R_0 = \kappa \int q_{\rm eff} d\eta - a\Delta\delta - \frac{a}{2}\int \delta^2 d\eta. \end{aligned} \tag{111}$$

The reference correction stays bounded. If the actual source has a long-time mean, the mean horizon entropy rate is 2π *times that mean. This concerns the source–clock sector, not an unlimited body energy reserve.*

Proof. At the upper barrier, the relative derivative is at most $-\kappa q_{\max}/a + 2(\kappa q_{\max})^2/R^2 < 0$. At the lower barrier it is at least $\kappa q_{\min}/(2a) + (\kappa q_{\min})^2/(8R^2) > 0$. Thus the strip is invariant. Integrate its upper bound using $R \geq R_0 + \kappa q_{\min}\eta/2$ to obtain the integral estimate. Integration of $R' + a\delta' + a\delta^2/2 = \kappa q_{\rm eff}$ gives the last identity. The strip also prevents finite-time blowup. □

A bounded clock-reference correction does not remove the program's work. If N disjoint contact changes each have angle increment at least $\Delta_{\min}$ and duration at most ν^{-1}, their total pump work is at least $N\nu\Delta_{\min}^2/(2\pi)$. At $N/\Delta\eta = \alpha\nu$ the mean work current is at least $\alpha\nu^2\Delta_{\min}^2/(2\pi)$, half in each output. This is an additional switching-charge requirement of this contact architecture, not a universal heat bound to be added again when that same output energy has already been counted in the entropy account.

The source is prepared by the contact, but the input reservoir is still prepared in a specified clock frame. The contact's local scattering law does not supply a microscopic moving material body or its controller. Its other port, reservoir preparation, switching work, clock conversion and recoil must be included before it can replace the complete source in the mass-10 table. Nor may a stationary entropy formula be applied instantaneously during the switch. For a stationary frequency cutoff Λ with the common vacuum retained, a separate current bound is

$$|\Delta q| \le \frac{1}{\pi} \sum_{T=T_H,T_o} T(\Lambda + T)e^{-\Lambda/T}.$$

This controls omitted thermal source energy, not the field's global trace distance or the stress of turning the contact on.

E Finite preparation and quantum material source certificates

E.1 Finite preparation on a predicted running clock

The finite interval need not be supplied by an infinite-energy KMS state. Q-RAIF C constructs a compact diffeomorphism of the vacuum that agrees with $\exp[2\pi T\eta(v_-)]$ on the regular operating interval [14, Section on thermal correlations from a finite-energy preparation]. Its full local covariance equals the thermal η covariance, while the global state is pure and has finite energy and its second moment. Two independent copies give the local contact's required inputs, retaining each copy's outside purification. Their stress is exactly Eq. (103) after the same Poincaré subtraction. A local bounded regularization and positive history clock approximate this preparation and its smooth stress to any fixed finite tolerance. The field-energy gain is paid by the clock-interaction energy; this is not a free reservoir renewal.

The prepared states and inertial contact now have a common fermion carrier [14, Proposition on a positive contact on the full CAR carrier]. A compact-gauge compensation proves the single-contact Hamiltonian positive and self-adjoint even when its static dressing is not implemented. The flavor rotation retains the original two fields' correlations instead of replacing them by independent current baths. On this carrier the actual reaction and the exterior smooth-source certificate apply together. Matching that quantum contact to the moving material action remains a separate requirement.

A record selected by earlier local operations can also choose the next thermal preparation on this same field [14, Proposition on record-selected local thermal preparation]. If its past field protocol lies in an interval of length a, the next preparation support has length b, and their gap is d, then the two complex chiral fields give $I_0 = \frac{1}{3}\log[1+ab/(d(d+a+b))]$. The probability-weighted conditional thermal error is at most $\sqrt{I_0/2}$. A smooth source test with actual and comparison second-moment norms M_a, M_i has error at most

$\sqrt{I_0}(M_a + M_i)$, in addition to the preparation-gate and clock errors. The outside purification and prior records remain in the device. The same relative-entropy bound includes a controller and its initially retained references obtained by the local past protocol; it passes through later joint channels on that controller and window. Additional incoming fields must be included rather than treated as independent. Separation, physical record transport and a new preparation are charged; the comparison state is not a physical reset of a correlated field. This permits a conditional finite update, while causal transport in the moving material and renewal of its clock, blanks and reference remain to be realized together.

The field contact can also acquire a finite-mass quantum position [14, Section on the contact position]. The same full-CAR Hamiltonian then conserves body-plus-field momentum; a finite anchor contributes its own recoil. Its moving-frame term is $(p - H_f)^2/(2M)$, so the fixed contact is a fixed-momentum large-mass limit, not a free prescribed trajectory. Position tails require a generator-weighted locality error before using the record-controlled thermal certificate. This support is nonrelativistic and does not yet provide the covariant material stress or the physical clock required below. Prescribing a constant velocity is a different limit: the harmonic support has a stiffness-dependent stability boundary, while the full finite-mass recoil Hamiltonian remains positive.

Energy conservation in the unfolded carrier does not identify its translation generator with bulk momentum. Folding its two flavors into right- and left-moving fields makes the finite-width mixing a mirror coupling. A local bidirectional current instead needs two flavors per direction, with $P = H_R - H_L$ and recoil $(p - P)^2/(2M)$ [14, Proposition on bidirectional current and physical recoil]. Its harmonic prescribed-velocity threshold is $\sqrt{a_0/(a_0 + d_R + d_L)}$, while the full finite-mass energy remains positive. The added preparations and purifications enter the source account, and each exhaust capacity counts its actually routed fields. For the same interval lengths a, b and gap $g > 0$, their product free vacuum gives $I_0 = \frac{2}{3}\log[1+ab/(g(g+a+b))]$, twice the two-field renewal value above; the conditional errors use that value. This supplies a physical momentum and local field density; deriving the propagating material's renormalized quantum stress is still needed before inserting that density as the complete matter in a JT apparatus.

For a flat-coordinate interval of length ℓ, the compact vacuum-orbit preparations at temperature T have infimum energy

$$E_{\min} = \frac{\pi c T^2 \ell}{12} - \frac{cT}{6}\tanh\frac{\pi T \ell}{2}.$$

On a changing clock the actual map, including its compact completion, sets the physical preparation cost. Its waveform cannot be redefined after a read changes the future trajectory.

Proposition E.1 (Source error from a predicted thermal clock)**.** *Prepare with $\widehat{\eta}$ while the actual clock is η. On a fixed regular source interval suppose both first derivatives are at least $m > 0$, both jth derivatives have absolute value at most M_j, and $\|\eta^{(j)} - \widehat{\eta}^{(j)}\|_\infty \le \epsilon_j$ for $j = 1, 2, 3$. Put $J_T = \pi c T^2/12$. The prepared one-point stress differs from the exact η-thermal stress by at most*

$$B_{\text{clock}} = 2J_T M_1 \epsilon_1 + \frac{c}{24\pi}\left(\frac{\epsilon_3}{m} + \frac{M_3\epsilon_1}{m^2} + \frac{3M_2\epsilon_2}{m^2} + \frac{3M_2^2\epsilon_1}{m^3}\right). \tag{112}$$

A constant two-port contact combines its two input bounds with weights $1-\chi, \chi$. A smooth source test f has additional clock error at most $\|f\|_1 B_{\text{clock}}$, separate from the autonomous execution's state and stress error.

Proof. Apply $|a^2 - b^2| \le 2M_1|a-b|$ to the squared Jacobian. The ratio of third to first derivative changes by at most $\epsilon_3/m + M_3\epsilon_1/m^2$. Factoring the difference of the squared second-to-first ratios gives $2M_2\epsilon_2/m^2 + 2M_2^2\epsilon_1/m^3$. The Schwarzian's factor $3/2$ yields Eq. (112). The same coordinate-vacuum subtraction cancels between the two preparations, not between different geometric backgrounds. □

This tolerance makes a record-selected prediction testable without an exactly known future geometry. It controls mean source, not complete state thermality or entropy. Learning a predictor that maintains the stated derivative tube, and embedding the local gate, clock and two-port stress in the moving material action, remain physical tasks. The thermal restriction has an exterior purification; it cannot be inserted unchanged into a finite-window entropy-reset argument.

E.2 A matched quantum material source

Positivity of an excitation Hamiltonian leaves its vacuum contribution to the source undetermined. A local Gaussian material makes the required comparison explicit. On a finite interval with Dirichlet boundaries, let $a \in C_c^\infty$ vanish near the endpoints and $m > 0$. The spatial form for an active field ϕ and a normalized angle Θ is

$$q_a = \int \left[|\phi' + a\Theta|^2 + |\Theta'|^2 + m^2|\Theta|^2\right] dx.$$

It is closed and coercive. Its operator, with the same boundary conditions for $a = 0$, is

$$K_a = \begin{pmatrix} -\partial_x^2 & -\partial_x a \\ a\partial_x & -\partial_x^2 + a^2 + m^2 \end{pmatrix}.$$

Here $-\partial_x a$ denotes $-\partial_x \circ a$, including the derivative of the field. A constant positive angle–storage mass matrix and free spectator fields leave the following coefficient unchanged.

Proposition E.2 (Matched material vacuum coefficient)**.** *For the proper-time cutoff comparison*

$$\Delta E_\Lambda = -\frac{1}{4\sqrt{\pi}} \int_{\Lambda^{-2}}^\infty t^{-3/2} \operatorname{Tr}(e^{-tK_a} - e^{-tK_0})\, dt,$$

and a fixed reference scale $\mu > 0$,

$$\Delta E_\Lambda = \frac{\log(\Lambda/\mu)}{8\pi} \int a^2 dx + C_\mu[a] + o(1). \tag{113}$$

The finite part requires a common material normalization condition.

Proof. Write $K_a = -\partial_x^2 + B\partial_x + C$, where $B = \left(\begin{smallmatrix} 0 & -a \\ a & 0 \end{smallmatrix}\right)$ and $C = \left(\begin{smallmatrix} 0 & -a' \\ 0 & a^2+m^2 \end{smallmatrix}\right)$. The antisymmetric connection $A = -B/2$ gives $K_a = -(\partial_x + A)^2 + P$, with

$$P = C - B'/2 + B^2/4 = \begin{pmatrix} -a^2/4 & -a'/2 \\ -a'/2 & m^2 + 3a^2/4 \end{pmatrix}, \qquad \operatorname{tr}(P - P_0) = a^2/2.$$

The heat coefficient for $-(\nabla^2 + E)$ uses $E = -P$ [3, Eq. (4.27)]. Matched boundary jets cancel, so the trace difference is $-\sqrt{t}\int a^2/(4\sqrt{\pi}) + O(t^{3/2})$. Its proper-time integral gives Eq. (113). Coercivity controls the large-t end of that integral. □

The coefficient can be assigned to a covariant material action. For the positive target metric

$$ds^2_{\rm mat} = (d\phi + \lambda\theta f(Z)dZ)^2 + m_Z^2 dZ^2 + I d\theta^2,$$

linearization about a static texture Z_0, with $\theta = 0$, gives $a = \lambda f(Z_0)Z_0'/\sqrt{I}$. Direct curvature calculation yields $\mathrm{Ric}_{ZZ} = -\lambda^2 f(Z)^2/(2I)$ there. The usual one-loop kinetic counterterm in the Euclidean action [4, Section 7.2] is, in the present normalization,

$$S_{\mathrm{ct},E} = \frac{\log(\Lambda/\mu)}{4\pi} \int \sqrt{g}\, \mathrm{Ric}_{AB}\, \partial_\mu X^A \partial^\mu X^B.$$

On the texture it cancels the logarithm in Eq. (113). Other potential, boundary and metric counterterms must be matched as well. The finite Gaussian model is not the full charged-fermion contact or an all-order continuum construction.

The charged carrier can nevertheless be retained at finite material resolution. Q-RAIF C's positive material construction [14] uses the full fixed-spin charge/current basis, including the unused oscillator modes, and keeps the material momentum squares. Its boson–fermion correspondence is an operator equivalence, not a neutral-sector replacement. This supplies a positive nonlinear regulated Hamiltonian. The local continuum, its thermal states and its renormalized source remain to be matched; the correspondence does not determine a new vacuum subtraction for each contact. Smoothing a permitted bounded gate under this interacting Hamiltonian also supplies the energy regularity needed for autonomous record-controlled operation. Its positive regulated energy tests have a source certificate, while their covariant continuum stress still requires a separate limit. A local positive binding potential also produces finite normalizable angle and storage modes with the finite contact's mechanical coefficients [14]. Their omitted material modes give an explicit self-energy and a filtered causal-response error. This avoids an infinite uniform angle, but an exact local light front and quantum thermal state are not supplied by the collective-mode projection. Local record-selected drives can instead be compared on the same complete Gaussian Hamiltonian and initial state [14], Proposition on local actuation with the initial state retained. The actual field's characteristic function controls the joint state error with references retained, and a centred Gaussian source changes by the classical full retarded response. Its force work is charged; an old mean contributes a cross term. This supplies a quantum actuation/source bridge in the fixed-texture sector, not a new vacuum for the projected body or a proof of the moving clock's stress and thermal renewal.

A source can be derived from this driven material rather than specified as a separate emitter. In the corresponding flat Gaussian sector, ξ_Θ is the complete record-driven angle response, including its coupling to the carrier. Outside the compact contact, the actual right and left profiles satisfy

$$\begin{aligned} \xi_R(u) &= -\frac{1}{2}\int a(y)\xi_\Theta(y, u+y)dy, & \xi_L(v) &= \frac{1}{2}\int a(y)\xi_\Theta(y, v-y)dy, \\ \Delta T_{uu} &= (\partial_u \xi_R)^2, & \Delta T_{vv} &= (\partial_v \xi_L)^2. \end{aligned} \tag{114}$$

Here $u = t - x$, $v = t + x$, the past displacement is zero and the initial Gaussian mean is zero. The carrier equation is $(\partial_t^2 - \partial_x^2)\xi_\phi = \partial_x(a\xi_\Theta)$. Its retarded Green function, integrated once in y, proves the two profiles. Unchanged covariance and common vacuum subtraction prove the stress identities. The integrated two-side carrier radiation is at most the actual force work; the rest is retained material energy. Different finite drive records

can therefore change the source through actual material motion. This is not injective: opposite drives have equal quadratic stress on centred input, and $a = 0$ emits no carrier source while material energy can remain. Putting these flat-background data into JT requires a common quantum-state and coordinate map plus the driver, moving support and clock's stress; the formulas alone do not supply that map.

Finite autonomous execution also needs a source certificate for this material, rather than the free-carrier estimate. On a fixed Dirichlet interval, Q-RAIF C derives a relative form bound for each compact-time quadratic energy or exterior null-stress test O:

$$\|(H+\Lambda)^{-1/2}O(H+\Lambda)^{-1/2}\| \le C_O, \qquad H = d\Gamma(\sqrt{K}), \quad \Lambda > 0.$$

The complete Gaussian mode tower and its common vacuum subtraction are retained. If actual and target output purifications have $e = \|\psi - \varphi\|$, $g_H = \|H(\psi - \varphi)\|$, then

$$|\langle O\rangle_\psi - \langle O\rangle_\varphi| \le C_O\sqrt{\Lambda e^2 + e g_H} \times \left[\sqrt{\Lambda + \|H\psi\|} + \sqrt{\Lambda + \|H\varphi\|}\right]. \tag{115}$$

Energy-regular return under this same H therefore certifies a common output source reading, with old references retained. Uniform physical rms bounds M_a, M_i give $g_H \le M_a + M_i$, so the certificate converges with its state error despite the growing history clock. The bound does not use ordinary trace distance on an unbounded stress. It is an output comparison under the agreed subsequent evolution, not an account of the driver and clock stress during their interaction. Its constants depend on the finite interval. Matched source tests can then enter the response estimate of this paper only after the state, coordinate and total-source identification is supplied; neither a uniform real-line limit nor the full moving JT apparatus follows from this Gaussian certificate. The unbounded-time weighted moments in the JT source hypothesis also need their own tail control; finite smooth tests alone do not certify them.

A physical change of the coupling or texture changes this fluctuation operator. It must use the same measured material parameters and vacuum normalization as the original configuration. Independently setting each contact's vacuum energy to zero would also discard its response to those changes. Conversely, the bare logarithm supplies no finite universal gravitational price. A covariantly renormalized material functional Γ, with $T_{\mu\nu} = -2\delta\Gamma/(\sqrt{-g}\,\delta g^{\mu\nu})$ and $E_i = \delta\Gamma/(\sqrt{-g}\,\delta X^i)$ for its scalar arguments, obeys

$$\nabla_\mu T^\mu{}_\nu = E_A\partial_\nu X^A + E_\Phi\partial_\nu\Phi,$$

where E_A, E_Φ are its variational equations. On the material equations this retains the compensating dilaton force. The same functional must be varied for the material and gravity; a static Euclidean vacuum comparison cannot replace the causal, initial-state response of a controller.

F Supported apparatus and an acquired-record source return

F.1 A material support with its own conserved core

A boundary action gives its conjugate energy but not the matter that holds the boundary. A metric-only local apparatus is also blind to the dilaton on a fixed AdS_2 patch. We now

change the matter model explicitly, without applying the preceding boundary readout laws to this new apparatus. In natural units add a timelike worldline Γ to the JT action:

$$I = \frac{1}{2\kappa}\int \sqrt{-g}\,\Phi(R+2/L^2)\,d^2x + I_{\rm outer} + I_{\rm fields} - \int_\Gamma M(\Phi)\,d\tau,$$
$$M = m_{\rm rest} + U(\Phi) + \langle H_{\rm core}\rangle. \tag{116}$$

The usual outer boundary terms are included. Take $m_{\rm rest} > 0$ and $U \geq 0$. The core has worldline quantum action $\int d\tau\,[i\psi^\dagger D_\tau\psi - \psi^\dagger(m_{\rm rest}+U+H_{\rm core})\psi]$; its expectation provides a semiclassical worldline source; the apparatus's internal spatial geometry is not resolved in this approximation. Q-RAIF C constructs a finite time-independent, nonnegative $H_{\rm core}$ containing the battery, clock-dependent interactions, detector, retained body and every thermal output cell [14]. Its mean energy is conserved. Output cells remain inside the apparatus until a release operation is supplied.

Let δ_Γ be the covariant line delta and $M_\Phi = \partial M/\partial\Phi$. Variation gives

$$R + 2/L^2 = 2\kappa M_\Phi \delta_\Gamma, \tag{117}$$
$$E_{\mu\nu} := g_{\mu\nu}\Box\Phi - \nabla_\mu\nabla_\nu\Phi - g_{\mu\nu}\Phi/L^2 = \kappa T^{\rm total}_{\mu\nu}. \tag{118}$$

The material stress is not separately divergence-free:

$$\nabla_\mu T^{{\rm total}\,\mu\nu} = -M_\Phi\delta_\Gamma\nabla^\nu\Phi, \qquad \nabla_\mu E^{\mu\nu} = -\tfrac{1}{2}(R+2/L^2)\nabla^\nu\Phi. \tag{119}$$

The identities agree by Eq. (117), including energy exchange with the dilaton. Before radiation forces are added, $Ma^\mu = -(g^{\mu\nu}+u^\mu u^\nu)\nabla_\nu M$. Direct dilaton coupling therefore changes the curvature equation at the wall.

Proposition F.1 (Positive static support)**.** *In a symmetric thin-wall limit choose a common normal toward increasing dilaton. At a static wall let $\Phi = \phi > 0$, with derivatives $d_\pm = \partial_n\Phi|_\pm$ and extrinsic curvatures $K_\pm$. The vacuum junction and constraint equations are*

$$[d] = -\kappa M, \quad [K] = -\kappa M_\Phi, \quad K_\pm d_\pm = \phi/L^2, \quad M\overline{K} + M_\Phi\overline{d} = 0. \tag{120}$$

For $M > 0$ and $D > \kappa M/2$, they have the solution

$$d_\pm = D \mp \kappa M/2, \qquad K_\pm = \frac{\phi}{L^2 d_\pm}, \qquad M_\Phi = -\frac{\phi M}{L^2 d_+ d_-}.$$

A positive quadratic material potential realizes these data whenever its stiffness a obeys $M - M_\Phi^2/(2a) > \langle H_{\rm core}\rangle$. If a positive inner invariant is required, also choose $\phi > Ld_-$. For every finite conserved core mean energy, a choice of M, D and sufficiently large a gives positive rest mass and linear radial stability.

Proof. In Gaussian normal coordinates $ds^2 = dr^2 - N(r)^2dt^2$, $R = -2N''/N$ gives the curvature jump. The singular tt metric equation gives $[d] = -\kappa M$, and the regular rr equation gives $Kd = \phi/L^2$. Its jump gives the averaged force equation. The displayed data satisfy all four conditions. Choose

$$\phi_* = \phi - M_\Phi/a, \quad U(\Phi) = \tfrac{a}{2}(\Phi-\phi_*)^2, \quad m_{\rm rest} = M - \frac{M_\Phi^2}{2a} - \langle H_{\rm core}\rangle > 0.$$

The regular patches near the wall are $N_\pm = \cosh(r/L) + LK_\pm\sinh(r/L)$ and $\Phi_\pm = \phi\cosh(r/L) + Ld_\pm\sinh(r/L)$. They have positive lapse there and solve the vacuum equations. □

For $L = 1$, $\kappa = .01$, $M = 10$, $\phi = 3$, $D = 2$ and $a = 100$,

$$d_+ = 1.95, \quad d_- = 2.05, \quad \mathcal{C}_+ = 5.1975, \quad \mathcal{C}_- = 4.7975, \qquad \mathcal{C}_\pm = \phi^2 - L^2 d_\pm^2.$$

Here $M_\Phi = -7.50469043$, $\phi_* = 3.07504690$, and $m_{\rm rest} + \langle H_{\rm core}\rangle = 9.71839811$. The two-cycle core with 39 battery levels has mean 2.34, including its clock-dependent interaction, so $m_{\rm rest} = 7.37839811 > 0$. The 135-level core has mean 7.14 and still fits with positive rest mass. Support and a running finite internal circuit thus have jointly satisfiable semiclassical source data, with clock and battery included in the mass.

The reference support is locally stable in radial vacuum dynamics. On the positive-normal branch let $\beta_\pm = \sqrt{\dot\Phi^2 + (\Phi^2 - \mathcal{C}_\pm)/L^2}$. The moving-wall junction $\beta_- - \beta_+ = \kappa M(\Phi)$ gives

$$\dot\Phi^2 + V(\Phi) = 0, \qquad V = \frac{\Phi^2 - \mathcal{C}_+}{L^2} - \left[\frac{\mathcal{C}_+ - \mathcal{C}_-}{2\kappa L^2 M(\Phi)} - \frac{\kappa M(\Phi)}{2}\right]^2. \tag{121}$$

At the reference, $V = V' = 0$, $V'' = 388805871/5681780 > 0$, and the proper-time linear frequency is 5.84937147. Increasing $\mathcal{C}_+$ by 10^{-5} gives a small oscillation between approximately 2.99755 and 3.00244. This proves linear radial stability at one vacuum support, not stability through an arbitrarily long export history.

The stability construction extends beyond the numerical example. At the supported point, differentiation of Eq. (121) with fixed $\mathcal{C}_\pm$ gives

$$V''(\phi) = \frac{2}{L^2} - \frac{2\phi^2}{L^4 d_+^2} - \frac{4D\phi^2}{L^4 d_+ d_-^2} + \frac{2d_+ d_-}{M}\, a. \tag{122}$$

Choose $M > \langle H_{\rm core}\rangle$ and $D > \kappa M/2$ first. The coefficient of a is positive, so finite stiffness can satisfy both $m_{\rm rest} > 0$ and $V'' > 0$. This supports any fixed finite compiled core; increasing accuracy or operating length may require a larger material device.

F.2 An output pulse must recoil against its source

A thermal cell becomes radiation only through a release mechanism. Even an idealized event must conserve momentum as well as energy. Suppose the initially static wall emits an inward null pulse of energy q in its initial local rest frame. Take an isolated instantaneous event with $0 < q < M/2$, with no external holding impulse. Local four-momentum conservation fixes

$$M_f = \sqrt{M^2 - 2Mq}, \quad v_f = \frac{q}{M-q}, \quad \dot\Phi_f = \frac{d_+ q}{M_f}, \quad \beta_{+,f} = \frac{d_+(M-q)}{M_f}. \tag{123}$$

Proposition F.2 (The source–support charge balance)**.** *Keeping $\mathcal{C}_+$ fixed and imposing the moving-wall junction after the event gives*

$$\mathcal{C}_{-,f} - \mathcal{C}_- = 2\kappa L^2 d_- q. \tag{124}$$

The wall's Killing charge $E_\Gamma = (\mathcal{C}_+ - \mathcal{C}_-)/(2\kappa L^2)$ loses exactly $d_- q$, the pulse's initial interior Killing energy. For an internal mass drop $\Delta = M - M_f$, $q = \Delta - \Delta^2/(2M)$ and recoil energy is $\Delta^2/(2M)$.

Proof. Initially $E_\Gamma = Md_+ + \kappa M^2/2$; after emission it is $M_f\beta_{+,f} + \kappa M_f^2/2$. Equation (123) gives their difference $(d_+ + \kappa M)q = d_- q$. Solving the mass shell for q gives the last claim. □

At $q = .06$ in this support, $M_f = 9.93981891$ and $\Delta\mathcal{C}_- = .00246$. Reducing mass while pinning the wall would introduce the spurious term $-\kappa^2 L^2 q^2$. The inner horizon entropy has infinitesimal response

$$\left.\frac{dS_{h,-}}{dq}\right|_{q=0} = \frac{2\pi L^2 d_-}{\sqrt{\mathcal{C}_-}}, \qquad S_{h,-} = \frac{2\pi}{\kappa}\sqrt{\mathcal{C}_-}.$$

The number $q = .06$ is an emission example, not the finite core's net heat automatically sent into the bulk. Its cells contain gross energy and an initial thermal contribution. The reference and all correlated ports must be transported before an excess source can be identified with charged heat. The finite energy-sector gates do not establish a local field release Hamiltonian or prepare a KMS exhaust on a running horizon clock. This branch supplies a positive support and its charge balance while leaving those transport operations explicit.

F.3 Continuous radiation moves the supported apparatus

The mass, recoil and radiation can be evolved together. On each side of the material wall use an ingoing Eddington–Finkelstein coordinate:

$$ds_\pm^2 = -f_\pm dv_\pm^2 + 2dv_\pm d\Phi, \qquad f_\pm = \frac{\Phi^2 - \mathcal{C}_\pm(v_\pm)}{L^2}.$$

Keep the normal toward increasing Φ and write $w = d\Phi/d\tau$, $\beta_\pm = \sqrt{w^2 + f_\pm}$. Let $P_{\rm in}$ be the proper power of inward radiation absorbed from the outer side, and $P_{\rm out}$ the inward power sent into the inner side. These are the actual renormalized source powers; an excess over a thermal reference is specified separately below. Put $M = m_{\rm rest} + U(\Phi) + e$ and $A = U'(\Phi)$, where e is usable internal mean energy. The mean worldline approximation is that of Eq. (116); junctions with dilaton-coupled surface matter are standard gravitational constructions [6, Sec. V].

Proposition F.3 (Radiating material junction)**.** *For smooth supplied powers and initial data with $M > 0$, $\beta_- - \beta_+ = \kappa M$ and $\beta_\pm - w > 0$, the coupled equations are*

$$\begin{aligned}
\dot e &= P_{\rm in} - P_{\rm out}, & \dot M &= Aw + P_{\rm in} - P_{\rm out},\\
\dot w &= \frac{\beta_+ P_{\rm out} - \beta_- P_{\rm in} - A\beta_+\beta_-}{M} - \frac{\Phi}{L^2}, & &\\
\dot{\mathcal{C}}_+ &= 2\kappa L^2 P_{\rm in}(\beta_+ - w), & \dot{\mathcal{C}}_- &= 2\kappa L^2 P_{\rm out}(\beta_- - w),\\
\dot v_\pm &= (\beta_\pm - w)^{-1}. &&
\end{aligned} \tag{125}$$

Dots denote proper time. These equations preserve the mass junction and obey $[K] = -\kappa A$. They have a local solution while the stated domain and energy reserves hold.

Proof. The regular JT equation gives $T_{v_\pm v_\pm} = \partial_{v_\pm}\mathcal{C}_\pm/(2\kappa L^2)$. Its contraction with the wall velocity gives the two invariant equations. The tangent and normal projections of

the material force balance give $\dot{M} - Aw = P_{\rm in} - P_{\rm out}$ and $M\overline{K} + A\overline{\beta} = P_{\rm out} - P_{\rm in}$. Direct evaluation of the extrinsic curvature gives

$$K_+ = \frac{\dot{w} + \Phi/L^2 + \kappa P_{\rm in}}{\beta_+}, \qquad K_- = \frac{\dot{w} + \Phi/L^2 + \kappa P_{\rm out}}{\beta_-}.$$

Their jump and the mass junction give Eq. (125). Differentiating $\beta_- - \beta_+ - \kappa M$ using these equations gives zero. The right-hand sides are smooth inside the domain, giving local existence and uniqueness. Positivity of quantum excess stress is not used in this algebraic consistency statement. □

For $P_{\rm in} = 0$, $R = \sqrt{\mathcal{C}_-} > 0$ and $d\eta/dv_- = R/L^2$, the actual clock conversion is

$$z := \frac{d\tau}{d\eta} = \frac{L^2(\beta_- - w)}{R}, \qquad \frac{dS_{h,-}}{d\eta} = 2\pi z^2 P_{\rm out}. \tag{126}$$

Recoil enters both the source and the clock; keeping the wall fixed while its mass decreases does not solve these equations.

F.4 Stored mechanical energy, radiation and the moving support

The angle need not be specified externally. Q-RAIF C gives a positive inertial current Hamiltonian with a storage coordinate, its retarded reaction and force spectrum [14, Section on an inertial contact]. A displacement selected by a record changes the actual excess current stress, not merely a reconstructed parameter. That local rest-frame model still needs its infrared representation, moving ports and complete quantum stress to be matched to the material action. In particular, $\gamma\langle\dot{\theta}^2\rangle$ alone is not its net heat: the noise–velocity correlation contributes to the work.

A distinct classical branch makes a common moving action explicit. Put a real massless scalar on each side of the wall, with actions $-\frac{1}{2}\int_{\mathcal{M}_\pm}\sqrt{-g}(\nabla\varphi_\pm)^2$. Append the mechanical worldline action

$$S_\Gamma = \int d\tau \left[-m_{\rm rest} - U(\Phi) + \frac{I}{2}\dot{\theta}^2 + \frac{J}{2}\dot{y}^2 - \frac{k}{2}(\theta - y)^2 - \frac{K}{2}y^2\right], \qquad I, J, k, K > 0, \tag{127}$$

and impose $\varphi_\pm|_\Gamma = g_\pm\theta$ with worldline Lagrange multipliers. The wall, storage and angle are all varied. These are full nonchiral scalar fields, not two auxiliary right movers reinterpreted as an anomaly-free bulk theory. With no classical incoming waves, their outgoing solutions are $\varphi_\pm(\xi_\pm) = g_\pm\theta(\tau_{\rm ret})$. Their proper powers and the mechanical equations are

$$\begin{aligned} P_\pm &= g_\pm^2\dot{\theta}^2, & \gamma &= g_+^2 + g_-^2, \\ I\ddot{\theta} + k(\theta - y) &= -\gamma\dot{\theta}, & J\ddot{y} + k(y - \theta) + Ky &= 0, \\ e &= \tfrac{I}{2}\dot{\theta}^2 + \tfrac{J}{2}\dot{y}^2 + \tfrac{k}{2}(\theta - y)^2 + \tfrac{K}{2}y^2, & \dot{e} &= -P_+ - P_-. \end{aligned} \tag{128}$$

Boundary variation gives the mechanical force; characteristics give the powers. There is no independent supplied radiation function. Incoming waves would add their force and work and are excluded in this branch. Quantizing a moving boundary also requires its vacuum stress and fluctuations; the classical formulas do not supply them.

The outer radiation now moves outward, unlike the outer input of Eq. (125). Use outgoing EF coordinate u_+ outside and ingoing v_- inside. Set $M = m_{\rm rest} + U + e$, adding any fixed probe offset, and keep $\beta_- - \beta_+ = \kappa M$. The same action gives

$$\begin{aligned} \dot w &= \frac{\beta_+ P_- - \beta_- P_+ - U'\beta_+\beta_-}{M} - \frac{\Phi}{L^2}, \\ \dot{\mathcal{C}}_+ &= -2\kappa L^2 P_+(\beta_+ + w), & \dot{\mathcal{C}}_- &= 2\kappa L^2 P_-(\beta_- - w), \\ \dot u_+ &= (\beta_+ + w)^{-1}, & \dot v_- &= (\beta_- - w)^{-1}. \end{aligned} \tag{129}$$

The opposite invariant signs follow from $T_{u_+u_+} = -\partial_{u_+}\mathcal{C}_+/(2\kappa L^2)$ and $T_{v_-v_-} = \partial_{v_-}\mathcal{C}_-/(2\kappa L^2)$. The tangent Ward equation is $\dot M = U'w - P_+ - P_-$ and the normal equation is $M\overline{K} + U'\overline{\beta} = P_- - P_+$. Here $K_\pm = (\dot w + \Phi/L^2 + \kappa P_\pm)/\beta_\pm$. These identities give both $[K] = -\kappa U'$ and conservation of the mass junction. They do not follow by putting a negative outer input into Eq. (125); the null direction also changes.

Proposition F.4 (A finite mechanical-source window)**.** *Equations* (128) *and* (129) *have a unique local solution in the domain* $M > 0$, $\beta_\pm > 0$, $\beta_+ + w > 0$, $\beta_- - w > 0$, $\mathcal{C}_- > 0$. *They preserve the mass junction,* $e \ge 0$, *and*

$$\int_0^\tau (P_+ + P_-)\, ds = e(0) - e(\tau) \le e(0). \tag{130}$$

For every fixed finite proper-time window, positive static-support data admit a sufficiently small positive coupling with nonzero radiation throughout a nontrivial subinterval and a solution inside that domain.

Proof. The right-hand side is smooth in the stated domain. Direct differentiation of the junction gives zero, and the positive quadratic mechanical energy obeys the displayed budget. At $g_\pm = 0$ the supported geometry is static and the internal oscillator energy is conserved. Continuous dependence on the couplings preserves its strict domain margins on any specified compact time interval for sufficiently small positive $g_\pm^2$. Nonstationary mechanical data give nonzero angle velocity. □

Balanced outgoing power also gives a necessary operating-lifetime bound for this same apparatus. If its fixed operating domain satisfies $0 < \Phi \le \Phi_{\max}$, $\Phi^2 > \mathcal{C}_-$ and $\beta_- - \dot\Phi \ge b_{\min} > 0$, then

$$\begin{aligned} \mathcal{C}_-(\tau) - \mathcal{C}_-(0) &\ge \kappa L^2 b_{\min} Q_{\rm out}(\tau), \\ Q_{\rm out}(\tau) &< \frac{\Phi_{\max}^2 - \mathcal{C}_-(0)}{\kappa L^2 b_{\min}}. \end{aligned}$$

Equation (129) integrates this inequality for $P_- = P_+ = P/2$. If each complete episode exports at least $q_{\min} > 0$, its count is below the latter bound divided by $q_{\min}$, as well as limited by the installed program and prepared bank. The two acquisitions in Foundation III are one episode, not two renewals. Enlarging the domain or changing its null-clock margin changes this necessary bound; it is not a general impossibility theorem.

The common energy and geometry can be held fixed across records. Use $I = 10$, $J = 20$, $k = 1$, $K = .1$, $g_+^2 = g_-^2 = 1/(4\pi)$ and $e(0) = .95$. Install $\theta(0) = \dot\theta(0) = 0$, $y(0) = A_R$ and $\dot y(0) = \sqrt{2[.95 - .55A_R^2]/J}$ for $A_R = .5, 1.3$. Both records then use the same mass-10

support, potential and initial invariants of Section F.1. With the positive probe offset .02 below, their common m_{rest} is 8.74839811. At proper time 80 the two solutions give

A_R	.5	1.3
$\int (P_+ + P_-)d\tau$	.428637558	.522051307
$e(80)$	.521362442	.427948693
$\mathcal{C}_-(80) - \mathcal{C}_-(0)$	.008793748	.010712134
$\Phi(80)$	3.003737133	3.004545378

(131)

Both remain outside the inner horizon with positive reserves. Integrating both invariants independently preserves the junction to 8.3×10^{-15}; another integrator agrees to 3.7×10^{-12}. The phase probe below, at $\lambda = .02$, yields Y-outcome probabilities .503428742 and .504187486. The difference is created during evolution, not by a different initial mass or an externally selected source. The numerical 80-window check is not an interval-arithmetic domain proof.

The inner rapidity clock is determined by the same solution: $\dot{\eta} = \sqrt{\mathcal{C}_-}/[L^2(\beta_- - w)]$ and $dS_{h,-}/d\eta = 2\pi(d\tau/d\eta)^2 P_-$. This source is coherent classical radiation, not an η-KMS heat output, so its horizon response cannot be substituted into the thermal entropy bound. Installing the mechanical record has a preparation cost. The finite return in Section F.6 instead lets an acquired quantum record exert force on the same storage coordinate. Preparing these initial mechanical data and renewing the whole apparatus are distinct from that return.

Proposition F.5 (A rest-frame mean-stress bridge)**.** *Use Q-RAIF C's quadratic current–mechanical contact on a fixed inertial line, with the same I, J, k, K and a common centered Gaussian state for mechanics and incoming currents. Displace only the mean mechanical data; take the mean initial angle to be zero. Keep all centered correlations and old references. For independent physical outgoing directions $s = \pm$, let the contact coefficients obey $\kappa_s = \sum_i q_{si}^2$ and $\gamma_s = \kappa_s/(4\pi)$, where q_{si} are the effective angle-coupling coefficients, not a change of the carrier's physical charge sectors. Assume the renormalized current products exist on the common domain. In the zero-width limit of the retarded mean solution, away from initial/contact singularities, the outgoing stress difference and mean mechanical equations are*

$$\Delta\langle T_{\mathrm{out},s}\rangle = \gamma_s v(\tau_{\mathrm{ret}})^2, \qquad I\ddot{\theta} + k(\theta - y) = -(\gamma_+ + \gamma_-)\dot{\theta}, \quad v = \dot{\theta}. \tag{132}$$

Thus $g_s^2 = \gamma_s$ identifies this excess mean stress and its mechanical energy loss with Eq. (128) *in that fixed local frame. The balanced scalar example requires $\kappa_+ = \kappa_- = 1$. Q-RAIF C's unscaled two-flavor bidirectional contact has $\kappa_+ = \kappa_- = 2$ and twice that total damping; scaling its angle coupling by $1/\sqrt{2}$ gives the balanced normalization.*

Proof. Take $f_a(x) = a^{-1}f(x/a)$ with nonnegative $f \in C_c^\infty(\mathbb{R})$ and $\int f = 1$. At finite width the displacement adds the classical linear solution without changing the common covariance. In direction s its current mean shift is $q_{si}h_a/(2\pi)$, where h_a is the retarded convolution of v with f_a along that characteristic. The Sugawara stress $\pi \sum_i : j_{si}^2 :$ has zero cross term with the centered baseline. Its difference is therefore $\gamma_s h_a^2$. Eliminating the mean currents gives the retarded kernel $\Gamma_{a,s}(t) = \kappa_s \int f_a(x) f_a(x - t)dx/(2\pi)$, whose one-sided integral is γ_s. For smooth mean histories its convolution error for a smoothly extended velocity is bounded by $\|v'\|_\infty \int_0^\infty t\Gamma_{a,s}(t)dt = O(a)$. For the initial-value solution, the initial kernel tail has integrated size $O(a)$ as well. The common kernel mass bounds

the linear Volterra solution uniformly on fixed windows; Gronwall then gives convergence to the displayed damped equation. Beyond the contact, h_a tends to the retarded velocity, proving the stress statement. The resulting positive quadratic displacement energy obeys $\dot{e} = -(\gamma_+ + \gamma_-)v^2$, the scalar-source balance. Incoming noise–velocity work and centered quantum stress remain in the common baseline, rather than being set to zero. □

This identifies an excess mean radiation channel. Absolute quantum stress, a quantum zero-width channel limit, dilaton-dependent mass, moving-frame work, physical support and recoil, acquired-record spring forces and stress fluctuations require further matching. In particular a statistical mixture of mechanical records has branch covariance and cannot be replaced by one Gaussian displacement in this proposition.

F.5 A quantum phase reads the moving dilaton

The material model also permits a direct finite sensor. For $\lambda > 0$, add a probe with

$$H_P(\Phi) = \epsilon_P I + \lambda(\Phi - \phi_0)Z_P, \qquad \epsilon_P > \lambda \sup_\Gamma |\Phi - \phi_0|, \tag{133}$$

initially in $|+\rangle$. This is the same kind of explicit dilaton coupling as the support, now in an internal quantum degree of freedom. It does not follow from metric-only matter. During the reading, $\langle Z_P \rangle = 0$ and Z_P is conserved. In the mean worldline approximation its energy is ϵ_P and its mean additional dilaton force is zero. Count ϵ_P in the mass, reducing m_{rest} by that amount in the displayed examples. The probe's phase gives

$$\Pr(Y_P = +1) = \frac{1 + \sin[2\lambda \int (\Phi - \phi_0) d\tau]}{2}. \tag{134}$$

With $\lambda = \epsilon_P = .02$, the $q = .01$ and $.02$ histories give .50617417 and .51230214 at the same proper duration. Their actual bit laws have total variation distance .00612797. No numerical dilaton value is supplied to the controller.

The terminal read itself can be implemented. Q-RAIF C gives a positive autonomous clock whose last two gates write the actual Y result into a record, then apply a true SWAP with a paid coherent blank [14]. The probe is renewed; all old correlations move into the blank and record. For a supplied finite $\Phi(\tau)$ trajectory, its joint error relative to the actual phase at $\tau_0 < T$ is bounded by

$$d_{\text{joint}} \le 2A\tau_0\sqrt{P_0} + 2\sqrt{P_0} + \lambda D(T - \tau_0), \quad D = \sup |\Phi - \phi_0|, \tag{135}$$

where P_0 is the clock's terminal binomial tail and A is the sum of its last two edge strengths. Finite resources suffice for each specified error; the clock mean energy is Jm, and must enter the supported mass. The sufficient $m = 10^5$, $T = 80$, $\tau_0 = 78$ example uses clock mean energy 1963.4954, not the mass-10 apparatus above. During writing and renewal the mean additional force need not vanish: the material equation uses $A_{\text{mat}} = U' + \lambda\langle Z_P \rangle$, with $|A_{\text{mat}} - U'| \le \lambda$. The probe, clock, blank and record belong in the same matter action. If e now includes their Φ-dependent internal energy, its equation is $\dot{e} = \lambda\dot{\Phi}\langle Z_P \rangle + P_{\text{in}} - P_{\text{out}}$; then $\dot{M} = A_{\text{mat}}\dot{\Phi} + P_{\text{in}} - P_{\text{out}}$ as required by the material Ward balance.

A smaller clock can be included directly in this mean material action, instead of using the large worst-case certificate. Take $m = 128$, $J = \pi/160$, $\epsilon_P = \lambda = .02$, and retain the complete 129-state clock, probe, record and blank throughout the proper-duration

80 evolution. Its initial internal mean energy is 2.5532741. Choose $m_{\mathrm{rest}} = 4.8251240$ so that the initial mass 10 already includes it and the declared source reserve 2.34. Evolve the joint Schrödinger equation for

$$H_{\mathrm{int}}(\Phi) = H_{\mathrm{hist}} + 2\epsilon_P I + \lambda(\Phi - \phi_0)Z_P$$

with the radiating junction and its actual η clock. The two contact outputs $\chi = .5$ and 1, at hot-input excess .02, give

q_χ	actual $\Pr(R = 1)$	final source reserve	probe reset distance
.010	.50529766	1.42403868	.00075428
.020	.51055894	.50256908	.00150041

The last column is the distance to the closest pure product of a renewed probe and a normalized retained internal reference, not an error certificate against the ideal meter's entire reference state. The actual record laws differ by .00526128. Norm, mass-junction and internal dilaton-work residuals are below 4×10^{-11} in this integration; a second integrator agrees on the displayed trajectory and record values within 1.3×10^{-9}. These are numerical checks, not interval-certified error bounds. For these gates and initial data, the vanishing mean force has an exact reason.

Proposition F.6 (A force-neutral read and replacement)**.** *Let K denote conjugation in the clock and computational bases and set*

$$\mathsf{C} = (-1)^{N_{\mathrm{clock}}} X_P X_B K, \qquad H_0 = H_{\mathrm{int}} - (Jm + 2\epsilon_P)I.$$

For the identity-edge clock with terminal C_Y and true SWAP, $\mathsf{C}H_0(\Phi)\mathsf{C}^{-1} = -H_0(\Phi)$ and $\mathsf{C}Z_P\mathsf{C}^{-1} = -Z_P$. With initial clock state 0 and data $|+\rangle_P|0\rangle_R|+\rangle_B$, every real dilaton trajectory has $\langle Z_P\rangle = 0$ and $\langle H_{\mathrm{int}}\rangle = Jm + 2\epsilon_P$ throughout the read.

Proof. Conjugation exchanges the two Y projections, and X_P exchanges them again. Thus $X_P X_B C_Y^* X_P X_B = C_Y$; the real SWAP commutes with $X_P X_B$. Clock parity reverses every off-diagonal edge, while X_P reverses Z_P. This proves both operator identities. Remove the constant Hamiltonian phase. Since C is antiunitary, it reverses both i and H_0 in the Schrödinger equation, and so maps its solution to another solution with the same initial state. Uniqueness makes that solution C-invariant. Expectations of the two odd Hermitian operators must vanish. □

The clock energy is therefore included from the start without changing the previously supported mean trajectory. This does not set the quantum force operator or its fluctuations to zero, and different blanks or gates need not have the symmetry. The general coupled force equation, not a zero-force constraint, was used in the numerical calculation. The thermal reservoir's quantum degrees, both-port material stress and its local preparation in η are not thereby included in that body.

The same finite meter also reads the mechanical-source branch of Section F.4. Its complete mean energy 2.5532741, rather than just the probe offset, is included from the start by using the common positive rest mass 6.21512398 for both equal-energy records. Proposition F.6 leaves their mean mechanical and geometric trajectories unchanged. Their actual acquired record probabilities are .502991099 and .503655046, a total variation distance .000663947. Keeping clock, blank and record gives distances

.000376618 and .000458340 to the closest renewed-probe pure product with its normalized internal reference. These are not errors against an ideal reference state. The full vector evolution preserves norm to 1.6×10^{-11} and internal mean energy to 2.6×10^{-12}; a second integration agrees on the record probability to 3.1×10^{-14}. This finite episode obtains the record through physical gates, rather than reading the numerical solution into a controller. In this force-neutral example the record does not act on the mechanics. The next section adds that interaction and includes its reaction during acquisition. The scalar source has no thermal reservoir whose regeneration could be inferred from either construction.

For input streams jointly independent of the record and prepared in a common clock frame, the record can select either of two contact angles in the next stationary window. If their transmissions are $\chi_0 \neq \chi_1$, its source-current mean is

$$q_{\mathrm{next}} = q_{\mathrm{hot}}[\chi_0 + (\chi_1 - \chi_0)\Pr(R = 1)].$$

It depends on the actual phase bit, not a supplied numerical dilaton. The angle change's pump work in Eq. (110) is also charged. This gives the conditional response-to-source map; it does not yet autonomously prepare the reservoirs or renew the entire controller. Quantum fluctuations of the probe force also lie outside the mean-source approximation. The phase sensor and the earlier fixed-boundary current sensor are different realizations, not two freely composable laws for one apparatus.

F.6 An acquired record drives a new material source

A record can act without first being converted into a supplied mechanical initial condition. Keep the two classical scalar fields and the material action of Section F.4, and append $i\langle\Psi, \dot{\Psi}\rangle - \langle\Psi, H_c\Psi\rangle$ to its proper-time Lagrangian. The normalized state Ψ contains a finite history clock, probe P, record R, replacement blank B and an old purification Q. Set

$$H_c(\Phi, y) = H_{\mathrm{hist}} + E_{\mathrm{off}}\mathbf{1} + \lambda(\Phi - \phi_0)Z_P + \frac{\chi_R}{2}P_R(y-d)^2, \qquad P_R = |1\rangle\langle 1|_R, \quad \chi_R \geq 0. \tag{136}$$

H_{hist} is the positive gate-conjugated clock of Q-RAIF C. Its first nonidentity edge performs C_Y, the next performs a true SWAP of P and B, and the remaining edges are identities. There is no proper-time switch. Initially $R = 0$, so the spring has zero expectation. The actual acquisition opens its force channel. $H_c \geq 0$ while $E_{\mathrm{off}} > |\lambda(\Phi - \phi_0)|$. The target d, couplings and initial clock are prepared hardware. The scalar fields, mechanical coordinates and geometry use an Ehrenfest mean-action approximation [61, Section 2]; the finite quantum core and its old reference are retained in full, without sampling or collapsing R.

Let $p = \langle P_R\rangle$, $z = \langle Z_P\rangle$ and $E_c = \langle H_c\rangle$. Variation of this one action gives

$$\begin{aligned} i\dot{\Psi} = H_c\Psi, \qquad & F_y = \chi_R(y-d)p, \qquad & F_\Phi = \lambda z, \\ J\ddot{y} + k(y-\theta) + Ky = -F_y, \qquad & \dot{e} = -P_+ - P_- - F_y\dot{y}, \qquad & \dot{E}_c = F_y\dot{y} + F_\Phi w. \end{aligned} \tag{137}$$

The angle equation and $P_\pm = g_\pm^2\dot{\theta}^2$ are unchanged. Here p is an expectation obtained by variation, not a decoded number fed to a separate control policy. The mass and its tangent balance are

$$M = m_{\mathrm{rest}} + U + e + E_c, \qquad \dot{M} = (U' + F_\Phi)w - P_+ - P_-. \tag{138}$$

Consequently Eq. (129) holds with U' replaced by $U' + F_\Phi$ in its acceleration equation; both invariant and null-clock equations retain their directions. Differentiating the mass junction with these equations gives zero. The separate work balances are

$$e(\tau) + Q_{\text{out}}(\tau) + \int_0^\tau F_y \dot{y}\, ds = e(0), \tag{139}$$
$$E_c(\tau) - E_c(0) = \int_0^\tau (F_y \dot{y} + F_\Phi w)\, ds.$$

The clock interaction and conditional spring are included in E_c. The spring breaks the force-neutral symmetry used above; F_Φ cannot be set to zero in this coupled calculation.

Proposition F.7 (A nonzero record-mediated source return)**.** *Take two initial geometries with the same $\Phi = \phi_0$, $w = 0$, mass, mechanical data and quantum input, but different $\mathcal{C}_-$; set $\mathcal{C}_+$ by the mass junction. In the positive exterior domain, either $\lambda = 0$ or $\chi_R = 0$ makes their proper emitted powers identical. With both interactions present, there are positive finite-clock data for which the emitted powers differ on a nonempty finite interval. Their record, mechanical response, source and geometry obey the same action and work account.*

Proof. For $\chi_R = 0$, the mechanical subsystem is independent of the reader and geometry. For $\lambda = 0$, the joint mechanical and quantum equations are independent of geometry. Either condition fixes identical mechanical sources for the two initial geometries.

For a nonzero example choose $\theta(0) = \dot{\theta}(0) = 0$, $y_0(y_0 - d) \neq 0$ and probe state $(I + \upsilon X_P)/2$, $\upsilon \neq 0$, with a blank R and its old purification held. Let $r \geq 1$ be the C_Y edge, $a_j > 0$ the clock edge strengths and $G_r = \prod_{j=1}^r a_j$. The initial acceleration difference δa follows from the junction equations, so that $\delta\Phi(t) = \delta a\, t^2/2 + O(t^3)$. The shortest-path Dyson expansion gives

$$\delta p(t) = K_r t^{2r+3} + O(t^{2r+4}), \qquad K_r = \frac{\upsilon G_r^2 \lambda \delta a}{6r!(r-1)!(r+3)}. \tag{140}$$

An insertion of $\lambda\delta a\, t^2 Z_P/2$ after $j < r$ forward edges contributes the time integral $t^{r+3}(j+1)(j+2)/(r+3)!$. Their sum is $t^{r+3}r/[3(r+3)r!]$. Using $C_Y^* P_R C_Y = (I + Y_P)/2$ in the interference with the unperturbed shortest path gives the coefficient above. The insertion after all r edges has Hermitian $P_R Z_P$ in that interference and zero real part. Further graph paths and spring insertions raise the order; the spring annihilates the blank record before acquisition.

Integrating the mechanical equations then yields

$$\delta(P_+ + P_-)(t) = -\frac{2\gamma k^2 y_0 \chi_R (y_0 - d) K_r}{I^2 J (2r+4)(2r+5)(2r+6)} t^{2r+7} + O(t^{2r+8}). \tag{141}$$

The baseline angle velocity is $k y_0 t/I + O(t^2)$. Different initial $\mathcal{C}_-$ give a nonzero δa for generic supported data, since initially $\dot{w} = -U'\beta_+\beta_-/M - \phi_0/L^2$. The displayed coefficient is nonzero for the stated nonzero parameters. The coupled finite-dimensional equations are analytic on their strict positive exterior domain. Local existence and continuity therefore give a nonempty admissible family, not just a numerically resolved difference. □

A complete integration uses the earlier I, J, k, K, $\gamma = 1/(2\pi)$, $e(0) = .95$, $y_0 = .5$ and $d = -.5$, with $\lambda = .2$, $\chi_R = .05$ and $E_{\mathrm{off}} = .04$. The 33-state clock has edge scale $\pi/160$, $r = 3$, mean energy .628318531 and spectral maximum 1.256637061. The initial probe has $\upsilon = .8$, its two-dimensional old purification is retained, and B starts in $|+\rangle$. The quadratic support has stiffness 100 and $U'(3) = -30/(1.95 \cdot 2.05)$, as in the static mass-10 support. Both geometries have initial $\Phi = 3$, $w = 0$, $M = 10$ and $\mathcal{C}_- = 4.7975 \pm .1$, with the outer invariant fixed by the junction. The initial core energy .668318531 is already in the mass; the common rest mass is 8.100079577. At proper time 100:

$\mathcal{C}_-(0) - 4.7975$	$\Pr(R = 1)$	Q_{out}	$\mathcal{C}_-(100) - \mathcal{C}_-(0)$
$-.1$	.506528009	.518808111	.010786169
$+.1$	.496292018	.517763845	.010480933

The largest proper-power difference on this window is 1.0350613×10^{-4}; with either interaction removed it is below 3.2×10^{-16}. With feedback removed the two record laws still differ, while their scalar sources agree. Thus acquisition and actuation are separately tested. The nonzero-source result concerns the emitted power, not the already different initial geometries.

The full core–reference vector is evolved along with both invariants. Norm, junction and the three work balances, including U's dilaton work, have residuals below 3.7×10^{-14} in the primary integration. A separate tensor-neighbour implementation with another integrator agrees on the record and source to 6.0×10^{-13} and 2.0×10^{-15}, respectively, and on the mass to 1.2×10^{-8}. The largest probe force is about .0082 and is retained. Clock energy changes during acquisition; it is not an uncharged timer. Only the C_Y edge changes P_R, giving

$$|\dot{p}| \le 2a_r \|\Psi_{r-1}\| \|\Psi_r\|.$$

This bound is checked on the interacting clock, not a free-clock tail. It supplies finite record retention when that edge is sparsely occupied, not permanent halt. The construction closes an actual finite geometry–record–force–source return in the mean material branch. Full quantum recoil, covariant thermal transport and renewal of the clock, blank and reference remain required for the sustainable loop below.

F.7 A spatial return with a physical clock

The geometry and support below are semiclassical mean fields. The internal oscillators and both retained spins are quantum. The regulator is part of the physical action; the statement does not remove its bandwidth cutoff.

Take JT gravity with radius R and coupling $\kappa > 0$, a full material label a, and a full synchronization scalar θ, with

$$S_b = -\int \sqrt{-g}\, M \left[mn + \frac{K}{2}(n - n_0)^2 \right], \quad n^2 = (\nabla a)^2, \qquad S_\theta = -\frac{J}{2} \int \sqrt{-g} (\nabla \theta)^2. \tag{142}$$

Require $m > K n_0 > 0$, n near n_0, and a time function $\theta_t > 0$ with timelike gradient. Use the unrestricted inverse embedding $\theta(X(\tau, \xi)) = \tau$, $a(X(\tau, \xi)) = \xi$ and $h = X^* g$. Only the internal expansions are finite: $\sqrt{I}\chi = \sum q_{c\alpha} e_\alpha(\xi)$ and $u = \sum q_{u\alpha} e_\alpha(\xi)$, for smooth supported real e_α. Substitute them into the two scalar kinetic actions and the potential

$$V = \frac{I \mu^2 \chi^2}{2} + \frac{1}{2} \left[\omega u + \frac{A\chi}{\omega} \right]^2 + E_o \rho_o - D_o, \quad A = \lambda + g w Y_P + \gamma w_2 Y_S, \tag{143}$$

before Legendre transform and metric/embedding variation. Here

$$D_o = \frac{C_\chi}{\omega^2}(\lambda g w Y_P + \lambda\gamma w_2 Y_S + g\gamma w w_2 Y_P Y_S), \qquad C_\chi(\xi) = e^T Q_c e / I.$$

Q_c is the fixed prepared clock covariance. Choose the paid scalar offset to dominate $|D_o|$. Quantize the resulting Hamiltonian with Weyl ordering and a joint-state Berry term on the actual θ leaves. Its kinetic matrices are

$$C = -\int \sqrt{-h}\, h^{\tau\tau} e e^T, \quad D = -\int \sqrt{-h}\, h^{\tau\xi} e (e')^T, \quad K_{\rm grad} = \int \sqrt{-h}\, h^{\xi\xi} e' (e')^T,$$

giving $(p - Dq)^T C^{-1}(p - Dq)/2 + q^T K q/2$ plus the paid offset/calibration terms. A proper-number sensor contributes

$$H_S = \int n_S(\xi) N [E_S + h_s \Phi(X) Z_S]\, \mathrm{d}\xi, \quad N = \sqrt{-h_{\tau\tau}}, \quad \int n_S \mathrm{d}\xi = 1. \tag{144}$$

n_S is smooth, symmetric, and has nonzero width. Include both spins, their Berry terms, all profile reactions, and any old reference in the same retained state. Restrict $|h_s| \sup|\Phi| < E_S/2$. No profile is an external time-dependent signal.

Proposition F.8 (Finite-bandwidth spatial return)**.** *There are finite positive I, M, J, R, smooth finite-width preparations, and nonzero finite h_s, γ for which this one mean-action system has a positive local solution and two actual histories with the same initial geometric constraints, differing only in their retained P record, whose next-program probabilities for $(1 + Y_S)/2$ differ at the same central proper time. The retained Y_S enters the subsequent actual material Hamiltonian, with a nonzero mixed sensing/return response of the material mean acceleration. All clock, support, interaction and probe sources are metric variations of this action, and satisfy the full Ward identity. This is an existence result, not a uniform gain, state-independent CPTP map, continuum-controller or thermal-renewal statement.*

Proof. Prepare $P = |\pm Y_P\rangle$, $S = |+X_S\rangle$, a common free product Gaussian field state with $\langle q\rangle = 0$, zero symmetrized qp covariance, and clock momentum $\sqrt{I}\Omega F$. References are retained, not reset. At $\gamma = h_s = 0$, calibration cancels the initial expected record-dependent potential as a functional of X. Thus the complete initial metric stresses coincide, not just their integrals. Take common static AdS data, $\theta = t$ on a finite prepared plateau, and hydrostatic bar density $n = n_0 + (m/K)[\cos(x/R) - 1]$ on a small causal patch. The common initial JT constraint is a linear spatial equation for Φ; adding a sufficiently large finite positive homogeneous solution $B\sec(x/R)$ makes $\Phi > 0$. This changes initial geometric data, not the charged positive source.

We first check the physical time equation. At rest, varying $k = X^t_\tau$ gives $C = \int x' e e^T/k$, $K_{\rm grad} = \int k e'(e')^T/x'$ and $K_{\rm mass} = \int e^{2\rho} k x' e e^T \mathcal{M}_\pm$. Hence the local time-momentum current $\delta H/\delta X^t_\tau$ contains only q^2, p^2 and commuting spin constants. Its commutator with the initial Hamiltonian is Weyl qp; its expectation, and the mixed time/spatial current expectation, vanish. The corresponding initial mixed principal block vanishes too. The full time EL equation therefore gives $\theta_{tt}(0) = 0$, or $\theta = t + O(t^3)$. It changes no potential trace coefficient of order t^2. Nevertheless, if $\theta = t + \theta_3 t^3/6 + \cdots$, the actual $T_{\theta,tt} = T_{\theta,xx} = J/2 + J\theta_3 t^2/2 + \cdots$ is retained; only its trace is identically zero.

Here is a nonempty analytic trace certificate. Choose one normalized even smooth mode e, constant and positive in an open central neighborhood, $w = 1$ on its support, and

$F = 1$. Put $k = \int (e')^2$, $\beta = \int \sec^2(x/R) e^2$, $Q_c = [2\sqrt{k + \mu^2 \beta}]^{-1}$, $Q_u = [2\sqrt{k + \omega^2 \beta}]^{-1}$. The exact static central trace-contrast coefficient is

$$d_2 = -D_0 + B_1/I + B_2/I^2, \quad D_0 = 4\lambda g \Omega^2 e(0)^2/\omega^2,$$
$$B_1/D_0 = \beta[\mu^2 Q_c + \omega^2(Q_c + Q_u)]/\Omega^2, \tag{145}$$
$$B_2/D_0 = 2\beta(\lambda^2 + g^2) Q_c/(\omega^2 \Omega^2). \tag{146}$$

These follow by $[t^2]Q = -\{K_\pm - K_0, Q_0\}/2$; the off-diagonal species covariance is included. The initial spatial momentum current is locally proportional to $ee' \operatorname{sym}(qp)$, so its force contrast and spatial derivatives vanish on the central plateau. No bar recoil trace term of order $1/I$ survives there. For $(\mu, \omega, \lambda, g, \Omega) = (.3, 1.4, .6, .2, .8)$, $B_1/D_0 \le 6.432292\beta$ and $B_2/D_0 \le 1.062926\beta$. Take finite R with $\beta < 1.01$, and $I = 100$. Then $d_2 < -0.9349 D_0$. Scale $M = \alpha L$, $J = \beta_b L$ together, with fixed positive ratios. Actual common-acceleration corrections to (146) are $O_I(L^{-1})$; hydrostatic preparation cancels the order-M force, not the quantum reaction. A sufficiently large finite L preserves $d_{2,\text{total}} < -D_0/2$.

For completeness the actual coupled equations have a local solution. Let $\mathcal{A} = 1 + \sum(q_j^2 + p_j^2)$. A degree-two Hamiltonian gives $\|\mathcal{A}^s U_z \psi\| \le e^{C_s T} \|\mathcal{A}^s \psi\|$ and

$$\|\mathcal{A}^2 (H_z - H_{\hat{z}})\psi\| \le C \|z - \hat{z}\|_{H^{r-1}} \|\mathcal{A}^3 \psi\|, \qquad r \ge 4. \tag{147}$$

Here z consists of first-order classical variables. Quadratic principal/source moments and quartic differentiated-current forces are Lipschitz in $D(\mathcal{A}^2)$ on this graph ball. The latter degree, which arises from noncommuting spin coefficients, is not suppressed. The full bar and synchronization principal forms have size L and strict positive kinetic/spatial margins; their finite quantum corrections, including smooth finite-rank kernels, have bounded norm. Large finite L preserves those margins.

To express geometry on actual leaves, use ambient-frame derivatives $P = (\rho_t, \rho_x)$ and $Q = (\Phi_t, \Phi_x)$ pulled back to X. Both first-order wave systems have the symmetric matrices

$$A_\tau = \begin{pmatrix} \theta_t & -\theta_x \\ -\theta_x & \theta_t \end{pmatrix} > 0, \qquad A_\xi = \begin{pmatrix} a_t & -a_x \\ -a_x & a_t \end{pmatrix}.$$

Their sources are respectively $(\kappa e^{2\rho} J_\Phi - e^{2\rho}/R^2, 0)$ and $(-2e^{2\rho}\Phi/R^2 - \kappa e^{2\rho} T, 0)$. No second embedding derivative enters this geometric principal block. The first-order classical energy in an H^r ball contracts in H^{r-1}; (147) contracts the state in $D(\mathcal{A}^2)$ using only a uniform $D(\mathcal{A}^3)$ bound. Both difference estimates have a factor T. Time derivatives of principal quadratic moments are at most quartic, so there is no higher-graph difference cycle. Picard iteration, followed by closedness and finite smooth-data bootstrap, gives the full unprojected EL solution. Smooth preparations with the additional finite set of graph moments give six time derivatives and the two coupling derivatives used below. Unrestricted X, including its time equation, gives $\nabla_\mu T^{\mu\nu} = -J_\Phi \nabla^\nu \Phi$; consequently the JT constraints propagate. The sources include the variation of the regulator and proper-number probe. At nonzero γ, retain $J_\Phi = h_s n_S \langle Z_S \rangle / \ell_\perp$ and the resulting $R(g) + 2/R^2 = 2\kappa J_\Phi$.

At $\gamma = 0$, Z_S is conserved with mean zero; the source evolution is independent of h_s, and $\langle Y_S \rangle = \sin(2h_s \Theta)$, where $\Theta = \int n_S N \Phi \, \mathrm{d}\tau \mathrm{d}\xi$. At the parity-fixed center, $X = (T(\tau), 0)$ and $N = T'$. Composing with each history's endpoint $\tau_\pm(t) = \theta_\pm(t, 0)$ therefore gives exactly $\Theta_\pm(t) = \int_0^t \Phi_\pm(v, 0) \mathrm{d}v$, with t central proper time. The wave equation with common initial data yields

$$\Delta\Phi(t, 0) = -\kappa d_{2,\text{total}} t^4/12 + O(t^5), \quad [t^5] \partial_{h_s} \Delta \langle Y_S \rangle \big|_0 = -\kappa d_{2,\text{total}}/30 \neq 0. \tag{148}$$

Finite-width proper-time phases converge in this fifth jet to (148). Scaling M, J together keeps background-dilaton times recoil jets bounded; the center endpoint contribution cancels exactly. Choose a small but nonzero symmetric width preserving the coefficient, then finite L preserving all margins. Lower-order recoil jets are allowed. Thus some finite short $t > 0$ and small nonzero h_s give nonzero next-program contrast. Smooth parameter dependence preserves it for sufficiently small finite $\gamma \neq 0$, when the retained program enters the actual material square (143).

One can also check a complete mean-force response, without attributing every term to geometry. Put $w_2 = 1$ on $\operatorname{supp} e$ and write $\bar{\Phi}_0 = \int n_S N \Phi(0) > 0$, $b = \beta$, $c = \langle q_c \rangle / \sqrt{I}$, $y = \langle q_u \rangle$, $a_\pm = \lambda \pm g$. At the static-support large-L limit the unperturbed means satisfy $\ddot{c} = -\nu_c^2 c - a_\pm^2 bc/(I\omega^2) - a_\pm by/I$, $\ddot{y} = -\nu_u^2 y - a_\pm bc$. For $\delta = \partial_\gamma \partial_{h_s}|_0$, independence at $\gamma = 0$ and $\partial_{h_s}\langle Y_S \rangle = 2\bar{\Phi}_0 t + O(t^3)$ give

$$\begin{aligned} \delta\ddot{c} = & -(\nu_c^2 + a_\pm^2 b/(I\omega^2))\delta c - a_\pm b \delta y/I - 2\bar{\Phi}_0 t(by/I + 2a_\pm bc/(I\omega^2)), \\ \delta\ddot{y} = & -\nu_u^2 \delta y - a_\pm b \delta c - 2b\bar{\Phi}_0 tc. \end{aligned}$$

The common $O(t^3)$ sensor term cancels in the branch fourth-jet. With $\dot{c}(0) = \Omega$, all other initial means zero, these equations include the clock/state variation and give

$$[t^4]\delta(\ddot{y}_+ - \ddot{y}_-) = \bar{\Phi}_0 \left[\frac{8\lambda g b^2 \Omega}{3I\omega^2} + O_I(L^{-1}) \right] \neq 0. \tag{149}$$

The finite-L correction includes proper-time pullback: mixed synchronization displacement can start at t^4, spatial displacement at t^5, while mixed curvature starts at t^5. Their fourth-jet mean-force contributions, divided by $\bar{\Phi}_0$, are $O_I(L^{-1})$ by the scaled positive EL inverse. Initial mixed source differences vanish. Ordinary phase terms are present: (149) is not a purely geometry-induced coefficient. Geometry's nonzero path was separately certified by (148). Choose finite L also preserving the strict sign in (149), then small finite h_s, γ and a short nonzero event. All parameters and the positive operating window are finite; no external read/reinstall is performed. □

F.8 Covariant bulk evolution of a retained quartet

The full-mode carrier requires a covariant source, not just a conserved finite-mode Hamiltonian. Use a physical metric $g = e^{2\rho}(-dt^2 + dx^2)$ and an independent unit material director $U = e^{-\rho}(\cosh\eta, \sinh\eta)$, with orthogonal unit V. The work field χ and director have actions

$$S_\chi = \frac{I_\chi}{2} \int \sqrt{-g}[(U\nabla\chi)^2 - C_\chi(V\nabla\chi)^2], \qquad S_\eta = \frac{I_\eta}{2} \int \sqrt{-g}[(U\Omega)^2 - C_\eta(V\Omega)^2], \tag{150}$$

where $\Omega = d\eta$ plus the physical spin connection and $0 < C_i < 1$. A positive elastic bar with coordinate a anchors this director through $-\Lambda \int \sqrt{-g}(U\nabla a)^2/2$. Prepare a strict bar domain $B - \Lambda > K_a > 0$ and finite alignment overcoming the work-clock orientation curvature $I_\chi(C_\chi - 1)$ and the quantum pressure debit. These are paid material parameters. Deriving U from ∇a instead would put second bar derivatives into the optical coefficients and lose the same-order estimate below.

The quartet comprises two nonchiral Majorana sectors of central charge $c_s = 1/2$, and all three spectator bosons of $c_s = 1$. Each sector has both chiralities and $\gamma_s = c_s/(24\pi)$; the total central charge is four. Its actual velocity $0 < v_s(\chi) < 1$ defines $\bar{g}_s = -v_s^2 U^\flat U^\flat +$

$V^\flat V^\flat$. Supply the original global representation, density operator and every old reference. Choose optical-metric covariant point splitting with no additional finite two-derivative medium counterterm. The anomaly normalization is the conventional one [2, Section 2]; the actual causal state, rather than a vacuum inverse-curvature functional, determines its homogeneous stress.

Proposition F.9 (Actual-state covariant quartet bulk)**.** *Supply a consistent global full-mode representation and a smooth actual initial state with the required energy moments and old references. Prepare smooth local clock/bar/director data with strict paid margins, the causal Hessian conditions* (160) *and the JT constraints* (163)*. The paid fields, retained quartet state and renormalized constrained JT equations then have a smooth common local-time bulk solution. Its mean solution and source are independent of admissible exterior data continuations inside the physical domain of dependence. Equality of its quantum restrictions additionally requires equality of the initial local joint-state restrictions, not merely stress means. This result does not prescribe arbitrary local quantum means or a physical moving boundary.*

Proof. For one sector, set

$$d = \cosh^2\eta - v^2\sinh^2\eta, \quad N = v/d, \quad \beta = (v^2-1)\sinh\eta\cosh\eta/d, \quad \omega = \rho + \tfrac{1}{2}\log d. \tag{151}$$

Thus $\bar{g} = e^{2\omega}\hat{g}$ with $\hat{g} = -N^2dt^2 + (dx+\beta dt)^2$. Its canonical chiral coefficients are $h_\sigma = N - \sigma\beta > 0$. Write $D_t = \partial_t - \beta\partial_x$, $k = \beta_x/N$, $E = T_+ + T_-$ and $P = T_+ - T_-$. The actual Virasoro evolution gives

$$D_tE + NP_x + 2N_xP - 2\beta_xE = -2\gamma\beta_{xxx}, \tag{152}$$

$$D_tP + NE_x + 2N_xE - 2\beta_xP = 2\gamma N_{xxx}. \tag{153}$$

The exact conversion to the orthonormal $\hat{g}$ frame is

$$\hat{E} = E - \gamma k^2, \quad \hat{P} = P + 2\gamma k_x, \quad \hat{S} = \hat{E} + \gamma\hat{R}, \qquad \hat{R} = -2(N_{xx} + D_tk - Nk^2)/N. \tag{154}$$

Substitution gives both covariant conservation equations with zero residual and trace $\gamma\hat{R}$. The lapse and shift variations of $-\gamma\int\beta_x^2/N$ give precisely this projective conversion. The Weyl cocycle $\gamma\int\sqrt{-\hat{g}}[(\hat{\nabla}\omega)^2 + \omega\hat{R}]$ then yields

$$\bar{T}_{ab} = \hat{T}_{ab} + 2\gamma\left[\hat{\nabla}_a\hat{\nabla}_b\omega - \omega_a\omega_b - \hat{g}_{ab}\hat{\Box}\omega + \tfrac{1}{2}\hat{g}_{ab}(\hat{\nabla}\omega)^2\right]. \tag{155}$$

Its trace is $\gamma\bar{R}$ and its $\bar{g}$ divergence vanishes. In the bulk, the combined conversion is the first-jet density

$$\mathcal{L}_{\mathrm{loc},s} = \gamma_s\left[-\frac{(D_t\omega_s - \beta_{s,x})^2}{N_s} + N_s\omega_{s,x}^2 + 2\omega_{s,x}N_{s,x}\right]. \tag{156}$$

The integrated-by-parts difference is $-2\gamma_s[\partial_x(\omega_sN_{s,x}) + \partial_t(\omega_sk_s) - \partial_x(\omega_s\beta_sk_s)]$. It cannot be dropped at a physical wall.

Define the canonical closed-time-path functional by $Z_{\mathrm{can}}[z_+, z_-] = \mathrm{tr}\big(U[z_+]\varrho_0U[z_-]^\dagger\big)$. Add $\Gamma_{\mathrm{loc}}[z_+] - \Gamma_{\mathrm{loc}}[z_-]$, with $\Gamma_{\mathrm{loc}} = \sum_s\int\mathcal{L}_{\mathrm{loc},s}$. Variation at coincident branches gives the actual canonical expectation and the local forces in (156). No new homogeneous Polyakov field, independent purifier or in-out state is introduced. Positive h_σ makes each canonical

Hamiltonian lower bounded, not nonnegative: diffeomorphism conjugation includes the debit $-\gamma_s \int h_{\sigma,x}^2/h_\sigma\,/2$ and the chosen Casimir term. On the prepared smooth coefficient ball these have a finite uniform bound. One common paid scalar offset covers it; its stress is included in both histories.

An exact local reduction supplies the same-order estimate. For each $u_j = \sigma_j h_j$, solve

$$\begin{aligned} w_{j,t} + u_j w_{j,x} + u_{j,x} w_j &= -u_{j,xx}, & w_j(0,x) &= 0, \\ A_{j,t} + u_j A_{j,x} + 2u_{j,x} A_j &= 0, & A_j(0,x) &= T_j(0,x). \end{aligned} \tag{157}$$

Differentiation gives $(\partial_t + u_j\partial_x + 2u_{j,x})(w_{j,x} - w_j^2/2) = -u_{j,xxx}$. Hence, with $Z_j = \gamma_j w_j$,

$$T_j = A_j - Z_{j,x} + Z_j^2/(2\gamma_j). \tag{158}$$

This is the exact Ward solution. If $f_{j,t} + u_j f_{j,x} = 0$, $f_j(0,x) = x$, then $w_j = \partial_x \log f_{j,x}$, $A_j = f_{j,x}^2 T_j(0, f_j)$ and $T_j = f_{j,x}^2 T_j(0,f_j) - \gamma_j \mathcal{S}(f_j)$. These are the actual characteristic connection and transported initial stress, not additional quantum fields or a new source action.

Put $q = (\chi, \eta)$, $P_q = q_t$, $Q_q = q_x$. Let K_{tt}, K_{tx}, K_{xx} be the q Hessians of the complete first-jet classical action plus (156), with $K_{tx} = \mathcal{L}_{P_q Q_q} + \mathcal{L}_{Q_q P_q}$, and let G have columns $g_j = \nabla_q h_j$. The matter principal matrix and its symmetrizer are

$$A_m = \begin{pmatrix} K_{tt}^{-1}K_{tx} & K_{tt}^{-1}K_{xx} & -K_{tt}^{-1}G \\ -1 & 0 & 0 \\ 0 & \operatorname{diag}(\gamma_j\sigma_j)G^T & \operatorname{diag}(\sigma_j h_j) \end{pmatrix}, \quad H_m = \begin{pmatrix} K_{tt} & 0 & 0 \\ 0 & -K_{xx} & G \\ 0 & G^T & \operatorname{diag}(h_j/\gamma_j) \end{pmatrix}. \tag{159}$$

Multiplication gives $H_m A_m = (H_m A_m)^T$. The Schur conditions for H_m and $H_m \pm H_m A_m$ are exactly

$$\begin{aligned} K_{tt} > 0, \qquad -K_{xx} - \sum_j \gamma_j g_j g_j^T/h_j > 0, \\ K_{tt} + K_{xx} \pm K_{tx} \pm \sum_j \frac{\gamma_j \sigma_j g_j g_j^T}{1 \pm \sigma_j h_j} > 0. \end{aligned} \tag{160}$$

At rest the director stiffness is corrected by $\Delta D = \sum_s 2\gamma_s(1 - v_s^2)^2/v_s$. Finite large inertias with strict $0 < C_i < 1$ give a nonempty domain; (160) uses its actual nonlinear coefficients, including nonzero director boost. These inequalities imply a uniform matter cone strictly inside the physical light cone.

The metric and dilaton each have light speeds $\lambda = \pm 1$. There is no resonant trace coupling between them. For the full (χ, η, ρ) Hessians of a quantum sector, direct substitution of (151) gives the exact identity

$$(K_{tt,s}^3 - \lambda K_{tx,s}^3 + K_{xx,s}^3) \begin{pmatrix} 0 \\ \lambda \\ 1 \end{pmatrix} = \sum_{\sigma=\pm 1} \frac{\gamma_s \sigma \lambda h_{\sigma,s,\eta}}{\lambda - \sigma h_{\sigma,s}} \nabla_{(\chi,\eta,\rho)} h_{\sigma,s}. \tag{161}$$

The right side has zero ρ component. The director connection also vanishes on this light mode; its clock component is zero. The corresponding stress is $Z_{\sigma,s} = \gamma_s \sigma h_{\sigma,s,\eta} \lambda p/(\lambda - \sigma h_{\sigma,s})$, with $P_\eta = -p$, $Q_\eta = \lambda p$ and $P_\chi = Q_\chi = 0$. Hence the complete trace residue is zero. The strict matter cone solves the matter-to-light Sylvester equations; the zero

resonant projection solves the remaining metric-to-dilaton equations. Pulling back the block positive form produces a smooth full symmetrizer. The ten A_j variables append the diagonal principal block $\mathrm{diag}(u_j)$ and positive form 1_{10}; they introduce no new light resonance.

For $m > 5/2$, use $q \in H^{m+1}$ and $P_q, Q_q, Z, A \in H^m$; (158) lies in H^{m-1}. The exact local split is

$$\begin{aligned} Z_{j,t} + u_j Z_{j,x} + \gamma_j \sigma_j g_j^T Q_{q,x} &= -\sigma_j (g_j^T Q_q) Z_j - \gamma_j \sigma_j h_{j,qq}[Q_q, Q_q], \\ A_{j,t} + u_j A_{j,x} &= -2\sigma_j (g_j^T Q_q) A_j. \end{aligned} \tag{162}$$

The force is $\sum_j g_j Z_{j,x} - \sum_j g_j [A_j + Z_j^2/(2\gamma_j)]$. All remainders are smooth local products in the same H^m algebra. The conversion contributes first-jet Hessian principal terms and quadratic first-jet remainders. No spatial inverse, omitted mean or higher derivative of the quantum state enters the iteration. Symmetric-hyperbolic iteration gives $d\mathcal{E}_m/dt \le C_R(1 + \mathcal{E}_m)$ on the declared operating ball. Tame higher-order estimates keep smooth data smooth on that same local time. The actual full-mode propagator therefore exists for the solved coefficients; uniqueness of the distributional Ward transport identifies its stress with the solved means on the interior patch, with the same initial state and all references retained. A norm-controlled exterior extension of the reduced initial variables can be chosen for the local iteration. It does not assign artificial means to that quantum state outside the initial interval, and no common-action solution is inferred there. Its bound fixes the local existence time; arbitrary exterior continuations are compared only on their common interval of existence.

This local reduction also proves physical bulk causality. Let two admissible continuations agree initially on (a, b), including scalar values and actual stress means. Their local difference energy on $(a + t, b - t)$ has endpoint fluxes $-V^T(H - HA)V$ on the left and $-V^T(H + HA)V$ on the right. They are nonpositive; the interior is bounded by its energy. Gronwall therefore gives identical mean solutions until the interval closes or the operating domain is left. The interval boundaries are the null curves of g. For quantum restrictions require the same initial joint local state on the causal interval and the compared old references. The actual sector diffeomorphisms pull local observables into that interval, giving the corresponding restriction equality. Stress means alone do not give quantum-state equality, nor is a Type-III local density matrix assumed.

Finally write the minimally coupled JT action as $(2\kappa)^{-1} \int \sqrt{-g}\Phi(R - 2K_{\mathrm{geo}})$. Its metric curvature equation is $\rho_{tt} - \rho_{xx} = K_{\mathrm{geo}} e^{2\rho}$. Set $w = \Phi_x$, $\pi = \Phi_t$. The initial constraints are

$$\Phi_x = w, \qquad w_x = \rho_x w + \rho_t \pi - K_{\mathrm{geo}} e^{2\rho}\Phi - \kappa T_{00}, \qquad \pi_x = \rho_x \pi + \rho_t w - \kappa T_{0x}. \tag{163}$$

Here T is the complete source just constructed, including initial conversion jets; needed accelerations come from its invertible time matrix. This spatial ODE has a local solution for three integration constants, with positive Φ on a sufficiently small interval. Conservation propagates its two traceless constraints locally along physical light directions. Arbitrary periodic positive-energy JT data are not assumed. A consistent global representation and state realize the quantum solution; the causal interior restriction no longer depends on an arbitrary exterior coefficient continuation. □

This closes the quartet's nonlinear bulk source–clock–director–JT interface. It supplies neither moving-wall boundary terms nor the other gate windows' errors. The original

mixed target and its purifier remain in the actual state, but clock and geometry changes alter gate areas. Their same-history finite-horizon comparison and the physical exhaust still have to be incorporated before the complete thermal apparatus is obtained.

F.9 A physical wall retains its thermal friction

A finite rest mass cannot repair a perfect mirror by ignoring its high-frequency reaction. Partial transmission has a causal passive linear-response formulation [1]. For the four actual Dirac flavours, the corresponding force can be derived from a smooth material mass layer rather than imported from a scalar mirror. In a static material frame let $\mathsf{h}_z = -iv\sigma_z\partial_x + M(x-z)\sigma_x$, with $M \in C_c^\infty$, $M \geq 0$, and positive bare mechanical mass m_b. The same mass action gives the translation force $F = \int M'(x-z)\psi^\dagger\sigma_x\psi\,dx + F_{\mathrm{ct}}$, distributed bar reaction and metric mass stress. Use one fixed local counterterm action, an actual static interacting Hadamard ground or KMS preparation, and smooth flat-start displacement tests.

Proposition F.10 (Passive Dirac-layer response)**.** *For the static ground-state linear response, the full-continuum induced vacuum inertia is finite. With positive bare mass and total positive elastic stiffness, the unreduced causal mechanical impedance is positive-real. In the static zero-charge KMS continuum the layer instead has positive ohmic friction. Its vacuum-style low-frequency inertia integral diverges, while the unreduced thermal impedance remains convergent and positive-real. These are linear statements about the specified passive preparation, not arbitrary driven quartet states.*

Proof. Let $u_{E,a}$, $a = L, R$, be energy-normalized scattering modes with incoming amplitude $(2\pi v)^{-1/2}$. Define $A = \int M/v$, $Q = \|M'\|_1$ and $B = e^A/\sqrt{\pi v}$. The exact transfer equation $u_x = (iE\sigma_z + M\sigma_y)u/v$ bounds $\|u\| \leq B$. Integration by parts gives

$$f_{ab}(E,E') := \int M' u^\dagger_{E,a}\sigma_x u_{E',b} = -(E+E')\int (M/v)u^\dagger_{E,a}\sigma_y u_{E',b},$$
$$|f_{ab}(E,E')| \leq B^2 \min\{Q, A|E+E'|\}. \tag{164}$$

The mass anticommutator vanishes and M removes the endpoint term. For $N = 4$ flavours the filled-negative-state force spectrum obeys

$$\chi_I(\nu) = \pi N \sum_{a,b}\int_0^\nu |f_{ab}(E, E-\nu)|^2\,dE \leq 4\pi N B^4 \min\{Q^2\nu, A^2\nu^3/3\},$$
$$\mu = \frac{2}{\pi}\int_0^\infty \frac{\chi_I(\nu)}{\nu^3}\,d\nu \leq \frac{16NB^4AQ}{\sqrt{3}}. \tag{165}$$

At zero energy the exact transfer gives reflection $\tanh^2 A$, and hence $\chi_I(\nu) = N\tanh^2 A\,\nu^3/(6\pi v^2) + O(\nu^4)$. Retain the complete causal memory. For $\mathrm{Re}\,s > 0$ its impedance is

$$Z(s) = m_b s + k/s + \frac{2}{\pi}s\int_0^\infty \frac{\chi_I(\nu)}{\nu(\nu^2+s^2)}\,d\nu. \tag{166}$$

The measure is positive and $s/(s^2+\nu^2) = \frac{1}{2}[(s-i\nu)^{-1} + (s+i\nu)^{-1}]$ has positive real part. Thus $\mathrm{Re}\,Z(s) > 0$; the response has no right-half-plane pole. The observed vacuum inertial mass is $m_{\mathrm{phys}} = m_b + \mu$, not the action's bare m_b. For two separated layers the same supportwise identity gives a positive matrix spectral measure. Adding their

actual finite static polarization/Casimir Hessian to the paid elastic stiffness extends this argument when that total stiffness and the bare mass matrix are positive.

In the KMS continuum, with Fermi function $n_T(E)$,

$$\chi_{I,T}(\nu) = \pi N \sum_{a,b} \int_{\mathbb{R}} [n_T(E) - n_T(E+\nu)] |f_{ab}(E+\nu, E)|^2 \, dE. \tag{167}$$

Its weights are positive. On shell, $\sum_{a,b} |f_{ab}(E,E)|^2 = 2E^2 R(E)/(\pi^2 v^2)$, where $R(E)$ is the actual reflection probability. Therefore

$$\chi_{I,T}(\nu) = \gamma_T \nu + o(\nu), \qquad \gamma_T = \frac{2N}{\pi v^2} \int E^2 R(E) [-n_T'(E)] \, dE > 0. \tag{168}$$

The vacuum μ integral now diverges in the infrared. Equation (166) itself stays finite: its infrared integrand uses this ohmic law, while $\int [n_T(E) - n_T(E+\nu)] dE = \nu$ and (164) give an ultraviolet linear bound. Its positive-real proof is unchanged. A finite prepared exterior has the corresponding discrete positive Kubo measure; this continuum friction limit is not an uncharged infinite thermal reservoir. □

All transmitted exterior channels, old purifiers, static polarization and motion work remain in this construction. Smooth spatial profiles alone do not justify applying an instantaneous force to the old free sea as a finite-energy vector; the valid response tests are also smooth in time. More importantly, the actual driven body need not be passive. Its opposite-chirality mass mixing and non-Gaussian quartet require a joint local-source and task-error estimate. The common-velocity Ward closure inside a massless gate does not close that boundary hierarchy. The passive wall result establishes a physical linear margin, without substituting that margin for the retained mixed-target reset proof.

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Measurement as Accessibility: Measurement Permissions, Stable Records and Public Readout

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Abstract

A measurement context supplies probes, environmental dynamics and physical record ports. These determine different objects: implementable effects, persistent content and readable records. We compute their relations in a fresh-memory collision model. For two pure conditional fragment states of overlap r, the exact pointer-effect permission set is $\sqrt{ab} + \sqrt{(1-a)(1-b)} \geq r$; the closest permitted sharp pointer test has error $(1 - \sqrt{1-r^2})/2$. Its operator span does not determine this resolution. Holevo and single-measurement information count different numbers of readable records: one environment has 12 Holevo records but only 8 certified readable ones. Conditional product fragments give explicit public-agreement laws, and pooled one-sided local tests attain the joint pure-state discrimination exponent. Recording does not require a pure memory. We derive a purity-deficit budget and exhibit partially mixed qubits whose exact permissions form a parallelogram, whose local measurement saturates Holevo information, and whose reduced source channel is the same complete dephasing for every memory polarization. At zero polarization this channel produces no record. Reusing a spent port can erase its record and restore coherence, so freshness cannot be inferred from a channel label. For a common unknown reader orientation, agreement alone cannot certify correctness; prepared-pointer references give a finite calibration bound and an explicit residual common-error term. The same actual fragment readout supports Paper A's finite correlation-geometry certificate. The construction accounts for preparation, readout and shared standards without selecting a unique global outcome or deriving the Born rule.

1 Introduction

1.1 Three questions about measurement

Standard quantum mechanics gives rules for outcome probabilities but does not say how definite records arise from states that assign amplitudes to several possibilities. Collapse models change the dynamics, many-worlds accounts deny that one outcome is selected, and decoherence explains the suppression of interference without selecting a result. This

paper does not add to that list. It asks a set of narrower questions that every account has to answer in some form, and it answers them inside a declared physical context.

The first question is one of *permission*. A probe is a physical channel. Which effects on the system can the probe actually implement, given its coupling and initial state? The second concerns *stability*: which effects, or which record values, survive the environmental dynamics over the relevant time window? The third concerns *public readout*: when can several readers, each with access to a different part of the environment, obtain the same value, and how often is a shared value wrong?

The three answers are different kinds of object. Each probe's implementable effects form a convex set inside an operator system; a family of probes gives their union, which is convex only if the permitted family also includes the required randomized choices. Exact fixed points of a channel form an operator system, and under further hypotheses a von Neumann algebra. Records concern a classical label and the information that environmental fragments carry about it. Treating the three as one "accessible algebra" obtained by intersection loses the information that matters, and the paper keeps them apart.

The central claim of the paper is that measurement, in the operational sense used here, is a property of an interaction context: what a declared probe family can implement, what the declared dynamics leaves fixed, and what declared fragments record. We do not explain why one global outcome occurs, and we assume the Born rule.

1.2 Results

Section 2 fixes the typed data of a context. Section 3 states the three tests and separates two notions of redundancy, one based on Holevo information and one based on what a measurement can read. Section 4 introduces a record model in which fresh memories collide once with a pointer, and computes:

- the exact set of pointer effects that a fragment can implement (Theorem 4.2) and the distance from a sharp pointer projection to that set (Corollary 4.3);
- a nesting of the three tests: fragment effects lie in the pointer algebra, which equals the fixed algebra of the induced dephasing (Proposition 4.4);
- the gap between Holevo and accessible information, with closed-form fragment lengths for both notions of redundancy (Propositions 4.5 and 4.6);
- a purity-deficit budget and exact nonzero readout from partially mixed memories (Propositions 4.7 and 4.8);

Section 5 turns to several readers: conditional independence, the error exponents of joint, pooled local and majority readout, and a finite calibrated public-readout bound when the readers share an unknown orientation. Section 6 collects the model's conditions and interfaces. Section 6.2 compares the approach with decoherence theory, quantum Darwinism and other frameworks, and Section 6.3 identifies the actual record interface consumed elsewhere.

Several ingredients are standard: the exact Holevo quantity of a branching state, the Helstrom bound, and the role of redundancy in quantum Darwinism [10, 11, 12]. What the paper adds is the typed separation, the exact permission set of a fragment, the nesting of the tests in one dynamics, the counting difference between Holevo and readable records, the mixed-memory realization and a calibration-sensitive public-readout certificate.

2 Typed accessibility data

2.1 Algebraic background

In the algebraic formulation of quantum theory a system is described by an algebra of observables, and states are positive normalized functionals on it [8, 9]. In algebraic quantum field theory local algebras are attached to regions without a global tensor product, and tensor product structures can be induced by the accessible observables [20, 22]. HAFF builds on this view and asks which operator structures a physical context selects. Paper A treats the reduction of a global state by physical channels and shows that two channels can generate the same algebra while resolving different things [1]. The present paper applies the same caution to measurement.

2.2 The objects of a context

Let $\mathcal{M}_U$ be the ambient von Neumann algebra. A context $\mathbf{c}$ specifies allowed probe channels, an environmental dynamics, a record algebra, time scales and resolution thresholds.

Definition 2.1 (Probe channel and implementable effects)**.** *For each allowed probe α, let*

$$\mathcal{R}^*_{\alpha,\mathbf{c}} : \mathcal{B}(\mathcal{K}_\alpha) \longrightarrow \mathcal{M}_U$$

be its normal unital completely positive Heisenberg map. The set of ***implementable effects*** *of the context is*

$$\mathrm{Eff}(\mathbf{c}) := \{\mathcal{R}^*_{\alpha,\mathbf{c}}(F) : 0 \le F \le \mathbf{1},\ \alpha \text{ allowed}\},$$

and the ***probe-reachable operator system*** *is its ultraweakly closed linear span,*

$$\mathrm{Eff}_{\mathrm{span}}(\mathbf{c}) := \overline{\mathrm{span}}^{\,uw}\,\mathrm{Eff}(\mathbf{c}).$$

The family $\mathfrak{R}_{\mathrm{probe}}(\mathbf{c}) := \{\mathcal{R}_{\alpha,\mathbf{c}}\}_\alpha$ is part of the context, not an observable algebra.

For a single probe Eff is convex, contains 0 and $\mathbf{1}$, and is closed under $E \mapsto \mathbf{1} - E$. The span is the natural object for linear reconstruction of expectation values from many runs. It is not a list of effects that one run of an allowed probe implements: Section 4 gives probes whose span is the whole pointer algebra while the sharp pointer projections are not implementable. The same distinction appears for reductions in Paper A, where the effects a channel reads out, $\Phi^*([0, \mathbf{1}])$, are in general fewer than the positive contractions in the range of Φ^* [1].

Definition 2.2 (Environmental fixed set)**.** *For a normal UCP semigroup $\{\mathcal{E}_t\}_{t\ge 0}$ on $\mathcal{M}_U$, set*

$$\mathrm{Fix}(\mathcal{E}) := \{O \in \mathcal{M}_U : \mathcal{E}_t(O) = O \text{ for every } t \ge 0\}.$$

This is always an ultraweakly closed operator system. It is a von Neumann subalgebra, for example, when the semigroup admits a faithful normal invariant state: equality in the Schwarz inequality then places every fixed element in the multiplicative domain. A discrete-time channel is treated in the same way with $t \in \mathbb{N}$.

Definition 2.3 (Pointer and record algebra)**.** *A* ***pointer/record algebra*** $\mathcal{P}_{\mathrm{env}} \subset \mathcal{M}_U$ *is a designated commutative von Neumann algebra generated by a classical record PVM* $Z(B)$*, or by a finite-resolution coarse graining. For a finite label set, a fragment record channel acts on label states by*

$$\Gamma_{k,*}\big((p_x)_x\big) = \sum_x p_x \rho^x_{E_k}.$$

Its normal UCP Heisenberg dual is

$$\Gamma_k^* : \mathcal{B}(\mathcal{H}_{E_k}) \longrightarrow \mathcal{P}_{\mathrm{env}}, \qquad \Gamma_k^*(F) = \sum_x \mathrm{Tr}(\rho^x_{E_k} F) Z_x.$$

More generally the state map acts on the predual of the commutative record algebra. Approximate pointer observables are treated by their distance to $\mathcal{P}_{\mathrm{env}}$*.*

There is no default equality among Eff, $\mathrm{Fix}(\mathcal{E})$ and $\mathcal{P}_{\mathrm{env}}$. The declared maps are the probe maps, the dynamics, the inclusions of chosen subalgebras into $\mathcal{M}_U$, and the record channels. A model may establish relations among the objects, for example that every implementable effect lies in $\mathcal{P}_{\mathrm{env}} \cap \mathrm{Fix}(\mathcal{E})$; such relations are results or assumptions of that model.

Definition 2.4 (Measurement-accessibility profile)**.** *The* ***measurement-accessibility profile*** *of a context is the typed datum*

$$\mathbf{Acc}(\mathbf{c}) := \big(\mathrm{Eff}(\mathbf{c}),\ \mathrm{Eff}_{\mathrm{span}}(\mathbf{c}),\ \mathfrak{R}_{\mathrm{probe}}(\mathbf{c}),\ \mathcal{P}_{\mathrm{env}}(\mathbf{c}),\ \mathrm{Fix}(\mathcal{E}_{\mathbf{c}});\ \Gamma_{k,*}, T_{\mathbf{c}}, \epsilon, \delta\big). \tag{1}$$

The operator-system and fixed-set statements above hold for general von Neumann algebras under the hypotheses shown. Entropic statements are made for finite alphabets or finite-resolution labels and specified states.

3 Three tests

3.1 Probe reachability

Constraint 3.1 (Probe reachability)**.** *An effect* $M(B) \in \mathcal{M}_U$ *is implementable in context* $\mathbf{c}$ *if* $M(B) = \mathcal{R}^*_{\alpha,\mathbf{c}}(F)$ *for some allowed probe* α *and some probe effect* $0 \le F \le \mathbf{1}$*. A POVM* M *is implementable if one probe and one probe POVM* F *give* $M(B) = \mathcal{R}^*_{\alpha,\mathbf{c}}(F(B))$ *for every outcome set* B*.*

Implementability is a property of a channel and an effect. A probe Hamiltonian may realize the channel, but the condition $[O, H_{\mathrm{probe}}] \neq 0$ is neither necessary nor closed under sums and products; ideal nondemolition couplings commute with the measured pointer observable. Effects outside $\mathrm{Eff}(\mathbf{c})$ are not ruled out as physical quantities. The context simply supplies no implementation of them, and enlarging the context, for example by allowing more fragments or repeated runs, is a change of context.

3.2 Dynamical stability

Constraint 3.2 (Stability)**.** *An effect* O *passes the* ***finite-window stability test*** *if*

$$\sup_{0 \le t \le T_{\mathbf{c}}} \|\mathcal{E}_t(O) - O\| < \epsilon. \tag{2}$$

Exact stability is the separate statement $O \in \mathrm{Fix}(\mathcal{E})$*.*

In decoherence theory the pointer observables are those left stable by the system–environment interaction [10]. The threshold ϵ and the window $T_{\mathbf{c}}$ are set by the recording mechanisms and time scales of the context. An implementable effect need not be stable, and an exact fixed point need not be implementable by any allowed probe.

3.3 Environmental redundancy: Holevo and readable records

Let X be the classical label of a pointer PVM, or of a finite coarse graining of a POVM, with distribution (p_x). For each environmental fragment E_k form the classical–quantum state $\rho_{XE_k} = \sum_x p_x |x\rangle\langle x| \otimes \rho^x_{E_k}$. Two quantities measure what E_k carries about X. The Holevo information

$$\chi(X : E_k) = S\Big(\sum_x p_x \rho^x_{E_k}\Big) - \sum_x p_x S(\rho^x_{E_k})$$

bounds what any measurement on E_k can extract. The accessible information

$$I_{\mathrm{acc}}(X : E_k) = \sup_{\text{POVM } N \text{ on } E_k} I(X : Y_N)$$

is what a single measurement on the fragment does extract at best, where Y_N is the outcome. Always $I_{\mathrm{acc}} \le \chi$ [14].

Constraint 3.3 (Redundancy)**.** *Fix $0 < \delta < 1$ and a family of disjoint fragments. The label X has R **Holevo records** if R disjoint fragments satisfy $\chi(X : E_k) \ge (1-\delta)H(X)$, and R **readable records** if R disjoint fragments satisfy $I_{\mathrm{acc}}(X : E_k) \ge (1-\delta)H(X)$. For a declared measurement N_k on each fragment one may use $I(X : Y_{N_k})$ in place of I_{acc}.*

Readable records are Holevo records, but not conversely, and the two counts can differ (Proposition 4.6). For continuous spectra X denotes a declared finite-resolution partition. No von Neumann entropy is assigned to an operator, and no expression such as $I(\hat{O} : E_k)$ is used.

Redundancy is what allows several observers to read the same label from different fragments without disturbing one another. It does not by itself say how often their common value is wrong, nor which global outcome is realized. Section 5 computes the former in the model.

4 A record model in which the three tests can be computed

4.1 Fresh memories colliding once with a pointer

The system S is a qubit with pointer projections $P_x = |x\rangle\langle x|$, $x \in \{0, 1\}$. The environment consists of n memory qubits $E_1, \ldots, E_n$, each prepared in $|0\rangle$. The k-th memory collides once with S through

$$U_k = P_0 \otimes \mathbf{1} + P_1 \otimes R_k, \qquad R_k = \begin{pmatrix} c_k & -\sqrt{1-c_k^2} \\ \sqrt{1-c_k^2} & c_k \end{pmatrix}, \qquad 0 \le c_k \le 1, \tag{3}$$

and does not interact again. Starting from $\sqrt{p}\,|0\rangle + \sqrt{1-p}\,|1\rangle$ with $0 < p < 1$, the state after n collisions is

$$|\Psi_n\rangle = \sqrt{p}\,|0\rangle|e_0\rangle + \sqrt{1-p}\,|1\rangle|e_1\rangle, \qquad |e_x\rangle = \bigotimes_{k=1}^{n} |e_{x,k}\rangle, \tag{4}$$

with $|e_{0,k}\rangle = |0\rangle$ and $|e_{1,k}\rangle = R_k|0\rangle$, so that $\langle e_{0,k}|e_{1,k}\rangle = c_k$. A fragment F is a declared set of memories; write

$$r_F = \prod_{k\in F} c_k, \qquad H_p(r) = h\left(\frac{1+\sqrt{1-4p(1-p)(1-r^2)}}{2}\right),$$

where $h(x) = -x\ln x - (1-x)\ln(1-x)$. All entropies are in nats. The label X is the value a pointer measurement on S would return.

The inputs are explicit: the pointer basis, the split into system and memories, the order of collisions, the fragment partition, and the purity of the memories. None of them is derived here. The model is a version of the standard branching model of quantum Darwinism [11, 12]; its role in this paper is to make all components of the profile of Definition 2.4 computable at once.

Proposition 4.1 (Record quantities)**.** *For every fragment F,*

$$\chi(X : E_F) = H_p(r_F), \qquad (\rho_S)_{01} = \sqrt{p(1-p)}\, r_{[n]},$$
$$S(\rho_S) = H_p(r_{[n]}), \qquad I(S : E_{[n]}) = 2H_p(r_{[n]}).$$

Proof. Given $X = x$ the fragment is in the pure state $|e_x^F\rangle = \bigotimes_{k\in F} |e_{x,k}\rangle$, and $\langle e_0^F|e_1^F\rangle = r_F$. The average state $p|e_0^F\rangle\langle e_0^F| + (1-p)|e_1^F\rangle\langle e_1^F|$ has two nonzero eigenvalues with sum 1 and product $p(1-p)(1-r_F^2)$, so its entropy is $H_p(r_F)$, and the conditional entropies vanish. Tracing the memories out of (4) multiplies the off-diagonal element by $\langle e_1|e_0\rangle = r_{[n]}$, and the global state is pure, so $S(\rho_S) = S(\rho_{E_{[n]}}) = H_p(r_{[n]})$ and $I(S : E_{[n]}) = 2S(\rho_S)$. □

The reduced dynamics of S is the dephasing channel $\mathcal{D}_k(\rho) = P_0\rho P_0 + P_1\rho P_1 + c_k(P_0\rho P_1 + P_1\rho P_0)$ at each step. Its fixed algebra, for any $c_k < 1$, is the diagonal algebra $\mathcal{P} = \mathrm{span}\{P_0, P_1\}$, which is also the pointer algebra.

4.2 Measurement permissions of a fragment

A reader with access to fragment F implements effects on S through the Heisenberg map

$$\Phi_F^*(A) = \langle e_0^F|A|e_0^F\rangle\, P_0 + \langle e_1^F|A|e_1^F\rangle\, P_1, \qquad A \in \mathcal{B}(\mathcal{H}_F), \tag{5}$$

obtained by coupling S to F through the collisions and tracing out the rest. The span of the image is all of $\mathcal{P}$ as soon as $0 < r_F < 1$, since it contains $\mathbf{1}$ and the operator $\Phi_F^*(|e_0^F\rangle\langle e_0^F|) = P_0 + r_F^2 P_1$. The implementable effects are fewer.

Theorem 4.2 (Permission set of a fragment)**.** *Let $|e_0\rangle, |e_1\rangle$ be unit vectors with $|\langle e_0|e_1\rangle| = r$. Then*

$$\big\{(\langle e_0|A|e_0\rangle, \langle e_1|A|e_1\rangle) : 0 \le A \le \mathbf{1}\big\}$$
$$= L_r := \big\{(a,b) \in [0,1]^2 :\ \sqrt{ab} + \sqrt{(1-a)(1-b)} \ge r\big\}.$$

Hence $\mathrm{Eff}_F := \Phi_F^*([0,\mathbf{1}]) = \{aP_0 + bP_1 : (a,b) \in L_{r_F}\}$.

Proof. Inclusion. For an effect A put $a = \langle e_0|A|e_0\rangle$ and $b = \langle e_1|A|e_1\rangle$. By the Cauchy–Schwarz inequality applied to $A^{1/2}e_0$ and $A^{1/2}e_1$, $|\langle e_0|A|e_1\rangle| \le \sqrt{ab}$, and likewise $|\langle e_0|(\mathbf{1}-A)|e_1\rangle| \le \sqrt{(1-a)(1-b)}$. Adding and using the triangle inequality gives $r \le \sqrt{ab} + \sqrt{(1-a)(1-b)}$.

Equality. The function $f(a,b) = \sqrt{ab} + \sqrt{(1-a)(1-b)}$ is a sum of geometric means, hence concave, so L_r is a compact convex set. The set of attained pairs is convex as the image of the convex set of effects under an affine map. At $r = 0$ the states are orthogonal and all pairs in the square are attained by diagonal effects; at $r = 1$ the states are identical up to phase and the attained set is the diagonal. For $0 < r < 1$, a compact convex subset of the plane is the convex hull of its boundary, so it suffices to attain every boundary point of L_r. Choose phases and an orthonormal basis of $\mathrm{span}\{e_0, e_1\}$ with $e_0 = (1, 0)$ and $e_1 = (\cos\theta, \sin\theta)$, $\cos\theta = r$, $0 \le \theta \le \pi/2$; effects supported on this span suffice. For $u = (\cos\varphi, \sin\varphi)$ with $\theta - \pi/2 \le \varphi \le 0$, the projection $|u\rangle\langle u|$ gives $a = \cos^2\varphi$ and $b = \cos^2(\theta - \varphi)$, and since $\cos\varphi \ge 0$, $\sin\varphi \le 0$, $\cos(\theta-\varphi) \ge 0$ and $\sin(\theta-\varphi) \ge 0$,

$$f(a,b) = \cos\varphi\cos(\theta-\varphi) - \sin\varphi\sin(\theta-\varphi) = \cos\theta = r.$$

This traces the arc of $\{f = r\}$ from $(1-r^2, 0)$ to $(1, r^2)$. The map $A \mapsto \mathbf{1} - A$ sends it to the arc from $(r^2, 1)$ to $(0, 1-r^2)$. The remaining boundary consists of segments of the square: $b = 0$ with $0 \le a \le 1 - r^2$, attained by $A = t|e_1^\perp\rangle\langle e_1^\perp|$; $a = 1$ with $r^2 \le b \le 1$, attained by $A = |e_0\rangle\langle e_0| + t|e_0^\perp\rangle\langle e_0^\perp|$, $0 \le t \le 1$; and their images under $A \mapsto \mathbf{1} - A$. On each edge of the square f reduces to $\sqrt{1-a}$, $\sqrt{b}$, $\sqrt{1-b}$ or $\sqrt{a}$, so these segments are exactly the parts of the edges in L_r. For $r < 1$, $f(a,a) = 1 > r$, so the level set $\{f = r\}$ inside the square consists of the two arcs. Every boundary point is therefore attained. □

For $0 < r < 1$ the permission set is bounded by two curved arcs and four square-edge segments. The corners $(0,0)$ and $(1,1)$ lie on those segments, not on the arcs $f = r$. It fills the square when $r = 0$ and collapses to the diagonal $\{a = b\}$, the trivial effects $a\mathbf{1}$, when $r = 1$. For $0 < r < 1$ it contains no sharp pointer projection: $(1,0) \notin L_r$ because $f(1,0) = 0$.

Corollary 4.3 (Closest implementable pointer test)**.** *For $0 \le r \le 1$,*

$$\min_{E \in \mathrm{Eff}_F} \|E - P_0\| = \varepsilon(r) := \frac{1 - \sqrt{1-r^2}}{2},$$

with $r = r_F$, attained at $E = (1-\varepsilon)P_0 + \varepsilon P_1$. The value $\varepsilon(r)$ is the minimal error probability for distinguishing $|e_0^F\rangle$ from $|e_1^F\rangle$ with equal priors.

Proof. For $E = aP_0 + bP_1$ one has $\|E - P_0\| = \max(1-a, b)$. If $\max(1-a, b) \le t < 1/2$, then $a \ge 1 - t > 1/2 > t \ge b$. In that region $\partial f/\partial a = \frac{1}{2}\big(\sqrt{b/a} - \sqrt{(1-b)/(1-a)}\big) < 0$ and $\partial f/\partial b = \frac{1}{2}\big(\sqrt{a/b} - \sqrt{(1-a)/(1-b)}\big) > 0$, so $f(a,b) \le f(1-t,t) = 2\sqrt{t(1-t)}$. Since $2\sqrt{\varepsilon(1-\varepsilon)} = r$ and $t \mapsto 2\sqrt{t(1-t)}$ is increasing on $[0, 1/2]$, the region contains points of L_r only if $t \ge \varepsilon(r)$, and $(1-\varepsilon, \varepsilon) \in L_r$. The identification with the Helstrom error is the formula $(1 - \sqrt{1 - 4p(1-p)r^2})/2$ at $p = 1/2$ [15]. □

For a fragment of m memories with equal overlaps c, $r_F = c^m$ and $\varepsilon(c^m) = c^{2m}/4 + O(c^{4m})$. Measurement permission for the pointer grows with the number of memories a reader controls, and an η-approximate pointer test becomes available once $c^{2m} \le 4\eta(1-\eta)$. Enlarging F is a change of context, in the sense of Constraint 3.1.

4.3 The three tests in one dynamics

Proposition 4.4 (Nesting)**.** *Let a probe act through a pointer-controlled unitary* $U = \sum_x P_x \otimes V_x$ *on* $\mathcal{H}_S \otimes \mathcal{K}$ *with probe state* $|\phi\rangle$*. Then for every* $A \in \mathcal{B}(\mathcal{K})$

$$\mathcal{R}^*(A) = \sum_x \langle\phi|V_x^* A V_x|\phi\rangle\, P_x \in \mathcal{P}_{\text{env}}.$$

In the record model, for every fragment F *with* $0 < r_F < 1$ *and every* $c_k < 1$*,*

$$\text{Eff}_F \subsetneq \text{Eff}_{\text{span},F} \cap [0,\mathbf{1}] = \mathcal{P} \cap [0,\mathbf{1}] = \text{Fix}(\mathcal{D}_k) \cap [0,\mathbf{1}].$$

Proof. $U^*(\mathbf{1}\otimes A)U = \sum_{x,x'} P_x P_{x'} \otimes V_x^* A V_{x'} = \sum_x P_x \otimes V_x^* A V_x$, and the partial expectation in $|\phi\rangle$ gives the formula. In the model $\text{Eff}_{\text{span},F} = \mathcal{P}$ as noted before Theorem 4.2, the fixed algebra of $\mathcal{D}_k$ is $\mathcal{P}$, and Eff_F is the lens of Theorem 4.2, which omits P_0. □

In this model every effect that the environment can report is a stable pointer effect, and the stable pointer effects are exactly the positive contractions of the span. The tests still differ. The lens measures resolution, the fixed algebra measures persistence, and redundancy, below, counts how many disjoint readers have access. A probe that is not pointer-controlled, for example one that rotates the system before recording, can reach effects outside $\text{Fix}(\mathcal{D}_k)$; those effects are then not stable under the environment that produced the records.

4.4 Holevo records and readable records

Proposition 4.5 (Accessible information of a fragment)**.** *Let*

$$P_e(r) = \frac{1 - \sqrt{1 - 4p(1-p)r^2}}{2}$$

be the Helstrom error for two pure states with overlap r *and prior* $(p, 1-p)$*. For every fragment,*

$$I_{\text{acc}}(X : E_F) = h(p) - h(P_e(r_F)) \le \chi(X : E_F) = H_p(r_F).$$

For $p = 1/2$*,*

$$I_{\text{acc}}(X : E_F) = \ln 2 - h\big(\varepsilon(r_F)\big), \qquad \chi(X : E_F) = h\Big(\frac{1 + r_F}{2}\Big),$$

and the two differ whenever $0 < r_F < 1$*.*

Proof. This binary pure-state optimum is standard [16, Sec. 3.5.4]; the following argument proves it for all POVMs. Write their conditional outcome laws as a_y, b_y, put $q_y = pa_y + (1-p)b_y$, and omit zero-weight outcomes. Their posterior probabilities are $\pi_y = pa_y/q_y$; set $C_y = 2\sqrt{\pi_y(1-\pi_y)}$ and

$$E(C) = h\Big(\frac{1 - \sqrt{1 - C^2}}{2}\Big).$$

Then $H(X \mid Y) = \sum_y q_y E(C_y)$. For $t = \sqrt{1-C^2}$, $E'(C) = C\,\text{artanh}(t)/t \ge 0$ and $E''(C) = (\text{artanh}(t) - t)/t^3 \ge 0$ for $0 < C < 1$; continuity extends convexity to the endpoints. The outcome-wise Cauchy–Schwarz argument of Theorem 4.2 gives

$$\sum_y q_y C_y = 2\sqrt{p(1-p)} \sum_y \sqrt{a_y b_y} \ge 2\sqrt{p(1-p)}\, r_F.$$

Jensen and monotonicity imply $H(X \mid Y) \geq E(2\sqrt{p(1-p)}r_F) = h(P_e(r_F))$. Diagonalizing $p|e_0^F\rangle\langle e_0^F| - (1-p)|e_1^F\rangle\langle e_1^F|$ shows that its Helstrom measurement has posterior error $P_e(r_F)$ on every nonzero outcome, so it attains the bound. The cases $r_F = 0, 1$ follow directly. The Holevo bound and value are those of Proposition 4.1. At equal priors the displayed forms agree only at $r_F = 0, 1$. □

At $p = 1/2$ and $r_F = 0.8$, $\chi = 0.325083$ while $I_{\text{acc}} = 0.192745$. With $\delta = 0.6$ the threshold $(1-\delta)\ln 2 = 0.277259$ lies between them: the fragment is a Holevo record and not a readable one. The threshold for a Holevo record and the threshold for a measurement-based record are therefore not equivalent, even when the measurement is optimal; the difference concerns one measurement on one fragment, and joint coding over many independent preparations is a different task.

Proposition 4.6 (Fragment lengths)**.** *Let all $c_k = c \in (0,1)$ and $0 < \delta < 1$, and let h^{-1} denote the inverse of h on $[0, 1/2]$. Put*

$$q_\delta = h^{-1}\big((1-\delta)h(p)\big), \qquad a_\chi = \frac{(1-2q_\delta)^2 - (2p-1)^2}{4p(1-p)},$$

$$e_\delta = h^{-1}\big(\delta h(p)\big), \qquad a_{\text{op}} = \frac{e_\delta(1-e_\delta)}{p(1-p)}.$$

A fragment of m memories is a Holevo record if and only if $c^{2m} \leq a_\chi$, and it is a readable record if and only if $c^{2m} \leq a_{\text{op}}$. The smallest such lengths are

$$m_\chi = \left\lceil \frac{\ln a_\chi}{2\ln c} \right\rceil, \qquad m_{\text{op}} = \left\lceil \frac{\ln a_{\text{op}}}{2\ln c} \right\rceil,$$

so m_{op} is the smallest readable length for every $0 < p < 1$.

Proof. $H_p(r)$ is a decreasing function of r^2, and $H_p(r) = (1-\delta)h(p)$ exactly when the larger eigenvalue $\lambda = (1+\sqrt{1-4p(1-p)(1-r^2)})/2$ equals $1 - q_\delta$, that is when $r^2 = a_\chi$. For the readable case, Proposition 4.5 gives $I_{\text{acc}} \geq (1-\delta)h(p)$ exactly when $h(P_e(r)) \leq \delta h(p)$, that is when $P_e(r) \leq e_\delta$, which rearranges to $4p(1-p)r^2 \leq 4e_\delta(1-e_\delta)$. Since $e_\delta \leq \min(p, 1-p)$, $a_{\text{op}} \leq 1$. □

p	c	n	δ	m_χ	m_{op}	Holevo records	certified readable
0.5	0.80	60	0.10	5	7	12	8
0.2	0.80	60	0.10	5	7	12	8
0.5	0.95	100	0.05	27	38	3	2

Table 1: Fragment lengths and the number of disjoint equal-length records in n memories. The binary pure-state optimum in Proposition 4.5 makes every listed m_{op} minimal over all POVMs. At $p = 0.2$, I_{acc} is .439321 for $m = 6$ and .458263 for $m = 7$, across the threshold .450362.

When fragments are unequal, the counts depend on the chosen partition. With equal overlaps, m_χ gives the largest Holevo count. m_{op} gives the largest readable count for the same pure binary ensemble at every prior.

4.5 The preparation resource for a record

Proposition 4.7 (Purity-deficit budget)**.** *Replace the memories by systems of dimension d_k in product states $\bigotimes_k \sigma_k$, and let the collisions be pointer-controlled unitaries $P_0 \otimes \mathbf{1} + P_1 \otimes W_k$. Then for every fragment*

$$\chi(X : E_F) = S\Big(p \bigotimes_{k\in F} \sigma_k + (1-p) \bigotimes_{k\in F} W_k \sigma_k W_k^*\Big) - \sum_{k\in F} S(\sigma_k) \le \sum_{k\in F} \big[\ln d_k - S(\sigma_k)\big],$$

while each collision multiplies the off-diagonal element $(\rho_S)_{01}$ by $\mathrm{tr}(\sigma_k W_k^*)$. If $\sigma_k = \mathbf{1}/2$ and $W_k = R_k$, no fragment carries any information about X, and the coherence of S is still multiplied by c_k.

Proof. The two conditional states of the fragment are unitarily related products, so both have entropy $\sum_{k\in F} S(\sigma_k)$; the average state has entropy at most $\sum_{k\in F} \ln d_k$. The block $P_0\rho_S P_1 \otimes \sigma_k$ is mapped to $P_0\rho_S P_1 \otimes \sigma_k W_k^*$, so tracing out the memory multiplies $(\rho_S)_{01}$ by $\mathrm{tr}(\sigma_k W_k^*)$. For $\sigma_k = \mathbf{1}/2$ both conditional states equal $\mathbf{1}/2$, and $\mathrm{tr}(R_k^*)/2 = c_k$. □

The initial purity deficits bound the information that these unitary ports can record. They do not require every memory to be pure. Dephasing is not evidence of a readable record: a maximally mixed port can generate the same source channel as a pure port and carry no label information. Paper F uses this resource budget as a quantitative low-record boundary condition [5]. Preparation, correlation and renewal must then be accounted for separately; this entropy bound alone is not a universal work price.

4.6 Partially mixed memories: exact permission and readout

Let each fresh qubit memory start in $\eta_\alpha = (I + \alpha Z)/2$, $0 \le \alpha \le 1$, and use

$$U = P_0 \otimes I + P_1 \otimes X.$$

The conditional states are $\eta_0 = \eta_\alpha$ and $\eta_1 = X\eta_\alpha X = (I - \alpha Z)/2$. Put $\varepsilon_\alpha = (1-\alpha)/2$.

Proposition 4.8 (A mixed-memory record)**.** *For one fresh memory, its exact effect permission set is*

$$\mathrm{conv}\{(0,0), (1,1), (1-\varepsilon_\alpha, \varepsilon_\alpha), (\varepsilon_\alpha, 1-\varepsilon_\alpha)\}. \tag{6}$$

For $\alpha > 0$ the effect span is the full pointer algebra, but the closest permitted P_0 test has distance ε_α. At equal label priors,

$$\chi(X : E) = I_{\mathrm{acc}}(X : E) = \ln 2 - h((1+\alpha)/2) = \ln 2 - S(\eta_\alpha). \tag{7}$$

Measurement of Z_E attains this value and has symmetric error ε_α. For every α, tracing the memory completely dephases the source.

Proof. Only the diagonal entries $u, v \in [0,1]$ of a memory effect matter. The reported pair is $((1-\varepsilon_\alpha)u + \varepsilon_\alpha v, \varepsilon_\alpha u + (1-\varepsilon_\alpha)v)$. Diagonal effects attain every such pair, proving the parallelogram. For any reported pair (a,b), $a - b = \alpha(u - v) \le \alpha$; hence $\max(1-a, b) \ge (1-\alpha)/2$, with equality at $u = 1, v = 0$. The conditional states commute, have entropy $h((1+\alpha)/2)$ and average to $I/2$. Their common eigenbasis measurement attains the Holevo quantity. Finally the source coherence multiplier is $\mathrm{Tr}(\eta_\alpha X) = 0$. □

Records in partially mixed spin environments are already part of quantum Darwinism [12]. For $0 < \alpha < 1$ this construction gives their exact effect permissions and an explicit readout saturating the one-port purity-deficit budget. At $\alpha = 0$ its effects collapse to the trivial diagonal (a, a) and it carries no record, although its source channel is unchanged. At $\alpha = 1$ it records perfectly. Thus neither an effect span nor the complete source channel fixes record quality.

After m fresh collisions, the conditional memory states are $\eta_x^{\otimes m}$. Measuring every Z_E reads the entire commuting ensemble and still saturates Holevo information. The reports are binary independent errors conditional on x. Majority vote has exact error

$$e_m = \Pr\{\mathrm{Bin}(m, \varepsilon_\alpha) > m/2\} + \tfrac{1}{2}\Pr\{\mathrm{Bin}(m, \varepsilon_\alpha) = m/2\} \le e^{-m\alpha^2/2}. \tag{8}$$

The bound is Hoeffding's for the sample mean of signs in $[-1, 1]$. Fano's inequality then gives a sufficient readable-record length: choose m so that the bound is at most $1/2$ and $h(e^{-m\alpha^2/2}) \le \delta \ln 2$. This constructive mixed-state extension needs no classification of all mixed conditional-state permission sets.

4.7 A spent port is not a fresh channel input

The same controlled-X unitary also makes the port distinction exact. Since $U^2 = I$, applying it a second time to the same memory, without intervening measurement, export or dynamics, returns the complete state to its pre-collision value. The first collision dephases the source and writes the α-dependent record; the second removes that record and restores the original source coherence. Two independent fresh collisions instead preserve complete source dephasing and leave two conditional memory copies. This reversal does not erase an external measurement record or a copy already transferred to another port.

The one-step reduced channel cannot be applied again to the source while silently replacing the spent memory by η_α. After the first collision the source and memory are correlated. A second actual interaction acts on that joint state, including its old purification. Paper F studies repeated contact and joint memory updating [5]; the distinction also applies to the retained-port acquisition used by Q-RAIF C [4]. Storage, overwrite, reset and acquisition are different operations even when their isolated channel labels are similar.

5 Public readout

5.1 Agreement, correctness and common error

Proposition 5.1 (Conditional independence of readers)**.** *Let readers $j = 1, \dots, J$ apply POVMs $N^{(j)}$ to disjoint fragments F_j. Then the joint distribution of the pointer value and the outcomes is*

$$\Pr(x, y_1, \dots, y_J) = p_x \prod_{j=1}^{J} \langle e_x^{F_j} | N_{y_j}^{(j)} | e_x^{F_j} \rangle.$$

For two readers who each report a value $\hat{x}_j(y_j) \in \{0, 1\}$ with error probabilities $\varepsilon_j(x) = \Pr(\hat{x}_j \ne x \mid x)$, the agreement rate A, the correctness rates C_j and the common error rate $E = \Pr(\hat{x}_1 = \hat{x}_2 \ne x)$ are

$$E = \sum_x p_x \varepsilon_1(x)\varepsilon_2(x), \qquad A = \sum_x p_x \big[(1 - \varepsilon_1(x))(1 - \varepsilon_2(x)) + \varepsilon_1(x)\varepsilon_2(x)\big],$$

and $C_1 + C_2 = 1 + A - 2E$*, so that* $\Pr(\textit{correct} \mid \textit{agree}) = 1 - E/A$.

Proof. Tracing S out of (4) gives $\rho_E = \sum_x p_x |e_x\rangle\langle e_x|$ with $|e_x\rangle$ a product over memories. A pointer measurement on S commutes with the fragment measurements and returns x with probability p_x, leaving the memories in $|e_x\rangle$. The product form of $|e_x\rangle$ over disjoint fragments gives the factorization. The formulas for E and A follow by summing over the four combinations of right and wrong reports, and the identity $C_1 + C_2 = 1 + A - 2E$ holds event by event: an agreeing correct pair contributes two correct reports, an agreeing wrong pair none, and a disagreeing pair one. □

With equal priors, $c = 0.8$ and two five-memory fragments read with their Helstrom measurements, each reader errs with probability $\varepsilon(0.8^5) = 0.02761$, the readers agree with probability $A = 0.94631$, and an agreeing pair is wrong with probability $E/A = 8.05 \times 10^{-4}$. In this model agreement certifies correctness up to the product of private errors. The identity $C_1+C_2 = 1+A-2E$ and the separation of agreement from correctness are the subject of Foundation II, where a contamination shared by two instruments raises the common error far above the product of private errors [7]. Independent private errors already produce a nonzero common error. An excess beyond the conditional-product formula requires an additional shared influence or a failure of that record law, such as shared instrument contamination or calibration error. T-DOME III makes this precise: under a symmetry acting on all readers at once, their relative sign is detectable from agreement and a common inversion is not, so the error that agreement can never expose is the direction shared by the readers' standards [21].

5.2 Calibrated public readout with a common orientation

Conditional independence is a property of the prepared ports and their actual instruments. It is not implied by agreement or commuting effects. The mixed-memory family gives an explicit calibration problem. Encode the pointer value as $z = (-1)^x$. Suppose all readers share an unknown fixed orientation $C \in \{-1, 1\}$, so each reported elementary sign has conditional mean $C\alpha z$. The physical response $\alpha > 0$ is supplied and stable across reference and target runs. A common inversion leaves every agreement rate unchanged. With equal target priors and no independently anchored reference, the transformations $(C, z) \mapsto (-C, -z)$ give the same observable target-record law, so the absolute orientation is unidentifiable. A known asymmetric source prior would be additional orientation information and a different calibration problem.

The context must supply a preparation of the physical projection P_0 whose label is anchored independently of the unknown reader orientation. Its verification and preparation are resources. If the preparation label is itself inverted by the same unknown C, the reference cannot remove the ambiguity. Given that permission, prepare $z = +1$ in n_{ref} independent reference trials and measure fresh mixed ports with the same physical C response used for target readout. Estimate C by the sign of their mean, breaking a tie at random. Hoeffding gives

$$\Pr\Big(\widehat{C} \neq C\Big) \leq \delta_{\mathrm{cal}} := \min\{1, e^{-n_{\mathrm{ref}}\alpha^2/2}\}. \tag{9}$$

A change of orientation or response between reference and target runs violates the premise; the bound does not detect arbitrary shared contamination.

Proposition 5.2 (Finite public correctness after calibration)**.** *Let J readers each control m fresh mixed-memory ports of the same target pointer. They measure the memory Z values, take a majority with independent fair tie coins, and multiply their decision sign by $\widehat{C}$. The reference trials are independent of the target ports. With e_m from* (8)*,*

$$A_J := \Pr(\textit{all readers agree}) = (1-e_m)^J + e_m^J, \qquad \Pr(\textit{all agree wrongly}) \le \delta_{\mathrm{cal}} + e_m^J. \tag{10}$$

Consequently

$$\Pr(\textit{wrong} \mid \textit{all agree}) \le \min\{1, (\delta_{\mathrm{cal}} + e_m^J)/A_J\}.$$

Proof. Correcting by a common sign does not change agreement. Conditional on the pointer and the true orientation, the fresh ports give independent symmetric majority errors of probability e_m. On the event $\widehat{C} = C$, all reports are wrong with probability e_m^J; on its complement this probability is at most one. Reference–target independence gives the stated bounds. □

For $\alpha = 0.6$, $m = 5$, $J = 2$, the exact private error is $e_5 = 0.05792$ and $A_2 = 0.8908694528$. With $n_{\mathrm{ref}} = 100$ the calibration failure bound is e^{-18}. Even after calibration, the conditional common-error bound is about 0.003766. Increasing the number of readers suppresses their independent mistakes, but a fixed shared calibration error does not disappear through agreement.

The protocol consumes n_{ref} prepared-pointer references and fresh mixed ports, Jm target ports, all corresponding readout bits, access to a common orientation standard, and the communication needed for calibration and agreement. These are prepared resources, not an inference of correctness from consensus. The preparation cost is not identified with the entropy deficit alone. T-DOME III analyzes the remaining common-sign ambiguity [21]; Foundation II distinguishes a shared state or instrument from an independently checked source [7]. Here the reference permission and its finite error are part of one explicit record protocol.

5.3 Many readers: error exponents

Proposition 5.3 (Joint, pooled and majority readout)**.** *Let R disjoint fragments each have overlap $r \in (0,1)$, and let the prior be $(p, 1-p)$.*

1. *Any measurement on the union of the fragments errs with probability at least $P_e(r^R) = p(1-p)r^{2R} + O(r^{4R})$, and a joint measurement attains this value.*

2. *Let each reader measure $\{|e_0^{F_j}\rangle\langle e_0^{F_j}|,\ \mathbf{1} - |e_0^{F_j}\rangle\langle e_0^{F_j}|\}$ on its own fragment, and decide $\hat{x} = 1$ if at least one reader obtains the second outcome. This pooled local rule errs with probability $(1-p)\, r^{2R}$.*

3. *Let $p = 1/2$, let each reader use its single-fragment Helstrom measurement, and let the decision be the majority of the R reports, with ties broken by a fair coin. The error probability is $r^{R+o(R)}$, and no rule that uses only the R reports has a better exponent.*

Proof. (1) The conditional states of the union are pure with overlap r^R, and the Helstrom bound is attained by the projective measurement onto the positive part of $p|e_0\rangle\langle e_0| - (1-p)|e_1\rangle\langle e_1|$ [15]. (2) Given $x = 0$, every reader obtains the first outcome with certainty. Given $x = 1$, reader j obtains it with probability $|\langle e_0^{F_j}|e_1^{F_j}\rangle|^2 = r^2$, independently by

Proposition 5.1, so all R readers obtain it with probability r^{2R}. (3) Each report is wrong with probability $\varepsilon = \varepsilon(r)$ independently. With equal priors and symmetric errors the likelihood ratio of the reports depends only on the number of reports for each value, so the majority rule is optimal among rules that use the reports. Its error is $\Pr(\mathrm{Bin}(R,\varepsilon) > R/2)$ plus half the probability of a tie, which equals $\exp\big[-R\,D(\frac{1}{2}\|\varepsilon) + O(\ln R)\big]$ with $D(\frac{1}{2}\|\varepsilon) = -\ln\big(2\sqrt{\varepsilon(1-\varepsilon)}\big)$ by the standard large-deviation estimate for binomial tails. Finally $4\varepsilon(1-\varepsilon) = 1-(1-r^2) = r^2$. □

Local readers who only pool their results afterwards thus reach the joint error exponent $2|\ln r|$ per fragment, with a prefactor larger by $1/p$. At $p = 1/2$ and $r = 0.8$, twenty fragments give a joint error of 3.32×10^{-5}, a pooled one-sided error of 6.65×10^{-5}, and a majority error of 1.58×10^{-3}. Readers who are allowed to choose each measurement after learning the earlier outcomes can attain the joint Helstrom error exactly for pure conditional states [17].

The pooled rule requires a matched implementation of the projector onto $|e_0^{F_j}\rangle$. Knowledge of the coupling is one way to supply it; a trusted prepared reference or a previously calibrated fixed reader is another. An incorrect coupling model generally sets up the wrong test. When that relation is unknown, calibration records can change the actual readout permission. T-DOME I computes the value of a readout model in a specified thermodynamic task [6]; the present exponent does not require every successful fixed reader to carry an explicit self-model.

6 Conditions and interfaces

6.1 Preparation and reading permissions

The carrier, pointer basis, collision order, preparation maps and fragment partition enter explicitly. The pure-memory model gives a noncommuting two-state permission lens; the mixed-memory model gives a commuting parallelogram and readable records without complete purity. Both use independent fresh ports. Retained ports must instead be evolved with their actual old correlations, as Section 4.7 demonstrates.

Reader permissions also matter. A one-sided pure-state test requires access to its conditional state vector. The orientation protocol requires an independently anchored pointer preparation and the same stable response in reference and target trials. Neither capability follows from membership of an effect in a generated algebra. Arbitrary shared errors and preparation–reader co-inversions remain outside that calibration certificate.

The global branching state remains a superposition. Born probabilities determine the label distribution used in the readout laws; no unique global outcome is selected by these calculations. A reader can be any physical device controlling a fragment. The model does not privilege a conscious observer.

Einselection and Paper B's stability analysis address how a given interaction chooses a pointer or protected sector [10, 2]. Paper B's formation constructions add real shared recording ports and Hamiltonian links. Locality and the history of those links must still explain why a particular fragment partition is available. Paper F addresses updating and repeated contact [5]; Q-RAIF C supplies finite acquisition on a carrier retaining quantum history [4]. These are complementary realizations, not permission to replace one model's initial states by another's without a physical bridge.

6.2 Relation to existing approaches

Decoherence and quantum Darwinism. Decoherence theory explains the suppression of interference through entanglement with an environment and identifies pointer observables by their stability [10]. Quantum Darwinism adds that objectivity requires redundant records in many fragments [11], and the exact information content of branching states in spin environments has been analysed in detail [12]. The present paper uses these results and adds the typed distinctions: the permission set of a fragment, the nesting of permission, stability and redundancy in one dynamics, and the difference between Holevo and readable records.

Spectrum broadcast structures. The conditional product form used in Proposition 5.1 is one half of a spectrum broadcast structure, the class of states in which many observers can find the same value independently [13]. The other half, perfect distinguishability of the conditional fragment states, holds here only in the limit $r_F \to 0$. Proposition 5.3 measures how fast readers approach that limit as fragments are added.

QBism and relational quantum mechanics. QBism reads quantum states as an agent's degrees of belief [18], and relational quantum mechanics takes states to be relative to a reference system [19]. Here accessibility is fixed by the interaction structure of a context, and a single objective global state is assumed. The emphasis on relations is shared with relational quantum mechanics; the stable records computed here are candidates for the relata that its relations require.

Algebraic quantum field theory. AQFT treats observable algebras as primary and states as functionals on them [8]. The profile of Definition 2.4 adds typed selection data to that setting. Its entropic statements are finite-dimensional, and no Type III version of the redundancy counts is claimed.

6.3 The actual record interface

Paper A uses these pure fragments as actual readers of its noisy GHZ source [1]. Their visibility $v_i = \sqrt{1 - r_i^2}$ gives the collision-derived law

$$\mathbb{E}(Y_i Y_j) = v_i v_j c_{ij}, \qquad c_{ij} = (1 - 2p_i)(1 - 2p_j).$$

The source sites are correlated; no independence of their answers is assumed. Calibrated finite samples then certify non-isometric shortest-path metrics for two equal-flip-budget noise allocations. This supplies the physical interface a dictionary statement alone lacks.

Paper B's block-constant recorder depends only on sector weights, under its specified commuting noise and recording permissions [2]. Its protected content, fixed algebra and finite channel range differ from this fragment-effect set. The mixed-memory family sharpens the distinction: one unchanged dephasing channel has permissions ranging from trivial to perfect. Paper D and Q-RAIF C add record-controlled physical sources and retained-port acquisition [3, 4]; they grant neither free sharp effects nor fresh inputs in place of spent ports.

The measured record law is also the output consumed by Paper F and T-DOME I's joint memory accounts and Foundation II's distinction between agreement and correctness [5, 6, 7]. Its entropy, preparation cost and action value retain their different resource

conditions. Shared geometry further requires source preparation, site labels, calibrated readers and a reconstruction rule; the finite orientation certificate here supplies one of those permissions.

7 Conclusion

Measurement permissions, stable content and public records are different physical data. In the fresh pure-memory model, the permission lens gives the sharp-test error, while the fixed pointer algebra gives persistence and fragment discrimination gives readable redundancy. Holevo and single-measurement thresholds differ. Pooled local tests attain the joint pure-state error exponent when their readers know the conditional fragment state.

The mixed-memory construction shows why preparation must be accounted for without demanding purity. Its actual readout saturates a purity-deficit budget, its exact permissions form a parallelogram, and its source channel remains complete dephasing even when no record is produced. Reusing the same port reverses its record rather than supplying another independent sample.

Public correctness additionally needs a physical standard. Conditional independence suppresses private common errors, but agreement cannot discover a shared orientation inversion. Prepared-pointer references calibrate that inversion with a finite failure probability; this probability remains in the public error bound and its resources remain in the protocol. Together with Paper A's actual record-to-geometry construction, these results identify a finite operational route from interaction to readable shared structure. They leave outcome selection, elementary factorization and general spacetime reconstruction as independent explanatory questions.

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Records, Dissipation and Intervention: Conditions for a Temporal Direction

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Abstract

Records, entropy production and intervention need not point in the same direction. We bound disjoint fragments' Holevo information by the purity supplied at a product boundary, bath work and a possible system contribution. Fair pointer copying into independent partly mixed ports saturates the pointer-controlled bound: each port's readable information equals its initial purity deficit. The same controlled operations yield an exact joint ledger, a posterior for action and, for one port, a transfer of its purity into the source without a bath. Retained correlations provide another resource: a conditional bound prices new ports relative to an old reference even when their marginal purity is zero. Fresh ports suppress coherence and accumulate records along a CP-divisible source dynamics. Reusing a port can instead return correlations and restore coherence. Reset, reversible relabelling and reversal of the complete recording history have different balances. For a bank degraded by independent flips, restoration of the complete retained state removes a specified joint entropy increment. Label-controlled thermal swaps realize the correction for nonzero target error and approach its entropy price; a declared thermal chiral exhaust adds a conditional finite-rate heat bound. Time, coupling order, gates and preparation are inputs. The results explain how records support later intervention and what must be restored for that capacity to continue.

1 Introduction

1.1 The problem

The fundamental dynamical laws are invariant under time reversal, or nearly so, while records, memories and the reach of actions are not: we have records of the past and can influence only the future. Since Boltzmann the standard explanation has been a special initial condition, a low-entropy past, together with the statistical tendency of entropy to grow [6, 7, 8, 9]. The explanation leaves several questions open. What exactly must be special about the initial condition for records to form? Is the growth of records the same arrow as the growth of entropy, or as the dissipation of a system's free energy? Why can an intervention affect later records and not earlier ones, and what would it take to undo a record?

The paper answers these questions for record-forming processes in which records are well defined. We do not derive a time parameter. The dynamics is given with an external parameter and an order of interactions. What the paper derives is a set of quantitative relations among four directions that are often treated as one: the growth of records, the growth of entropy, the dissipation of free energy, and the reach of interventions. It shows when they coincide, when they separate, and what each costs.

1.2 Results

Section 3 recalls the record model of Paper E [3], in which fresh memories collide once with a pointer, and the two counts of records it supports. The results are:

- a purity law valid for every record-forming process that starts from a product of system, memories and thermal bath: the Holevo information of disjoint fragments about a pointer label is at most the initial purity of the memories, plus the work not stored in the system in units of k_BT, plus the difference between the label entropy and the initial system entropy (Theorem 4.1). The records version of a low-entropy past is a stock of purity or free energy, and the bound is attained;

- in the collision model, an exponential bound on the system's coherence in terms of the number of records and the corresponding loss of free energy, which disappears when the system starts diagonal (Proposition 5.1); record growth along a CP-divisible reduced dynamics (Proposition 5.2); and a gap between local and global entropy that grows linearly with the number of records (Proposition 5.3);

- thermal reset prices: at least k_BT times local entropy for separate fragment erasers in the pure-reset limit, a joint-reset infimum at most k_BT times label entropy, and zero net endpoint work for complete dynamical reversal (Proposition 6.1);

- partly mixed ports that attain the pointer-controlled purity bound, an exact acquisition–readout–action ledger and a conditional record budget with a retained reference (Propositions 4.3 and 4.4);

- an exact intervention order: operations on the system cannot change memories that no longer interact, and change records formed later (Proposition 7.2); repeated interaction breaks the alignment (Proposition 7.3).

- a separate price for renewing a noisy record bank: a complete cyclic correction removes $R[h(q) - h(\varepsilon)]$ nats, while a protected old record can require no renewal. A thermal chiral exhaust gives a conditional finite-rate capacity for such a bank (Propositions 8.1 and 8.2).

Section 9 collects the comparisons, including the two models of T-DOME I in which correlations return to a system without records or no correlations return at all [5]. Section 10 relates the results to the thermodynamic arrow, the past hypothesis and timeless formulations, and Section 10 places the paper in the HAFF programme.

2 Time in physical theory

In Newtonian mechanics and in the Schrödinger equation time is an external parameter, and the equations have no intrinsic arrow. In canonical quantum gravity the Wheeler–DeWitt equation $\hat{H}|\Psi\rangle = 0$ contains no time parameter [11]. The Page–Wootters construction recovers an effective time from correlations between a clock subsystem and the rest of a stationary universal state [12], but it explains an ordering rather than its asymmetry. The thermodynamic arrow supplies the asymmetry through a coarse-graining and a low-entropy boundary condition [9, 10].

The model below uses an external parameter and an externally given order of interactions. It asks how records become asymmetric relative to a boundary condition, and it states that boundary condition as a resource. It is not an input-free derivation of time.

3 Records and how to count them

3.1 The record model

The system S is a qubit with pointer projections P_x, $x \in \{0, 1\}$, and initial state $\sqrt{p}\,|0\rangle + \sqrt{1-p}\,|1\rangle$. Memory qubits $E_1, \ldots, E_n$ start in $|0\rangle$, and the k-th collides once with S through $U_k = P_0 \otimes \mathbf{1} + P_1 \otimes R_k$, where R_k is a real rotation with $\langle 0|R_k|0\rangle = c_k \in [0, 1]$. After n collisions

$$|\Psi_n\rangle = \sqrt{p}\,|0\rangle|e_0\rangle + \sqrt{1-p}\,|1\rangle|e_1\rangle, \qquad |e_x\rangle = \bigotimes_k |e_{x,k}\rangle, \qquad \langle e_{0,k}|e_{1,k}\rangle = c_k. \tag{1}$$

For a fragment F put $r_F = \prod_{k\in F} c_k$ and

$$H_p(r) = h\left(\frac{1 + \sqrt{1 - 4p(1-p)(1-r^2)}}{2}\right), \qquad h(x) = -x\ln x - (1-x)\ln(1-x).$$

Paper E shows that $\chi(X : E_F) = H_p(r_F)$, that the coherence of S is $\sqrt{p(1-p)}\,r_{[n]}$, and that $S(\rho_S) = S(\rho_{E_{[n]}}) = H_p(r_{[n]})$ [3]. Here X is the value a pointer measurement on S returns. Entropies are in nats.

This pure-port model will be compared with the partly mixed controlled-X model below. The inputs are the pointer basis, the split into system and memories, the fragment partition, the order of collisions with its external parameter, and the preparation of the memories. The collisions can be implemented by couplings $g_k(t)P_1 \otimes \sigma_y$ that vanish at the start and end of each pulse and commute with a pointer Hamiltonian $H_S = \varepsilon P_1$; with degenerate memory Hamiltonians the recording unitary has zero net endpoint work. Preparation of the ports, calibrated pulse control and later bank renewal are separate resources. T-DOME I gives the full thermodynamic account of this protocol [5].

3.2 Redundancy and protocol-relative comparisons

Definition 3.1 (Redundancy index)**.** *For a classical label X, a classical–quantum state ρ_{XE} and a declared family of disjoint fragments $\{E_k\}$, define*

$$\mathcal{R}_\delta^{\chi} = \#\big\{k : \chi(X : E_k) \geq (1-\delta)H(X)\big\}, \qquad \mathcal{R}_\delta^{\mathrm{acc}} = \#\big\{k : I_{\mathrm{acc}}(X : E_k) \geq (1-\delta)H(X)\big\}.$$

The first counts Holevo records and the second records that a permitted single measurement can read; $\mathcal{R}_\delta^{\mathrm{acc}} \le \mathcal{R}_\delta^{\chi}$, and Paper E gives closed-form fragment lengths for both in the collision model. With $p = 1/2$, $c_k = 0.8$ and $\delta = 0.1$, blocks of five memories are Holevo records and blocks of seven are readable, so after n collisions $\mathcal{R}_\delta^{\chi} = \lfloor n/5 \rfloor$ and $\mathcal{R}_\delta^{\mathrm{acc}} = \lfloor n/7 \rfloor$. High redundancy supports public readability of the label; it does not select one global outcome. Here the quoted pure-port counts permit arbitrary fragment POVMs, as in Paper E. A restricted readout menu needs its own accessible-information optimization; ideal algebra closure does not grant those measurements.

Definition 3.2 (Protocol-relative comparisons)**.** *Fix a dynamics, a fragment family and a label X, and let α, β be configurations along the resulting history. Write $\alpha \preceq_{\mathcal{R}} \beta$ if $\mathcal{R}_\delta(\rho_\alpha) \le \mathcal{R}_\delta(\rho_\beta)$, and $\alpha \preceq_{\mathrm{inc}} \beta$ if the algebra of memories that have interacted at α is contained in that at β. Each relation is a preorder on its declared domain.*

In the collision model both comparisons follow the collision order. They need not agree in general: decoherence may increase the number of records while the coherence-sensitive effects of the system, whose expectation values scale with $r_{[n]}$, become unreadable on S itself.

3.3 Temporal inputs

The construction uses temporal orientation at several points: the parameter of the dynamics, the order of collisions, the comparison of records at different parameter values, and a boundary condition of fresh memories at one end of the history. The comparisons above therefore compress an already timed process. The aim of the following sections is to identify what that boundary condition consists of, what record growth costs and implies, and which other directions follow it.

4 The purity of the past

Bath balances below use finite entropies, relative entropies and mean energy changes. An infinite reservoir requires a controlled limit of finite reservoirs preserving those balances, not subtraction of divergent global entropies.

Theorem 4.1 (Purity law for records)**.** *Let a system S, memories $E_1, \dots, E_n$ of dimensions d_k with degenerate Hamiltonians, and a bath B with Hamiltonian H_B start in the product state $\rho_S \otimes \bigotimes_k \sigma_k \otimes \tau_B$, where τ_B is the Gibbs state of H_B at inverse temperature β. Let them evolve under a time-dependent Hamiltonian whose interaction terms vanish at the initial and final times and whose bath and memory terms are then H_B and the given degenerate Hamiltonians, let W be the work done on the whole and ΔE_S the change of the energy of S. Let X be the outcome of a final measurement of a PVM on S with rank-one projections, and let $F_1, \dots, F_R$ be disjoint sets of memories. Then*

$$\sum_{j=1}^{R} \chi(X : E_{F_j}) \le \sum_{k=1}^{n} \big[\ln d_k - S(\sigma_k)\big] + \beta\,(W - \Delta E_S) + \big[H(X) - S(\rho_S)\big]. \tag{2}$$

Without a bath the middle term is absent, and the bound holds for an arbitrary joint unitary on S and the memories, with no condition on their Hamiltonians.

Proof. Let ρ be the final state, and $\rho' = \sum_x (P_x \otimes \mathbf{1})\rho(P_x \otimes \mathbf{1})$ its pinching in the pointer basis. Pinching does not decrease entropy and unitary evolution preserves it, so $S(\rho') \geq S(\rho_S) + \sum_k S(\sigma_k) + S(\tau_B)$. Since the P_x have rank one, $S(\rho') = H(X) + S(E_{[n]}B \mid X)$, where the conditional entropy refers to the classical–quantum state of X, the memories and the bath. Write G for the memories outside $\bigcup_j F_j$. Subadditivity of conditional entropy gives

$$S(E_{[n]}B \mid X) \leq \sum_j S(E_{F_j} \mid X) + S(G \mid X) + S(B \mid X) \leq \sum_j S(E_{F_j} \mid X) + \ln d_G + S(\rho_B),$$

with ρ_B the final state of the bath. Nonnegativity of $D(\rho_B \| \tau_B)$ gives $S(\rho_B) \leq S(\tau_B) + \beta \Delta E_B$. Hence

$$\sum_j S(E_{F_j} \mid X) \geq S(\rho_S) + \sum_k S(\sigma_k) - H(X) - \ln d_G - \beta \Delta E_B.$$

Since $\chi(X : E_{F_j}) = S(E_{F_j}) - S(E_{F_j} \mid X) \leq \ln d_{F_j} - S(E_{F_j} \mid X)$, summing over j gives (2) with $\beta \Delta E_B$ in the middle. The memory Hamiltonians are degenerate and the interactions vanish at the endpoints, so the first law gives $\Delta E_B = W - \Delta E_S$. Without a bath the same argument applies with B omitted. □

The bound is attained without a bath. If S starts in $|0\rangle$, a single memory starts maximally mixed, and a controlled-NOT with the memory as control and S as target is applied, then the memories have no purity, $H(X) - S(\rho_S) = \ln 2$, and $\chi(X : E_1) = \ln 2$. In that example the label is created by copying the memory's randomness into a pure system, which is why the system can contribute one record's worth. For a bath-free pointer-controlled unitary, $U = \sum_x P_x \otimes V_x$, the system term is absent: the conditional states of the memories are unitary images of their initial state, and the argument of the proof gives $\sum_j \chi(X : E_{F_j}) \leq \sum_k [\ln d_k - S(\sigma_k)]$. If in addition each V_x is a product over memories, as in the collision model, each fragment is bounded separately by its own purity, $\chi(X : E_F) \leq \sum_{k \in F} [\ln d_k - S(\sigma_k)]$ (Paper E's purity-deficit bound). Without the product form the per-fragment bound can fail: with E_1 maximally mixed, E_2 pure, $p = \frac{1}{2}$ and V_1 the swap of E_1 and E_2, the fragment $\{E_1\}$ holds 0.216 nats about X.

Corollary 4.2 (Purity budget of a record history)**.** *If R disjoint fragments are Holevo records at level δ, then*

$$R(1-\delta)H(X) \leq \beta \sum_k \big[F(\sigma_k) - F(\mathbf{1}/d_k)\big] + \beta\,(W - \Delta E_S) + H(X) - S(\rho_S),$$

where F is the nonequilibrium free energy at temperature T. In particular, memories that start in thermal equilibrium can hold R records only if the work not stored in the system satisfies $\beta(W - \Delta E_S) \geq R(1-\delta)H(X) - [H(X) - S(\rho_S)]$. In a pointer-controlled process without a bath, $R(1-\delta)H(X) \leq \sum_k [\ln d_k - S(\sigma_k)]$.

Proof. Each record contributes at least $(1-\delta)H(X)$ to the left side of (2). For a degenerate Hamiltonian the thermal state of a d-dimensional memory is $\mathbf{1}/d$, and $\beta[F(\sigma) - F(\mathbf{1}/d)] = D(\sigma \| \mathbf{1}/d) = \ln d - S(\sigma)$. The pointer-controlled statement is the summed bound stated after Theorem 4.1. □

The acquisition boundary is thus a quantitative resource, not a requirement that every memory be pure. Stable old records can also serve as standards across time, provided their couplings and orientation connect the old and present instruments [15]. The bound counts simultaneously present information in declared disjoint fragments, not all records that could ever be made, read, erased and remade. With product maximally mixed ports, no bath and pointer-controlled recording, it gives zero; an old correlation can change that boundary, as the conditional bound below shows. The theorem specifies what the acquisition history must supply. It does not explain the primitive origin of that stock.

4.1 Partly mixed ports: acquisition, reading and action

Let a fair classical pointer X be held in S, initially $I/2$. Each of R fresh ports starts independently in $\eta_\alpha = (I+\alpha Z)/2$, with $0 \le \alpha \le 1$. Put $\varepsilon = (1-\alpha)/2$ and $i_\alpha = \ln 2 - h(\varepsilon)$. Apply controlled X from S to each port, using the same physical mixed-port coupling as Paper E. The resulting state is diagonal:

$$P(x, \boldsymbol{y}) = \tfrac{1}{2} \prod_{j=1}^{R} (1-\varepsilon)^{\mathbf{1}_{y_j = x}} \varepsilon^{\mathbf{1}_{y_j \neq x}}. \tag{3}$$

An actual Z reading obtains Y_j. The stated calibration fixes which port value labels which pointer; a common unknown reversal would not be removed by collecting more ports.

Proposition 4.3 (A saturated acquisition and action ledger)**.** *Every port has Holevo and accessible information i_α, equal to its initial purity deficit. Write $I_R = I(X : \boldsymbol{Y})$. Then*

$$\begin{aligned} S(SE) &= \ln 2 + Rh(\varepsilon), & S(E \mid X) &= Rh(\varepsilon), \\ S(E_j) &= \ln 2, & S(E) &= Rh(\varepsilon) + I_R, \\ \sum_j S(E_j) - S(E) &= Ri_\alpha - I_R, & H(X \mid \boldsymbol{Y}) &= \ln 2 - I_R. \end{aligned} \tag{4}$$

For $R = 1$, feedback controlled by the actual port, $D = X \oplus Y$, leaves the source in η_α and the port in $I/2$, independently. For degenerate source and port Hamiltonians, the source free energy increases by $k_B T i_\alpha$, at zero net endpoint work for the ideal recording and feedback unitaries.

Proof. Conditional ports have entropy $h(\varepsilon)$, while their fair-label averages are $I/2$. Their states commute, so Z reading attains Holevo information. Thus $\sum_j \chi = Ri_\alpha$ saturates the pointer-controlled bound for any number of independent ports, including partly mixed ones. Conditional product structure gives $S(SE)$ and $S(E \mid X)$; the definition of I_R gives $S(E)$. For one port, $Y = X \oplus N$, where N is independent with $\Pr(N = 1) = \varepsilon$. Fair X makes Y independent of N; feedback gives $D = N$. Both gates are basis permutations and preserve the complete joint entropy. The source entropy falls from $\ln 2$ to $h(\varepsilon)$, giving the free-energy increment. □

The posterior is available from the actual record, not just from the algebra it generates. For $k = \sum_j y_j$,

$$\Pr(X = 1 \mid \boldsymbol{y}) = \frac{(1-\varepsilon)^k \varepsilon^{R-k}}{(1-\varepsilon)^k \varepsilon^{R-k} + (1-\varepsilon)^{R-k} \varepsilon^k}, \tag{5}$$

with the deterministic limits interpreted on their support. A reversible gate controlled by $\boldsymbol{y}$ can send X to $X \oplus g(\boldsymbol{y})$. Majority is the minimum-error choice here; ties may be resolved by a fixed label. All original ports remain. Computing and retaining an extra classical copy or a decision register would require its own physical stock, so the ideal implementation uses the ports themselves as controls.

The one-port result locates the resource after action: initial memory purity has moved into the source, not disappeared into an unpriced bath. It supplies free energy for a subsequent task, not work already delivered to a work store. T-DOME I implements such a task and accounts for its reset. Neither the source nor the bank has completed a thermodynamic cycle. Paper C's blank-body transfer is another use of finite stock, with different dynamics; its resource account must be retained when the two mechanisms are composed.

4.2 Old references change the acquisition boundary

Proposition 4.4 (Conditional budget with a retained reference)**.** *Let X be classical, K a retained reference, and $E = E_1 \otimes \cdots \otimes E_R$, with $\dim E_j = d_j$. The initial state may correlate all of them. Recording applies $\sum_x |x\rangle\langle x| \otimes I_K \otimes V_x$, with V_x acting on E only. Then*

$$\sum_j I(X : E_j \mid K)_{\text{out}} \leq \sum_j \ln d_j - S(E \mid XK)_{\text{in}}. \tag{6}$$

For independent fresh product ports the right side is their summed purity deficit. This bounds final conditional information, not only its increment during recording.

Proof. The controlled unitary preserves $S(E \mid XK)$. Strong subadditivity gives $\sum_j S(E_j \mid XK)_{\text{out}} \geq S(E \mid XK)_{\text{out}}$. Also $S(E_j \mid K) \leq S(E_j) \leq \ln d_j$. Subtracting these inequalities proves the claim. □

Take independent fair bits X, L, an old reference $K = L$ and a new port $E = L$. Each port is marginally maximally mixed. Controlled X writes $Y = L \oplus X$ into E; E alone carries no information about X, but reading K, E recovers it exactly. Here $S(E \mid XK)_{\text{in}} = 0$ and $I(X : E \mid K)_{\text{out}} = \ln 2$, attaining Eq. (6). The usable stock is the old correlation and permission to read the reference jointly, not a hidden pure marginal. Creating and retaining the two correlated copies of L belongs to the earlier history's account.

For classical readable K, the conditional information is the Holevo quantity averaged over its records, and in this commuting example it is actually attained. A quantum reference need not admit that interpretation: the entropy inequality remains valid, but its conditional mutual information is not by itself a measurement permission or accessible information. This distinction connects acquisition to T-DOME I's retained side information without identifying unlike tasks.

5 Records and dissipation

Proposition 5.1 (Records bound the coherence that survives)**.** *In the collision model with $0 < p < 1$, let $0 < \delta < 1$, let q_δ satisfy $h(q_\delta) = (1-\delta)h(p)$ with $q_\delta \leq 1/2$, and put*

$$a_\chi = \frac{(1-2q_\delta)^2 - (2p-1)^2}{4p(1-p)} \in (0,1).$$

If R disjoint fragments, of any sizes and any collision overlaps c_k, are Holevo records, then

$$r_{[n]}^2 \le a_\chi^R, \qquad |(\rho_S)_{01}| \le \sqrt{p(1-p)}\, a_\chi^{R/2}.$$

If the pointer Hamiltonian is diagonal in the pointer basis, the system's nonequilibrium free energy falls by

$$F_S(0) - F_S(n) = k_B T\, H_p(r_{[n]}) \ge k_B T\, H_p\big(a_\chi^{R/2}\big).$$

If instead S starts in $pP_0 + (1-p)P_1$, every record quantity is unchanged and $F_S(n) = F_S(0)$.

Proof. A fragment is a Holevo record exactly when $r_F^2 \le a_\chi$ (Paper E's Holevo fragment criterion). The overlaps multiply over disjoint fragments, and the memories outside them contribute factors at most 1, which gives the first two bounds. The collisions preserve the pointer populations, so the energy of S is unchanged and $F_S(0) - F_S(n) = k_B T[S(\rho_S(n)) - S(\rho_S(0))] = k_B T\, H_p(r_{[n]})$; H_p decreases in r. For the diagonal initial state the conditional fragment states are the same, so all $\chi(X : E_F)$ are unchanged, and ρ_S does not change at all. □

Records and dissipation in the system therefore coincide only when the system has coherence to lose. As R grows, $H_p(a_\chi^{R/2}) \to h(p)$ and the system loses all the free energy carried by its coherence. A system that is already diagonal acquires the same records at no cost to itself. In the pure-port model, the marginal purity of written memories is transferred into correlations. The complete unitary preserves joint entropy; the mixed-port ledger makes that transfer explicit.

Proposition 5.2 (Records along a CP-divisible dynamics)**.** *The reduced dynamics of S after n collisions is $\mathcal{D}_n \circ \cdots \circ \mathcal{D}_1$, where $\mathcal{D}_k$ keeps the populations and multiplies the coherence by c_k. Every intermediate map $\mathcal{D}_m \circ \cdots \circ \mathcal{D}_{l+1}$ is completely positive and trace preserving, and for equal overlaps $c \in (0,1)$ applied at intervals Δt the family embeds in the dephasing semigroup generated by $\mathcal{L}(\rho) = \frac{\gamma}{2}(Z\rho Z - \rho)$ with $\gamma = -\ln c/\Delta t$. Along this dynamics $\mathcal{R}_\delta^\chi$ and $\mathcal{R}_\delta^{\mathrm{acc}}$ grow without bound in n.*

Proof. Each $\mathcal{D}_k$ is a dephasing channel, and compositions of channels are channels. The semigroup multiplies the coherence by $e^{-\gamma t}$, which equals c^m at $t = m\Delta t$. The record counts are $\lfloor n/m_\chi \rfloor$ and $\lfloor n/m_{\mathrm{op}} \rfloor$ for equal overlaps. Unbounded record counts refer to a family with growing fresh-port supply, not an unbounded yield from a fixed finite bank. □

Record growth thus does not require memory effects in the reduced dynamics of the system. The memories, once written, have a mixture-of-product marginal correlated through X and are independent conditional on a pointer reading; a Markovian system dynamics says nothing about whether the environment's outputs are correlated. The converse separation also holds. T-DOME I gives an autonomous two-qubit model in which correlation built between a system and an environment qubit returns to the system at a recurrence time while the environment holds no information at all about the pointer, and it shows that the pure-dephasing spin–boson model with a Lorentz–Drude spectral density has no backflow at any temperature [5]. Backflow and records are logically independent.

Proposition 5.3 (Local and global entropy)**.** *For disjoint fragments $F_1, \dots, F_R$ of the collision model,*

$$\sum_{j=1}^{R} S(\rho_{F_j}) = \sum_j \chi(X : E_{F_j}), \qquad S\big(\rho_{\cup_j F_j}\big) \le h(p).$$

If the F_j are Holevo records, the total correlation among them, $\sum_j S(\rho_{F_j}) - S(\rho_{\cup_j F_j})$, is at least $[R(1-\delta)-1]h(p)$.

Proof. The conditional states of every fragment are pure, so $S(\rho_{F_j}) = \chi(X : E_{F_j})$. The union of the fragments has conditional states that are pure as well, so $S(\rho_{\cup F_j}) = H_p(r_{\cup F_j}) \le H_p(0) = h(p)$. Subtracting gives the last claim. □

With $p = 1/2$, $c = 0.8$ and $n = 60$ memories in twelve blocks of five, the local entropy is 7.661 nats while the entropy of all sixty memories is 0.693147 nats, equal to $\ln 2$ to nine digits. A reader with access to single fragments sees entropy grow with every record. The global entropy of the environment saturates at the entropy of the label. Which of the two grows without bound is fixed by the access structure, not by the dynamics.

6 Undoing records

Proposition 6.1 (Local and joint reset)**.** *Let the pure-port model's memories have degenerate Hamiltonians. Each eraser couples the memories it controls to its own independent Gibbs bath at inverse temperature β, with interactions off at endpoints. A reset with final memory entropy tending to zero has:*

1. *for separate fragment erasers,*

$$\liminf W^{\text{local}} \ge k_B T \sum_j S(\rho_{F_j}) = k_B T \sum_j \chi(X : E_{F_j}) \ge R(1-\delta) k_B T h(p)$$

   when those fragments are Holevo records;

2. *for one eraser controlling all memories but not S, reset work infimum $k_B T S(\rho_{E_{[n]}}) = k_B T H_p(r_{[n]})$, approached with vanishing reset error and increasingly large, engineered thermal reservoirs;*

3. *for an agent controlling S and every memory, the inverse $U_n^{-1} \cdots U_1^{-1}$ restores the initial product state and source coherence at zero net endpoint work in the ideal recording model.*

Proof. For one controlled memory M and its initially independent bath, unitarity and subadditivity give $\Delta S_B \ge S(\rho_M) - S(\rho'_M)$. The Gibbs identity gives $\beta W = \beta Q = \Delta S_B + D(\rho'_B \| \tau_B)$, since the degenerate memory energy is unchanged. Hence $\beta W \ge S(\rho_M) - S(\rho'_M)$. Apply this separately, or to the entire bank. Separate baths cannot exploit correlations between fragments; their marginal inputs fix these entropy decreases. The pure-port bank marginal has rank at most two. A unitary acting only on the degenerate bank maps its two eigenvectors to a logical qubit tensored with blank remaining bits. The logical qubit has eigenvalues $(1-z, z)$, $0 \le z \le 1/2$, and entropy $h(z) = H_p(r_{[n]})$. For $z > 0$, thermal swaps through decreasing positive error probabilities to $\epsilon > 0$ approach

$k_B T[h(z)-h(\epsilon)]$, as in Proposition 8.2. Taking $\epsilon \to 0$ gives the joint reset infimum; for $z = 0$ the bank unitary suffices. The same thermal-swap method applies more generally [14]. The inverse recording unitaries have the same unchanged endpoint energy as the forward gates. □

Exact purification of a mixed memory is not attainable with a finite-dimensional, full-rank Gibbs bath at finite energies and temperature. Its input joint rank is $\mathrm{rank}(\rho_M)d_B > d_B$, whereas a pure final marginal would force the joint state to have rank at most d_B. Unitary evolution preserves rank. Thus the work infimum is not an exact finite-bath reset guarantee; finite reset error uses the entropy decrease in the proof, with nonnegative bath relative-entropy and correlation terms.

For $p = 1/2, c = 0.8, n = 60$, the corresponding asymptotic reset bounds are $19.505\, k_B T$ for sixty separate ports, $7.661\, k_B T$ for twelve five-port fragments, $5.368\, k_B T$ for eight seven-port readable fragments, and $0.693\, k_B T$ for the joint bank. The local price reflects the access structure, not a larger amount of distinct label content.

Lemma 6.2 (Reset with an old reference)**.** *Let M, K have zero endpoint Hamiltonians and the bath be thermal and independent of MK. Allow a joint unitary on MKB that leaves the marginal of K unchanged. Then*

$$\beta W = \beta Q \geq S(M \mid K)_{\mathrm{in}} - S(M').$$

In the pure-reset limit the lower bound is $S(M \mid K)_{\mathrm{in}}$.

Proof. The decrease of the retained joint entropy is $S(MK)_{\mathrm{in}} - S(M'K')_{\mathrm{out}}$. Since $S(M'K') \leq S(M') + S(K') = S(M') + S(K)_{\mathrm{in}}$, it is at least $S(M \mid K)_{\mathrm{in}} - S(M')$. The independent-bath argument above applies to the joint retained state. □

The target here preserves the declared reference marginal. If other reference correlations must also be restored, they belong to the retained target and its ledger. Quantum conditional entropy is not a universal attained work price without an allowed protocol. For two classical copies $M = K = L$ of a fair bit, controlled XOR from K resets M exactly at zero endpoint work and leaves K unchanged. Its conditional entropy is zero, although $S(M) = \ln 2$. Permission to use the reference changes the reset task.

Changing a record is not resetting it. A unitary bit flip reverses its labels at zero endpoint work for a degenerate memory and retains all information. Swapping a record into an independent maximally mixed port makes the old memory uninformative at zero endpoint work, but transfers the record into the other port instead of destroying the joint history. Local thermal reset, such transfer, error correction and inversion of the complete source–bank unitary cannot be assigned one erasure bill. Exact history reversal is allowed when all required gates and systems are accessible. Neither its probability nor a suppression of reversed histories follows without an ensemble and measure [16].

7 Intervention and the agent's time order

Definition 7.1 (Protocol-relative causal-influence diagnostic)**.** *Fix a record-forming protocol with external parameter t, labels X_A, X_B, and a family of interventions on X_A. Write $X_A \rightsquigarrow_{\mathsf{P}} X_B$ when (i) both labels have records in the declared fragments, (ii) the X_A records form no later than the X_B records according to t, and (iii) interventions on X_A, with the rest of the protocol held fixed, change the distribution of X_B or its records.*

The diagnostic is not asserted to be transitive, antisymmetric or independent of the intervention family. Redundancy makes an influence readable across fragments; the intervention structure carries the causal content. The next two propositions compute the diagnostic in the collision model.

Proposition 7.2 (Interventions reach only later records)**.** *In the collision model let an arbitrary channel $\mathcal{T}$ act on S between the j-th and $(j+1)$-th collisions.*

1. *The final reduced state of $E_1, \dots, E_j$ is the same as without $\mathcal{T}$.*
2. *If $\mathcal{T}$ is the pointer flip $\rho \mapsto \sigma_x \rho \sigma_x$, the final state is $\sum_x \sqrt{p_x}\,|1-x\rangle|e_x^{\le j}\rangle|e_{1-x}^{>j}\rangle$: the earlier memories record the value x and the later ones record $1-x$.*
3. *If the Kraus operators of $\mathcal{T}$ are diagonal in the pointer basis, the joint distribution of the pointer value and all records is unchanged.*

Proof. (1) After the j-th collision, $\mathcal{T}$ and the later collisions act on $S \otimes E_{j+1} \otimes \cdots \otimes E_n$, and the partial trace over these systems is invariant under any operation on them. (2) σ_x maps the branch $\sqrt{p_x}|x\rangle|e_x^{\le j}\rangle$ to $\sqrt{p_x}|1-x\rangle|e_x^{\le j}\rangle$, and the later collisions then write the value $1-x$. (3) Diagonal Kraus operators commute with the pointer-controlled collisions and with the pointer measurement, and they leave the populations unchanged. □

For a flip after the second of four collisions with overlaps $(0.3, 0.5, 0.2, 0.4)$ and $p = 1/2$, the reduced states of the first two memories and of the last two are each unchanged, while the joint state of the first and third memories changes by 0.437 in its largest matrix element: the flip is visible only in the relation between earlier and later records. In this fresh-port protocol, interventions on the source can alter later records, while detached earlier marginals remain fixed. The flip example makes the change visible in joint relations even when single-fragment marginals do not change.

Proposition 7.3 (Repeated interaction lets a later intervention rewrite an earlier record)**.** *Let a memory collide with S twice through the same rotation with $c = \cos(\pi/4)$, and let $p = 1/2$. After the first collision the memory holds $H_{1/2}(2^{-1/2}) = 0.4165$ nats about the initial pointer value. If nothing happens between the collisions, the second collision makes the record perfect, $\ln 2$ nats. If S is flipped between the collisions, the second collision leaves the memory with no information about the initial pointer value.*

Proof. Write R for the rotation by $\pi/4$, so $R^2|0\rangle = |1\rangle$. Without the flip, the memory ends in $|0\rangle$ if $x = 0$ and in $R^2|0\rangle = |1\rangle$ if $x = 1$. With the flip, it is in $|0\rangle$ after the first collision if $x = 0$ and is then rotated once, ending in $R|0\rangle$; if $x = 1$ it is in $R|0\rangle$ after the first collision and is not rotated again. Both branches end in $R|0\rangle$. □

For the coherent source $|+\rangle$, the same two-qubit protocol also gives the entropy and population ledger:

$$|\Psi_1\rangle = (|0,0\rangle + |1\rangle R|0\rangle)/\sqrt{2}, \quad |\Psi_{2,\text{noflip}}\rangle = (|00\rangle + |11\rangle)/\sqrt{2}, \quad |\Psi_{2,\text{flip}}\rangle = |+\rangle \otimes R|0\rangle.$$

Source populations stay $1/2, 1/2$, and joint entropy stays zero. Put $a = h[(1+2^{-1/2})/2] = 0.4164955$. At first contact, $S(S) = S(E) = a$ and $I(S:E) = 2a$. The no-flip continuation gives $\ln 2, \ln 2, 2\ln 2$. The flip continuation gives $0, 0, 0$: source coherence and both marginal purities return while the earlier pointer information disappears. For a pointer-diagonal source Hamiltonian, energy stays fixed; source free energy first falls by $k_B T a$

and then recovers it on the flip branch. In fact $U(X_S \otimes I)U = X_S \otimes R$, so the echo decouples every source input. Reading or discarding the port after the first contact would change this coherent protocol. The source trace distance of the inputs $|+\rangle, |-\rangle$ falls to $2^{-1/2}$ at first contact and returns to 1 after the echo: this is distinguishability backflow, not merely an entropy change.

The alignment of the intervention order with the record order thus rests on a specific feature of the dynamics: memories that have recorded do not interact with the source again. Under repeated interaction, whether an early memory ends as a perfect record or as no record can be decided by an intervention made after it first recorded.

An agent can therefore date an intervention by its effect on records within the stated one-contact architecture. It cannot change a detached port by acting only on the source. Acting directly on that port can relabel, transfer or reset it, with the different balances above; reconnecting it to the source can also restore coherence. The alignment comes from the actual interaction menu, fresh stock and external order, not from a universal prohibition on changing the past.

8 Renewing records instead of making new ones

A written distinction and a renewable distinction are different resources. The purity law prices acquisition from its declared product boundary. It does not make an old record disappear when fresh blank memories run out. For example, a copied Z label remains unchanged under Z dephasing, although each fair-label record has zero marginal purity. Maintaining it costs no entropy removal in that noise model. Independent bit flips give a different, quantitative renewal problem.

Proposition 8.1 (The entropy price of a cyclic record bank)**.** *Let a classical label X be retained unchanged, and let R degenerate binary registers C_i be conditionally independent copies of X with error $\varepsilon \in [0, 1/2)$. Between corrections each undergoes an independent flip of probability $p \in (0, 1/2]$, whose noise port is inaccessible. The controller starts independent of XC and is unchanged by this noise. A correction returns the complete joint state of X, the bank and its controller to the pre-noise state. Its bath starts thermal and independent, and its interactions vanish at the endpoints. Put*

$$q = p + (1 - 2p)\varepsilon, \qquad d_{p,\varepsilon} = h(q) - h(\varepsilon) > 0. \tag{7}$$

The entropy removed from the retained state is exactly $Rd_{p,\varepsilon}$ per correction, and its bath heat obeys $\beta Q \geq Rd_{p,\varepsilon}$.

If corrections occur at frequency ν and the sole exhaust has the thermal, zero-charge, extensive-current properties of HAFF-D's chiral exhaust [2], the mean heat current obeys

$$\overline{\dot{Q}} \geq k_B T \nu R d_{p,\varepsilon} + \frac{3\hbar}{\pi c_{\mathrm{ex}}} (\nu R d_{p,\varepsilon})^2. \tag{8}$$

Thus an exhaust-power budget $\overline{\dot{Q}} \leq J$ permits at most

$$R \leq \frac{\sqrt{(k_B T)^2 + 4\alpha J} - k_B T}{2\alpha \nu d_{p,\varepsilon}}, \qquad \alpha = \frac{3\hbar}{\pi c_{\mathrm{ex}}}. \tag{9}$$

Here c_{ex} is dimensionless chiral capacity, and $\nu R d_{p,\varepsilon}$ is the dimensionless entropy removed per unit time, in nats per time. The current bound is a long-window extensive average,

not an individual pulse identity. No inference from vanishing budget to vanishing records follows when $d_{p,\varepsilon} = 0$.

Proof. Before noise $S(XC) = H(X) + Rh(\varepsilon)$, and after it $S(XC) = H(X) + Rh(q)$. Cyclicity of the entire retained state prevents a controller or a correlation with the label from absorbing that entropy without charge. For its independent Gibbs bath B, a joint unitary and subadditivity give $\Delta S_B \geq Rd_{p,\varepsilon}$. The Gibbs identity $\beta Q = \Delta S_B + D(\rho'_B \| \tau_B)$ proves the heat bound. The actual net extensive entropy current is at least $\nu R d_{p,\varepsilon}$. Substitution into HAFF-D's increasing finite-rate bound gives Eq. (8); inversion gives Eq. (9). □

Proposition 8.2 (Thermal correction with an accessible label)**.** *Under the renewal proposition's conditional-iid error law, assume* $0 < \varepsilon < q \leq 1/2$*, and allow retained* X *to control reversible XOR gates on each bank register. Supply an independent Gibbs bath qubit with gap* $\Delta_\varepsilon = k_B T \ln[(1-\varepsilon)/\varepsilon]$ *for each register. XOR by* X*, swap with that bath qubit, and XOR by* X *again restore the complete retained* X, C*, controller target. Per register the bath heat and mean driven work are*

$$Q = W = k_B T (q - \varepsilon) \ln[(1-\varepsilon)/\varepsilon]. \tag{10}$$

With $q = t_0 > t_1 > \cdots > t_m = \varepsilon$*, successive swaps with independent Gibbs qubits of gaps* $\Delta_i = k_B T \ln[(1-t_i)/t_i]$ *give*

$$\beta Q = \beta W = \sum_i (t_{i-1} - t_i) \ln[(1-t_i)/t_i] \longrightarrow h(q) - h(\varepsilon) \tag{11}$$

as the partition mesh tends to zero. All stages have finite dimensions and finite gaps for a fixed positive target error. The finite-step excess is exactly

$$\beta Q - [h(q) - h(\varepsilon)] = \sum_i D(\mathrm{Bern}(t_{i-1}) \| \mathrm{Bern}(t_i))$$

per register; the spent baths end independent of the retained target.

Proof. The first XOR converts $C_i = X \oplus N_i$ into the independent error bit N_i of probability q, without changing X. The thermal swap replaces it by an independent error bit of probability ε; the second XOR restores the intended conditional law. The bath output has probability q, so its energy gain is $\Delta_\varepsilon(q - \varepsilon)$. Controllers remain untouched, and conditional independence is restored. For several swaps, the output of bath qubit i has probability t_{i-1}. Summing its energy gain gives Eq. (11). The right-endpoint Riemann sums tend to $\int_\varepsilon^q \ln[(1-t)/t]\, dt = h(q) - h(\varepsilon)$. The degenerate retained registers have no endpoint energy change, so the first law gives $W = Q$. Each swap's heat equals its error-bit entropy decrease plus $D(\mathrm{Bern}(t_{i-1}) \| \mathrm{Bern}(t_i))$. Telescoping gives the stated finite-step excess. All initial error and bath bits are independent after the first XOR, and the swaps merely permute them; hence the spent baths are independent of the final target. □

This is a label-preserving use of the thermal swap construction of Ref. [14]. These swaps are driven gates between degenerate bank bits and nondegenerate bath qubits, not spontaneous energy-conserving collisions. The label's control permission, thermal preparations, clock and switching belong to the implementation. Spent bath qubits retain the exported entropy and are not reset by the construction. It realizes cyclicity of

the declared retained bank, not an autonomous battery-complete apparatus or an indefinitely recycled bath. The vanishing-error limit $\varepsilon \to 0$ also requires growing gaps or an appropriate limiting reservoir. This thermal-qubit implementation does not establish the separate sole-chiral-exhaust conditions or sharpness of its finite-rate bound.

For $p = .12$ and $\varepsilon = .05$, one correction removes .2378232484 nat per register. Eight correlated copies of one fair label require 1.9025859874 nats of renewal, despite carrying only one bit of distinct label content. Their independent degradation, not the number of distinct labels, sets this maintenance bill. The cyclic target and the independent noise are substantive inputs. Jointly correlated errors, an accessible noise record, a noncyclic pure controller or a protected bank can change the cost. The heat-current claim additionally needs the actual exhaust assumptions; it is not a Landauer bound on every control pulse.

This gives the record side of a maintained observer. T-DOME III and Brain Anchor let retained history choose which access to repair [17]; Eq. (9) prices one implementation of the retained bank on which such a choice relies. Foundation's exact fixed algebras describe passive protection, not a requirement to export positive entropy indefinitely. Under HAFF-D's further matched horizon-flux conditions, the bank's removal rate can enter the conditional horizon-entropy bound. Those conditions connect a real renewal process to gravity; record existence alone does not.

9 Which arrows coincide

Table 1 collects the directions studied above in four specified settings: the collision model with a coherent or a diagonal initial system state, the collision model with one repeated collision, and the autonomous two-qubit model of T-DOME I, in which a system qubit and an environment qubit build and return correlation without any pointer record [5].

Direction	collision, coherent S	collision, diagonal S	one repeated collision	two-qubit recurrence
Record count grows	yes	yes	can fall	no records
Local entropy of fragments grows	yes	yes	can fall	constant E
Marginal memory purity	falls	falls	falls, then returns	no fresh bank
System free energy falls	yes	no	falls, then returns	falls, then returns
Interventions reach only later records	yes	yes	no	no records
Local thermal reset bound $\propto R$	yes	yes	no final record	—
Correlation flows back to S	no	no	yes, with flip	yes

Table 1: Specified protocols, not universal arrows. The repeated-contact column uses the coherent, unmeasured flip continuation of Proposition 7.3. Reset prices refer to blank targets, not arbitrary changes of record content.

With pure fresh ports interacting once, record accumulation, local entropy growth and decreasing marginal port purity align; detached marginals also fix which source

interventions can still reach them. Local thermal reset prices that bank relative to a blank target. These relations do not make every record update an entropy-producing operation. The mixed-port ledger shows that acquisition and action redistribute finite stock, while the echo reverses the local entropy and correlation changes. Source free-energy loss depends on initial coherence. Reduced-system backflow, record acquisition and bank renewal remain separate questions.

10 Relations and interfaces

The purity law gives the record boundary of the past hypothesis a quantitative form, while separating record accumulation from a source's loss of free energy [9, 10]. Mixed-environment redundancy [13] is one instance; the theorem also prices stock drawn from work and a bath. Page–Wootters correlations can supply ordering within a stationary state [12]. A direction of recording additionally needs the acquisition boundary and actual contact architecture specified here.

Complete-access reversal is compatible with retrocausal programmes [16]; restricted thermal reset is a different task. The causal-influence diagnostic computes supplied interventions in a physical recording process. It does not turn them into a derived causal partial order.

Paper E supplies the recording instruments [3]. The present posterior, intervention order and reference-dependent budgets are the record-side interfaces to T-DOME I's thermodynamic agent [5]; Paper C implements a separate maintained sensing body [1]. Paper D's heat and horizon bounds apply only under their matched source conditions [2]. The purity law supplies a physical input for Paper G's question of how selection permissions form [4]; it does not derive those permissions, clocks or a geometric algebra.

11 Conclusion

The product-boundary purity law bounds simultaneously present fragment information. Independent partly mixed ports attain the pointer-controlled bound, and their actual readout supplies a posterior for action. The complete ledger tracks where purity and correlations go; a one-port feedback transfers source free energy without already extracting work. A retained reference can supply conditional stock even when a new port is marginally maximally mixed.

Fresh one-contact ports align record acquisition, loss of coherence and restricted source intervention in a particular architecture. Recontact can instead erase the earlier record and recover source coherence. Reversible relabelling, transfer, thermal reset and exact history reversal are distinct operations. Their prices depend on the target, retained references and accessible gates, not on the word *undoing*.

A noisy record bank poses a further task. Independent errors increase its complete retained entropy; label-controlled thermal swaps restore that target for positive error and approach the entropy-removal bound. A specified chiral exhaust additionally limits the correction rate. This gives a nonempty maintenance mechanism without claiming a renewable autonomous universe or the origin of its clocks and couplings. A broader theory must explain how fragment and interaction structures form, how correlated noise changes renewal, and how actual entropy export participates in source and geometric response.

A Joint-state checks

The mixed-port check constructs the complete diagonal density operator and applies the recording CNOTs, not separately chosen conditional laws. For $\alpha = 0, .2, .8, 1$ and $R = 1, 2, 3, 5$, it verifies Eq. (4), the actual Z-record posterior and one-port feedback. The maximally mixed keyed example is an explicit three-bit permutation. Forty full-density examples with a quantum reference and two initially correlated ports test the conditional bound; the entropy proof establishes it generally.

The coherent echo check applies both collisions and the intermediate flip to the same pure state, verifying $UX_SU = X_S \otimes R$, all marginal and joint entropies, and the recovery of coherence. For renewal, an eight-input XOR–SWAP–XOR permutation verifies the full label–bank target. Thermal chains with $1, 4, 16, 64, 256, 1024$ stages verify the heat and its relative-entropy excess. At $p = .12, \varepsilon = .05$, one stage costs $0.31799941\, k_B T$ per register; 1024 stages cost $0.23789030\, k_B T$, approaching $0.23782325\, k_B T$. These checks certify the finite-dimensional specified circuits, not the physical chiral exhaust.

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Internal Selection and Revisable Organization: How History Changes Permissions and Physical Sources

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Abstract

A physical system can use its records to revise the operations through which it reads and acts. We construct this process with an internal program, a fixed executor and a record-driven update. Controlled couplings select protected algebras; overlap of their spectra permits coherent internal choices without a central program label. The same choice changes an outgoing matter source. A qubit loop has two configurations whose records select their own protection, and different initial histories produce late source weights 1/3 and 2/3. With rotation noise these choices persist for a calculable window and eventually lose their contrast. An arbitrary quantum record can also be converted into the two sustained source configurations: the largest achievable source-weight contrast equals the trace distance between the record states. Finite execution errors shorten the history window by their accumulated operational error. A separate price concerns sharp readouts. Selecting every sharp qubit measurement on a great circle to worst-case basis-state error δ requires program dimension of order $\delta^{-1/2}$; pure and mixed programs have the same feasible dimensions for this criterion. These results identify what history can form inside a supplied architecture. They support a broader mechanism in which learned relations become revisable permissions, while leaving primitive couplings and their origin as a distinct generation problem.

1 Introduction

A record can change what a system is able to do next. A damaged body may install a different protective encoding; a reader may move its probe; an internal estimate may lower the authority given to an unreliable response. These changes concern actual operations. To explain them, a theory of selected structure needs a physical account of how a history selects and revises the relations that carry its own activity.

Papers A–F separate parts of this problem. Channels determine which distinctions are read in Paper A; couplings and observation windows determine which survive in Paper B; instruments create records in Paper E; their retention and renewal have resource prices in Paper F [1, 2, 4, 5]. Paper C connects sensing and intervention to a body's viability, while Paper D asks how selected structure and physical sources constrain geometry [25, 3].

Here the question is whether the choice of operation can itself arise from records inside the system.

An operation, a stored description and the device that executes it are different physical objects. We specify an internal program register, a fixed executor and an update instrument. Under these inputs the system selects its couplings, maintains a choice through the records it protects, and changes the matter it emits. The choice need not be a central environmental record. Its origin in a distinguishable history and its effect on a source can be tested separately.

The main construction is a qubit with two record-selected protective configurations. Their history-dependent source weights remain different over a finite noisy window. We then identify the exact information that can pass from a quantum record to this source choice, including a construction that reaches the trace-distance bound. An error bound makes the same finite window robust to imperfect execution. These statements supply a causal interface from history to permission and source; they do not use a reconstructed geometry as a substitute for an actual change of matter.

The cost of selecting operations is separate from the cost of retaining their records. Sharp alternatives require orthogonal program supports. For the one-parameter great-circle qubit family the required register dimension grows as $\delta^{-1/2}$, and mixing cannot improve it under the two basis-state error criterion. Section 6 states this result; Appendix A contains the full proof and numerical bounds. Section 7 places the finite target-register boundary in its proper scope. Section 8 develops the broader formation mechanism and its failure conditions.

Breuer's state self-measurement limit [12] and Wolpert's physical inference limits [13] concern what an embedded system can know. Here we ask what it can choose and execute. The existence of self-reference is not in dispute: self-reproducing automata, recursive programs and fixed-point constructions provide positive examples [14, 21, 22]. What requires explanation is the physical relation between a description, its execution and the history that revises it.

2 A typed model of internal selection

An element of an observable algebra is an operator, not a description of an algebra, and a formula naming a selection rule is syntax until an encoding maps it into physical degrees of freedom. The following model makes that encoding explicit.

Definition 2.1 (Internal selection model)**.** *A finite internal selection model consists of*

1. *a finite-dimensional system S, on which selected operations act;*
2. *a program register P of dimension d_P and program states $|p\rangle \in P$, $p \in \mathcal{P}$, unit vectors that need not be orthogonal;*
3. *an executor, a fixed physical operation on $P \otimes S$, which for each program state determines an operation on S: a channel $\Phi_p(\rho) = \mathrm{Tr}_P\, \mathcal{G}(|p\rangle\langle p| \otimes \rho)$, a measurement E^p obtained by following $\mathcal{G}$ with a fixed measurement, or a program-controlled coupling to an environment;*
4. *records, the classical outcomes of instruments, and an update rule, a fixed instrument that prepares the next program state from records.*

A selection rule is internalized if the choice of operation is made by program states under a fixed executor and a fixed update rule. A configuration (p, ρ) is self-grounding if ρ is left stable by the operation selected by p and the update rule, applied to the records that ρ produces, returns p.

The space of descriptions is the set of program states, the encoding is the register, the evaluation map is the executor, and the admissible fixed points are the self-grounding configurations. The executor's dispatch map, $p \mapsto \Phi_p$, is physical dynamics; it is not itself a program. Whether it can be made one is the question of Section 7.

3 Selecting one's own accessible algebra

Paper B shows that a system coupled to its environment through commuting operators $A_1, \ldots, A_K$ at positive rates γ_k has, as its stable algebra, the commutant $\{A_k\}'$, and that environmental records can carry only the label of the block of that algebra. We now let the system choose its own couplings. Take an orthonormal program basis and program-controlled couplings

$$A_k = \sum_{p \in \mathcal{P}} |p\rangle\langle p| \otimes A_k^{(p)}, \tag{1}$$

where for each program p the operators $A_1^{(p)}, \ldots, A_K^{(p)}$ on S commute. Let $|p, x\rangle$ be the product of $|p\rangle$ with a joint eigenvector of the $A_k^{(p)}$, with eigenvalues $a_k^{(p)}(x)$.

Proposition 3.1 (Program selection and central records)**.** *For the couplings* (1)*:*

1. *they commute, and the algebra fixed by the generated dephasing semigroup is* $\{A_1, \ldots, A_K\}'$*.*

2. *Compressed to the subspace* $|p\rangle \otimes S$ *of a program* p*, this algebra is* $|p\rangle\langle p| \otimes \{A_1^{(p)}, \ldots, A_K^{(p)}\}'$*: each program selects the algebra that Paper B assigns to its own couplings.*

3. *The matrix element between* $|p, x\rangle$ *and* $|q, y\rangle$ *is multiplied by* $\exp[-\frac{t}{2}\sum_k \gamma_k (a_k^{(p)}(x) - a_k^{(q)}(y))^2]$*. Coherence between two programs survives exactly between joint eigenvectors with equal eigenvalue tuples. Disjoint joint spectra for* p *and* q *make every operator connecting those two programs decay. The label projector* $|p\rangle\langle p| \otimes \mathbf{1}_S$ *is central in the stable algebra if and only if the joint spectrum of* p *is disjoint from that of every other program. In that case environmental fragments can record its label.*

Proof. The operators are block diagonal in p and commute within each block, so they commute; the basis $|p, x\rangle$ diagonalizes all of them, with eigenvalues $a_k^{(p)}(x)$. Paper B's proposition on commuting couplings then gives the Schur-multiplier form in (3) and the fixed algebra in (1). Its compression to $|p\rangle \otimes S$ consists of the operators there that commute with the $A_k^{(p)}$, which is (2). A program label is central precisely when it is constant on each joint eigenspace, which gives the disjointness condition in (3). The record statement is Paper B's result that fragments carry only block labels, applied to the joint system. □

When the programs have disjoint joint spectra, their labels belong to the centre of the stable algebra and can become redundant environmental records. Shared eigenvalue tuples protect coherence between the corresponding configurations; the spectra need not coincide completely for this exception to occur. With $A = |0\rangle\langle 0| \otimes Z + |1\rangle\langle 1| \otimes X$ on a program qubit and a system qubit, both programs have eigenvalues ± 1, and the superposition $(|0,0\rangle + |1,+\rangle)/\sqrt{2}$ is an eigenvector of A. Replacing X by $X + 3$ makes the spectra disjoint, and the same superposition decays to an equal mixture of the two selections. Both cases are computed in the accompanying check script.

For more than two programs, disjointness from just one alternative is not enough. With $A = |p\rangle\langle p| \otimes Z + |q\rangle\langle q| \otimes (Z + 3\mathbf{1}) + |r\rangle\langle r| \otimes Z$, the spectra of p and q are disjoint, but $|p,0\rangle\langle r,0|$ commutes with A and not with $|p\rangle\langle p| \otimes \mathbf{1}_S$. Thus p is not a central label. The condition in Proposition 3.1 concerns the whole program family.

Internal selection can also change a material source. Use the field normalization of Paper D, in which a coupling through a projector R obeys $U^\dagger j(u) U = j(u) + R g'(u)/2$. The program controls that projector while the executor, drive and switching schedule stay fixed.

Proposition 3.2 (A program-selected material source)**.** *Let R_p be projectors on S and put*

$$A = \sum_p |p\rangle\langle p| \otimes R_p, \qquad U = \exp(-i\lambda A \otimes Q), \qquad Q = b + b^\dagger.$$

Take an initial state ρ_{PS} independent of the field state σ. Set $r = \mathrm{Tr}(\rho_{PS} A)$. The output field marginal is exactly

$$\sigma_{\mathrm{out}} = (1 - r)\sigma + r\, e^{-i\lambda Q} \sigma e^{i\lambda Q}.$$

When the displayed current moments exist, its vacuum-subtracted stress is

$$T_{\mathrm{out}}(u) = T_{\mathrm{in}}(u) + r \left[g'(u) \langle j(u) \rangle_\sigma + \frac{g'(u)^2}{4} \right].$$

In a fixed classical program branch p, a non-atomic quadrature spectral law and $\lambda \neq 0$ give the exact multiplicative domain $\{R_p\}'$ of the induced system channel. Thus changing the program can change both the represented algebra and the field source under the same executor.

Proof. Since $A^2 = A$, write $U = (1 - A) \otimes 1 + A \otimes e^{-i\lambda Q}$. The off-diagonal terms have zero partial trace over PS, because $\mathrm{Tr}(A \rho_{PS} (1 - A)) = 0$. This gives the field mixture even when the program is coherent or correlated with S. Squaring $j + A g'/2$ gives the stress formula: initial independence supplies the mixed expectation $\langle A j \rangle = r \langle j \rangle$, and $\langle A^2 \rangle = r$. Within branch p, the channel is the two-block dephasing channel of Paper D; a non-atomic spectral law makes its nonzero-gap characteristic value strictly smaller than one, so equality in the Schwarz inequality holds exactly on $\{R_p\}'$. □

For $R_0 = (1 + Z)/2$, $R_1 = (1 + X)/2$ and $\rho_S = (1 + 0.8Z)/2$, a program bit prepared in 0 or 1 gives $r = 0.9$ or 0.5, respectively. Their conditional pointer algebras differ, although the two coupling spectra are identical. The field stress therefore need not reveal the program label as a central sector to respond to its choice. With a classical program mixture π, $r = 0.9\pi_0 + 0.5\pi_1$. A record-driven program update can change this weight without an externally assigned emission probability. The global algebra $\{A\}'$ stays fixed; the selected branch algebra on S and the outgoing source vary. This separates internal

selection from a change of the executor. Paper D supplies the corresponding JT response for this same wavepacket with matched initial coefficient data; the current statement identifies its physical source, not the origin of the JT action or of a Type III geometric sector. The program, field preparation and switching remain resources of the executor.

4 A self-grounding selection

A selection rule can take as input a record that the selection itself protects. A small model gives two self-grounding fixed points and their history-dependent selection.

A memory qubit M is subject to rotation noise $H = \frac{\omega}{2}Y$. A classical program bit p selects a dephasing coupling of M with rate γ: through Z for $p = 0$ and through X for $p = 1$. Strong dephasing in a basis protects a record in that basis against rotation, by the quantum Zeno mechanism [19, 20]. Every time τ the memory is read in the protected basis, and the update rule sets p equal to the outcome. The rule is internalized in the sense of Definition 2.1, and it is self-applied: the record that sets the selection is a record in the algebra that the selection protects.

After a read the configuration is one of four: $F_0 = (p = 0, |0\rangle)$, $T_1 = (p = 1, |1\rangle)$, $F_1 = (p = 1, |-\rangle)$ and $T_0 = (p = 0, |+\rangle)$. In F_0 and F_1 the protected record agrees with the program it sets; these are the self-grounding configurations. In T_1 and T_0 the record has just switched the program to a basis in which that record is not sharp.

Proposition 4.1 (Self-grounding configurations)**.** *Without rotation noise, F_0 and F_1 are absorbing, and from T_1 and T_0*

$$\Pr(\textit{absorbed in } F_1 \mid T_1) = \tfrac{2}{3}, \qquad \Pr(\textit{absorbed in } F_1 \mid T_0) = \tfrac{1}{3}.$$

From any initial program and memory state the loop reaches F_0 or F_1 with probability one.

Proof. A read in the basis that is already sharp returns the same outcome, so F_0 and F_1 repeat. In T_1 the memory $|1\rangle = (|+\rangle - |-\rangle)/\sqrt{2}$ is read in the X basis, giving F_1 or T_0 with probability $\frac{1}{2}$ each; in T_0 the memory $|+\rangle$ is read in the Z basis, giving F_0 or T_1 with probability $\frac{1}{2}$ each. Writing a and b for the absorption probabilities into F_1 from T_1 and T_0, $a = \frac{1}{2} + \frac{1}{2}b$ and $b = \frac{1}{2}a$, so $a = \frac{2}{3}$, $b = \frac{1}{3}$. Every read from a transient configuration ends in an absorbing one with probability $\frac{1}{2}$, and the first read from an arbitrary initial state produces one of the four configurations. □

Corollary 4.2 (History selects a sustained source)**.** *In the zero-rotation loop, insert before each read a fresh field interaction of Proposition 3.2, using $R_0 = (1 + Z)/2$ and $R_1 = (1 + X)/2$. Each cycle uses a distinct tensor-factor port, in the same state and initially independent of the program, memory and all previous ports. The four-state transition law and its absorption probabilities are unchanged. In F_0 the output field is $e^{-i\lambda Q}\sigma e^{i\lambda Q}$; in F_1 it is σ. Thus the two self-grounding configurations carry different steady field sources under one executor. Starting from T_1 or T_0, the late-cycle mean source weight tends to $1/3$ or $2/3$, respectively.*

Proof. Within either program branch the field coupling is diagonal in the basis of the ensuing read. It changes no outcome probabilities, so the transition matrix of Proposition 4.1 is unchanged. The configurations $F_0 = (0, |0\rangle)$ and $F_1 = (1, |-\rangle)$ are eigenvectors of A with eigenvalues 1 and 0. They remain product states of program and memory, and

each pass gives the claimed field output. The source weight therefore converges to the indicator of absorption in F_0. Its expectation is one minus the corresponding probability of absorption in F_1, which gives the stated fractions. □

The read/update history generates the source difference under one executor. Fresh ports and memory operations sustain it. Each cycle supplies a fresh product-state field port, a protected-basis read, the program update and its protecting environment. For a centered field input the coupling's mean switching work is $r_n E_g$, where $E_g = \lambda^2 \bar{\omega} = \int g'^2 du/4$ and r_n is the actual source weight. Noncentered inputs add the displayed current cross term. Incoming beam preparation, memory retention and eventual reset have their own charges. The 1/3 and 2/3 weights arise from absorption of the two transient histories, not a continuing supply of new source information. Distinct modes of a common bosonic field must be orthogonal and prepared in a product state to realize these ports; mere time translation does not suffice. For Paper D's exact packet, a translation by Δ has overlap modulus $4a^2/(4a^2 + \Delta^2)$. Source accounting for such overlapping pulses requires their joint field state. The full-line packet is a scattering idealization; a record decided at a finite time needs a compact switch scheduled after that decision, as constructed in Paper D. With rotation, the following lifetime uses the stated continuous dephasing generator.

The dephasing plays no role without noise, because the reads are made in the protected basis. With noise it decides whether the self-grounding configurations last.

Proposition 4.3 (Lifetime of a self-grounding selection)**.** *With rotation noise, the probability of leaving F_0 (or F_1) in one cycle is $\varepsilon = \frac{1}{2}(1-z(\tau))$, where z solves $\ddot{z}+2\gamma\dot{z}+\omega^2 z = 0$, $z(0) = 1$, $\dot{z}(0) = 0$. For $\gamma > |\omega|$,*

$$z(\tau) = \frac{\lambda_+ e^{\lambda_- \tau} - \lambda_- e^{\lambda_+ \tau}}{\lambda_+ - \lambda_-}, \qquad \lambda_\pm = -\gamma \pm \sqrt{\gamma^2 - \omega^2},$$

so $\varepsilon \approx \omega^2 \tau/(4\gamma)$ when $\gamma \gg |\omega|$, $\gamma\tau \gg 1$ and $\omega^2\tau/\gamma \ll 1$. Without protection $\varepsilon = \sin^2(\omega\tau/2)$; with protection in the wrong basis ε approaches $\frac{1}{2}$ in the strong-dephasing regime.

Proof. In F_0 the Bloch vector starts at $z = 1$. The rotation gives $\dot{x} = \omega z$, $\dot{z} = -\omega x$, and dephasing through Z adds $-2\gamma x$ to $\dot{x}$. Eliminating x gives the stated equation; a Z read returns 1 with probability $\frac{1}{2}(1 - z)$. The case F_1 is the same with x and z exchanged, since rotation about y treats both records alike. For $\gamma \gg \omega$, $\lambda_+ \approx -\omega^2/(2\gamma)$ dominates. Without dephasing $z = \cos\omega\tau$. Dephasing in the wrong basis destroys the record at rate 2γ. □

For $\omega = 0.1$ and $\tau = 1$, the escape probability per cycle is 1.42×10^{-3}, 2.37×10^{-4} and 2.49×10^{-5} for $\gamma = 1$, 10 and 100, against 2.50×10^{-3} without protection. Protection in the wrong basis gives 0.433 at $\gamma = 1$ and approaches 0.5 at the larger rates. At $\gamma = 100$ a self-grounding selection lasts on average 4.0×10^4 cycles before the loop passes through a transient configuration and settles again, in F_0 or F_1.

The residence time alone does not determine how long two histories produce different sources. The same four-state process gives that distinction exactly.

Proposition 4.4 (History-dependent source window)**.** *Let $\gamma > 0$, $\omega \neq 0$ and $\tau > 0$. In each cycle first evolve for time τ with the selected rotation–dephasing generator, then*

apply the fresh field interaction of Corollary 4.2, and then read and update. Put

$$k = \sqrt{\gamma^2 - \omega^2}, \quad f = e^{-\gamma\tau}\left(\cosh k\tau + \gamma\frac{\sinh k\tau}{k}\right), \quad b = \omega e^{-\gamma\tau}\frac{\sinh k\tau}{k},$$

with analytic continuation when $\gamma < |\omega|$ and $\sinh(k\tau)/k = \tau$ at $k = 0$. Set $\varepsilon = (1-f)/2$, $q_+ = (1+b)/2$, $q_- = (1-b)/2$. In the post-read order (F_0, T_1, F_1, T_0) the transition matrix and the source-weight vector are

$$T = \begin{pmatrix} 1-\varepsilon & \varepsilon & 0 & 0 \\ 0 & 0 & q_+ & q_- \\ 0 & 0 & 1-\varepsilon & \varepsilon \\ q_- & q_+ & 0 & 0 \end{pmatrix}, \qquad r = \begin{pmatrix} 1-\varepsilon \\ q_- \\ \varepsilon \\ q_- \end{pmatrix}. \tag{2}$$

For a post-read distribution μ_n, the next field output has mean stress

$$T_{\mathrm{in}}(u) + (\mu_n r)\left[g'(u)\langle j(u)\rangle_\sigma + g'(u)^2/4\right].$$

The chain has the unique stationary law

$$\pi = \frac{(q_-, \varepsilon, q_+, \varepsilon)}{1+2\varepsilon}, \qquad r_\infty = \pi r = \frac{1}{2} - \frac{b}{2(1+2\varepsilon)}.$$

For initial states T_1 and T_0, define

$$R_1^\circ = \varepsilon + f\frac{q_-^2}{1-q_+q_-}, \qquad R_0^\circ = \varepsilon + f\frac{q_-}{1-q_+q_-}.$$

Their actual source weights satisfy, for every integer $n \geq 0$,

$$|\mu_n r - R_i^\circ| \leq n\varepsilon + (q_+q_-)^{\lfloor n/2\rfloor}, \qquad \mu_0 = T_i. \tag{3}$$

Thus a small escape rate preserves different history-selected sources over $1 \ll n \ll \varepsilon^{-1}$, although both histories ultimately have the same stationary mean.

Proof. The two Bloch generators on (x, z) are

$$\begin{pmatrix} -2\gamma & \omega \\ -\omega & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & \omega \\ -\omega & -2\gamma \end{pmatrix}.$$

Exponentiating them gives (2). The source weight is the population of the selected projector immediately before the read, not its value in the preceding post-read state. The intervening field coupling preserves that population. It equals the probability that the next program is 0. For nonzero rotation and positive dephasing, the Bloch norm strictly decreases from each of the four pure initial states over any positive interval. Hence $0 < \varepsilon < 1$ and $0 < q_\pm < 1$. The graph of T is connected and has self-loops, giving uniqueness and convergence. Direct multiplication gives $\pi T = \pi$ and the stated source weight.

For the window bound, replace the two F rows by absorbing rows, retaining $q_\pm$ in the transient rows and retaining the same observable r. In this comparison chain the absorption probabilities into F_0 are $q_-^2/(1-q_+q_-)$ and $q_-/(1-q_+q_-)$; the corresponding limiting expectations of r are R_1° and R_0°. Transient survival after n steps is at most $(q_+q_-)^{\lfloor n/2\rfloor}$. Couple the true and comparison chains until their first escape from an F state. That event has probability at most $n\varepsilon$, and $0 \leq r \leq 1$, proving (3). □

The history window and the lifetime of one configuration are different. The characteristic polynomial is

$$(z-1)\left[z^3+(2\varepsilon-1)z^2+(\varepsilon^2-q_+q_-)z+\varepsilon^2-\varepsilon+q_+q_-\right].$$

For small ε the slow root is $1-\varepsilon/(1-q_+q_-)+O(\varepsilon^2)$, so the late history contrast decays on a scale of about $(1-q_+q_-)/\varepsilon$ cycles. In the strong-protection regime of Proposition 4.3, this is about $3\gamma/(\omega^2\tau)$, whereas a single configuration has mean residence $1/\varepsilon \approx 4\gamma/(\omega^2\tau)$. At $\gamma = 100$, $\omega = 0.1$, $\tau = 1$, the source weights from T_1 and T_0 are respectively 0.33844 and 0.66090 after 1000 cycles, and 0.49370 and 0.50579 after 10^5 cycles. Both approach 0.4997500.

The loop selects a protection because the record it protects says so, and its history selects a material source through the same program-controlled executor. Finite noise makes that selection metastable rather than permanent. The rotation–dephasing stage, field interaction and read are sequential operations here; simultaneous rotation during a finite field pulse would require a different transition law. The protecting environment, fresh packets, read instrument and update map are supplied. The loop generates the history-dependent choice, not these resources.

5 When a history changes the next source

The four-state loop starts with explicit configurations. An earlier history need not already be a classical choice among them. Let R be a retained physical record, and let $\Lambda : R \to P \otimes M$ be a fixed preparation channel acting after that history has formed. It can include a measurement, an outcome record and a controlled preparation. The next fresh field port remains independent of R, P, M and old emitted ports. This time ordering gives a direct criterion for what the history can cause.

Proposition 5.1 (Record-to-source distinguishability)**.** *For two record states ρ_0, ρ_1 and the projector A in Proposition 3.2, the two source weights satisfy*

$$r_0 - r_1 = \mathrm{Tr}[(\rho_0-\rho_1)\Lambda^*(A)], \qquad |r_0-r_1| \le D(\rho_0,\rho_1) := \tfrac{1}{2}\|\rho_0-\rho_1\|_1.$$

For the zero-rotation loop, the largest contrast over fixed preparation channels Λ equals $D(\rho_0,\rho_1)$. A channel attaining it prepares F_0 or F_1 from an actual binary record measurement. Both subsequent histories remain in their chosen configuration, so the same contrast occurs in every later fresh source port.

Proof. The first equality is the adjoint-channel identity. Complete positivity and unitality of Λ^* give $0 \le \Lambda^*(A) \le 1$. Writing the trace-zero Hermitian operator $\rho_0-\rho_1$ as its positive part minus its negative part bounds the expectation of any effect by D. To attain the bound, let H be the spectral projector onto its positive part. Measure $\{H, 1-H\}$ and prepare F_0 on outcome H, F_1 otherwise. The resulting source weight is $\mathrm{Tr}(\rho_i H)$ because F_0, F_1 have A-eigenvalues $1, 0$. Their difference is D. Proposition 4.1 preserves these configurations and Corollary 4.2 preserves their source weights. □

The discrimination identity is a standard trace-distance fact; its role here is to determine exactly how much record information can become a difference in protected configuration and actual source. A nonzero source contrast is possible precisely when the two record states differ, provided this binary measurement and the two preparations are

available. A specified executor can fail to use that difference: if $\Lambda^*(A)$ is orthogonal to $\rho_0 - \rho_1$ in trace pairing, its outgoing source is the same. Distinguishable histories need not yield distinguishable sources. Conversely, changing a conditional pointer algebra need not change r; a maximally mixed M gives $r = 1/2$ for both the Z and X selections.

This construction does not preserve an arbitrary coherent input record under measurement. For a classical history stored in orthogonal states the preparation can be controlled without deleting it. An independent old content–reference state Ω_{CE} is preserved jointly, even conditioned on the new outcome, when all preparation and loop operations act on R, P, M and leave C, E untouched. If R is already correlated with CE, the old marginal is preserved by the nonselective channel but conditioning can change its state. This is the distinction required when selection is claimed to retain an earlier body or history.

Ideal execution is not required for a finite history effect. Suppose each complete read/update instrument still outputs the four listed configurations, its actual sampled transition in cycle j is $\widetilde{T}_j$, and its actual source weight is $\widetilde{r}_n$, with

$$\max_x \mathrm{TV}(\widetilde{T}_j(x,\cdot), T(x,\cdot)) \le a_j, \qquad \|\widetilde{r}_n - r\|_\infty \le c_n.$$

These are errors of the stated four-state instrument and observable, not an identification of every physical error with a flip.

Corollary 5.2 (A robust finite source window)**.** *From either transient initial configuration T_i, the actual source weight obeys*

$$|\widetilde{\mu}_n \widetilde{r}_n - R_i^\circ| \le n\varepsilon + (q_+q_-)^{\lfloor n/2 \rfloor} + \sum_{j=1}^{n} a_j + c_n.$$

Consequently the contrast between the two histories is at least

$$\frac{|f| q_+ q_-}{1 - q_+q_-} - 2n\varepsilon - 2(q_+q_-)^{\lfloor n/2 \rfloor} - 2\sum_{j=1}^{n} a_j - 2c_n,$$

whenever the right-hand side is positive.

Proof. At each step, couple the true row and its ideal counterpart with mismatch probability at most a_j. After n steps their distributions differ in total variation by at most $\sum_j a_j$. Expectations of $r \in [0,1]$ differ by at most that amount, and changing the source observable costs at most c_n. Apply Eq. (3). Finally, $|R_0^\circ - R_1^\circ| = |f| q_+q_-/(1 - q_+q_-)$, and use the triangle inequality for the two source weights. □

For sufficiently small escape and execution errors, this bound certifies a nonempty interval of different physical consequences. Lower escape rates extend that interval only if execution errors and fresh-port correlations remain controlled. Holding a noisy selection indefinitely would need renewed resources or a stronger protection mechanism. The present bound supplies no free infinite stream.

6 The price of selecting sharp readouts

The fixed executor may be followed by a fixed measurement, with a fixed ancilla. A program selects effects E_z^p on a labelled target basis $\{|b_z^p\rangle\}$. We use the worst-case basis-state error

$$\delta_p = \max_z \left(1 - \langle b_z^p | E_z^p | b_z^p \rangle\right).$$

The program dimension counts the selectable register; a fixed ancillary state does not enlarge it. This criterion concerns the target outcome probabilities, and does not compare complete instruments in diamond norm.

Theorem 6.1 (Cost of sharp selection)**.** *1. If two pure program states select two distinct labelled sharp measurements exactly, they are orthogonal. Exact mixed programs for distinct sharp measurements have orthogonal supports. A register of dimension d_P selects at most d_P distinct labelled sharp measurements exactly.*

2. Let S be a qubit, and suppose that for every Bloch angle θ on a great circle some pure program state selects the sharp measurement along θ with error at most δ, where $0 < \delta < 1$. Then

$$d_P \;\geq\; L(\delta) := \max_{M\geq 1} \frac{M}{1 + \frac{1}{3}\big(2\sqrt{\delta} + \delta\big)^2 (M^2 - 1)}, \tag{4}$$

and $L(\delta)\sqrt{\delta} \to \sqrt{3}/4 \approx 0.433$ as $\delta \to 0$. Conversely, $N = \lceil \pi/(2 \arcsin\sqrt{\delta}) \rceil$ orthogonal program states with a controlled rotation achieve error δ, and $N\sqrt{\delta} \to \pi/2$.

3. Part (2) holds unchanged for mixed program states, with $0 < \delta < 1$. For each target a pure program in the same register has exactly the same two basis-state errors. Consequently $d_P \geq L(\delta)$ and $\liminf_{\delta\to 0} d_P\sqrt{\delta} \geq \sqrt{3}/4$; the same grid gives the upper bound. Mixing cannot reduce the required program dimension under this criterion.

The proof and representative numerical bounds are in Appendix A. The constants $\sqrt{3}/4$ and $\pi/2$ are lower and constructive upper bounds; the result establishes the exponent, not an optimal leading constant. Exact orthogonality is the measurement counterpart of the no-programming theorem for unitaries [15]. The equality of the pure and mixed costs here follows from the convex joint numerical range of two Hermitian operators [18]. Three independent errors need not admit the same pure replacement. The asymptotic exponent applies to this great-circle family and this criterion; universal processors have other resource bounds [17].

A two-axis reader needs only two orthogonal choices [11]. Tracking a continuum of axes to increasing accuracy needs a growing register. Storing and refreshing the chosen programs has a further purity and free-energy account [4, 5]. Coherently returning a program does not, by itself, assign it a reset heat. Conversely, a large register does not supply the entropy outlet needed to renew noisy records.

7 What the executor leaves specified

Rules, programs, their revision and self-grounding selections can be internal. Within Definition 2.1, a full copy of an executor acting on $P \otimes S$ acts on dimension $d_P d_S$, whereas each selected operation acts on S or part of it, of dimension at most $d_S < d_P d_S$ when $d_P \geq 2$. It cannot be that full executor. A meta-register P' can select among executors on $P \otimes S$ under a fixed meta-executor. At the top of each stipulated finite hierarchy the model still supplies an executor as physical dynamics. This is a feature of the architecture, not a general prohibition on a system containing its own control law.

The top dispatch map is specified dynamics in this architecture. A meta-register can make the current executor selectable, at a resource cost set by the family to be selected. For the one-parameter great-circle qubit family, Theorem 6.1 gives order $\delta^{-1/2}$ register dimension; it assigns no common exponent to every continuous family. Stored classical choices also use the record resources described above. The dimension argument excludes a full executor acting on the larger space from being an operation on the smaller target; it permits storing a description of that executor. In infinite dimension the counting argument fails, since a separable Hilbert space is isomorphic to its tensor product with $\mathbb{C}^2$. Kleene and Lawvere fixed points [21, 22], and the order-theoretic results of Scott and Tarski [23, 24], also permit self-reference. Existence of a self-representing physical executor in that setting remains open.

Every stipulated finite hierarchy specifies its top dispatch map. Self-application is different: the loop of Section 4 applies a rule to its own protected records and reaches fixed points without adding levels. A claim that every operation is selected faces a separate resource test. By Theorem 6.1, an executor selecting every sharp readout exactly would require infinitely many orthogonal program states.

The distinction between structure and knowledge has a concrete meaning here. The dispatch map is physical dynamics; program contents are records. When their coupling spectra are disjoint they can be public environmental records, while overlapping eigenvalue tuples can hide some coherence. Proposition 3.2 adds a further distinction: a program need not become a central label to change the field source through the operation it selects.

8 From selectable rules to formed permissions

The loop forms a choice among supplied couplings. A self-frame needs more: history must establish a relation between a task and the body's own reading or action channels. T-DOME II separates an aligned axis from a signed relation that can actually predict [26]. If opposite couplings produce the same retained axis, the axis can be perfectly aligned while its posterior prediction is zero. Outcome-bearing records supply the missing sign. A useful program must encode whichever distinctions its future task consumes.

Self-readout then becomes a problem of attribution. A changed residual can come from the world, from a damaged body, or from a miscalibrated interface. T-DOME III uses identifiable references and paid maintenance to decide among these causes [9]. Its action is based on retained value and body condition; confidence alone need not determine whether repair is worthwhile. Imprint W supplies a complementary mechanism in which the system uses its own reading history to choose a later inquiry [11]. These are mechanisms for revising programs, with their own record laws and budgets.

Brain Anchor locates formed relations in a body: diagnostic outcomes can alter routes and install a different protected encoding [29]. Such relations can be distributed among components. Calling their aggregate a program register does not imply that a single place stores a full description, or that every relation becomes a central environmental label. Proposition 3.1 already gives a finite counterexample to the latter inference. A mathematical encoding is an interface for analysis; a physical implementation must say where its records, actuators and update couplings reside.

Q-RAIF C gives a further, independent source implementation [27]. Its actual quantum history enters a complete positive Hamiltonian and changes material motion and

interacting field source. The fresh-port projector mixture used here is simpler and has different inputs. The two constructions support the same causal question, but their clocks, controllers and source objects cannot be exchanged merely because both have an internal program. Returning a field response to the same body requires the actual sensor and finite-resource operations, as studied in Foundation III [28]. A reconstructed geometry, a field stress and the total apparatus stress remain distinct.

The more ambitious generation problem concerns how these operational relations become available in the first place. A fixed meta-register only relocates a supplied dispatch map. A more informative mechanism is adaptation of the body: records change local couplings, routes or encodings, and the new relations alter which later records and acts are reliable. This can generate a permission from primitive interactions without generating the primitive interactions themselves.

Conjecture 8.1 (Formation of revisable permissions)**.** *Consider a body with locally adjustable couplings and a finite family of reversible encodings of the same logical content. Let actual diagnostic instruments act on relational degrees of freedom outside that logical algebra, and distinguish configurations with different task values. Suppose exploratory operations reach each necessary diagnostic relation, the best configuration has a nonzero value gap on each stationary interval, and paid update and maintenance resources suffice over that interval. There exist local record-driven update dynamics for which task-relevant relations become usable permissions from initially uncalibrated couplings, while previously encoded content remains within a stated error budget. When the task or body changes slowly enough, the same retained relations support detection and revision, rather than only repeating a fixed choice.*

The proposed mechanism is concrete: exploration creates diagnostic records; local retained relations select a route or encoding; an actuator installs it; subsequent errors and maintenance outcomes modify those relations. History-selected protection, signed self-frames and local body learning support different parts of this mechanism. Its quantitative test compares formed relations with supplied ones under the same preparation, diagnostic, installation and content-loss budgets. Formation must lose its advantage when diagnostics alias task-distinct configurations, exploration cannot reach the relevant coupling, or noise destroys old content before learning and migration finish. A distributed learner that changes only labels while all implemented instruments remain fixed also fails this test.

The conjecture asks for a common finite construction with these inputs and outcomes. It is stronger than the supplied two-choice loop, and is not used in its proof. An infinite stationary guarantee would additionally need a sustainable resource supply and control of rare failures. A finite family chosen by the theorist can demonstrate a formation mechanism; it does not explain the origin of every physical interaction or privilege one universal observable algebra.

9 Conclusion

History can select the relations through which a system continues to act. In the finite construction, records choose their own protection and alter the next field source under one executor. Shared coupling spectra show that an internal choice need not become a central label. The exact noisy transition law distinguishes residence in one configuration, a finite history-dependent source window and eventual loss of source contrast. Record-

to-source transfer reaches the trace-distance limit, while accumulated execution errors bound how much of that window remains usable.

Selecting accurate sharp readouts has its own dimensional cost. The one-parameter great-circle family requires order $\delta^{-1/2}$ dimension, with the same feasibility for pure and mixed programs under two target errors. This price and the finite target/executor distinction constrain a specified architecture. They allow stored descriptions, self-application and stronger models of executor formation.

The resulting account joins internal description to a physical consequence without making them identical. Its broader question is how a body's learned relations become revisable permissions while content, references and resources survive. The formation conjecture sets out a mechanism and interventions for that question; primitive dynamics, general geometric generation and first-person experience require their own arguments.

The selection registers and target systems are finite-dimensional; the source realization uses the specified bosonic field port of Paper D. The counting observation concerns this target-register architecture, not every formalism for physical self-reference, and fails in infinite dimension. Internal program changes select actual couplings and can alter their output stress. They do not by themselves explain why this executor exists, reconstruct a geometric sector or establish reflective experience.

Paper D's sustained selection–response conjecture additionally prices thermal entropy export and its gravitational reading [3]. The fresh-port program loop here supplies neither that complete thermal account nor its joint sharp-exhaust and causal-return implementation.

A Proofs and constructions for sharp selection

Here we give the full argument for the readout cost. A readout here is a measurement on S implemented by a fixed executor: a unitary on $P \otimes S \otimes A$, with an ancilla A in a fixed state, followed by a fixed measurement. Purifying the fixed ancilla and dilating the measurement gives projections F_z pulled back to the input space. The embedding $|p\rangle \mapsto |p\rangle|a\rangle$ keeps program overlaps and has span dimension d_P; the ancillary dimension is not a selectable program resource. Write G_z for the induced POVM effects on the original $P \otimes S$, after compression to the fixed ancilla state. A pure program $|p\rangle$ selects effects $E_z^p = (\langle p| \otimes \mathbf{1})G_z(|p\rangle \otimes \mathbf{1})$. Its target is a labelled orthonormal basis $\{|b_z^p\rangle\}$ of S, with selection error $\delta_p = \max_z\big(1 - \langle b_z^p|E_z^p|b_z^p\rangle\big)$. For a mixed program σ, use $E_z^\sigma = \mathrm{Tr}_P[(\sigma \otimes \mathbf{1})G_z]$ and the same error on the target basis states. This is a basis-state error criterion, not a diamond-norm distance between instruments.

Lemma A.1 (Near-orthogonality)**.** *For programs p, q and outcomes $z \neq z'$,*

$$|\langle q|p\rangle|\,|\langle b_{z'}^q|b_z^p\rangle| \leq \sqrt{\delta_p} + \sqrt{\delta_q} + \sqrt{\delta_p\delta_q}.$$

Proof. Put $v = |p\rangle|b_z^p\rangle|a\rangle$ and $w = |q\rangle|b_{z'}^q\rangle|a\rangle$. Then $\langle v|F_z|v\rangle = \langle b_z^p|E_z^p|b_z^p\rangle \geq 1 - \delta_p$, so $\|(\mathbf{1} - F_z)v\| \leq \sqrt{\delta_p}$, and likewise $\|(\mathbf{1} - F_{z'})w\| \leq \sqrt{\delta_q}$. Write $v = F_z v + (\mathbf{1} - F_z)v$ and $w = F_{z'}w + (\mathbf{1} - F_{z'})w$. Since $F_{z'}F_z = 0$, the inner product $\langle w|v\rangle = \langle q|p\rangle\langle b_{z'}^q|b_z^p\rangle$ consists of three cross terms bounded by $\sqrt{\delta_p}$, $\sqrt{\delta_q}$ and $\sqrt{\delta_p\delta_q}$. □

The two-expectation fact used below is the two-Hermitian-operator case of joint numerical-range convexity, equivalent to the Toeplitz–Hausdorff theorem [18]. A rank-reduction proof makes clear why the replacement stays in the original register.

Lemma A.2 (Matching two error rates)**.** *For any two Hermitian operators B_0, B_1 on a finite program register P and any density operator σ on P, there is a unit vector $|p\rangle \in P$ such that*

$$\langle p|B_z|p\rangle = \mathrm{Tr}(\sigma B_z), \qquad z = 0, 1.$$

Proof. If σ has rank $r \geq 2$, the Hermitian operators on its support form a real vector space of dimension $r^2 \geq 4$. The three real linear constraints

$$\mathrm{Tr}\, H = \mathrm{Tr}(HB_0) = \mathrm{Tr}(HB_1) = 0$$

therefore have a nonzero Hermitian solution H supported there. Its zero trace implies that it has both positive and negative eigenvalues. Since σ is positive definite on its support, the interval of t for which $\sigma + tH \geq 0$ contains a neighbourhood of zero and has finite endpoints. At either endpoint the rank strictly decreases, while the trace and both expectations remain unchanged. Repeat on the smaller support until the density operator has rank one. □

Theorem 6.1 is proved below with the same fixed ancilla and basis-state error criterion.

Proof. (1) With $\delta_p = \delta_q = 0$ the lemma gives $\langle q|p\rangle\langle b^q_{z'}|b^p_z\rangle = 0$ for all $z \neq z'$. If $\langle q|p\rangle \neq 0$, each $|b^p_z\rangle$ is orthogonal to every $|b^q_{z'}\rangle$ with $z' \neq z$, hence proportional to $|b^q_z\rangle$, and the two labelled measurements coincide. Each mixed program's errors are averages of nonnegative pure-component errors. Zero average forces every vector in its spectral decomposition to select the same measurement exactly. The pure result makes the supports for distinct measurements orthogonal. Their nonzero dimensions sum to at most d_P.

(2) Choose M targets at angles $\theta_j = 2\pi j/M$ with program states $|p_j\rangle$. For $z = 0$, $z' = 1$ and Bloch angle φ between the targets, $|\langle b^q_1|b^p_0\rangle| = |\sin(\varphi/2)|$, so the lemma gives $|\langle p_j|p_l\rangle| \leq (2\sqrt{\delta} + \delta)/\sin(\pi|j-l|/M)$. The Gram matrix Γ of the $|p_j\rangle$ has $\mathrm{Tr}\,\Gamma = M$ and, by the identity $\sum_{k=1}^{M-1} \csc^2(\pi k/M) = (M^2-1)/3$,

$$\mathrm{Tr}\,\Gamma^2 \leq M + M\,\tfrac{1}{3}(2\sqrt{\delta} + \delta)^2(M^2 - 1).$$

Then $d_P \geq \mathrm{rank}\,\Gamma \geq (\mathrm{Tr}\,\Gamma)^2/\,\mathrm{Tr}\,\Gamma^2$, which is (4). With $c = \frac{1}{3}(2\sqrt{\delta} + \delta)^2$ the maximum over M lies near $M = c^{-1/2}$ and equals $\frac{1}{2}c^{-1/2}(1 + O(\sqrt{c}))$, which gives the limit. For the converse, the executor $\sum_j |j\rangle\langle j| \otimes R(\theta_j)^\dagger$ followed by a measurement in the computational basis selects the sharp measurement along $\theta_j = 2\pi j/N$ exactly; using the nearest θ_j for a target θ misaligns the axes by at most π/N, which costs error $\sin^2(\pi/(2N)) \leq \delta$.

(3) Fix a target basis $\{|b_0\rangle, |b_1\rangle\}$ and define positive operators on the original register P by

$$B_z = \mathbf{1}_P - (\mathbf{1}_P \otimes \langle b_z|)G_z(\mathbf{1}_P \otimes |b_z\rangle), \qquad z = 0, 1.$$

A mixed program σ has target errors $\mathrm{Tr}(\sigma B_z)$. Lemma A.2 supplies a pure program in the original P matching both errors exactly. Choose it for each target angle. The fixed executor then has a pure-program family with the same error bound δ and dimension d_P, so part (2) applies. Conversely, pure programs are included among mixed programs. This equivalence concerns the two target errors, not preparation costs or equality of the complete instruments; arbitrary additional error constraints need not admit a rank-one replacement. □

For k positive error operators $B_1, \ldots, B_k$, a mixed program whose errors are all at most δ always has a pure spectral component with every error at most $k\delta$: some component has the sum of its errors no larger than the mixed sum. Exact matching of all errors need not hold when $k \geq 3$: the three positive operators $(I+\sigma_x)/2$, $(I+\sigma_y)/2$, $(I+\sigma_z)/2$ all have expectation $1/2$ in the maximally mixed qubit, while no pure state matches all three. The two-error equality is therefore stronger than this general factor-k bound.

δ	$L(\delta)$	$N(\delta)$	$L\sqrt{\delta}$	$N\sqrt{\delta}$
10^{-2}	4.15	16	0.415	1.600
10^{-4}	43.1	158	0.431	1.580
10^{-6}	432.8	1571	0.433	1.571

Table 1: Common pure- and mixed-program lower bound (4) and the grid construction for selecting every sharp qubit measurement on a great circle to worst-case basis-state error δ.

D'Ariano and Perinotti give finite exact programming for covariant measurements and polynomial approximate programming for general measurements [16]. This does not remove the sharp-selection constraint.

For N grid programs and a uniformly distributed angle the mean error is $\frac{1}{2} - \frac{N}{2\pi} \sin \frac{\pi}{N} \approx \pi^2/(12N^2)$. Doubling the register dimension therefore divides the leading mean error by four.

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Postscript: Persistence, Participation and an Unfinished Self

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A world can leave a record without leaving anyone able to use it. A reader can continue without knowing everything about itself. These two distinctions frame the question of closure in the Holographic Alaya–Field Framework: what can end, what can remain, and what would be required for participation to begin again?

The framework studies physical access rather than treating intelligibility as an attribute of matter in isolation. A channel resolves some differences and loses others. A stable relation can preserve classical labels or quantum content. A body can use retained history to change its next action, its own channels and the physical sources it produces [1, 2, 3, 4, 7, 5]. These claims connect world formation with agency, but they do not make the universe depend for its existence on a particular observer. Nor do they identify a stable observable algebra, a finite reconstruction of geometry and a spacetime metric. Each reaches a different part of the question.

1 A trace is not a continuing reader

Consider a fair classical label X copied perfectly into R two-level memories. Its joint state is

$$\rho_{XC} = \frac{1}{2}\sum_{x=0}^{1} |x\rangle\langle x| \otimes (|x\rangle\langle x|)^{\otimes R}.$$

Every memory alone is maximally mixed; the bank has no marginal purity. Nevertheless each copy determines X. Dephasing in the Z basis leaves this joint record unchanged. The product state with the same maximally mixed marginals carries no such record. The difference is in the correlations, not in the local inventories.

An archive of this kind may outlast the machinery that formed it. A later body with appropriate access can still read it. Without that machinery, however, the archive does not acquire a new observation, test an interpretation or correct its own readout. Passive protection and active participation are different physical achievements. Foundation's distinction between held content and return into another operation makes this difference constructive [9]. The continuing operation must actually be available; a comparison state used to bound an error is not a free preparation of another program.

This is also a philosophical distinction. What persists as a trace need not persist as an agent. The past can remain physically relevant through future interventions even after its

original reader is gone. Conversely, continuity of an agent need not mean preserving every past datum or every microscopic component. In the framework's agency account it means maintaining an organization that can carry usable relations into later choices, repair its channels and revise what those relations warrant [3, 8]. This is a proposed criterion of causal participation, not a claim that causal organization alone explains experience.

2 Acquisition and renewal have different limits

HAFF F bounds record acquisition at a declared initial resource boundary [6]. Its assumptions include initially independent, unused memories and a specified thermal work account. The initial purity

$$B = \sum_{k} [\ln d_k - S(\sigma_k)]$$

is then part of what can pay for new correlations, together with supplied resources and the system's own contribution. The example above explains why one cannot impose this product boundary anew at each time. Old correlations are resources even when $B = 0$. Running out of fresh purity does not erase them.

Renewal asks another question: what must be done when an existing useful relation degrades? In F's conditional-independent bank, $C_i = X \oplus N_i$, with error probability $\varepsilon < 1/2$, independent flips of probability $0 < p \le 1/2$ give

$$q = p + (1 - 2p)\varepsilon, \qquad d_{p,\varepsilon} = h(q) - h(\varepsilon) > 0,$$

where $h(u) = -u \ln u - (1 - u) \ln(1 - u)$. Returning the complete retained label, bank and controller to their joint target removes $R d_{p,\varepsilon}$ nats. An independent Gibbs bath at temperature T must receive

$$Q \ge k_B T R d_{p,\varepsilon}.$$

Here the flip record is unavailable, the bath is initially independent of the noisy retained state, and the controller cannot absorb the entropy without being restored. For $p = .12$ and $\varepsilon = .05$, eight copies of one fair label require 1.9025859874 nats of correction. The number of distinct labels is one; the independent degradation of eight carriers still has a price.

A sole thermal chiral exhaust with the additional current assumptions in F and D has a quadratic finite-rate price. That result does not apply to every bath or every control pulse. Exact protection gives $d = 0$ in the bank model and defeats a blanket assertion that records require positive upkeep. Accessible noise records, correlated errors and a controller allowed to consume its own purity alter the boundary as well. The distinction matters before any cosmological extrapolation.

3 A finite-body extension

The bank is not the only setting in which a retained relation has a renewal cost. The following result treats an entire finite memory as one object, allows correlated and quantum content, and makes the noise assumption explicit. Entropies and mutual informations are in nats.

Proposition 1 (Renewing a useful relation under partial reset)**.** *Let θ_{XMC} be a target state of a retained reference X, a memory M of dimension d, and a controller C. Suppose $I_\theta(M:X) \geq \iota > 0$. During each degradation interval the unavailable noise acts as*

$$\mathcal{D}_\lambda(\theta) = (1-\lambda)\theta + \lambda\, \theta_{XC} \otimes \tau_M, \qquad \tau_M = I_M/d, \quad 0 < \lambda \leq 1.$$

A correction restores the complete target θ_{XMC}, using a bath initially independent of the noisy retained state and Gibbs at temperature T. If the correction is a possibly driven unitary on the retained system and bath, with no other entropy sink and vanishing endpoint interaction energy, then

$$Q \geq k_B T \lambda \iota. \tag{1}$$

If the total retained-system energy, including internal couplings, is equal immediately before and after the correction, and driven work and the counted bath heat are the only energy flows, then $W = Q$. A degenerate memory with additive, unchanged apparatus energies is one sufficient realization. For N renewals with $T_j \geq T_* > 0$, $\lambda_j \geq \lambda_* > 0$, and $I_{\theta_j}(M_j : X_j) \geq \iota_* > 0$, a finite total supplied work $\sum_j W_j \leq E_{\mathrm{av}}$ implies

$$N \leq \frac{E_{\mathrm{av}}}{k_B T_* \lambda_* \iota_*}. \tag{2}$$

For the last statement each interval must satisfy the preceding cyclic correction and energy conditions, even when its target differs.

Proof. Write $Z = XMC$ and $\eta = \mathcal{D}_\lambda(\theta)$. Entropy concavity gives

$$\begin{aligned} S(\eta) - S(\theta) &\geq \lambda[S(\theta_{XC}) + \ln d - S(\theta)] \\ &= \lambda[\ln d - S_\theta(M|XC)] \\ &= \lambda[\ln d - S_\theta(M) + I_\theta(M:XC)] \\ &\geq \lambda I_\theta(M:X) \geq \lambda\iota. \end{aligned}$$

The last line uses $S(M) \leq \ln d$ and monotonicity of mutual information under discarding C. The proof is valid for quantum X and for a controller correlated with the target.

Initially Z and the Gibbs bath B are independent. Entropy conservation under the correction and subadditivity at its end give $\Delta S_B \geq S(\eta) - S(\theta)$. For $\tau_B = e^{-\beta H_B} / \operatorname{Tr} e^{-\beta H_B}$,

$$\beta Q = \Delta S_B + D(\rho'_B \| \tau_B), \qquad \beta = (k_B T)^{-1}.$$

This proves (1). With vanishing endpoint bath coupling, the first law reads $W = \Delta E_{XMC} + Q$. The total-energy and energy-flow assumptions give $\Delta E_{XMC} = 0$ and hence $W = Q$; summing the lower bounds proves (2). □

The reset mixture models a lost alternative, not an accessible branch flag the controller can undo for free. It acts on the whole designated memory, so no protected subsystem within that memory escapes the stated channel. The cost is paid for restoring the joint relation, not for restoring a local marginal. A fair classical copy has a maximally mixed memory marginal and still satisfies the bound. An initially independent maximally mixed memory has $\iota = 0$ and does not.

This extends the renewal argument beyond independent copies. It does not establish that every physical body experiences this noise, that every update needs a fixed amount

of information, or that every target can be restored under a body's allowed access. The bound is necessary where such a correction exists. A consumed controller, an accumulating waste register or an accessible auxiliary noise record must be included among the resources. If the noise environment is retained, its correlations cannot be discarded when defining the target. Calling only the memory cyclic would hide the expenditure. The entropy proof permits an externally scheduled unitary; it is not a proof of autonomous apparatus closure. A physical battery that takes part in the entropy account must be included or separately shown not to supply an additional entropy sink.

Equation (2) bounds successful renewals, not necessarily calendar time. A system can postpone a last update arbitrarily far. A finite upper bound on an uninterrupted active interval additionally requires a positive minimum update rate and a bound on gaps. Hibernation, cooling toward zero temperature, decreasing task quality and improving protection are substantive alternatives, not semantic exceptions.

4 An unfinished self can continue

A demand for complete self-description makes participation seem impossible for the wrong reason. A self-model need only support the inquiries and actions on which continuation depends. T-DOME II gives a concrete example: history must retain the signed relation between the task and the body's channels. An accurately learned but unsigned axis can leave the prediction useless [8]. The useful self-frame therefore consists in how a system can reach its task, not in an inventory of everything it is.

Some distinctions are inaccessible to a body's legitimate references. Foundation II separates that failure of identifiability from disagreement, and separates agreement from correctness [9]. A hidden symmetry can make one parameter unknowable while leaving a valuable intervention completely identifiable. In other cases inquiry resolves the ambiguity, but its preparation and maintenance consume resources. A continuing body must decide which error matters and which probe can repair it before the inquiry destroys the content it was meant to help. Brain Anchor's learning-versus-survival tradeoff supplies one physical version of this problem [12].

Finite self-knowledge is consequently not equivalent to paralysis. A body may track residual errors, revise its references and retain a margin of viable action without ever obtaining a complete account of itself. The stronger philosophical claim is that an imperfect, historically formed organization can be a continuing subject precisely through this revisability. Its identity lies partly in the maintained relation between history and possible action, not in an immutable perfect image.

A rival account could identify continuation with persistence of the stored pattern. It then misses a relevant difference between two copies of the same archive: one has working probes and can revise its routing after an error, while the other's channels have permanently failed. Their stored descriptions agree; their possible next histories do not. The active criterion explains that difference without demanding exact microscopic identity. Two functioning copies may both inherit a causal history, so this criterion does not by itself select a unique first-person successor. That further identity question, like experience, cannot be decided by the survival of a data pattern alone.

That claim still needs an account of first-person experience. Consciousness III's explanatory gap cannot be crossed by renaming calibration as awareness [10]. A candidate bridge must explain why these particular internal relations matter to experience, dis-

tinguish alternatives that make the same reports, and expose a relevant intervention. Neither Proposition 1 nor a finite self-frame proves that a body feels anything. The thermodynamic argument addresses the support for continuation; the experience argument addresses what, if anything, that continuation is like.

5 A cosmological participation-window conjecture

The universe's duration and a body's active history need not have the same limit. Dyson's cooling and scaling scenario explicitly allows indefinitely continued life and communication from finite energy under its assumed laws [13]. Finite resources alone therefore do not establish a finite active window. In asymptotically de Sitter models a detector can encounter cosmological-horizon thermal radiation [14]. Krauss and Starkman examine how horizon restrictions, recoverable energy and memory requirements change the prospect of indefinite activity [15]. These are competing conditional mechanisms, not observational proof of the far future.

The framework suggests a more specific conjecture.

Conjecture 1 (Resource-limited participation)**.** *In a late-time causal domain with finite accessible usable work, a positive environmental temperature floor and unavoidable loss of task-relevant relations, any fixed-capacity body that continues at a fixed nonzero task quality can perform only finitely many successful inquiry–revision–action updates, even while passive traces and spacetime structure persist.*

The proposed mechanism is loss of correlations on which usable permissions depend. A retained history supports diagnosis; diagnosis changes routing or protection; the revised body supports another action and its physical source. Mixing damages the relations required to repeat that loop. Restoration exports entropy or consumes a fresh resource. A horizon may restrict replenishment and a thermal floor may prevent the cost per restoration from tending to zero. Exhaustion then ends this participation without erasing every trace.

The bank and partial-reset propositions establish this mechanism in two specified memory models. G supplies finite history-dependent source differences, while Imprint and Brain connect retained relations to subsequent inquiry and bodily change [7, 11, 12]. The conjecture reaches further: an interacting adaptive body can compress memories, migrate codes and replace its task. To prove it for a proposed body class one must obtain a uniform positive resource cost per successful update after optimizing those operations. Neither a count of bit erasures nor a total marginal purity inventory supplies that result automatically. The temperature floor, accessible work bound and irreducible relation loss must also be established for the actual cosmological domain. They are not consequences of selecting an algebra.

The conjecture is weakened or falsified by a body at the stipulated quality with an indefinitely usable protected relation, by vanishing renewal cost after admissible compression or migration, by a cooling law permitting a summable sequence of positive costs, or by unbounded replenishment. An accessible noise record can break the proposed loss mechanism. Discovery of one such regime would identify which part of the mechanism fails, rather than merely extending a numerical lifetime. A static surviving archive is not a counterexample to this active claim, but a continuing reader with the stated resources would be.

The sustained selection–response conjecture in D has different resource quantifiers [4]: it asks for families of longer runs whose installed energy, size and operating domain may grow, while one body is reused within each run. The participation conjecture here fixes capacity and limits the total accessible work. Neither quantifier refutes the other. Foundation III's finite semiclassical mean-action source–response construction [5] concerns a specified operating window; it does not establish indefinitely renewed thermal operation.

6 Origin, return and what closure would mean

An unused record port and an exhausted one can be locally indistinguishable and have different futures. The former may be independent of the source and backed by a usable gradient. The latter can carry correlations with an old history or lack the resources needed for another operation. Allowed future interventions can distinguish them. Equal local states therefore do not identify origin with terminus.

The stronger origin proposal in G is physical formation of permissions: a body with locally adjustable relations encounters diagnostic tasks, retains their results and uses them to install a useful coupling [7]. Exploration, a distinguishable value difference and enough survival time must be supplied by the primitive dynamics. This moves the problem beyond an external program name, without proving that every supplied executor arises in this way. Enlarging an executor introduces another joint interaction to explain; it does not prove a universal final boundary or forbid a finite system from carrying a useful description of itself.

A cosmological return would require more still. One concrete mechanism candidate is that contraction or another geometric transition creates newly accessible nonequilibrium gradients; interactions then allocate part of that work to fresh record ports and bodies, while old correlations and exported entropy remain accounted for in other degrees of freedom. A repeated active phase would be a renewal of resources and formation, not necessarily recurrence of the global state. The physical transition must have its own dynamics, show that the gradients can be used, and explain why old entropy does not obstruct the new phase. Local thermal swaps or recurrence of a finite register do not establish any of these cosmological steps. A bounce that restores density but no usable work would fail the mechanism; so would a transition requiring an uncharged deletion of the old history. No such return is forced by the participation-window conjecture.

Closure thus has several meanings that must not be confused. A record can become unreadable to one body; a body can lose the capacity to revise itself; a cosmic domain can cease to support new active readers. None entails the nonexistence of its remaining physical structures. Nor must an imperfect self finish knowing itself before it can live. The framework's most consequential prospect is that history changes what a body can do next, and that this relation can be maintained and revised for a time without being complete.

A history can remain after the means of continuing it have gone. A continuing self need not wait to become a finished account of itself.

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Lorentzian Read-out from Local Unitary Propagation: Clifford Relations, Causal Cones and Spectral Mass

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Abstract

A positive control metric cannot become Lorentzian by a change of variables. We instead derive the relative sign from the propagation of an internal quantum state. For a first-order Hermitian symbol with linearly independent spatial coefficients, every band lies on one quadratic shell if and only if the coefficients obey the corresponding Euclidean Clifford relations. Operator-norm bounds on deviations give a uniform squared-band certificate without asserting repair to an exact representation. The resulting form $\omega^2 - Q(q)$ has signature $(1, d)$; it describes the full characteristic cone when both frequency branches occur. The evolution parameter is supplied, while unitarity makes its conjugate frequency a hyperbolicity direction. The internal dimension bounds $d + [\mu \neq 0]$ by $2\nu_2(s) + 1$, so a qubit supports a massless shell in at most three spatial dimensions. On a lattice, the strict support cone, maximum group speed and low-energy read-out cone differ. For several species, off-diagonal mixing with fixed diagonal principal blocks cannot produce a single quadratic shell. Interactions can align speeds: in a one-loop mediator model, divergent accumulated coupling along every edge is sufficient, even when their active ranges of renormalization scale are disjoint. Near a supplied Lorentz-invariant conformal fixed point, additional conserved spin-2 currents mark the marginal exceptions to speed alignment. A finite record protocol for the same coined walk certifies its mass-setting angle, separating the ideal rest-and-curvature and group speeds from each other and from the supplied support speed. The geometric dictionary, actual noisy transport and physical inputs remain separate.

Keywords: Lorentzian signature, Clifford algebra, hyperbolic polynomials, quantum cellular automata, quantum walks, Weyl and Dirac equations, causal cone, spectral mass, Q-RAIF

1 Introduction

Programmes that reconstruct geometry from quantum data usually take the Lorentzian signature as given. Holographic entanglement formulas and thermodynamic derivations of the Einstein equation relate entropies to areas and energy fluxes in a spacetime whose causal

structure is already Lorentzian [2, 3]. The Holographic Alaya-Field Framework (HAFF) studies which distinctions a physical reduction can resolve and what geometric data can be reconstructed from them [42, 43]; a spacetime metric needs a further dictionary there as well. This paper asks where the indefinite direction of such a read-out can come from in a model whose dynamics is given explicitly.

Two facts fix the starting point. A positive definite form stays positive under every linear change of variables (Proposition 1), so no control metric, information metric or work form becomes Lorentzian by reparameterization. The companion paper on persistence makes the same point physically: its descent metric and its field-derived work form are both positive, and the Lorentzian form is a third structure [45]. Second, imposing Clifford relations $\{e_\mu, e_\nu\} = 2q_{\mu\nu}\mathbf{1}$ for a chosen Lorentzian q puts the signature into the premise; the universal property of $Cl(V, q)$ then supplies an encoding, rather than a signature-selection mechanism [4, 5].

The model class is local unitary propagation: a particle with an internal Hilbert space $\mathbb{C}^s$ hops on a lattice by a translation-invariant, finite-range unitary step, as in quantum walks and the one-particle sector of quantum cellular automata [9, 10, 11, 12, 13]. Its low-energy generator is a first-order Hermitian matrix symbol. In this setting four objects can be defined separately: a positive norm on generators, which plays the role of a control metric; the causal cones of the dynamics; the algebra of the symbol's coefficients; and the spectral mass. For a first-order Hermitian symbol, unitarity alone makes the characteristic polynomial hyperbolic in the frequency (Theorem 2). For linearly independent spatial coefficients, all bands lie on one quadratic shell if and only if the coefficients satisfy the shell relations of Theorem 4; the characteristic form is then ω^2 minus the Hilbert–Schmidt norm of the spatial symbol and has signature $(1, d)$, and the Lorentzian Clifford generators are $\gamma^0 = \beta$, $\gamma^i = \beta\alpha_i$ (Theorem 4, Corollary 5). The internal dimension bounds the spatial one, $d + [\mu \neq 0] \leq 2\nu_2(s) + 1$. For a qubit carrier three is the largest dimension with an isotropic massless cone, and the only one in which every internal rotation axis is coupled to a direction of motion (Theorem 8, Proposition 10).

On the lattice, group velocities lie in the convex hull of the neighbourhood, so the speed of any massless read-out cone is at most the inradius of that polytope per step (Proposition 12). Exact formulas for the split-step walk and computations for the Weyl automaton then show that the read-out map is not fixed by the dynamics: a band point, a momentum window and the observable that defines the speed all have to be chosen (Section 5).

The mathematical ingredients are standard: Dirac's square root of the Klein–Gordon operator [6], Gårding's hyperbolic polynomials [7], the representation theory of complex Clifford algebras [8], and the derivations of Weyl and Dirac equations from quantum walks and automata [9, 11, 13]. Hyperbolicity as the defining property of a physical dispersion relation is developed in Ref. [14], and an information-theoretic argument linking three spatial dimensions to the qubit is given in Ref. [15]. This paper separates the four objects in one model, identifies which inputs of a Lorentzian read-out follow from propagation, and calculates how the remaining read-out choices enter. The several-species analysis adds a quantitative fixed-block mixing obstruction and applies cut-balanced convergence to interacting speeds that need not couple simultaneously.

The argument proceeds from the symbol and its shell to dimension, lattice windows and several-species dynamics. A residual bound connects the exact shell criterion to calibrated

approximate symbols; a finite angle certificate then connects one such dictionary to actual records. The concluding sections distinguish these contributions from the physical selection of a carrier, an interacting fixed point and a complete source-and-return apparatus.

2 Local unitary propagation and four objects

2.1 The model

Let $\Lambda \subset \mathbb{R}^d$ be a lattice of physical positions and let the one-particle Hilbert space be $\ell^2(\Lambda) \otimes \mathbb{C}^s$. A translation-invariant step with finite neighbourhood $N \subset \Lambda$ acts as

$$(W\psi)(x) = \sum_{h \in N} A_h \, \psi(x - h), \qquad A_h \in M_s(\mathbb{C}), \tag{1}$$

and is unitary exactly when its symbol

$$W(k) = \sum_{h \in N} A_h \, e^{-ik \cdot h} \tag{2}$$

is unitary for every k in the Brillouin zone. One step takes time τ. The eigenvalues of $W(k)$ are $e^{-i\omega_b(k)\tau}$ with quasi-energies $\omega_b(k)$ defined modulo $2\pi/\tau$, and the group velocity of band b is $\nabla_k \omega_b$. With this convention the pure shift $A_h = \mathbf{1}$ for a single h moves every state by $+h$ per step.

Near a point k_* where a gap closes or nearly closes, the step is described by an effective generator. If a parameter θ controls the gap, we say that the walk has *first-order symbol*

$$H(q) = \sum_{i=1}^{d} q_i \alpha_i + \mu\beta, \qquad \alpha_i, \beta \in \mathrm{Herm}(s), \tag{3}$$

at k_* if $(i/\tau) \operatorname{Log} W_\theta(k_* + q) = H(q) + O(|q|^2 + |\mu|^2)$ with μ proportional to θ. This is the scaling limit studied for quantum walks and automata [11, 12, 13]: H generates the continuum dynamics $i\partial_t \psi = H(-i\nabla)\psi$ when lattice spacing and step time go to zero at fixed ratio. For example, the split-step walk of Section 5.2 has $W_\theta(k) = \mathrm{diag}(e^{-ika}, e^{ika}) R(\theta)$ with $R(\theta) = \cos\theta - i \sin\theta \, \sigma_y$, and at $k_* = 0$ its first-order symbol is $H(q) = (a/\tau) q \sigma_z + (\theta/\tau)\sigma_y$.

2.2 Four objects

A control metric. The generators of the internal space form the real vector space $\mathrm{Herm}(s)$, which carries the positive Hilbert–Schmidt norm

$$G_{\mathrm{HS}}(A) = \frac{1}{s} \operatorname{tr}(A^2). \tag{4}$$

Restricted to the span of the coefficients of H, it is the natural positive metric on the controls that the symbol couples to translations. Other positive forms of the same type appear in the series: the descent metric G of a gradient controller and the work form K derived from the stress of a field-coupled actuator [45], and monotone information metrics on states [29].

Causal cones. The step (1) has a strict causal region. After n steps a state supported in S is supported in $S + nN$, the n-fold Minkowski sum, so all influence lies in the cone over the polytope $C_N = \tau^{-1} \operatorname{conv} N$. A first-order symbol has a second cone, its characteristic cone, defined in Section 3. The two need not agree.

A representation. The coefficients α_i, β generate a real algebra inside $M_s(\mathbb{C})$. Whether they satisfy Clifford relations, and with which quadratic form, is a property of the dynamics, not a choice of encoding.

Spectral mass. For a selected logarithm and energy zero, the positive band's value at k_* is the rest energy $E_0 = \hbar|\omega(k_*)|$, and its curvature defines an inertial mass m_* through $\omega(k_* + q) \simeq \omega(k_*) + \hbar q^2/(2m_*)$. A measured phase difference alone does not fix an unknown common energy offset. A Lorentzian read-out identifies $E_0 = m_* c^2$, which defines a speed c; whether this c agrees with either causal cone is a question about the model.

Proposition 1 (Positive forms stay positive)**.** *Let G be a positive definite form on a real vector space V and $T : U \to V$ linear. Then T^*G, given by $(T^*G)(u, u') = G(Tu, Tu')$, is positive semidefinite, and positive definite if T is injective. In particular no positive definite form is congruent to a form with a negative direction.*

Proof. $(T^*G)(u, u) = G(Tu, Tu) \geq 0$, with equality only if $Tu = 0$. The last statement is Sylvester's law of inertia. □

The Lorentzian form of a read-out is therefore not a pullback or a reparameterization of G_{HS}, G or K. In the model it arises as a different kind of object, a form on frequency–momentum space built from the spectrum of the evolution.

3 Where the indefinite direction comes from

For a first-order symbol (3) write $P(\omega, q) = \det(\omega \mathbf{1} - H(q))$ for its characteristic polynomial and $P_0(\omega, q) = \det(\omega \mathbf{1} - \sum_i q_i \alpha_i)$ for the principal part. A homogeneous polynomial p on $\mathbb{R}^n$ is *hyperbolic with respect to e* if $p(e) \neq 0$ and $t \mapsto p(\xi + te)$ has only real roots for every ξ [7].

Theorem 2 (Hyperbolicity in the evolution frequency)**.** *For Hermitian α_i, β, every root in ω of $P(\omega, q)$ is real for every real q, and P_0 is hyperbolic with respect to the frequency direction $e = (1, 0, \ldots, 0)$. The propagation speed of the principal part in a unit direction u is the largest eigenvalue of $\sum_i u_i \alpha_i$.*

Proof. The roots of $P(\cdot, q)$ are the eigenvalues of the Hermitian matrix $H(q)$. For P_0, $P_0(e) = 1$ and the roots of $t \mapsto P_0(\xi_0 + t, \xi')$ are $-\xi_0$ plus the eigenvalues of $\sum_i \xi'_i \alpha_i$, which are real. The characteristic speeds of $i\partial_t \psi = \sum_i \alpha_i (-i\partial_i)\psi$ in direction u are the eigenvalues of $\sum_i u_i \alpha_i$. □

The evolution parameter is supplied by the model. Unitarity makes its conjugate frequency a hyperbolicity direction; it does not choose how many fundamental time parameters

a theory should have. The next lemma determines the signature when the characteristic form is quadratic.

Lemma 3 (Hyperbolic quadratic forms are Lorentzian)**.** *A real quadratic form p on $\mathbb{R}^n$ with $p(e) > 0$ is hyperbolic with respect to e if and only if it has exactly one positive eigenvalue. Then $p = \ell^2 - r$ with ℓ linear and r positive semidefinite, and if p is non-degenerate its signature is $(1, n-1)$.*

Proof. Decompose $\xi = se + \zeta$ with ζ in the p-orthogonal complement of e. Then $p(\xi + te) = p(e)(s+t)^2 + p(\zeta)$, which has real roots in t for every ξ if and only if $p(\zeta) \le 0$ on the complement. That is the statement that p has one positive direction and is negative semidefinite on a complementary hyperplane. □

Theorem 4 (Single shell, Clifford relations and the Lorentzian form)**.** *Let $\alpha_1, \dots, \alpha_d, \beta \in \mathrm{Herm}(s)$ with $\alpha_1, \dots, \alpha_d$ linearly independent, and fix $\mu \in \mathbb{R}$. The following are equivalent.*

(a) There is a quadratic form Q on $\mathbb{R}^d$ such that every eigenvalue ω of $H(q)$ satisfies $\omega^2 = Q(q) + \mu^2$ for all q.

(b) $H(q)^2 = (Q(q) + \mu^2)\mathbf{1}$ for all q.

(c) $\{\alpha_i, \alpha_j\} = 2Q_{ij}\mathbf{1}$, $\mu\{\alpha_i, \beta\} = 0$ and $\mu^2(\beta^2 - \mathbf{1}) = 0$.

If they hold, then:

1. *Q is positive definite and $Q(q) = G_{\mathrm{HS}}\big(\sum_i q_i \alpha_i\big)$;*
2. *the read-out form $\eta(\omega, q) = \omega^2 - Q(q)$ on frequency–momentum space has signature $(1, d)$, and the characteristic set of P_0 lies on its null cone; the sets coincide if $d \ge 2$, if $\mu \neq 0$, or if $d = 1$ and α_1 has both positive and negative eigenvalues;*
3. *if $d + [\mu \neq 0] \ge 2$, then s is even and $P(\omega, q) = (\omega^2 - Q(q) - \mu^2)^{s/2}$;*
4. *every group velocity $v = \nabla_q \omega$ satisfies $Q^{-1}(v, v) = Q(q)/(Q(q) + \mu^2) \le 1$, with equality only for $\mu = 0$.*

Proof. (a)⇒(b): $H(q)$ is Hermitian, hence diagonalizable, and all its eigenvalues square to $Q(q) + \mu^2$. (b)⇒(c): expanding,

$$H(q)^2 = \sum_{i,j} q_i q_j \tfrac{1}{2}\{\alpha_i, \alpha_j\} + \mu \sum_i q_i \{\alpha_i, \beta\} + \mu^2 \beta^2.$$

Comparing the parts of degree two, one and zero in q with $(Q(q) + \mu^2)\mathbf{1}$ gives the three relations. (c)⇒(a): the same expansion gives (b), and (b) gives (a).

For item 1, $Q(q)\mathbf{1} = (\sum_i q_i \alpha_i)^2 \ge 0$; if $Q(q) = 0$ the Hermitian matrix $\sum_i q_i \alpha_i$ squares to zero and so vanishes, and linear independence gives $q = 0$. Taking the normalized trace gives $Q(q) = G_{\mathrm{HS}}(\sum_i q_i \alpha_i)$. For item 2, η is ω^2 minus a positive definite form, and every root of $P_0(\cdot, q)$ belongs to $\{\pm\sqrt{Q(q)}\}$. When $d \ge 2$, the principal symbol at nonzero q anticommutes with an invertible symbol in a Q-orthogonal direction, pairing its two signs.

If $\mu \neq 0$, β provides the pairing. In the remaining one-dimensional case both signs occur exactly when α_1 has both signs. For item 3, choose coordinates with $Q = c^2\mathbf{1}$. The matrices α_i/c together with β (if $\mu \neq 0$) pairwise anticommute and square to $\mathbf{1}$. If at least two are present, each x anticommutes with an invertible g, giving $\operatorname{tr} x = \operatorname{tr}(g^{-1}xg) = -\operatorname{tr} x$. Thus $H(q)$ is traceless, and its eigenvalues $\pm\sqrt{Q+\mu^2}$ have equal multiplicity. For item 4, at a differentiable band point with $\omega \neq 0$, $\nabla_q \omega = Qq/\omega$ and $Q^{-1}(v,v) = Q(q)/\omega^2$. At a massless node the same statement applies to directional band limits. □

Corollary 5 (Lorentzian Clifford generators)**.** *Under Theorem 4 with $\mu \neq 0$, the matrices*

$$\gamma^0 = \beta, \qquad \gamma^i = \beta\alpha_i \tag{5}$$

satisfy $\{\gamma^0,\gamma^0\} = 2$, $\{\gamma^0,\gamma^i\} = 0$ and $\{\gamma^i,\gamma^j\} = -2Q_{ij}$, so they generate a representation of $Cl(1,d)$ on $\mathbb{C}^s$; γ^0 is Hermitian and the γ^i are anti-Hermitian. The eigenvalue equation $H(q)\psi = \omega\psi$ is equivalent to $(\omega\gamma^0 - \sum_i q_i\gamma^i - \mu)\psi = 0$, and $(\omega\gamma^0 - \sum_i q_i\gamma^i)^2 = \eta(\omega,q)\mathbf{1}$. For $\mu = 0$ and $d \geq 1$, the matrices

$$\Gamma^0 = \begin{pmatrix} 0 & \mathbf{1} \\ \mathbf{1} & 0 \end{pmatrix}, \qquad \Gamma^i = \begin{pmatrix} 0 & \alpha_i \\ -\alpha_i & 0 \end{pmatrix} \tag{6}$$

generate a representation of $Cl(1,d)$ on $\mathbb{C}^s \oplus \mathbb{C}^s$.

Proof. $(\beta\alpha_i)^2 = -\beta^2\alpha_i^2 = -Q_{ii}$ and $\{\beta,\beta\alpha_i\} = \alpha_i - \alpha_i = 0$ use $\{\alpha_i,\beta\} = 0$ and $\beta^2 = 1$; similarly $\{\beta\alpha_i,\beta\alpha_j\} = -\{\alpha_i,\alpha_j\}$. Multiplying $H(q) - \omega$ by β gives the second statement. The block matrices are checked by direct multiplication. □

What has been derived. In the model the indefinite direction of the read-out form is the frequency ω, the spectral variable of the one-parameter group of time steps. The positive part is the Hilbert–Schmidt norm of the spatial symbol, a control-type metric. The Lorentzian form is the difference $\omega^2 - G_{\mathrm{HS}}(\sum_i q_i\alpha_i)$, which by Proposition 1 could not be obtained from any positive form alone. Theorem 2 makes the supplied evolution frequency a hyperbolicity direction for every first-order Hermitian symbol; Lemma 3 says that a quadratic cone with one time direction has signature $(1,d)$; and Theorem 4 says that the cone is quadratic, with all internal states on it, exactly when the propagation coefficients satisfy Euclidean Clifford relations. The Lorentzian Clifford algebra of Corollary 5 is then constructed, not supplied: the time-like generator is the mass term β.

The signature is no longer an input. The Clifford relations are equivalent to an observable property, a single shell, which means no birefringence and no internal state that propagates differently from the others. That property, isotropy of Q (which can be absorbed into the choice of spatial coordinates for one species) and the existence of a band point with a linear symbol remain inputs.

Remark 6 (Two boundary cases). For $d = 1$ and $\mu = 0$, $\alpha_1^2 = Q_{11}\mathbf{1}$ allows unequal multiplicities of $\pm Q_{11}^{1/2}$, and for $s = 1$ the symbol $H(q) = cq$ has a single branch. Its characteristic set is one chiral branch, with one future null direction. This is the chiral field used as an actuator in Ref. [45]: it supplies one null direction, and a full $(1,1)$ cone needs both chiralities. In the other direction, a quadratic cone can occur without a single shell. The Maxwell

field in Riemann–Silberstein form has the symbol $c\,\mathbf{S}\cdot q$ with spin-one matrices; its characteristic polynomial is $\omega(\omega^2 - c^2|q|^2)$, and the flat factor is removed by the Gauss constraint. Theorem 4 concerns symbols for which every internal state lies on the cone.

3.1 A forward certificate for approximate shells

The exact operator identity has a finite-error consequence. It requires bounds on implemented coefficients or their calibration errors, not merely access to effects whose ideal algebra contains those coefficients.

Proposition 7 (Squared-band certificate)**.** *Choose a positive quadratic form Q and Hermitian coefficients α_i, β. Let*

$$R_{ij} = \tfrac{1}{2}\{\alpha_i, \alpha_j\} - Q_{ij} I, \qquad R_{i\beta} = \{\alpha_i, \beta\}, \qquad R_\beta = \beta^2 - I.$$

Suppose their operator norms are bounded by calibrated numbers $\epsilon_{ij}, \epsilon_{i\beta}, \epsilon_\beta$. If the implemented symbol is $\widetilde{H}(q) = H(q) + E(q)$ with Hermitian E and $\|E(q)\| \le e(q)$, put

$$\Delta(q) = \sum_{i,j} |q_i q_j| \epsilon_{ij} + |\mu| \sum_i |q_i| \epsilon_{i\beta} + \mu^2 \epsilon_\beta,$$
$$L(q) = \sum_i |q_i| \|\alpha_i\| + |\mu| \|\beta\|, \qquad B(q) = \Delta(q) + 2L(q)e(q) + e(q)^2.$$

Every eigenvalue $\widetilde{\omega}$ obeys

$$\left|\widetilde{\omega}^2 - Q(q) - \mu^2\right| \le B(q).$$

With $S = Q(q) + \mu^2 \ge 0$, this gives $\left||\widetilde{\omega}| - \sqrt{S}\right| \le \sqrt{B(q)}$. If $S > B(q)$, a sharper radius is $B(q)/(\sqrt{S} + \sqrt{S - B(q)})$. The bounds hold uniformly on any specified window where their input bounds hold.

Proof. Expanding $H^2 - SI$ gives $\sum q_i q_j R_{ij} + \mu \sum q_i R_{i\beta} + \mu^2 R_\beta$, whose norm is at most Δ. The additional square error is $HE + EH + E^2$, of norm at most $2Le + e^2$. Evaluate on an eigenvector of the Hermitian $\widetilde{H}$. Finally, $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|}$ for nonnegative x, y; away from zero use the difference of squares. □

This is a forward dispersion certificate. It does not say that nearly anticommuting coefficients can be repaired to an exact Clifford representation, that the two frequency multiplicities are balanced, or that a finite effect menu implements operator-norm tomography. A calibration valid on a restricted state set is not a norm bound on the entire carrier. The distinction matters: a Hilbert–Schmidt trace identity can hold exactly while different internal states retain different propagation speeds.

4 Spatial dimension and the internal space

Theorem 8 (Dimension bound)**.** *Suppose Theorem 4 holds on $\mathbb{C}^s$, and let $n = d + [\mu \ne 0]$. Write $s = 2^\nu m$ with m odd, $\nu = \nu_2(s)$. Then $n \le 2\nu + 1$. Conversely, for every s and every $n \le 2\nu_2(s) + 1$ there are coefficients satisfying Theorem 4.*

Proof. Choose spatial coordinates in which $Q = c^2\mathbf{1}$; the coefficients transform linearly, so the relations are preserved. Then $e_i = \alpha_i/c$ and, if $\mu \neq 0$, $e_{d+1} = \beta$ are n pairwise anticommuting Hermitian involutions, which define a representation of the complex Clifford algebra $\mathbb{C}l_n$ on $\mathbb{C}^s$. This algebra is isomorphic to $M_{2^{n/2}}(\mathbb{C})$ for even n and to $M_{2^{(n-1)/2}}(\mathbb{C}) \oplus M_{2^{(n-1)/2}}(\mathbb{C})$ for odd n [8]. Every finite-dimensional representation is a direct sum of irreducible ones of dimension $2^{\lfloor n/2 \rfloor}$, so $2^{\lfloor n/2 \rfloor}$ divides s and $\lfloor n/2 \rfloor \leq \nu$. For the converse take an irreducible representation of $\mathbb{C}l_{2\nu+1}$, which has dimension 2^ν, tensor it with $\mathbf{1}_m$, and use any n of its generators. □

Corollary 9 (Qubit carriers)**.** *For $s = 2$: an isotropic single-shell symbol exists for massless propagation in $d \leq 3$ and for massive propagation in $d \leq 2$. Massive single-shell propagation in $d = 3$ requires $4 \mid s$; the minimal case $s = 4$ is the Dirac symbol $\alpha_i = \sigma_x \otimes \sigma_i$, $\beta = \sigma_z \otimes \mathbf{1}$, a pair of Weyl qubits coupled by the mass term.*

For $s = 2$ and $d = 3$ the symbol is $H(q) = c\,\sigma \cdot q$ up to a unitary and a rotation, and

$$\det\bigl(\omega\mathbf{1} - c\,\sigma \cdot q\bigr) = \omega^2 - c^2|q|^2. \tag{7}$$

This is the Lorentzian form of signature $(1, 3)$, obtained from the Pauli relations $\{\sigma_i, \sigma_j\} = 2\delta_{ij}$, which are Euclidean. No operator on $\mathbb{C}^2$ anticommutes with all three Pauli matrices, so $Cl(1, 3)$ has no representation on $\mathbb{C}^2$; the Lorentzian form exists anyway, as the characteristic form of the evolution.

Two forms on the same carrier. The identity $\det(x_0 I + x \cdot \sigma) = x_0^2 - |x|^2$ is preserved by determinant-one filtering, not by the positive Hilbert–Schmidt norm. For the massive Dirac carrier, the time-like generator in Corollary 5 is the mass coefficient, not a modular generator.

Proposition 10 (Full coupling of a qubit)**.** *Let $s = 2$, $\mu = 0$, and let the symbol satisfy Theorem 4 with traceless coefficients. The map $q \mapsto \sum_i q_i\alpha_i$ is an injective linear map from $\mathbb{R}^d$ into the three-dimensional space of traceless Hermitian 2×2 matrices. It is onto, so that every internal rotation generator is coupled to some direction of motion, if and only if $d = 3$.*

Proof. Injectivity is the linear independence assumed in Theorem 4; the target has dimension three. □

Proposition 10 is a precise version of a familiar observation: momentum space of a Weyl particle is identified with the Lie algebra of internal rotations of its qubit. It is a characterization under a stated premise, not a derivation of three dimensions. The premise, that no internal axis is left uncoupled, is a physical input. The following conjecture states what would have to be shown to turn it into a selection principle.

Conjecture 11 (Minimal carriers and three dimensions)**.** *In local unitary dynamics whose elementary internal carrier is a qubit and whose low-energy propagation is isotropic and single-shell, propagation that uses the full internal space selects $d = 3$, and massive propagation requires two carriers coupled by a chirality-exchanging term.*

Theorem 8 and Proposition 10 reduce this selection proposal to a physical reason for coupling every internal axis to momentum. Irreducibility alone does not supply it: $H = q_x\sigma_x + q_y\sigma_y$ generates the full matrix algebra, while the omitted σ_z is not a conserved charge. The three-dimensional Weyl automaton [13] and the minimal-directional-system argument of Ref. [15] motivate the proposal, but do not select three dimensions within the present class. A selection principle must distinguish the two-dimensional counterexample by stability, resources or formation, and must explain why a qubit is the relevant elementary carrier.

5 The lattice level: causal polytope, three speeds and nodes

5.1 Group velocities lie in the causal polytope

Proposition 12 (Group velocities and the neighbourhood)**.** *For the step* (1)*, let b be a band and k_0 a point where the eigenvalues of $W(k_0)$ are distinct. Then $\nabla_k\omega_b(k_0) \in \tau^{-1}\operatorname{conv} N$.*

Proof. Take $\tau = 1$. Let X be the position operator and ψ finitely supported. Then $\langle X\rangle_n = \langle W^n\psi, XW^n\psi\rangle$ satisfies $\langle X\rangle_n - \langle X\rangle_0 = \sum_{j<n}\langle W^j\psi, VW^j\psi\rangle$ with $V = W^*[X,W]$. From Eq. (1), $([X,W]\psi)(x) = \sum_h hA_h\psi(x-h)$, so V is bounded and translation invariant with symbol $V(k) = W(k)^*\, i\nabla_k W(k)$. Write $W(k) = \sum_b e^{-i\omega_b(k)}P_b(k)$ wherever the eigenvalues are distinct, which is almost everywhere once it holds at one point, since the discriminant of the characteristic polynomial is real-analytic in k. For each such k the Cesàro means $n^{-1}\sum_{j<n}W(k)^{-j}V(k)W(k)^j$ converge to $\sum_b P_bVP_b$, and differentiating $W = \sum_b e^{-i\omega_b}P_b$ with $P_b(\nabla P_b)P_b = 0$ gives $P_bVP_b = \nabla\omega_b\, P_b$. By bounded convergence,

$$\frac{\langle X\rangle_n - \langle X\rangle_0}{n} \longrightarrow \bar v(\psi) = \sum_b \int \nabla\omega_b(k)\, \langle\hat\psi(k), P_b(k)\hat\psi(k)\rangle\, \frac{dk}{|B|},$$

where B is the Brillouin zone. On the other hand $W^n\psi$ is supported in $\operatorname{supp}\psi + nN$, so for every unit vector u, $u\cdot\langle X\rangle_n \le \max_{x\in\operatorname{supp}\psi} u\cdot x + n\max_{h\in N} u\cdot h$. Hence $u\cdot\bar v(\psi) \le h_N(u) := \max_{h\in N} u\cdot h$. The map $\hat\psi \mapsto \bar v(\psi)$ is continuous in L^2 because $|\nabla\omega_b| \le \|V(k)\|$ is bounded, and finitely supported states are dense, so the inequality holds for all ψ. Taking $\hat\psi = P_b(k)\varphi$ with φ concentrating at k_0 gives $u\cdot\nabla\omega_b(k_0) \le h_N(u)$ for every u, which is membership in $\operatorname{conv} N$. □

The ballistic scaling used in the proof is the expectation-value form of the weak limit theorem for quantum walks [16].

Corollary 13 (Speed of a massless read-out cone)**.** *If a band has a first-order massless symbol at k_* whose cone is round with speed c, then $c \le \tau^{-1}r_{\mathrm{in}}(\operatorname{conv} N)$, where r_{in} is the inradius.*

Proof. Near k_* the group velocity tends to $c\,\hat q$ along every direction $\hat q$, so $c\hat q \in \tau^{-1}\operatorname{conv} N$ for all $\hat q$. □

5.2 The split-step walk

Consider the coined walk that applies a coin rotation and then a conditional shift,

$$W_\theta(k) = \begin{pmatrix} e^{-ika}\cos\theta & -e^{-ika}\sin\theta \\ e^{ika}\sin\theta & e^{ika}\cos\theta \end{pmatrix}, \qquad 0 < \theta < \frac{\pi}{2}, \tag{8}$$

with neighbourhood $N = \{a, -a\}$ and step time τ.

Proposition 14 (Three speeds of one walk)**.** *For the walk* (8)*:*

1. $\cos(\omega\tau) = \cos\theta\cos(ka)$*;*
2. *the support speed is* a/τ*, and the largest group velocity is* $(a/\tau)\cos\theta$*, attained at* $ka = \pm\pi/2$*;*
3. *near* $k = 0$*,* $\omega(k)^2\tau^2 = \theta^2 + \theta\cot\theta\,(ka)^2 + O((ka)^4)$*; the rest energy is* $E_0 = \hbar\theta/\tau$*, the inertial mass* $m_* = \hbar\tau\tan\theta/a^2$*, and the Lorentzian read-out speed defined by* $E_0 = m_* c_{\text{eff}}^2$ *is* $c_{\text{eff}} = (a/\tau)(\theta\cot\theta)^{1/2}$*;*
4. $\cos\theta < (\theta\cot\theta)^{1/2} < 1$*.*

Proof. Item 1: $\det W_\theta = 1$ and $\operatorname{tr} W_\theta = 2\cos\theta\cos(ka)$. Item 2: the support statement is Eq. (1) with $N = \{\pm a\}$. Differentiating item 1, $v = (a/\tau)\cos\theta\sin(ka)/\sin(\omega\tau)$, and $v^2 \le (a/\tau)^2\cos^2\theta$ is equivalent to $\cos^2\theta\cos^2(ka) \le \cos^2(ka)$, with equality only at $\cos(ka) = 0$. Item 3: differentiating item 1 twice at $k = 0$, where $\omega\tau = \theta$ and $\omega' = 0$, gives $\tau\sin\theta\,\omega'' = a^2\cos\theta$; then $\frac{1}{2}(\omega^2)'' = \omega\omega''$, $m_* = \hbar/\omega''$, and $E_0/m_* = \omega(0)\omega''(0)$. Item 4 is $\sin\theta\cos\theta < \theta < \tan\theta$. □

The Lorentzian speed read from the rest energy and the inertia of slow particles lies strictly between the largest transport speed and the support speed. No wave packet reaches it, because $v \le (a/\tau)\cos\theta < c_{\text{eff}}$; and the strict causal cone is wider. In the scaling limit $a, \tau \to 0$ with $a/\tau = c$ and $\theta/\tau = \mu$ fixed, all three speeds tend to c and the dispersion tends to $\omega^2 = c^2q^2 + \mu^2$ [11, 12]. At $\theta = 0$ the walk is a pair of exact shifts, with $\omega = \pm k$ and one speed. The deviations at finite mass are $1 - c_{\text{eff}}^2\tau^2/a^2 = \theta^2/3 + O(\theta^4)$ and $1 - \cos^2\theta = \theta^2 + O(\theta^4)$. The consequences for mass read-outs are developed in the companion paper on mass and energy [47].

5.3 The Weyl automaton on the body-centred cubic lattice

The minimal three-dimensional automaton of D'Ariano and Perinotti [13] has $s = 2$, neighbourhood $N = \{\ell(\pm1, \pm1, \pm1)/\sqrt{3}\}$ of eight vectors of length ℓ, and symbol

$$W(k) = e^{-i\kappa_x\sigma_x}\, e^{-i\kappa_y\sigma_y}\, e^{-i\kappa_z\sigma_z}, \qquad \kappa = \frac{\ell k}{\sqrt{3}}, \tag{9}$$

with $\cos(\omega\tau) = c_x c_y c_z - s_x s_y s_z$, where $c_i = \cos\kappa_i$, $s_i = \sin\kappa_i$. Expanding, $c_x c_y c_z = 1 - |\kappa|^2/2 + O(|\kappa|^4)$ and $s_x s_y s_z = \kappa_x\kappa_y\kappa_z + O(|\kappa|^5)$, so

$$\omega\tau = |\kappa| + \frac{\kappa_x\kappa_y\kappa_z}{|\kappa|} + O(|\kappa|^3). \tag{10}$$

At leading order $W(k) \simeq \mathbf{1} - i\sigma \cdot \kappa$, the symbol is $c\,\sigma \cdot k$ with $c = \ell/(\sqrt{3}\tau)$, and Eq. (7) gives a round Lorentzian cone. Isotropy fails at relative order $|\kappa|$ through the parity-odd term in Eq. (10). Table 1 collects analytic properties and numerical unitary wave-packet diagnostics. It is not a table of records obtained from a physical measuring instrument.

Property	**Value**
Causal polytope $\tau^{-1}\,\mathrm{conv}\,N$	cube $\lvert x_i\rvert \le \ell/(\sqrt{3}\tau)$ per unit time (analytic)
Inradius of the cube	$\ell/(\sqrt{3}\tau) = c$: Corollary 13 is saturated (analytic)
Group velocity components	at most c, attained on the cube faces
Largest group speed	$(3/2)^{1/2}c$; the cube corners, at speed $\sqrt{3}c$, are not reached by group velocities
Band-touching points in the Brillouin zone	four: at quasi-energy 0, $k = 0$ (chirality $+1$) and $\kappa = -\frac{\pi}{2}(1,1,1)$ (chirality -1); at quasi-energy π/τ, one of each chirality
One-site initial state	probability outside the cube: zero to machine precision; outside the round cone of radius ct: 0.50 after 16 steps, 0.47 after 24
Gaussian packets near $k = 0$, 24 steps	probability outside radius $ct + R_0$ (R_0 the 99% radius at $t = 0$): 0.084, 0.023, 0.008 for momentum widths $\sigma_\kappa = 0.4$, 0.25, 0.15

Table 1: Weyl automaton (9). Chirality is the sign of the determinant of the linear map $q \mapsto$ coefficient of $-i\sigma$ in $\pm W(k_0 + q)$. The packets carry a fixed spinor and therefore populate both bands.

5.4 What the read-out map adds

The lattice results show three ways in which the dynamics leaves the Lorentzian read-out underdetermined.

First, at finite mass the speed of the read-out depends on the observable that defines it. In the walk (8), rest energy and inertia give c_{eff}, the fastest wave packets give $(a/\tau)\cos\theta$, and the support front gives a/τ (Proposition 14).

Second, the Lorentzian cone governs low-momentum states only. In the Weyl automaton a state prepared on one site contains all momenta and leaves almost half of its probability outside the round cone, while it never leaves the cube; the leakage falls as the momentum width shrinks (Table 1).

Third, the Brillouin zone contains several band-touching points. In the Weyl automaton there are two at quasi-energy zero with opposite chiralities, as the doubling theorem for lattice Hamiltonians would lead one to expect [17], although that theorem does not apply directly to automata. A low-energy observer reads a Weyl cone of one chirality or the other depending on which point the prepared states are near. The dynamics does not choose; the preparation and the instruments do. In one dimension a single chiral cone is possible, since the shift by one site is a legitimate automaton; whether a strictly local isotropic automaton in three dimensions can carry a single node is a question we leave open.

So in this model the read-out map cannot be removed. What can be said precisely is what it consists of: a band point, a momentum window around it, and an observable defining the speed, each with computable consequences. This matches the treatment of access in HAFF, where what a physical reduction resolves depends on the channel and not only on the algebra it generates [42]. Here the Lorentzian geometry read by an observer is the characteristic geometry of the momentum window that its couplings reach.

5.5 Finite records certify one walk's dictionary

The choice of a read-out need not leave all its parameters inaccessible. The companion mass–energy paper realizes the same coin rotation by a fresh Gaussian oscillator, computes its exact open-channel linewidth, and reads its rest phase through controlled mixed-memory instruments [47]. On a supplied finite periodic ring, a normalized zero mode is preserved by a spatially uniform field coupling. Independent X and Z record trials at one and ten steps resolve the phase alias and certify θ. The ring, uniform support, axes, gain, clock, fresh ports and record archive are priced inputs.

Let that actual protocol give $\theta \in [\theta_-, \theta_+]$ at its stated confidence, with supplied a, τ. Monotonicity of $\tan\theta$, $\cos\theta$ and $C(\theta) = \theta\cot\theta$ gives simultaneous intervals

$$E_0 \in \frac{\hbar}{\tau}[\theta_-, \theta_+], \qquad m_* \in \frac{\hbar\tau}{a^2}[\tan\theta_-, \tan\theta_+],$$
$$c_{\text{eff}}^2 \in \frac{a^2}{\tau^2}[C(\theta_+), C(\theta_-)], \qquad v_{\max} \in \frac{a}{\tau}[\cos\theta_+, \cos\theta_-].$$

At $\theta_+ = \pi/2$ the inertial upper bound is infinite. With the priced angle radius $b = .004296823$ at 95% confidence derived in Ref. [47], true angle .3 and $a = \tau = 1$, every confidence interval on that event lies inside $[.3 - 2b, .3 + 2b]$. The implied intervals are

$$c_{\text{eff}} \in [.9838969, .9856636], \qquad v_{\max} \in [.9527616, .9578408], \qquad c_{\text{support}} = 1.$$

The support speed is supplied, not measured by the phase protocol. The other two are model implications of its angle certificate. The full preparation, pulse, read-memory and reference budget is given with that protocol.

These are parameter inferences for the prescribed ideal walk, informed by an actual open-drive experiment. They are not its noisy long-time transport velocities: nonzero-momentum transport requires the real Bloch channel and inter-momentum coherences. Nor does calibration prove that the dictionary is Lorentzian throughout the band. The companion finite-angle window theorem gives precisely where a Lorentzian mass reading becomes negative and must be rejected. Thus an observer's interface can be calibrated, can distinguish otherwise conflated speeds, and can fail at a testable boundary.

6 Several species: when do the cones become one?

Theorem 4 gives every single-shell species its own form Q. For isotropic species in one frame, a common light cone requires their speeds to agree. In a free model nothing makes them

agree: each speed is a parameter of the step. This section asks whether coupling the species can make the speeds equal, so that one cone is an output of the dynamics. The answer depends on the kind of coupling. Off-diagonal mixing cannot produce a single quadratic principal shell while its distinct diagonal shells are held fixed. Interactions can change those diagonal propagation laws: in the mediator model their speeds converge when each coupling has divergent accumulated strength toward low energy.

6.1 Mixing does not merge cones

Write the principal symbol of two species in a direction u as $A_1(u)$ on $\mathbb{C}^{s_1}$ and $A_2(u)$ on $\mathbb{C}^{s_2}$, both single-shell, $A_a(u)^2 = Q_a(u)\mathbf{1}$. A one-particle coupling adds a block off-diagonal term, which may itself depend linearly on q.

Theorem 15 (Mixing and the single shell)**.** *For every $s_1 \times s_2$ matrix V, the symbol $A = \begin{pmatrix} A_1 & V \\ V^* & A_2 \end{pmatrix}$ satisfies*

$$\min_{\lambda\in\mathbb{R}} \big\|A^2 - \lambda\mathbf{1}\big\|_{\mathrm{HS}}^2 \;\geq\; \tfrac{1}{2}\min(s_1, s_2)\,\big(Q_1(u) - Q_2(u)\big)^2. \tag{11}$$

Hence the coupled system has a single shell only if $Q_1 = Q_2$. Moreover the largest eigenvalue of A is at least the largest eigenvalue of either block and the smallest is at most the smallest of either block.

Proof. The diagonal blocks of $A^2 - \lambda$ are $Q_1 - \lambda + VV^*$ and $Q_2 - \lambda + V^*V$, and the Hilbert–Schmidt norm of a block matrix is at least that of its diagonal blocks. If V has rank r, then VV^* and V^*V share the nonzero eigenvalues $\sigma_1^2, \ldots, \sigma_r^2$ and have $s_1 - r$ and $s_2 - r$ zero eigenvalues. Using $(x + a)^2 + (x + b)^2 \geq \frac{1}{2}(a - b)^2$ for each shared eigenvalue, and for $k = \min(s_1, s_2) - r$ pairs of zeros, the squared norm is at least $\frac{1}{2}(r + k)(Q_1 - Q_2)^2$. The last statement is Cauchy interlacing for the principal submatrices A_1 and A_2. □

The bound (11) does not depend on the strength of the coupling. A mass-type coupling, independent of q, leaves the principal symbol untouched, but can change low-momentum band slopes. For example,

$$H(q) = \begin{pmatrix} v_1 q & g \\ g & v_2 q \end{pmatrix}, \qquad \omega_\pm(q) = \tfrac{1}{2}(v_1 + v_2)q \pm \sqrt{g^2 + \tfrac{1}{4}(v_1 - v_2)^2 q^2}.$$

For $g \neq 0$ both slopes at $q = 0$ equal $(v_1 + v_2)/2$, although the two distinct principal speeds remain v_1, v_2. Equal slopes at one band point are not a common quadratic characteristic cone. An on-site lattice coin can likewise change the effective logarithm of a step; the fixed-block theorem applies only after its principal blocks have been specified. For a chiral principal velocity matrix with fixed diagonal entries, the Schur–Horn theorem makes the eigenvalues majorize those entries, so off-diagonal mixing cannot reduce their variance. The interacting route below instead changes the propagation coefficients themselves.

6.2 Loops average speeds

Lemma 16 (Loops of one speed)**.** *Let a self-energy be a loop whose internal lines all have Lorentzian propagators of speed c and whose vertices are covariant under the Lorentz group of speed c. Then the coefficient of* $\ln\Lambda$ *in it is Lorentz covariant with speed c, for any fixed regular cutoff shape enclosing the origin and scaled by Λ.*

Proof. The substitution $k \mapsto (k_0, k/c)$ makes the integrand covariant under the standard Lorentz group. For a homogeneous integrand f of degree $-D$, the coefficient of $\ln\Lambda$ is the flux of the divergence-free field $xf(x)$ through a surface enclosing the origin, and it is the same for every such surface. □

The smallest interacting model is a real scalar of speed u coupled to N Dirac fermions of speed v by $g\phi\bar\psi\psi$ in $3+1$ dimensions. We integrate out a shell of spatial momenta with the frequency unrestricted; the logarithmic coefficients are scheme independent at this order. By Lemma 16 the fermion loop adds $\frac{Ng^2}{4\pi^2 v^3}(p_0^2+v^2p^2)\,dl$ to the inverse scalar propagator $p_0^2+u^2p^2$, where $l=\ln(\Lambda/E)$, so the squared scalar speed is replaced by a convex combination of u^2 and v^2. The fermion self-energy contains one fermion and one scalar line. Its frequency integrals are

$$J_0=\int\frac{t^2\,dt}{(t^2+v^2)(t^2+u^2)^2}=\frac{\pi}{2u(v+u)^2},\qquad J_s=\int\frac{dt}{(t^2+v^2)(t^2+u^2)^2}=\frac{\pi(v+2u)}{2vu^3(v+u)^2},\tag{12}$$

and the ratio of the spatial to the temporal coefficient gives the speed toward which v moves.

Proposition 17 (Speed flow in the Yukawa model)**.** *At one loop,*

$$\frac{dv}{dl}=a\Big(\frac{v+2u}{3}-v\Big),\quad a=\frac{g^2}{4\pi^2u(u+v)^2},\qquad \frac{du^2}{dl}=\frac{Ng^2}{4\pi^2v^3}\,(v^2-u^2),\tag{13}$$

and the coupling obeys $dg^2/dl=-(g^4/2\pi^2)\big[1/(vu(v+u))+N/(2v^3)+1/(u(v+u)^2)\big]$. At $u=v$ the linearized flow of $u-v$ has rate $(6N+1)g^2/(24\pi^2v^3)$, and $u+6Nv$ is conserved to linear order.

For $N=1$ these equations coincide with those of Anber and Donoghue [18], who derived them by dimensional regularization. The momentum-shell calculation gives the same logarithmic coefficients. Each speed moves toward a target that lies between the speeds of the lines in its loop: the fermion toward $(v+2u)/3$, the scalar toward v. The scalar moves $6N$ times faster, so the common speed lies close to the initial fermion speed. Since the Yukawa coupling is marginally irrelevant, $g^2\simeq 8\pi^2c_*^3/[(2N+3)l]$, where c_* is the common limiting speed and

$$\frac{|u-v|}{v}\;\propto\;\Big[\ln\frac{\Lambda}{E}\Big]^{-(6N+1)/(3(2N+3))},\tag{14}$$

with exponent $7/15$ for one Dirac fermion. Integrating the full flow to $l=10^9$ gives log–log slopes -0.4667 for $N=1$ and -0.7576 for $N=4$, as predicted.

Theorem 18 (Speeds converge through a common mediator)**.** *Let fermion species of speeds $v_1, \dots, v_n$ couple to one scalar of speed u with prescribed nonnegative, locally integrable functions $g_1^2(l), \dots, g_n^2(l)$, each fermion obeying the speed equations in* (13) *and u^2 receiving the sum of the fermion loops. The theorem is conditional on these coupling histories; a multi-species beta function must separately establish their accumulated strengths. Assume positive initial speeds. Let D be the difference between the largest and the smallest of $u, v_1, \dots, v_n$, and M_0 the initial largest speed. Then the largest speed never increases, the smallest never decreases, and*

$$D(l) \le D(0) \exp\Big(-\frac{1}{24\pi^2 M_0^3} \int_0^l \min_a g_a^2 \, dl'\Big). \tag{15}$$

If $\int g_a^2 \, dl < \infty$, the speed v_a changes in total by at most $D(0) \int g_a^2 \, dl/(24\pi^2 m_0^3)$, with m_0 the initial smallest speed.

Proof. Each v_a moves toward u at rate $\frac{2}{3}a_a$, and u^2 moves toward a weighted mean of the v_a^2, so u decreases when it is the largest speed and increases when it is the smallest. Consider the upper Dini derivative D^+ of the maximum minus the minimum. If both extremes are fermions $v_a \ge v_b$, then $D^+ \le \frac{2}{3}[a_a(u - v_a) - a_b(u - v_b)] \le -\frac{2}{3}(\min_c a_c)D$; if the maximum is u, then $D^+ \le -\frac{2}{3}a_b(u - v_b)$; the case of u minimal is symmetric. All speeds stay in $[m_0, M_0]$, where $\frac{2}{3}a_a \ge g_a^2/(24\pi^2 M_0^3)$, and Grönwall's inequality gives (15). The last statement follows from $|v_a'| \le \frac{2}{3}a_a D(0)$ and $a_a \le g_a^2/(16\pi^2 m_0^3)$. □

The estimate gives a rate when the weakest simultaneous coupling has divergent integral. Convergence itself needs less.

Corollary 19 (Intermittent interactions)**.** *In the one-loop star model of Theorem 18, suppose the nonnegative functions g_a^2 are locally integrable and initial speeds are positive. Every speed converges. Species connected to the mediator by edges with $\int_0^\infty g_a^2 \, dl = \infty$ have the same limiting speed as the mediator. In particular, all speeds converge to one value if each such integral diverges, even when the couplings are never active simultaneously.*

Proof. Write $v_a' = A_a(u - v_a)$ and $u' = \sum_a B_a(v_a - u)$, with

$$A_a = \frac{g_a^2}{6\pi^2 u(u + v_a)^2}, \qquad B_a = \frac{g_a^2(u + v_a)}{8\pi^2 u v_a^3}.$$

For an active edge, $B_a/A_a = \frac{3}{4}(1 + u/v_a)^3$. All speeds stay in $[m_0, M_0]$, so the two directional rates are uniformly comparable, and their integrals diverge exactly when $\int g_a^2$ does. This is a type-symmetric continuous-time consensus system. The cut-balanced convergence theorem [1], applied to the coefficients along the solution, gives convergence and equality of limits within each component of the infinite-integral interaction graph. The theorem allows state-dependent coefficients. It does not provide the same rate as (15) when $\min_a g_a^2$ vanishes. □

The accumulated interaction across renormalization scales decides the outcome. The variable $l = \ln(\Lambda/E)$ is scale, not laboratory time; intermittent edges here are active on different scale intervals. An identically uncoupled species keeps its cone; a finite integrated

coupling need not erase an initial difference. Marginally irrelevant couplings remove differences as a power of $\ln(\Lambda/E)$; near an interacting fixed point they fall as a power of E/Λ, with the one-loop exponent $7g_*^2/(24\pi^2 c_*^3)$ for $N = 1$. In $4-\varepsilon$ dimensions the Yukawa fixed point gives $(6N+1)\varepsilon/[3(2N+3)]$ at this order. The weak-coupling model must remain applicable along this flow. A mediator that breaks rotations can instead drive speeds apart, as in nematic criticality of d-wave superconductors [20].

The common speed lies between the smallest and the largest initial speed. Each initial speed of a massless round cone is at most the inradius of conv N per step (Corollary 13), so the emergent cone lies inside the causal polytope.

6.3 Fixed points and the number of cones

Near an interacting fixed point the question has an answer that does not rely on perturbation theory.

Theorem 20 (Speed splittings at a conformal fixed point)**.** *Let the infrared fixed point be a unitary conformal field theory in $D = d+1 \geq 3$ dimensions, Lorentz invariant for one cone, and consider local perturbations invariant under translations, spatial rotations and CPT. Up to total derivatives the Lorentz-violating perturbations are $\int d^D x\, O_{0\cdots 0}$ for symmetric traceless primaries of even spin $\ell \geq 2$. Their couplings satisfy $D-\Delta_O \leq 2-\ell \leq 0$; equality in the first inequality means that the current is conserved, while a marginal Lorentz-violating coupling requires a conserved spin-2 current. If the stress tensor is the only conserved spin-2 current, all independent spin-2 speed perturbations are irrelevant. When the excited non-conserved spin-2 modes have a smallest gap $\gamma_2 = \min(\Delta_O - D) > 0$, their slowest linearized decay is $(E/\Lambda)^{\gamma_2}$.*

Proof. Descendants integrate to zero. A primary transforms in a finite-dimensional irreducible representation of the Lorentz group, whose complexification is that of $SO(D)$, and because $(SO(D), SO(D-1))$ is a Gelfand pair only the symmetric traceless representations contain a rotation-invariant component, once each. CPT acts on a Hermitian rank-ℓ tensor by $(-1)^\ell$, which removes odd ℓ, and scalars are Lorentz invariant. The unitarity bound $\Delta \geq \ell + D - 2$ for $\ell \geq 1$ holds with equality exactly for conserved currents [21, 22]. The perturbation $\int T_{00}$ rescales time and changes no relative speed. □

The stability argument is established in Refs. [23, 24]; it assumes the Lorentz-invariant fixed point and does not prove that every microscopic dynamics reaches it. Additional conserved spin-2 currents are the marginal exceptions. In three dimensions, for exactly two such currents, a suitable basis generates separate translations and the two families of current correlators factorize [25, Section 8]. This is dynamical separation of the current sectors, not necessarily a tensor product of the full theories: a shared singlet constraint can remain. Nor does counting currents count sectors in general; two free scalar fields already have three conserved spin-2 currents. The result needed here is the local criterion for suppressing relative-speed perturbations, while identifying all independently propagating sectors requires further structure.

In two dimensions several propagation sectors can retain different speeds, as in spin–charge separation. A Sugawara decomposition gives an additional sector only when $T - T_J$

is nonzero. Under its positivity, spectrum and finite-signal-speed assumptions, the scale-to-Lorentz theorem permits either one Lorentz factor or several [26]; it does not forbid a common cone from being selected dynamically. Establishing one cone therefore requires information about the actual sectors and their interactions, rather than the dimensionality alone.

6.4 What the lattice adds

For number-conserving automata whose interaction preserves the supplied neighbourhood, each distinguishable particle stays in $x_j + nN$ after n steps. The centre-of-mass version of Proposition 12 therefore bounds every isolated bound-state band by $\operatorname{conv} N$ per step. Changing the neighbourhood changes that bound. Equalization of low-energy cones does not itself change the support cone.

For a loop dominated by one linear band point, the logarithmic coefficient carries that point's read-out speed by Lemma 16. Finite ultraviolet matching terms need not have the same ratio of temporal and spatial coefficients. A Lorentz-violating regulator can leave corrections of order $g^2/16\pi^2$ [28], subsequently reduced only at the infrared rates above.

Composites need not inherit a constituent's read-out cone. Take distinguishable walkers with $\theta_A = .3$, $\theta_B = .5$, $a = \tau = 1$, first apply both free coined steps, and then multiply the coincident-position sector by $e^{+.4i}$. With total momentum K conjugate to $(x_A + x_B)/2$, the isolated positive band at $K = 0$ has $\omega_0 = .7614121$ and $m_* = 1/\omega_0'' = .9023$, compared with the free threshold .8 and constituent mass sum .8556. Thus $c_{\text{eff}}^2 = \omega_0\omega_0'' = .8439$, below the constituent values $\theta_A \cot\theta_A = .9698184$ and $\theta_B \cot\theta_B = .9152439$. Periodic relative-coordinate calculations with 91 and 131 sites agree at the quoted precision. They also track a localized band beyond $|K| = .68$; no such endpoint or global maximum speed is inferred from this local mass calculation. Contact interactions can produce bound molecules in quantum walks [19]. In the example above, the separately computed quasienergy and curvature do not obey the relativistic binding-energy–inertia relation.

A common cone can also be kinematic. The minimal isotropic Weyl automaton is unique on the body-centred cubic lattice [13], so all massless species built from it share $c = \ell/(\sqrt{3}\tau)$; a Dirac automaton built from two of them has first-order speed $c(1 - m^2)^{1/2}$ in lattice units, and photons built from pairs of Weyl automata have the dispersion $2|n_{k/2}|$ and the same low-energy cone [27]. For an electron on a lattice of Planck spacing, $m \approx 4 \times 10^{-23}$ and the mass correction is of order 10^{-45}. Loop matching can generate relative-speed corrections of order $g^2/16\pi^2$ unless a symmetry forbids them. Contact binding also changes the composite cone in the example above, but its size is model dependent and carries no general loop suppression factor.

6.5 How much running is needed

With marginal couplings the logarithmic rate (14) is far too slow to produce the observed equality of speeds. Reducing a difference by a factor 10^{-13} requires $\ln(\Lambda/E) \approx 10^{28}$ for one Dirac fermion with $g^2 = 4\pi$ at the cutoff, and about 10^{12} even for a hundred fermions, while $\ln(M_{\text{Pl}}/m_e) = 51.5$. Anber and Donoghue reached the same qualitative conclusion [18].

Conjecture 21 (One cone needs a strongly coupled regime)**.** *In local unitary dynamics with several interacting gapless species, the relative speed difference of two sectors at scale E is $\delta c(E) \simeq \delta c(\Lambda) \exp(-\int_E^\Lambda \gamma \, d\ln\mu)$, where γ is the anomalous dimension of their relative stress tensor and $\delta c(\Lambda)$ is of order $g^2/16\pi^2$ from lattice matching even when the step is common. Equalities of order 10^{-15} then require $\int \gamma \, d\ln\mu \gtrsim 30$ above accessible energies, that is, a strongly coupled regime with γ of order one over many decades, or a symmetry that removes $\delta c(\Lambda)$. The largest residual differences sit between the most weakly coupled sectors.*

Theorems 18 and 20 tie equalization to interaction strength; lattice matching regenerates relative-speed differences, while marginal running removes them only logarithmically. The quantitative estimates above and the power-law emergence in strongly coupled holographic models [24] support the conjecture. Anber and Donoghue's observation that the weakest interactions leave the largest differences also makes gravitational waves against light a sensitive comparison [18].

A viable ultraviolet realization must place the strongly coupled range consistently with the observed weakly coupled particle description and experimental bounds. A composite realization is one possibility, not a necessary consequence of the conjecture. A symmetry route, such as non-relativistic supersymmetry protecting dimension-four terms, can avoid strong coupling but must survive symmetry breaking. Gravity is shared by all species yet is irrelevant at low energy; this flow cannot align its cone there. The relation of the gravitational-wave and matter cones therefore requires an ultraviolet mechanism or a protecting symmetry.

6.6 Consequences

The mediator model makes a common speed a conditional output of divergent accumulated interactions. The conformal argument establishes stability against relative-speed perturbations near a supplied Lorentz-invariant fixed point; it does not establish formation of that fixed point. This separates three inputs to the higher-dimensional Consciousness construction: each species' shell and signature, the alignment of their cones, and the actual infrared regime. Its boosted-ball family requires a common cone. In the two-dimensional HAFF gravity model [44], identifying the matter and Jackiw–Teitelboim cones remains an input; the two-dimensional sector theorem neither supplies nor forbids that identification.

7 Relations to other frameworks and papers

7.1 Algebraic frameworks

Sequential operations are represented here by linear maps under ordinary composition, which is associative by definition. Jordan and octonionic models need additional bracketing and composition rules before tensor products, subsystems or error correction can be transferred [30, 31, 32]; this is a statement of scope, not a no-go theorem. Von Neumann and C^*-algebras are compatible with everything above, since the step (1) and its symbol live in a concrete C^*-algebra; such algebras do not single out a quadratic form until a dynamics is

given. Clifford algebras are distinguished in this paper by a dynamical property, the single shell of Theorem 4, rather than by being chosen as an encoding.

7.2 Thermodynamic gravity, modular flow and spectral triples

Jacobson's derivation of the Einstein equation assumes local Lorentzian causal structure, local equilibrium and an entropy–area relation [3], and entanglement derivations of linearized gravity work on a Lorentzian background [33, 34]. The present model supplies local Lorentzian cones for the low-momentum sector of a local unitary dynamics. It supplies neither equilibrium nor the entropy–area relation, and Section 5.4 shows that its cones depend on the momentum window.

Local algebras in relativistic quantum field theory are of Type III [35, 36]. For the vacuum and wedge regions the Bisognano–Wichmann theorem identifies modular flow with the boost subgroup [37, 48]; the identification presupposes the Lorentzian structure it describes. Whether the modular data of states of a lattice automaton reproduce the boosts of its low-energy cone in the scaling limit is open and could be computed in the walk of Section 5.2.

In Connes' noncommutative geometry a Dirac operator with Euclidean Clifford symbol encodes Riemannian geometry [38], and Lorentzian versions are an active problem [39, 40, 41]. Theorem 4 says that using such a Dirac operator as a Hamiltonian, $i\partial_t\psi = D\psi$, produces the Lorentzian characteristic cone of signature $(1, d)$ on time times space. We have not worked out the relation to the Krein-space approaches cited.

7.3 The Q-RAIF papers and HAFF

The persistence paper of the series [45] distinguishes three quadratic structures: a designer's positive descent metric G, a positive work form K derived from field stress, and the Lorentzian read-out form. The present paper adds the source of the third, $\eta = \omega^2 - G_{\mathrm{HS}}$, and shows why neither G nor K could supply it. The actuator field of that paper is the chiral case of Remark 6; the stress it carries propagates along a single null ray, which is the matter source used in the HAFF paper on gravity [44].

The representation paper [46] asks for a quadratic-space morphism before signatures can be compared. Under the single-shell condition of Theorem 4, the spatial symbol $q \mapsto \sum_i q_i\alpha_i$ supplies an isometry from $(\mathbb{R}^d, Q)$ into the generators with G_{HS}. The converse fails: the trace identity $G_{\mathrm{HS}}(H(q)) = Q(q)$ does not imply the operator identity $H(q)^2 = Q(q)\mathbf{1}$. For example, the local unitary walk $W(k) = \mathrm{diag}(e^{-ik}, e^{-2ik})$ has $H(q) = q\,\mathrm{diag}(1, 2)$. Its symbol is an isometry for $Q(q) = 5q^2/2$, but its two branches have different speeds and do not share a quadratic shell. The operator identity, rather than its trace, is the condition that connects this metric to a common propagation cone.

The mass–energy paper [47] constructs a covariant positive-energy representation of the continuum single-shell symbol, closing the spectral interface used for its mass theorem. Selection of that sector remains physical input, and a finite lattice window is not invariant under arbitrary boosts. Its field pulse and actual phase protocol connect occupation, lifetime and finite records; Section 5.5 imports that certificate rather than treating an ideal band as a measurement.

8 Discussion

Ingredient of a Lorentzian read-out	**Status in local unitary propagation**
Evolution time	supplied; unitarity makes its frequency a hyperbolicity direction (Theorem 2)
Signature $(1, d)$	derived for any quadratic cone (Lemma 3)
Clifford relations	equivalent to a single shell (Theorem 4); Lorentzian generators constructed (Corollary 5)
Single shell, isotropy, linear band point	inputs; observable and testable in a given model
Spatial dimension d	input; bounded by the internal space (Theorem 8)
Speed c	a property of the dynamics; at finite mass it depends on the defining observable (Proposition 14)
Read-out map	input: band point, momentum window and speed observable (Section 5.4)
Common speed of several species	conditional output of the mediator model; stable against irrelevant relative-speed perturbations near a supplied Lorentz-invariant fixed point, whose formation is not derived (Theorems 18, 20)
Lattice, neighbourhood, step	inputs

Table 2: Inputs of a Lorentzian read-out in local unitary propagation.

Table 2 separates the derived signature from the remaining inputs: lattice data, absence of birefringence and the observer's read-out map. Deriving the first two requires a formation model; no property of the dynamics alone specifies the third.

The problems that follow have the most leverage. Section 6 answers the question of a common cone for several species under stated conditions; what remains there is the anomalous dimension γ_2 in a strongly coupled lattice model, which Conjecture 21 needs. Another is chirality: whether a strictly local, isotropic three-dimensional automaton can have a single node, which would remove one component of the read-out map. The mass-setting interface now has a nonempty field realization and finite certificate, but renewed drives, full noisy spatial transport and covariant support remain physical tasks. A fourth is the modular route: computing the modular flow of a lattice state and comparing it with the boosts of the low-energy cone. Position-dependent coins, which produce effective curved metrics, would connect the model to the gravitational questions of HAFF.

9 Conclusion

Propagation supplies the relative sign in $\omega^2 - G_{\mathrm{HS}}$; a positive control metric alone cannot. The single-shell criterion, dimension bound and calibrated residuals connect this form to a specified carrier. Finite angle records imply separated ideal dictionary-speed intervals, while

noisy transport remains a different observable. For several species, the mediator model aligns cones under accumulated-coupling conditions; the conformal argument tests their stability, not the creation of a fixed point. These are the respective interfaces to mass–energy read-out, the common-cone geometry of Consciousness II and the still-supplied matter cone of HAFF-D. Formation of the carrier, clock and accessible momentum window remains a dynamical question.

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Thermodynamic Stability Constraints on Persistent Open Quantum Subsystems: Control Algebra, Physical Actuation and Energy Cost

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Persistent control has to be realized by an interaction, with an error and a cost. We construct one such actuator: a degenerate bit coupled to a mode of a free chiral field through $U = \exp[-i\lambda X(b+b^\dagger)]$ and a smooth switch g. For independent diagonal bit inputs, the same unitary fixes the flip error, added stress $(g')^2/4$, switching work $\lambda^2\bar{\omega}$ and a positive quadratic cost on switch profiles. A flip with failure ϵ needs mean drive occupation at least $\theta_\epsilon(1-2\epsilon)/[4\sqrt{\epsilon(1-\epsilon)}]-\frac{1}{2}$, where $\theta_\epsilon = \arccos(2\epsilon-1)$; the leading cost is $\pi/(4\sqrt{\epsilon})$. Displaced squeezed states nearly attain it, whereas classical drives need order ϵ^{-1}. A calibrated feedback task with scheduled fresh independent drives has an exact stationary loss, coupling rate and stability proof, with one acceptance condition joining calibration records to drive energy. A second bound prices response readout: if two branches differ only by probe delays bounded in difference by $\Delta\tau$, binary error ϵ requires $E\Delta\tau \geq \hbar[1-2\sqrt{\epsilon(1-\epsilon)}]/\zeta$, with $\zeta = \max_{x>0}(1-\cos x)/x$. Retained ancillas and joint decoding do not remove this bound, and a three-level construction saturates it across the allowed signed-delay families. A retained oscillator also supports a thermal branch: a coherent echo protects held quantum content, a mixed qubit port reads the coupling through one binary outcome, and a finite Gibbs bank restores the oscillator while retaining every spent output. These energy and memory accounts do not identify exported energy with removed entropy. The descent metric, physical work form and Lorentzian propagation form remain distinct.

I. INTRODUCTION

Interaction with a large environment drives a quantum system towards decoherence and thermalization [1, 6]. A subsystem that keeps a non-equilibrium steady state (NESS) does so through a control process paid for by its surroundings, and such processes can be modelled as feedback [7–10]. This paper asks what persistence requires of the controls. The question has two parts. One concerns the operator structure the control dynamics needs in order to remain Lyapunov stable. The other concerns how a control operation is physically realized, what error it carries and what it costs.

Persistence alone does not select a Clifford algebra or an indefinite metric. Gradient descent requires a positive control metric, and standard dissipative and coherent control stabilizes target states without Clifford structure [15–18]. Section III distinguishes associativity of operator composition, positivity of a descent metric and the optional Clifford encoding of a chosen quadratic form.

The physical part is the main subject. In an abstract control model a pulse is a free exact unitary. In a physical model it is an interaction with a field, and the field brings quantum fluctuations, energy and stress with it. We work out one such actuator exactly. A degenerate two-level system couples to the derivative of a free chiral scalar field through a smooth switch. The same unitary then fixes the flip channel and its failure probability, the outgoing stress and switching work, and a positive quadratic form on switch profiles that is derived from the field rather than chosen.

Three results follow from this construction. Theorem 6 bounds the energy cost of an accurate flip for every input state of the driving mode; squeezed Gaussian drives nearly attain the bound, while Proposition 8 shows that classical drives cannot approach it. Theorem 11 places the actuator in a calibrated feedback task and gives a complete closed-loop persistence theorem, including bath noise, reading noise and gate failure. Proposition 12 turns the task requirement into one condition on calibration records and drive energy together, so a shortage of either resource can make the task infeasible.

An interface can also be queried while it carries unknown quantum content. Section VII gives the corrected parity instrument and its collection-dependent damage. A coherent controlled-parity echo protects the body for arbitrary mode inputs and returns a finite meter; its mode access, control and readout are separate physical resources. Resonant contacts realize a retained receiver and a finite packet collector, with explicit occupation, pulse-duration and finite-area-error accounts. A positive-frequency finite-band field supplies a continuous counterpart: a finite prescribed drive prepares its matched packet, with certified collection loss and separately counted receiver renewal.

For the retained oscillator, Section VII C also supplies a thermal alternative to vacuum inquiry. A mixed qubit port produces an actual binary record after the echo, and a finite bank of Gibbs oscillators restores the active port without discarding its old correlations. The record gain, read backreaction and pulse durations are derived from this same protocol. Its bank and clocks are finite resources; it is a different implementation from travelling-field collection.

Control also needs the response to come back through an instrument. A bound based on mean energy, in the quantum-speed-limit tradition [34], prices a specified translation-only readout in Theorem 14. It permits mixed probes, retained ancillas and joint decoding. Its role is to constrain the physical response-to-program route of HAFF-D, not to identify every sensing channel with a time delay.

Within the Q-RAIF series this paper carries the control and persistence side of the physical interface. Its outgoing stress is the matter source whose gravitational response is studied in the HAFF paper on gravity [29], and its feedback task is the one analysed in the Imprint Chain [36], where the corrective flip was a supplied exact pulse. Nothing here makes an observer the author of geometry. Persistence is analysed inside a supplied field theory, and geometric reconstruction in HAFF depends on its own algebraic and dictionary inputs [28].

Section II states the feedback problem. Section III records the algebraic constraints. Section IV constructs the actuator, Section V proves the energy–accuracy results, and Section VI treats persistence, calibration, response readout and their resource accounts. Section VIII compares the quadratic structures, Section IX relates the results to other work, and Section X discusses what remains open.

II. PERSISTENCE AS A FEEDBACK PROBLEM

Consider $\mathcal{H}_{\mathrm{tot}} = \mathcal{H}_R \otimes \mathcal{H}_E$ with reduced dynamics

$$\dot{\rho}_R = -i[H_{\mathrm{eff}}, \rho_R] + \mathcal{D}[\rho_R]. \tag{1}$$

The controls must be specified in the composed system dynamics, rather than inferred from a formal stationarity condition on a protocol cost.

Standard quantum control writes $H_{\mathrm{ctrl}} = \sum_k u_k(t) G_k$ with generators of a Lie algebra $\mathfrak{g}$, $[G_i, G_j] = i f_{ijk} G_k$ [14]. The Lie algebra fixes the accessible directions. The size of a correction is quantified by a positive cost, norm or information metric supplied separately. This separation is standard and is not a defect of Lie-algebraic control. Section IV shows where such a cost can come from physically: from the stress of the field that implements the control.

III. WHAT THE OPERATOR ALGEBRA MUST SUPPLY

A. Associativity as a boundary of the model

Remark 1 (Associativity as a control-model boundary)**.** *The master-equation model represents sequential controls by linear maps under ordinary composition, which is associative by definition [1]. This is a declared scope condition, not a theorem excluding every non-associative dynamics. Jordan or octonionic models would need additional bracketing, subsystem and composition rules before a control calculation could be transferred [12, 13]. Q-RAIF Paper A [30] uses the same operator-composition boundary for propagation.*

B. A positive metric for Lyapunov descent

For an auxiliary gradient controller on parameters λ consider the descent rule

$$\dot{\lambda} = -\Gamma\, G^{-1} \nabla_\lambda V, \qquad \Gamma > 0. \tag{2}$$

Proposition 2 (Positive descent metric)**.** *The controller contribution to $\dot{V}$ under Eq. (2) is $-\Gamma(\nabla_\lambda V)^T G^{-1} (\nabla_\lambda V)$. It is nonpositive for every gradient exactly when G^{-1}, equivalently the invertible symmetric matrix G, is positive definite. If G has a negative direction, a gradient along it makes the contribution positive.*

Proof. The first statement is the chain rule. A symmetric quadratic form is nonnegative on all vectors precisely when it is positive semidefinite, and an invertible positive semidefinite matrix is positive definite. □

The descent mechanism therefore selects positive definiteness, not Lorentzian signature. Channel types such as unitary, dephasing and dissipative can be labelled by a discrete index, a direct-sum decomposition or supplied weights; their qualitative difference does not need a negative norm.

Parameter descent is not yet the physical state equation. With drift H_0, dissipator $\mathcal{D}$ and controls $H_{\mathrm{ctrl}}(\lambda) = \sum_i u_i(\lambda) G_i$, the closed loop is

$$\dot{\rho} = -i[H_0 + H_{\mathrm{ctrl}}(\lambda), \rho] + \mathcal{D}[\rho], \qquad \dot{\lambda} = -\Gamma G^{-1} \nabla_\lambda V, \tag{3}$$

and for differentiable $V(\rho, \lambda)$

$$\dot{V} = D_\rho V[-i[H_0 + H_{\rm ctrl}, \rho] + \mathcal{D}[\rho]] - \Gamma(\nabla_\lambda V)^T G^{-1}(\nabla_\lambda V). \tag{4}$$

A stability proof must control the first term as well. Setting a control amplitude equal to a gradient component gives no general sign.

Dissipative and coherent control can stabilize target states with positive costs and Lie or operator-algebraic controls [15–18]. Von Neumann and C^*-algebras are compatible with this model; neither supplies a preferred quadratic form without a dynamics. These examples also show why persistence alone imposes no indefinite Clifford structure.

C. Clifford algebra as a parameterization

If one chooses a quadratic space (V, q) and requires control generators to satisfy $v^2 = q(v)\mathbf{1}$, the universal property identifies $Cl(V, q)$ as the minimal algebra for that presentation [11]. This is a definitional fact. It does not show that persistence, channel discrimination or Lyapunov descent imposes those relations. Pauli controls of a qubit satisfy $\{\sigma_i, \sigma_j\} = 2\delta_{ij}\mathbf{1}$, the relations of $Cl(3, 0)$, so the Clifford package is a convenient description of qubit controls. A particular feedback model may favour some representation because of its controls and costs, but that preference has to be shown in its closed-loop dynamics.

a. State stability. Equation (4) is the continuous-time obligation: a parameter descent controls only its own contribution, while drift, dissipator and controlled state evolution remain. For full-rank states and a fixed full-rank target, $\dot{D}(\rho\|\rho_*) = \mathrm{tr}[\dot{\rho}(\log\rho - \log\rho_*)]$. Theorem 11 establishes stability instead for a discrete composed channel with a physically realized, imperfect correction.

IV. A CONTROL FLIP REALIZED BY A FIELD COUPLING

A. The coupling and its channel

The field-coupling and drive bounds in this and the next section use $\hbar = 1$: $\bar{\omega}N$ and $\bar{\omega}\lambda^2$ are energies in these units. In physical units they are $\hbar\bar{\omega}N$ and $\hbar\bar{\omega}\lambda^2$; the field Hamiltonian and stress carry the same overall $\hbar$. Quadratures, occupation and pulse areas are dimensionless. The later delay, entropy and retained-oscillator accounts display $\hbar$ explicitly.

The controlled system is a two-level system with zero free Hamiltonian, Pauli operators X and Z. A bosonic mode with $[b, b^\dagger] = 1$ and quadratures $Q = b + b^\dagger$, $P = i(b^\dagger - b)$, $[Q, P] = 2i$, interacts through

$$U = \exp[-i\lambda XQ], \qquad \lambda > 0. \tag{5}$$

In the eigenbasis $X|x\rangle = x|x\rangle$, $U = \sum_x |x\rangle\langle x| \otimes e^{-i\lambda xQ}$, and $e^{-i\lambda xQ}$ is the displacement $D(-i\lambda x)$. Write $\chi_\sigma(k) = \mathrm{tr}(\sigma e^{-ikQ})$ for the characteristic function of Q in a mode state σ.

Proposition 3 (Channel of the coupling)**.** *Let the mode start in a state σ independent of the bit. The bit channel multiplies the matrix element $\rho_{xx'}$ in the X basis by $\chi_\sigma(\lambda(x - x'))$. For bit states diagonal in Z, its output Z statistics agree with a flip having failure probability*

$$\epsilon = \tfrac{1}{2}\big(1 + \mathrm{Re}\,\chi_\sigma(2\lambda)\big). \tag{6}$$

For a coherent input $|\alpha\rangle$, $\alpha > 0$, the factor is

$$e^{-\lambda^2(1 - xx') - 2i\alpha\lambda(x - x')}. \tag{7}$$

If $2\lambda\alpha = \pi/2$ and $N = \alpha^2$, the channel on every bit state is

$$\mathcal{F}_N(\rho) = s_N X\rho X + (1 - s_N)\rho, \qquad s_N = \frac{1 + e^{-2\lambda^2}}{2}, \qquad \lambda^2 = \frac{\pi^2}{16N}, \tag{8}$$

so that $\epsilon_N = 1 - s_N = \big(1 - e^{-\pi^2/(8N)}\big)/2$.

Proof. Tracing the mode gives the factor $\mathrm{tr}(\sigma\, e^{i\lambda x' Q} e^{-i\lambda x Q}) = \chi_\sigma(\lambda(x-x'))$. A Z-diagonal state $(I+zZ)/2$ has $\rho_{+-} = z/2$ in the X basis, so the output mean of Z is $z\,\mathrm{Re}\,\chi_\sigma(2\lambda)$, which is $(1-2s)z$ for a flip with success probability s. For the coherent input, $D(\beta)D(\alpha) = e^{(\beta\alpha^* - \beta^*\alpha)/2} D(\alpha+\beta)$ gives $e^{-i\lambda x Q}|\alpha\rangle = e^{-i\alpha\lambda x}|\alpha - i\lambda x\rangle$. The overlap is the coherent-state inner product $\langle \alpha - i\lambda x' | \alpha - i\lambda x\rangle = e^{-\lambda^2(1-xx') - i\alpha\lambda(x-x')}$ times the Weyl phase $e^{-i\alpha\lambda(x-x')}$ from composing the displacements; this is Eq. (7). At $2\lambda\alpha = \pi/2$ the off-diagonal factor is $-e^{-2\lambda^2}$, and $1 - 2s_N = -e^{-2\lambda^2}$ yields Eq. (8) on all inputs. □

The channel $\mathcal{F}_N$ differs from the ideal Ad_X by $\epsilon_N(\mathrm{id} - \mathrm{Ad}_X)$, so its half diamond-norm distance from the ideal flip is ϵ_N, and its average gate infidelity is $2\epsilon_N/3$. These are different error conventions. The coherent family above needs infinite occupation for an exact flip; Section V shows that this holds for every input state.

B. A local switch on a free field

Equation (5) becomes a local operation once the mode is defined by a switch. Take a right-moving free massless scalar in $1+1$ dimensions with null coordinate u,

$$\phi(u) = \int_0^\infty \frac{d\omega}{\sqrt{4\pi\omega}} \big(a_\omega e^{-i\omega u} + a_\omega^\dagger e^{i\omega u}\big), \qquad [a_\omega, a_{\omega'}^\dagger] = \delta(\omega - \omega'), \tag{9}$$

current $j = \partial_u \phi$ and stress $T_{uu} =: j^2 :$. The current obeys $[j(u), j(v)] = \frac{i}{2}\delta'(u-v)$. For a real smooth compactly supported switch g define

$$Q_g = \int g(u)\, j(u)\, du = \lambda(b + b^\dagger), \qquad \lambda^2 = \int_0^\infty \frac{\omega}{4\pi} |\hat g(\omega)|^2 d\omega, \qquad \hat g(\omega) = \int g(u) e^{-i\omega u} du, \tag{10}$$

where $b = \int_0^\infty f^*(\omega) a_\omega d\omega$ with $f(\omega) = i\sqrt{\omega/(4\pi)}\,\hat g^*(\omega)/\lambda$ is a normalized mode. Its mean frequency is $\bar\omega = \int \omega |f|^2 d\omega$, and the current contains it as $j(u) = \psi(u) b + \psi^*(u) b^\dagger + j_\perp(u)$ with $\psi(u) = -i\int_0^\infty \sqrt{\omega/(4\pi)}\, f(\omega) e^{-i\omega u} d\omega$.

A probe at fixed position with interaction Hamiltonian $g(t) X j(t)$ evolves by $U = e^{i\vartheta} \exp(-iXQ_g)$ for a real phase ϑ. The reason is that $[g(t)Xj(t), g(t')Xj(t')]$ is a multiple of the identity, because $X^2 = 1$ and the current commutator is a c-number, so the Magnus series stops at second order and the second-order term is a phase. This uses three conditions: a single probe, a fixed coupling operator X, and a probe free Hamiltonian that is zero or commutes with X. Nonzero field frequencies are harmless. A probe Hamiltonian that does not commute with X makes $X(t)$ time dependent, and several probes keep relative phases; neither case is covered. A point-like limit of the switch needs regularization, so we use smooth profiles. This exact Weyl evolution of a degenerate detector is a known method [21, 22]; what we add is its use as a priced feedback actuator.

C. Stress, work and a derived quadratic cost

Proposition 4 (Stress of one encounter)**.** *For the unitary $U = \exp(-iXQ_g)$,*

$$U^\dagger j(u) U = j(u) + \tfrac{1}{2} X g'(u), \qquad U^\dagger T_{uu}(u) U = T_{uu}(u) + g'(u)\, X j(u) + \tfrac{1}{4} g'(u)^2. \tag{11}$$

The expected added stress is $g'(u)\langle X j(u)\rangle + g'(u)^2/4$. The cross term vanishes if the bit and the fresh field input are initially independent and $\mathrm{tr}(\rho X) = 0$, or if their joint state has no coherence in the bit's Z basis. A Z-diagonal bit marginal alone is not sufficient.

Proof. $[Q_g, j(u)] = \int g(v)[j(v), j(u)] dv = -\frac{i}{2} g'(u)$ is a c-number, so $e^{iXQ_g} j(u) e^{-iXQ_g} = j(u) + \frac{1}{2} X g'(u)$ exactly. Squaring, using $X^2 = 1$, and noting that a c-number shift leaves normal ordering unchanged gives the stress identity. Under independence $\langle Xj\rangle = \mathrm{tr}(\rho X)\langle j\rangle = 0$, even when the input beam has $\langle j\rangle \neq 0$. If the joint state is block diagonal in the bit's Z basis, $X \otimes A$ has zero expectation for every field operator A. For the last claim, $(|\uparrow\rangle|0\rangle + |\downarrow\rangle|1_b\rangle)/\sqrt{2}$ has bit marginal $I/2$ but $\langle Xb\rangle = 1/2$, hence $\langle X j(u)\rangle = \mathrm{Re}\,\psi(u) \neq 0$; with a relative phase i the integrated cross term equals $-\lambda\bar\omega$ and changes the work below. □

Under the stated joint condition the added stress is

$$F_g(u) = \tfrac{1}{4} g'(u)^2 = 4\lambda^2 [\mathrm{Im}\,\psi(u)]^2, \tag{12}$$

the second form following from $2\lambda \operatorname{Im}\psi = g'/2$. If the interaction is off before and after the encounter and the bit Hamiltonian is zero, the energy delivered by the external switching agent equals the change of field energy,

$$W_{\text{switch}} = \frac{1}{4}\int g'(u)^2 du = \frac{1}{4\pi}\int_0^\infty \omega^2 |\hat{g}(\omega)|^2 d\omega = \lambda^2 \bar{\omega}, \tag{13}$$

by Parseval's identity and $|f|^2 = \omega|\hat{g}|^2/(4\pi\lambda^2)$. This holds for every input state of the field that satisfies the joint condition.

Proposition 5 (Derived quadratic cost)**.** *For $g = \sum_i a_i g_i$ with real smooth compactly supported g_i, $W_{\text{switch}} = \sum_{ij} a_i K_{ij} a_j$ with $K_{ij} = \frac{1}{4}\int g_i' g_j' du$. If the g_i are linearly independent, K is positive definite.*

Proof. $a^T K a = \frac{1}{4}\int (\sum_i a_i g_i')^2 du$ vanishes only if $\sum_i a_i g_i$ is constant; compact support makes the constant zero, and linear independence gives $a = 0$. □

D. Coherent drive

Let the mode b carry a coherent state $|\alpha\rangle$, $\alpha = \sqrt{N}$, with all orthogonal modes in vacuum, independent of a Z-diagonal bit. The output field marginal is then

$$\rho_F^{\text{out}} = \tfrac{1}{2}|\alpha - i\lambda\rangle\langle\alpha - i\lambda| + \tfrac{1}{2}|\alpha + i\lambda\rangle\langle\alpha + i\lambda|, \tag{14}$$

independently of the bit's bias, although bit and field are correlated afterwards. The incoming field energy is $E_{\text{in}} = N\bar{\omega}$ and the outgoing energy is $(N+\lambda^2)\bar{\omega}$. A wave packet is not an eigenmode of the field Hamiltonian, so these are mean values; the field Hamiltonian is not $\bar{\omega} b^\dagger b$. The incoming stress is $4N[\operatorname{Re}\psi(u)]^2$ and the encounter adds F_g of Eq. (12). The gravitational use of these profiles is developed in Ref. [29].

V. THE ENERGY COST OF AN ACCURATE FLIP

How much drive is needed for a flip of given accuracy? The coherent family gives $N = \pi^2/(16\ell)$ with $\ell = -\frac{1}{2}\log(1-2\epsilon)$, about $\pi^2/(16\epsilon)$. The following theorem bounds every input state.

Theorem 6 (Energy–accuracy bound)**.** *Let σ be any state of the mode b with finite mean occupation $N = \operatorname{tr}(\sigma b^\dagger b)$, let $\lambda > 0$, and let the coupling* (5) *flip Z-diagonal bit states with failure probability $\epsilon \in (0, \frac{1}{2})$ as in Eq.* (6)*. With $\theta_\epsilon = \arccos(2\epsilon - 1) \in (\pi/2, \pi)$,*

$$N \geq \frac{\theta_\epsilon(1-2\epsilon)}{4\sqrt{\epsilon(1-\epsilon)}} - \frac{1}{2}, \tag{15}$$

$$N + \lambda^2 \geq \frac{\theta_\epsilon}{4\sqrt{\epsilon(1-\epsilon)}} - \frac{1}{2}. \tag{16}$$

Both right-hand sides equal $\pi/(4\sqrt{\epsilon}) - 1 + O(\sqrt{\epsilon})$. The second inequality is a bound on an occupation-plus-activation budget. It equals the outgoing single-mode energy in units of $\bar{\omega}$ only with the independent zero-cross-term field inputs of Section IV C; correlations can change the actual switching work. An exact flip, $\epsilon = 0$, is impossible at finite occupation.

Proof. Write $C = \cos(2\lambda Q)$ and $U_2 = e^{-2i\lambda Q}$, so that $\langle C\rangle = \operatorname{Re}\chi_\sigma(2\lambda) = 2\epsilon - 1$ and $b^\dagger b = (Q^2 + P^2 - 2)/4$. Finite N makes both second moments finite.

Position quadrature. The function $G(c) = \arccos^2 c$ is convex on $[-1, 1]$: with $c = \cos y$, $y \in (0, \pi)$, one finds $G''(c) = 2(1 - y\cot y)/\sin^2 y \geq 0$, since $y\cot y \leq 1$. For every real q, $(2\lambda q)^2 \geq G(\cos 2\lambda q)$, with equality when $|2\lambda q| \leq \pi$ and strict inequality otherwise because $G \leq \pi^2$. Jensen's inequality for the spectral distribution of Q gives

$$\langle Q^2\rangle \geq \frac{G(\langle C\rangle)}{4\lambda^2} = \frac{\theta_\epsilon^2}{4\lambda^2}. \tag{17}$$

Momentum quadrature. Since $U_2^\dagger P U_2 = P - 4\lambda$, we have $[U_2, P] = 4\lambda U_2$ on the domain of P. Writing $\delta U_2 = U_2 - \langle U_2\rangle$, $\delta P = P - \langle P\rangle$ and applying Cauchy–Schwarz to the Hilbert–Schmidt products $\operatorname{tr}(\sigma\, \delta U_2\, \delta P)$ and $\operatorname{tr}(\sigma\, \delta P\, \delta U_2)$

gives $|\langle[U_2, P]\rangle| \le 2\,\Delta U_2\,\Delta P$ with $(\Delta U_2)^2 = 1 - |\langle U_2\rangle|^2$. Hence $\Delta P \ge 2\lambda|\chi|/\sqrt{1-|\chi|^2}$ with $\chi = \chi_\sigma(2\lambda)$. Because $|\chi| \ge |\operatorname{Re}\chi| = 1 - 2\epsilon$ and $x/\sqrt{1-x^2}$ increases,

$$\langle P^2\rangle \ge (\Delta P)^2 \ge \frac{\lambda^2(1-2\epsilon)^2}{\epsilon(1-\epsilon)}. \tag{18}$$

In particular $|\chi| < 1$ at finite N, which excludes $\epsilon = 0$.

Combination. Adding the two bounds,

$$N \ge \frac{1}{4}\left[\frac{\theta_\epsilon^2}{4\lambda^2} + \frac{\lambda^2(1-2\epsilon)^2}{\epsilon(1-\epsilon)}\right] - \frac{1}{2}, \tag{19}$$

and minimizing over λ by the arithmetic–geometric mean inequality gives Eq. (15). Adding λ^2 changes the coefficient of λ^2 to $\frac{1}{4}[(1-2\epsilon)^2/(\epsilon(1-\epsilon)) + 4] = 1/[4\epsilon(1-\epsilon)]$, and the same step gives Eq. (16). The expansion uses $\theta_\epsilon = \pi - 2\sqrt{\epsilon} + O(\epsilon^{3/2})$. □

The bound is nearly attained by Gaussian drives.

Proposition 7 (Coherent and Gaussian families)**.** *(a) The coherent family with $2\lambda\alpha = \pi/2$ needs $N = \pi^2/(16\ell)$, $\ell = -\frac{1}{2}\log(1-2\epsilon)$. (b) For a displaced pure Gaussian state whose Q has mean 2α and variance $V > 0$, $\chi(2\lambda) = e^{-4i\lambda\alpha}e^{-2\lambda^2 V}$ and $N = \alpha^2 + (V + V^{-1} - 2)/4$. At $2\lambda\alpha = \pi/2$ and failure ϵ,*

$$\min N = \frac{\sqrt{1+\pi^2/(4\ell)} - 1}{2}, \qquad \min(N+\lambda^2) = \frac{1}{2}\sqrt{\Big(1+\frac{\pi^2}{4\ell}\Big)(1+4\ell)} - \frac{1}{2}, \tag{20}$$

attained at $V = (1+\pi^2/(4\ell))^{-1/2}$ and $V = [(1+4\ell)/(1+\pi^2/(4\ell))]^{1/2}$ respectively. Both equal $\pi/(4\sqrt{\epsilon}) - \frac{1}{2} + O(\sqrt{\epsilon})$, so their gap to Theorem 6 tends to half a quantum.

Proof. The Gaussian characteristic function gives χ. The failure condition reads $\lambda^2 V = \ell$, so $\alpha^2 = \pi^2/(16\lambda^2) = aV$ with $a = \pi^2/(16\ell)$ and $N = (1+4a)V/4 + 1/(4V) - \frac{1}{2}$, minimized at $V = (1+4a)^{-1/2}$. Adding $\lambda^2 = \ell/V$ replaces $1/(4V)$ by $(1+4\ell)/(4V)$. □

Squeezing is what makes the better scaling possible.

Proposition 8 (Classical drives)**.** *If $\sigma = \int P(\beta)|\beta\rangle\langle\beta|\,d^2\beta$ with a probability density $P \ge 0$, then a flip with failure ϵ needs*

$$N \ge \frac{(\pi - \arccos(1-2\epsilon))^2}{16\ell} = \frac{\pi^2}{16\epsilon}\big(1 - 4\sqrt{\epsilon}/\pi + O(\epsilon)\big). \tag{21}$$

Proof. Each coherent component has $\chi_\beta(2\lambda) = e^{-2\lambda^2}e^{-4i\lambda\operatorname{Re}\beta}$, so $\int P\cos(4\lambda\operatorname{Re}\beta) = -(1-2\epsilon)e^{2\lambda^2}$. This must be at least -1, hence $\lambda^2 \le \ell$. The convexity argument of Theorem 6 applied to $y = 4\lambda\operatorname{Re}\beta$ gives $16\lambda^2\int P(\operatorname{Re}\beta)^2 \ge (\pi - \arccos[(1-2\epsilon)e^{2\lambda^2}])^2 \ge (\pi - \arccos(1-2\epsilon))^2$. Finally $N = \int P|\beta|^2 \ge \int P(\operatorname{Re}\beta)^2$ and $\lambda^2 \le \ell$. □

At fixed λ, a constrained minimum can be certified by a supporting ground-state problem. If ψ_μ is the ground state of $b^\dagger b + \mu\cos(2\lambda Q)$ and $\langle C\rangle_{\psi_\mu} = 2\epsilon - 1$, then every σ with the same $\langle C\rangle$ satisfies $\operatorname{tr}(\sigma b^\dagger b) \ge \langle b^\dagger b\rangle_{\psi_\mu}$, because $\operatorname{tr}\sigma(b^\dagger b + \mu C)$ is at least the ground energy. Table I lists numerical variational minima on a position grid, further minimized over λ, alongside the analytic bounds and the explicitly attainable Gaussian half-turn family. Halving the grid checks resolution stability, not the continuous global minimum. The computed states are close to displaced squeezed states, while the coherent half-turn family uses substantially more occupation.

Remark 9 (Scope of the occupation)**.** *(i) For a multimode field state, Cauchy–Schwarz applied to $b = \int f^* a_\omega d\omega$ gives $\langle b^\dagger b\rangle \le \omega_h^{-1}\langle H_F\rangle$ with $\omega_h^{-1} = \int |f|^2\omega^{-1}d\omega$, so the field energy is at least ω_h times the bound* (15)*; under the zero-cross-term condition of Section IV C, $\omega_h \le \bar\omega$ then makes field energy plus switching work at least ω_h times the bound* (16)*. (ii) The occupation is the drive level present at the encounter, not energy absorbed by the bit. The bit is degenerate; the beam leaves with energy $(N+\lambda^2)\bar\omega$ in the coherent case. Whether drive energy is lost depends on whether the outgoing beam is recovered. (iii) Squeezed inputs can have locally negative normal-ordered stress, and their preparation and phase reference are additional resources.*

Recovering the beam is not free of consequences. Two bits placed one after the other on the same chiral packet, with the same switch profile in u, experience $U = \exp[-i\lambda(X_1 + X_2)Q]$.

ϵ	Classical bound (21)	Coherent, $2\lambda\alpha = \pi/2$	Bound (15)	Grid variational	Gaussian half-turn (20)
0.1	3.496	5.529	1.165	1.847	1.904
0.05	8.589	11.709	2.278	2.921	2.958
0.01	53.53	61.07	6.742	7.314	7.330
0.001	591.7	616.2	23.80	24.32	24.33

TABLE I. Mean drive occupation needed for a flip with failure probability ϵ. Including one activation's switching work at $\epsilon = 0.01$: bound (16) 6.890, grid variational minimum 7.470, Gaussian half-turn 7.487, coherent 61.08. The grid minima change by less than 2×10^{-3} when the spacing is halved; this is not a certified global error bound.

Proposition 10 (Shared drive correlates errors)**.** *For a shared drive each bit's marginal channel is the single-encounter channel. For two bits starting at $Z = +1$, $\langle Z_1 Z_2 \rangle = \frac{1}{2}[1 + \mathrm{Re}\,\chi(4\lambda)]$. For the coherent family of Proposition 3 this equals $(1 + e^{-8\lambda^2})/2$, which exceeds the product $\langle Z_1 \rangle \langle Z_2 \rangle = e^{-4\lambda^2}$ obtained with independent drives.*

Proof. In the X basis the joint element with indices $(x_1 x_2), (x_1' x_2')$ is multiplied by $\chi(\lambda(x_1 + x_2 - x_1' - x_2'))$; tracing one bit recovers Proposition 3. $Z_1 Z_2$ connects (x_1, x_2) with $(-x_1, -x_2)$, which gives arguments $\pm 4\lambda$ and 0. For the coherent family $\chi(4\lambda) = e^{-8\lambda^2} e^{-2\pi i}$, and $(1 + y^2)/2 > y$ for $0 < y < 1$. □

At $N = 16$ the single failure probability is 0.0371, and the probability that both flips fail is 0.00393, against 0.00138 for independent drives. Drive energy can be shared only at the price of correlated failures. The persistence analysis below uses a fresh drive at each opportunity.

VI. PERSISTENCE WITH A FINITE-ENERGY ACTUATOR

A. The task

We use the feedback task of the Imprint Chain [36] with the corrective flip replaced by the coupling of Section IV. The persistent subsystem is a bit restricted to Z-diagonal states, with value $W = \pm 1$ and target $W = +1$. A bath step flips W with probability $d \in (0, \frac{1}{2})$; put $c = 1 - 2d$. A reading prepares a fresh probe, copies W by a CNOT, rotates the probe by U_a with $a \in \{0, 1\}$, and measures it with effects $E_{y|s} = \frac{1}{2}(I + y\nu\sigma_s)$, $\sigma_0 = Z$, $\sigma_1 = X$, where the instrument label S is hidden and fixed. Then

$$\Pr(Y = y \mid W = w, S = s, a) = \begin{cases} (1 + yw\nu)/2, & a = s, \\ 1/2, & a \neq s. \end{cases} \tag{22}$$

A calibration stage selects a; its mismatch probability $q = \Pr(a \neq S)$ is fixed along a trajectory. In each task step the system reads Y, accepts feedback with an independent probability $h \in (0, 1]$, switches on the coupling if accepted and $Y = -1$, and then the bath acts. The loss is the occupation probability $L_t = \Pr(W_t = -1)$.

Two declarations fix the resource accounting. Each opportunity has its own drive, initially independent of the bit and of earlier records, and the outgoing field never returns to the controller's reading access. The drive is scheduled: the same incoming beam arrives at every opportunity and the command only decides whether g is switched on. On-demand emission of a whole packet would need a router and a timing scheme that we do not model.

B. Closed-loop stability

Theorem 11 (Persistence with a finite-energy actuator)**.** *Let each coupling act on the bit as the flip channel $sX\rho X + (1 - s)\rho$, which is the case exactly when $\chi_\sigma(2\lambda)$ is real, for instance for the coherent and Gaussian families at $2\lambda\alpha = \pi/2$. Put $\Lambda = c(1 - hs)$ and $B = chs\nu$, and let the task start from W uniform.*

(a) Conditional on $a = S$, the bit evolves by a fixed channel $\mathcal{T}$ on Z-diagonal states, $m \mapsto \Lambda m + B$ for the mean m of W. Conditional on $a \neq S$, $m \mapsto \Lambda m$.

(b) $\mathcal{T}$ has the unique fixed point $\rho_* = (I + m_* Z)/2$, $m_* = B/(1 - \Lambda)$. The trace distance to ρ_* contracts by exactly Λ per step, and $D(\rho_t \| \rho_*)$ is nonincreasing and tends to zero.

(c) The stationary loss is

$$L_\infty(q) = \frac{1}{2} - \frac{(1-q)B}{2(1-\Lambda)}. \tag{23}$$

(d) The stationary rate of actual couplings is

$$p(q) = \frac{h}{2}\left[1 - \frac{(1-q)\nu B}{1-\Lambda}\right] = h\nu L_\infty(q) + \frac{h(1-\nu)}{2}, \tag{24}$$

of which a fraction s succeed. Over a window of T steps the expected number of couplings is $\frac{h}{2}\big[T - (1-q)\nu\frac{B}{1-\Lambda}\big(T - \frac{1-\Lambda^T}{1-\Lambda}\big)\big]$.

(e) For two calibrations with $q' < q$, $L_\infty(q) - L_\infty(q') = (q-q')B/[2(1-\Lambda)]$ and $p(q) - p(q') = h\nu[L_\infty(q) - L_\infty(q')]$.

Proof. With $a = S$ a reading has $\mathbb{E}[WY] = \nu$, and the coupling is attempted with probability $h(1-Y)/2$ and flips with probability s. Thus the mean after feedback is $m - 2s\,\mathbb{E}[Wh(1-Y)/2] = (1-hs)m + hs\nu$, and the bath multiplies it by c. With $a \neq S$, Y is independent of W and the mean after feedback is $(1-hs)m$. Fresh probes and fresh drives make each step the same channel on the bit marginal. Trace distance between Z-diagonal states is $|m-m'|/2$, which the affine map multiplies by $\Lambda < 1$. Data processing gives $D(\mathcal{T}\rho\|\rho_*) \le D(\rho\|\rho_*)$, and $m_t \to m_*$ gives convergence. Starting from $m_0 = 0$, the branch means are $m_*(1-\Lambda^t)$ and 0, weighted by $1-q$ and q; this gives (c). The attempt probability is $h(1-\nu m)/2$ for $a = S$ and $h/2$ otherwise, which gives (d) after averaging and using $(1-q)m_* = 1 - 2L_\infty$. Part (e) follows from (c) and (d). □

Gate failure enters the recursion through hs, while the attempt rate keeps h: a failed gate uses the whole encounter. Stability here does not rely on a gradient rule or a metric G. It follows from the contraction of the composed channel, and the relative entropy to the stationary state is a Lyapunov function of the full closed loop, which is what Section III C 0 a required of a proof. Whether persistence is useful is a separate matter: the stationary polarization is positive only on the calibrated branch and grows with s.

Table II uses $\nu = 0.8$, $c = 0.9$, $h = 0.4$ and coherent drives. The calibrated column uses 32 passive two-read blocks, $q_{32} = 0.0404888$ (Section VI C). With an exact flip the calibrated loss would be 0.199631 [36].

N	ϵ_N	Loss, $q = \frac{1}{2}$	Loss, q_{32}	Rate p, $q = \frac{1}{2}$	Rate p, q_{32}	Added energy per opportunity
1	0.354394	0.360165	0.231654	0.155253	0.114129	0.070401
4	0.132699	0.348517	0.209300	0.151525	0.106976	0.016497
16	0.037104	0.344779	0.202127	0.150329	0.104680	0.004036
64	0.009546	0.343806	0.200259	0.150018	0.104083	0.001003

TABLE II. Stationary loss and coupling rate with a coherent drive of occupation N. The last column is $p\lambda^2$ at q_{32} in units of $\bar{\omega}$; the incoming energy $N\bar{\omega}$ is paid at every opportunity in addition. Reading, probe and calibration costs are not included.

C. Calibration with the same imperfect gate

The Imprint Chain separates calibration blocks by an exact randomization $\mathcal{R}(\rho) = \frac{1}{2}(\rho + X\rho X)$. With an imperfect gate, triggering it with probability $\frac{1}{2}$ leaves a fraction $1-s$ of the bit's mean. Triggering with probability $1/(2s)$, which is at most one because $s \ge \frac{1}{2}$, and discarding the coin gives

$$\Big(1 - \frac{1}{2s}\Big)\rho + \frac{1}{2s}\big(sX\rho X + (1-s)\rho\big) = \tfrac{1}{2}(\rho + X\rho X) \tag{25}$$

exactly. This presumes that s is known. If the trigger probability is rounded to k binary digits, each randomization has flip probability within $s2^{-k-1}$ of $\frac{1}{2}$. Since $\mathrm{id} - \mathrm{Ad}_X$ has diamond norm 2, the record distributions of a protocol with $n+1$ randomizations differ from the exact ones by at most $(n+1)s2^{-k-1}$ in total variation, or 1.52×10^{-5} for $N = 16$, $n = 32$, $k = 20$.

Passive two-read blocks randomize, read, let the bath act and read again, without a feedback flip. With the compensated randomization they keep the Imprint Chain likelihood: the product of the two readings is positive with probability $p_{\mathrm{cal}} = (1 + c\nu^2)/2$ if $a = S$ and $\frac{1}{2}$ otherwise. After n blocks the optimal mismatch probability is

$$q_n = \frac{1}{2} \sum_{k=0}^{n} \binom{n}{k} \min\{p_{\mathrm{cal}}^k (1 - p_{\mathrm{cal}})^{n-k}, 2^{-n}\}. \tag{26}$$

Active blocks, with a feedback flip after the first reading, change once the flip is imperfect. For $a = S$,

$$\Pr(y_1, y_2) = \tfrac{1}{4}\big[1 + c\nu^2 y_2\big(s + (1 - s)y_1\big)\big], \tag{27}$$

and the pair is uniform for $a \neq S$. This law is obtained from the passive pair by an S-independent post-processing that keeps the first reading and relabels with probability s, so gate error creates no calibration information about the sensor. The success probability s itself is readable from the closed loop: in the persistence task the lag-one correlation of readings on gated steps differs from that on ungated steps by a term proportional to s, so the agent's own records determine s, the bath contraction and the squared gain without references, and never the sign [33]. Calibration with n blocks uses $2n$ readings and $n + 1$ randomizations with an expected $(n + 1)/(2s)$ couplings.

D. Records and drive in one acceptance condition

Proposition 12 (Joint record–drive condition)**.** *If $chs\nu > 0$, the stationary loss satisfies $L_\infty(q) \leq L_*$ exactly when*

$$q \leq 1 - \frac{(1 - 2L_*)(1 - c + chs)}{chs\nu}. \tag{28}$$

If the right-hand side is negative, no amount of calibration achieves L_* with that actuator.

Proof. Rearrange Eq. (23) using $1 - \Lambda = 1 - c + chs$. □

Take $L_* = 0.2$. A coherent drive with $N = 1$ fails even with a known interface, whose loss is 0.220330. For $N = 4$, 16 and 64 the minimal numbers of passive blocks are 57, 36 and 33; since q_n is nonincreasing, checking that $n - 1$ blocks fail and n succeed establishes the minimum. With 32 blocks fixed, Eq. (28) requires $s \geq 0.9943781$, that is $\epsilon \leq 0.0056219$. Coherent half-turn drives then need $N \geq 109.10$, classical drives need $N \geq 98.93$ by Proposition 8, and by Theorem 6 every single-mode drive acting as a flip channel needs $N \geq 9.390$ (and $N + \lambda^2 \geq 9.503$). The Gaussian half-turn family reaches 9.957 and the computed grid variational minimum is 9.945. These minima refer to the declared block protocol and feedback rule, not to every possible experiment.

Corollary 13 (Scaling of the persistence cost)**.** *Among drives acting as flip channels, holding the stationary loss within δ of its exact-flip value at a known interface requires a drive occupation per activation that grows as $\delta^{-1/2}$ for optimal drives and as δ^{-1} for classical drives.*

Proof. At $q = 0$, L_∞ is a smooth decreasing function of s with nonzero derivative at $s = 1$, so δ is proportional to ϵ to first order. Theorem 6 and Proposition 7 give occupation of order $\epsilon^{-1/2}$ for optimal drives, and Proposition 8 together with the coherent family gives order ϵ^{-1} for classical drives. □

E. Resource accounting

The resources of the task are kept as a vector. Each task opportunity uses one reading, one probe and CNOT, a gate coin, a bath interval and one incoming drive of occupation N, prepared whether or not it is used. Each actual coupling adds switching work $\lambda^2\bar{\omega}$ at rate p. For $N = 16$ and 32 blocks, calibration uses 64 readings and 33 randomization opportunities carrying incoming energy $528\bar{\omega}$, with an expected 17.14 couplings and switching work $0.6606\bar{\omega}$. Over a task window of 60 steps the expected number of couplings is 6.494. The occupations 61.07 and 7.33 at $\epsilon = 0.01$ are input levels, and 7.49 minimizes input plus one activation's work; none of these is the total energy of an experiment, which would also include reading, preparation, records and the switching apparatus.

F. Finite-energy response readout

An outgoing stress can change a boundary trajectory, which can in turn change signal delays. Consider the particular readout in which these delays are the only branch-dependent signal. Use one calibrated proper clock, nonnegative probe Hamiltonians H_j on separate ports, and a joint probe–ancilla state independent of the unknown branch. The ancilla may be correlated with the probes but is not acted on by the signal. In branch $k = 0, 1$ the probes undergo $U_k = \exp[-i \sum_j H_j \tau_{kj}/\hbar]$. Preparation and final joint measurement can be arbitrary. The charged probe input is $E = \sum_j \operatorname{Tr}(\rho H_j)$, above the declared zero levels.

Theorem 14 (Sharp signed-delay energy bound)**.** *Let* $\Delta\tau = \max_j |\tau_{1j} - \tau_{0j}| > 0$ *and define*

$$\zeta = \max_{x>0} \frac{1 - \cos x}{x} = 0.724611\ldots, \qquad z = \frac{\zeta E \Delta\tau}{\hbar}. \tag{29}$$

The trace distance of the two output states, including the retained ancilla, is at most $\sqrt{z(2-z)}$ *for* $0 \le z \le 1$*, and at most one otherwise. For equal priors, error at most* $\epsilon < 1/2$ *therefore requires*

$$E \ge \frac{\hbar}{\zeta \Delta\tau} \left[1 - 2\sqrt{\epsilon(1-\epsilon)}\right]. \tag{30}$$

This lower bound is sharp across the permitted probe families, including oppositely signed delay differences. It need not be sharp for a specified Hamiltonian or a readout with all delay differences of the same sign. If $\Delta\tau = 0$*, no such protocol distinguishes the branches.*

Proof. The scalar maximum occurs at the unique $x_* \in (0, \pi)$ satisfying $\tan(x_*/2) = x_*$; for $x \ge \pi$, $(1-\cos x)/x \le 2/\pi$, whereas the value at $2\pi/3$ gives $\zeta \ge 9/(4\pi) > 2/\pi$. Thus $1 - \cos x \le \zeta |x|$ for every real x. Purify the probe–ancilla input. The overlap of the two resulting pure states is

$$A = \operatorname{Tr} \rho \exp(-iK), \qquad K = \sum_j H_j (\tau_{1j} - \tau_{0j})/\hbar.$$

The Hamiltonians commute strongly, and their positivity gives $|K| \le (\Delta\tau/\hbar) \sum_j H_j$ by joint spectral calculus. Consequently $\operatorname{Re} A \ge 1 - z$. For $z \le 1$, the pure-state trace distance is $\sqrt{1 - |A|^2} \le \sqrt{z(2-z)}$; discarding the purifying reference cannot increase it. Equal-prior error is at least $(1 - D)/2$ for trace distance D. Solving this inequality gives Eq. (30).

For saturation, take two probes with equal excited energy $e = x_* \hbar/\Delta\tau$, and delay differences $+\Delta\tau, -\Delta\tau$. On the three states $|00\rangle, |e0\rangle, |0e\rangle$ prepare

$$|\psi\rangle = \sqrt{1-p}\,|00\rangle + \sqrt{p/2}\,(|e0\rangle + |0e\rangle), \qquad p = \frac{1 - 2\sqrt{\epsilon(1-\epsilon)}}{1 - \cos x_*}.$$

Here $0 \le p < 1$, $E = pe$, and the branch overlap is exactly $1 - p(1 - \cos x_*) = 2\sqrt{\epsilon(1-\epsilon)}$. The optimal binary measurement attains error ϵ and Eq. (30). □

Mean-energy quantum speed limits constrain distinguishable evolution, including arbitrary-fidelity targets [34, 35]. Here the relevant resource is the sum of the physical probe energies, not the signed generator's ground-state shift; this accounts for opposite delay differences without assigning a negative probe cost. A readable branch marker, changed output frequency or amplitude, branch-dependent preparation, or a signal interaction other than translation falls outside the theorem. For example, an unpriced orthogonal marker would distinguish the branches at zero probe energy. Receiver, clock and preparation costs remain additional resources.

In HAFF-D, three echoes determine the coefficients of a supplied JT solution. Its source-compensated inverse permits known stresses in both null directions and propagates timing and source-calibration errors to the next coupled projection [29]. This connects a selection-dependent source, its response and a subsequent operation, conditional on the declared instrument. For its compact-pulse example the largest proper-delay difference is 0.05001338; a translation-only instrument with error 0.1 requires input energy at least $11.0374\hbar$ in that clock's units. The geometric inverse does not construct such a probe or its reflecting boundary. Its supplied bulk source-compensation test is not an energy–resolution demonstration. A joint apparatus must meet the readout bound, the actuator bound and its actual stress and entropy accounts without counting the same outgoing energy twice.

G. Finite-rate entropy export

Drive occupation and exported heat obey different bounds. If the maintained subsystem and its finite reused controller remove a net dimensionless entropy current $\dot{s} > 0$, and all of it leaves through a chiral conformal exhaust of capacity $c_{\rm ex}$ above a thermal input at temperature T, HAFF-D gives [29]

$$\dot{Q} \geq k_B T \dot{s} + \frac{3\hbar}{\pi c_{\rm ex}} \dot{s}^2.$$

The current must include retained side information and controller consumption; the number of attempted flips alone does not determine it. The bound assumes an extensive long-window entropy-current account, zero chemical potential or the matched conserved-charge account, and one common clock. It is not an exact bound for every short packet. It remains positive at $T = 0$ for a thermal input, where ordinary Landauer heat is zero. For independent parallel exhausts at one temperature, $c_{\rm ex}$ is their total capacity.

A nonthermal drive changes this comparison. If its incoming energy and dimensionless entropy currents are $e_{\rm in}$ and $\sigma_{\rm in}$, the same Gibbs argument gives

$$\dot{Q} + e_{\rm in} - E_{\rm th}(\sigma_{\rm in}) \geq k_B T_\sigma \dot{s} + \frac{3\hbar}{\pi c_{\rm ex}} \dot{s}^2, \qquad E_{\rm th}(\sigma) = \frac{3\hbar\sigma^2}{\pi c_{\rm ex}}, \quad T_\sigma = \frac{6\hbar\sigma_{\rm in}}{\pi c_{\rm ex} k_B}.$$

The incoming energy above its entropy-matched thermal value can pay for export. In particular, a coherent beam with zero entropy is not a zero-energy reservoir. Its drive occupation and the additional heat must be accounted for separately.

This heat is neither the incoming occupation $N\bar{\omega}$ nor an extra charge to add to an outgoing energy already counted. It is a constraint on that energy when it carries the removed entropy. The scheduled drive calculation above does not establish its exhaust-capacity assumptions. A complete apparatus must price reading, recovery, sensing and entropy transport in the same cycle.

VII. CALIBRATING AN INTERFACE WHILE IT CARRIES QUANTUM CONTENT

Preparing test states before storage is not the only way to learn an interface. The outgoing field can carry information about the coupling without carrying the logical content that the apparatus is trying to keep. We give the instrument, its loss under incomplete collection, and a coherent implementation.

On the two-qubit code $\mathcal{C} = \mathrm{span}\{|01\rangle, |10\rangle\}$, let Z_L have eigenvalues $+1, -1$. The coupling $Z_1 + \alpha a Z_2$ restricts to mZ_L, $m = 1 - \alpha a$, with unknown $\alpha > 0$ and chosen command a. Write

$$Q_f = \sqrt{k}(b + b^\dagger), \qquad U_m = e^{-imZ_L Q_f}, \qquad k > 0, \quad \Pi = (-1)^{b^\dagger b}.$$

The independently prepared incoming mode is vacuum. A beam splitter collects fraction ξ of this mode, leaving fraction $1 - \xi$ inaccessible.

Proposition 15 (A content-preserving calibration instrument)**.** *Measure the collected mode's parity and apply Z_L for an odd outcome. The corrected instrument, including an arbitrary retained reference, is*

$$\mathcal{I}_\pm(\rho) = p_\pm(a)\mathcal{D}_{\xi,m}(\rho), \qquad p_\pm(a) = \frac{1 \pm e^{-2k\xi m^2}}{2}, \tag{31}$$

where $\mathcal{D}_{\xi,m}$ fixes the populations and multiplies logical coherence by $e^{-2k(1-\xi)m^2}$. At full collection each outcome is a scalar multiple of the identity channel. Fresh-port queries chosen from earlier records can therefore calibrate the interface without replacing its held state.

Proof. The accessible displacement has the expansion $I_L \cos(m\sqrt{\xi}Q_A) - iZ_L \sin(m\sqrt{\xi}Q_A)$. On the vacuum the two field vectors have opposite parity, with squared norms p_+, p_-. The conditional pulse makes both logical factors I_L. The independent inaccessible displacement commutes with this correction and leaves $\mathcal{D}_{\xi,m}$. Each branch is thus Eq. (31), not merely an equality of its unconditional channel. Composition proves the adaptive statement; the same computation tensors with any reference. □

Environment-assisted correction of unitary Kraus branches is a known general mechanism [3, Proposition 2]. Here its records are also the data for learning the same interaction's gain and collection fraction. Repeated ancilla-based

parity measurements have been implemented for a cavity mode [5]; collecting the full mode of a travelling chiral field is an additional requirement, not supplied by that experiment.

Partial collection prices the calibration itself. For N queries at each of $a = \pm h$, with fixed α, ξ, the held coherence is multiplied by

$$\exp[-4Nk(1-\xi)(1+\alpha^2 h^2)]. \tag{32}$$

This loss occurs before any subsequent operating benefit. With $k = .2$, $\alpha = 1.1$, $h = .5$ and $N = 4096$, keeping the calibration's dephasing error below .01 needs $1 - \xi \le 4.7335 \times 10^{-6}$. At $\xi = .99$ its error is essentially $1/2$. More records are not automatically better for held content.

A. A resonant collection port

The collection fraction can itself be realized by a finite-duration interaction. This construction starts with a retained source oscillator b and a receiver oscillator c of the same frequency ω, with c initially in vacuum. It does not assume that a broadband travelling mode has already been trapped by this interaction. In the interaction picture put

$$H_{\mathrm{cap}}(t) = \hbar g(t) C, \quad C = b^\dagger c + c^\dagger b, \quad \theta = \int g(t)dt, \quad U_{\mathrm{cap}} = e^{-i\theta C}.$$

Proposition 16 (Finite resonant collection)**.** *For $0 \le \theta \le \pi/2$, the receiver collects $\xi = \sin^2\theta$ of the source mode. Corrected receiver-parity readout realizes Eq.* (31) *with this ξ. If $|g| \le g_{\max} < \omega$, the total Hamiltonian $\hbar\omega(n_b + n_c) + H_{\mathrm{cap}}$ is bounded below and the pulse needs duration at least $\theta/g_{\max}$. A pulse starting and ending at zero does no net prescribed-switch work in this isolated two-mode model. For the independently prepared vacuum encounter, the mean occupations after collection are ξkm^2 and $(1-\xi)km^2$. Their sum is km^2.*

A pulse-area error δ changes ξ by at most $|\delta|$. For any input with $\langle n_b + n_c\rangle \le \bar{n}$, including a retained reference, its half-trace error is at most $\min\{1, \sqrt{2|\delta|\bar{n}}\}$.

Proof. The Heisenberg transforms are $b \mapsto b\cos\theta - ic\sin\theta$ and $c \mapsto c\cos\theta - ib\sin\theta$. The conditional coherent states produced by U_m therefore split into receiver and remainder amplitudes with squared fractions $\xi, 1-\xi$. The receiver's known phase does not change parity. Applying Proposition 15 proves the instrument and the occupation account, for any logical input.

The normal modes $(b \pm c)/\sqrt{2}$ have instantaneous frequencies $\omega \pm g$, which proves the lower bound. The amplitude bound proves the duration statement. Both C and total occupation commute with the Hamiltonian, so $\int \langle \partial_t H_{\mathrm{cap}}\rangle dt = \hbar\langle C\rangle[g(t_{\mathrm{end}}) - g(t_{\mathrm{start}})] = 0$. Finally $|\sin^2(\theta+\delta) - \sin^2\theta| \le |\delta|$ and $|C| \le n_b + n_c$. The purified-vector estimate $2\langle 1 - \cos(\delta C)\rangle \le 2|\delta|\bar{n}$ proves the energy-constrained error bound. □

This account includes transferred energy and a gate clock. Receiver parity control, reading and retaining a syndrome, preparing the next empty receiver, and supplying the switching apparatus remain separate resources. Emptying the receiver exports energy; it is not included in the zero net switch work above. Restoring a degraded collection setting also needs a stated physical permission and price. Brain Anchor [4] uses such a permission in a finite model where the same content survives inquiry, operation and maintenance. Resonant port capture, local capture of a continuous field and sustained thermal export are three different implementation questions.

Whole-mode control can be reduced to successive port contacts in a finite packet model. Take n source oscillators of frequency ω and one vacuum receiver c. For real $f_j \ge 0$, $\sum_j f_j^2 = 1$, form the packet mode $b_f = \sum_j f_j b_j$. Sequential source interactions $e^{-im\sqrt{k} f_j Z_L (b_j + b_j^\dagger)}$ produce the same packet displacement as $e^{-im\sqrt{k} Z_L (b_f + b_f^\dagger)}$. After contacting port j, couple only that port to c through the resonant Hamiltonian above, with area $0 \le \theta_j \le \theta_{\max} \le \pi/2$. Released ports are inaccessible. The receiver's packet amplitude, up to its known common phase, obeys

$$y_0 = 0, \qquad y_j = y_{j-1}\cos\theta_j + f_j \sin\theta_j.$$

Proposition 17 (Sequential local-port collection)**.** *The receiver collects fraction $\xi = y_n^2$ of the packet, and its corrected parity instrument is Eq.* (31)*. Among these permitted real resonant contacts, final collection is maximized by*

$$\theta_j = \min\{\theta_{\max}, \arctan(f_j / y_{j-1})\},$$

with the first angle $\pi/2$ before applying the cap when $y_{j-1} = 0 < f_j$. When $y_{j-1} = f_j = 0$, choose $\theta_j = 0$. For a uniform packet and $\theta_{\max} = .5$, the first five contacts are capped and the remaining ones are unconstrained. For $n \ge 5$ the exact optimum is

$$\xi = 1 - \frac{d}{n}, \qquad d = 5 - \cot^2(.25)(1 - \cos^5(.5))^2 = 1.47392688\ldots.$$

Thus 148 uniform ports suffice for collection at least .99, and fewer do not under the same cap. With no cap below $\pi/2$, the angles $\theta_1 = \pi/2$, $\theta_j = \arctan(1/\sqrt{j-1})$ collect a uniform finite packet completely.

Proof. Passive mode mixing sends the packet's conditional coherent displacement to receiver amplitude $-iy_n$ and a remainder of squared norm $1 - y_n^2$. Orthogonal input modes are vacuum. Their overlaps prove the instrument. Each step maximizes $y\cos\theta + f_j \sin\theta$ at the stated angle. The maximum is nondecreasing in $y \geq 0$, because all permitted cosines are nonnegative. Induction therefore makes this stepwise optimum a final optimum. For a uniform packet, the first capped steps give $y_j = n^{-1/2}\cot(.25)(1 - \cos^j(.5))$. After the fifth, the interior optimum has $y_j^2 = y_{j-1}^2 + 1/n$. This proves the formula and threshold. The uncapped recursion gives $y_j = \sqrt{j/n}$. □

Each contact has a Hamiltonian bounded below when $g_{\max} < \omega$, needs duration at least $\theta_j/g_{\max}$ and conserves total occupation. The source prepares mean total occupation km^2, the receiver holds ξkm^2 and the released ports hold $(1-\xi)km^2$. Parity control is needed only on the retained receiver. For contact-area errors δ_j, the half-trace error is at most $\min\{1, \sum_j \sqrt{2|\delta_j|\bar{n}}\}$ when total occupation is bounded in mean by $\bar{n}$; telescope the preceding energy-constrained bound. Source switching, receiver reuse and detector errors require their own account. These are local contacts in a finite collision architecture, not a proof that a specified relativistic wave packet admits that architecture or that its clocks and references can be sustained indefinitely.

B. Positive-frequency continuous collection

A specified continuous field can also supply the collection port. Let $\omega = \Omega + \delta$, $|\delta| \leq B < \Omega$, and couple it to a retained oscillator c. With $[a_\delta, a_{\delta'}^\dagger] = \delta(\delta - \delta')$, take a real even form factor g, $J = g^2$ and $G = \|g\|_2$. Put

$$H(h) = \hbar\Omega n_c + \hbar\int_{-B}^{B} (\Omega+\delta) a_\delta^\dagger a_\delta d\delta + \hbar h(t)\int_{-B}^{B} g(\delta)(c^\dagger a_\delta + a_\delta^\dagger c)d\delta, \qquad |h| \leq 1. \tag{33}$$

This is an excitation-preserving effective Hamiltonian with a specified band-smeared coupling. Its frequency form factor gives spatial tails; it is not a strictly point-local relativistic detector. No negative physical frequencies or Markov extension of the band are used. Matched pulse absorption by a tunable resonator is an established mechanism [2]; the question here is how its finite-band realization enters the held-content instrument and source account.

In the frame rotating at Ω, the one-particle operator is

$$A = \begin{pmatrix} 0 & \langle g| \\ |g\rangle & \delta \end{pmatrix}, \quad S = e^{-iAT}, \quad r(T) = \langle c|S|c\rangle.$$

The capture pulse has $h = 1$ on $[0, T]$ and is otherwise off. Deviations from this prescribed pulse are bounded below.

Proposition 18 (Matched finite-band capture)**.** *If $\Omega > B + G$, every Hamiltonian in Eq. (33) is bounded below by $\hbar(\Omega - B - G)N$. Write $\xi_0 = 1 - |r(T)|^2 > 0$. Among normalized incoming field modes, the mode*

$$f = -P_{\text{field}} S^\dagger |c\rangle / \sqrt{\xi_0}$$

maximizes collection and is collected with fraction ξ_0. It can be prepared from independent vacuum by a finite prescribed body-controlled linear drive with envelope $u(s) = r(T-s)$, $0 \leq s \leq T$. For even J, r and this envelope are real. The prepared field amplitudes are

$$m\beta w_u(\delta), \qquad w_u(\delta) = -ig(\delta)\int_0^T u(s)e^{i\delta s}ds, \quad w_u = -P_{\text{field}} S^\dagger|c\rangle, \quad \beta = \sqrt{k/\xi_0}. \tag{34}$$

For an arbitrary prepared envelope and $w_u \neq 0$, replace k by $k_{\text{eff}} = \beta^2\|w_u\|^2$ and use $\xi = |\langle c|S(0, w_u)\rangle|^2/\|w_u\|^2$. Corrected receiver-parity readout then realizes Eq. (31) with these actual parameters.

Proof. $\|A\| \leq B + G$ and second quantization prove the lower bound. The norm of the field part of the receiver row is $\sqrt{\xi_0}$; Cauchy–Schwarz proves the optimum. Its orthogonal complement has zero collection, so the statement is not uniform over unknown packets.

Eliminating the field component of $S^\dagger|c\rangle$ gives Eq. (34). Reflection $\delta \mapsto -\delta$, field sign reversal and complex conjugation define an antiunitary taking A to $-A$ and leaving its propagator invariant. Hence r is real. One explicit preparation places the band-smeared source at $x = -T$ and the receiver at $x = 0$, with propagation speed one. Drive

the source during $t \in [-T, 0]$ with envelope $u(t+T)$ and a prescribed carrier phase Ω. Explicitly its interaction-picture Hamiltonian is

$$H_s^I(t) = \hbar m \beta Z_L u(t+T) \int_{-B}^{B} g(\delta)[a_\delta^\dagger e^{i\delta(t+T)} + a_\delta e^{-i\delta(t+T)}]d\delta.$$

Its displacement is $m\beta w_u$; the receiver is coupled only after this drive. The source Hamiltonian is linear in the band field and proportional to mZ_L. Its time-ordering commutator is a scalar times Z_L^2, so it changes only an overall phase. Completing the square bounds its instantaneous energy below since $\Omega - B > 0$ and $\int J/\omega$ is finite.

Passive field evolution splits the conditional coherent amplitudes into a receiver component of squared fraction ξ and a remainder of fraction $1-\xi$. Their vacuum overlaps give the claimed instrument for arbitrary logical input and retained reference. Initial independence and parameters fixed within an encounter are required. □

The propagation prescription is within the band-smeared model: a strictly band-limited field operator has no compact spatial support. The construction supplies a drive and a receiver, not their formation from unknown interactions. For real envelopes and even g, the free input and output energies are

$$E_{\text{source}} = \hbar\Omega k_{\text{eff}} m^2, \qquad E_c = \hbar\Omega\xi k_{\text{eff}} m^2, \qquad E_{\text{free,out}} = \hbar\Omega(1-\xi)k_{\text{eff}} m^2. \tag{35}$$

The same reflection symmetry makes the mean detuning energy zero before and after collection. Consequently the net prescribed capture-switch work is zero in this family. The complete source drive supplies E_{source}. For other packets the capture work need not vanish: at mean occupation $\bar{n}$, its magnitude is at most $2\hbar B\bar{n}$. Detector, carrier, clock and support costs are separate from these accounts.

Proposition 19 (A finite waveform certificate)**.** *Let $J(\delta) \le J_0$. Approximate $r(T-s)$ by linear interpolation of node values on L uniform intervals, each rounded on a prescribed grid of spacing q. Include a further uniform envelope error η_u. With $\nu = \sqrt{\xi_0}$,*

$$\sigma = \sqrt{2\pi J_0 T}\left[\frac{G^2(T/L)^2}{8} + \frac{q}{2} + \eta_u\right]$$

bounds the prepared field-vector error relative to Eq. (34)*. An interval enclosure error for node values is added inside the brackets. Let duration and arrival errors be bounded by d_T, d_t, and the capture pulse have $\int |h_{\text{actual}} - h|dt \le d_h$. Put $\varepsilon = (B+G)d_T + Bd_t + Gd_h$. If $\sigma < \nu$, then*

$$1 - \xi \le \left[\frac{|r|\nu + \sigma + \varepsilon(\nu+\sigma)}{\nu - \sigma}\right]^2, \qquad \left(1 - \frac{\sigma}{\nu}\right)^2 \le \frac{k_{\text{eff}}}{k} \le \left(1 + \frac{\sigma}{\nu}\right)^2. \tag{36}$$

Arrival and duration bounds here concern collection and the logical parity instrument, which are insensitive to the common carrier phase.

Proof. $|r''(s)| = |\langle Ac|e^{-iAs}|Ac\rangle| \le G^2$ gives the interpolation bound. Plancherel over the entire frequency line, followed by restriction to the band, gives $\|w_u - w_{\text{ideal}}\| \le \sigma$. Duhamel bounds the carrier-referenced propagator error by ε. The ideal outgoing field remainder has norm $|r|\nu$. Adding preparation and propagator errors to this norm and dividing by the actual input norm, at least $\nu - \sigma$, proves the first bound. The second follows by the same input-norm bounds. □

For a concrete continuum certificate use

$$J(\delta) = \frac{1}{2\pi}\left[1 - (\delta/B)^2\right]^2, \quad \Omega = 20, \quad B = 5, \quad T = 16.$$

Frequencies are measured in units of $2\pi J_0$ and times in its reciprocal. Then $G^2 = 8/(3\pi)$. The exact even bath moments are $\mu_{2j} = 16J_0B^{2j+1}/[(2j+1)(2j+3)(2j+5)]$. If $M_n = \langle c|A^n|c\rangle$, then

$$M_0 = 1, \qquad M_{2n} = \sum_{j=0}^{n-1} \mu_{2j} M_{2(n-1-j)}, \qquad r(t) = \sum_{n=0}^{\infty} \frac{(-1)^n M_{2n} t^{2n}}{(2n)!}.$$

Decomposing each departure and return to c proves the recurrence. The degree-P Taylor remainder is at most $[(B+G)T]^{P+1}/(P+1)!$ by self-adjoint functional calculus. Directed intervals at 120 decimal digits and $P = 400$ give $r(16) = 0.000061743299929498335 73\ldots$ with remainder below 1.53×10^{-79} and $\xi_0 > 0.99999999618$.

Choose $L = 512$, $q = 1/65536$, $\eta_u = d_T = d_t = d_h = 10^{-5}$. The 513 rounded node values fit the grid $[-1/64, 1]$, whose 66561 labels need 17 bits each. The fixed table uses 8721 logical bits, apart from its decoder, interpolator, carrier and clock. A separately encoded normalization amplitude rounded on a 2^{-24} grid contributes less than 10^{-7} relative error at nominal $k = .2$; it fits within η_u and uses 24 more label bits. Eq. (36) gives $1 - \xi < 4.42935 \times 10^{-7}$ and $k_{\text{eff}}/k \in [.999030268, 1.000970203]$. For $\alpha = 1.1$, $h = .5$, $N = 4096$ probes at each of $\pm h$ and nominal $k = .2$, the calibration's collection-dependent dephasing error is below .000946. This contrasts with the loss of content at $\xi = .99$ above; it does not include parity-gate or detector errors.

A second numerical representation replaces frequency multiplication by cell midpoints while retaining g itself. The invariant vector in each cell is $g\mathbf{1}_{\text{cell}}/\|g\mathbf{1}_{\text{cell}}\|$, with coupling $\sqrt{\int_{\text{cell}} J}$. Its finite matrix approximates the continuum propagator with operator error at most TB/M for M cells. These cells are a numerical resolution, not fresh physical ports. Dense and sparse matrix calculations agree with the moment enclosure and the prepared waveform. The recurrence and rounding rule specify the waveform without a fitted infinite-precision function.

After an encounter, swapping c with an independent same-frequency vacuum oscillator returns c exactly to vacuum, even when its old state is correlated with a reference. The swap exports E_c and has the finite duration and occupation account of Proposition 16. It supplies a reusable receiver for a finite bank of encounters, not a free thermal sink. Repeating the query law needs independent field preparations as well as independent vacuum exhausts. Independent field sectors selected one at a time give a finite-bank realization whose Hamiltonian remains bounded below. 8192 queries consume 8192 such preparations and exhausts, while sharing the waveform program. In this calibration example the nominal source energy is $E_{\text{probes}}/[\hbar(2\pi J_0)] = 4096(.2)(20)[(1 - .55)^2 + (1 + .55)^2] = 42680.32$. It is predominantly collected and later exported, not eliminated by the small content-damage bound. Serial preparation and capture take $8192(2T) = 262144$ time units before parity control and receiver reset. Time windows in one band-limited waveguide are not automatically independent; that alternative requires a return/reset bound. Thermal exhaust resets c to a thermal state and changes the readout law. For example, for the exactly matched mode, a thermal receiver of mean occupation n_T gives even-parity probability

$$\frac{1}{2}\left[1 + \frac{e^{-2\xi k m^2/[1+2(1-\xi)n_T]}}{1 + 2(1-\xi)n_T}\right]$$

after mixing with the vacuum source packet, rather than the vacuum formula. For an unmatched packet, the thermal occupation term instead uses $|\langle c|S|c\rangle|^2 n_T$, which need not equal $(1 - \xi)n_T$.

Proposition 20 (Coherent parity echo)**.** *Suppose the relevant mode remains available for joint control. The bounded involution*

$$R = |0_L\rangle\langle 0_L| \otimes I + |1_L\rangle\langle 1_L| \otimes \Pi = I_L \otimes P_{\text{even}} + Z_L \otimes P_{\text{odd}}$$

satisfies, on the quadrature domain,

$$R(Z_L \otimes Q_f)R = I_L \otimes Q_f, \qquad RU_m R = I_L \otimes e^{-imQ_f}. \tag{37}$$

This protects the body's marginal for every mode input and even for an initially correlated body–mode state. For an independent vacuum input, the first R acts trivially; the final R alone disentangles the body and mode.

Proof. $\Pi Q_f \Pi = -Q_f$. In the $+1$ logical block the conjugated generator is Q_f, and in the -1 block it is $-\Pi Q_f \Pi = Q_f$. Functional calculus gives the unitary identity. A unitary on the mode alone cannot change the body's reduced state. On vacuum $\Pi|0\rangle = |0\rangle$. □

One qubit meter implements R coherently. Prepare it in $|0\rangle_C$, apply

$$M = \exp[-i(\pi/2)P_{\text{odd}} \otimes Y_C], \qquad C_Z = I_L \otimes |0\rangle\langle 0|_C + Z_L \otimes |1\rangle\langle 1|_C,$$

and then apply $M^\dagger C_Z M$, with the tensor identities understood. The meter computes parity, controls the body phase and uncomputes parity; it returns exactly to $|0\rangle_C$. Bounded Hamiltonians $g(t)P_{\text{odd}} \otimes Y_C$ and $\gamma(t)(I_L - Z_L)/2 \otimes |1\rangle\langle 1|_C$ suffice, with pulse areas $\pi/2$ and π. Amplitude bounds impose corresponding nonzero durations. No classical syndrome bit has to be erased in this coherent correction. Producing a retained classical calibration record is a separate operation. Pulse area is not a heat bill: the control field, capture, clock and support still need their physical energy account. Half-diamond errors of supplied primitive channels accumulate at most by their sum, by telescoping compositions; such a bounded-channel error alone does not bound an unbounded stress observable.

A retained oscillator offers a direct alternative to the spectral meter. In its interaction picture, with $n = b^\dagger b$ and $P_1 = |1_L\rangle\langle 1_L|$, the dispersive Hamiltonian $H_R(t) = \hbar\chi(t)P_1 \otimes n$ implements R when $\int \chi(t)dt = \pi$. Such number-controlled parity gates are established cavity operations [5, pp. 2–3]. For a degenerate logical body, $\chi \geq 0$ and

oscillator energy $\hbar\omega n$, the total Hamiltonian is bounded below. If χ starts and ends at zero and the gate is isolated, $P_1 n$ is conserved, so its net prescribed-switch work is exactly zero:

$$W_R = \hbar \int \dot{\chi}\, \langle P_1 n\rangle dt = 0.$$

The oscillator's stored energy, gate duration and controller are not zero. The integrated expected interaction energy is $\pi\hbar\langle P_1 n\rangle$; $\chi \le \chi_{\max}$ gives duration at least $\pi/\chi_{\max}$. If the pulse area has error δ, its half-trace error, with any reference and an input occupation bound $\langle n\rangle \le \bar{n}$, is at most $\min\{1, \sqrt{2|\delta|\bar{n}}\}$. Indeed the relative unitary is $e^{-i\delta P_1 n}$, and the purified vector error squared is bounded by $2|\delta|\langle P_1 n\rangle$, using $1-\cos x \le |x|$. This is an energy-constrained error, not an unrestricted diamond bound. It gives an explicit retained-oscillator gate account; it does not price trapping a broadband travelling mode or operating the classical switch.

C. Thermal inquiry with a binary record and a finite bank

The echo gives an actual thermal inquiry protocol on the retained oscillator. Let its input be the centered Gibbs state τ_T of $H_b = \hbar\omega n$, with $n_T = (e^{\hbar\omega/(k_B T)} - 1)^{-1}$ and $\nu_T = 2n_T + 1$. This input is independent of the held body, its old reference and earlier records. Both parity gates in $RU_m R$ are used: the first is generally consequential on a thermal input. The result is $\rho_{L\mathrm{old}} \otimes D_m \tau_T D_m^\dagger$, where $D_m = e^{-im\sqrt{k}Q}$. Thus the body and its entire old reference are preserved, while the oscillator has

$$\langle P\rangle = -2m\sqrt{k}, \qquad \mathrm{Var}(P) = \nu_T, \qquad E_b - E_{b,\mathrm{in}} = \hbar\omega k m^2.$$

The displaced Gibbs state has the same entropy as τ_T.

A quadrature mean is not yet a saved record. Prepare a fresh degenerate qubit C in $\eta_v = (I + vZ_C)/2$, $0 < v \le 1$, independently of the oscillator and all earlier outputs. Couple it through

$$V_\kappa = e^{-i\kappa P \otimes Y_C}, \qquad \kappa > 0,$$

then measure X_C and store the outcome $Y = \pm 1$. The preparation, orientation, measurement and storage are declared operations.

Proposition 21 (Thermal binary inquiry)**.** *For this protocol the instrument on the held body and its old reference is*

$$\mathcal{J}_y(\rho_{L\mathrm{old}}) = \frac{1 + y\mu_T(m)}{2}\rho_{L\mathrm{old}}, \qquad \mu_T(m) = -ve^{-2\kappa^2\nu_T}\sin(4\kappa m\sqrt{k}). \tag{38}$$

The read interaction adds oscillator energy $\hbar\omega\kappa^2$ before conditioning on the record. If $m = \pm m_0$ with known $m_0 > 0$, $0 < \kappa \le \pi/(8m_0\sqrt{k})$ gives sign gain $\gamma_T = ve^{-2\kappa^2\nu_T}\sin(4\kappa m_0\sqrt{k}) > 0$. For J fresh trials, the sign estimate from $-\sum_j Y_j$ has error at most $e^{-J\gamma_T^2/2}$, including ties as errors.

Proof. The echo acts only on the independent oscillator, so all subsequent oscillator–port operations factor from $\rho_{L\mathrm{old}}$. Conjugation gives $V_\kappa^\dagger X_C V_\kappa = \cos(2\kappa P)X_C + \sin(2\kappa P)Z_C$. The Gibbs characteristic function is $\langle e^{itP}\rangle = e^{-t^2\nu_T/2}$, and displacement changes it by $e^{-2itm\sqrt{k}}$. Taking the qubit expectation proves Eq. (38). Also $V_\kappa^\dagger Q V_\kappa = Q + 2\kappa Y_C$, while P is fixed. Since the independent port has $\langle Y_C\rangle = 0$ and $Y_C^2 = I$, $n = (Q^2 + P^2 - 2)/4$ increases in mean by κ^2. Each sign-aligned binary variable has mean γ_T; Hoeffding's inequality gives the stated bound. Independence here follows from fresh inputs and the renewal below, rather than from reusing a disturbed mode. □

This record reads a sign that parity cannot resolve. Its orientation reference is an additional permission. The sine is periodic, so one setting does not identify arbitrary continuous m; an unknown initial quadrature mean also confounds it. If the mode initially shares correlations with the body, the echo still preserves the unconditional body state, but measuring the mode may steer it. That case does not give Eq. (38).

The pulse can have bounded amplitude and finite duration. In the laboratory frame choose a smooth real $g(t)$ of compact time support, with $\int g(t)dt = \kappa$, and set

$$H_{\mathrm{read}}(t) = \hbar\omega n + \hbar g(t)[P\cos(\omega t) + Q\sin(\omega t)] \otimes Y_C. \tag{39}$$

Its interaction-picture generator is exactly $\hbar g(t) P \otimes Y_C$. The final laboratory evolution includes a known free rotation; subsequent pulses and read axes use the same clock. Completing the square gives $H_{\mathrm{read}}(t) \ge -\hbar g(t)^2/\omega$. Adding the

scalar $\hbar g(t)^2/\omega$ makes it nonnegative and changes only an overall phase; the scalar vanishes at both endpoints. For $|g| \le g_{\max}$ the duration is at least $\kappa/g_{\max}$. The net prescribed-switch work is the endpoint free-Hamiltonian energy increase $\hbar\omega\kappa^2$ in this isolated, degenerate-port model.

The echo's displacement admits the same finite pulse construction, with laboratory force $\hbar m\sqrt{k}f(t)[Q\cos(\omega t) - P\sin(\omega t)]Z_L$ and $\int f(t)dt = 1$. Completing the square bounds its Hamiltonian below by $-\hbar k m^2 f(t)^2/\omega$; a finite command range keeps this bound uniform. The source pulse supplies $\hbar\omega k m^2$ on the Gibbs input. The two parity gates have the finite dispersive realization above. These rotating profiles supply phase and clock access explicitly; their implementation by an autonomous controller is a further physical question.

Proposition 22 (Finite thermal renewal with retained outputs)**.** *Take a bank of J independent same-frequency oscillators in τ_T, independent mixed read ports, and storage for J binary records. After each echo and read, couple b to the next fresh bank oscillator through $\hbar h(t)(b^\dagger c + c^\dagger b)$ with area $\pi/2$. The active oscillator returns exactly to τ_T, independent of all earlier retained outputs. Its entire post-read state and correlations are transferred to the spent bank oscillator, up to known phase rotations. After summing over the new outcome, the entire old body/reference/history state is preserved, including for commands selected from earlier records. If $|h| \le h_{\max} < \omega$, the total contact Hamiltonian is bounded below and its duration is at least $\pi/(2h_{\max})$. A pulse off at both endpoints has zero net prescribed-switch work. For fixed m and κ, the spent oscillator's mean excess energy above its incoming Gibbs value is*

$$\Delta E_{\text{spent}} = \hbar\omega(km^2 + \kappa^2). \tag{40}$$

This is a finite J-trial realization of Eq. (38)*.*

Proof. The contact sends $b \mapsto -ic$, $c \mapsto -ib$. For any joint post-read state ρ_{bA}, where A contains the held body, old reference, port and records, the output is $\tau_{T,b} \otimes \rho'_{cA}$. Gibbs states are invariant under the known phase rotation. This identity transfers correlations instead of tracing them out. The two instantaneous normal frequencies are $\omega \pm h$. Total occupation and $b^\dagger c + c^\dagger b$ are conserved during the contact, which proves the stability and zero switch-work statements as in Proposition 16. The preceding two displacement accounts prove Eq. (40) after averaging over the retained record. For an old classical history r, the new record appends the kernel $p(y \mid m_r)$ to its existing conditional body/reference state. Summing over y leaves each old block, and hence their full joint state, unchanged. Induction proves renewal and this nonselective preservation for adaptive commands. Conditioning only on a new outcome can reweight old history blocks and change the reader's body/reference state; it need not preserve that conditional state. For a fixed unknown sign and fixed settings the new inputs give independent trials. □

For example, $k = .2$, $m_0 = .7$, $v = .6$ and $\kappa = \min\{\pi/(8m_0\sqrt{k}), 1/(2\sqrt{\nu_T})\}$ give sufficient trial counts 178, 247 and 593 for error .05 at $n_T = .2$, .5 and 2. The respective excess oscillator energies per trial, in units of $\hbar\omega$, are .276571, .223 and .148. Higher temperature weakens the record even though this choice reduces the read-pulse energy. A bank consumes one fresh Gibbs oscillator and one mixed read port per trial, retains the spent oscillator and a binary record, and pays both parity gates, the source/read profiles and the resonant contact duration. The initial active Gibbs oscillator is needed in addition to the J fresh bank inputs. Preparation, transport, readout, storage, orientation and the clock/support must be charged in addition.

Neither term in Eq. (40) is an entropy-export certificate. The source displacement preserves entropy; the read operation changes correlations with its port and record. Replenishing the bank or erasing spent outputs requires their joint state, including side information. This construction supplies repeated thermal inquiry for each finite bank. It does not supply the chiral capacity of Section VI, an infinite thermal line or the total covariant stress of a controller.

For a centered independent input the original outgoing source is $m^2 F_f$, $F_f = (f')^2/4$. The coherent echo leaves this coupling-added stress in the field while preserving content. Its implementation requires an available mode and controlled parity; it supplies neither a local capture apparatus for a whole continuous field nor a sustained thermal exhaust. The finite oscillator identity and the moving-field apparatus are different levels of the construction.

VIII. THREE QUADRATIC STRUCTURES

The analysis now contains three quadratic objects, and they should not be identified.

The descent metric G of Proposition 2 is a positive form on changes of controller parameters. A designer of a gradient controller chooses it, and its positivity is what descent requires. The work form K of Proposition 5 is a positive form on switch profiles. It is derived from the field stress and prices a control setting. The Lorentzian form used in Q-RAIF Paper A [30] represents spacetime read-out; it is indefinite and derived from a single-shell propagation symbol with independently supplied evolution time and spatial carrier.

K and G live on different spaces: K on profiles, G on tangent vectors of the parameter space. If the controller parameters are the amplitudes a of $g = \sum_i a_i g_i$, the parameter space is linear and one may choose $G = K$. The descent

direction $-K^{-1}\nabla V$ then maximizes the first-order decrease $-\nabla V \cdot v$ among parameter changes with $v^T K v = 1$, by Cauchy–Schwarz in the K inner product. This is a design option that measures parameter changes in the units that price settings. Persistence does not force it, and the feedback of Section VI is not a gradient controller: its stability was proved without any G.

Positive quadratic drive charges with supplied coefficients appear in other Q-RAIF protocols. The field coupling shows how such a coefficient can be fixed by the stress of the field that carries the control. Both G and K are positive, so the actuator does not supply the negative direction of a Lorentzian form.

IX. RELATIONS TO OTHER WORK

A. Q-RAIF Papers A and C

Paper A [30] looks for the relative sign in translation-invariant local unitary propagation. When all bands share one quadratic shell, the propagation symbol obeys Euclidean Clifford relations. Its form $\eta = \omega^2 - Q(q)$ has one positive frequency direction and d negative spatial directions; the mass term together with the spatial generators yields $Cl(1, d)$. Reversing the overall metric convention reverses these signs, not their relative opposition. By Sylvester's law neither positive G nor K of Section VIII can be brought to that indefinite form.

The two papers use different quadratic structures, and Section VIII keeps them apart. Paper C [31] asks what a coupling supplies in place of an assumed representation map. For the coupling XQ_g of this paper the algebra represented exactly is the commutant of X, for every drive state, and the defect on the remaining controls reproduces the energy–accuracy bound of Theorem 6.

The mass–energy paper [32] now uses the same quadrature interaction to realize a quantum-walk coin. With a fresh Gaussian drive each step, its mean angle fixes the rest resonance, while drive fluctuations set its decay rate. At input occupation N the smallest Gaussian rate is $\theta^2/[2\tau N(N+1)]$, attained by a displaced squeezed state. This connects the price of control to the persistence of a spectral readout. The rate is an open-channel linewidth, not a new spectral mass, and shift, preparation and port reset costs are separate.

B. Quantum control theory

Lie-algebraic control characterizes reachability through the algebra generated by drift and control Hamiltonians [14]. Stabilization against perturbations also needs magnitudes and costs. In the realized actuator these come from the physical coupling: the rotation angle from the drive amplitude, the error from the drive's quadrature fluctuations, and the cost from the field stress. A Clifford quadratic form is not needed to supply them.

C. Decoherence-free subspaces

Decoherence-free subspaces [19, 20] are protected passively by symmetry. The feedback task gives the active counterpart. The held-code instrument, coherent echo and finite collection protocols also join passive encoding to priced inquiry and correction: their exact identities protect a supplied code, while collection loss and gate error determine what survives. Brain Anchor [4] uses records of this interface for finite-window inquiry and maintenance decisions. Forming the code and its physical permission from a prior history is a further mechanism, not a consequence of encoding it.

D. The Imprint Chain and HAFF

The Imprint Chain [36] supplies the instrument, the calibration and the task, with an exact supplied flip. Replacing that flip by the field coupling raises the calibrated loss from 0.199631 to 0.202127 at $N = 16$ and makes the drive energy part of the feasibility condition. Each coupling also adds the stress F_g to the outgoing field. The HAFF gravity paper [29] uses this outgoing field as a matter source in a Jackiw–Teitelboim model; there, calibration changes the mean source through the coupling rate p of Eq. (24). Selecting the coupled projection changes its represented algebra and source even at fixed activation schedule. A boundary readout can then update the next projection, with known probe stresses and bounded errors included in the inverse. Section VI F supplies one energy constraint on a specified realization of that return.

E. Quantized control fields

The effect of field quantization on gate accuracy has a substantial literature. For Jaynes–Cummings-type couplings, conservation of excitation number leads to error bounds controlled by photon-number fluctuations, of order $1/\bar{n}$ for coherent fields [24]; coherent-field control limits for such models were analysed in Ref. [25], and simultaneous driving of several atoms by one quantized pulse in Ref. [26]. The coupling studied here is a quadrature coupling of a degenerate bit and conserves no excitation number, which is why squeezing changes the scaling from ϵ^{-1} to $\epsilon^{-1/2}$. Theorem 6 gives the quadrature-specific bound and explicit near-attainment under these assumptions. A covariant stress tensor for a finite detector, including its localization, is constructed in Ref. [23]; it shows why the apparatus stress cannot simply be omitted.

X. DISCUSSION

For gradient-type feedback the algebra must compose associatively and the descent metric must be positive. When control is realized by a field, the structure that limits persistence is the canonical commutation relation of the drive quadratures: $[Q, P] = 2i$ fixes the trade-off between flip accuracy and drive occupation in Theorem 6, and it is the reason squeezed drives do better than classical ones. The Pauli and Clifford relations of the controlled bit label the operation; they do not set its cost. Persistence has a closed-loop proof in this model, with an explicit Lyapunov function, and the drive energy and calibration records enter one feasibility condition.

Several inputs remain supplied. The free chiral field, the degenerate bit, the clock and switch, the drive ports and the instrument family are given. The feedback task also supplies the success probability for compensated randomization and scheduled beams. Its analysis concerns Z-diagonal bit states and flip channels; complex $\chi_\sigma(2\lambda)$ instead rotates Z into Y. The held-content construction concerns arbitrary logical inputs. Its finite-band realization prepares and collects a specified packet, with a finite waveform, clock tolerances and independent vacuum exhausts. It returns a coherent meter; the carrier, detector, port selection and vacuum supply remain separate inputs. The stress of the switch and its support is not computed; outgoing-field stress is used outside the apparatus.

The thermal bank identifies what renewal has to preserve. A fresh input restores the oscillator only because the previous state, including its record correlations, moves to a spent output. Bank replenishment and memory erasure must therefore be priced on that joint output, as in the information thermodynamic account [9, 10]. The chiral capacity bound constrains a specified entropy flow; the finite oscillator construction gives an implementable inquiry and transfer, rather than that flow's attainability.

Repeated propagation in one field remains a stronger problem. Proposition 10 proves that shared drives can preserve each marginal channel while correlating failures. A persistence calculation on that joint history may therefore have a different energy scaling from the fresh-drive task. The other unresolved connection is formation: which interactions and retained records produce the code, access permissions and controller used here? T-DOME, Imprint and Brain Anchor address that mechanism; the present results constrain the resources of the operations it must supply.

XI. CONCLUSION

Persistence of an open subsystem needs controls, and controls are physical. At the algebraic level, associativity and a positive descent metric are what gradient feedback requires, and Clifford algebra remains an optional parameterization. At the physical level, a flip realized by a field coupling has an error, an outgoing stress, a switching work and a derived quadratic cost. Every drive state needs occupation of order $\pi/(4\sqrt{\epsilon})$ for failure ϵ, squeezed drives nearly reach this, and classical drives need order $\pi^2/(16\epsilon)$. In a calibrated feedback task the same actuator gives a complete stability proof, exact loss and coupling rates, and a single condition that ties calibration records to drive energy. Returning a response through a translation-only instrument has a separate sharp mean-energy bound, even with retained ancillas and joint decoding. Together these results price both directions of a control interface; the finite thermal oscillator branch realizes content-preserving inquiry and physical port renewal, but does not supply the chiral exhaust capacity or a reusable geometric readout. The work form, the descent metric and the Lorentzian read-out form stay distinct.

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Retained Quantum History, Physical Sources and Receiver Renewal in Q-RAIF

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Abstract

Acquired quantum records can drive, sense and act again on the same complete material without replacing its old history by independent baths. We construct a positive autonomous Gaussian interface with physical energy-moment and smooth interacting-source bounds, retaining its clock and all old references. A paid Gibbs bank renews a specified receiver. Acquisition, decision, force, renewal and the next reading share one finite cursor path with arbitrarily small joint error as clock resources grow. Exact algebraic transport is a separate question: a reduced coupling supplies a completely positive map whose multiplicative domain is its exactly represented algebra. Discarded non-atomic ports limit recovery; retained oscillator loops and closed propagation permit exact operations.

A full-mode fermion-ring realization renews the same body to uniform per-cycle tolerance through four thermal outlets, then reinstalls the mixed receiver and current purifier by neutral pumps. Physical gates compress noisy multi-cycle records, load overlapping packets and sort the actual spectrum into a paid finite-volume thermal carrier. With every atypical label and old purification retained, the complete output Gibbs deficit per unit time can approach zero in a prescribed finite layout. A separate massive-outside body supplies full-angle local gates and weighted physical-source control.

A spatial work-clock field executes a finite smooth schedule using the actual retained field state. Matrix Ward equations and the delayed incoming kernel close at the same derivative order; physical energy moments control the complete joint gate/reset output. Gap work, clock baseline stress and its finite work correction remain charged. This uses static endpoints and a supplied material frame, with horizon-dependent clock resources. The distinct receiver, ring and massive-body constructions do not yet supply one dynamically supported, covariant apparatus with sustained fixed-capacity sharp exhaust.

Keywords: Q-RAIF, multiplicative domain, quantum records, autonomous control, material response, local stress, thermal renewal

1 Introduction

A physical interface must say what happens to the degrees of freedom that acquire its records. Are they lost, coherently returned, or used in another interaction? Q-RAIF A supplies Lorentzian read-out under its propagation-symbol conditions; Q-RAIF B gives actuation, energy and record permissions; the mass–energy paper identifies momentum and mass read-outs [25, 26, 27]. Here the task is retained-history control and physical renewal, including the source produced by those operations.

The main argument starts with discarded versus retained ports. One oscillator closes exact Weyl loops; one continuum has a compact-time closure obstruction but admits finite adaptive return with its old correlations retained. The multiplicative-domain and propagation results in Appendices A–D state precisely which algebra a coupling transports exactly [29, 30, 31]. This is not a universal conservation-law no-go: Wigner–Araki–Yanase bounds address different specified measurement architectures [34, 35, 36].

Sections 5–6 build the material interface: charged fields and their spin sectors, a finite mode with its remainder, same-Hamiltonian state/source certificates, and actual acquisition–force–sensing–action under one positive autonomous Hamiltonian. The record remains an operator in the old joint history. Its connected fluctuations contribute to the source even when mean trajectories coincide with a classical mixture. The interacting Gibbs reference is defined on that same Hamiltonian, but equilibrium alone supplies no running cursor or blank records.

Section 7 separates port replacement, later free-field thermal windows and whole interacting-equilibrium preparation. Its full-ring construction adds uniform body renewal, neutral block compression, a finite-width packet interface and charge-sector thermal sorting on the actual carrier. The covariance and energy need not commute. Complete-output sharpness retains every bad label; it does not prepare an indefinitely renewed inlet or controller. Section 8 supplies the retained-state field–clock evolution and full joint-state error for a finite gate/reset schedule. The massive-outside gate is a separate physical layout, not the massless ring's reset with a different name.

Three clock/resource statements must be kept distinct. Corollary 16 holds clock mean energy and norm fixed by taking $J = A/m$, while receiver-renewal duration and space grow. Proposition 17 holds J and duration fixed while clock energy mJ grows, placing all finite rounds on one cursor. Proposition 24 evolves a spatial work field χ, retaining its stress and finite work, with static endpoints/frame and inertia chosen for the finite horizon. None establishes an indefinitely running fixed-resource apparatus.

The auxiliary battery, outlet, thermal-preparation, recoil, diagnostic and stress-flow proofs remain in the appendices. Their carriers and supports are explicitly distinct. Section 9 finally supplies a dynamical-support/clock action and identifies the quantum gate and total-source matching still needed. A finite cursor is not thereby a transported relativistic controller, nor is a free thermal window the interacting material's Gibbs state.

Unless stated otherwise, $\hbar = c = k_B = 1$. Energies expressed as frequencies acquire a factor $\hbar$ when units are restored, times and pulse areas use the corresponding frequency units, and entropy is in nats. The combinations $b + b^\dagger$, $i(b^\dagger - b)$ and occupation numbers are dimensionless; material smearings use the normalizations specified with their Hamiltonians below. All joint-state assertions include the stated old references; a preserved nonselective

marginal need not remain unchanged after conditioning on a new record.

2 Discarded ports and coherent return

A discarded record port and a retained environment permit different controls. Write $\chi(t) = \mathrm{Tr}(\sigma_E e^{-itE})$ for the port's spectral characteristic function. Appendix B derives the single-coupling channel and its multiplicative domain; the following bound survives arbitrary later controls that cannot reach the lost port.

Proposition 1 (A fresh port leaves an irreversible defect)**.** *Suppose a finite-dimensional system first undergoes ideal unitary controls and then one coupling $e^{-iX\otimes E}$ to an initially independent port, with X having at least two distinct eigenvalues and the environment's spectral distribution non-atomic. Let F collect every auxiliary port accessible later; just before this coupling the joint state is $\rho_S \otimes \sigma_E \otimes \tau_F$ for every system input. If E is thereafter inaccessible, no subsequent system channel can make the complete channel unitary on the full system algebra. More precisely, let Λ include all the preceding and subsequent controls, and let $\mathcal{V}$ be any unitary channel. For every pair of distinct eigenvalues x, x',*

$$\frac{1}{2}\|\Lambda - \mathcal{V}\|_\diamond \;\geq\; \frac{1 - |\chi(x' - x)|}{2}. \tag{1}$$

Proof. Choose eigenvectors $|x\rangle, |x'\rangle$ and the orthogonal states $|\pm\rangle = (|x\rangle \pm |x'\rangle)/\sqrt{2}$. Their trace distance is one before the coupling and $|\chi(x' - x)| < 1$ afterwards. All subsequent operations induce a system channel and cannot increase this distance. Pull the two inputs back through the preceding unitaries and call them $\rho_\pm$. If $e_\pm = D(\Lambda\rho_\pm, \mathcal{V}\rho_\pm)$, the triangle inequality gives $1 \leq e_+ + |\chi(x' - x)| + e_-$. Thus $\max e_\pm \geq (1 - |\chi|)/2$, which also bounds half the diamond norm from below. □

The independence concerns the entire later-accessible auxiliary system jointly, including any purification of the discarded port. A later register correlated with that port can reveal which phase was applied and allow its reversal; the system's reduced channel alone then does not describe the available recovery operation. For a qubit and no later noise, the bound is attained: remove the phase of χ, and the remaining dephasing is the mixture $(1 + |\chi|)\mathcal{I}/2 + (1 - |\chi|)\mathcal{Z}/2$. Its half diamond distance from $\mathcal{I}$ is $(1 - |\chi|)/2$.

Proposition 2 (An exact interface from a retained oscillator)**.** *Let $Q = b + b^\dagger$, $P = i(b^\dagger - b)$ satisfy the Weyl relations, so $[Q, P] = 2i$, and let X be Hermitian with finite spectrum. Then*

$$e^{-i\lambda XQ} e^{-i\mu XP} e^{i\lambda XQ} e^{i\mu XP} = e^{-2i\lambda\mu X^2} \otimes I. \tag{2}$$

Every initial oscillator state is returned exactly. The induced system map has multiplicative domain $M_s(\mathbb{C})$, including operators that do not commute with the coupling observable X.

Proof. In an X eigenblock, the two pulse generators have scalar commutator

$$[-i\lambda x Q, -i\mu x P] = -2i\lambda\mu x^2.$$

The Weyl group commutator is therefore that scalar phase. Summing the spectral blocks gives (2). □

For a qubit choose $X = (I + \sigma_x)/2$. With $\lambda\mu = \pi/2$, the loop is $e^{-i\pi X} = -\sigma_x$: an exact corrective flip, even with a vacuum or thermal oscillator input. The intermediate system–oscillator correlations are returned, not discarded. Conditional displacement loops of this kind are established geometric gates [24]; the point here is their implication for the interface. Continuous spectrum alone cannot prohibit exact control of a retained port.

Corollary 3 (One retained oscillator gives universal qubit control)**.** *Suppose the four pulses of Proposition 2 can be applied through either $P_z = (I + \sigma_z)/2$ or $P_x = (I + \sigma_x)/2$, with arbitrary real pulse areas. Every qubit unitary, up to global phase, is a sequence of at most three closed loops, and the oscillator returns after each loop.*

Proof. A loop with $2\lambda\mu = \alpha$ implements $e^{-i\alpha P_z} = e^{-i\alpha/2}e^{-i\alpha\sigma_z/2}$, and similarly for P_x. The z–x–z Euler decomposition of a qubit unitary therefore gives three loops. For example the three angles $\pi/2, \pi/2, \pi/2$ give a Hadamard up to global phase. Each identity holds on the full system–oscillator tensor product, so composition returns even an oscillator initially correlated with another system. □

For a larger system the same construction supplies the one-parameter groups e^{-itP_j}. Full unitary control follows when the real Lie algebra generated by iP_j is $\mathfrak{u}(s)$; generation of the associative matrix algebra alone is not the required controllability condition. The pulse permissions, including the choice between noncommuting projectors, remain physical inputs.

The loop is not costless. For a centred input and a mode of frequency ω, the largest conditional occupation increase along the four corners is $(\lambda^2 + \mu^2)x^2$. Minimizing it for the qubit flip gives $\lambda = \mu = \sqrt{\pi/2}$ and peak mean stored energy $\pi\omega\langle X\rangle \leq \pi\omega$. The mode's final energy is unchanged; an ideal reversible switch can recover the intermediate work. This is a stored-energy statement, not a total-device dissipation estimate. It assumes access to both quadratures, their controlled pulse order, and coherent retention until closure. A travelling field that has left the apparatus need not supply these permissions. An implementation with smooth local spacetime couplings must establish them separately.

Thus a record carried away by an inaccessible port constrains future control, while a coherently returned record can disappear without thermal erasure. Retaining an interface can replace a preparation-accuracy requirement by a closure requirement. The oscillator, its routing, clock and switching controls are part of that retained interface; they are not supplied by an abstract algebra homomorphism.

3 Finite return through one thermal field

The oscillator loop closes at one frequency. A propagating field has an interval of frequencies, and a sequence timed for its carrier does not close at every detuning. We now compute what remains, and how a single thermal line can be used over a finite adaptive protocol without resetting it between operations. The field theory, phase controls and reading instruments are still inputs.

3.1 Finite-width forces and the outgoing record

Let $\omega = \Omega + \delta$, $|\delta| \le B < \Omega$, and let g be real, even, square-integrable, and nonzero on an interval. In units $\hbar = 1$, the field Hamiltonian is $H_F = \int (\Omega + \delta) a_\delta^\dagger a_\delta \, d\delta$. For a system projector P with both eigenvalues $0, 1$, apply

$$H_s^I(t) = iP \int g(\delta) \left[u(t) e^{i\delta t} a_\delta^\dagger - \overline{u(t)} e^{-i\delta t} a_\delta \right] d\delta. \tag{3}$$

This is the interaction picture of a prescribed carrier-modulated force. It supplies two field quadratures and a clock; it is band-smeared, not compactly localized in space. At finite $u(t)$ the laboratory Hamiltonian is bounded below by $-|u(t)|^2 \int g^2/(\Omega+\delta)\, d\delta$. A scalar shift makes it nonnegative without altering the gate.

Use four disjoint rectangular pulses of width $\sigma < \tau$, centred at $j\tau$, with complex labels $z_j = (1, i, -1, -i)$ and heights $r z_j/(\sigma\sqrt{Z})$, where

$$Z = \int g(\delta)^2 \operatorname{sinc}^2(\delta\sigma/2) d\delta, \qquad f(\delta) = g(\delta) \operatorname{sinc}(\delta\sigma/2)/\sqrt{Z}, \quad \operatorname{sinc} x = \sin x / x.$$

Thus $\|f\|_2 = 1$. The individual field displacements are $\alpha_j = r z_j f e^{ij\delta\tau}$. Put

$$M(t) = \int |f|^2 \cos(\delta t) d\delta, \qquad C = 3M(\tau) - M(3\tau), \qquad F = rf(1 - e^{2i\delta\tau})(1 + ie^{i\delta\tau}).$$

Proposition 4 (Continuum loop and its thermal defect)**.** *The complete four-pulse unitary is*

$$U_{\text{loop}} = e^{ir^2 CP} D(PF), \qquad D(PF) = \exp \int P(Fa^\dagger - \overline{F}a) d\delta. \tag{4}$$

If $C > 0$, choose $r^2 = \theta/C$ for a target phase $e^{i\theta P}$, $\theta > 0$. For an initially independent thermal field with inverse temperature β, define

$$D_\beta = \int |F(\delta)|^2 \coth[\beta(\Omega + \delta)/2] \, d\delta. \tag{5}$$

The thermal state is specified by its Weyl characteristic; a continuum KMS state need not be a trace-class density matrix on vacuum Fock space. After removal of the target phase the system channel is dephasing between the two P blocks, with coherence factor $e^{-D_\beta/2}$. Its half diamond distance from the target unitary is $(1 - e^{-D_\beta/2})/2$, and its representation defect norm on a unit-norm off-block Pauli control is $1 - e^{-D_\beta}$.

No nonzero compact-time force through this single profile can close the continuum exactly. This last statement concerns the source (3)*, not all multimode control architectures.*

Proof. The source commutator is a scalar field commutator times $P^2 = P$, so the time-ordered exponential is a conditional displacement and a commuting P phase. Within each constant-phase pulse the self-phase vanishes after integration over the even band: its sine kernel is odd in δ. The six inter-pulse Weyl phases add to $r^2[3M(\tau) - M(3\tau)]$. Summing the four displacement vectors gives F. This derivation includes finite pulse width; the pulses are not delta kicks.

The thermal Weyl characteristic is $\langle D(F)\rangle_\beta = e^{-D_\beta/2}$. Tracing the field gives the stated channel, a mixture of identity and conjugation by $2P - I$ after the target phase. Orthogonal off-block superpositions attain the half diamond distance; multiplying its Heisenberg off-block factor gives the defect.

For a general integrable compact-time u the displacement is $g(\delta)\int u(t)e^{i\delta t}dt$. Exact unitary control after discarding the thermal field requires this displacement to vanish. The Fourier transform of compact-time u is entire. Vanishing on an interval where $g \neq 0$ forces it to vanish identically, and Fourier uniqueness gives $u = 0$. At one oscillator frequency only a single zero is required, which is why Proposition 2 has no such obstruction. □

For $\theta = \pi$ and $P = (I + X)/2$, the target is a corrective flip up to phase. With $\mu_2 = \int \delta^2 |f|^2 d\delta$,

$$C \geq 2 - \tfrac{3}{2}\tau^2\mu_2, \tag{6}$$

$$\|F\|_2^2 \leq 8r^2\tau^2(1 + B\tau)\mu_2, \tag{7}$$

$$D_\beta \leq \coth[\beta(\Omega - B)/2]\|F\|_2^2. \tag{8}$$

The first uses $M(\tau) \geq 1 - \tau^2\mu_2/2$ and $M(3\tau) \leq 1$. For the second, $|1 - e^{2i\delta\tau}| \leq 2|\delta|\tau$ and $|1 + ie^{i\delta\tau}|^2 = 2 - 2\sin(\delta\tau) \leq 2(1 + B\tau)$. Temperature multiplies the residual record; it does not remove it.

3.2 One line throughout an adaptive sequence

Multiplying the reduced one-loop channels would assume a fresh field. That is not the evolution we consider. The same field and its correlations remain present, and subsequent projectors need not commute.

Proposition 5 (Finite reuse without independence)**.** *Start with one thermal line independent of system, controller and retained reference. A protocol uses at most L loops, interspersed with arbitrary system–controller instruments whose dilations do not act on the line. Projectors may be selected from acquired records and may differ between loops. Suppose every permitted branch has $D_\beta \leq D_{\max}$ and its target phase is implemented as in* (4)*. The complete actual process, including final records, system and field, is at operational state distance at most*

$$\min\{1, L\sqrt{D_{\max}}\}$$

from the process with ideal target gates and the same initial line. Every terminal loss in $[0, 1]$ *has an expectation difference within this bound, as does the total variation of the complete record laws. No reset or independent-error hypothesis is used. Here operational distance is half the norm distance of the states on their joint observable algebra; it is half trace distance whenever a density-matrix description applies.*

Proof. Purify the initial thermal state, or use its thermal GNS cyclic vector. For a loop acting on the ideal-prefix state,

$$\|(D(PF) - I)|\Psi\rangle\|^2 = 2\langle P\rangle(1 - e^{-D_\beta/2}) \leq D_\beta.$$

This is uniform over system inputs, retained entanglement and the record-controlled choice of projector. In the ideal prefix the line remains thermal and independent; the actual prefix need not. Telescope the actual unitary dilations against the ideal dilations, placing the ideal prefix on each difference and leaving the actual suffix as a norm-preserving factor. The vector difference is at most $L\sqrt{D_{\max}}$. The norm distance of vector states, followed by restriction to physical observables or processing records, gives the stated operational bound. The proof does not replace any actual correlated prefix by a reset state. □

For constant per-pulse relative amplitude error at most e_a, centre error d_t and carrier-phase error e_φ, with unchanged pulse widths and still disjoint pulses, each displacement changes in vacuum norm by at most

$$e = r[e_a + (1 + e_a)(Bd_t + e_\varphi)].$$

The six Weyl phases change by at most $6(2re + e^2)$. The same proof therefore gives the finite-setting bound

$$L\left[\sqrt{D_{\max}} + 4\sqrt{\coth[\beta(\Omega - B)/2]}\, e + 6(2re + e^2)\right]. \tag{9}$$

It concerns these specified setting errors, not arbitrary time-dependent source noise or heating added during the force.

As a concrete family take $g^2 = 15[1 - (\delta/B)^2]^2/(16B)$, $\Omega = 10$, $B = 1$, $\beta = .2$ and $\sigma = \tau/2$. The unfiltered second moment is $B^2/7$; since $\operatorname{sinc}^2(\delta\sigma/2) \le 1$, $\mu_2 \le B^2/(7Z)$ and $Z \ge \operatorname{sinc}^2(B\sigma/2)$ in the stated small-width range. At $\tau = 5 \times 10^{-5}$, $e_a = e_\varphi = 10^{-8}$ and $d_t = 10^{-9}$, conservative rational bounds in (9) put an entire 64-loop adaptive protocol below .00512 operational distance. Tight numerical integration gives one-loop error 1.48×10^{-9}. The sequence statement comes from the joint bound, not from treating those tiny errors as independent. The clock's frequency units are supplied by B; these numbers are not unlabelled laboratory seconds.

3.3 Source energy and control resources

For one loop and an independent zero-mean thermal input, the excess free-field energy is

$$\Delta E_F = \hbar\langle P\rangle \int (\Omega + \delta)|F(\delta)|^2 d\delta. \tag{10}$$

For the chiral current normalization of Paper B, its corresponding free outgoing excess stress is $4\langle P\rangle[\operatorname{Re}\psi_F(u)]^2$, where $\psi_F(u) = \int -i\sqrt{\omega/(4\pi)}F(\omega)e^{-i\omega u}d\omega$. The chosen g vanishes at band edges and gives finite first two time moments. This is the free field source, not the stress of its support, phase reference or switching apparatus.

A reused line has extra cross terms. Let $E_1 = \int \omega|F|^2 d\omega$ bound the unit-projector displacement energy of each loop, and let $n_{\max} = [e^{\beta(\Omega - B)} - 1]^{-1}$. For loops sharing the stated profile, with arbitrary projector choices,

$$|\Delta E_F^{(L)}| \le \hbar\left[2L\sqrt{(\Omega + B)n_{\max}E_1} + L^2E_1\right]. \tag{11}$$

Indeed the final annihilation field is its initial value plus $\sum_j F_j P_j^{\mathrm{past}}$, with $\|P_j^{\mathrm{past}}\| \le 1$. The quadratic term is at most $L^2 E_1$ by the weighted L^2 triangle inequality. Each initial thermal cross term is at most $\sqrt{(\Omega + B) n_{\max} E_1}$ by Cauchy–Schwarz. Thus the single-loop energy cannot simply be multiplied by L. This bound prices an excess over the supplied thermal background, not that background itself.

Small final radiation does not mean small driving demands. With $\|g\|_2 = 1$, the four rectangular sources have

$$\int |u(t)|^2 dt = \frac{4r^2}{\sigma Z}, \qquad |H_{\min}| \le \frac{\hbar r^2}{\sigma^2 Z(\Omega - B)}$$

during a pulse. The first is a declared quadratic envelope charge, not heat; the second is the magnitude of a lower energy bound, not consumed work. Narrowing the loop reduces its escaped record but increases these source and clock requirements. At the 64-loop settings the single-loop envelope charge is about 2.51×10^5, although its unit-projector radiated energy is about $4.49 \times 10^{-8}\hbar$. Finite pulse heights are allowed; ideal switching of the external apparatus is still a supplied control.

For fixed band, temperature and target phase, $D_\beta = O(\tau^2)$ as $\tau \to 0$. Taking $\sigma = \tau/2$ and choosing τ proportional to ε/L therefore suffices for joint error ε, with setting errors also reduced to order ε/L. Within this family the total quadratic envelope charge scales as L^2/ε. This is a constructive resource scaling, not a lower bound for every field-control architecture.

Finite reuse removes the independent-port premise; the initially thermal line is never reset. Stored records, controller renewal, exhaust and total source/geometric response still require a compatible apparatus, not merely a small reduced-channel error.

4 What must be retained by a material interface

A material interface must let the next operation use the record without replacing the held system or its old references. The appendices supply a guarded battery and clock (E), local export (F), actual thermal sources with switching work (G), and contact reaction/recoil (H). Their lattice, co-propagating and bidirectional carriers are distinct.

The main construction now fixes the complete material K. Its mode projection retains the omitted reaction, its local drive uses the actual initial state, and its source certificate controls an unbounded observable. These inputs support actual quantum acquisition, force and later sensing on the same Hamiltonian.

5 From a charged material to a certified finite interface

A material action can retain the charged fields rather than replace them by neutral oscillators. The operator boson–fermion correspondence gives a convenient construction on a circle of circumference L. For one Dirac flavor with antiperiodic spatial fermions, its current oscillators and integer chiral charges have free energy

$$H_0 = \frac{\pi}{L}(N_R^2 + N_L^2) + \sum_{k \ge 1} \frac{2\pi k}{L}(m_{R,k} + m_{L,k}).$$

The current descendants of the charge vacua form a complete orthonormal basis of the fermion Hilbert space [7, Sections 2.5–2.6]. The correspondence includes Klein factors and charged odd operators; it is stronger than equality of partition functions.

For a scalar ϕ of period $\sqrt{\pi}$, put $w = N_R + N_L$ and $n = N_R - N_L$. Its zero-mode description is

$$\bigoplus_{w\in\mathbb{Z}} L^2\big([0,\sqrt{\pi}],\, \psi(\phi_0+\sqrt{\pi}) = (-1)^w\psi(\phi_0)\big) \otimes \mathcal{F}_{\mathrm{osc}}.$$

Here $p_0 = \sqrt{\pi}n$, $n \equiv w \pmod 2$, and the charge energy is $\pi(w^2+n^2)/(2L)$. The state $w = 1, n = 0$ would have $N_R = N_L = 1/2$ and is excluded. An ordinary compact scalar with periodic zero-mode wavefunctions has different sectors. Fixed-spin bosonization requires this additional spin information [8, Section 3.5].

Take two such flavors and real material fields Z, θ, y. The local classical coframe is $d\phi_a + \lambda_a\theta f(Z)dZ$, together with $m_Z dZ, \sqrt{I}\,d\theta, \sqrt{J}\,dy$, where $m_Z^2, I, J > 0$. Its positive canonical energy is

$$H_{\mathrm{mat}} = \frac{1}{2}\int\left[\sum_a \Pi_a^2 + \frac{(\Pi_Z - \sum_a \lambda_a\theta f(Z)\Pi_a)^2}{m_Z^2} + \frac{\Pi_\theta^2}{I} + \frac{\Pi_y^2}{J}\right]dx$$
$$+ \frac{1}{2}\int\left[\sum_a (\phi_a' + \lambda_a\theta f(Z)Z')^2 + m_Z^2 Z'^2 + I\theta'^2 + Jy'^2 + 2V\right]dx.$$

Assume smooth f and a smooth nonnegative V independent of the scalar zero modes. A Fourier regulator retains modes $0 < |k| \le K$ of the interaction and material fields, all winding and charge sectors, and the complete free oscillator tail above K. The real Fourier coordinates are not separately wrapped angles.

Proposition 6 (Positive charged material at finite resolution)**.** *For each finite K, the displayed ordered sum of squares has a nonnegative self-adjoint Friedrichs realization. With the free tail retained, the known boson–fermion unitary transports it to the full four-chiral-field CAR Hilbert space tensor the material space. It strongly commutes with all four canonical chiral charges. No small-coupling bound on the finite λ_a is needed.*

Proof. Work first in each winding sector on smooth zero-mode line-bundle sections and compactly supported smooth wavefunctions of the finite real coordinates. The kinetic factor

$$D_Z(x) = \Pi_Z(x) - \sum_a \lambda_a\theta(x)f(Z(x))\Pi_a(x)$$

is symmetric: its extra coefficient is independent of every ϕ_a coordinate, so its formal divergence vanishes. The other momentum factors are symmetric and the spatial factors are real multiplication operators. Their integrated squares and V define a densely defined nonnegative symmetric operator on the common core. Its Friedrichs extension exists. Finite sums of winding sectors give a dense core for their direct sum. Add the positive free-tail Hamiltonian and use the complete current/charge basis to transport this operator to the fermion space. The smeared fermion operators themselves are transported, not replaced by cutoff vertex exponentials, so their CAR are retained. Translations of each $\phi_{a,0}$ leave the core and closed form invariant, as do winding phases. The associated charge unitaries therefore commute with the self-adjoint realization. □

Low-mode canonical squares retain their common zero-point constant; the free tail uses one fixed normal ordering. No coupling-dependent vacuum subtraction is made. The material momenta, including the shift of Π_Z, remain in the operator. Fourier resolution acts on the coupling, not on the carrier's charges or occupation numbers. It is spatially nonlocal at finite K. Removing this cutoff while controlling propagation, renormalized stress and interacting thermal preparations is a separate limit.

The local current also fixes which coupling is intended. With signature $(-,+)$ and $\epsilon^{01}=1$, let $j^\mu=\epsilon^{\mu\nu}\partial_\nu\phi/\sqrt{\pi}$ and $a_\mu=\lambda\theta f(Z)\partial_\mu Z$. Then

$$-a_\mu\partial^\mu\phi=A_\mu j^\mu,\qquad A_\mu=\sqrt{\pi}\,\epsilon_{\mu\nu}a^\nu.$$

For a static texture, $A_0=-\sqrt{\pi}\lambda\theta f(Z)Z'$ and $A_1=0$, giving the required density potential. The coframe's $-a_\mu a^\mu/2$ term and momentum shifts accompany it. The local dictionary does not by itself fix global holonomies or spin weights [9, Eq. (9) and Section III]; the construction above uses the specified operator correspondence.

Rotating (ϕ_1,ϕ_2) to active and spectator Gaussian modes must also rotate their charge lattice. For instance $(w_1,w_2)=(1,0)$ becomes $(1/\sqrt{2},1/\sqrt{2})$, not two independent integer windings. This scalar normal-mode change is different from rotating the actual fermion flavors within each direction. A Gaussian diagonalization cannot discard their spin constraint or their charged operators.

This material also admits finite autonomous record-controlled operation. Extend its Hamiltonian H_K trivially over finite degenerate record registers. For a bounded self-adjoint permitted operator C and real $h\in C_c^\infty$, set

$$A=\int h(s)e^{isH_K}Ce^{-isH_K}ds,\qquad V=e^{i\ell A}.$$

Integration by parts gives $\|[H_K,A]\|\le\|h'\|_1\|C\|$ and $\|[H_K,[H_K,A]]\|\le\|h''\|_1\|C\|$. The energy-change estimates preceding Proposition 52 now apply with the *interacting material* H_K, so finite products meet Proposition 49. Record-controlled C are finite block sums of the permitted operators; all previous reference systems remain in its input purification.

For $0\le g\in C^\infty(S^1)$, weighting the regulated coupled-sector energy density above by g gives a positive form Q_g with $0\le Q_g\le\|g\|_\infty H_K$. It is a closable sum of the same weighted differential and multiplication squares. Its actual expectation during autonomous execution obeys Eq. (114), with $a=\|g\|_\infty$ and $d=0$. This supplies a state and energy certificate on the charged material itself, without a hard physical energy cutoff. It concerns the regulated energy test: a renormalized covariant stress, spatially local history clock and interacting thermal preparation need their own estimates. Gate permission and the finite-rms initial state are specified inputs, not products of this smoothing argument.

5.1 A finite material mode and its remainder

The finite mechanical coordinates need not arise from fields uniform over the whole line. A local binding potential supplies normalizable angle and storage modes, with the discarded modes still present. Here we specify a fixed-texture Gaussian realization; its state, source

and thermal estimates are not an equivalence theorem for the preceding nonlinear charged regulator. In normalized fields $u = (\sqrt{I}\theta, \sqrt{J}y)$, let

$$A_{\mathrm{m}} = \begin{pmatrix} k/I & -k/\sqrt{IJ} \\ -k/\sqrt{IJ} & (k+K)/J \end{pmatrix}, \qquad \alpha = \lambda_{\min}(A_{\mathrm{m}}) > 0.$$

Choose $0 < s < 1$ and $\mu > 0$ with $\alpha > s\mu^2$. The material operator

$$L_{\mathrm{m}} = -\partial_x^2 + A_{\mathrm{m}} + s^2\mu^2 I - s(s+1)\mu^2 \operatorname{sech}^2(\mu x) I$$

has a strictly positive pointwise mass matrix. For a supplied kink $Z_0 = z_0 \tanh(\mu x)$, the well is $1 - Z_0^2/z_0^2$. Using that polynomial in the material potential keeps it nonnegative for all real Z, since $1 - Z^2/z_0^2 \le 1$. The well can therefore belong to the same positive local action; the kink's preparation and position fluctuations remain physical inputs.

Proposition 7 (Finite modes and their reaction bound)**.** *The normalized $b(x) = N \operatorname{sech}^s(\mu x)$ gives two material modes with squared-frequency matrix A_{m}. The orthogonal material space has gap $m_{\mathrm{gap}}^2 = \alpha + s^2\mu^2$. For compact real $a(x)$, the complete Gaussian form is*

$$q[\phi, u] = \|\phi' + a u_\Theta\|^2 + \langle u, L_{\mathrm{m}} u\rangle.$$

Retaining the carrier and $u = b q_{\mathrm{m}}$ gives

$$q_P[\phi, q_{\mathrm{m}}] = \|\phi' + ab q_\Theta\|^2 + q_{\mathrm{m}}^T A_{\mathrm{m}} q_{\mathrm{m}}.$$

For $z < m_{\mathrm{gap}}^2$, elimination of the orthogonal material space has self-energy $\Sigma(z) \ge 0$, with

$$\langle \chi, \Sigma(z)\chi\rangle \le \frac{\|Q(a\phi' + a^2 b q_\Theta, 0)\|^2}{m_{\mathrm{gap}}^2 - z}.$$

Here Q projects off b in both material components, and the exact retained operator is $K_{PP} - \Sigma(z)$.

Proof. For $D_s = \partial_x + s\mu \tanh(\mu x)$, $L_{\mathrm{m}} = D_s^* D_s + A_{\mathrm{m}}$ and $D_s b = 0$. Its partner is

$$D_s D_s^* = -\partial_x^2 + s^2\mu^2 + s(1-s)\mu^2 \operatorname{sech}^2(\mu x) \ge s^2\mu^2.$$

The nonzero spectra of the two products agree; thus b is the only spatial bound profile, and its orthogonal space has the stated gap. The two retained frequencies need not both lie below that gap. Normalization gives $N^2 = \mu\Gamma(s+1/2)/(\sqrt{\pi}\Gamma(s))$ and $\|b'\|^2 = s^2\mu^2/(2s+1)$, so its gradient cost is finite and retained in A_{m}. Projection gives q_P exactly. The omitted block is $K_{QQ} = Q[L_{\mathrm{m}} + \operatorname{diag}(a^2, 0)]Q \ge m_{\mathrm{gap}}^2 Q$, and $K_{QP}\chi = Q(a\phi' + a^2 b q_\Theta, 0)$. Solving its equation gives $\Sigma(z) = K_{PQ}(K_{QQ} - z)^{-1} K_{QP}$ and the bound. □

The projected contact has profile ab and its complete square. The original k, K, I, J are recovered, rather than fitted after the projection. The remaining material field has not been reset or removed. For a response bound put

$$d_{\mathrm{eff}} = \|ab\|^2, \qquad \beta_a = 4\|a\|_\infty^2(1 + d_{\mathrm{eff}}/\alpha) + 2\|a^2 b\|^2/\alpha.$$

The inequality $\|\phi'\|^2 \le 2(1 + d_{\text{eff}}/\alpha)q_P$ yields $\Sigma(z) \le \beta_a K_{PP}/(m_{\text{gap}}^2 - z)$ as forms. For $\sigma > 0$, let $\epsilon = \beta_a/(m_{\text{gap}}^2 + \sigma^2) < 1$, $R_{\text{ex}} = P(K + \sigma^2)^{-1}P$ on the retained space and $R_P = (K_{PP}+\sigma^2)^{-1}$. Normalizing the Schur complement by $K_{PP}+\sigma^2$ gives $I-D$, $0 \le D \le \epsilon I$, hence

$$\|R_{\text{ex}} - R_P\| \le \frac{\epsilon}{(1-\epsilon)\sigma^2}.$$

This certifies the exponentially weighted causal response: $\int_0^\infty e^{-\sigma t}\sin(t\sqrt{K})/\sqrt{K}\,dt = (K + \sigma^2)^{-1}$.

For finite-time linear-response forcing $F(t) = K_{QP}\chi(t)$ and zero omitted response displacement and velocity, its exact reaction is

$$u_Q(t) = -\int_0^t K_{QQ}^{-1/2}\sin((t-r)\sqrt{K_{QQ}})F(r)\,dr, \qquad \sqrt{2E_Q(t)} \le \int_0^t \|F(r)\|dr.$$

Here E_Q is the energy of that response, not its vacuum fluctuations. The force uses the actual retained history, including backreaction. An initial omitted excitation adds its homogeneous evolution. Near the material threshold, or near a retained resonance, a small projection alone does not bound unfiltered scattering or rapid gates. The full local Gaussian propagator retains its light cone; a single collective coordinate generally does not preserve that sharp front. The estimates concern linear response and omitted energy, not a complete quantum-state or interacting thermal equivalence. They give a finite-body reduction with an explicit remainder while the physical clock, nonlinear recoil and thermal renewal are still to be matched.

5.2 A local record drive on the same quantum material

A finite record can drive the material without replacing its vacuum by that of the projected oscillator. Keep the complete Gaussian operator K of Section 5.1, let $\omega = \sqrt{K}$ and $H = d\Gamma(\omega)$, and use one common material vacuum calibration. For a real material smearing $h = (0, h_\Theta, h_Y)$ define

$$Q(h) = \frac{a(\omega^{-1/2}h) + a^*(\omega^{-1/2}h)}{\sqrt{2}}.$$

The bound $q \ge \alpha\|u\|^2$ gives $\langle h, K^{-1}h\rangle \le \|h\|^2/\alpha$ by the inverse-form variational formula. Spectral Cauchy–Schwarz then gives $\|K^{-1/4}h\|^2 \le \|h\|^2/\sqrt{\alpha}$. Thus these material smearings are defined even when the carrier has no mass gap. No static infrared dressing unitary is assumed.

Let $f \in C_c^\infty(0,T)$ be real and $F_1 = \int |f|$. An acquired classical bit $r \in \{0,1\}$ selects the drive $-rf(t)Q(h)$. In the interaction picture its exact unitary is

$$U_{I,r} = e^{ir^2\Phi_h}W(rz_h), \qquad z_h = \frac{i}{\sqrt{2}}\omega^{-1/2}\int_0^T f(t)e^{it\omega}h\,dt,$$
$$\Phi_h = \frac{1}{2}\int_{s>t} f(s)f(t)\langle h, \omega^{-1}\sin((s-t)\omega)h\rangle\,dt\,ds. \tag{12}$$

The commutator is a scalar, so the Magnus expansion stops here. The Weyl convention and coherent-state overlap are standard [6]. In the Schrödinger picture the mean displacement is the full retarded response

$$\xi_h(t) = r \int_0^t \frac{\sin((t-s)\omega)}{\omega} h f(s)\, ds.$$

For centred Gaussian input, its added energy is

$$E_h = \|\omega^{1/2} r z_h\|^2 = \frac{r^2}{2} \left\| \int f(t) e^{it\omega} h \, dt \right\|^2 \le \frac{r^2 F_1^2}{2} \|h\|^2.$$

It equals the actual force work $\int r f \langle h, \dot{\xi}_h \rangle dt$. The instantaneous forced form is bounded below by $-r^2 f(t)^2 \langle h, K^{-1} h \rangle / 2$. Adding the opposite scalar makes it nonnegative; that offset is not net work, heat or a construction of the driver.

To address the angle mode, choose smooth $0 \le \chi_R \le 1$, equal to one on $|x| \le R$ and zero outside a finite larger interval. Set

$$A_R = \int b^2 \chi_R, \quad h_R = (0, b\chi_R / A_R, 0), \quad h_b = (0, b, 0), \quad d_R = \|h_R - h_b\|.$$

Both have retained force $\langle b, h_\Theta \rangle = 1$. Only h_R is the permitted compact drive; h_b is a comparison smearing in the same full model. If $t_R = \int_{|x|>R} b^2$, then $d_R^2 \le t_R/(1-t_R)$ and $\|h_R\|^2 \le (1-t_R)^{-1}$. For $s = 1/2$, $t_R = 1 - (2/\pi)\arctan(\sinh(\mu R))$. Reducing this tail grows the actual drive support.

Proposition 8 (Local actuation with the initial state retained)**.** *Let $\Delta z = z_{h_R} - z_{h_b}$, with both drives evolved under full K. Then*

$$\|\Delta z\|^2 \le \frac{F_1^2 d_R^2}{2\sqrt{\alpha}}, \qquad \Delta E := \|\omega^{1/2} \Delta z\|^2 \le \frac{F_1^2 d_R^2}{2}.$$

For an actual initial field marginal σ_r with its reference retained, the half trace distance of the two complete outputs obeys

$$D_r \le \sqrt{1 - |\operatorname{Tr} \sigma_r W(r\Delta z)|^2}.$$

A classical distribution of records gives joint distance at most $\sum_r p_r D_r$. On the same Gaussian ground state this becomes $\sqrt{1 - e^{-r^2\|\Delta z\|^2}}$. On a finite Dirichlet interval, for the actual full-K Gibbs state $\sigma_\beta = e^{-\beta H}/Z$, the two displaced field states satisfy

$$S(\sigma_{h_R} \| \sigma_{h_b}) = \beta r^2 \Delta E, \qquad D_{\mathrm{field},r} \le \sqrt{\beta r^2 \Delta E / 2}.$$

Proof. The bound on $K^{-1/4}$ and the spectral integral in Eq. (12) prove the first two inequalities. Writing $B_R = \int b^2 \chi_R^2 \le A_R$ gives $d_R^2 = B_R/A_R^2 - 1 \le A_R^{-1} - 1 \le t_R/(1-t_R)$. Purify the actual field and reference state. The overlap of the two outputs is its Weyl characteristic function, up to a scalar phase. The pure-state trace-distance formula and partial-trace contraction give the retained-reference bound. For the Gibbs identity, conjugate both outputs by $W(-r z_{h_b})$. Displacement preserves entropy, $\log \sigma_\beta = -\beta H - \log Z$, and the initial mean is zero. Its energy increment is $r^2 \Delta E$; Pinsker gives the field-only bound. □

The Gibbs energy bound concerns the reduced field states. A retained thermal purification still uses the characteristic-function bound; its covariance can make that distance larger. Here ΔE is the energy of the difference response, not $E_{h_R} - E_{h_b}$, which has a cross term. No independent ground state of K_{PP} is substituted. Its dynamic reduction still has the reaction error of Proposition 7. An actual old field state must use its own characteristic function; initial correlation is not replaced by a fresh thermal packet. Coherent record controls also retain the Magnus and Weyl phases and are not covered by the classical-record distance bound.

For a centred Gaussian input the covariance is unchanged. A quadratic source, renormalized with the same fixed vacuum subtraction in both experiments, therefore changes by the classical value $T[\xi_h]$. Nonzero initial mean adds its bilinear cross term. The local energy balance has force input $rf(t)h \cdot \dot{\xi}_u$; the driver must supply that work. It uses the complete response, including omitted material modes, not merely the isolated mode's energy. This is the fixed-texture Gaussian source, not the full moving quantum apparatus's stress. The emitted carrier is determined by that material motion. Its equation is $(\partial_t^2 - \partial_x^2)\xi_\phi = \partial_x(a\xi_\Theta)$. For zero past displacement, outside the compact contact the right and left outgoing profiles are

$$\xi_R(u) = -\tfrac{1}{2}\int a(y)\xi_\Theta(y, u+y)dy, \qquad \xi_L(v) = \tfrac{1}{2}\int a(y)\xi_\Theta(y, v-y)dy, \quad u = t-x,\ v = t+x.$$

Integrating the one-dimensional retarded Green function against $\partial_y(a\xi_\Theta)$ gives these endpoint integrals directly. The extra stresses of centred Gaussian input are $(\xi_R')^2$ and $(\xi_L')^2$. They use the complete material response, not an independent emitter. With no incoming drive-induced radiation, the total emitted carrier energy is at most the force work E_h; the remaining material energy is retained. A zero contact $a = 0$ emits no carrier source despite possible stored material energy. Opposite drives also have equal quadratic stress on centred input, so a command label alone does not identify the emitted source.

A finite menu of these record-selected unitaries admits the bounded prepared-bit dilation of Appendix I.1. For compact smooth h_R on the Gaussian ground state, its Weyl amplitudes have all positive frequency moments, so the required twice differentiable H graph orbits exist. Smoothing and autonomous return then use this same H. Other initial states need their own graph-moment certificate. The extra bit, clock, battery, program and retained outputs remain resources. The record must reach the entire drive support by physical transport before the operation; the dilation does not prove that transport or the clock's covariant stress. Approximating a smooth local source during autonomous execution also needs its relative source bound, not trace distance alone. The following estimate provides it for finite-interval output readings of this material.

5.3 A source certificate for the quantum material

Fix a Dirichlet interval of length L and keep all modes of the complete Gaussian K. Write its pointwise mass matrix as $mI \le M(x) \le M_{\max}I$, $m > 0$. For $X = (\phi, \Theta, Y)$ its spatial energy form is

$$q[X] = \|\phi' + a\Theta\|^2 + \|u'\|^2 + \langle u, Mu\rangle, \qquad u = (\Theta, Y).$$

With $C_g = 2 + 2\|a\|_\infty^2/m$ and $V = 2\|a\|_\infty^2 + M_{\max}$, Cauchy–Schwarz gives $\|X'\|^2 \le C_g q[X]$ and $q[X] \le 2\|X'\|^2 + V\|X\|^2$. Min–max on three Dirichlet components bounds the ordered frequencies:

$$c_- j \le \omega_j \le c_+ j, \qquad c_- = \frac{\pi}{3L\sqrt{C_g}}, \quad c_+ = \sqrt{2\pi^2/L^2 + V}.$$

The full tower, not a finite Fock cutoff, enters the next statement.

Let $\mathcal{Y} = (\Pi, \phi' + a\Theta, u', \sqrt{M}u)$ be the energy-factor fields, so their squared norm divided by two is the energy density. For a real symmetric bounded spatial matrix $B(x)$, $\|B(x)\| \le G$, and real $h \in C_c^\infty$, set

$$O = \frac{1}{2}\int h(t) \int : \mathcal{Y}(t,x)^T B(x) \mathcal{Y}(t,x) : \, dx\, dt.$$

Normal ordering uses the same K vacuum in both compared states. $B = gI$ tests the local energy. In a free exterior carrier region a rank-one choice tests $g(\Pi_\phi \mp \phi')^2/4$, the corresponding null stress. These tests may have negative quantum expectations.

Proposition 9 (A same-Hamiltonian quadratic-source bound)**.** *Put $H = d\Gamma(\omega)$, $H_N = \|h^{(N)}\|_1$, $N \ge 2$. The source is a Hermitian form on* $\operatorname{Dom} H^{1/2}$*,*

$$O = d\Gamma(A) + P + P^*, \qquad P = \sum_{ij} b_{ij} a_i a_j.$$

It obeys $|d\Gamma(A)| \le a_0 H$, $a_0 = G\|h\|_1$. Define

$$\kappa^2 = \sum_{ij} \frac{|b_{ij}|^2}{\omega_i \omega_j}, \qquad \nu^2 = \sum_{ij} \frac{|b_{ij}|^2}{\omega_i}.$$

Both converge, with the explicit sufficient bounds

$$\kappa \le \frac{G H_N}{2c_-^N} \sqrt{\zeta(2N-1) - \zeta(2N)},$$
$$\nu^2 \le \frac{G^2 H_N^2 c_+}{8 c_-^{2N}} [\zeta(2N-2) - \zeta(2N-1)]. \tag{13}$$

For any reference energy $\Lambda > 0$, let $C_O = a_0 + 2\kappa + \nu/\sqrt{\Lambda}$. Then $(H+\Lambda)^{-1/2} O (H+\Lambda)^{-1/2}$ is a bounded Hermitian form of norm at most C_O.

Proof. Choose real eigenmodes e_j of K. If $L_s^* L_s = K$ is its spatial energy factor, the annihilation coefficients of $\mathcal{Y}$ are

$$v_j = (-i\sqrt{\omega_j/2}\, e_j, L_s e_j/\sqrt{2\omega_j}), \qquad \langle v_i, v_j \rangle = \omega_j \delta_{ij}.$$

Spatial multiplication by B is bounded by G on this factor space. Time evolution preserves the energy norm; integration against h proves the number-form estimate. The pair coefficient is $b_{ij} = \frac{1}{2}\int v_i^T B v_j dx\, \widehat{h}(-\omega_i - \omega_j)$, where $\widehat{h}(\xi) = \int h(t) e^{i\xi t} dt$. Integrating N times by parts gives

$$|b_{ij}| \le \frac{G\sqrt{\omega_i \omega_j} H_N}{2(\omega_i + \omega_j)^N}.$$

Summing over $i + j = k$ uses $k - 1$ pairs and $\sum_{j=1}^{k-1} j = k(k-1)/2$; the frequency bounds give Eq. (13).

For a unit finite-particle spectral vector of energy E, Cauchy–Schwarz applied to $\sqrt{\omega_i}a_i\psi$ and $\sum_j \overline{b_{ij}}a_j^*\psi/\sqrt{\omega_i}$ gives

$$|\langle P\rangle| \le \sqrt{E(\kappa^2 E + \nu^2)} \le \kappa E + \nu\sqrt{E}.$$

The creation-vector square is a vacuum term ν^2 plus a number form bounded by $\kappa^2 H$: the matrix $\omega^{-1/2} b \omega^{-1/2}$ has Hilbert–Schmidt norm κ. Thus $|\langle O\rangle| \le (a_0 + 2\kappa)E + 2\nu\sqrt{E} \le C_O(E + \Lambda)$. Pair tails converge in these two sums; the uniformly bounded number forms converge on the finite-energy core. Their normalized forms therefore extend to the stated bounded Hermitian form. □

This estimate concerns a spacetime-smeared source form, not an unsmeared density. Its constants depend on L; they do not prove a uniform real-line limit. Compact $W^{2,1}$ time tests also satisfy the $N = 2$ proof. No free chiral energy inequality has been assigned to the coupled material.

Corollary 10 (Output source error with references retained)**.** *For normalized actual and target vectors $\psi, \varphi \in \operatorname{Dom} H$, including their actual references, let $e = \|\psi - \varphi\|$ and $g_H = \|H(\psi - \varphi)\|$. Then*

$$\begin{aligned} |\langle O\rangle_\psi - \langle O\rangle_\varphi| &\le C_O\sqrt{\Lambda e^2 + e g_H} \\ &\quad \times \left[\sqrt{\Lambda + \|H\psi\|} + \sqrt{\Lambda + \|H\varphi\|}\right]. \end{aligned} \tag{14}$$

Proof. Tensor the bounded normalized form with the reference identity and polarize. Its difference is bounded by C_O times $\|(H + \Lambda)^{1/2}(\psi - \varphi)\|$ times the sum of the two weighted norms. The first squared norm is at most $\Lambda e^2 + e g_H$; each remaining squared norm is at most $\Lambda + \|H\psi\|$ or $\Lambda + \|H\varphi\|$. □

The energy-regular return theorem supplies e, g_H for a specified same-H graph-domain operation. It consequently certifies these output source readings under common subsequent H evolution, not the missing driver or clock stress during its interaction. In fact its uniform physical rms bounds M_a, M_i suffice: $g_H \le M_a + M_i$ in Eq. (14). The resulting source error tends to zero with the autonomous state error even as the history clock grows; its physical rms bounds do not grow with that clock. This does not add a permanent halt or decouple an always-running controller for free. A mixed input uses its retained purification. On this finite interval a full-K Gibbs purification has finite energy moments: with $q_\beta = e^{-\beta c_-}$,

$$\langle H\rangle \le \frac{c_+ q_\beta}{(1 - q_\beta)^3}, \qquad \operatorname{Var} H \le \frac{c_+^2 q_\beta(1 + q_\beta)}{(1 - q_\beta)^5}.$$

The bosonic product partition function converges. Bounding occupations by $q_\beta^j/(1 - q_\beta)$ and summing $j q_\beta^j$ and $j^2 q_\beta^j$ proves the two estimates. All higher polynomial moments are finite as well. This does not prepare the Gibbs state, prove a particular gate's graph orbit, or renew a correlated reservoir.

For the vacuum-driven outputs $|z\rangle, |w\rangle$ of two actual forces on this same material, put $c = e^{-\|z-w\|^2/2}$, $E_z = \|\omega^{1/2} z\|^2$ and $\Delta E = \|\omega^{1/2}(z-w)\|^2$. Choosing only a comparison phase to make their overlap c gives the exact identity

$$\|(H+\Lambda)^{1/2}(|z\rangle - |w\rangle)\|^2 = c\Delta E + (1-c)(E_z + E_w + 2\Lambda).$$

Indeed $\langle z|H|w\rangle = \langle z|w\rangle\langle z, \omega w\rangle$. Their source difference is at most C_O times the square root of this expression times $\sqrt{E_z+\Lambda} + \sqrt{E_w+\Lambda}$. Proposition 8 bounds its two difference terms for compact forces. This comparison keeps full K; changing the box or using the projected oscillator changes the problem. For coherently controlled records the actual relative phases remain, rather than being physically aligned branch by branch.

Trace distance alone cannot replace these estimates: a trace-small admixture can have arbitrarily large energy. Nor does matching mean energy suffice. A small vacuum–two-particle amplitude changes a pair-source reading to first order while its energy changes to second order. The source certificate retains both the state and its energy norm. Matching this finite-interval Gaussian readout to a charged nonlinear continuum and the complete moving apparatus remains a separate physical construction.

6 Acquired quantum records on the complete material

The preceding force calculation conserved a classical record during its drive. An acquired record need not be classical in the full old history. Keep the complete finite-interval K and $H_F = d\Gamma(\omega)$, $\omega = \sqrt{K}$, of Section 5.3. Let the finite controller include probes, record cells and program cursor. Its nonnegative Hamiltonian H_D need not commute with the force amplitudes. The following construction therefore includes acquisition and a later record-controlled response within one autonomous dynamics.

6.1 A positive common Hamiltonian and a uniform physical moment

Let B_a be finitely many commuting bounded self-adjoint controller operators and h_a real smooth compact material smearings. Define

$$F = \sum_a B_a h_a, \qquad C_{\min} = \frac{1}{2}\sum_{ab} B_a B_b \langle h_a, K^{-1} h_b\rangle,$$
$$H_{\mathrm{tot}} = H_D + H_F - \sum_a B_a Q(h_a) + C_{\min}. \tag{15}$$

The full canonical material fields obey $[Q(h), P(k)] = i\langle h, k\rangle$, with $Q(h) = [a(\omega^{-1/2}h) + a^*(\omega^{-1/2}h)]/\sqrt{2}$. This is the canonical normalization of the material Hamiltonian, distinct from the earlier dimensionless $[b+b^\dagger, i(b^\dagger - b)] = 2i$ convention. The amplitudes commute at equal Schrödinger time, not necessarily at different Heisenberg times. In particular a cursor-controlled record force and a probe force can commute while acquisition edges in H_D change those same records.

Proposition 11 (Positive quantum interface and physical rms bound)**.** *The Hamiltonian* (15) *is self-adjoint and nonnegative. Its domain is equivalent to* $\operatorname{Dom} H_F$*. For a normalized initial vector* ψ *in that domain, including all old references, put*

$$L_F = \sqrt{2}\sum_a \|B_a\|\|\omega^{-1}h_a\|, \qquad c_F = \frac{1}{\sqrt{2}}\sum_a \|B_a\|\|\omega^{-1/2}h_a\|,$$

$$A_F = \|H_{\rm tot}\psi\| + \|H_D\| + \|C_{\rm min}\| + c_F, \qquad M_F = \left[\frac{L_F + \sqrt{L_F^2 + 4A_F}}{2}\right]^2.$$

Then $\|H_F e^{-itH_{\rm tot}}\psi\| \le M_F$ *for every* t*. If the initial cursor lies in a force-idle region and* $\|H_D\|$*, coupling norms and initial physical rms stay fixed, this certificate does not grow with the finite program length.*

Proof. Resolve the common spectrum of the B_a into projections P_b and write $f_b = \sum_a b_a h_a$, $z_b = -\omega^{-3/2} f_b/\sqrt{2}$. The finite interval has a gap; hence the finite controlled Weyl unitary $D = \sum_b P_b \otimes W(z_b)$ exists and $D^* H_F D = H_F - \sum B_a Q(h_a) + C_{\rm min}$. It follows that $H_{\rm tot} = D^* H_F D + H_D \ge 0$. The linear field bound

$$\|Q(h)\zeta\| \le \sqrt{2}\|\omega^{-1}h\|\|H_F^{1/2}\zeta\| + \|\omega^{-1/2}h\|\|\zeta\|/\sqrt{2}$$

also makes the force infinitesimally H_F-bounded, giving the graph domain statement by the closed operator realization. For $\psi_t = e^{-itH_{\rm tot}}\psi$ and $x = \|H_F\psi_t\|$, conservation of $\|H_{\rm tot}\psi_t\|$ gives $x \le A_F + L_F\sqrt{x}$, since $\langle H_F\rangle_t \le x$. Solving this quadratic inequality proves the bound. Spectral-core approximation extends the calculation to the stated domain. In a force-idle initial region, $\|H_{\rm tot}\psi\| \le \|H_F\psi\| + \|H_D\|$. Every step tensors with the actual reference identity. □

The displacement D proves positivity; it is not a free preparation of a dressed vacuum. This finite-interval argument is not an infinite-line static-dressing theorem. A positive nearest-neighbor Feynman cursor with bounded edge gates can have $\|H_D\| \le 4J + \|H_{\rm data}\|$ independently of its site count [4]. This removes a program-length factor from the physical moment bound, not from cursor space, execution time or preparation cost. For the uniform-J cursor, Foundation III's lemma "Wholly idle input and a finite completion window" in the appendix "A Finite Arrival Bound at Fixed Hopping Energy" bounds arrival for a packet supported wholly in the input idle region of the pure gate-dressed hopping comparison [50]. Cell gaps and forces require the subsequent Duhamel control; this is neither the spin-clock nor the spatial work-clock estimate below. Linear finite-system/field Hamiltonians and their domain analysis are established tools [5]; the issue here is their common retained-record interface.

The field obeys the operator retarded equation

$$Q(t) = \cos(\omega t)Q(0) + \omega^{-1}\sin(\omega t)P(0) + \int_0^t \omega^{-1}\sin[\omega(t-s)]F(s)\,ds. \tag{16}$$

It follows by solving $\dot{Q} = P$, $\dot{P} = -KQ + F(t)$ as weak smeared identities on the energy core. The $B_a(s)$ in $F(s)$ are the actual joint Heisenberg operators. Their expectations are not an

externally supplied force history. With material-only force, the exterior carrier formula of Section 5.2 holds with the complete operator $\Delta\Theta$:

$$\Delta\phi_R(u) = -\tfrac{1}{2}\int a(y)\Delta\Theta(y, u+y)\,dy, \qquad \Delta\phi_L(v) = \tfrac{1}{2}\int a(y)\Delta\Theta(y, v-y)\,dy.$$

An additional carrier-smeared probe contributes its own retarded source. Quadratic stress uses correlations of these operators, not only the square of their mean.

6.2 Actual acquisition, sensing and the next action

For a probe P and a blank record R, the unitary $C_Y = P_-^Y \otimes I_R + P_+^Y \otimes X_R$ acquires the probe's Y reading. Retain P, R and their old references. During a subsequent block that conserves R, the force of Section 5.2 acts as $V = \sum_{r=0,1} |r\rangle\langle r| \otimes e^{ir^2\Phi_h} W(rz_h)$. For actual initial material/reference blocks $\sigma_{rr'}$, the output is

$$\sum_{rr'} |r\rangle\langle r'| \otimes e^{i(r^2-r'^2)\Phi_h} W(rz_h)\sigma_{rr'}W(r'z_h)^*. \tag{17}$$

This is not a sampled classical force. For a pure complete input with both record branches present, dephasing R changes the full retained state by half trace distance $\sqrt{p(1-p)}$, where $p = \Pr(R=1)$. Indeed the deleted off-diagonal block connects two orthogonal vectors of norms $\sqrt{p}, \sqrt{1-p}$ and has two eigenvalues $\pm\sqrt{p(1-p)}$.

Now let a fresh sensor S start in $|+x\rangle$, couple by $e^{igZ_S Q(b)}$ to the material after its actual propagation, and acquire its Y value into a cell E. The latter starts with thermal occupation ν, independently prepared with its purifier retained. For a centered Gaussian material initially independent of the first probe, let μ_r be the actual conditional mean of $Q(b)$ and V_b its conditional variance after the force. Then

$$\Pr(E=1 \mid R=r) = \frac{1}{2} - \frac{1-2\nu}{2} e^{-2g^2V_b}\sin(2g\mu_r). \tag{18}$$

The sensor's Y mean is $-\operatorname{Im}\langle e^{2igQ(b)}\rangle$; the Gaussian characteristic function and the mixed cell's visibility $1-2\nu$ give the formula. The acquisition acts on the same disturbed material, and the actually acquired n_E can control the next force. Its energy and source are part of that dynamics. For a correlated or non-Gaussian input use its actual characteristic function, not this Gaussian specialization.

Before sensing, a conserved source record and centered product input give $\langle\Delta j\rangle = pj_h$ but $\Delta\langle{:}\, j^2 \,{:}\rangle = pj_h^2$, where j_h is the complete driven carrier current. The connected contribution is $p(1-p)j_h^2$, not an extra energy charge. Similarly the full added material/carrier energy is pE_h, of which p^2E_h is the energy of the mean displacement. Record dephasing leaves this immediate field marginal unchanged while changing the old joint history. Equal mean source therefore does not establish equal later control.

The formulas describe the sequential block. Under the autonomous Hamiltonian (15), acquisition, force and sensing instead occupy cursor regions that a dispersive cursor can revisit. Its deadline statistics require the actual joint evolution, without postselection or assignment of the sequential probabilities.

A one-mode check retains the old probe reference and the second cell's purifier: $p = .2766071$, $\Pr(E = 1 \mid R = 1) = .4267590$, dephasing distance .4473205, and first-acquisition probe/reference change below 4.1×10^{-16}. Cutoffs $6, 10, 14$ reduce Gaussian-law error from 1.04×10^{-7} to 1.11×10^{-16}. Field energy .0418368 splits into mean .0115724 and connected .0302645; sensing adds .04805 $= g^2/2$ at $g = .31$. This checks the sequence, not the full spatial K theorem.

A separate 31-site positive cursor diagnostic executes actual first acquisition, record force, sensing, second acquisition, receiver/bank SWAP and a later force controlled by the spent cell's actual record. With both cell purifiers retained, its unpostselected deadline $t = 40$ has field energy .0300724, physical rms at most .186211 against the uniform bound 3.01330, and energy residual below 1.9×10^{-13}. But only .857524 of its cursor probability is after the SWAP region; its receiver occupation is .157712, exceeding the prepared bank value .13 by .027712. Unfinished and returning cursor branches retain pre-SWAP data and can revisit acquisition and force regions. The program therefore fails an exact deadline-renewal claim. The force-idle window condition and its completion error must be certified separately, as done below; the sequential probability formula is not assigned to this autonomous output.

6.3 The local field source during interaction

The free-output qualification in Corollary 10 is unnecessary for a specified smooth test under the same H_{tot}. Let $B(x)$ be a real symmetric smooth matrix, supported strictly inside the Dirichlet interval, with $\|B\|_\infty = G$; let $w \in C_c^\infty$ be real and supported in $[0, T]$. Use the same free-K vacuum subtraction and energy factors $\mathcal{Y} = (P, L_s Q)$, $L_s^* L_s = K$. Define

$$O_{\mathrm{int}} = \int w(t) \int [\mathcal{Y}(t,x)^T B(x) \mathcal{Y}(t,x)/2 - \text{free vacuum}_K]\, dx\, dt.$$

This is the material/carrier field source, not the controller's stress. By (16),

$$\mathcal{Y}(t) = \mathcal{Y}_0(t) + \Delta\mathcal{Y}(t), \qquad \Delta\mathcal{Y}(t) = \sum_a \int_0^t M_{t-s} h_a B_a(s)\, ds,$$

where $M_\tau h = (\cos(\omega\tau)h, L_s \omega^{-1} \sin(\omega\tau) h)$ and $\|M_\tau h\| = \|h\|$. As a vector of joint-state operators, $\|\Delta\mathcal{Y}(t)\psi\| \le W(t)\|\psi\|$, $W(t) = t \sum_a \|B_a\| \|h_a\|$.

Proposition 12 (An interacting-window field-source bound)**.** *Assume smooth coefficients of K and the smooth compact h_a above. For each a, define finite constants*

$$D_{P,a} = \sup_{0 \le \tau \le T} \|\omega^2 [B M_\tau h_a]_P\|,$$
$$D_{S,a} = \sup_{0 \le \tau \le T} \|\omega^2 L_s^* [B M_\tau h_a]_S\|.$$

With $\omega_i \ge c_i$ as in Section 5.3, set

$$C_0(t) = \frac{t}{\sqrt{2}} \sum_a \|B_a\| [D_{P,a} c_-^{-3/2} \zeta(3/2) + D_{S,a} c_-^{-5/2} \zeta(5/2)],$$
$$C_1(t) = \frac{t}{\sqrt{2}} \sum_a \|B_a\| [D_{P,a} c_-^{-2} \zeta(2) + D_{S,a} c_-^{-3} \zeta(3)].$$

For any $\Lambda > 0$, O_{int} is a Hermitian form with

$$\|(H_F + \Lambda)^{-1/2} O_{\mathrm{int}} (H_F + \Lambda)^{-1/2}\| \le C_{\mathrm{int}}, \tag{19}$$

where

$$C_{\mathrm{int}} = C_O + \int |w(t)| \left[\frac{C_1(t)}{\sqrt{\Lambda}} + \frac{C_0(t) + GW(t)^2/2}{\Lambda} \right] dt.$$

Here C_O is the free source constant with time test w. All references and the finite controller are retained. For two initial joint vectors of rms at most M_F and vector distance e, their actual interacting-window source difference is at most

$$2C_{\mathrm{int}} \sqrt{\Lambda e^2 + 2M_F e} \sqrt{\Lambda + M_F}. \tag{20}$$

Proof. The free part is Proposition 9. The response-square term is bounded by $GW(t)^2/2$ without assuming commuting response components. For the symmetrized cross term, let $v_i(t,x)$ be the free annihilation coefficient of mode i and $C_i(t) = \int v_i(t,x)^T B(x) \Delta\mathcal{Y}(t,x)\, dx$. Projection of the two response components onto e_i and insertion of ω^2 give

$$\|C_i(t)\| \le \frac{t}{\sqrt{2}} \sum_a \|B_a\| [D_{P,a} \omega_i^{-3/2} + D_{S,a} \omega_i^{-5/2}].$$

Interior multiplication by smooth B removes boundary compatibility conditions. Iterating the elliptic operator $K = -I\partial_x^2 +$ smooth lower-order terms controls the necessary Sobolev orders of propagated compact h_a, proving finiteness of the displayed graph norms uniformly over the compact time interval. Thus $\sum_i \|C_i(t)\| \le C_0(t)$ and $\sum_i \|C_i(t)\|/\sqrt{\omega_i} \le C_1(t)$.

For $E = \langle H_F \rangle_\psi$, $\|a_i \psi\| \le \sqrt{E/\omega_i}$ and $\|a_i^* \psi\| \le \sqrt{E/\omega_i} + 1$. Bounding $\langle a_i C_i \rangle$ by $\|a_i^* \psi\| \|C_i \psi\|$, and $\langle a_i^* C_i^* \rangle$ by $\|a_i \psi\| \|C_i^* \psi\|$, bounds the cross expectation by $2C_1(t)\sqrt{E} + C_0(t)$. No commutation between C_i and a_i has been used. The C_0 term is the creation contribution; treating the record as a c-number would incorrectly remove it. Apply $2\sqrt{E} \le (E + \Lambda)/\sqrt{\Lambda}$ and add the free and response bounds. Absolute convergence of the cross terms, the bounded response vector and the established free form give spectral-core convergence. Polarization proves (19). The weighted-form difference estimate of Corollary 10, with $\|H_F(\psi - \varphi)\| \le 2M_F$, proves (20). □

The constants do not depend on cursor site count if its coupling norms stay fixed. They do depend on the box, smooth test, profiles and time window. Rough spatial multipliers, boundary distributions and unsmeared stress are not covered. This result compares retained inputs under one common H_{tot}; it does not compare arbitrary different apparatus Hamiltonians or supply the controller's missing stress.

6.4 A local contact energy and the interacting thermal reference

The inverse-K counterterm in (15) is useful for proof but is not a local physical density. For smooth compact material profiles $w_a(x)$ choose $h_a = (0, M(x) w_a(x))$ and replace it by

$$C_{\mathrm{loc}} = \frac{1}{2} \sum_{ab} B_a B_b \int w_a^T M w_b \, dx.$$

The actual mass-potential density becomes

$$V_{\mathrm{loc}}(x) = \tfrac{1}{2}[u - \sum_a B_a w_a]^T M[u - \sum_a B_a w_a]. \tag{21}$$

All gradient and current squares of the complete material remain. For each joint spectral branch $w = \sum_a b_a w_a$, the inverse-form variational formula gives

$$\langle (0, Mw), K^{-1}(0, Mw)\rangle = \sup_X \{2\langle w, Mu\rangle - q[X]\} \le \langle w, Mw\rangle,$$

because $q[X] \ge \langle u, Mu\rangle$. Consequently $C_{\mathrm{loc}} - C_{\min} \ge 0$ is a bounded controller operator. The actual local-contact Hamiltonian is $D^* H_F D + H'_D$ with $H'_D = H_D + C_{\mathrm{loc}} - C_{\min} \ge 0$. The positivity, rms and source proofs therefore apply to this physical contact, using its actual C_{loc} and the adjusted controller norm. No local stress is assigned to a nonlocal inverse operator.

Proposition 13 (Interacting equilibrium and retained record capacity)**.** *For either contact, put $d = \dim \mathcal{H}_D$, $d_0 = \|H'_D\|$, and $Z_F(\beta) = \operatorname{Tr} e^{-\beta H_F}$. Then the actual interacting Gibbs state exists and*

$$\begin{aligned} d e^{-\beta d_0} Z_F(\beta) \le Z_{\mathrm{tot}}(\beta) \le d Z_F(\beta), \\ \langle H_{\mathrm{tot}}^2 \rangle_\beta \le \frac{16 e^{\beta d_0}}{e^2 \beta^2} \frac{Z_F(\beta/2)}{Z_F(\beta)}. \end{aligned} \tag{22}$$

If independent uniform input bits $X_1, \dots, X_N$ are retained in the accessible controller and material with field mean energy at most M, and each is later readable with error $\epsilon_j \le 1/2$, then

$$\sum_j [\log 2 - h_2(\epsilon_j)] \le \log d + \inf_{\beta > 0} [\beta M + \log Z_F(\beta)]. \tag{23}$$

Any decoder-accessible outside reference is an additional resource.

Proof. The eigenvalues of $D^* H_F D + H'_D$ lie between those of $D^* H_F D$ and those eigenvalues plus d_0, by min–max. Summing scalar exponentials proves the partition bounds; operator monotonicity of the exponential is neither available nor needed. For $x \ge 0$, $x^2 e^{-\beta x/2} \le 16/(e^2\beta^2)$. The partition ratio then proves the moment bound; its controller multiplicity cancels. Subadditivity and the field Gibbs variational inequality give $S(\rho) \le \log d + \beta M + \log Z_F(\beta)$. Binary Fano, conditional entropy subadditivity for the independent classical inputs, and data processing give $\sum_j [\log 2 - h_2(\epsilon_j)] \le I(X^N : \text{apparatus}) \le S(\rho)$. Take the infimum. The frequency bound also yields $\log Z_F(\beta) \le q/(1-q)^2$, $q = e^{-\beta c_-}$, so the capacity is finite for fixed resources. □

The same-H Gibbs purification is a graph-domain input, not the product Gaussian in (18). Stationary equilibrium supplies neither directed cursor nor blank records; $\beta^{-1} D(\rho_{\mathrm{prep}} \| \rho_\beta)$ is a finite running state's free-energy excess, not its preparation protocol. At fixed energy, cursor dimension increases record capacity only logarithmically. Any outside history bank or entropy channel remains a resource.

7 Finite renewal and continuing return

7.1 Replacing a port is not renewing the interacting material

There are three distinct thermal operations below: replacement of a specified finite port; local thermal restriction on a free-carrier window; and preparation of the whole interacting Gibbs state. Only the first two have the stated finite operating protocols.

Proposition 14 (Universal held-content preservation forces a product extension)**.** *Let a channel $\mathcal{E}$ append a finite-dimensional port B to a held d-dimensional system A, whose marginal is the identity channel on every input, including all reference extensions. Then $\mathcal{E}(\rho) = \rho \otimes \tau$ for one fixed port state τ. Thus such a universal nondisturbing preparation cannot reproduce a correlated interacting Gibbs state. For $H = JZ_AZ_B$, $x = \beta J$, its Gibbs state is $\gamma = (I - \tanh(x)Z_AZ_B)/4$. At held input $I/2$ every such extension has half trace distance from γ at least $|\tanh x|/2$, attained by $\tau = I/2$.*

If instead only the classical Z_A label is conserved while conditional port Gibbs states $\tau_\pm = (I \mp \tanh(x)Z_B)/2$ are installed, the largest retained coherence factor is $\operatorname{sech} x$*. After the best held-system phase correction, an input $|+\rangle$ therefore suffers half trace distance at least $(1 - \operatorname{sech} x)/2$.*

Proof. Dilate the channel and purify its fixed ancilla. A pure held input must have a pure unchanged held marginal; hence the isometry sends $|\psi\rangle$ to $|\psi\rangle|e_\psi\rangle$. Applying this to basis states and their superpositions forces one common environment vector. Tracing the environment gives the product extension. Equivalently the identity channel has rank-one Choi state. For the Gibbs example the observable Z_AZ_B has norm one, expectation zero in $(I/2)\otimes\tau$ and $-\tanh x$ in γ, giving the distance bound; direct diagonal subtraction proves attainment. In a Z-nondemolition conditional preparation the two environment/port vectors purify τ_+, τ_-. Their overlap is at most their root fidelity, $\sum_z \sqrt{p_+(z)p_-(z)} = \operatorname{sech} x$, with equal Schmidt bases attaining it. That overlap multiplies the held off-diagonal matrix element. The qubit trace-distance formula proves the last assertion. □

This is a universal identity-channel condition, not a prohibition on preparing one specified state or maintaining a selected classical content. It supplies no universal constant for arbitrary approximate identity channels. An entire-body SWAP can prepare a fresh correlated copy, but it moves the old body's quantum content and correlations to the spent copy; it is not that identity channel.

For an attainable finite update, specify instead one receiver cell E with $H_E = \delta n_E$ and a fresh bank cell B with the same gap, Gibbs state η_β, and retained purifier. All held material, earlier records and old references form $\mathcal{O}$; the previous state of E may be correlated with all of them. The true S_{EB} commutes with $H_E + H_B$. The positive pulse $g(t)(I - S_{EB})$, $g \geq 0$, $\int g = \pi/2$, implements the true SWAP at finite duration, at least $\pi/(2g_{\max})$. At its endpoint E is η_β independently of the old $\mathcal{O}E$ state. That complete old state now belongs to $\mathcal{O}B$, with its correlations retained. The new cell remains correlated with its own Gibbs purifier. The $\mathcal{O}$ marginal is unchanged nonselectively; old records stored in E have moved to the spent slot rather than being erased. For an isolated prescribed pulse its switch work is zero because the bare cell energy is conserved and the pulse vanishes at the endpoints. This is not zero total controller or preparation cost.

Such a SWAP is also a bounded edge gate in H_D of the positive common material Hamiltonian. The cell must be force-idle while it is replaced. If n_E is still coupled to the material, $[S_{EB}, n_E Q(h)] \neq 0$; the matched gap alone gives no exact reset for that interacting contact. A finite-window estimate can retain this imperfection explicitly.

Proposition 15 (Finite-window port renewal with a live material)**.** *Compare the actual autonomous interface with an ideal renewal evolution having the same H_D, matched cell gaps and SWAP edge gate, but no material force in its renewal window $[0, T_R]$. Suppose their difference on the reference trajectory is $-\sum_a \Delta B_a Q(h_a) + \Delta C$, where $\Delta B_a = P_{\rm leak}\Delta B_a$, $\Delta C = P_{\rm leak}\Delta C$ and these controller operators commute with H_F. Here $P_{\rm leak}$ is a controller orthogonal projection; the differences are self-adjoint, so their displayed one-sided support is also two-sided support. Write $b_a = \|\Delta B_a\|$, $c = \|\Delta C\|$, $p(s) = \|P_{\rm leak}\psi_s^i\|^2$, and assume $\|H_F\psi_s^i\| \leq M$. Define*

$$L_\Delta = \sqrt{2}\sum_a b_a \|\omega^{-1} h_a\|, \qquad c_\Delta = c + \sum_a b_a \|\omega^{-1/2} h_a\|/\sqrt{2}.$$

Including the bank and every old reference, the vector error is at most

$$\epsilon_R \leq \int_0^{T_R} [L_\Delta \sqrt{M}\, p(s)^{1/4} + c_\Delta \sqrt{p(s)}]\, ds. \tag{24}$$

Any specified incomplete ideal-SWAP or clock error is added separately. The same bound controls half trace distance of the renewed cell and held-reference marginal from their ideal outputs.

Proof. Duhamel bounds the vector difference by the integrated norm of the Hamiltonian difference on ψ_s^i. Apply the linear-field estimate to $P_{\rm leak}\psi_s^i$. Its norm is $\sqrt{p}$ and

$$\|H_F^{1/2} P_{\rm leak}\psi_s^i\|^2 \leq \|P_{\rm leak}\psi_s^i\| \|H_F P_{\rm leak}\psi_s^i\| \leq M\sqrt{p}.$$

The field and counterterm contributions give (24). Pure-vector distance bounds half trace distance, which contracts under the stated restrictions. □

Corollary 16 (A constructive finite receiver-renewal family)**.** *Fix $A > 0$, $0 < p < 1/2$, a matched receiver/bank pair and a material smearing h. For even m, use the positive spin-history clock with $J = A/m$, a true receiver/bank SWAP on the first edge and identities thereafter. Put $\chi_m = \mathbf{1}_{\{j \geq m/2\}}$ and*

$$H_m = H_{\rm hist} + H_E + H_B + H_F - \chi_m n_E Q(h) + \tfrac{1}{2}\chi_m n_E \langle h, K^{-1} h\rangle.$$

Its initial cursor is $j = 0$; the material/old receiver state and all old references may be correlated, with initial physical rms $M < \infty$. The independent bank is Gibbs with its own purifier retained. This is a positive autonomous Hamiltonian. At $t_R = (m/A)\arcsin\sqrt{p}$ its joint half trace distance from a completed receiver renewal, with an explicitly specified comparison clock, is at most

$$2(1-p)^{m/2} + \frac{m \arcsin\sqrt{p}}{A}[L_\Delta \sqrt{M}\, e^{-mD/4} + c_\Delta e^{-mD/2}], \quad D = D_{\rm Ber}(1/2\|p) > 0. \tag{25}$$

It tends to zero while the clock mean energy A, its norm bound $2A$ and the common-H physical rms certificate remain bounded independently of m. Clock space and duration grow; the bank is not cycled for free.

Proof. The field/contact terms are the positive controlled-displacement Hamiltonian of Proposition 11; the history clock and matched cell energies are nonnegative. For the force-free reference, each prefix is either the true SWAP or identity and commutes with $H_E + H_B + H_F$. Prefix conjugation therefore gives the same binomial cursor law for every correlated input. For $s \le t_R$ its binomial parameter is at most p; Chernoff gives $\Pr(j \ge m/2) \le e^{-mD}$. The ideal physical rms is the unchanged M. Proposition 15 proves the second term. The no-SWAP probability is $(1-p)^m$. As comparison use the same untrimmed force-free cursor with the completed SWAP and freely evolved material/reference state. Only its $j = 0$ branch differs from the ideal joint state; the retained-purification vector error is at most $2\sqrt{(1-p)^m}$. Half trace distance is no larger. No clock outcome is postselected. Triangle inequality proves (25). Finally $\|H_{\mathrm{hist}}\| \le 2A$ and its initial mean is A. The cell norms and force/counterterm norms are fixed, so the initial total rms and the physical bound of Proposition 11 are uniform. □

This condition has a finite nonempty realization. Take the positive spin-history clock with $m = 64$, $J = .02$, the matched SWAP on its first edge, and a material force $-.2\,\mathbf{1}_{\{j\ge 32\}} n_E Q(h)$ with $\omega = 1$ and unit one-mode smearing. At $T_R = \arcsin\sqrt{.1}/J = 16.08753$ the force-free reference cursor has $\mathrm{Bin}(64, .1)$ law; throughout this window its source tail is at most 7.04652×10^{-16}. Its uncompleted first-edge probability is $.9^{64} = .00117902$. For vacuum material input $M = 0$, (24) is below 6.9×10^{-8}. Completing the ideal SWAP on the same untrimmed cursor costs at most $2\sqrt{.9^{64}} = .068674$ in vector norm. The clock's mean energy is 1.28. Direct sparse evolution of the full force-on Hamiltonian at Fock cutoffs $6, 10, 14$, retaining a correlated held-body/receiver input, its old reference and the fresh bank purifier, gives vector difference 6.54×10^{-10} from the same force-free reference. The receiver's half trace distance from its bank state is .00043624; the held-body/reference marginal changes by less than 2.5×10^{-13} and energy conservation residual is below 4.1×10^{-13}. These are finite-regulator diagnostics of the nonempty renewal window, not a continuum proof. This is a paid receiver-renewal block with given correlated input, not an assertion that a previous program produces its cursor for free.

Actual leakage and completion require paid force-idle padding. The receiver certificate preserves held material and old content, but neither prepares its interacting Gibbs state nor implements an entropy outlet. A J-cell bank retains J spent slots, outputs and purifiers; each isolated replacement transfers excess energy $\delta(\langle n_E\rangle - \nu)$ of either sign. Preparation, clock, storage and removal remain charged. Q-RAIF B's oscillator-bank formulas [26] concern its different material. Repeated noisy readings of one command also do not justify $J \log 2$ in the independent-bit capacity bound (23).

7.2 Several rounds on one running material

The next reading can be installed on the same cursor that performs replacement. Fix a finite sequence of permitted data gates U_b, including receiver–bank swaps, and a nonnegative data energy H_0 with $[U_b, H_0] = 0$. Degenerate logic and matched-cell exchanges meet this condition. Keep the body, all banks, their purifiers, previous records and reversible workspace in the input. Between gates use the linear material forces and completing terms of Eq. (15).

Choose rational $0 < a_1 < \cdots < a_q < 1$ and integers m for which $k_b = ma_b$ are integral. Put U_b on edge k_b of the spin clock; other edges are identities. Each cursor interval has its own fixed force amplitudes, commuting within that interval. Orthogonal cursor supports make the full amplitudes commute. With J fixed, the resulting $H_m \geq 0$ is time independent and has common domain $\operatorname{Dom} H_F$. Its comparison schedule is

$$p(t) = \sin^2(Jt), \qquad t_b = J^{-1} \arcsin \sqrt{a_b}, \qquad T = \pi/(2J).$$

It applies the specified gates at t_b while the same complete material evolves throughout. Only the comparison uses these switching times; the actual apparatus has its gates and forces fixed in space along its cursor.

Proposition 17 (A finite sequence with its history retained)**.** *Let the smooth force profiles satisfy $h_\ell, \omega^{1/2} h_\ell \in L^2$, and fix an input purification with $\|H_F \psi\| \leq M_0$, including arbitrary old references. The actual H_m approximates the scheduled sequence jointly with its complete, untrimmed cursor, at every fixed checkpoint away from the t_b. At T all data gates have been included. The joint vector error tends to zero as $m \to \infty$, uniformly on this input class. Both actual and comparison material rms energies are bounded by a constant independent of m.*

More explicitly, set

$$L_P = \sqrt{2} \sum_\ell \|B_\ell\| \|h_\ell\|, \qquad c_P = \frac{1}{\sqrt{2}} \sum_\ell \|B_\ell\| \|\omega^{1/2} h_\ell\|.$$

For any energy $E_* > 0$ a common rms bound is

$$M_T = \left[\sqrt{M_0 + E_*} + \frac{T}{2} \left(L_P + \frac{c_P}{\sqrt{E_*}} \right) \right]^2 - E_*. \tag{26}$$

In the prefix gauge, let H_b be the material/data Hamiltonian in interval b, and choose K_b so that $\|(H_b - H_{b-1})\zeta\| \leq K_b$ for normalized vectors with $\|H_F \zeta\| \leq M_T$. These constants are finite by the linear field bound. Then the gauge-state error satisfies

$$e_m(t) \leq \sum_{b=1}^{q} K_b \int_0^t e^{-m[p(s) - a_b]^2} \, ds. \tag{27}$$

At a physical checkpoint add $2 \sum_b e^{-m[p(t) - a_b]^2}$; this additional term is zero at T. For $0 < \alpha_b < \min(a_b, 1 - a_b)$ and $v_b = 2J \sqrt{(a_b - \alpha_b)(1 - a_b - \alpha_b)}$,

$$e_m(T) \leq \sum_b K_b \left[\frac{\sqrt{\pi}}{v_b \sqrt{m}} + T e^{-m\alpha_b^2} \right]. \tag{28}$$

Proof. The prefix conjugation used for the history clock transforms its hopping term to $mJ + 2JJ_x$ and each force to the corresponding prefix-conjugated data amplitude. The transformed material Hamiltonian is piecewise constant in j. Compare it with the product

of the untrimmed spin-clock state and the data/material sequence H_b. The clock distribution is $\mathrm{Bin}(m, p(t))$. Writing both piecewise Hamiltonians as sums of their jumps gives

$$H_m^{\mathrm{gauge}} - H_{\mathrm{cmp}}(t) = \sum_b (\mathbf{1}_{\{j \geq k_b\}} - \mathbf{1}_{\{t \geq t_b\}})(H_b - H_{b-1}).$$

The norm of each clock factor on the comparison state is the square root of its wrong-side probability. Hoeffding's bound is $e^{-m[p(t)-a_b]^2}$. Duhamel's formula therefore proves (27). On $|p - a_b| \leq \alpha_b$ one has $p' \geq v_b$; integrate the Gaussian there and bound its complement by $Te^{-m\alpha_b^2}$ to obtain (28). Undoing the gauge at a checkpoint costs at most twice the square root of the probability of an incorrect prefix. At T the reference cursor is exactly $|m\rangle$, giving no prefix error.

For the energy estimate, all controller terms commute with H_F, even though their norms grow with m. On the energy core, $[H_F, Q(h)] = -iP(h)$ and

$$\left\| \sum_\ell B_\ell P(h_\ell) \zeta \right\| \leq L_P \| H_F^{1/2} \zeta \| + c_P \| \zeta \|.$$

For $x(t) = \|H_F \psi_t\|$ this gives $D^+ x \leq L_P \sqrt{x} + c_P$. Hence $D^+ \sqrt{x + E_*} \leq (L_P + c_P/\sqrt{E_*})/2$, which proves (26). The data gates commute with H_F and cause no jump in this bound. Finite-mode energy-core regularization gives the same estimate before removing the cutoff; the common-domain propagator limit and closed-operator lower semicontinuity extend it to $\mathrm{Dom}\, H_F$. The linear field bound then supplies the K_b used above. Every estimate acts on the full purification and tensors with the old reference identity. □

For prescribed finite N, compile acquisition, decision, record force and bank swap on this path. Proposition 57 supplies one decision circuit from actual meter records and retained posterior weights; its repair flag can drive the force. Registers and thermal pairs are prepared initially, and spent correlations remain. At checkpoints, (20) and (26) certify a later reading and source test under the same subsequent H_m.

As a finite diagnostic, two probe readings of one oscillator, an intervening receiver–bank swap and a later force controlled by both records give endpoint joint vector errors $.0701594, .0355814, .0192136, .0107206, .00393182$ at $m = 16, 32, 64, 128, 512$, with $J = 1$. An old probe reference and the spent bank are retained. At $p = 7/16$, after the swap and before the second probe interval, receiver excitation is .0321012 for $m = 128$ and .00089552 for $m = 512$; endpoint accuracy alone would miss this timing error. The $m = 128$ final coordinate is .0393364 against the scheduled .0393304; removing the record-controlled forces gives -2.89×10^{-6} against zero. Fock cutoffs 8 and 10 change the endpoint vector error by 1.1×10^{-11}. These are regulator checks of the construction.

The clock uses $m + 1$ states, mean energy mJ, variance mJ^2, norm $2mJ$ and maximal edge coefficient at most $J(m+1)/2$. Fixed duration thus costs growing energy; fixed-energy renewal instead costs growing duration. Neither is indefinite fixed-capacity operation, and this material-source bound excludes controller covariant stress.

Appendix K gives a further use of this program: local linear-current pulses implement stress flows in the neutral $c = 1$ carrier, uniformly on bounded mean-energy inputs. The

construction supplies a physical permission used in finite modular-geometry tasks, with explicit control and coherence conditions.

The comparison clock need not have the actual clock's energy. For one oscillator take $H_F = \omega a^\dagger a$, $Q = (a + a^\dagger)/\sqrt{2}$, $V = -bQ + C$ and $C = b^2/(2\omega)$. Use identity clock edges and turn on V for cursor sites $j \geq m/2$, with even m. The positive Hamiltonian is $H_m = H_{\mathrm{clock,m}} + H_F + \mathbf{1}_{j \geq m/2} V$. Starting in $|0\rangle_{\mathrm{clock}} |0\rangle_F$, its conserved total energy is mJ. The scheduled endpoint is $|m\rangle \otimes \zeta$ up to phase, where $\zeta = e^{-i(T/2)(H_F+V)}|0\rangle$ and $\langle H_F + V\rangle_\zeta = C$. Consequently

$$\langle H_m \rangle_{\mathrm{scheduled}} - \langle H_m \rangle_{\mathrm{actual}} = C \quad \text{for every even } m. \tag{29}$$

The joint-state error still tends to zero. The finite work is paid by the actual clock and missed by its unperturbed product comparison.

This defect can be repaired for the total-energy observable. Rotate the terminal spin toward negative S_x by an angle θ. Its clock energy is $mJ - mJ \sin\theta$, and its vector distance from $|m\rangle$ is at most $|\theta|\sqrt{m}/2$. If $\epsilon_m(\theta)$ is its probability below $m/2$, the product state's total-energy excess is $-mJ \sin\theta + C - \epsilon_m(\theta)\langle V\rangle_\zeta$. It equals C at zero and is negative at $\theta = \arcsin[2C/(mJ)]$ for sufficiently large m: the last probability is at most $e^{-3m/8}$ once $\theta \leq \pi/6$. Continuity therefore supplies an exactly energy-matched comparison state with an additional $O(m^{-1/2})$ vector error. This is a physical comparison vector, not an extra gate supplied by the original dynamics. Matching one energy expectation does not match an extended stress distribution. HAFF-D uses the mass-work identity of its common action to keep this debit in a finite source comparison [23].

7.3 Local thermal windows and a continuing diagnostic return

Port replacement is distinct from using a later region of a correlated field. Appendix I.1 executes a local thermal preparation with a paid finite autonomous clock. Appendix I.2 proves its record-selected same-field conditional error from vacuum mutual information and controls the source with second moments. The complete outside purification stays present; a spatially separate window is not declared globally fresh. These free-CAR certificates require a new thermal matching proof before use on the nonlinear interacting material.

Appendix J.1 gives a separate finite-horizon return in which the actual meter word, not an inaccessible fault label, selects repair of the held body and a thermal exchange. Appendix J.2 supplies an actual terminal record and replacement of its specified probe. The full input bank, posterior arithmetic, reversible garbage, clock energy and retained output history belong to that horizon. The actual mixed-port quantum-material update above does not inherit their different deadline or mean-support guarantees.

7.4 Repeated renewal of a complete ring body

Use a metric-graph carrier with four complex chiral body fields on rings of length w, each coupled to its own thermal line over an arc $0 < \ell < w$. All spatial modes remain. Their charges are $(+1, -1, +1, -1)$; the first pair is the receiver, the second its current purifier. Two colocated, copropagating rails on each arc obey

$$i(\partial_t + \partial_x)\Psi = h(x)W(t - x)\Psi, \quad h \in C_c^\infty(0, \ell), \quad \int h = 1. \tag{30}$$

The exact ray transfer is $e^{-iW(u)}$, $u = t - x$. Take $W(u) = \theta(u)\sigma_y$ and include a physical ring connection with round-trip phase $e^{-i\Delta w}$. The overlap transit is included in w; the voltages and their switching work are supplied controls.

Choose $\tau = Kw$, $K \in \mathbb{N}$, a real $\lambda \in C_c^\infty(0,\tau)$, and $\theta = \sqrt{w}\lambda$, $|\theta| < \pi/4$. Iteration gives

$$A_w(u) = \prod_{j=0}^{K-1} \cos[\sqrt{w}\lambda(u + jw)], \qquad 0 \le u < w, \tag{31}$$

$$F_w(s) = \frac{\sin[\sqrt{w}\lambda(s)]}{\sqrt{w}} e^{i\Delta s} \prod_{j\ge 1:\, s+jw<\tau} \cos[\sqrt{w}\lambda(s + jw)]. \tag{32}$$

The old-ring multiplier is A_w; F_w is the captured zero-mode envelope. Put $\eta^2 = w^{-1}\int_0^w |A_w|^2$. Retaining the full complementary outgoing field gives $\|F_w\|_2^2 = 1 - \eta^2$ by CAR. Endpoint flatness makes the last product smooth. With $\Omega = 2\pi/w$, the captured nth mode has carrier $\Delta + n\Omega$.

The reference sea fills $n < 0$ and leaves $n \ge 0$ empty. Sum over all four species in

$$N_{\mathrm{ex}} = \sum_{n\ge 0} d_n^\dagger d_n + \sum_{n<0} d_n d_n^\dagger, \tag{33}$$

$$P = \Omega\left(\sum_{n\ge 0} n d_n^\dagger d_n + \sum_{n<0} |n| d_n d_n^\dagger\right), \qquad Q = N_{\mathrm{ex}} + P/\Omega. \tag{34}$$

Let $r(E)$ be the occupation of the normalized captured envelope with carrier E in the actual temperature-T KMS line. Its real envelope gives $r(-E) = 1 - r(E)$. For $A_w(w\xi) = \sum_k a_k e^{2\pi i k\xi}$ set

$$\alpha = \sum_k \sqrt{1 + |k|}\, |a_k|, \tag{35}$$

$$V_A = \tfrac{1}{2}\sum_{k>0} k(k+1)|a_k|^2 + \tfrac{1}{2}\sum_{k<0} |k|(|k|+3)|a_k|^2, \tag{36}$$

$$S_Q = (1-\eta^2)\left[r(\Delta) + \sum_{n\ge 1}(1+n)\{r(\Delta + n\Omega) + r(n\Omega - \Delta)\}\right], \tag{37}$$

$$b = 2(\sqrt{V_A} + \sqrt{S_Q}). \tag{38}$$

Proposition 18 (Uniform renewal with retained history)**.** *Suppose $\langle Q\rangle < \infty$, $0 < \Delta < \Omega$, and the next incoming supports have their unconditional KMS marginals. Correlations with the body and all earlier records are allowed. The actual reset satisfies*

$$\sqrt{\langle Q\rangle_{\mathrm{reset}}} \le \alpha\sqrt{\langle Q\rangle_{\mathrm{old}}} + b. \tag{39}$$

Between resets allow the translation-invariant pumps below and supplied logical noise on the four zero modes. For post-reset $x_j = \sqrt{\langle Q\rangle_j}$, if $\sqrt{2}\alpha < 1$, then

$$x_{j+1} \le \sqrt{2}\alpha x_j + 2\alpha + b, \quad \sup_j x_j \le \max\left\{x_0, \frac{2\alpha + b}{1 - \sqrt{2}\alpha}\right\}. \tag{40}$$

Against the actual outside marginal, including all old references,

$$\tfrac{1}{2}\|\rho_{\mathrm{body},R} - |\mathrm{sea}\rangle\langle\mathrm{sea}| \otimes \rho_R\|_1 \le 2x_j. \tag{41}$$

At fixed cycle duration and T, one finite choice of pulse strength, gap and ring length makes this error arbitrarily small uniformly in the number of cycles, with four active outlet species.

Proof. Apply all weighted wrong-occupation annihilators to a purification of the actual joint state. Weighted convolution, using $1 + |n| \le (1 + |m|)(1 + |n - m|)$, bounds their old excitation-annihilation part by $\alpha\sqrt{\langle Q\rangle}$. CAR bounds the creation covariance by the identity; its squared norm is at most V_A per species. The incoming vector has squared norm S_Q. Minkowski's inequality on the four-species direct sum proves (39), without factoring the old state from the line. Pumps and logical noise preserve P, and $N_{\mathrm{ex}} \le P/\Omega + 4$, so $Q_{\text{after gate}} \le 2Q_{\text{before gate}} + 4$ in expectation. This gives (40). The nonsea projection is bounded by $N_{\mathrm{ex}} \le Q$. Projection of a joint purification onto the pure sea and trace-distance contraction give (41).

The full sums are finite before imposing any numerical cutoff. Set $\Gamma = \int \lambda^2$ and $g_w = \log\cos(\sqrt{w}\lambda)$, extended by zero. For each $m \ge 3$, $G_m = \sup_{w\le w_0} w^{-1}\|g_w^{(m)}\|_1 < \infty$ if $\sqrt{w_0}\|\lambda\|_\infty < \pi/4$. Poisson summation bounds the nonconstant Fourier coefficients of $\log A_w$ by $G_m w^m/(2\pi|k|)^m$. Weighted Wiener exponentiation therefore gives

$$\alpha \le e^{-\Gamma/2+b_m}, \quad V_A \le 2e^{-\Gamma}(e^{b_m} - 1)^2, \quad b_m = \frac{2G_m w^m}{(2\pi)^m}[\zeta(m) + \zeta(m-1)]. \tag{42}$$

Write $F_w = e^{i\Delta s} f_w$. Every derivative of its logarithmic product is a sum of $O(1/w)$ terms of size $O(w)$, with flat endpoints. Thus $D_m = \sup_{w\le w_0} \|f_w^{(m)}\|_2^2 < \infty$. Splitting the Fourier integral at $E/2$ bounds the exact Fermi occupation by

$$r(E) \le e^{-E/(2T)} + \frac{(2/E)^{2m} D_m}{1 - \eta^2}, \qquad E > 0. \tag{43}$$

For $\Omega \ge 2\Delta$ and $z = e^{-\Omega/(4T)}$,

$$\begin{aligned} S_Q \le{}& (1 - \eta^2) r(\Delta) + 2(1 - \eta^2)\frac{z(2 - z)}{(1 - z)^2} \\ &+ 2D_m(4/\Omega)^{2m}[\zeta(2m) + \zeta(2m - 1)]. \end{aligned} \tag{44}$$

Choose Γ large, then Δ large, then $w = \tau/K$ small. All errors become small simultaneously. A finite initial reset puts x_0 at the same tolerance. Earlier nonselective even operations on disjoint line characteristic supports leave the next unconditional KMS marginal unchanged; guards longer than the overlap transit provide that separation. □

The mixed target is restored by a local neutral quartet pump

$$V_4 = \int_0^w (D_1^\dagger D_2^\dagger D_3^\dagger D_4^\dagger + \mathrm{h.c.})\, dx. \tag{45}$$

It commutes with P. On the sea, creation requires four nonnegative mode indices summing to zero: only the four zero modes occur. The reverse transition obeys the same argument. Their two-dimensional subspace is exactly invariant, with matrix element $1/w$.

For the full operator, bosonize the periodic fields with $q_i \in \mathbb{Z}+1/2$. The charge coordinate $\phi_c = \sum_i \phi_i/2$ gives the weight-two vertex $e^{2i\phi_c}$. At fixed transverse charges, $q_c = q_0 + 2\mathbb{Z}$ with $q_0 \in \frac{1}{2}\mathbb{Z}$, a fixed parity sector of a charge fermion; the other three bosons and the original Klein phases remain. The weight-two vertex splits this stress into two positive $c = 1/2$ Virasoro stresses with velocities $1 \pm \kappa/\pi$ [1]. The four Ramond ground weights sum to $1/2$, hence

$$\|V_4\psi\| \le \pi^{-1}\|(P + \Omega/2)\psi\|. \tag{46}$$

Finite-dimensional P eigenspaces are invariant; their direct sum defines the self-adjoint pump. The pair pump $B = \int (D_1^\dagger D_2^\dagger + \text{h.c.})dx$ obeys $|B| \le P/\Omega + 1$: every active nonzero pair costs at least Ω in P. For sea-normal-ordered number $N \ge -P/\Omega$,

$$P + \Delta N + \kappa V_4 + b_2 B \ge \left(1 - \frac{\Delta}{\Omega} - \frac{|\kappa|}{\pi} - \frac{|b_2|}{\Omega}\right) P - \frac{|\kappa|}{w} - |b_2|. \tag{47}$$

Small enough pump peaks give a common finite physical ground offset and positive Hamiltonian. This offset is part of the apparatus source.

A quartet pulse with phase $e^{-4i\Delta t}$ and area $w^{-1}\int \kappa(t)dt = \arcsin\sqrt{\epsilon}$ prepares

$$|\Phi_\epsilon\rangle = \sqrt{1-\epsilon}|00\rangle_A|00\rangle_R + \sqrt{\epsilon}|11\rangle_A|11\rangle_R. \tag{48}$$

The pair pump selects logical Z or X, retaining spectrum $(1-\epsilon, \epsilon)$ and the actual current purifier. Unitary contraction preserves (41); the next reset exports that purifier and its correlations through the same lines. Fixed gate duration requires $\kappa_{\text{peak}} = O(w)$, compatible with (47) and the ordered reset parameters.

The body has finite energy-constrained storage capacity rather than finite Hilbert dimension. Its positive-energy Gibbs partition function is finite for every inverse temperature $\beta > 0$, giving $S \le \beta\langle H\rangle + \log Z(\beta)$. Four complex outlets, any anomaly-cancelling partners, gap energy, drive work and stored outputs are charged. Fixed per-cycle error does not imply vanishing error for the whole growing history; that stronger task requires decreasing error per cycle and increasing resources. Autonomous covariant driving is treated below. Direct renewal exports the current purifier's two correlated occupation records into different species. Against a thermal reference that factorizes across species, their redundancy gives a positive complete-output relative-entropy rate whenever the reset and full-joint clock errors are small enough [23, Proposition "Output deficit and direct renewal"]. Improving this reset alone therefore does not approach the sharp heat-current limit. That task requires a physical output converter with its own work, memory and charge account.

Proposition 19 (Neutral block compression by physical gates)**.** *Buffer L original noisy A/R quartets in L equal-circumference slots of each physical charge species $(+,-,+,-)$. Permit local current phases, the pair and quartet pumps above, and same-charge quadratic mixers between connected slots. These prescribed gates can approximate a reversible code which confines their 4^L-dimensional support to m_L active zero modes and fixes the remaining*

modes, retaining arbitrary old references and intercycle correlations, where

$$m_L = \min\left\{ r : \binom{r}{\lfloor r/2 \rfloor} \geq 4^L, \quad 0 \leq 2L - \lfloor r/2 \rfloor \leq 4L - r \right\} = 2L + \tfrac{1}{2}\log_2 L + O(1). \quad (49)$$

No additional pure working register or outgoing species is used. The buffer, initialization, control, offsets and eventual discharge are physical resources.

Proof. Relabel only the existing negative-charge zero modes by $f_i = d_i^\dagger$ and the positive-charge ones by $f_i = d_i$. Neutrality is $\sum f_i^\dagger f_i = 2L$ on $4L$ modes; this relabeling leaves the full sea and original charge representation unchanged. Pair pumps and same-charge mixers become one-body rotations. Connected rotations and currents generate $U(4L)$. The quartet becomes, with its calibrated Klein phase, $Q_{ab;cd} = f_a^\dagger f_b^\dagger f_c f_d + \text{h.c.}$. Permutations move it to any four distinct labels.

Set $n_i = f_i^\dagger f_i$. Balanced rotations in $(a, c), (b, d)$, followed by averaging over their sixteen current sign phases, give $-\frac{1}{2}(n_a - n_c)(n_b - n_d)$. This average is implemented by a deterministic Trotter word, without a discarded random record. Quadrilateral differences span all two-body diagonal terms modulo currents on the fixed-number sector: their missing coefficient arrays are $\beta_{ij} = u_i + u_j$, for which $\sum_{i<j} \beta_{ij} n_i n_j = (2L - 1) \sum_i u_i n_i$. Choose a diagonal $H_D = \sum \alpha_i n_i + \sum \beta_{ij} n_i n_j$ whose finitely many hopping transition frequencies are nonzero and distinct up to sign. Such coefficients avoid finitely many proper hyperplanes. Vandermonde interpolation in iterated commutators with H_D isolates the real and imaginary generators of each individual transition between occupied $2L$-subsets. These subsets form a connected Johnson graph. Its edge generators give the full special unitary group on the neutral sector; finite group-commutator and Trotter words approximate each required Givens rotation to the stated tolerance.

Enumerate the original 4^L pair-code basis states and inject them into the first 4^L occupation strings on m_L active modes with $\lfloor m_L/2 \rfloor$ particles. Fix the inactive pattern to $2L - \lfloor m_L/2 \rfloor$ particles and extend the injection to a neutral-sector unitary. Its action is $U_L \otimes I$ on every old reference. Stirling's formula gives (49). For the physical embedding, the translation-invariant equal-ring integrals commute with P. Its complete $P = 0$ sector is precisely the zero-mode Fock space tensored with the same spectator sea; the control word never projects out the full field tower. The previous relative-energy margins, with finite offsets and small primitive amplitudes, define the full-field gates. Unitary contraction carries a certified input-reset error through the word. □

For example, two quartets occupy eight zero modes but need only six active modes after this encoding. The fixed occupations result from reversible redistribution, not from a recurring purity supply. Two paid banks can hold old batches while new states arrive; both must subsequently be discharged and reused through the same four lines. The compiler may be very long and expensive. It proves finite prescribed controllability, not a sustainable autonomous clock or an efficient fixed-rate layout. It also preserves the data spectrum: concentrating it does not prepare thermal occupations of the outlet translation generator. The next two propositions supply finite physical loading, sorting and return; their clock, source and repeated-inlet accounts remain distinct.

Proposition 20 (A finite-width packet interface)**.** *Fix a physical charge species, m paid equal-ring memories and one of its original outlet lines. Let $v(t)$ be a smooth row of m overlapping envelopes, flat outside an interior guard of $(0,\tau)$, with $\int v^\dagger v\,dt = (1-\eta^2)I_m$, $0<\eta<1$. A local Hermitian mixer on a finite common arc emits these memory columns with $O(w)$ packet error and terminal amplitude $\eta I_m + O(w)$ as the ring round-trip time w decreases. An analogous mixer captures any such incoming overlapping packet span. All original body modes and the complementary outgoing field remain in the canonical map.*

For a finite physical carrier of length V with no return during the window, let the joint input have physical excitation energy at most E, retaining arbitrary old references. With $\Omega = 2\pi/w$, $M_\Lambda = O(V\Lambda) + m$ low modes and any fixed sufficiently large smoothness order r, the actual and comparison full outputs have half trace distance at most

$$C\left(\sqrt{E/\Lambda} + \sqrt{E/\Omega}\right) + C_{m,V,\tau,\eta,r}\left[(M_\Lambda+1)w(1+\Lambda)^2 + (1+\Lambda)^{1-r} + w^{r-1}\right]. \tag{50}$$

The constants include the supplied envelope norms. First choose Λ, then w. This is a finite-carrier, prescribed-control result; it is not a global trace theorem for an infinite KMS rail.

Proof. Set $K(t) = I - \int_0^t v^\dagger v\,ds$, $\Phi = K^{1/2}$ and

$$\lambda^\dagger = v\Phi^{-1}, \qquad A = \dot\Phi\Phi^{-1}, \qquad H = i\left(A + \tfrac{1}{2}\lambda\lambda^\dagger\right).$$

Since $K \geq \eta^2 I$ and $A + A^\dagger = \Phi^{-1}\dot K\Phi^{-1} = -\lambda\lambda^\dagger$, H is smooth and Hermitian. The comparison equations

$$\dot d = Ad - \lambda\psi_{\mathrm{in}}, \qquad \psi_{\mathrm{out}} = \psi_{\mathrm{in}} + \lambda^\dagger d$$

have homogeneous propagator $\Phi(t)$ and emitted columns $\lambda^\dagger\Phi = v$. For capture, choose a column F with $\int FF^\dagger = (1-\eta^2)I$, and use

$$R(t) = \left[I - \int_t^\tau FF^\dagger\,ds\right]^{1/2}, \quad \lambda = R^{-1}F, \quad A = -R^{-1}\dot R.$$

The same identity gives Hermitian H; the future propagator is $G(\tau,t) = R(t)$, so the capture column is $-F(t)$. Its sign is an ordinary current phase, not reversed chiral transport.

In units of propagation speed one, choose a smooth arc profile $h_w(x) = w^{-1}h_0(x/w)$ of integral one and use the actual local potential

$$\mathcal{F}_w(t,x) = h_w(x)W_w(t-x), \qquad W_w = \begin{pmatrix} wH & -i\sqrt{w}\lambda \\ i\sqrt{w}\lambda^\dagger & 0\end{pmatrix}. \tag{51}$$

The internal gap is lowered for this window, with its work charged. Along a characteristic the exact arc transfer is $S_w = e^{-iW_w} = \left(\begin{smallmatrix} B_w & C_w \\ D_w & E_w\end{smallmatrix}\right)$. Thus the full fields obey

$$b(t) = B_w(t-w)b(t-w) + C_w(t-w)\psi_{\mathrm{in}}(t-w), \qquad \psi_{\mathrm{out}}(t) = D_w(t)b(t) + E_w(t)\psi_{\mathrm{in}}(t).$$

In each fixed derivative norm, $B_w = I + wA + O(w^2)$, $C_w = -\sqrt{w}\lambda + O(w^{3/2})$ and $D_w = \sqrt{w}\lambda^\dagger + O(w^{3/2})$. Telescoping the noncommuting ordered products over τ/w returns gives

the $O(w)$ comparison. The complete star is unitary, so its emitted Gram matrix plus its residual-body Gram matrix is exactly I_m.

The full tower also contributes. Each positive derivative of B_w is $O(w)$; differentiating an ordered product sums $O((\tau/w)^j)$ terms with j differentiated factors, each supplying $O(w^j)$. Its derivatives are therefore bounded uniformly in small w. Flat guards make the residual multiplier periodic on the ring. Its nonconstant Fourier coefficients satisfy $\|a_k\|_{\mathrm{HS}} \le C_r w^r/|k|^r$. The emitted and captured envelopes have the corresponding uniform smooth bounds; their nth body aliases retain the exact factors $e^{\pm in\Omega t}$.

The complementary line kernel is the direct term $E_w(t)\delta(t-s)$ plus its causal delayed returns

$$\sum_{j\ge 1} D_w(t)B_w(t-w)\cdots B_w(t-(j-1)w)C_w(t-jw)\delta(s-t+jw).$$

Each coefficient and each fixed center derivative is $O(w)$, with $O(\tau/w)$ terms. The center derivative $\partial_t+\partial_s$ leaves both these delayed diagonals and the comparison Volterra factor $\Theta(t-s)$ unchanged. Fourier integration by parts therefore bounds its nonidentity kernel by $C_r(1+|\omega-\omega'|)^{-r}$, uniformly in w. Across opposite sea polarizations, $|\omega-\omega'| = |\omega|+|\omega'|$; the squared tail above Λ is $O((1+\Lambda)^{2-2r})$. On a fixed low-frequency subspace, the causal sum is a Riemann approximation with one-particle error $Cw(1+\Lambda)^2$. Together with the body alias bounds this controls the relative transformation's opposite-polarization Hilbert–Schmidt block.

For the state estimate, a proof-only energy cutoff excludes field excitations above Λ and nonzero body excitations with probability at most $E/\Lambda + E/\Omega$. Gentle projection works with the old reference retained. The remaining finite excitation space has M_Λ modes, so its Fock lift costs at most M_Λ times the one-particle error. The canonical pair-rotation decomposition bounds the vacuum-lift error by the opposite-polarization Hilbert–Schmidt norm [48, Section 7.4, Theorem 31]. The actual normal-ordered lifts may differ by an overall phase, irrelevant for this prescribed channel. Combining the cutoff, low-band and full-tower estimates gives (50). □

Capture retains the actual covariance $C_{\mathrm{cap}} = \int F_w(t)C_{\mathrm{in}}(t,s)F_w(s)^\dagger\,dt\,ds$, including off-diagonal packet correlations and old body–line terms. Orthogonal wavefunctions need not have independent thermal occupations. A quantum program coherently selecting different loaders also requires their actual relative lift phases to be calibrated and charged. For the same finite graph, the source comes from its full covariance and local point splitting, and the work is $\int\langle\partial_t H_{\mathrm{graph}}\rangle dt$, including gap changes, contact energy and physical ground offsets. With $C_h = \int h_0^2$, the local quadratic scale satisfies

$$\int \operatorname{tr}\mathcal{F}_w^2 dx = C_h\left[w\operatorname{tr}H^2 + 2\|\lambda\|^2\right].$$

This does not bound all higher spatial derivatives or clock resources uniformly. Nor does (50) alone control heat: an old spectator occupied with probability E/Ω can export energy E while disappearing in trace norm. Heat or source convergence requires propagated uniform energy integrability, such as a controlled higher energy moment or explicit weighted output tails. The packet map supplies the loading step for the following thermal converter. Its

energy estimate uses the actual carrier rather than packet occupation numbers assigned independently of it.

Proposition 21 (Charge-sector thermal sorting on a retained carrier)**.** *Use the original four charge species and neutral gates, and retain four physical carrier loops of round-trip time $V = L/\nu$ in their complete finite-volume Gibbs state $\tau_{T,V}$, $T > 0$. Include its purification and all old references. Supply L independently realized noisy original quartets with marginal $\rho_S = \rho_0^{\otimes L}$, where*

$$\rho_0 = \begin{pmatrix} (1-q)(1-\epsilon) & 0 & 0 & (1-2q)\sqrt{\epsilon(1-\epsilon)} \\ 0 & q\epsilon & 0 & 0 \\ 0 & 0 & q(1-\epsilon) & 0 \\ (1-2q)\sqrt{\epsilon(1-\epsilon)} & 0 & 0 & (1-q)\epsilon \end{pmatrix}, \qquad h_\lambda = S(\rho_0).$$

The source and complete carrier initially have a product marginal; their external purifications are retained. There are prescribed finite words of the original smooth gates that return the source near its original blank and achieve, in natural units,

$$\lim_{\Lambda\to\infty} \limsup_{L\to\infty} \frac{D(\rho_{B,\mathrm{out}} \| \tau_{T_*,V})}{V} = 0, \qquad T_* = T + \frac{6\nu h_\lambda}{\pi c_{\mathrm{ex}}}, \quad c_{\mathrm{ex}} = 4. \tag{52}$$

Implementation error can be made arbitrarily small after choosing each finite block and cutoff Λ, with physical energy tails controlled. The entire output B, its uncaptured complement and every atypical label are present in this relative entropy. Finite paid internal pipeline stages permit the specified throughput without adding external outlets. This is a finite-carrier converter; an actual stationary inlet, repeated preparation and an autonomous reused controller require further physical constructions.

Proof. On each loop let $D = -i\partial_x$, P_Λ its Fourier subspace with $|\omega| \le \Lambda$, and χ a smooth cutoff flat zero in endpoint guards of length g. If $P_\Lambda \chi^2 P_\Lambda u_j = g_j u_j$, retain $g_j \ge 1 - e$ and set $f_j = \chi u_j/\sqrt{g_j}$. These are orthonormal smooth envelopes for Proposition 20. For their isometry F, the actual captured matrices are

$$C_T = F^*(1 + e^{D/T})^{-1} F, \qquad h_F = F^* D F.$$

They need not commute. A physical one-body rotation diagonalizes C_T for its spectral probabilities, whereas the final occupation states use the eigenbasis of h_F. Relabel existing negative-charge modes by $f = d^\dagger$ as in the neutral code, retaining their energy $-h_F^T$ and the physical sea constant.

Here is an exact finite count. In total charge sector Q, let $E_{\beta,Q}, S_{\beta,Q}, \sigma^2_{\beta,Q}$ be the energy, entropy and energy variance of the actual carrier-sector Gibbs matrix. For $b_Q > \sigma_{\beta,Q}$ there are at least

$$\left(1 - \frac{\sigma^2_{\beta,Q}}{b_Q^2}\right) e^{S_{\beta,Q} - \beta b_Q} \tag{53}$$

occupation states below $E_{\beta,Q} + b_Q$. Chebyshev gives this shell's mass; each state in it has probability at most $e^{-S_{\beta,Q}+\beta b_Q}$. Map the charge-resolved good input eigenlabels injectively

into these low-energy states with the source blank, then complete the injection to a sector bijection. Every bad label occupies a remaining physical state. If their probability is $p_{\rm bad}$,

$$\Pr(S_{\rm out} \neq b) \leq p_{\rm bad}, \qquad S(S_{\rm out}) \leq h(p_{\rm bad}) + 4Lp_{\rm bad} \log 2.$$

For the depolarizing source specified in the proposition, exact blank restoration would violate the full-rank finite input's rank count. No label is postselected away.

At fixed sufficiently large Λ, the source eigenvalue strings and the Gaussian carrier's actual spectral probabilities have information variance $O(L)$. An information margin $L^{9/10}$ and charge window $|Q| \leq L^{5/8}$ lose $o(1)$ probability. Choose the finite-bank target entropy larger than the input entropy by $3L^{19/20}$ and put $b_Q = L^{17/20}$. For these charges exponential tilting of the bounded occupation Bernoullis gives $\log \Pr_\beta(Q) = -O(Q^2/L + \log L)$. Fourier inversion bounds the tilted probability at its mean below by $cL^{-1/2}$: its characteristic function is bounded by $e^{-cL\sin^2(\theta/2)}$, and its scaled central part converges to a nonzero Gaussian integral. Conditioning its $O(L)$ energy variance therefore gives $\sigma^2_{\beta,Q} = O(L^{3/2})$ and $|E_{\beta,Q} - E_\beta| = O(|Q| + L^{3/4})$. The count loss in (53) is smaller than the entropy margin. Hence $p_{\rm bad} \to 0$, source entropy is $o(L)$, and the carrier energy excess is $o(L)$.

The bijection uses the original gates. For a hop $a \to b$, a quadrilateral interaction $(n_a - n_c)(n_b - n_d)$ has difference $n_d - n_c$. Two distinct spectator subsets have the same differences for every c, d only when one is empty and the other full. Generic current and quadrilateral coefficients therefore separate all other oriented edges; finite Vandermonde filtering and the connected occupation graphs give independent special-unitary actions in sectors $2 \leq k \leq m-2$. The linked $k = 1, m-1$ sectors are unnecessary: the original source has $k_S = 2L$ on $4L$ modes, so every occupied source–carrier sector is interior. Spectral diagonalization and the chosen bijection are finite commutator/Givens words. Their phases are retained.

Capture and return act on the same complete carrier as $U^*_{\rm load} U_{\rm sort} U_{\rm load}$. For commensurate loops $U_0(V) = 1$ up to its declared sea phase. Thus the inverse of a free interval δ is the positive wait $V - \delta$. Smooth signed current kicks invert the local mixing steps: evolution with $H_0 + (a/\varepsilon)J(F)$ for time ε approaches $e^{-iaJ(F)}$ on the smooth energy core, with error at most $\varepsilon \int_0^1 \|H_0 e^{-isaJ(F)}\psi\| ds$. Higher weighted versions follow from smooth multiplication and the summable opposite-sea tails. Reverse a finite Trotter word using these waits and kicks. Their work, peak ground shifts, relative phases and contact sources are paid. Its G whole-loop waits require G internal storage stages; narrowing a contact does not remove that latency. All stages have the original species and only the original four external outlets.

To pass from the bank to the complete physical carrier, let d be the number of discarded Fourier vectors and M the bank size. Then $d \leq C(\Lambda g + 1)/e$. The retained Fourier span need not be invariant under D; its discarded cross block contributes $C\Lambda d$, in addition to the cutoff error $CM(\Lambda\sqrt{e} + 1/g)$. Taking $g = V^{1/2}$ and $e = V^{-1/4}$ makes these and the entropy compression error $o(V)$ at fixed Λ. The unselected thermal tails have energy and entropy per V at most $C(1+\Lambda)^2 e^{-\Lambda/T_*}$. The ideal conjugation preserves the entropy of source plus full carrier, so Araki–Lieb and subadditivity give

$$|S(B_{\rm out}) - S(\tau_{T,V}) - Lh_\lambda| \leq S(S_{\rm out}).$$

Together with the physical energy estimate and Gibbs identity, this bounds the deficit in (52) by $o_\Lambda(1) + C(1+\Lambda)^2 e^{-\Lambda/T_*}$. Full finite-carrier Gibbs moments and the smooth gate

graphs control implementation tails; Gibbs energy-constrained entropy continuity then gives the stipulated finite accuracy. Trace error alone would not give this energy or entropy conclusion. □

Remark 22 (Other supplied neutral source spectra). The proof of Proposition 21 also applies to a product source $\bigotimes_{j=1}^{L} \sigma_j$ from a fixed finite alphabet of known states in the same neutral quartet code, with the same joint source–carrier product marginal, paid gates and controlled physical energy tails. Joint operations include the receiver and its current, resettable purifier; external old references remain untouched. Put $s_L = \sum_j S(\sigma_j)$ and suppose $s_L/L \to s$ and the source information variance is $O(L)$. Replace h_λ by s in T_* and Lh_λ by s_L in the entropy identity. The finite-bank target uses the actual s_L plus the same subextensive entropy margin; its target temperature tends to T_*. No rate of convergence of s_L/L is needed. Eigenlabel concentration uses this variance and entropy. The source still has $k_S = 2L$, so the neutral-sector compiler remains in its interior sectors. The final full-carrier entropy bound uses s_L; thermal counts, energy tails and paid loading/unloading are unchanged. HAFF-D's pointer-aligned dephasing supplies such rank-two sources at fixed noise parameters [23]. This is supplied-state renewal, not recovery of arbitrary unknown content against a forbidden purifier or a proof of the product marginal for an adaptive source.

The source debit is the complete quartet entropy h_λ. For $0 < q, \epsilon < 1/2$, its difference from the bare receiver change is the correlation lost in the logical depolarizing step:

$$h_\lambda - [h(\epsilon_q) - h(\epsilon)] = I(A:R)_{\text{before}} - I(A:R)_{\text{after}} \geq 4q\, h(\epsilon) > 0.$$

The equality uses the pure corrected logical quartet and its noisy marginals; joint convexity for the depolarizing mixture gives the bound. With a conserved classical axis P, the source entropy conditional on P is Lh_λ. Its near-blank return removes $s^{PS} = Lh_\lambda - o(L)$ from the complete source–program system. Entropy conservation with the independent thermal carrier gives

$$S(B_{\text{out}}|P) - S(\tau_{T,V}) = s^{PS} + I(S_{\text{out}} : B_{\text{out}}|P).$$

No noise purifier is granted as usable side information. An autonomous controller's sensing and renewal must also enter its complete ledger. HAFF-D gives the corresponding conditional deficit account [23].

The fixed Z and X noisy quartets are isospectral. Their entropy-optimal outlet heat agrees to leading order, although their retained correlations and recovered work can differ. The direct source contrast below belongs to its specified unsorted reset. Its causal intervention remains valid, but does not certify persistent heat gain for the sharp converter.

Corollary 23 (The selected axis changes the maintenance source)**.** *Fix* $0 < \epsilon < 1/2$ *and* $0 \leq q < 1/2$. *On the same complete post-reset state, compare the two physical axis preparations above, followed by the logical depolarizing channel with* $\epsilon_q = \epsilon + q(1-2\epsilon)$ *and the same reset. Do not undo the* X *rotation. There are fixed finite reset parameters for which, uniformly after the initial preparation cycles,*

$$\int \zeta(t)(\langle T_{\text{out}}\rangle_X - \langle T_{\text{out}}\rangle_Z)dt \geq \frac{\Delta(1-\eta^2)}{2}(1-2q)(1-2\epsilon) > 0, \tag{54}$$

where $\zeta \geq 0$ is smooth and equals one on the emitted pulse. This is a paired causal intervention on one pre-preparation state, not a comparison of histories selected by conditioning on different acquired outcomes.

Proof. The ideal total zero occupations are $2(\epsilon_q + \epsilon)$ and $1 + 2\epsilon$. Their difference is $D = (1 - 2q)(1 - 2\epsilon)$. Let v_w be the real emitted zero-mode envelope. The emitted nth mode is $v_n = e^{-i(\Delta+n\Omega)t}v_w$, and $J_k = \int \zeta v_w^2 e^{-ik\Omega t}$, $J_0 = 1 - \eta^2$. The exact old-body matrix of the smeared stress is

$$B_{mn} = [\Delta + \Omega(m + n)/2]J_{n-m}.$$

Its diagonal is $J_0(P + \Delta N_s + \Delta N_0)$ after sea normal ordering. Both interventions preserve P; $|N_s| \leq P/\Omega$ and $\|N_0\| = 4$. If d bounds the half-trace reset error and M_E bounds $\langle P/\Omega\rangle$, their diagonal contrast is at least $\Delta J_0(D - 16d - 2M_E)$.

These errors can be controlled independently of the zero occupation. Separating the old zero mode in the excitation-vector proof gives

$$\sqrt{\langle P/\Omega\rangle_{j+1}} \leq \alpha\sqrt{\langle P/\Omega\rangle_j} + 2(\sqrt{Z_A} + \sqrt{V_P} + \sqrt{S_P}),$$

where $Z_A = \sum_{n\neq 0} |n||a_n|^2$, $V_P \leq V_A$ is the nonzero-mode creation leakage, and S_P is S_Q with its zero term removed and weight n instead of $1 + n$. Thus $M_E \to 0$ as $w \to 0$ at fixed Γ, Δ. Before reset take $M_Q = 4 + 2M_E$ as a bound on $\langle Q\rangle$.

Set $R_2 = \sum_{k\neq 0}(1 + |k|)^2|J_k|$. Weighted Schur estimates for equal sea polarizations and a Hilbert–Schmidt estimate across the sea give a two-branch off-diagonal error at most $2(\Delta + \Omega)R_2(M_Q + 4\sqrt{M_Q})$. For $m \geq 5$, integration by parts gives

$$R_2 \leq 2\|(\zeta v_w^2)^{(m)}\|_1\Omega^{-m}[\zeta_R(m) + 2\zeta_R(m - 1) + \zeta_R(m - 2)],$$

where ζ_R denotes the Riemann zeta function. The envelope derivatives are uniformly bounded as w decreases.

Old body–line cross terms also remain. Let S_f be the exact line-to-line scattering operator and $g_n = S_f^*(i\zeta\partial_t + i\zeta'/2)v_n$. Its delay kernel contains shifts kw. Since $e^{-in\Omega kw} = 1$, direct substitution gives $g_n = e^{-iE_n t}(E_n a_w + b_w)$, $E_n = \Delta + n\Omega$, with all derivatives of a_w, b_w uniformly bounded independently of n, Δ. The wrong KMS occupation of this unnormalized mode is bounded by

$$2(E_n^2\|a_w\|_2^2 + \|b_w\|_2^2)e^{-|E_n|/(2T)} + 2(2/|E_n|)^{2m}(E_n^2\|a_w^{(m)}\|_2^2 + \|b_w^{(m)}\|_2^2).$$

Sum this over all integers n and the four species to obtain G_T. The sum converges for $m \geq 2$. Its zero term tends to zero with Δ; its nonzero terms tend to zero with $\Omega \geq 2\Delta$. CAR and Cauchy–Schwarz bound the difference of the actual cross stresses by $4\sqrt{4 + M_E}\sqrt{G_T}$, without a product-state premise. The incoming-only term cancels because body-only interventions leave the same incoming marginal unchanged. The full lower bound is therefore

$$\begin{aligned}\Delta J_0(D - 16d - 2M_E) - 2(\Delta + \Omega)R_2(M_Q + 4\sqrt{M_Q}) \\ - 4\sqrt{4 + M_E}\sqrt{G_T}.\end{aligned} \tag{55}$$

Choose Γ large, then Δ large, then w small. The reset bounds and the displayed convergent tails make the error less than $\Delta J_0 D/2$. These choices remain compatible with the pump positivity condition (47). □

The energy difference is paid by preparing and maintaining different physical axes at a fixed gap. It is emitted by the reset itself, not by an additional program-tagged source. Conditional histories can have different incoming backgrounds; the causal result makes no claim that those observed conditional means obey the same formula.

8 A spatial work clock for the retained thermal body

The waveforms above can be generated by a physical clock field over a specified finite horizon. Its source depends on the retained quantum state, including the body–line correlations created by earlier gates. For this construction supply a static material frame (U, V) and reflecting endpoints. Let

$$S_\chi = \frac{I}{2} \int \sqrt{-g}[(U \cdot \nabla\chi)^2 - C(V \cdot \nabla\chi)^2], \qquad 0 < C < 1, \tag{56}$$

with Neumann data, initially $\chi = 0$, $\chi_t = 1$, $\chi_x = 0$. Gate profiles are smooth functions of χ and the material coordinate. Use flat guard intervals between a zero-gap, fixed real quartet quadrature and the quadratic gap, pair and mixer gates. The actual gap-lowering and restoration work, offsets and finite normal-order terms are included in the Hamiltonian. Internally mixed components have a common positive free speed v_0.

Proposition 24 (Retained-state field–clock evolution)**.** *Fix a finite list of such windows and a finite horizon H. Prepare the original mixed four-mode target and its purifier, the empty-zero body reference and an actual finite-volume Gibbs reservoir; retain every subsequent state and reference. Assume smooth stress and kernel remainders, sufficiently many finite energy moments, and the strict quartet inequalities* (60) *below. Choose the reservoir long enough to prevent a return before H. Then sufficiently large finite physical inertia I_H gives a timelike work clock and the same sequence of gate types throughout H. Its force is that of the actual evolved full-CAR state. The quadratic windows and their delayed matrix boundary close at the same derivative order. The quartet window admits a reflected symmetrizer at that order. In each specified lower source norm the field and the paid clock-work source have a comparison error $O_H(I^{-1/2})$; a quadratic window with identical incoming data has $O_H(I^{-1})$ error. These estimates retain the clock's $O(I)$ baseline stress and its finite work contribution.*

Proof. During a quadratic window the Heisenberg multiplet obeys $(\partial_t + \sigma v_0 \partial_x)\Psi = -iF\Psi$, where $F = F(\chi, a)$ is Hermitian and $\sigma = \pm 1$ is chirality. Pair gates use constrained Majorana components; their doubling adds no physical outlets. Subtract the fixed symmetric spatial singularity $-i\sigma\kappa_J/(x-y)$, $\kappa_J = 1/(2\pi)$, from $\Gamma(x,y) = \langle \Psi^\dagger(y)\Psi(x)\rangle$. Let $J = \Gamma_{\rm reg}(x,x)$ and $K = -i\sigma(\partial_x - \partial_y)\Gamma_{\rm reg}/2|_{y=x}$. Point splitting gives the full matrix equations

$$\begin{aligned} J_t + \sigma v_0 J_x &= -i[F, J] - \sigma\kappa_J F_x, \\ K_t + \sigma v_0 K_x &= -i[F, K] - \frac{\sigma}{2}\{F_x, J\}, \qquad F_x = F_\chi \chi_x + F_a a_x. \end{aligned} \tag{57}$$

The force in (56) is $-\sum_\sigma \langle F_\chi, J_\sigma\rangle - \partial_\chi E_o$. For complex components the contraction is a trace; for constrained Majoranas it is half the trace. Finite sea shifts are retained: the

periodic empty-zero convention has $[L_n^{\rm sea}, J_m^{\rm sea}] = -mJ_{n+m}^{\rm sea} - (n/2)\delta_{n+m,0}$. Passing to canonical currents uses $J_n = J_n^{\rm sea} - \delta_{n0}/2$ and $L_n = L_n^{\rm sea} + \delta_{n0}/12$, with the corresponding fixed Hamiltonian subtraction. It leaves the physical zero occupation unchanged.

For a mixing arc of length $\ell < w$, set $d_a = \ell/v_0$, $d_{\rm out} = (w-\ell)/v_0$ and $d = w/v_0$. The arc Wilson matrix $G(u)$ maps entry time u to exit $u + d_a$. With its body/line blocks G_{BB}, G_{BF} and the actual outside phase Q, the incoming body satisfies

$$b(u) = A(u)b(u-d) + B(u)f(u-d), \quad A(u) = QG_{BB}(u-d), \quad B(u) = QG_{BF}(u-d). \tag{58}$$

Finite iteration expresses its cross kernel as

$$\Gamma_{BF}(u,v) = R_M \Gamma_{B_0F}(u - Md, v) + \sum_{j=0}^{M-1} R_j B(u - jd)\Gamma_{FF}(u - (j+1)d, v),$$

where $R_j = A(u)\cdots A(u-(j-1)d)$ and $R_0 = 1$. These are kernels of the original joint state. At $v = u$ every reservoir separation is at least d; all clock-dependent incoming data precede u by at least $d_{\rm out} > 0$.

The incoming cross stress requires covariant relative time differentiation. For constant exterior connections $F_{0,B}, F_{0,F}$,

$$K_{BF} = \frac{i}{2v_0}(\partial_u - \partial_v)\Gamma_{BF}|_{v=u} - \frac{F_{0,B}\Gamma_{BF} + \Gamma_{BF}F_{0,F}}{2v_0}.$$

For the full right-moving transfer put $S = (iG'G^\dagger + GF_{0,\rm in}G^\dagger - F_{0,\rm out})/v_0$. The complete spatial point-split conversion is

$$\begin{aligned} J_{\rm out} &= GJ_{\rm in}G^\dagger + \kappa_J S, \\ K_{\rm out} &= GK_{\rm in}G^\dagger + \tfrac{1}{2}\{S, GJ_{\rm in}G^\dagger\} + \tfrac{\kappa_J}{2}S^2. \end{aligned} \tag{59}$$

Distinct body/line finite-circle backgrounds are added before and subtracted after this conversion. Differentiating unitarity twice in the symmetric split cancels the anti-Hermitian second derivative, including noncommuting connection terms. A scalar total stress would lose the BF data in this equation.

Differentiate the arc Wilson matrix by Duhamel's formula. Its $(m+1)$st entry-time derivative needs only derivatives through m of χ_t, a first-order clock variable. The initial two-time kernel remainders and their required derivatives are bounded away from the declared diagonal singularity. Finite history products therefore give a locally Lipschitz C^m boundary functional in (χ_t, χ_x, J, K), $m \geq 3$. On time steps shorter than $d_{\rm out}$ its incoming data are already fixed. Characteristic iteration of (57) and the acoustic-clock wave equation uses these same jets and yields the quadratic-window estimate. The constants depend on H/d and the initial norms.

For a quartet window write $v_s = v_0 + sa(\chi)$, $g_s = sa_\chi$, $s = \pm$, and $\gamma_s = 1/(48\pi)$. The actual chiral stress satisfies $T_{\sigma s,t} + \sigma v_s T_{\sigma s,x} + 2\sigma v_{s,x} T_{\sigma s} = \sigma\gamma_s v_{s,xxx}$. Use $Y_{\sigma s} = (1-\partial_x^2)^{-1}T_{\sigma s}$ and $Z_{\sigma s} = \partial_x Y_{\sigma s}$, retaining their means. For $r = \chi_t$, $p = \chi_x$, the principal matrix has $A_{rp} = -C$, $A_{rZ} = -g_s/I$, $A_{pr} = -1$, $A_{Zp} = \sigma\gamma_s g_s$, $A_{ZZ} = \sigma v_s$. Its symmetrizer has $H_{rr} = I$, $H_{pp} = IC$, $H_{pZ} = g_s$, $H_{ZZ} = v_s/\gamma_s$. Direct Schur elimination gives

$$IC > 2\gamma_s a_\chi^2 \sum_s v_s^{-1}, \qquad I(1-C) > 2\gamma_s a_\chi^2 \sum_s \frac{v_s}{1-v_s^2}. \tag{60}$$

Thus H and $H \pm HA$ are positive. The exact remainder follows by commuting $1-\partial_x^2$ through the Ward equation; its terms with $v_{s,xx}Z$ gain one derivative under $\partial_x(1-\partial_x^2)^{-1}$. They are locally Lipschitz at the same Sobolev order. At a reflected endpoint $p = 0$, $Y_{Rs} = Y_{Ls}$, $Z_{Rs} = -Z_{Ls}$, so the energy flux vanishes. Smooth higher classical energies recover the desired lower-norm actual stress. Increasing finite I preserves the flat guard intervals and a timelike clock over the fixed finite list.

For the solved smooth profiles, the full-tower propagator carries the actual state through each seam. Its distributions satisfy these same linear Ward and kernel equations; uniqueness identifies the mean force with that retained state. Finitely higher initial graph and classical norms can be selected for the finite list. In a gate controlled by a conserved record projector use unnormalized weighted kernels; each c-number anomaly carries that projector's actual weight. The controlled unitary also retains off-diagonal record coherence. Register updates occupy their separately paid slots.

Finally the actual clock work is

$$\frac{d}{dt}\frac{I}{2}\int (r^2 + Cp^2) = -\int r\,[\sum_{\sigma}\langle F_\chi, J_\sigma\rangle + \partial_\chi E_o],$$

with the quartet force replacing the bracket during its slot. Write $\chi = t + \eta_I/I$ and solve $\eta_{0,tt} - C\eta_{0,xx} = -\text{force}_{\text{ref}}$ with the same corrected initial data. The source estimate controls $\eta_I - \eta_0$; the finite retarded work term and the $O(I^{-1})$ quadratic clock remainder remain in the comparison. No growing clock baseline multiplies an uncontrolled state error. □

Corollary 25 (Full joint accuracy of the finite schedule)**.** *Use the original charge-glued field space, its finite reservoir and all retained quantum registers and references in Proposition 24. In fixed energy units put $G_{\text{en}} = 1 + \sum_{\text{rails}} v_0 P_{\text{rail}}$, the positive physical excitation energy, including holes and every spectator. The finite zero-mode multiplicity is retained. Suppose the joint initial state has $M_1^2 = \operatorname{Tr}(\rho_0 G_{\text{en}}^2) < \infty$, with the finitely higher smooth input norms required for the schedule. Compare its solved clock history with the prescribed history on the same field space and in the same initial state. If*

$$a_I(t) = \|(H_I(t) - H_0(t))G_{\text{en}}^{-1}\|, \qquad b(t) = \|[G_{\text{en}}, H_0(t)]G_{\text{en}}^{-1}\|,$$

then, for every $t \le H$, the half trace distance of their full joint quantum outputs is bounded by

$$e_H = M_1 \int_0^H a_I(s) \exp\left(\int_0^s b(r)\,dr\right) ds = O_H(I^{-1/2}). \tag{61}$$

At a reset checkpoint whose prescribed-history whole-body error is d_{ring} in (41), the actual output satisfies

$$\tfrac{1}{2}\|\rho^I_{\text{body},R} - |\text{sea}\rangle\langle\text{sea}| \otimes \rho^I_R\|_1 \le d_{\text{ring}} + 2e_H. \tag{62}$$

Here R is the actual outside system, including every old reference. The same joint bound applies after preparing the original mixed receiver with its current purifier and before its next specified readout. It is an energy-constrained state estimate, not an unconstrained diamond-norm assertion.

Proof. The local quartet is a stress smearing in its original sector representation. The positive-energy bound [3, Sections 2.2–2.3]

$$\|L_n\psi\| \le \sqrt{1+c/12}(1+|n|^{3/2})\|(1+L_0)\psi\|$$

and Fourier Cauchy–Schwarz control a profile difference by its H^m norm, $m > 2$. The energy commutator needs one further derivative. For a quadratic current, pair or mixer, separate the same-sea number part from the pair part. The first is bounded relative to excitation number and hence to G_{en} on the fixed finite hardware. A smooth compact-arc profile gives pair matrix entries bounded by $C_s(1+E_n+E_m)^{-s}$. A sum-frequency shell has $O(q)$ entries: absolute summation requires $s > 2$, and the energy commutator requires $s > 3$. Three and four integrable spatial derivatives, respectively, are safe integer choices. Different circumferences enter through their physical energies and finite spacing constants; an unweighted sum of L_0's would not supply this mixer commutator. The estimates act across the original CAR sectors, including record coherences and their scalar phases. They use no Gaussian replacement at a seam.

The smooth propagators preserve the common energy core. For a purification Ψ_0 of the actual initial state, the commutator estimate gives $\|G_{\mathrm{en}}U_0(s)\Psi_0\| \le M_1\exp(\int_0^s b)$. Duhamel's formula then bounds $\|(U_I(t)-U_0(t))\Psi_0\|$ by (61); every reference carries the identity. The clock comparison at the chosen lower coefficient norm gives $\int_0^H a_I \le C_H I^{-1/2}$, after supplying the required higher norms. Its existence interval already covers the entire schedule. Triangle inequality with the prescribed reset costs e_H; replacing its outside marginal by the actual one costs another e_H. Re-preparation and the same subsequent instrument contract this error. □

The delay in reset bounds is the round-trip time d, whereas w is spatial length. For example, $v_0 = .5$, $w = 10^{-8}$, $\Delta = 10^7$, $T = \tau = 1$ and $\Gamma = 32$ admit a smooth flat reset pulse with $D_0 < 9.9\times10^8$ and $r_d < 4\times10^{-5}$. The four-mode joint reset bound is 0.026503. Finite additional clock accuracy is obtained by increasing I_H; this argument does not provide its numerical value. Restoring the gap preserves the original X/Z work difference $\Delta(1-2q)(1-2\epsilon)$. For a constant arc background $F_0 = \mathrm{diag}(\Delta, 0)$ the actual local connection

$$F = F_0 + v_0 h(a)e^{-iF_0a/v_0}W(\chi - a/v_0)e^{iF_0a/v_0}, \qquad \int h\,da = 1,$$

gives the full circuit phase $e^{-i\Delta d}$, retaining its spatial force and emitted carrier energy.

The evolving work clock still has a supplied frame/endpoints; traction and geometric reaction need their own equations. Equal mixer speeds, smooth inputs and separated quartet/current slots are required. Horizon-dependent inertia and reservoir length leave fixed-capacity sharp exhaust open.

8.1 A local gate on a massive body

A spatial body need not have the massless ring's energy graph. Pay instead for a finite physical circle with a mass-zero central interval of length ℓ and a positive smooth Dirac mass throughout the rest of its return arc. Its four thermal outlet fields remain separate,

with the original charges and capacity. Both mass layers, the exterior polarization, support pressure and initial preparation belong to the apparatus. This layout replaces a compact barrier followed by a long gapless body exterior; it does not give that old layout a gap by assumption.

Proposition 26 (A guarded local gate with full physical energy control)**.** *Let $h_M = -iv\sigma_z\partial_x + M(x)\sigma_x$ on this circle. There is a nonempty smooth mass-profile class with a simple lowest positive mode E_0, no zero mode and next positive energy $E_1 > E_0$. Set $H_M = d\Gamma(|h_M|)$, with particles and holes above the actual filled mass sea, and $A = 1 + H_M$. Let $h \geq 0$ be smooth, supported strictly inside the mass-zero region. On the original charge-glued space define the local vertex*

$$Q_+(h) = \int h(x)[R_1^\dagger R_2^\dagger R_3^\dagger R_4^\dagger + L_1^\dagger L_2^\dagger L_3^\dagger L_4^\dagger]dx, \qquad Q = Q_+ + Q_+^\dagger.$$

All modes and old references are retained. For each fixed graph order s, sufficiently weak smooth resonant drives

$$H_{\rm lab}(t) = H_M + \lambda b(\lambda t)[e^{-4iE_0t}Q_+(h) + e^{4iE_0t}Q_+(h)^\dagger] \tag{63}$$

implement a prescribed full logical angle on the four-ground-mode code P_0, in physical time $T = U/\lambda$, with

$$\|A^s(U_{\rm lab}(T) - U_{\rm target,lab}(T))P_0\| \leq C_s\lambda. \tag{64}$$

The target includes its calibrated laboratory phase. The error is uniform in external old references, not an unrestricted-energy diamond bound. Smooth physical mass-force and stress tests satisfy $\|A^{-4}S_fA^{-4}\| < \infty$, so weighted state errors also control their actual renormalized expectations at this fixed mass.

Proof. The original orthogonal boson rotation and re-fermionization give, for each chirality, the operator identities

$$Q_\sigma(h) = \pi^{-1}(T_{\sigma,1}(h) - T_{\sigma,2}(h)), \qquad T_{\sigma,1} + T_{\sigma,2} + \sum_{a=1}^{3} T_{\sigma,a} = \sum_{i=1}^{4} T_{\sigma,{\rm Dirac},i}.$$

The individual rotated stresses need not be quadratic in the massive species. Their total on the right is quadratic. The commuting sector stresses have positive shifted forms by the chiral QEI [12, Theorem 4.1 and Remarks 2–3]. Their joint spectral calculus therefore bounds $\|Q(h)\psi\|$ by $C\|(K_h + C_h)\psi\|$, where K_h is the actual original localized kinetic stress. Global charge gluing is retained; no independently prepared sector vacua are supplied.

In the physical mass polarization its one-particle operator is

$$D_h = -iv\sigma_z(h\partial_x + h'/2) = hh_M - iv\sigma_z h'/2,$$

since $hM = 0$. For a number block b and $\omega = |h_M|$, $b^2 \leq C^2\omega^2$ and hence $|b| \leq C\omega$. On an n-particle tensor space, different-particle factors commute, so

$$(\sum_i b_i)^2 \leq \sum_i b_i^2 + 2\sum_{i<j}|b_i||b_j| \leq C^2(\sum_i \omega_i)^2.$$

This gives a uniform Fock bound without a spurious $\sqrt{n}$. The number-block closure follows from the commutator criterion with $1+\omega^2$ on the smooth elliptic core. On the guard all mass jets vanish. The complete symbols of the spectral projections there agree with the massless chiral projections; consequently $P_+ D_h P_-$ is smoothing. Its actual mass-eigenbasis coefficients decay faster than every fixed power of $E_n + E_m$. Number and pair estimates give QA^{-1} bounded. For signed derivatives use a wider guarded positive cutoff, not a bound on h'/h. Local translation gives $\mathrm{ad}^j_{H_M} Q(h) = (-iv)^j Q_R(h^{(j)}) + (iv)^j Q_L(h^{(j)})$. Thus every fixed required energy commutator is relatively bounded and the smooth weak-drive propagator preserves the corresponding graphs over its entire gate time.

The physical spectral class can be constructed, rather than postulated. For a step reference with mass m on the exterior arc of length d, the subthreshold parity equations are

$$\tan\frac{E\ell}{2v} = \sqrt{\frac{m-E}{m+E}}\tanh\frac{d\sqrt{m^2-E^2}}{2v} \quad (+),$$

with the negative reciprocal square-root on the right for parity $(-)$. If $m > 3\pi v/(2\ell)$, the first positive root is simple, $0 < E_0 < \pi v/(2\ell)$, while $E_1 > \pi v/\ell$. At zero the transfer is $\exp((\int M/v)\sigma_y)$ and has no unit eigenvalue. Smooth the two steps just outside the body. Transfer Duhamel bounds the perturbation on fixed complex contours by $C\|M_{\mathrm{smooth}} - M_{\mathrm{step}}\|_1$; choosing it below the positive minimum of the explicit step characteristic function preserves these root counts and a strict gap. This gives a real smooth class, without treating a sampled eigengrid as its proof. In the body its normalized ground mode has $R = e^{iE_0x/v}/\sqrt{2Z}$ and $L = e^{-iE_0x/v}/\sqrt{2Z}$. For $\operatorname{supp} h \subset (-\ell/8, \ell/8)$,

$$g_4 = \frac{1}{2Z^2}\int h\cos(4E_0x/v)dx > 0, \qquad g_2 = \frac{1}{Z}\int h\cos(2E_0x/v)dx > 0.$$

Let N be signed flavour number, particles minus holes. It is not excitation number or total electric charge. Then $K_0 = H_M - E_0 N \geq 0$ has the sixteen-dimensional kernel P_0 and gap $\delta = \min(E_1 - E_0, 2E_0) > 0$; its graph is equivalent to A. Because $[N, Q_+] = 4Q_+$, e^{-iE_0tN} transforms (63) exactly to $K_0 + \lambda b(\lambda t)Q$. This is a proof coordinate, not a physical global-number pulse. The compression is $P_0QP_0 = g_4C_4^\dagger + \overline{g}_4C_4$; all off-code terms remain in the laboratory Hamiltonian.

On $|z| = \delta/2$ the weighted resolvent norms $\|A^jQ(K_0 - z)^{-1}A^{-j}\|$ are finite. Choose $\lambda\|b\|_\infty$ within their Neumann tube and the physical relative-bound margin. The contour projection and its Kato transport give an effective code matrix $aP_0QP_0 + O_s(a^2)$, $a = \lambda b$. Its integrated remainder is $O_s(\lambda)$ because $\int_0^T a^2dt = \lambda\int_0^U b^2du$. The transport residual costs $\int_0^T |\dot{a}|dt = \lambda\,\mathrm{TV}(b)$. Energy propagation costs $\exp(C_s\int|b|du)$, independent of λ. Choosing $\int bdu = \theta/g_4$ proves (64). Pair resonance at $2E_0$ and guarded currents supply the paired-code rotations and phase. For $\theta = \arcsin\sqrt{\epsilon}$ the vacuum preparation is the original mixed A/R target, at final energy $4E_0\epsilon$ up to the weighted error, supplied by the pump.

Finally a smeared physical mass bilinear has order zero and a metric stress order one. A safe pair coefficient bound is $C_f(1 + E_n + E_m)^d$, $d \leq 1$. Weighting both sides by A^{-4} bounds each creation term by $(1 + E_n + E_m)^{-4}$. One-dimensional energy-shell counting has $O(q)$ pairs, so the absolute sum converges. Number terms obey the previous estimate. With one fixed Hadamard subtraction scheme, for $d_4 = \frac{1}{2}\|A^4(\rho - \sigma)A^4\|_1$ this yields $|\operatorname{Tr} S_f(\rho - \sigma)| \leq$

$2\|A^{-4}S_f A^{-4}\|d_4$. Counterterm changes between backgrounds are not cancelled by this fixed-background comparison. □

This gate acts on the full massive carrier, with the actual pump phase, work and wall source. It does not use a hardware projector onto four smeared modes. Its source certificate is for smooth tests at a fixed mass, not pointwise source jets on moving walls. The massive carrier's outlet mixer and whole-sea renewal must be computed on its own spectral generator; the earlier massless ring's reset coefficients cannot be reused. Nor are its ground modes already the massless short-ring storage modes in Proposition 21. Physical transfer, moving support and a reused work clock must join these layouts before they form one sustained device.

9 A common covariant ledger: controller, clock and support

The local potential (21) has potential stress $-g_{\mu\nu}V_{\rm loc}$ if its variables are Lorentz scalars. That term alone omits the mechanism which moves its profile and changes its amplitude. A common local action makes this omission explicit and offers a candidate completion at the classical action and regulated canonical levels.

Take dynamical scalars $X = (\phi_1, \phi_2, Z, \theta, y, \chi, r_1, \dots, r_d)$ on a $(1+1)$-dimensional background of signature $(-,+)$. Here Z is a support texture, χ a clock coordinate, and r_j memory/control coordinates. Specify the positive target coframe

$$\begin{aligned} e_{\phi_a} &= d\phi_a + \lambda_a \theta f(Z) dZ, & e_Z &= m_Z dZ, \\ e_\theta &= \sqrt{I}\, d\theta, & e_y &= \sqrt{J}\, dy, \\ e_\chi &= \sqrt{I_\chi}\, d\chi, & e_{r_j} &= \sqrt{I_j}\, dr_j. \end{aligned}$$

with all diagonal coefficients strictly positive, and put $G = \sum e \otimes e$. Let $b_a(\chi, r)$ be smooth bounded functions, $w_a(Z)$ smooth profiles compact in the texture value, and $M(Z)$ a smooth strictly positive material matrix. With $u = (\sqrt{I}\theta, \sqrt{J}y)$ and $v = u - \sum_a b_a(\chi, r) w_a(Z)$, choose

$$S = -\int \sqrt{-g}\, [\tfrac{1}{2} G_{AB}(X) g^{\mu\nu} \partial_\mu X^A \partial_\nu X^B + V(X)]\, d^2x, \tag{65}$$

$$V = V_Z(Z) + V_{\rm mem}(r) + V_{\rm clock}(\chi) + \tfrac{1}{2} v^T M(Z) v \geq 0. \tag{66}$$

The last square replaces the material mass potential rather than adding a second copy of it. Z, χ, r are actual fields in this candidate, not prescribed positions, waveforms or cursor amplitudes. A prepared finite-energy texture can localize the contact; localization is not produced by writing $w_a(Z)$.

Proposition 27 (Common-action stress and clock reaction)**.** *For smooth finite-energy classical data of* (65)*, metric variation gives*

$$T_{\mu\nu} = G_{AB} \partial_\mu X^A \partial_\nu X^B - g_{\mu\nu} [\tfrac{1}{2} G_{AB} \partial X^A \cdot \partial X^B + V]. \tag{67}$$

The Euler–Lagrange equations are

$$\nabla_\mu(G_{AB}\partial^\mu X^B) - \tfrac{1}{2}(\partial_A G_{BC})\partial X^B \cdot \partial X^C - \partial_A V = 0. \tag{68}$$

They imply $\nabla^\mu T_{\mu\nu} = 0$ for the complete matter action. In specified Minkowski coordinates its canonical energy is

$$H = \int [\tfrac{1}{2}\Pi^T G^{-1}\Pi + \tfrac{1}{2}X'^T G X' + V]\, dx \geq 0.$$

At finite Fourier resolution, the following specified canonical ordering has a nonnegative Friedrichs realization on the supplied charge sectors and finite real coordinate space.

Proof. Vary the inverse metric with all scalar variables held fixed and use $T_{\mu\nu} = -2(\delta S/\delta g^{\mu\nu})/\sqrt{-g}$. Integration by parts in a field variation gives (68); contraction with $\partial_\nu X^A$ gives the stress divergence. In Minkowski coordinates $\Pi_A = G_{AB}\dot{X}^B$, so the Legendre transform has the displayed positive squares. The coframe is nonsingular and triangular. For a spatial Fourier regulator, let q denote its finite real coordinates and compact scalar zero modes. The reduced Lagrangian has positive kinetic matrix $G_K(q)$; put $A_K = G_K^{-1}$ and let $W_K(q) \geq 0$ be its integrated gradient and potential energy. Use Lebesgue measure on the real coordinates and the scalar zero-mode line bundles of Proposition 6, with their charge/winding direct sum. On smooth compactly supported sections choose the divergence-form kinetic ordering

$$\mathfrak{q}_K[\psi] = \tfrac{1}{2}\int (D_i\psi)^*(A_K)^{ij}(D_j\psi)\, dq + \int W_K|\psi|^2\, dq.$$

The D_i are coordinate derivatives with the stated zero-mode boundary conditions. Smooth locally elliptic coefficients make this densely defined nonnegative form closable; its Friedrichs realization is the specified operator. If the interaction acts only on retained low modes, the complete free carrier tail is added positively, as in the charged regulator. This choice of canonical measure and ordering is not a claim of regulator-independent covariant quantization. □

In particular,

$$I_\chi \Box\chi = \partial_\chi V = V'_{\text{clock}} - \sum_a (\partial_\chi b_a) w_a^T M v.$$

The amplitude change reacts on the clock. The memory equations have the corresponding $\partial_{r_j} b_a$ force, and the support equation includes derivatives of w_a, M and the target coframe. Dropping these terms returns an externally driven model. For a free massless clock initially having $\partial_t\chi = \Omega$ over a proper-length interval ℓ, its initial kinetic energy is at least $I_\chi\Omega^2\ell/2$. Smooth spatial tapers and contact reaction add costs and change that free trajectory. A locally timelike clock gradient, texture, memory preparation and boundary conditions are specified initial data; they are not guaranteed throughout an arbitrary run.

The action includes clock/support reaction; canonical positivity is only in the chosen coordinates and finite regulator. Curved-space quantum positivity, renormalized continuum stress and causal propagation after cutoff removal require further proofs. Matching (15)

requires quantum acquisition and transported gates, the prepared texture/charge sectors, and a common finite-cursor/source estimate. The Gaussian K needs static supplied Z and controlled freezing or small perturbations of r, χ; dynamical cross terms and backreaction otherwise change H. That approximation needs an error bound, as does matching HAFF D's dilaton coordinates and full supported quantum state.

10 Discussion

An abstract representation, an exact reduced operation and a renewable physical interface are different requirements. Image closure follows from a homomorphism. A coupling supplies a completely positive map, whose exact algebra is its multiplicative domain. A discarded non-atomic control port therefore limits exactness, but a retained oscillator or closed propagation does not obey that discarded-port premise.

Acquisition is a physical change of the old history, not a supplied source label. The actual record drives the same complete material and produces another reading. Its connected fluctuations contribute to the source; dephasing can preserve an immediate field marginal while changing later joint control. Physical moments and smooth-source forms certify that return without replacing an unbounded stress by a trace-distance estimate.

Renewal depends on what is held. Universal identity-channel preservation cannot append a correlated interacting equilibrium. A specified receiver can nevertheless be replaced by a paid Gibbs bank, with spent correlations retained; the full ring can be reset and its mixed target reinstalled through four thermal outlets. Neutral compression, packet loading and spectral sorting supply a complete-output converter, not merely thermal mean energy. Isospectral axes need not retain a heat-source contrast after optimal sorting, and accurate direct reset alone leaves its purifier redundancy. Persistent sharp-exhaust gain remains a joint implementation question.

The clock distinctions in the introduction fix the resource claims: bounded-energy slow receiver renewal, fixed-duration energy-growing finite rounds, and a spatial work field with horizon-dependent inertia. The full joint-state estimate for the latter retains every old reference and charges finite clock work in the source comparison. Static frame/endpoints and dynamical support are still different problems. The neutral-current stress compiler gives another finite physical control permission, with coherence and precision budgets; it is not noise-protected control over every charged sector.

Foundation III supplies a complementary finite semiclassical mean-action scalar/JT return at common initial constraints: a learned frame protects old content, its record changes the material source, and a later X-basis reading of the old probe carries the geometric response into the spent cell [50]. Its first-order signal arises after acquisition; the cursor and old purifications remain. Matching that run to the complete quantum material here requires a physical state/source bridge. HAFF D's rest-frame mean-stress proposition identifies one excess-radiation sector under fixed covariance and linear contact assumptions, not the entire acquired-record apparatus. Neither construction treats a spent port as a fresh bath.

A world model must retain the carrier and conserved total source; an interface must specify actual records and executable access; a continuing observer must maintain the content and resources those records serve. The common-action candidate connects these respon-

Interface question	Established construction	Required input or remaining bridge
Exact operations	Multiplicative domain; propagation symbol; retained loops	Spectral law, access and quadratic-space conditions
Actual quantum return	Acquisition, full material response and second record under one positive H	Prepared ports, allowed gates, physical transport and finite cursor
Local source	Same-H smooth interacting field bound; local positive contact potential	Smooth interior tests and moments; controller/support stress is separate
Finite maintenance	Receiver-bank SWAP; correlated-window and diagnostic protocols	Paid bank, storage and clock; not full interacting-material Gibbs renewal
Repeated body renewal	Full ring contraction, mixed-target preparation, neutral code, packet loading and finite-carrier sorting	Four thermal outlets, paid buffer, prescribed gates and energy-constrained capacity
Massive spatial body	Guarded local full-angle gate and weighted physical stress tests	Paid mass profile and state; actual outlet renewal and moving support remain
Field–clock evolution	Retained-state finite schedule with matrix Ward and delayed boundary	Static walls/frame; smooth input norms and horizon-dependent inertia
Geometric flow tasks	Local linear-current synthesis of neutral $c = 1$ stress gates	Circle recurrence, coherent control and growing precision budgets
Covariant apparatus	Common local action with dynamic clock and support reaction	Quantum gate matching, charged continuum and renormalized total stress

Table 1: The main interfaces and their distinct burdens. Auxiliary apparatus proofs are collected in the appendices.

sibilities at the classical and regulated canonical levels. Its quantum acquisition, charged continuum, moving supports and total renormalized stress still need matching. Neither exact multiplicativity nor finite repair derives experience or public geometric agreement. Conservation-law and multimode bounds [36, 38, 39] do not settle all these architectures at once. Split/modular inclusions and virtual subsystems [43, 44, 42] organize retained access; formation and maintenance of its permissions remain questions for T-DOME and Imprint.

A The algebraic statement

Let (V, q) be a finite-dimensional real vector space with a non-degenerate quadratic form and $Cl(V, q)$ the real Clifford algebra generated by V with $v^2 = q(v)\mathbf{1}$. When generators are to act as operators, those with $q(e) > 0$ are taken Hermitian and those with $q(e) < 0$ anti-Hermitian; this fixes an involution with $e^* = \operatorname{sign}(q(e))\, e$ on an orthogonal basis.

Definition 28 (Representation input)**.** A *representation* of a control algebra $\mathcal{O}_{\text{int}} \cong Cl(V, q)$ in an environmental C^*-algebra $\mathcal{A}_{\text{ext}}$ is a unital $*$-homomorphism $\phi : \mathcal{O}_{\text{int}} \to \mathcal{A}_{\text{ext}}$.

Lemma 29 (Semisimplicity)**.** *$Cl(V, q)$ is semisimple. If $\dim V$ is even it is simple, and every nonzero algebra homomorphism out of it is injective.*

Proof. By the classification of real Clifford algebras [40, 41], $Cl(V, q)$ is a matrix algebra over $\mathbb{R}$, $\mathbb{C}$ or $\mathbb{H}$, or a direct sum of two such algebras, according to signature and dimension modulo eight. For even $\dim V$ it is a single matrix algebra. The kernel of a homomorphism is a two-sided ideal, and a simple algebra has none besides 0 and itself. □

Proposition 30 (Image closure)**.** *Let ϕ be as in Definition 28. Then $\operatorname{Im}\phi$ is a $*$-subalgebra of $\mathcal{A}_{\text{ext}}$ in which the Clifford relations hold. If $Cl(V, q)$ is simple, in particular when $\dim V$ is even, ϕ is injective. If it is the sum of two simple algebras, its image is isomorphic to the full algebra or to one of the two summands.*

Proof. Images of $*$-homomorphisms are $*$-subalgebras and relations descend. Simplicity gives injectivity. The only proper ideals of a sum of two simple algebras are zero and either summand; a unital representation has a proper kernel. □

Non-degeneracy is needed: for $q(x, y) = x^2$ on $\mathbb{R}^2$, the map $e_1 \mapsto \operatorname{diag}(1, -1)$, $e_2 \mapsto 0$ is a nonzero unital homomorphism into $M_2(\mathbb{R})$ that is not injective. An algebra homomorphism also does not compare signatures. That requires a quadratic-space morphism $j : (V, q) \to (W, Q)$ with $Q(jv) = q(v)$, and, where relevant, preservation of the involution and grading by the induced map of Clifford algebras. Proposition 30 provides neither the map ϕ nor such a j. Sections B–D ask where they can come from.

B Couplings give channels

Let a finite-dimensional system S interact with an environment $\mathcal{E}$ through a unitary U on $\mathcal{H}_S \otimes \mathcal{H}_\mathcal{E}$, with the environment in a normal state σ. The induced Heisenberg map on system observables is

$$\Phi^*(A) = \operatorname{tr}_\mathcal{E}\big[(\mathbf{1} \otimes \sigma)\, U^*(A \otimes \mathbf{1})U\big]. \tag{69}$$

It is unital and completely positive, and by Stinespring's theorem every such map arises in this way [29]. It is in general not multiplicative. Its *multiplicative domain*

$$\mathcal{M}_\Phi = \{A :\ \Phi^*(A^*A) = \Phi^*(A)^*\Phi^*(A),\ \Phi^*(AA^*) = \Phi^*(A)\Phi^*(A)^*\} \tag{70}$$

is a C^*-subalgebra on which Φ^* is a $*$-homomorphism, and $\Phi^*(AB) = \Phi^*(A)\Phi^*(B)$ whenever $A \in \mathcal{M}_\Phi$ [30, 31]. It is the largest part of the system's algebra that the coupling represents exactly. For other elements we measure the failure by the *defect*

$$\delta(A) = \Phi^*(A^*A) - \Phi^*(A)^*\Phi^*(A) \ \geq 0. \tag{71}$$

Theorem 31 (Exactly represented algebra of a dephasing coupling)**.** *Let $X = \sum_x xP_x$ have finite spectrum, let E be self-adjoint on $\mathcal{H}_\mathcal{E}$, and $U = \exp(-iX \otimes E)$. Write $\chi(t) = \operatorname{tr}(\sigma e^{-itE})$ and $A_{xx'} = P_x A P_{x'}$. Then*

$$\Phi^*(A) = \sum_{x,x'} \chi(x'-x)\, A_{xx'}, \tag{72}$$

the relation $x \sim x'$ defined by $|\chi(x'-x)| = 1$ is an equivalence relation on the spectrum of X, and

$$\mathcal{M}_\Phi = \{A :\ A_{xx'} = 0 \text{ whenever } |\chi(x'-x)| < 1\}. \tag{73}$$

On $\mathcal{M}_\Phi$, Φ^* is an inner automorphism $A \mapsto W^*AW$ with $W = \sum_x u_x P_x$ for phases u_x. If $|\chi(t)| < 1$ for every nonzero difference t of eigenvalues of X, then $\mathcal{M}_\Phi = \{X\}'$ and Φ^* is the identity on it.

Proof. $U = \sum_x P_x \otimes e^{-ixE}$, so $U^*(A \otimes \mathbf{1})U = \sum_{x,x'} A_{xx'} \otimes e^{ixE}e^{-ix'E}$, and taking the expectation in σ gives Eq. (72).

If $|\operatorname{tr}(\sigma e^{-itE})| = 1$, write $\sigma = \sum_i p_i |\psi_i\rangle\langle\psi_i|$; each $|\langle\psi_i|e^{-itE}|\psi_i\rangle| \leq 1$, so equality forces every ψ_i to be an eigenvector of e^{-itE} with one common eigenvalue $\chi(t)$. Hence e^{-itE} acts as the scalar $\chi(t)$ on the support of σ. If this holds for t_1 and t_2, it holds for $t_1 + t_2$ with $\chi(t_1+t_2) = \chi(t_1)\chi(t_2)$. Together with $\chi(0) = 1$ and $\chi(-t) = \overline{\chi(t)}$ this shows that $\sim$ is an equivalence relation, and that within a class with base point x_0 the phases $u_x = \chi(x - x_0)$ satisfy $\chi(x'-x) = \overline{u_x}u_{x'}$.

The diagonal blocks of the defect are

$$P_y\big[\Phi^*(A^*A) - \Phi^*(A)^*\Phi^*(A)\big]P_y = \sum_x \big(1 - |\chi(y-x)|^2\big) A_{xy}^* A_{xy}, \tag{74}$$

a sum of positive operators with nonnegative coefficients. It vanishes for all y only if $A_{xy} = 0$ whenever $|\chi(y-x)| < 1$; the condition with AA^* is the same with the roles of rows and columns exchanged. Conversely, if A has this block pattern, Eq. (72) gives $\Phi^*(A) = \sum \overline{u_x}u_{x'}A_{xx'} = W^*AW$, and these A form a $*$-algebra on which Φ^* is multiplicative. If no nonzero difference has $|\chi| = 1$, the classes are single eigenvalues, the algebra is $\bigoplus_x P_x B(\mathcal{H}_S) P_x = \{X\}'$, and W is scalar on each block. □

Corollary 32 (Continuous spectrum forbids exact coupled control)**.** *If the spectral distribution of E in σ has no atoms, then $|\chi(t)| < 1$ for every $t \neq 0$, so $\mathcal{M}_\Phi = \{X\}'$. In particular this holds for every normal state when E has purely continuous spectrum. Every exactly represented element A commutes with X, and hence with the interaction on its natural domain. No nonzero rescaling of the coupling changes this conclusion.*

Proof. The equality $|\int e^{-ite}\,d\mu(e)| = 1$ requires the phase e^{-ite} to be constant μ-almost everywhere. For $t \neq 0$ its level sets are translates of the countable lattice $(2\pi/t)\mathbb{Z}$, and a non-atomic probability measure gives each such set measure zero. Absolute continuity is sufficient but unnecessary. □

A field quadrature, or the smeared current Q_g of Paper B, has purely absolutely continuous spectrum, so Corollary 32 applies to every normal drive state, whatever its energy. For other couplings, $|\chi(t)| = 1$ requires support in an eigenspace of e^{-itE}, not necessarily an eigenstate of E. Periodic spectra can give exact revivals: with $X = \pi Z$, $E = \mathrm{diag}(0, 1)$ and $\sigma = |+\rangle\langle+|$, the interaction unitary is $I \otimes \mathrm{diag}(1, -1)$ and the system channel is identity, although σ is not an E eigenstate. A Fock state in a number-type coupling gives another finite-energy possibility. The quadrature obstruction concerns the specified single coupling; extending it to composite local controls requires a separate argument.

C The field-coupled flip

Paper B couples a degenerate bit to one mode of a free chiral field through $U = \exp(-i\lambda XQ)$, $\lambda > 0$, with $Q = b + b^\dagger$, $P = i(b^\dagger - b)$ and $[Q, P] = 2i$ [26]. Here $E = \lambda Q$ and the eigenvalues of X are ± 1. Write $\chi = \mathrm{tr}(\sigma e^{-2i\lambda Q})$ and $\sigma_\pm = e^{\mp i\lambda Q}\sigma e^{\pm i\lambda Q}$ for the field states conditioned on $X = \pm 1$. A flip of Z-diagonal states fails with probability $\epsilon = (1 + \mathrm{Re}\,\chi)/2$.

Theorem 33 (Representation defect of the field-coupled flip)**.** *Let the drive mode be in a normal state σ with mean occupation $N = \mathrm{tr}(\sigma b^\dagger b) < \infty$.*

1. *In the X eigenbasis, Φ^* multiplies the two off-diagonal entries by χ and $\overline{\chi}$, and $\mathcal{M}_\Phi = \{X\}'$, the algebra generated by X.*

2. *For every A with vanishing diagonal in the X basis, in particular Z and Y, $\delta(A) = (1 - |\chi|^2)\,\mathrm{diag}(|A_{-+}|^2, |A_{+-}|^2)$; thus $\delta(Z) = \delta(Y) = (1 - |\chi|^2)\mathbf{1}$.*

3. $1 - |\chi|^2 \;\geq\; \dfrac{\arccos^2(\mathrm{Re}\,\chi)\,|\chi|^2}{(2N+1)^2}$.

4. $1 - |\chi|^2 \geq D(\sigma_+, \sigma_-)^2$, *with equality when σ is pure, where D is the trace distance.*

Proof. Item 1 is Theorem 31 with eigenvalue differences ± 2 and Corollary 32. Item 2 follows from Eq. (72) by multiplying out 2×2 matrices.

Item 3 uses Paper B's two quadrature bounds, valid for a normal (density-operator) drive state of finite occupation and $[Q, P] = 2i$ [26]: with $G(c) = \arccos^2 c$,

$$\langle Q^2\rangle \geq \frac{G(\mathrm{Re}\,\chi)}{4\lambda^2}, \qquad (\Delta P)^2 \geq \frac{4\lambda^2|\chi|^2}{1 - |\chi|^2}.$$

The first is spectral Jensen; the second is the unitary uncertainty bound. Since $4N + 2 = \langle Q^2 + P^2\rangle$,

$$\frac{4\lambda^2|\chi|^2}{1 - |\chi|^2} \leq \langle P^2\rangle = 4N + 2 - \langle Q^2\rangle \leq 4N + 2 - \frac{G(\mathrm{Re}\,\chi)}{4\lambda^2}.$$

With $u = 4\lambda^2$, $a = G(\mathrm{Re}\,\chi)$ and $A = 4N + 2$, this reads $1 - |\chi|^2 \geq u^2|\chi|^2/(Au - a)$ for some $u > a/A$. The right side is minimized over u at $u = 2a/A$, where it equals $4a|\chi|^2/A^2$.

Item 4. If $\sigma = |\psi\rangle\langle\psi|$, the conditioned states are pure with overlap $\langle e^{i\lambda Q}\psi, e^{-i\lambda Q}\psi\rangle = \chi$, and $D^2 = 1 - |\chi|^2$. For mixed σ, applying $e^{\mp i\lambda Q} \otimes \mathbf{1}$ to a purification of σ gives purifications of $\sigma_\pm$ with overlap χ, so Uhlmann's theorem gives fidelity $F(\sigma_+, \sigma_-) \geq |\chi|$, and $D \leq (1 - F^2)^{1/2}$ [32, 33]. □

Corollary 34 (The energy–accuracy bound as a representation statement)**.** *For a flip with failure probability $\epsilon \in (0, \frac{1}{2})$ and $\theta_\epsilon = \arccos(2\epsilon - 1)$,*

$$N \;\geq\; \frac{\theta_\epsilon(1 - 2\epsilon)}{4\sqrt{\epsilon(1-\epsilon)}} - \frac{1}{2}. \tag{75}$$

Proof. $\mathrm{Re}\,\chi = 2\epsilon - 1$ gives $|\chi|^2 \geq (1 - 2\epsilon)^2$, hence $1 - |\chi|^2 \leq 4\epsilon(1 - \epsilon)$. Combining with item 3 of Theorem 33 gives $4\epsilon(1 - \epsilon) \geq \theta_\epsilon^2(1 - 2\epsilon)^2/(2N + 1)^2$. □

This is exactly the occupation bound of Paper B. Read in the present terms, an accurate flip requires the coupling to represent Z almost multiplicatively, and the defect $1-|\chi|^2$ cannot vanish at finite occupation. Item 4 identifies the defect with information: for pure drives it is the squared distinguishability of the two field states that record the value of X. The field is then a fragment in the sense of the HAFF paper on measurement, with conditional overlap $|\chi|$; that paper determines which pointer effects such a fragment can implement [28], and the same overlap fixes the representation defect here. The exactly represented algebra $\{X\}'$ is the pointer algebra of the decoherence literature [37]: the control interface transports exactly the part of the system that the environment records.

For the phase-fixed half-turn squeezed family, Paper B gives occupations 2.958, 7.330 and 24.329 at $\epsilon = .05, .01, .001$, against bound values 2.28, 6.74 and 23.80. Its separately optimized finite quadrature-grid values are variational minima, not certified global optima; the continuous supporting ground-state argument has its own assumptions. Mixtures of coherent states require occupation of order $\pi^2/(16\epsilon)$ [26].

C.1 Same clock as mode matching

The requirement that control and environment share a clock, so that their operational windows overlap, has a precise form in the field-coupled flip. The switch g defines a normalized mode function f and the mode operator $b_f = \int \overline{f(\omega)}\, a_\omega\, d\omega$ with $Q_g = \lambda(b_f + b_f^\dagger)$ [26]. A drive is prepared in some normalized mode f', by a source with its own clock.

Proposition 35 (Mode overlap)**.** *Let the drive occupy mode f' with mean occupation N', all modes orthogonal to f' being in the vacuum, and let $c = \int \overline{f(\omega)} f'(\omega)\, d\omega$. Then the bit channel equals the channel of Theorem 33 for the reduced state of mode f, which is the output of a pure-loss channel of transmissivity $|c|^2$ and has mean occupation $|c|^2 N'$. A flip with failure ϵ therefore needs*

$$N' \;\geq\; \frac{1}{|c|^2}\left[\frac{\theta_\epsilon(1 - 2\epsilon)}{4\sqrt{\epsilon(1-\epsilon)}} - \frac{1}{2}\right]. \tag{76}$$

If $f'(\omega) = f(\omega)e^{i\omega\Delta}$, a copy of f delayed by Δ, then $c = \int |f(\omega)|^2 e^{i\omega\Delta} d\omega$. For a positive-frequency field this integral is over $\omega > 0$. Extending a narrowband Gaussian spectral density of width Ω to the full real line gives $|c| \simeq e^{-\Omega^2\Delta^2/2}$ and an occupation penalty $e^{\Omega^2\Delta^2}$ in the range where this approximation controls the overlap. The exact positive-frequency integral, not this Gaussian approximation, determines the cost at long delays.

Proof. Write $f = \bar{c}f' + r$ with $\int \overline{f'}\, r\, d\omega = 0$ and $\|r\|^2 = 1 - |c|^2$. Since $g \mapsto b_g$ is antilinear, $b_f = c\, b_{f'} + b_r$, where $[b_r, b_r^\dagger] = 1 - |c|^2$ and b_r commutes with $b_{f'}$ and $b_{f'}^\dagger$. The mode $r/\|r\|$ is orthogonal to f' and in the vacuum, independent of mode f', so the state of mode f is that of a beam splitter of transmissivity $|c|^2$ with vacuum in the other port, and its occupation is $|c|^2 N'$. Q_g involves only mode f, so the bit channel depends only on this reduced state, and Corollary 34 applies to it. Translation gives the exact Fourier overlap. The Gaussian specialization follows only after the stated extension of the frequency support. □

The prepared drive has energy $N'\bar{\omega}'$, while only the fraction $|c|^2$ of its occupation reaches the selected mode. This is a charged preparation resource, not automatically dissipated heat. The source/switch clock offset therefore enters Paper B's feedback acceptance as a mode-matching loss.

D A representation generated by propagation

The situation is different when the "environment" is part of the same closed dynamics. In the local propagation model of Paper A, a particle with internal space $\mathbb{C}^s$ moves on a lattice Λ by a unitary step $W = \sum_{h\in N} T_h \otimes A_h$, where T_h translates by h [25]. The internal algebra is represented by

$$\phi : M_s(\mathbb{C}) \to B\big(\ell^2(\Lambda) \otimes \mathbb{C}^s\big), \qquad \phi(A) = \mathbf{1} \otimes A, \tag{77}$$

a faithful unital $*$-homomorphism, and $[\phi(A), W] = \sum_h T_h \otimes [A, A_h]$ is nonzero unless A commutes with every A_h. The requirement of Definition 28 together with nontrivial coupling is met exactly, because nothing is traced out: the translational degrees of freedom and the internal ones evolve under one unitary. The contrast with Section C is the content of Corollary 32: exactness is lost when the controlling degrees of freedom are an environment in a state with continuous spectral support.

The propagation also supplies the quadratic-space morphism that Section A found missing. Near a band point the step has a first-order symbol $H(q) = \sum_i q_i\alpha_i + \mu\beta$ with Hermitian coefficients [25].

Proposition 36 (The propagation symbol as a quadratic-space morphism)**.** *Let $j(q) = \sum_i q_i\alpha_i$ with $\alpha_1, \ldots, \alpha_d$ linearly independent. Then $j(q)^2 = Q(q)\mathbf{1}$ for a quadratic form Q if and only if, for $\mu = 0$, every eigenvalue of $H(q)$ lies on one shell $\omega^2 = Q(q)$. In that case Q is positive definite, j is an isometry from $(\mathbb{R}^d, Q)$ into $\mathrm{Herm}(s)$ with the normalized Hilbert–Schmidt form, and j extends to a unital $*$-homomorphism $Cl(\mathbb{R}^d, Q) \to M_s(\mathbb{C})$. If moreover $\mu \neq 0$, $\beta^2 = \mathbf{1}$ and $\{\alpha_i, \beta\} = 0$, then*

$$J(\omega, q) = \omega\beta - \sum_i q_i\beta\alpha_i \tag{78}$$

satisfies $J(\omega,q)^2 = (\omega^2 - Q(q))\mathbf{1}$ *and extends to a unital homomorphism* $Cl(\mathbb{R}^{1+d},\eta) \to M_s(\mathbb{C})$ *with* $\eta(\omega,q) = \omega^2 - Q(q)$*, which preserves the involution in which* β *is Hermitian and the* $\beta\alpha_i$ *are anti-Hermitian.*

Proof. The first statement is the single-shell theorem of Paper A [25] with $\mu = 0$; isometry follows from $\frac{1}{s}\operatorname{tr} j(q)^2 = Q(q)$, and the extension from the universal property of Clifford algebras. For J, expand J^2 using $\beta^2 = \mathbf{1}$ and $\beta\alpha_i\beta = -\alpha_i$: the cross terms $\omega q_i\{\beta, \beta\alpha_i\}$ vanish and $\sum q_i q_j \frac{1}{2}\{\beta\alpha_i, \beta\alpha_j\} = -Q(q)$. Hermiticity of β and anti-Hermiticity of $\beta\alpha_i = -\alpha_i\beta$ follow from $(\beta\alpha_i)^* = \alpha_i\beta$. □

J is the map $j : (V,q) \to (W,Q)$ of Section A with $(V,q) = (\mathbb{R}^{1+d},\eta)$, a Lorentzian quadratic space, and W the real span of β and the $\beta\alpha_i$ in $M_s(\mathbb{C})$, with the form defined by $Y^2 = Q_W(Y)\mathbf{1}$. Here it is supplied by the dynamics, and its existence is equivalent to an observable property of the propagation: no birefringence.

On the lattice the map is exact only at leading order. For $s = 2$ and $W(k) \in SU(2)$, the effective generator $H_{\text{eff}}(k) = (i/\tau)\operatorname{Log} W(k)$ is traceless, so $H_{\text{eff}}(k)^2 = \omega(k)^2\mathbf{1}$ holds at every k; what fails is linearity and isotropy of $k \mapsto H_{\text{eff}}(k)$. In the Weyl automaton of Paper A, $H_{\text{eff}}(k) = c\,\sigma\cdot k$ up to corrections of relative order $|k|\ell$, which include a parity-odd anisotropic term. In the split-step walk the quadratic identity $\omega^2 = c_{\text{eff}}^2 q^2 + \mu^2$ fails at order $(qa)^4$, with coefficient $-\theta^2/45 + O(\theta^4)$ in lattice units [27]. These orders are the precise sense in which a lattice propagation gives an approximate representation.

E A finite conserved control core

A finite prescribed circuit, energy supply and thermal ports can share a time-independent Hamiltonian. We combine energy-ladder control [20] with a finite clock [22], using complete energy sectors and an exactly solvable history clock. Register, gate permissions and initial resources are supplied inputs.

E.1 A guarded battery for the whole circuit

Let $H_D = \delta\sum_a q_a|a\rangle\langle a|$, with integers $0 \le q_a \le Q$, and let $H_B = \delta\sum_{n=0}^{K} n|n\rangle\langle n|$, $K \ge 2Q$. The register D includes its program, retained system and all thermal ports; an inert reference can be appended. Define

$$V(U)|a, E - q_a\rangle = \sum_b U_{ba}|b, E - q_b\rangle, \qquad Q \le E \le K, \tag{79}$$

for any unitary U, and take the identity on incomplete energy sectors.

Proposition 37 (Finite energy-conserving circuit lift)**.** *The operator* $V(U)$ *is unitary, commutes with* $H_0 = H_D + H_B$*, and* $V(U)V(W) = V(UW)$*. Put* $K = 2Q + L - 1$ *and prepare*

$$|\beta_L\rangle = \sqrt{\frac{2}{L+1}}\sum_{j=1}^{L}\sin\frac{\pi j}{L+1}|Q + j - 1\rangle, \qquad e_B = 2\sin\frac{\pi}{2(L+1)}.$$

For any input of D entangled with a reference and initially independent of this battery, the joint half trace distance between the lifted circuit $V(U_m)\cdots V(U_1)$ and $U_m\cdots U_1\otimes I_B$ is at most $\min\{1, 2Qe_B\}$. The comparison retains the battery and all correlations.

Proof. Every complete energy sector carries a copy of the entire data space; the others carry the identity. This proves unitarity, conservation and composition. Extend the battery vector by zero to a bilateral ladder with shift S. Its adjacent overlap is $\cos[\pi/(L+1)]$, so $\|(S^k - I)\beta_L\| \le |k|e_B$. With $D_q = \sum_a |a\rangle\langle a|\otimes S^{-q_a}$, $V(U) = D_q(U\otimes I)D_q^*$ on the guarded inputs. Each dressing difference has vector norm at most Qe_B, even with a retained reference. Their sum bounds the joint state distance after purification. The guards make this bilateral calculation agree with the finite-sector action; no shift wraps through an endpoint. □

The bound concerns the whole compiled circuit at a fixed register energy range. Fresh thermal units in a growing protocol enlarge Q. Battery coherence, mean energy and preparation remain resources; the battery is not an exact catalyst.

E.2 Autonomous timing and a usable output window

Let $V_j = V(U_j)$ and add clock states $|0\rangle, \dots, |m\rangle$. For $J > 0$ and $\Lambda = Jm$, set

$$H = H_0 + \Lambda I + \sum_{j=1}^{m} J\sqrt{j(m-j+1)}\big(|j\rangle\langle j-1|\otimes V_j + \text{h.c.}\big). \tag{80}$$

The clock-dependent interaction is part of the apparatus energy.

Proposition 38 (Positive autonomous execution)**.** *The Hamiltonian is nonnegative and commutes with H_0. From clock state $|0\rangle$, at $T = \pi/(2J)$ it gives*

$$e^{-iHT}(|0\rangle\otimes|\psi\rangle) = e^{-i\Lambda T}(-i)^m|m\rangle\otimes e^{-iH_0T}V_m\cdots V_1|\psi\rangle.$$

If only the first $r \le m$ edges perform gates and the rest are identities, then throughout $[T/2, 3T/2]$ the data's half trace distance from the completed lifted circuit, including its free evolution, is at most

$$P_{m,r} = \Pr\{\mathrm{Bin}(m, 1/2) < r\}. \tag{81}$$

The joint data–clock distance to the completed data and the normalized completed-tail clock state is at most $\sqrt{P_{m,r}}$.

Proof. Put $W_j = V_j\cdots V_1$, $W_0 = I$, and $\mathcal{W} = \sum_j |j\rangle\langle j|\otimes W_j$. Conjugation by $\mathcal{W}^*$ reduces the hopping term to the spin-$m/2$ chain $2JJ_x$, commuting with H_0. Its spectrum lies in $[-Jm, Jm]$; the shifted block lies in $[0, 2\Lambda]$. The spin rotation transfers 0 to m at T, as in perfect-transfer chains [21]. Its position law is $\mathrm{Bin}(m, \sin^2 Jt)$. Every prefix $j \ge r$ has the same completed data. Tracing the clock leaves at most the incomplete probability as error. Projection onto the completed tail has squared overlap one minus this probability, giving the square-root joint bound. On the stated window $\sin^2 Jt \ge 1/2$. □

The joint target uses the normalized tail, not the unchanged full clock wavefunction; the latter comparison instead has bound $2\sqrt{P_{m,r}}$. For $r = 10$, $m = 64$ and $\Lambda = .4$, $P_{m,r} \approx 1.77111\times10^{-9}$ and $T \approx 251.327$ in the declared energy units. The clock has 65 states. This finite window is not permanent halting: the closed clock eventually returns and reverses its gates. Its initialization, idle states and any output receiver remain charged resources.

E.3 Two cycles through the same thermal contact

The core has seven qubits: detector S, retained body Q, contact F, output cells E_1, E_2 and dephasing cells D_1, D_2. The body and dephasing cells have zero bare gap. The other four have gap δ, so the register has maximum charge 4. Prepare S, F, E_1, E_2 independently in $\tau = \mathrm{diag}(1-\epsilon, \epsilon)$, $\beta\delta = \ln[(1-\epsilon)/\epsilon]$, and each D_i in its zero-gap Gibbs state $I/2$. The body may be entangled with a retained reference and is never re-prepared.

Cycle i consists, in order, of

$$e^{-i\theta_i X_S P_Q}, \quad P_Q = |1\rangle\langle 1|_Q, \quad \mathrm{CZ}_{S,D_i}, \quad \mathrm{CX}_{S,Q}, \quad \mathrm{SWAP}_{S,F}, \quad \mathrm{SWAP}_{F,E_i}\,.$$

The dephased detector directly controls the body operation. Each gate is lifted and placed on an edge of Eq. (80); no external switching times enter. The first swap transfers the old detector and all its correlations to F, putting an independent Gibbs state in S. The second moves the old state to E_i and restores F from the unused Gibbs cell. Both conserve bare energy. Replacement is exact against every retained reference, including the battery, and returns a thermal state rather than a pure blank.

The first recorded bit has probability ϵ when the body is $|0\rangle$ and $\epsilon + (1-2\epsilon)\sin^2\theta_1$ when it is $|1\rangle$. This is a genuine body reading. The command acts on the already retained body and can damage its initial coherence; it is not a coherence-preservation claim. For $\epsilon = .1$, $\delta = .1$, $\theta_1 = \pi/3$ and $\theta_2 = \pi/2$, with the body initially half of a Bell pair, output populations are $.4, .26$. Net heat $Q_E = .046$ equals the battery's mean energy loss. The joint output obeys

$$\beta Q_E = S(\rho_{E_1E_2}) - 2S(\tau) + D(\rho_{E_1E_2}\|\tau\otimes\tau). \tag{82}$$

This follows from $-\ln\tau = \beta H_E + \ln Z$. Other ports, the body, reference, clock and battery remain in the full entropy account.

At $L = 31$ the battery has 39 levels and mean initial energy 1.9. The two sensing rotations give a joint error bound $(\sin\theta_1 + \sin\theta_2)e_B = .183123$; the full purification gives the smaller upper bound .070347. For $\Lambda = .4$, the complete Hamiltonian has cutoff 5.0 and initial mean 2.34. At $L = 127$, the purification upper bound is .017610, mean energy 7.14 and cutoff 14.6. These are executions retaining the complete correlated state.

The cells hold a final energy .066, including .02 initially present; net heat .046 is not automatically emitted radiation. Spatial transport, release recoil and a stationary thermal outlet are further operations. HAFF D gives a material-support branch compatible with the retained core's mean energy [23]. It changes the support's dilaton coupling; the spin cells are not identified with the preceding continuum.

F A local outlet with an autonomous clock

Heat stored in a cell can leave through a local Hamiltonian. Let the output cell be site 0 of a semi-infinite fermion chain,

$$H_{\mathrm{lead}} = \omega\sum_{j\geq 0} n_j + g\sum_{j\geq 0}(c_j^\dagger c_{j+1} + c_{j+1}^\dagger c_j), \qquad \omega > 2g > 0. \tag{83}$$

Finite open chains have a nonnegative many-body Hamiltonian and spatially nearest-neighbour coupling. Equivalently they are open spin chains with XY exchange under the Jordan–Wigner map; the reference-uniform cell channel below uses this tensor-product realization. The infinite formula denotes its local dynamics, not a finite total-energy thermal bath. Initially site 0 has population p and may be correlated with old references. The β-KMS bath on $j \geq 1$ is initially independent of that entire cell–reference state. Use as comparison the same bath and the site population $\epsilon = (1 + e^{\beta\omega})^{-1}$.

Proposition 39 (Spatial export and reduced thermalization)**.** *Write $w_j(t)$ for the one-particle propagator from site 0. The excess covariance and on-site energies are*

$$\Delta\Gamma(t) = (p - \epsilon)|w(t)\rangle\langle w(t)|, \qquad \Delta E_j = \omega(p - \epsilon)|w_j(t)|^2. \tag{84}$$

For the homogeneous chain the excess hopping energy is zero, and the energy beyond the cell is $\omega(p - \epsilon)(1 - |w_0|^2)$. On the semi-infinite chain,

$$\alpha(t) := w_0(t) = e^{-i\omega t}\frac{J_1(2gt)}{gt}. \tag{85}$$

Let $b(t)$ be the population produced at site 0 from the bath alone and $n_{\text{base}}(t) = |\alpha|^2\epsilon + b(t)$. Against every old external reference, the reduced cell's half trace distance from its Gibbs replacement is at most

$$\min\{1,\ |\alpha|^2 + |\alpha|/2 + |n_{\text{base}}(t) - \epsilon|\}. \tag{86}$$

The retained lead and its initial purification are excluded from that product replacement claim.

Proof. Quadratic fermion evolution sends a covariance to $\Gamma(t) = U(t)\Gamma(0)U(t)^*$, giving the rank-one difference. On a bipartite chain successive components of w have relative phase $-i$ after removing $e^{-i\omega t}$, so $\mathrm{Re}(w_j^* w_{j+1}) = 0$. The surface spectral density is semicircular; its Fourier transform gives Eq. (85). This is standard tight-binding propagation with non-Markovian returns [19, Sec. III]. The thermal bath is even in fermion parity. The cell channel therefore sends population p to $|\alpha|^2 p + b$ and off-diagonal entry ρ_{01} to $\alpha^*\rho_{01}$. For a state with reference blocks ρ_{ab}, the diagonal difference from replacement at population n_{base} is $|\alpha|^2[(1 - \epsilon)\rho_{11} - \epsilon\rho_{00}]$, of trace norm at most $|\alpha|^2$. Positivity gives $\|\rho_{01}\|_1 \leq 1/2$. Adding the change from n_{base} to ϵ proves the bound. □

The lead need not be exactly Gibbs for the bare cell gap. The bath band has width $4g$, and the derivative of the Fermi function is bounded by $\beta/4$, so its occupation differs from ϵ by at most $\beta g/2$ throughout that band. Narrowing g controls this mismatch but lengthens transport. Equation (85) decays as $t^{-3/2}$ with returns and zeros; energy export is not monotone. Finite leads eventually recur. For a chain with sites $0, \ldots, N$, the finite and infinite wavevectors agree through Taylor order N. For $N + 1 > 2gt$, a norm bound on their difference is

$$\frac{2(2gt)^{N+1}}{(N+1)!}\left(1 - \frac{2gt}{N+2}\right)^{-1}.$$

This bounds propagation, not a global distance between finite and infinite thermal states.

For the two output cells with populations $.4, .26$, use two identical leads with $\omega = .1$, $g = .01$, $\beta = \ln 9/.1$. Their total initial excess is .046. At the first surface-amplitude zero, $t = 191.5853$, this excess is outside the cells. A finite calculation with 256 bath sites per lead gives $n_{\text{base}} = .10142988$. At $t = 2000$, excess energy .04599817 is in the leads and .04422294 is beyond site 16. Each bath initially contains energy 2.64854457. These values are transport after circuit completion; they do not discard the output correlations or identify a lattice energy profile with a null CFT stress tensor.

Reduced thermalization also differs from the exact SWAP of Appendix E. A two-site passive exchange at its perfect-transfer time is an iSWAP. Acting on $|+\rangle\langle+| \otimes \operatorname{diag}(.9, .1)$, it gives the cell exactly that Gibbs marginal, but the joint half trace distance from the product of its two marginals is .18. The bath cannot be called fresh for later cell–bath encounters.

F.1 The outlet can remain on during computation

Let H_{hist} be Eq. (80), with only its first r edges active, and let $H_B \geq 0$ be a finite bath Hamiltonian. For a local outlet interaction V_o with $\|V_o\| \leq v$, take $H_{\text{tot}} = H_{\text{hist}} + H_B + V_o + vI$. It is nonnegative and time independent. The offset v is part of its mass and energy account.

Proposition 40 (Always-on outlet bound)**.** *Put $T = \pi/(2J)$, choose $0 < t_0 < T$, and define*

$$P_0 = \Pr\{\text{Bin}(m, \sin^2 Jt_0) \leq r\}, \qquad A_r = J \sum_{j=1}^{r} \sqrt{j(m-j+1)}.$$

Compare the actual joint state at $t \in [t_0, T]$ with completed lifted data, including free evolution to t_0, and the free clock wavefunction at t_0, subsequently evolved with the outlet and identity clock edges. Keeping every reference, their half trace distance is at most

$$\min\{1,\ vt_0 + 2\sqrt{P_0} + 4A_r(t - t_0)\sqrt{P_0}\}. \tag{87}$$

Proof. Removing the always-on outlet only up to t_0 costs at most vt_0 by Duhamel's formula. Replacing all prefixes by the completed data costs at most $2\sqrt{P_0}$ in purification norm. After t_0, compare with the same outlet and free clock; only the first r history edges differ. The free-clock probability of positions $\leq r$ is at most P_0 through T. The norm of the differing edges on that state is at most $4A_r\sqrt{P_0}$. A second Duhamel bound gives the last term. No independence of the output cells or later fresh-bath composition is used. The index $\leq r$ includes the last active edge. □

For any finite transport window the bound can be made small by increasing the idle clock and its energy. For example $r = 10$, $m = 10^{10}$, $J = .001$, $\sin^2 Jt_0 = 128/m$ and two outlets with $v = .02$ give $t_0 = .1131371$, $T = 1570.7963$ and the sufficient joint bound .002262742. This uses $10^{10} + 1$ clock states and clock mean energy 10^7; it proves finite compatibility, not an efficient clock. A direct smaller Hamiltonian with 17 clock states, five battery levels and outlet norm 10^{-5} gives joint purification distance .00015225 below its certificate .00025067. Thus spatial export need not be externally switched, but a fixed-size indefinitely operating exhaust is not obtained.

G Thermal sources and their preparation

A conformal fermion defect gives a local relativistic alternative to the lattice outlet. The real-time defect construction and its thermal current are established results [17, Sections 2, 3 and 5]. Take two independent copies of the Majorana construction, so each port is a complex chiral fermion with $c = 1$. In the folded description, a constant contact angle acts on the two fields at the same contact time:

$$\begin{pmatrix}\psi_1^{\mathrm{out}}\\ \psi_2^{\mathrm{out}}\end{pmatrix} = S_\theta \begin{pmatrix}\psi_1^{\mathrm{in}}\\ \psi_2^{\mathrm{in}}\end{pmatrix}, \qquad S_\theta = \begin{pmatrix}\cos\theta & -i\sin\theta\\ -i\sin\theta & \cos\theta\end{pmatrix}, \quad \chi = \sin^2\theta. \tag{88}$$

The inputs are jointly independent, zero-charge quasifree thermal streams in one specified time coordinate, at $T_H < T_o$. This assumes a prepared reservoir, rather than an already prepared outgoing field.

Proposition 41 (Thermal transfer and its retained correlations)**.** *For $\omega > 0$ put $n_T = (1 + e^{\omega/T})^{-1}$. The source-port occupation and transferred energy current are*

$$n_\chi = (1-\chi)n_{T_H} + \chi n_{T_o}, \qquad q_\chi = \frac{\pi\chi}{12}(T_o^2 - T_H^2). \tag{89}$$

Its cross covariance with the return port has modulus $\sqrt{\chi(1-\chi)}(n_{T_o} - n_{T_H})$. For $0 < \chi < 1$ the source is not a KMS state at any single temperature; at $\chi = 1$ it is exactly the supplied hot state. Defining its stationary marginal entropy-current excess, with $h(x) = -x\ln x - (1-x)\ln(1-x)$, by

$$s_\chi = \frac{1}{\pi}\int_0^\infty [h(n_\chi) - h(n_{T_H})]d\omega,$$

one has

$$q_\chi \geq T_H s_\chi + \frac{3s_\chi^2}{\pi}, \tag{90}$$

strictly for $0 < \chi < 1$.

Proof. Rotate the full input covariance with S_θ. The factor $1/\pi$ in the integrals includes particles and holes of a complex chiral fermion; $\pi^{-1}\int \omega n_T d\omega = \pi T^2/12$. At high frequency $n_\chi \sim \chi e^{-\omega/T_o}$ for $\chi > 0$, whereas a single-temperature Fermi occupation has tail coefficient one. For fixed energy the Gibbs variational principle maximizes entropy at $T_{\mathrm{eff}} = \sqrt{T_H^2 + 12q_\chi/\pi}$. Hence $s_\chi \leq \pi(T_{\mathrm{eff}} - T_H)/6$, with equality only for that thermal state. Eliminating T_{eff} gives the bound and its strictness. □

The return occupation is $\chi n_{T_H} + (1-\chi)n_{T_o}$. Both outputs must be kept when accounting for entropy and correlations. This is not thermal replacement of an arbitrary qubit against every retained reference. If the inputs have a cross covariance, even their thermal marginals do not imply Eq. (89). A finite matched-frequency check uses the positive Hamiltonian $H = \omega(n_1 + n_2) + g(c_1^\dagger c_2 + c_2^\dagger c_1)$, $0 < g < \omega$. It commutes with the free energy and transfers equal energy out of the reservoir and into the source. This mode check does not establish a continuum limit for a microscopic moving contact.

Changing the contact has a cost in the same output fields. For a smooth prescribed $\theta(t)$ the full unequal-time covariance is $C_{\mathrm{out}}(t,t') = S(t)\,\mathrm{diag}(C_H, C_o)(t-t')S(t')^\dagger$. The

short-distance kernel is $C_T(s) = iT/[2\sinh(\pi T s)]$. At $t \pm s/2$, its common vacuum term is multiplied by $\cos[\theta(t+s/2) - \theta(t-s/2)]$. Point splitting therefore gives an added current $\dot\theta^2/(4\pi)$ in each output, so the total work supplied to the fields is

$$W_{\text{pump}} = \frac{1}{2\pi}\int \dot\theta^2 dt \geq \frac{(\Delta\theta)^2}{2\pi\Delta t}. \tag{91}$$

This agrees with the energy-shift dissipation form in adiabatic scattering [18, Eq. (8)]; here point splitting derives it exactly for the specified energy-independent contact. The inequality is Cauchy–Schwarz. A cosine ramp gives $W_{\text{pump}} = \pi(\Delta\theta)^2/(16\Delta t)$. Both ports' pump energy is charged; it is separate from the static net thermal transfer. A small angle error alone is no source certificate: $\theta = \theta_0 + \alpha\sin\Omega t$ has mean total pump current $\alpha^2\Omega^2/(4\pi)$. Instantaneous thermal entropy formulas do not apply during this switch. The controller that supplies this work and the contact's material stress still require their own realization. The input preparation has the following finite-window alternative.

G.1 Thermal correlations from a finite-energy preparation

Fix a bounded interval I of length L and $T > 0$. The geometric KMS construction is locally a vacuum pullback by the exponential map [10, Section 2.8, Proposition 2.4 and Theorem 2.5]. Its compact completion permits a finite global preparation. Use the state convention

$$C_\rho(x,y) = \frac{i\sqrt{\rho'(x)\rho'(y)}}{2\pi[\rho(x) - \rho(y) - i0]}, \qquad \langle T(x)\rangle_\rho = -\frac{c}{24\pi}\{\rho, x\}.$$

Here $c = 1$ for each complex-fermion stream, and the inverse diffeomorphism implementer acts on the vacuum.

Proposition 42 (A thermal restriction with finite global energy)**.** *There is a smooth increasing $\rho : \mathbb{R} \to \mathbb{R}$, equal to the identity outside a compact interval, whose vacuum-orbit state has the complete zero-charge thermal restriction on I. Put $h = \log\rho'$ and $g(y) = \rho'(\rho^{-1}(y))$. Its energy and variance are*

$$\begin{aligned} E_\rho &= \frac{c}{48\pi}\int h'(x)^2\,dx, \\ \operatorname{Var}_\rho(H_f) &= \frac{c}{48\pi^2}\int_0^\infty \omega^3 |\widehat{g-1}(\omega)|^2\,d\omega < \infty, \end{aligned} \tag{92}$$

where $\widehat{k}(\omega) = \int e^{-i\omega y}k(y)dy$. The global state is pure, not globally KMS. Its outside correlations purify the thermal restriction.

Proof. Set $\alpha = 2\pi T$ and prescribe $\rho = e^{\alpha x}$ on a neighborhood of I. Extend $\log\rho'$ smoothly with compact support. A disjoint bump can adjust $\int(e^h - 1)$ to zero: a positive bump has unbounded positive integral, while a negative bump on sufficiently large support has negative range exceeding any fixed surplus. Then $\rho(x) = x + \int_{-\infty}^x (e^h - 1)$ is a compact completion, up to an irrelevant translation on I. Its covariance on I is $iT/[2\sinh\pi T(x - y - i0)]$. Wick's rule identifies all local fermion correlations; the cited geometric construction gives the

corresponding observable-net restriction. Integrating the Schwarzian gives the first moment in Eq. (92). Conjugating H_f into the vacuum gives $T(g) + E_\rho$. Since $H_f\Omega = 0$, the variance is the vacuum second moment of $T(g-1)$, with the normalization of [11, Eq. (4.11)]. The test $g - 1$ is smooth and compact, so that moment is finite. □

Proposition 43 (Preparation cost in the vacuum orbit)**.** *Every compact-diffeomorphism vacuum preparation with this thermal restriction obeys*

$$E_\rho \geq \frac{\pi c T^2 L}{12} - \frac{cT}{6} \tanh \frac{\pi T L}{2}. \tag{93}$$

The right side is the infimum in this class when outside support is unrestricted, and is not attained for $T, L > 0$.

Proof. Thermal stress gives $\{\rho, x\} = -(2\pi T)^2/2$ on I. Thus $w = h'$ satisfies $w' = (w^2 - \alpha^2)/2$. The branches $w = \pm\alpha$ and the nonsingular coth branches have $w^2 \geq \alpha^2$. The remaining branches are $-\alpha \tanh[\alpha(x-s)/2]$; their derivative-square integral on I is minimized when s is its midpoint. That integral is $\alpha^2 L - 4\alpha \tanh(\alpha L/4)$. For its compact completions, flatten the endpoint derivatives in collars of width ϵ, at cost $O(\epsilon)$, and join the endpoint h-values to zero over length D, at cost $O(D^{-1})$. A disjoint bump of scale D corrects the displacement integral with bounded amplitude and cost $O(D^{-1})$. Taking $\epsilon \to 0$ and then $D \to \infty$ reaches the infimum. Zero exterior cost would force $h' = 0$ outside I, incompatible with its endpoint derivatives. □

This is a preparation-class bound, not a minimum energy for all states or complete devices. Its asymptotic slope is the thermal current. Local negative stress outside I is included; counting only the positive plateau gives a different quantity. For $L = 8$ and $T = 1/(2\pi), .31894305$, explicit smooth completions cost $.11257242, .60868371$, above the respective infima $.02748002, .15992956$. Increasing support or sharpening a boundary can approach the infimum without keeping the drive or rms resources uniformly bounded.

For a regular increasing clock $\eta(v)$ on a finite interval, complete $\rho(v) = \exp[2\pi T \eta(v)]$ instead. The entire local covariance is the thermal η covariance pulled back by that clock, and

$$\langle T(v) \rangle_\rho = \eta'(v)^2 \frac{\pi c T^2}{12} - \frac{c}{24\pi} \{\eta, v\}. \tag{94}$$

Two independently prepared copies therefore supply the two-port input correlations on the operating interval, including the output cross covariance and switching stress. Their global purity and inside–outside correlations remain in the ledger. Thermal local correlations alone do not establish a finite-window erasure bound. The cost in Eq. (92) uses the actual physical coordinate; Eq. (93) cannot be applied to a rapidity interval without its clock conversion. The local diffeomorphism has a driven stress-unitary preparation path. Its autonomous bounded approximation is developed in Appendix I.1.

G.2 A record-selected source is not its average covariance

Suppose a retained classical program R has probabilities p_r, is jointly independent of the two prepared incoming streams, and selects θ_r. Conditional on r, the preceding contact law holds.

For a finite frequency regulator with independent input mode pairs, write $\rho_{r,N} = \bigotimes_{j=1}^{N} \rho_{r,j}$ for the reduced source and $\bar{\rho}_N = \sum_r p_r \rho_{r,N}$ for its unconditional state. The return stream and program are retained in the joint process.

Proposition 44 (The source carries the program's relations)**.** *For distinct frequencies j, k, with $\Delta n_j = n_o(\omega_j) - n_H(\omega_j)$,*

$$\mathrm{Cov}(n_j, n_k) = \mathrm{Var}_p(\chi_r)\, \Delta n_j \Delta n_k. \tag{95}$$

Furthermore,

$$S(\bar{\rho}_N) = \sum_r p_r S(\rho_{r,N}) + \mathcal{I}_N, \qquad 0 \le \mathcal{I}_N \le H(R). \tag{96}$$

Here $\mathcal{I}_N = I(R : \text{source})$ in the classical–quantum record state. The product state with the same one-mode occupations has entropy $\sum_j h(\sum_r p_r n_{r,j})$, generally larger. If $H(R) = o(\mathcal{T})$ on a growing observation window and the conditional stationary source entropy rates exist, the unconditional excess rate is $\bar{s} = \sum_r p_r s_{\chi_r}$. Its mean current obeys

$$\bar{q} \ge T_H \bar{s} + \frac{3\bar{s}^2}{\pi}.$$

This limit assumes the same program is retained across the window, not a fresh independently randomized choice at each mode or time.

Proof. Conditional occupations are independent and have means $n_{r,j} = n_H(\omega_j) + \chi_r \Delta n_j$. Conditional covariance therefore vanishes; covariance of their means gives Eq. (95). The entropy identity is the classical–quantum mutual information; its upper bound is the classical program entropy. Subadditivity gives the product-marginal upper bound. Divide the identity by the window length. Averaging Eq. (90) and using $\sum_r p_r s_r^2 \ge \bar{s}^2$ proves the current bound. □

A single persistent program bit can thus create correlations across an arbitrarily long source without contributing an extensive new entropy of its own. Replacing that state by an independent Gaussian state at its averaged covariance destroys those relations. If a new program is acquired at every interval, its entire retained sequence belongs in R; $H(R)$ can then be extensive. Angle switches have the separate two-port work of Eq. (91), and neither entropy identity makes those switches free. These distinctions apply to a gravitational source selected by internal records: mean stress alone does not specify the source state.

There is an exact finite autonomous realization of the static record-controlled exchange. For matched frequencies let $K_j = c_{H,j}^\dagger c_{o,j} + c_{o,j}^\dagger c_{H,j}$ and use

$$V = \sum_r |r\rangle\langle r| \otimes \prod_j e^{-i\theta_r K_j}, \qquad H_0 = \sum_j \omega_j (n_{H,j} + n_{o,j}).$$

It commutes with H_0. Put V on one edge of the positive history clock and identities on the remaining edges. The full Hamiltonian $H_0 + H_{\mathrm{hist}}$ is nonnegative, time independent and conserves H_0. Its endpoint evolution executes the exchange uniformly over all retained references. This is a finite-mode control realization; its clock initialization and finite thermal preparations are inputs. It does not implement the moving conformal contact's material stress or remove its continuum switching work.

Proposition 45 (Two reservoirs and the complete controller ledger)**.** *Let finite reservoirs start at Gibbs states τ_H, τ_o with inverse temperatures $\beta_H > \beta_o$, jointly independent of a complete zero-bare-gap controller M. Its internal correlations and retained references are included in M, but not purifiers of the reservoirs: those would violate the stated initial independence. Under any joint unitary conserving reservoir bare energy, put $Q = \Delta E_H = -\Delta E_o$, $D_i = D(\rho_i' \| \tau_i)$ and $I' = S(\rho_H') + S(\rho_o') + S(\rho_M') - S(\rho_{HoM}')$. Then*

$$(\beta_H - \beta_o)Q = D_H + D_o + I' - \Delta S_M. \tag{97}$$

If the complete controller entropy is unchanged, the right side is $D_H + D_o + I' \geq 0$. Returning its marginal does not mean eliminating its output correlations.

Proof. The Gibbs identity gives $D_i = \beta_i \Delta E_i - \Delta S_i$. Initial independence and unitary invariance of total entropy give $\Delta S_H + \Delta S_o + \Delta S_M = I'$. Add the Gibbs identities and use energy conservation. This is the multi-reservoir form of the relative-entropy ledger underlying the sharpened Landauer principle [16, Theorem 3]. □

At frequencies $1, 2$, temperatures $.7, 1.4$, program probability .37 and angles $.2, 1.1$, direct joint evolution transfers .13174972 energy into the colder source. Its cross-frequency covariance is .00249817. Source entropy is .86881190, conditional entropy .84854079, and the independent state at averaged occupations has entropy .86900230. A five-state clock has mean energy .12 and interaction cutoff .24; its endpoint matrix and the thermal ledger agree with direct evolution. Both reservoirs have total bare cutoff 6 in these units. None of these cutoffs describes an infinite KMS field.

An operational error also needs an energy qualification before it certifies a source. For $0 \leq H_E \leq E_{\mathrm{cut}} I$, $|\operatorname{tr} H_E(\rho - \sigma)| \leq E_{\mathrm{cut}} D(\rho, \sigma)$. Without a common energy constraint, states at trace distance ϵ can differ by fixed or unbounded energy: mix the vacuum with probability ϵ in occupation $N = 1/\epsilon$. A joint clock or gate-distance certificate alone therefore does not certify stress of an unbounded field. The continuum source and its support require energy-weighted control as well as retained records.

H A physical contact: reaction, source and recoil

A prescribed scattering angle charges its work but leaves its actuator outside the model. Give the angle θ a momentum and inertia $I > 0$, and couple it to a second coordinate y with inertia $J > 0$ and potential

$$V(\theta, y) = \frac{k}{2}(\theta - y)^2 + \frac{K}{2} y^2, \qquad k, K > 0. \tag{98}$$

The coordinates are noncompact; the potential pins them. In the σ_x basis the two-port contact has charges $q_1 = 1, q_2 = -1$. For a smooth, nonnegative profile f_a of integral one, use the current Hamiltonian

$$H = \frac{p_\theta^2}{2I} + \frac{p_y^2}{2J} + V(\theta, y) + \pi \sum_{i=1}^{2} \int : \left(j_i(x) + \frac{q_i \theta f_a(x)}{2\pi} \right)^2 : \, dx. \tag{99}$$

The currents obey $[j_i(x), j_l(x')] = -i\delta_{il}\delta'(x-x')/(2\pi)$. At oscillator cutoffs, normal ordering uses the positive completed-square Hamiltonian. Its θ^2 counterterm must not be dropped. The continuum identities below apply on a domain where the current and stress products and their regulated limits exist. The positive form constructions below supply both a neutral-current Hamiltonian and the contact on the full fermion carrier. A static current displacement can still have infinite particle number; its dressing and asymptotic state need separate representation control. An oscillator cutoff is not a certificate for the full two-temperature fermion channel. The anomaly-induced reaction is the mechanism of the fermion–rotor model [13, Sections 2.1–2.2]. Here it is applied to the pinned two-coordinate actuator and its stored energy.

Proposition 46 (Current reaction with finite contact width)**.** *Set* $J_i = j_i + q_i\theta f_a/(2\pi)$. *Hamiltonian* (99) *gives*

$$(\partial_t + \partial_x)J_i = \frac{q_i}{2\pi}\dot\theta f_a, \qquad I\ddot\theta + V_\theta = -\int f_a \sum_i q_i J_i\,dx, \qquad J\ddot y + V_y = 0. \tag{100}$$

With the free initial currents defined at $t = 0$*, elimination of the field gives*

$$I\ddot\theta + V_\theta + \int_0^t \Gamma_a(t-s)\dot\theta(s)\,ds = -\xi_a(t) - \Gamma_a(t)\theta(0), \tag{101}$$
$$\Gamma_a(s) = \frac{1}{\pi}\int f_a(x) f_a(x-s)\,dx,$$
$$\xi_a(t) = \int f_a(x)\sum_i q_i j_i(0, x-t)\,dx.$$

The integrated retarded coefficient is $\gamma = 1/(2\pi)$*. For a Gaussian profile of standard deviation* a*,*

$$\Gamma_a(s) = \frac{\gamma}{\sqrt{\pi}a}e^{-s^2/(4a^2)}, \qquad \operatorname{Re}\widetilde\Gamma_a(\Omega) = \gamma e^{-a^2\Omega^2}. \tag{102}$$

Proof. The current commutator gives the first equation in (100); canonical commutators give the other two. Solve the transport equation along its characteristics. The initial $q_i\theta(0)f_a/(2\pi)$ term gives the initial slip in (101), and the source integral gives its convolution. The autocorrelation of f_a is even and has integral one on the whole line, hence $\int_0^\infty \Gamma_a(s)\,ds = 1/(2\pi)$. Fourier transformation gives (102). □

Thus damping is derived from the current algebra, not appended to an externally specified pump. For a velocity history extended into the past with bounded derivative,

$$|(\Gamma_a * v)(t) - \gamma v(t)| \le \|v'\|_\infty \int_0^\infty s\Gamma_a(s)\,ds. \tag{103}$$

The last integral is $2\gamma a/\sqrt{\pi}$ for the Gaussian. If f_a is supported in $[-a, a]$, it is at most $2\gamma a$. This controls smooth histories, not the ultraviolet error of an entire quantum channel. The initial slip is also physical: setting the mean angle to zero does not set its operator fluctuations to zero.

For independent zero-charge thermal inputs with Fermi occupations n_H, n_o, Wick contraction in the original port basis gives the unsymmetrized incoming force spectrum

$$S_\xi(\Omega) = \frac{|\widehat{f}_a(\Omega)|^2}{2\pi} \int_{\mathbb{R}} [n_H(E)(1 - n_o(E+\Omega)) + n_o(E)(1 - n_H(E+\Omega))]\, dE. \tag{104}$$

Changing variables in $S_\xi(-\Omega)$ yields

$$S_\xi(\Omega) - S_\xi(-\Omega) = 2\gamma\Omega|\widehat{f}_a(\Omega)|^2. \tag{105}$$

Indeed each occupation integrates $n(E) - n(E+\Omega)$ to Ω over the filled sea. The retarded response is temperature independent; its noise is not. Only at equal temperature T does the spectrum reduce to $\Omega|\widehat{f}_a|^2/[\pi(1 - e^{-\Omega/T})]$ and obey the single KMS fluctuation–dissipation relation. Different-temperature inputs are generally correlated in the charge basis and cannot be replaced by two independent thermal boson baths.

In a point-contact limit with well-defined renormalized products and the symmetric contact prescription, the total field power and mechanical energy satisfy

$$P_{\text{field}} = \tfrac{1}{2}\{\dot\theta, \xi\} + \gamma\dot\theta^2, \qquad \dot E_{\text{mech}} = -P_{\text{field}}. \tag{106}$$

This follows by substituting the incoming–outgoing current jump into the Sugawara stress and using the canonical mechanical equation. At finite width the full Hamiltonian supplies the energy identity, including retained current energy. The positive prescribed-angle pump term alone is not the power of a quantum actuator. Its velocity correlates with the incoming current. In a stationary interacting Gibbs state $\langle\dot\theta^2\rangle$ can be positive while the net energy transfer is zero.

Proposition 47 (A mechanical record changes the actual field source)**.** *Take a common initial state with zero first moments of the two mechanical coordinates, their momenta and the currents, and finite stress expectations as above. In branch R, displace y by A_R and leave the other first moments unchanged. Relative to the undisplaced evolution under the same Hamiltonian, let $v_R(t)$ be the deterministic mean angle velocity and define*

$$h_R(t,x) = \int_0^t v_R(s) f_a(x - t + s)\, ds, \qquad \Delta\langle T_{\text{field}}(t,x)\rangle = \gamma h_R(t,x)^2. \tag{107}$$

Then

$$\Delta E_{\text{mech}}(t) + \gamma \int h_R(t,x)^2\, dx = \frac{k+K}{2} A_R^2. \tag{108}$$

The actual excess stress scales as A_R^2 for fixed hardware.

Proof. The Heisenberg equations are linear. Displacing the initial y adds the classical solution to the undisplaced operator solution; it leaves every centered correlation unchanged. Characteristics give $\Delta\langle J_i\rangle = q_i h_R/(2\pi)$. The zero first moments remove the cross term in the stress difference, giving (107). Differentiating its spatial integral gives $v_R(\Gamma_a * v_R)$, the negative derivative of the mechanical displacement energy. The initial displacement cost is $(k+K)A_R^2/2$. □

This cancellation compares displaced and undisplaced evolutions; it does not discard the quantum noise term in (106). The common baseline stress still has to be counted. Energy retained inside the contact profile is not yet energy that has reached an external outlet. An acquired record can select A_R through a paid preparation. That preparation, its storage and its physical copying are additional operations, not free changes of a target potential.

For $I = 10, J = 20, k = 1, K = .1, a = .2$, a 192-mode positive oscillator regulator initially displaced by $A_R = 1.3$ has excess energy .9295. At $t = 80$, .5103950 is in its shifted field coordinates and .4191050 remains mechanical. Direct time-domain integration of the continuum memory equation and separate spatial integration give .5103953 and .4191052, with total energy residual 4.8×10^{-7} at step .005. This is a check of the continuum energy identity, not a cutoff channel certificate. The Gaussian profile has spatial tails. In the regulator's displaced interacting Gibbs state at $T = .02$, $t = 8$, the two branches $A_R = .5, 1.3$ give $\langle \sin^2 \theta \rangle = .426644, .716636$. These are coordinate observables, not transmission probabilities for the two-temperature fermion problem. A finite regulated bath eventually recurs.

H.1 Installing a program on the running carrier

Proposition 48 (Autonomous installation on an always-running carrier)**.** *Let $0 \le H_{\mathrm{phys}} \le E_{\mathrm{cut}} I$ be a finite actuator–field regulator, and let a prefix of r gates prepare its mechanical state from an already acquired record. Use a positive history clock with $m \ge r$, strength J, transfer endpoint $T = \pi/(2J)$ and identity edges after the prefix. The common, always-on Hamiltonian*

$$H = H_{\mathrm{hist}} + I_{\mathrm{clock}} \otimes H_{\mathrm{phys}} \ge 0 \tag{109}$$

has, for $t_0 \le t \le T$, joint distance from the completed-prefix state freely continued under H_{phys} for $t - t_0$ at most

$$d_{\mathrm{joint}} \le \min\{1, E_{\mathrm{cut}} t_0 + 2\sqrt{P_0} + 4A_r(t - t_0)\sqrt{P_0}\}, \tag{110}$$

where $P_0 = \Pr\{\mathrm{Bin}(m, \sin^2 Jt_0) \le r\}$ and $A_r = J \sum_{j=1}^{r} \sqrt{j(m - j + 1)}$. All prior references are retained. For each finite continuation window and specified error, finite sufficiently large clock resources attain that error.

Proof. During the prefix, Duhamel comparison with H_{hist} gives $E_{\mathrm{cut}} t_0$. Completing the prefix on its clock tail costs $2\sqrt{P_0}$. Thereafter compare with the identity-edge clock and the same H_{phys}. Their reference state factors across clock and data, so the nonidentity edges have norm at most $4A_r\sqrt{P_0}$; Duhamel gives the last term. The binomial lower tail decreases up to T. Set $\sin^2 Jt_0 = \mu/m$ with μ a sufficiently large multiple of $\log m$. Then $t_0 \to 0$ and the tail beats the polynomial growth of A_r. Choose T to contain the specified continuation window. □

For a bounded physical source observable $0 \le B \le B_{\mathrm{cut}} I$, the expectation error is at most $B_{\mathrm{cut}} d_{\mathrm{joint}}$. Thus a finite-cutoff record can install a mechanical program while the same carrier and field continue to evolve; no external switch at t_0 is required. A three-level actuator and three-level field, with the old record reference retained, give joint distance .00204248 below

the bound .04082928 in one small-clock check. The clock mean energy is 960, so that check is not an installation inside the mass-10 gravitational body. Larger clocks improve the guarantee at larger energy and dimension. The continuum displacement, infrared sector, spatial transport and matching to a moving action need separate control.

The upper energy cutoff is unnecessary for a class of permitted gates. The relevant condition concerns how a gate changes energy, rather than the spectrum of the carrier itself.

Proposition 49 (Energy-regular return on an unbounded carrier)**.** *Let $H_p \geq 0$ be self-adjoint, and let the finite prefix unitaries $W_j = V_j \cdots V_1$, $W_0 = I$, preserve* $\operatorname{Dom} H_p$*. Suppose*

$$B_j = W_j^* H_p W_j - H_p, \qquad b = \max_{j \leq r} \|B_j\|, \qquad c = \max_{j \leq r} \|[H_p, B_j]\|$$

extend to bounded operators, with B_j preserving $\operatorname{Dom} H_p$*. Let ψ be the normalized full input purification, including prior references, with $E = \|H_p \psi\| < \infty$. Use the same positive history clock as in Proposition 48. For $t_0 \leq t \leq T$, the joint vectors under $H_{\mathrm{hist}} + I \otimes H_p$ and the reference completed-prefix evolution differ by at most*

$$\varepsilon_m(t) = (E + b)t_0 + 2\sqrt{P_0} + 4A_r(t - t_0)\sqrt{P_0}. \tag{111}$$

Their physical rms energies obey

$$\|H_p \Psi_a(t)\| \leq M_a = E + tc + b, \qquad \|H_p \Psi_i(t)\| \leq M_i = E + b. \tag{112}$$

These bounds do not grow with the padded clock. If a self-adjoint source observable Q has $\operatorname{Dom} H_p \subseteq \operatorname{Dom} Q$ *and $\|Q\zeta\| \leq a\|H_p\zeta\| + d\|\zeta\|$, then*

$$|\langle Q \rangle_a - \langle Q \rangle_i| \leq \varepsilon_m(t)\,[a(M_a + M_i) + 2d]. \tag{113}$$

In particular, this certifies the unbounded physical energy $Q = H_p$. For any fixed finite continuation window, $\varepsilon_m \to 0$ with sufficient clock padding.

Proof. The ideal history branches are orthogonal in the clock, and $\|H_p W_j \psi\| = \|(H_p + B_j)\psi\| \leq E + b$. The early Duhamel comparison therefore costs $(E + b)t_0$; the prefix tail and late comparison are as in Proposition 48. This gives a vector bound on the retained purification, not only a distance between reduced states.

Put $G = \sum_j |j\rangle\langle j| \otimes W_{\min(j,r)}$ and write the ungated clock as K_m. In the history gauge the total Hamiltonian is

$$G^* H_{\mathrm{tot}} G = K_m \otimes I + I \otimes H_p + \sum_j |j\rangle\langle j| \otimes B_{\min(j,r)}.$$

The last term and its commutator with H_p are bounded by b, c. The propagator commutator consequently has norm at most tc, so the gauge-state rms energy is at most $E + tc$. Transforming back adds at most b. The same domain-preserving commutator argument applies to the reference. Finally write the expectation difference as $\langle \Psi_a - \Psi_i, Q\Psi_a \rangle + \langle Q\Psi_i, \Psi_a - \Psi_i \rangle$ and apply Cauchy–Schwarz. The large-clock limit follows from the same binomial tail choice. □

A positive quadratic-form source needs a weaker estimate. If $0 \le Q \le aH_p + dI$ as forms, Cauchy–Schwarz in the form norm gives

$$|\Delta\langle Q\rangle| \le (\sqrt{aM_a + d} + \sqrt{aM_i + d})\sqrt{a\varepsilon_m(M_a + M_i) + d\varepsilon_m^2}. \tag{114}$$

Here $\|H_p^{1/2}(\Psi_a - \Psi_i)\|^2 \le \|\Psi_a - \Psi_i\|\,\|H_p(\Psi_a - \Psi_i)\|$ proves the bound. A normally ordered local stress is not automatically a positive form or H_p-bounded; its relative estimate must be supplied.

Proposition 50 (A smooth chiral stress certificate)**.** *Consider a positive-energy chiral CFT satisfying the Fewster–Hollands hypotheses, with $H_f = \int T(x)\,dx$ and central charge c_{ch}. Let $0 \le f \in C_c^\infty$ and $\Lambda_f > \|f\|_\infty$. Define*

$$\begin{aligned} q_f &= \frac{c_{\mathrm{ch}}}{12\pi}\int |(\sqrt{f})'|^2\,dx, \\ q_{\mathrm{comp}} &= \frac{c_{\mathrm{ch}}}{48\pi}\int \frac{f'(x)^2}{\Lambda_f - f(x)}\,dx, \qquad d_f = q_f + q_{\mathrm{comp}}. \end{aligned} \tag{115}$$

Then, as quadratic forms,

$$0 \le T(f) + q_f I \le \Lambda_f H_f + d_f I. \tag{116}$$

If $H_f \le a_0 H_p + b_0 I$ as forms for the physical Hamiltonian of Proposition 49, its expectation certificate (114) *applies to the actual local stress $T(f)$ with $a = \Lambda_f a_0$ and $d = d_f + \Lambda_f b_0$. The constant shift cancels between the two states.*

Proof. The chiral quantum energy inequality gives $T(f) \ge -q_f I$ [12, Theorem 4.1]. To obtain the upper bound, one cannot apply that theorem directly to the non-Schwartz function $\Lambda_f - f$. Choose $k \in C_c^\infty$, $0 \le k \le 1$, equal to one on $[-1, 1]$, and set $\chi_L(x) = k(x/L)^2$. For L enclosing $\operatorname{supp} f$, the admissible test $g_L = \chi_L(\Lambda_f - f)$ has QEI constant

$$q(g_L) = q_{\mathrm{comp}} + \frac{c_{\mathrm{ch}}\Lambda_f}{12\pi L}\int |k'|^2.$$

Its two derivative supports are disjoint. On the common CFT core, $T(\chi_L)$ tends in expectation to H_f, using the integrable stress and the translation charge. Thus $T(g_L) = \Lambda_f T(\chi_L) - T(f) \ge -q(g_L)I$ gives the upper bound. The positive form then extends continuously to $\operatorname{Dom} H_f^{1/2}$. Combining it with the stated physical form inequality and (114) proves the certificate. □

For the inertial contact there is a useful conditional transfer of this field bound to the interacting carrier. In its positive completed-square regulator, write the shifted field energy as $\sum_l \omega_l(a_l + s_l\theta)^*(a_l + s_l\theta)$ and put $d_a = \sum_l \omega_l|s_l|^2 = k_{\mathrm{cur}}\int f_a^2/(4\pi)$, where $k_{\mathrm{cur}} = 2$. Since $V \ge kK\theta^2/[2(k + K)]$, the elementary square inequality gives

$$H_f \le 2H_{\mathrm{shifted}} + 2d_a\theta^2 \le \left[2 + \frac{4d_a(k + K)}{kK}\right] H_p. \tag{117}$$

In the neutral current Fock representation, that positive Hamiltonian can be constructed without a static dressing unitary. Set $g_i(\omega) = q_i\sqrt{\omega}\,\widehat{f_a}(\omega)/(2\pi)$, so $\sum_i \int |g_i|^2/\omega = d_a$. On the form domain of $H_0 = H_{\rm mech} + H_f$, define

$$\mathfrak{q}_H[\psi] = \mathfrak{q}_{\rm mech}[\psi] + \sum_i \int_0^\infty \left\| \sqrt{\omega}\, a_i(\omega)\psi + \frac{g_i(\omega)}{\sqrt{\omega}}\theta\psi \right\|^2 d\omega. \tag{118}$$

This includes the counterterm. Put $A_a = 2+4d_a(k+K)/(kK)$. The same square inequalities give $\mathfrak{q}_0 \le (1+A_a)\mathfrak{q}_H$ and $\mathfrak{q}_H \le A_a\mathfrak{q}_0$. Thus the two form norms, with $\|\psi\|^2$ added, are equivalent. The closed positive form defines a self-adjoint Hamiltonian, and (117) holds for it. Coupling cutoffs converging in $L^2(d\omega/\omega)$ converge in the common form norm and hence in norm resolvent on this field space. This does not require $g_i/\omega \in L^2$: when $\widehat{f_a}(0) \neq 0$, the static dressing can fail to be a Fock unitary even though the Hamiltonian exists.

For a compact contact profile, a stress test outside its support reads the physical outgoing current stress itself. A Gaussian profile has tails and is not covered by that exact exterior assertion. The gates must be energy-regular for this interacting H_p; free-field regularity alone does not establish that condition when the gate overlaps the contact. The exterior case admits an exact transfer below. This neutral-current construction does not establish a dressed ground state or asymptotic charged representation. The full fermion carrier and its prepared thermal inputs can instead be treated directly. A uniform zero-width limit does not follow from the form inequality.

The contact on the full fermion carrier. The two complex fermions need not be replaced by two neutral current Fock spaces. Work in their full Dirac-sea CAR representation, with positive translation generator H_f. Smooth local gauge implementers and the current anomaly act on this whole space, including its charged sectors [45, Sections 3.2–3.3 and Appendices A–B]. In the σ_x basis write $J(f) = \sum_i q_i j_i(f)$, $\kappa_c = \sum_i q_i^2 = 2$ and $d_f = \kappa_c\|f\|_2^2/(4\pi)$.

Proposition 51 (A positive contact on the full CAR carrier)**.** *For real $f \in C_c^\infty(\mathbb{R})$ and the pinned mechanical potential* (98)*, the form*

$$\mathfrak{q}_p = \mathfrak{q}_{\rm mech} + \mathfrak{q}_f + \langle\theta J(f)\rangle + d_f\langle\theta^2\rangle$$

is closed and nonnegative on the form domain of $H_0 = H_{\rm mech} + H_f$. It defines the single-contact self-adjoint H_p on the full fermion space. Put $a_0 = kK/[2(k+K)]$ and $L = \max\{2, 1 + 2d_f/a_0\}$. Then $L^{-1}\mathfrak{q}_0 \le \mathfrak{q}_p \le L\mathfrak{q}_0$. The assertion does not require a static dressing unitary.

For $b \in C_c^\infty((1,2))$ of integral one, set $b_D(x) = D^{-1}b(x/D)$ and $f_D = f - (\int f)b_D$, with disjoint supports. The corresponding compact-gauge contacts converge in norm resolvent to H_p. If $d_b = (\int f)^2\kappa_c\|b\|_2^2/(4\pi D)$ and $r_D = \sqrt{d_b/a_0} + d_b/a_0$, then, for $Lr_D < 1$,

$$\|(H_{p,D}+1)^{-1} - (H_p+1)^{-1}\| \le \frac{Lr_D}{1-Lr_D}.$$

Proof. First let $f = F'$ with $F \in C_c^\infty$. Multiplication by $e^{\pm i\lambda F}$ has a Hilbert–Schmidt commutator with the one-particle Hardy projection: its squared norm is $\iint |g(x)-g(y)|^2/(x-$

$y)^2\,dxdy/(4\pi^2)$, finite by smoothness and compact support of $g-1$. Thus $W_\lambda = e^{-i\lambda J(F)}$ is implemented on the full CAR space. On the common smooth core, $[J(F), H_f] = iJ(F')$ and $[J(F), J(F')] = i\kappa_c \int F'^2/(2\pi)$ give

$$W_\lambda H_f W_\lambda^* = H_f + \lambda J(f) + d_f\lambda^2 \geq 0.$$

For general f, use f_D above. On a finite smooth particle–hole core, $J(b_D) \to 0$ strongly: its vacuum norm is $O(D^{-1})$, and commuting it past each fermion changes the test function by multiplication with b_D, of supremum norm $O(D^{-1})$. Meanwhile $d_{f_D} \to d_f$. Passing to the limit and minimizing over λ gives the form inequality

$$|\langle\psi, J(f)\psi\rangle| \leq 2\sqrt{d_f}\,\|\psi\|\,\|H_f^{1/2}\psi\|.$$

Polarization extends the form to $\operatorname{Dom} H_f^{1/2}$. For each mechanical coordinate the shifted field form $A_f(\theta) = H_f + \theta J(f) + d_f\theta^2$ is nonnegative and obeys $A_f \leq 2H_f + 2d_f\theta^2$ and $H_f \leq 2A_f + 2d_f\theta^2$. Integrate over the mechanical coordinates and use $V \geq a_0\theta^2$. The displayed equivalent form norms establish closedness and positivity.

For the compensator, the same current bound and Cauchy–Schwarz give

$$|\langle\theta J((\int f)b_D)\rangle| \leq \sqrt{d_b/a_0}\,\mathfrak{q}_0.$$

The counterterm difference is $d_b\langle\theta^2\rangle$. Hence

$$|\mathfrak{q}_{p,D} - \mathfrak{q}_p| \leq r_D\mathfrak{q}_0 \leq Lr_D\mathfrak{q}_p.$$

The resolvent identity in the $(H_p+1)^{1/2}$ form scale gives the stated estimate. □

The finite-rms preparations of Proposition 42 now belong to the same fermion carrier as the inertial contact. Their outside purifications remain present. A constant flavor rotation preserves the vacuum but generally gives off-diagonal covariance for unequal temperatures; the rotated currents are not two independent KMS baths. The force law therefore uses the original quasifree input correlations, not a substitute product of bosonic thermal states. Finite form energy does not by itself supply the graph regularity required by the autonomous return theorem. On the stated smooth input core, exterior even local gates have the same free and interacting commutators by CAR locality, and the local-source certificate uses $H_f \leq LH_p$.

The compensator is a proof and approximation device, not a free heat outlet. For $\int f = 1$, $\operatorname{supp} f \subset [-a, a]$ and $0 \leq s \leq T < D - a$, its retarded kernel differs from the single contact's by at most $2d_b$. For every fixed D, however, $\int f_D = 0$ makes the integrated kernel vanish. Its dc response is zero, whereas the single contact's is $1/(2\pi)$. Norm-resolvent convergence is consequently not a uniform infinite-time heat-transport statement. The single-contact Hamiltonian itself only couples in $\operatorname{supp} f$; it does not require a physical compensator extending to infinity.

These estimates certify smooth, finite-window source charges. To use a sharp or unbounded gravitational kernel, one must prove the relevant source support or tail control.

An accurate field gate now has a route to an accurate local source; the source state, its correlations and its gravitational realization still have their own inputs.

The gate class is nonempty inside an accessible algebra. Let $A = \int h(s) e^{isH_p} C e^{-isH_p} ds$ for a bounded self-adjoint C and a real $h \in C_c^\infty$. Its first two commutator norms are bounded by $k_1 = \|h'\|_1 \|C\|$ and $k_2 = \|h''\|_1 \|C\|$. For $V = e^{i\lambda A}$, the energy change $B = V^* H_p V - H_p$ satisfies

$$\|B\| \le |\lambda| k_1, \qquad \|[H_p, B]\| \le |\lambda| k_2 + \lambda^2 k_1^2.$$

Both follow by differentiating $B = \int_0^\lambda e^{-isA} i[H_p, A] e^{isA} ds$. Products retain the bounded-commutator property. Consciousness II's bounded source preparation and shared-pulse result supply this smoothed gate class in a selected shared algebra for its stated translation generator and vacuum [15]. The exterior case below transfers it to the interacting current Hamiltonian. Record-controlled versions act in the record algebra tensored with that selected algebra. Thus an acquired record can install a permitted source without truncating the whole physical spectrum. An exact Weyl displacement need not satisfy the bounded energy-change condition; it is not covered by this assertion. Nor does an infinite KMS field have the finite global rms energy required here.

Proposition 52 (An exterior source gate for the interacting contact)**.** *Use the positive current Hamiltonian* (118) *or the full fermion contact of Proposition 51 with compact profile f_a. Let C be bounded, self-adjoint and localized in a bounded current interval I. Assume it and the free translates $\alpha_s^f(C)$ for $s \in \operatorname{supp} h$ belong to the selected accessible algebra, where $h \in C_c^\infty$ is real. Suppose their localization intervals are disjoint from* $\operatorname{supp} f_a$*. For $A = \int h(s) \alpha_s^f(C) ds$ and $V = e^{i\lambda A}$,*

$$[H_p, A] = [H_f, A], \qquad [H_p, [H_p, A]] = [H_f, [H_f, A]].$$

Both commutators are bounded, with the k_1, k_2 estimates above. The gate is therefore energy-regular for H_p, not only for H_f. Its bounded energy change is the same for both Hamiltonians. Record-controlled versions also meet the autonomous return conditions.

Proof. Write the interaction form as $H_p - H_f = H_{\mathrm{mech}} + \theta \sum_i q_i j_i(f_a) + d_a \theta^2$. Disjointly supported current smearings have zero symplectic pairing; their Weyl algebras commute. Thus A, and the first two derivative smearings defining its free commutators, strongly commute with every $j_i(f_a)$ and act trivially on the mechanical coordinates. The displayed equalities first hold as polarized forms on the common form domain. Free translation smoothing gives bounded commutators; these identities extend through the closed form to $\operatorname{Dom} H_p$ and imply its invariance under V. Integrating the first identity along e^{isA} gives $V^* H_p V - H_p = V^* H_f V - H_f$. The second identity gives the required bounded commutator of this energy change. A finite record control is a block sum of these same identities with a zero-gap record register. □

This class contains a genuine source preparation. Choose a non-scalar localized C in a common accessible algebra, with the local vacuum separating, and a narrow nonnegative h of integral one whose translates stay exterior. Consciousness II's bounded-source proposition then gives $V\Omega$ finite, strictly positive free-field energy for $0 < |\lambda| \|A\| < \pi$. On a finite-rms mechanical state tensored with the current vacuum, that energy is also the change of H_p:

the gate commutes with the contact interaction. With the usual local stress domain and integrability assumptions, the source perturbation vanishes outside the translated interval, and a smooth test equal to one there reads its full positive energy. The autonomous return and QEI bounds certify this preparation on the actual carrier while retaining the acquired record and prior references. A non-scalar shared algebra by itself need not contain such an exterior localized observable; the localization and permission assumptions are part of this construction. It supplies a finite source, not a thermal reservoir, a covariant local history clock or a rule that forms its own permissions.

For an explicit return, take $H_p = N$ on an infinite oscillator and retain three degenerate bits: a held source bit, its acquired record, and an output meter. A controlled NOT first acquires the source bit; that record then rotates oscillator levels $0, 1$ by .2 or .9; a final controlled NOT reads level 1 into the output meter. All higher oscillator levels form a reducing sector. This is an exact unbounded model, not a cutoff approximation. With initial source probabilities $.4, .6$ and its old purification retained, a 513-state clock of mean energy 512 gives vector error .118943 below .199530, and source energy .3839746172 against the reference .3839484296. The energy difference is 2.62×10^{-5}. The actual output-record probability is .3839746172. Clock energy decreases by the source gain; total-energy residual is below 4×10^{-12} in this check. A separate two-level block calculation agrees with the full-register evolution to 2.2×10^{-15} for the smaller clocks. Spatial transport, thermal preparation and the moving support remain distinct from this acquisition–preparation–reading realization.

H.2 The contact position can recoil

The contact profile need not follow a trajectory specified from outside. Let Y be its quantum position, $P_Y = -i\partial_Y$, and $M > 0$ its mass. On the full fermion carrier set

$$U_f(a) = e^{-iaH_f}, \qquad J_Y = U_f(Y) J(f) U_f(Y)^*.$$

The mechanical angle and storage coordinate remain internal to the body. This field translation exists on the full CAR space; it is not the static infrared dressing of the current.

Proposition 53 (Finite-mass contact and momentum conservation)**.** *The form*

$$\mathfrak{q}_{\mathrm{mov}} = \frac{\|P_Y \Psi\|^2}{2M} + \mathfrak{q}_{\mathrm{mech}} + \mathfrak{q}_f + \langle \theta J_Y \rangle + d_f \langle \theta^2 \rangle$$

is closed and nonnegative on the form domain of $P_Y^2/(2M) + H_{\mathrm{mech}} + H_f$. Its self-adjoint Hamiltonian H_{mov} strongly commutes with $P_{\mathrm{tot}} = P_Y + H_f$. A finite anchor Z of mass M_0 can be added with $P_Z^2/(2M_0) + k_s(Y - Z)^2/2$, $k_s > 0$; the conserved momentum then includes P_Z.

Proof. For almost every Y, field translation preserves the current form bound and H_f of Proposition 51. Integration over Y gives the same equivalent form norms after the positive position kinetic energy is added. The relative-anchor form is positive and closed on its natural domain as well. Simultaneous body and field translation sends J_Y first to J_{Y-a} and then to J_Y. All remaining terms, including the relative-anchor potential, are invariant. Uniqueness of the closed form realization implies commutation of H_{mov} with this translation group and therefore strong commutation with its generator. □

Thus the entire total-momentum distribution is conserved. These Hamiltonians measure energy above the body and anchor rest energies; those constant costs must be added in an apparatus ledger. The translation law is unchanged. On a common differentiable core the reaction is

$$\dot{P}_Y = -k_s(Y-Z) - \theta\partial_Y J_Y, \qquad \dot{P}_Z = k_s(Y-Z), \qquad \dot{H}_f = \theta\partial_Y J_Y.$$

The field-energy change is balanced by the body and anchor. Its instantaneous sign is not a certificate of exported heat. A positive record-controlled internal spring can change this dynamics without breaking translation symmetry; its preparation and gate regularity remain separate requirements.

The direct-integral unitary $\mathcal{W}(Y) = e^{-iYH_f}$ gives $\mathcal{W}^* P_Y \mathcal{W} = P_Y - H_f$ and $\mathcal{W}^* J_Y \mathcal{W} = J(f)$. With no anchor, the fixed total-momentum fiber is

$$H_M(p) = H_p + \frac{(p-H_f)^2}{2M}. \tag{119}$$

This is the recoil term of a mobile-field canonical transformation, as in the polaron setting [49]; the current form here is the full-CAR contact above. For a finite anchor put $M_t = M + M_0$, $\mu = MM_0/M_t$ and $r = Y - Z$. Its kinetic terms in a fiber become

$$\frac{(p-H_f)^2}{2M_t} + \frac{[p_r - (M_0/M_t)H_f]^2}{2\mu}.$$

Dropping either recoil square changes the model.

For fixed p, Eq. (119) converges in strong resolvent to H_p as $M \to \infty$. Its positive forms decrease to $\mathfrak{q}_p$ on $\operatorname{Dom} H_p^{1/2} \cap \operatorname{Dom} H_f$. This intersection contains a core for $\mathfrak{q}_p$, by equivalence with the free mechanical/field form. Resolvent variational minimizers have the closed-form lower bound; approximating a target vector by that core gives a recovery sequence with recoil penalty tending to zero. Strict convexity gives norm convergence of the resolvent vectors. This is not an operator-norm smallness statement. For a static orbit with the indicated finite fourth energy moment, Duhamel gives

$$\|(e^{-itH_M(p)} - e^{-itH_p})\psi\| \le \frac{1}{2M}\int_0^t \|(p-H_f)^2 e^{-isH_p}\psi\|\, ds.$$

Finite mean field energy alone does not meet this orbit condition.

For the unbounded harmonic coordinates used here, there is a sharper limit on a prescribed constant-velocity replacement. Take $p = Mv$ and remove the free kinetic energy $Mv^2/2$. The fiber is $H_p - vH_f + H_f^2/(2M)$, not H_p. Put $v_* = a_0/(a_0 + d_f)$. For $v < v_*$, the form of $H_p - vH_f$ is closed and nonnegative. For $v_* < v < 1$, it is unbounded below on the smooth core. To see both directions, complete the current form with coefficient $1 - v$:

$$H_p - vH_f \;\ge\; H_{\text{mech,kin}} + [a_0 + d_f - d_f/(1-v)]\theta^2.$$

For the reverse statement use the compensated compact gauge of Proposition 51. Its coherent state has $\langle J(f)\rangle = 2d_f\lambda$ and $\langle H_f\rangle = (d_f + d_b)\lambda^2$. Minimizing over λ gives mechanical leading

coefficient $a_0+d_f-d_f^2/[(1-v)(d_f+d_b)]$. First choose finite D large enough to make it negative, then send the mechanical packet center to infinity, with $y = k\theta/(k+K)$. Fixed packet widths and the conditional gauge derivative have bounded kinetic cost independent of the center. The energy tends to minus infinity. At $v = v_*$ the core form is nonnegative, but the previous equivalent form norm is lost. The full finite-mass H_{mov} remains positive: its recoil square prevents treating body kinetic energy as an unlimited constant-velocity source.

Quantum position also changes the locality requirement for the next thermal preparation. A position wavepacket has tails; its classical center lying in an interval does not localize the whole protocol. Choose a smooth position cutoff χ whose support keeps the full contact profile in a bounded interval I. Let the past interval A contain all free translates of I over the operating time, together with the other field couplings in the interaction picture. Replacing its interaction by $\chi\theta J_Y + d_f\chi^2\theta^2$ defines a positive local comparison Hamiltonian H_A. On a common operator-domain trajectory put

$$\delta_{\mathrm{loc}} = \int_0^t \|(H_{\mathrm{mov}} - H_A)e^{-isH_{\mathrm{mov}}}\Psi\|\, ds.$$

The retained-reference vector error is at most δ_{loc}. The thermal comparison in Appendix I.2 with the *actual* record distribution then has trace error at most $2\delta_{\mathrm{loc}}+\sqrt{I_0/2}$: one locality error changes the joint state, the other changes its record marginal. For source second-moment norms M_a, M_l, M_{pl}, M_{pa} of the actual, local, local-product and actual-product outputs, respectively, its source error is at most

$$\delta_{\mathrm{loc}}(M_a + M_l + M_{pl} + M_{pa}) + \sqrt{I_0}(M_l + M_{pl}).$$

All interaction-picture field couplings of the comparison protocol must lie in A; position-tail probability alone does not bound the generator integral. The body has genuine quantum recoil, but its Schrödinger kinematics does not supply a relativistic causal support, a clock or a dilaton stress tensor.

H.3 Physical locality after unfolding

The preceding H_f translates two co-propagating unfolded fields. Writing physical fields as $\xi_R(x) = \psi_1(x)$ and $\xi_L(x) = \psi_2(-x)$ changes the bulk momentum to H_1-H_2. The finite-width flavor current before diagonalization becomes

$$\int f(x)[\xi_R^\dagger(x)\xi_L(-x) + \xi_L^\dagger(-x)\xi_R(x)]\, dx.$$

It is a mirror coupling, not a local bulk density. The CAR give $[J(f), \xi_L(y)] = -f(-y)\xi_R(-y)$, allowing an immediate response between different positions. Point contact avoids that spatial kernel, but a profile $f_a(x) = a^{-1}f(x/a)$ at fixed integral has $d_{f_a} = d_f/a$. The existing energy bounds are not uniform in that limit. Neither folding nor shrinking supplies a finite local material body for free.

A local current alternative uses two flavors in each physical direction, with charges $+1, -1$ after a within-direction flavor rotation. Let $H_R, H_L \geq 0$, $H = H_R + H_L$, $P = H_R - H_L$, and $d_s = \kappa_s\|f\|_2^2/(4\pi)$, $\kappa_R = \kappa_L = 2$. Both currents $J_R(x), J_L(x)$ now act at the same physical x.

Proposition 54 (Bidirectional current and physical recoil)**.** *The form*

$$H_p^{\mathrm{bi}} = H_{\mathrm{mech}} + H + \theta[J_R(f) + J_L(f)] + d\theta^2, \qquad d = d_R + d_L,$$

is closed and nonnegative on the free form domain. Translating its current by e^{-iYP} and adding $P_Y^2/(2M)$ gives a nonnegative Hamiltonian conserving $P_Y + P$. Its fixed-momentum fiber is $H_p^{\mathrm{bi}} + (p-P)^2/(2M)$. For the harmonic internal coordinates above and equal $d_R = d_L = d/2$, $H_p^{\mathrm{bi}} - vP$ has a positive closed form for $|v| < \sqrt{a_0/(a_0+d)}$ and is unbounded below on the smooth core for $\sqrt{a_0/(a_0+d)} < |v| < 1$.

Proof. In physical coordinates the left current has the opposite signs in both the current commutator and its commutator with H_L. For $f = F'$, $G = J_R(F) - J_L(F)$ therefore satisfies

$$e^{-i\theta G} H e^{i\theta G} = H + \theta(J_R(f) + J_L(f)) + d\theta^2.$$

Compact compensation in each direction handles arbitrary compact f, as in Proposition 51. The same equivalent form norms and translation proof then apply, with P replacing H_f. Completing the two current forms separately gives

$$H_p^{\mathrm{bi}} - vP \geq H_{\mathrm{mech,kin}} + \left[a_0 + d - \frac{d_R}{1-v} - \frac{d_L}{1+v}\right]\theta^2.$$

The bracket is $a_0 - dv^2/(1-v^2)$ in the balanced case. For necessity, use independent compact compensated coherent states in the two directions. Their minimized leading coefficient is $a_0 + d - d_R^2/[(1-v)(d_R + d_{b,R})] - d_L^2/[(1+v)(d_L + d_{b,L})]$. It is negative for fixed sufficiently spread compensators above the stated threshold. Harmonic packet centers tending to infinity then give the negative core sequence, with bounded kinetic cost. □

At equality only core nonnegativity is asserted. The finite-mass Hamiltonian remains positive with its recoil retained. This alternative has four complex chiral fields, $c_R = c_L = 2$; each actual outlet must count the fields routed through it. Four input preparations, their purifiers and the additional control permissions are real resources. For interval lengths a, b and separation $g > 0$ in the product free vacuum, Appendix I.2 uses $I_0 = \frac{2}{3}\log[1 + ab/(g(g + a + b))]$ for these four fields, twice the two-field value. Both the conditional preparation error and the source-moment certificate must use this larger I_0. The balanced free carrier and its physical recoil do not yet quantize a propagating material controller or its complete stress tensor.

I Local preparation and correlated-window renewal

I.1 A local preparation can be executed autonomously

The compact preparation unitary V of Proposition 42 is local, but its energy change need not be bounded. A preparation bit permits a bounded regularization that controls the field state in energy norm as well as Hilbert norm. Let H act trivially on that zero-gap bit and set

$$C = |1\rangle\langle 0| \otimes V + |0\rangle\langle 1| \otimes V^*, \qquad u_0 = |0\rangle\Omega, \quad u_1 = Cu_0.$$

Then $C = C^*$, $C^2 = I$ and $e^{-i\pi C/2}u_0 = -iu_1$. For a nonnegative even $h \in C_c^\infty(-1,1)$ of integral one, put $\mu_2 = \int s^2 h(s)ds$ and

$$C_\delta = \int h(s)e^{i\delta sH}Ce^{-i\delta sH}ds, \qquad W_\delta = e^{-i\tau C_\delta}, \quad \tau = \pi/2.$$

Proposition 55 (Local preparation in the energy graph norm)**.** *Suppose $F_l(t) = e^{itH}Ce^{-itH}u_l$, $l = 0, 1$, is twice continuously differentiable in the H graph norm for $|t| \le \delta_0$. Let $L_2 = \max_{l,t} \|F_l''(t)\|$ and $G_2 = \max_{l,t} \|HF_l''(t)\|$ there. For $0 < \delta \le \delta_0$ define*

$$e_\delta = \mu_2\delta^2 L_2/\sqrt{2}, \qquad g_\delta = \mu_2\delta^2 G_2/\sqrt{2}, \qquad k_1 = \|h'\|_1/\delta, \quad k_2 = \|h''\|_1/\delta^2.$$

The bounded local gate W_δ has energy-regular constants $b \le \tau k_1$ and $c \le \tau k_2 + \tau^2 k_1^2$. Moreover,

$$\begin{aligned} \|(W_\delta - e^{-i\tau C})u_0\| &\le \tau e_\delta, \\ \|H(W_\delta - e^{-i\tau C})u_0\| &\le \tau g_\delta + \tfrac{1}{2}\tau^2 k_1 e_\delta. \end{aligned} \tag{120}$$

For a compact diffeomorphism of the vacuum Virasoro module the stated graph condition holds for free translation $H = H_f$. For every finite source tolerance, a finite positive autonomous history clock can therefore execute this preparation while approximating its local thermal restriction and each specified smooth stress test.

Proof. Time smoothing gives $\|[H, C_\delta]\| \le k_1$ and $\|[H, [H, C_\delta]]\| \le k_2$, hence the gate constants by Proposition 49. Evenness of h cancels the first Taylor term. The second-order remainder bounds $(C_\delta - C)u_l$ by $\mu_2\delta^2 L_2/2$ and its H norm by $\mu_2\delta^2 G_2/2$. The ideal gate path lies in $\operatorname{span}\{u_0, u_1\}$ with coefficient sum at most $\sqrt{2}$. Duhamel gives the first estimate. For $\zeta \in \operatorname{Dom} H$,

$$\|He^{-itC_\delta}\zeta\| \le \|H\zeta\| + tk_1\|\zeta\|.$$

Inserting this in the same Duhamel integral gives the second estimate, which tends to zero as $O(\delta)$. The smooth diffeomorphism representation preserves $\bigcap_n \operatorname{Dom} L_0^n$, and its action is smooth there [12, Sections 5.1–5.2, Proposition 5.1]. Möbius translation generators act continuously on this domain; Ω, $V\Omega$ and the required orbit therefore have the stated graph differentiability. First make the preparation error small by choosing δ, then, for fixed δ, apply the autonomous return and smooth-stress certificates. The required clock is finite at each tolerance, although its resources need not stay bounded as that tolerance decreases. □

No global vacuum or energy projector enters C_δ. Its support is the preparation interval enlarged by the short translations. If this interval stays outside the pinned-current contact, the commutator equalities above make W_δ energy regular for the same interacting H_p. Graph convergence also needs its H_p-graph condition on the input; commutator equality alone does not give it. Schwartz mechanical vectors tensored with the smooth vacuum core supply an explicit class: the mechanical polynomials, smeared currents and twice-differentiated exterior orbit act continuously on that core. Overlapping intervals and more general mechanical states need separate domain control.

In the free-field execution, field-energy gain is exactly the loss of the clock-interaction energy $\langle H_{\text{hist}}\rangle$; their sum is conserved. The preparation bit, initial vacuum, allowed local

gate, clock coherence and waveform are supplied resources. Locality here is locality of a permitted continuum-algebra interaction, not a derived laboratory actuator or a covariant moving clock. A finite record can select among these gates while retaining its coherences and references. Online prediction of its changing thermal clock and physical renewal remain further requirements.

I.2 A retained record can select the next thermal window

A later interval of the same field is not an independent reservoir merely because it has not yet met the contact. The vacuum correlates disjoint regions; a record made in one can steer the conditional state in the other. Separation nevertheless gives a quantitative renewal condition without discarding those correlations.

Proposition 56 (Record-selected local thermal preparation)**.** *Consider c_f independent complex chiral fermions in their vacuum. Let A, B be split-separated intervals of lengths a, b and gap $d > 0$. The complete past field protocol and acquired record R are a normal local channel on A, with initial ancillas independent of the field. Here R denotes the commuting readout algebra of the retained register, not an extra physical dephasing. For each result r, let an even $V_r \in \mathcal{A}(B)$ prepare a specified thermal restriction τ_r on $J \subset B$ from the vacuum. Put*

$$I_0 = \frac{c_f}{6} \log \left(1 + \frac{ab}{d(d+a+b)} \right).$$

After record-controlled preparation,

$$\frac{1}{2} \left\| \rho'_{RJ} - \sum_r p_r |r\rangle\langle r| \otimes \tau_r \right\| \leq \sqrt{I_0/2}. \tag{121}$$

The norm is that of normal states on the local algebra. The bound is a probability-weighted conditional error, not a uniform bound for arbitrarily rare results.

Let Q_r be a smooth source test on J and $Q = \bigoplus_r V_r^* Q_r V_r$. On the split algebra set $\alpha = \rho_{RB}$, $\beta = \rho_R \otimes \omega_B$ and suppose $M_a^2 = \alpha(Q^2)$, $M_i^2 = \beta(Q^2)$ are finite. Then

$$|\alpha(Q) - \beta(Q)| \leq \sqrt{I_0}(M_a + M_i). \tag{122}$$

Neither assertion replaces the actual outside purification by a product state.

Proof. The past local channel leaves the B marginal equal to ω_B. Relative-entropy data processing gives $S(\alpha\|\beta) = I(R : B) \leq I_{\mathrm{vac}}(A : B)$. For free complex chiral fermions, the multi-interval vacuum entropy formula gives the displayed I_0 [46, Section 4.1, Eq. (63)]. For commuting even observable restrictions it supplies an upper bound by data processing; the graded fermion fields are not being replaced by bosonic tensor factors. Measuring an arbitrary binary effect and applying classical Pinsker gives $\|\alpha - \beta\|/2 \leq \sqrt{I_0/2}$. Controlled preparation and restriction are contractions, proving Eq. (121).

For the source bound use the natural-cone representatives ξ_α, ξ_β. Spectral Jensen for the relative modular operator gives [47, Section 2, Eqs. (2.1)–(2.6)]

$$S(\alpha\|\beta) \geq -2 \log \langle \xi_\alpha, \xi_\beta \rangle \geq \|\xi_\alpha - \xi_\beta\|^2.$$

Faithful regularization gives the same bound for finite-entropy normal states. Expanding the two expectations of Q and applying Cauchy–Schwarz gives Eq. (122). □

The estimate can also retain more than the classical record. If the past protocol outputs a larger controller algebra $\mathcal{K}$, the same data-processing argument gives $S(\rho_{\mathcal{K}B}\|\rho_{\mathcal{K}}\otimes\omega_B) \le I_0$. The controller may include its mechanics and references initially independent of the field. Any subsequent normal joint channel on $\mathcal{K}$ and B preserves the local trace bound. Its output source obeys Eq. (122) with the actual and comparison output moments. This also controls a later joint contact, provided its participating algebra is genuinely contained in this input. It does not include the entire field complement as an independent reference.

The source moments are explicit inputs. For the record-only estimate, if the past instrument has local effects $E_r \in \mathcal{A}(A)$ and $m_r^2 = \omega((V_r^* Q_r V_r)^2)$, then locality and $0 \le E_r \le I$ give $M_a^2 \le \sum_r m_r^2$ and $M_i^2 = \sum_r p_r m_r^2$. Smooth compact preparations and stress tests have finite m_r. A trace-distance bound alone would not control an unbounded source.

For a desired local trace error $\epsilon > 0$, a sufficient separation is

$$d \ge \frac{\sqrt{(a+b)^2 + 4ab/(e^{12\epsilon^2/c_f} - 1)} - (a+b)}{2}.$$

This is the gap required by the vacuum-information certificate, not a necessary distance for every instrument: an uninformative record can already be independent at a smaller gap.

For a specified finite sequence, use the actual hull of the past field operations at each step and a separated preparation support for the next window. The estimate applies to that actual past, without resetting it. The local gates can be smoothed and executed by the positive history clock under the graph-domain conditions above; its additional gate and source errors are charged separately. All prior registers and field purifications remain present. The gap consumes spatial reach and transport time. The record must reach the preparer through a permitted physical channel; a controlled unitary does not itself construct that transport. The growing supports, storage and clock therefore define a finite operating family, not a fixed-capacity perpetual reservoir. This local certificate also does not prove that the whole future field and every old quantum reference are jointly a product state. Nor does it alone certify the entropy rate of a correlated released field.

J Actual diagnostic return and terminal reading

J.1 The reading itself supplies the repair and contact program

The program need not be a label given to the apparatus. Use three zero-gap pointer bits for world, sensor and record frame, and three meter bits. Start the routine meter blank and the other two meters in independent diagonal states with bit probability .1. Three controlled NOTs copy body parity into the routine meter; two copy sensor and frame values into their respective noisy meters. All six bits and the actual held body remain in the core.

The acquired meter word o controls a fixed repair table $a(o)$. The table is the first two-step decision of Consciousness III's continuing-body model [14], with independent coordinate fault

probability .05. It depends on the acquired word and known model, not on the inaccessible true cause. On one history edge use the unitary

$$V = \sum_{o} |o\rangle\langle o| \otimes X_{a(o)} \otimes S_{\theta \mathbf{1}_{a(o)\neq 0}}, \tag{123}$$

where $X_0 = I$ and $X_{e_j} = X_j$. The exchange S acts on two matched-frequency fermion cells initially prepared at temperatures $.7, 1.4$, with $\omega = 1$ and $\theta = .7$. Thus the actual physical repair permission and actual energy transfer are performed by one program-controlled history edge. Whether the scorer later calls the repair correct has no effect on the interaction.

Put the five meter gates and this gate on the first six edges of an eight-edge clock, followed by two identities. Every gate commutes with reservoir bare energy. The joint positive conserved Hamiltonian has eight qubit registers, nine clock states and dimension 2304. At $J = .06$, clock mean energy is .48, interaction cutoff .96, and the total spectral cutoff is 2.96. Initial mean energy is 1.00197392. Direct evolution retaining 128 purification columns agrees with the full gate product to vector error 2.7×10^{-14}; bare energy conservation residual is 4.5×10^{-16} in this check. Actual restoration failure is .02891, repair probability .1355, and cold-port energy gain is .00761031. The latter equals $\Pr(a \neq 0)\sin^2\theta(n_o - n_H)$, not a probability of incorrect repair supplied as a source command. The contact angle is a hardware choice: setting it to zero leaves the logical repair intact. This construction connects actual repair to an actual source; it does not make that heat transfer a universal minimum cost of repairing a pointer. The complete six-bit controller entropy is unchanged. Reservoir relative entropies are .0001838782 and .0001317734, and joint output total correlation is .0051202854. They obey Eq. (97) with residual below 3×10^{-16}. Reservoir purification columns are retained for the vector certificate, not folded into its initially independent controller in the entropy ledger.

This is a first-interval conserved realization of acquisition returning to repair and a source. Initial meter purity, biased noise, reservoir preparation, reference labels and hardware are supplied and charged separately from the finite Hamiltonian's energy. The two cells store transferred energy; spatial release and a covariant supported contact are not implied by this finite calculation. Its value is that reading, decision, repair and source no longer need mutually incompatible clocks or an external selector within the demonstrated interval.

Proposition 57 (A finite continuing diagnostic return)**.** *For every specified horizon L, the rational pointer model with $h = 1/20$, $\epsilon = 1/10$ and the two-step decision admits one finite, time-independent, nonnegative Hamiltonian. Its completed circuit has the joint law of all L logical intervals: the same damaged and repaired body, all actual meter and action records, exact retained posterior weights, and one record-selected thermal exchange per interval. No inaccessible fault bit enters the decision circuit. With a fresh matched-frequency cell pair of gap ω per interval,*

$$Q_L = \omega \sin^2\theta(n_o - n_H) \sum_{t=1}^{L} \Pr(a_t \neq 0). \tag{124}$$

For a circuit with m padded history edges, clock mean energy is Jm, the total energy cutoff is $2L\omega + 2Jm$, and its exact completion time is $\pi/(2J)$ in units $\hbar = 1$.

Proof. Prepare $3L$ independent diagonal fault ancillas and let each control the corresponding held-body flip. Prepare $3L$ meter bits as above. The observer's reversible decision uses only the meter words, retained integer weights and known earlier actions. The rational update has denominator 800000 per interval; all weights and comparisons have finite integer size. A reversible Boolean circuit computes each update and action, keeping its inputs and garbage. The action controls the held-body flip and exchange with the next prepared thermal pair. Every gate commutes with the sum H_0 of cell energies, because the logic is degenerate and each exchange has matched gaps. Put the gates on a positive history clock. The prefix-unitary conjugation gives $H_0 + JmI + 2JJ_x$, proving positivity, bare-energy conservation and full-reference endpoint completion. Each unused pair is still jointly independent of the acquired program at its contact. Conditional energy transfer is zero or $\omega \sin^2 \theta(n_o - n_H)$, which proves Eq. (124). □

For the first three logical intervals, the expected number of repairs is .44635607 and $Q_3 = .02506944$ at the stated cell settings. There are nine readings, nine prepared fault bits and six thermal cells. Keeping all preceding weight vectors requires at most 984 bits in this case, before records, arithmetic workspace and reversible garbage. The general saved-weight cost grows quadratically with L. This construction removes the need for a new externally chosen circuit at every diagnostic return. Its logical intervals are not equally spaced proper-time samples of a continuously acting bath: only the completed joint law is asserted. All ancillas and cells are prepared before the finite run, all outputs remain in its ledger, and the clock can reverse after completion. It proves neither a fixed-capacity cyclic apparatus nor a covariant moving contact. These are distinct renewal and material implementation questions, not failures of the finite return.

J.2 An actual terminal record and probe replacement

Let probe P retain any old reference, record R start in $|0\rangle$, and a paid blank B start in $|+\rangle$. Write $C_Y = P_-^Y \otimes I_R + P_+^Y \otimes X_R$, then apply the true SWAP_{PB}. The record-one probability is $\mathrm{tr}(P_+^Y \rho_P)$. The final probe is $|+\rangle$, independent of all old references and the record; its old correlations have moved into B and R, not disappeared. Positive generators $gP_+^Y \otimes (I - X_R)$ and $g(I - \mathrm{SWAP}_{PB})$ implement these gates at duration $\pi/(2g)$. A fermionic exchange is not a substitute for this true SWAP.

The physical probe may continue to acquire a phase while the meter executes. Move the r nonidentity gates of the clock in Appendix E to its last r edges. All earlier edges are identities. Put $T = \pi/(2J)$, choose $t_0 < T$ and define

$$P_0 = \Pr\{\mathrm{Bin}(m, \sin^2 Jt_0) \geq m - r\}, \qquad A = J \sum_{j=m-r+1}^{m} \sqrt{j(m-j+1)}.$$

Proposition 58 (Terminal read under a retained phase coupling)**.** *Let the actual probe coupling be $\lambda d(t) Z_P$, with $|d(t)| \leq D$. Relative to writing the probe's actual phase at t_0 and replacing it at T, the full joint half trace distance obeys*

$$d_{\mathrm{joint}} \leq 2At_0\sqrt{P_0} + 2\sqrt{P_0} + \lambda D(T - t_0). \tag{125}$$

All old references, the blank, record and clock are included. For any specified finite trajectory and positive error tolerance, finite m and $t_0 < T$ suffice.

Proof. Before t_0, compare with the identity-edge clock and the actual free probe phase. Only the last r edges differ. Their action on that reference is bounded by $2A\sqrt{P_0}$, since the binomial upper tail increases up to T. Duhamel gives the first term. At t_0, dressing the reference with the gate prefixes changes its joint state by at most $2\sqrt{P_0}$. From there to T, omit only the bounded phase coupling, at cost $\lambda D(T-t_0)$; the compiled clock completes the meter and SWAP exactly. First choose t_0 near T, then increase m to suppress the tail. □

The positive clock has mean energy Jm, not zero. Add the probe/blank offset $2\epsilon_P I$, with $2\epsilon_P > \lambda D$, to keep the complete internal Hamiltonian positive. For $m = 10^5$, $r = 2$, $T = 80$, $t_0 = 78$ and $\lambda D = .00032$, a sufficient error is below .000641; clock mean energy is 1963.4954. This cannot be inserted into a mass-10 support without changing that support. A direct 17-state clock check gives joint distance .0034887 below its certificate .0046290. The construction renews the probe over a finite window. It consumes a coherent blank and a record slot and does not reset the entire reader or its clock. Its phase frame and allowed gate family remain inputs. HAFF D also solves a smaller, 129-state meter clock jointly with a source-matched mean moving support [23]. Its energy is included in the initial supported mass, and the actual record depends on the dilaton trajectory. That mean-field calculation does not include the contact reservoir's microscopic degrees or stress. For the specified C_Y, SWAP and coherent blank, an antiunitary symmetry makes the mean probe force vanish throughout the read, although the force operator and its fluctuations remain nonzero.

K Local linear-current control of stress flows

Consider the neutral current on a circle of length 2π, with

$$J_n = \sqrt{n}\, a_n, \quad J_{-n} = \sqrt{n}\, a_n^*, \quad J_0 = 0, \qquad H = L_0 = \sum_{n\geq 1} n a_n^* a_n.$$

Our Fourier convention is $f_n = (2\pi)^{-1}\int_0^{2\pi} f(\theta)e^{-in\theta}\,d\theta$. The same convention applies to the real current smearing h. The primitive interaction is $-Bj(h)$, where $j(h) = \sum_{n\geq 1}\sqrt{n}(h_{-n}a_n + h_n a_n^*)$ and B is a finite controller coefficient. Fix $h \in C_c^\infty(I,\mathbb{R})$ on a proper arc I, with $h_n \neq 0$ for $n \geq 1$. Such an h exists: in the Fréchet space of real smooth functions supported in a fixed compact subarc, each equation $h_n = 0$ defines a proper closed subspace, so the intersection of their open dense complements is nonempty. For a prescribed finite cutoff N, a nonnegative sufficiently narrow bump already gives $h_n \neq 0$ for $1 \leq n \leq N$; this alternative makes the probe depend on N. No uniform lower bound on $|h_n|$ is assumed.

Proposition 59 (A neutral-current stress compiler)**.** *Suppose signed local linear-current coupling, finite degenerate-helper rotations and record phase gates, calibrated clock preparation and finite readout are available, with coherent control intervals. For every real smooth f, $t \in \mathbb{R}$, $E < \infty$ and $\delta > 0$, a finite positive autonomous program implements $e^{itT(f)}$ within joint half-trace distance δ, uniformly over inputs with $\mathrm{tr}(H\rho) \leq E$ and arbitrary retained*

references. Its helpers and original clock are included in the comparison with their designated final states. Here

$$T(f) = \frac{1}{2} \sum_{p,q \in \mathbb{Z}} f_{-p-q} : J_p J_q :$$

is the $c = 1$ Sugawara stress tensor. The program uses no field reset, heralding or primitive quadratic coupling.

Proof. We first include the field's free propagation. For a finite quadrature $Q(v) = \sum_{n \le N} (v_n a_n^* + \overline{v_n} a_n)$ put

$$G_n = \frac{v_n}{\sqrt{n} h_n}, \qquad g_v(u) = \frac{1}{2\pi} \sum_{n \le N} (G_n e^{-inu} + \overline{G_n} e^{inu}), \quad 0 \le u \le 2\pi.$$

The Hamiltonian

$$H - s g_v(u) A j(h) + s^2 g_v(u)^2 C_h, \qquad C_h = \sum_{n \ge 1} |h_n|^2,$$

is positive by completion of the oscillator squares, with shifts $s g_v(u) A h_n / \sqrt{n}$ and $A = X$ or Y. In the interaction picture $[j_u(h), j_w(h)]$ is a scalar. Thus the Magnus series terminates after its scalar second term, and $e^{-2\pi i H} = 1$ gives the endpoint gate $e^{i\phi_s} e^{isAQ(v)}$. The phase ϕ_s is computed from the scalar commutator and the completion integral. Subsequent pulses start at integer periods and therefore act on the same $Q(v)$. Changing s to $-s$ pays for the inverse without reversing H. A Y pulse may instead use a paid Z rotation conjugating the X coupling. For coherent record branches, the known phases are removed by record phase gates.

The exact trigonometric comparator does not have smooth compact-time switching. Multiplying it by an all-order-flat endpoint taper χ_η makes the physical coupling smooth and gives $\|(1 - \chi_\eta) g_v\|_1 \le 2\eta \|g_v\|_\infty$. Writing $C_1 = \sum n |h_n|^2$ and $D_h = \sum n^2 |h_n|^2$, we have

$$\|j(h)\psi\| \le 2\sqrt{C_h} \|H^{1/2}\psi\| + \sqrt{C_1} \|\psi\|, \qquad \sqrt{\langle H \rangle_u} \le \sqrt{E} + \sqrt{D_h} |s| \|g_v\|_1.$$

The second bound follows directly from the controlled displacement. It includes the transient excitation of all modes, not just the selected endpoint modes. Duhamel's formula then makes each taper error arbitrarily small in joint vector norm, including old references. For a finite word of such pulses the energy bound is $(\sqrt{E} + \sqrt{D_h} L)^2$, where $L = \sum_{\text{slots}} |s| \|g_v\|_1$.

Next use the current-factor cutoff

$$T_N(f) = \frac{1}{2} \sum_{0 < |p|, |q| \le N} f_{-p-q} : J_p J_q : .$$

This is a selfadjoint finite oscillator quadratic. Finite occupation vectors are analytic for a finite quadratic, hence form an essential selfadjointness core. Let $P_M = 1_{[0,M]}(H)$, M an integer. It has finite rank. For $N > M$, the omitted number-preserving terms have annihilator index $m \le M$ and creator index $n > N$. Their norm on P_M is bounded by

$\sqrt{nM}|f_{n-m}|$. Omitted pair annihilators vanish; pair creators have norm at most $M+2$. Consequently

$$\|[T(f)-T_N(f)]P_M\| \le b_N(M) \longrightarrow 0,$$
$$b_N(M) = \sqrt{M}\sum_{m=1}^{M}\sum_{n>N}\sqrt{n}|f_{n-m}| + \frac{M+2}{4}\sum_{k>N}k^2|f_k|.$$

For the pair term we used $\sum_{n=1}^{k-1}\sqrt{n(k-n)} \le k^2/2$. These absolutely convergent sums identify the limit on the finite-energy core as the Sugawara $T(f)$. Carpi–Weiner's essential selfadjointness theorem for real smooth stress smearings [2, Theorem 4.4] supplies the self-adjoint closure of the limit; it does not substitute for the current-factor tail estimate. On this common core, the resolvent identity and density of $(T(f)-z)D_{\mathrm{fin}}$ give strong resolvent convergence. Hence $e^{itT_N(f)} \to e^{itT(f)}$ strongly, and

$$a_N(M,t) = \|[e^{itT_N(f)} - e^{itT(f)}]P_M\| \longrightarrow 0.$$

The uniform reference estimate is obtained only once, at the input of the whole program. For isometries V, W, purification and the P_M decomposition give

$$D_{\mathrm{joint}}(V\rho V^*, W\rho W^*) \le \|(V-W)P_M\| + 2\sqrt{\frac{E}{M+1}}.$$

It is independent of reference dimension. Thus the factor-cutoff unitaries converge uniformly over the stated mean-energy class. The core-tail bound is not itself a rate for $a_N(M,t)$, since a stress-evolved vector need not stay in P_M.

Write $r = (q_1, p_1, \dots, q_N, p_N)$. A real symmetric matrix K_N gives

$$T_N = \tfrac{1}{2}r^T K_N r - c_N, \quad c_N = \tfrac{f_0}{2}\sum_{n\le N} n, \quad T_N = \tfrac{1}{2}\sum_{\ell\le 2N}\kappa_\ell Q_\ell^2 - c_N$$

after real orthogonal diagonalization. This is an algebraic decomposition, not a free optical transformation: each Q_ℓ is selected directly by the local pulse above. Finite quadratic Trotter products converge on oscillator Schwartz vectors and hence strongly, including after restriction to P_M. Each square exponential is compiled with the same retained helper by four Pauli pulses. For an exponent of magnitude $|\alpha|$, n blocks and $s = \sqrt{|\alpha|/(2n)}$, the second-moment estimate is

$$D_{\mathrm{joint}} \le (13\sqrt{2})^{1/3}\sqrt{|\alpha| m_2}\, n^{-1/6}, \qquad m_2 = \mathrm{tr}(\rho Q_\ell^2) \le 4\,\mathrm{tr}(H\rho) + 1.$$

To see the moment bound, on a spectral value q set $C_\vartheta = e^{i\vartheta X}e^{i\vartheta Y}e^{-i\vartheta X}e^{-i\vartheta Y}$. Taylor's integral remainder gives $\|C_\vartheta - e^{-2i\vartheta^2 Z}\| \le 13|\vartheta|^3$: the third derivative of the four-factor product has norm at most 64, and the quadratic exponential contributes at most $2\vartheta^4$ for $|\vartheta| \le 1$; outside that interval the unitary distance is at most 2. Telescoping n blocks yields a spectral error $\min\{2, A|q|^3\}$, with $A = 13|\alpha|^{3/2}/(2\sqrt{2n})$. The inequality $\min\{4, A^2q^6\} \le 4^{2/3}A^{2/3}q^2$ and the spectral theorem give the displayed joint bound, retaining the helper and reference. The sign is chosen by the Pauli order; all inverse pulses are paid. Moments for

a finite word are evaluated on its ideal prefixes, whose oscillator moments are finite on P_M. Telescoping the complete word against those prefixes avoids both fresh-helper resets and a circular block-count estimate.

Finally approximate the smooth finite schedule in L^1 by constant slots with rational history-clock thresholds. Every slot has the positive field Hamiltonian $H - B_j j(h) + C_h B_j^2$. Install the finite rotations and record phase gates on the original clock path; these gates commute with the degenerate data bare energy. The resulting sum of the positive history clock, data energy and completed field squares is a positive autonomous Hamiltonian on the common field-energy domain. For $G = \max_j \|B_j\|$ and $D_3 = \sum n^3 |h_n|^2$, its field rms obeys

$$D^+ \|H\psi_u\| \le 2G\sqrt{D_h}\sqrt{\|H\psi_u\|} + G\sqrt{D_3}.$$

The bound is independent of clock size. The finite-boundary history-clock estimate therefore makes the complete endpoint error arbitrarily small on P_M, with the same original clock and all record branches retained. Choosing successively M, N, Trotter length, square-block counts, taper/slot precision and clock size allocates any prescribed total δ. □

The resources can grow sharply. With $\Gamma(v) = \|g_v\|_1 \le 2\sum_{n\le N} |v_n|/(\sqrt{n}|h_n|)$, a square exponent of size $|\alpha|$ uses field duration $8\pi n$, area $2\sqrt{2n|\alpha|}\Gamma(v)$ and positive completion integral

$$\frac{2C_h|\alpha|}{\pi} \sum_{n\le N} \frac{|v_n|^2}{n|h_n|^2}.$$

Helper rotations, phase gates and readout have their own finite budgets; rotation slots can last 2π so the field drift is identity. For an autonomous clock of duration T and size $m+1$, $J = \pi/(2T)$, its initial mean energy is mJ, norm $2mJ$ and largest edge coefficient at most $J(m+1)/2$. Preparation, drive bandwidth and retained storage are additional resources. These are sufficient finite bounds, not optimal costs or net-heat identities.

The uniform channel statement assumes only mean energy. Expectations of unbounded stress require separate graph-norm control: for smooth g, $\|T(g)\psi\| \le C_g\|(H+1)\psi\|$. Small joint channel error alone does not certify source convergence on the whole mean-energy class. The finite program's field moments also do not include a covariant stress tensor for clock and support.

The target preserves the Virasoro vacuum subspace, although the individual current pulses and finite T_N approximants can leave it; the endpoint leakage is part of the error above. The compiler concerns the neutral current sector. Its controls are not invariant under the $SO(3)$ action in the $SU(2)_1$ carrier, so an always-on selecting noise needs a separate control/noise certificate. The circle, calibrated port, coherent interval and controller preparation remain physical inputs.

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Memory as a Usable Resource: Correlations, Records and Complete Endpoint Accounting

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Abstract

A stored correlation has value only through an operation that can use it. We ask which correlations can fund a free-energy gain, who can read them and what reading, feedback and memory reset cost. The survival functional $\mathcal{S} = \Delta F - W$ obeys an exact balance retaining mutual information, environmental displacement from equilibrium, control energy and interaction energy. Three models distinguish their roles: Lorentz–Drude dephasing has no backflow at any temperature but a nonzero interaction energy; an autonomous two-qubit recurrence saturates the correlation budget without a pointer record; fresh-memory collisions create records available to several readers. A classical memory used for feedback is worth at most $k_B T I(X{:}M)$, with matched quasistatic feedback approaching the value. Every feedback protocol assumes a posterior and loses at least $k_B T$ times its expected divergence from the true posterior. A wrong readout model can therefore make a record worse than no memory. For a tail-controlled estimate from n references per sign, calibration has leading loss $k_B T(1+g^2/(1-b^2))/(4n)$ and pays through reuse. Within a reset menu with no added purity, positive read–harvest–reset work requires accessible correlated side information. Two partially mixed record ports realize this possibility: one drives matched feedback and the other reduces the reset cost through a reversible parity operation. These results establish the memory–self–introspection route's first functional link: relevant memory permits a gain, a model of one's readout realizes its full value, and calibration builds that model at a calculable cost. A separate finite-history classification bound proves that, for each fixed observation window, a longer finite episode can improve the attainable decision risk.

Keywords: open quantum systems, information thermodynamics, system–environment correlations, quantum records, feedback control, self-calibration, interaction energy, non-Markovian dynamics

1 Introduction

1.1 What memory can pay for

A system coupled to a thermal bath can lose the free energy needed for further action. An autonomous detailed-balance semigroup makes that loss precise; a structured environment can instead return correlations at a later time. Persistent structures, from molecular machines to organisms, maintain themselves by importing low-entropy energy and exporting entropy [2]. Within that general picture a narrower question can be answered exactly: when can correlations that a system has already built with its environment fund a later interval, and what does it take for an agent to turn a stored record into usable work?

Two errors are easy to make here. One is to treat any memory effect in the reduced dynamics as a resource. A memory kernel represents retained history; it does not guarantee that the history can be used, and the sign of the free-energy balance is fixed by the resources and the endpoint terms, not by Markovianity as such. The other is to treat information as free once it is stored. A record has value only for an agent that can read it, act on it with a correct model of its own reading, and dispose of the memory afterwards; each step has a price. The endpoint balance and the feedback task below specify which terms must be paid.

1.2 Position within the series

This paper is the first of the T-DOME (Thermodynamic Dynamics of Observer-Memory Entanglement) series, the third line of a programme whose other two lines are HAFF and Q-RAIF.

Line	**Question**		**Role here**
HAFF [25, 26]	How do stable, public records form?	Ocean	Supplies the collision model of records
Q-RAIF [27]	How does a subsystem read and act on its world?	Fish	Reading and control interfaces; costs of persistence
T-DOME I (this work)	What can stored correlations fund, and for whom?	Seed	Ledger, correlation types, value of memory and readout model
T-DOME II [28]	How can a bounded agent organize what it keeps?	Ego	Frame and feature selection under a budget
T-DOME III [29]	How does an agent keep its model of itself calibrated?	Loop	Detection, tracking and body-gated repair

The three T-DOME papers form a conditional sequence of models along what the programme calls the memory–self–introspection route: an agent keeps records of its world, models its own access to that world, and checks that model against references. The present paper supplies the first link and the thermodynamic currency for the other two. It shows

in one explicit task that a memory is informationally necessary for a work gain, that every protocol acts on an implied posterior and obtaining the full value requires that posterior to match the agent's readout, and that calibrating the model from references is an achievable mechanism whose cost and benefit can be computed. Paper II shows that under a reading budget a self-frame formed from records is informationally necessary for the task, prices it and identifies the blind spot it creates; Paper III develops drift detection, priced tracking and history-selected repair of body-gated access.

1.3 Results

- An exact accounting identity for the survival functional with every endpoint term retained (Theorem 8), a budget bound and a charge–discharge interval closure (Corollary 9, Proposition 10), and a restricted autonomous ceiling for a detailed-balance semigroup (Theorem 6).

- Three exact models with different roles (Section 5): no backflow at any temperature and a nonzero interaction-energy term in the Lorentz–Drude pure-dephasing model (Propositions 11 and 12); exact saturation of the correlation budget without any record in an autonomous two-qubit model (Proposition 13); the complete ledger of the collision model of records (Proposition 14).

- The work value of a record held in a classical memory and its information-theoretic ceiling (Theorem 16); the price of acting on a wrong posterior, which every protocol implicitly assumes (Theorem 17); the local metric of that price and the loss after calibration from references (Proposition 18), with the resulting amortization rule (Corollary 20) and a drift law for self-calibration, proved in Paper III (Proposition 21).

- A read–harvest–reset bound for a specified reset menu (Proposition 22), and a two-mixed-port realization with a constructive conditional reset (Proposition 23). The task restores the private memory and the retained record's marginal, not the complete record source.

Sections 8–10 distinguish temporal influence from kernel memory, price specified archives and compare retained history with fixed observation windows.

2 Thermodynamic setting

2.1 Open systems and memory kernels

Consider $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E$ with total Hamiltonian

$$H(t) = H_S(t) \otimes \mathbb{1}_E + \mathbb{1}_S \otimes H_E + H_{\mathrm{int}}(t), \qquad H_S(t) = H_R + H_{\mathrm{ctrl}}(t), \tag{1}$$

where H_R is the fixed bare system Hamiltonian, H_{ctrl} the control and H_{int} the coupling. Under the Born–Markov and secular approximations the reduced dynamics obeys a GKSL equation [3, 4]. For a fixed environmental reference σ_E, set $\mathcal{P}A = \mathrm{tr}_E(A) \otimes \sigma_E$, $\mathcal{Q} = \mathrm{id} - \mathcal{P}$

and $\mathcal{L}(t)A = -i[H(t), A]$. Eliminating the $\mathcal{Q}$ component gives the exact Nakajima–Zwanzig equation

$$\dot{\rho}_S(t) = -i[H_{\text{eff}}(t), \rho_S(t)] + \int_0^t ds\, \mathcal{K}(t,s)\, \rho_S(s) + \mathcal{I}_0(t), \tag{2}$$

where $H_{\text{eff}}(t) = H_S(t) + \text{tr}_E[H_{\text{int}}(t)(\mathbb{1}_S \otimes \sigma_E)]$ and

$$\mathcal{K}(t,s)A = \text{tr}_E[\mathcal{L}(t)\mathcal{G}_Q(t,s)\mathcal{Q}\mathcal{L}(s)(A \otimes \sigma_E)]\,, \qquad \mathcal{I}_0(t) = \text{tr}_E[\mathcal{L}(t)\mathcal{G}_Q(t,0)\mathcal{Q}\rho_{SE}(0)]\,.$$

Here $\mathcal{G}_Q$ propagates the $\mathcal{Q}$ subspace with generator $\mathcal{Q}\mathcal{L}(t)\mathcal{Q}$ [5, 6]. The inhomogeneous term vanishes for $\rho_{SE}(0) = \rho_S(0) \otimes \sigma_E$; correlated inputs allowed below generally retain it. Non-Markovian dynamics can show information backflow, a temporary increase of the trace distance between two evolving states [14]. Neither the kernel nor backflow is used as a resource below; the accounting identity applies directly to the joint unitary dynamics.

2.2 Conventions

Definition 1 (Thermodynamic setup)**.** *Let $\beta = (k_B T)^{-1}$. The reference state of the system is $\rho_{\text{eq}} = e^{-\beta H_R}/Z_R$, and its nonequilibrium free energy is*

$$F(\rho) = \text{tr}(\rho H_R) - \beta^{-1} S(\rho), \qquad D(\rho\|\rho_{\text{eq}}) = \beta\big(F(\rho) - F_{\text{eq}}\big) \geq 0. \tag{3}$$

The environmental reference is the fixed Gibbs state $\rho_E^{\text{th}} = e^{-\beta H_E}/Z_E$. The inclusive work over $[0,\tau]$ is $W = \int_0^\tau \text{tr}\big(\rho(t)\,\partial_t H(t)\big)dt$, so that $W = \Delta\langle H\rangle$ exactly.

Definition 2 (Standing assumptions)**.** *Unless stated otherwise:*

(A1) *$\mathcal{H}_S \otimes \mathcal{H}_E$ is finite dimensional and the total evolution is unitary, generated by* (1)*. Infinite baths are treated as limits of finite mode sets in which the quantities below have finite limits.*

(A2) *Every energy change is booked: $W = \Delta\langle H_R\rangle + \Delta\langle H_{\text{ctrl}}\rangle + \Delta\langle H_E\rangle + \Delta\langle H_{\text{int}}\rangle$. Dropping $\Delta\langle H_{\text{int}}\rangle$ is a declared approximation, never a consequence of autonomy.*

(A3) *H_E is time independent; the actual initial state of the environment need not be ρ_E^{th}.*

(A4) *The initial state $\rho_{SE}(0)$ may be correlated.*

(A5) *Relative entropies are finite (full rank, or restriction to supports).*

2.3 The survival functional

Definition 3 (Survival functional)**.** *For an interval $[a,b]$,*

$$\mathcal{S}[a,b] := F(\rho_S(b)) - F(\rho_S(a)) - W[a,b], \qquad \beta\,\mathcal{S} = -\Sigma, \quad \Sigma := \beta(W - \Delta F). \tag{4}$$

The name records the intended use: $\mathcal{S} > 0$ means the system gained more free energy than the external protocol invested, a thermodynamic profit; $\mathcal{S} < 0$ means it paid more than it gained. Mathematically $\mathcal{S}$ is a generalized entropy-balance functional. For an uncorrelated thermal initial environment and zero changes in the endpoint control and interaction energies, $\Sigma \geq 0$ and $\mathcal{S} \leq 0$. When those endpoint terms change, their signs enter the balance of Theorem 8. For correlated initial conditions Σ can be negative [15], and the next sections say what pays for it.

In measurement-based feedback, the generalized second law $\Sigma \geq -I_{\mathrm{fb}}$ allows $\mathcal{S} > 0$ at the expense of the information I_{fb} gained by measurement [7, 8, 9]. Pre-existing correlations enter the balance in an analogous role. A memory kernel is not itself a measurement record, and the two protocols are kept distinct; Section 6 returns to feedback with an explicit record.

3 A restricted autonomous baseline

Definition 4 (Autonomous Davies baseline)**.** *The autonomous baseline has $H_{\mathrm{ctrl}} = 0$, fixed H_R, and a time-independent GKSL generator $\mathcal{L}$ obtained as the weak-coupling limit of coupling to one thermal bath, satisfying quantum detailed balance with respect to the faithful Gibbs state ρ_{eq} [11, 1]. No measurement, feedback or initial system–bath resource is included.*

Lemma 5 (Spohn [11])**.** *For a GKSL semigroup $\Lambda_t = e^{\mathcal{L}t}$ with invariant state ρ_{eq}, $t \mapsto D(\Lambda_t\rho\|\rho_{\mathrm{eq}})$ is nonincreasing.*

Proof. Relative entropy is contractive under CPTP maps and $\Lambda_t\rho_{\mathrm{eq}} = \rho_{\mathrm{eq}}$. □

Theorem 6 (Restricted autonomous ceiling)**.** *Under Definition 4, $\mathcal{S}[0,\tau] = F(\rho(\tau)) - F(\rho(0)) \leq 0$.*

Proof. $W = 0$, and by (3) and Lemma 5, $\beta\Delta F = \Delta D(\rho\|\rho_{\mathrm{eq}}) \leq 0$. □

The theorem is a free-energy monotonicity statement for one autonomous detailed-balance class. It is not a ceiling for driven or feedback-controlled Markovian agents, which can persist under continuous driving; time-dependent protocols need an instantaneous detailed-balance hypothesis and a different theorem. The reduced semigroup omits explicit inter-epoch correlations, which is a modelling loss, not a proof that such correlations are unusable.

4 Initial-correlation accounting

4.1 The identity

The quantum mutual information $I(S{:}E) = S(\rho_S) + S(\rho_E) - S(\rho_{SE}) = D(\rho_{SE}\|\rho_S \otimes \rho_E)$ measures all correlations between system and environment. For any full-rank σ_E,

$$D(\rho_{SE}\|\rho_S \otimes \sigma_E) = I(S{:}E) + D(\rho_E\|\sigma_E), \tag{5}$$

an algebraic identity that holds for correlated ρ_{SE} [16].

Lemma 7 (Information–thermodynamic identity)**.** *For unitary evolution from any initial state,*

$$\Delta I(S{:}E) + \Delta D(\rho_E \| \rho_E^{\mathrm{th}}) = \Delta S_S + \beta\, \Delta\langle H_E \rangle. \tag{6}$$

Proof. By (5) with $\sigma_E = \rho_E^{\mathrm{th}}$ and $\ln \rho_E^{\mathrm{th}} = -\beta H_E - \ln Z_E$, the quantity $I + D(\rho_E \| \rho_E^{\mathrm{th}})$ equals $-S(\rho_{SE}) + S(\rho_S) + \beta\langle H_E \rangle + \ln Z_E$. Unitarity keeps $S(\rho_{SE})$ fixed, and taking differences gives (6). □

Theorem 8 (Survival balance)**.** *Under* (A1)–(A5)*, for any interval and any initial state,*

$$\boxed{\beta\, \mathcal{S} = -\Delta I(S{:}E) - \Delta D(\rho_E \| \rho_E^{\mathrm{th}}) - \beta\, \Delta\langle H_{\mathrm{ctrl}} \rangle - \beta\, \Delta\langle H_{\mathrm{int}} \rangle.} \tag{7}$$

Proof. By (A2) and $\Delta F = \Delta\langle H_R \rangle - \beta^{-1} \Delta S_S$, $\Sigma = \beta(W - \Delta\langle H_R \rangle) + \Delta S_S = (\Delta S_S + \beta \Delta\langle H_E \rangle) + \beta \Delta\langle H_{\mathrm{ctrl}} \rangle + \beta \Delta\langle H_{\mathrm{int}} \rangle$. Lemma 7 replaces the bracket by $\Delta I + \Delta D$, and $\beta\mathcal{S} = -\Sigma$. □

The identity is exact. It is the Esposito–Lindenberg–Van den Broeck decomposition [15] written with arbitrary initial correlations and with both endpoint terms kept. It is dynamics-agnostic: Markovianity does not enter, and a memory kernel matters only if a model shows that it preserves correlations across an interval boundary and later allows their consumption. Autonomy removes the control term but not the interaction term; Section 5.1 gives an autonomous model in which the interaction energy changes at every time.

4.2 Regimes and the budget

Corollary 9 (Regimes and budget)**.** *Suppose* $\Delta\langle H_{\mathrm{ctrl}} \rangle = \Delta\langle H_{\mathrm{int}} \rangle = 0$ *on the interval, as for a switched protocol whose control and interaction Hamiltonians are zero at both endpoints.*

1. *If the interval starts from a product state with a thermal environment,* $\beta\mathcal{S} = -I(S{:}E; b) - D(\rho_E(b) \| \rho_E^{\mathrm{th}}) \le 0$.
2. *If correlations are consumed,* $\Delta I < 0$*, and* $|\Delta I| > \Delta D(\rho_E \| \rho_E^{\mathrm{th}})$*, then* $\mathcal{S} > 0$.
3. *In every case* $\beta\mathcal{S}[a, b] \le I(S{:}E; a) + D(\rho_E(a) \| \rho_E^{\mathrm{th}})$.

Proof. Insert the hypotheses into (7) and use $I \ge 0$, $D \ge 0$ at the end of the interval. □

When the endpoint terms do not vanish they are added to the budget with their signs. The budget in (iii) is the profit ceiling: correlations present at the start of the interval plus the environment's displacement from equilibrium.

Proposition 10 (Charge–discharge interval closure)**.** *Under the hypotheses of Corollary 9 at the times* 0, t^* *and* τ*, and with* $I(S{:}E; 0) = 0$ *and* $\rho_E(0) = \rho_E^{\mathrm{th}}$*,*

$$\begin{aligned} \beta\mathcal{S}[0, t^*] &= -I(t^*) - D(t^*) \le 0, \\ \beta\mathcal{S}[t^*, \tau] &= -[I(\tau) - I(t^*)] - [D(\tau) - D(t^*)], \\ \beta\mathcal{S}[0, \tau] &= -I(\tau) - D(\tau) \le 0, \end{aligned}$$

with $D(t) = D(\rho_E(t) \| \rho_E^{\mathrm{th}})$.

Proof. ΔF and W are additive over concatenated intervals, and Theorem 8 applies to each interval. □

A positive second interval is funded by the correlations built in the first, and their concatenation cannot have positive $\mathcal{S}$. No restoration of the complete joint state is assumed in this accounting statement. This charge–discharge structure is realized exactly by the two-qubit model of Section 5.2. Under the Born approximation every step restarts from a product state, so the resource is removed by construction; a reduced semigroup law by itself does not say that every microscopic dilation is uncorrelated. BLP backflow and correlation consumption can coincide in particular models, but neither implies the other in general [10], so backflow is not a witness for the sign of $\mathcal{S}$ without the full balance.

5 Which correlations can be used: three exact models

5.1 Pure dephasing with a Lorentz–Drude bath: no backflow, nonzero interaction energy

Let $\hbar = 1$ and

$$H = \frac{\omega_0}{2}\sigma_z + \sum_k \omega_k b_k^\dagger b_k + \sigma_z \sum_k (g_k b_k + g_k^* b_k^\dagger), \qquad J(\omega) = \sum_k |g_k|^2 \delta(\omega - \omega_k) = \frac{2\lambda\gamma\omega}{\omega^2 + \gamma^2}, \tag{8}$$

with $\lambda, \gamma > 0$. With this normalization $\int_0^\infty J(\omega)\omega^{-1} d\omega = \pi\lambda$. The initial state is any system state times the thermal state of the bare bath at inverse temperature $\beta \in (0, \infty]$. Removing the free rotation, the coherence is multiplied by a factor of modulus

$$|p(t)| = e^{-G_\beta(t)}, \qquad G_\beta(t) = 4\int_0^\infty \frac{J(\omega)}{\omega^2}(1 - \cos\omega t)\coth\frac{\beta\omega}{2}\, d\omega, \tag{9}$$

with $\coth = 1$ at $\beta = \infty$. This is the standard exact solution: the two eigenvalues of σ_z displace each bath mode in opposite directions, the displacement difference is $2g_k(1-e^{-i\omega_k t})/\omega_k$, and the thermal characteristic function gives the factor $\exp[-|\delta\alpha_k|^2 \coth(\beta\omega_k/2)/2]$ per mode [1, 17]. The integral converges for every finite t, so $|p(t)| > 0$.

Proposition 11 (No backflow at any temperature)**.** *For every $\beta \in (0, \infty]$ and $\lambda, \gamma > 0$, $G'_\beta(t) > 0$ for all $t > 0$. Hence $|p(t)|$ decreases strictly, the trace distance of any two system states does not increase, the reduced dynamics is CP-divisible, and for $\beta < \infty$, $\mathcal{S}[a, b] < 0$ on every interval $0 \le a < b$ when the system starts with nonzero coherence.*

Proof. Differentiating under the integral sign, which is allowed because $|\sin\omega t| \le \min(1, \omega T)$ for $t \le T$ gives an integrable majorant, yields

$$G'_\beta(t) = 8\lambda\gamma \int_0^\infty f(\omega)\sin(\omega t)\, d\omega, \qquad f(\omega) = \frac{\coth(\beta\omega/2)}{\omega^2 + \gamma^2}.$$

The function f is positive and strictly decreasing on $(0, \infty)$ and tends to 0; near $\omega = 0$ it behaves as $2/(\beta\gamma^2\omega)$, and $f(\omega)\sin\omega t$ stays bounded there. Substituting $u = \omega t$ and splitting

$(0,\infty)$ into half-periods,

$$\int_0^\infty f(\omega)\sin(\omega t)\,d\omega = \frac{1}{t}\sum_{j=0}^{\infty}(-1)^j a_j, \qquad a_j = \int_0^\pi f\Big(\frac{j\pi+x}{t}\Big)\sin x\,dx.$$

The a_j are positive, strictly decreasing and tend to 0, so the alternating sum is at least $a_0 - a_1 > 0$. Between times $s < t$ the coherence factor is multiplied by $p(t)/p(s)$ with modulus less than 1, which is a CPTP dephasing map, and qubit trace distances shrink under it. Populations are conserved and $W = 0$, so $\beta\mathcal{S}[a,b] = -[S(\rho_S(b)) - S(\rho_S(a))]$, which is negative because the entropy of a qubit with fixed populations increases as its coherence decreases. □

For $\lambda = 1$ and $\gamma = 0.5$ at zero temperature, $|p| = 0.313$, 0.0360 and 2.2×10^{-4} at $t = 0.5$, 1 and 2; at $\beta = 2$ the values are 0.138, 1.43×10^{-3} and 1.0×10^{-9}. Changing γ/λ or the temperature does not produce a backflow interval in this model.[1]

Proposition 12 (Interaction energy)**.** *For any initial system state and a bath that starts in its bare thermal state with vanishing first moments,*

$$\langle H_I\rangle_t = -2\int_0^\infty \frac{J(\omega)}{\omega}(1-\cos\omega t)\,d\omega = -2\pi\lambda\big(1-e^{-\gamma t}\big), \qquad \Delta E_B[0,t] = 2\pi\lambda\big(1-e^{-\gamma t}\big),$$

independently of the temperature, while $\Delta E_S = 0$ *and* $W = 0$*.*

Proof. The Heisenberg equations give $b_k(t) = e^{-i\omega_k t}b_k(0) + (g_k^*/\omega_k)\sigma_z(e^{-i\omega_k t} - 1)$. With $\sigma_z^2 = \mathbb{1}$ and $\langle b_k(0)\rangle = 0$, $\langle\sigma_z(g_k b_k(t) + g_k^* b_k^\dagger(t))\rangle = -2|g_k|^2(1-\cos\omega_k t)/\omega_k$. Summing with the spectral density and using $\int_0^\infty \cos(\omega t)/(\omega^2+\gamma^2)\,d\omega = \pi e^{-\gamma t}/(2\gamma)$ gives the formula. The total energy is conserved and σ_z commutes with H, so $\Delta E_B = -\Delta\langle H_I\rangle$. □

The balance on $[a,b]$ is therefore $\beta\mathcal{S} = -\Delta I - \Delta D_B - \beta\Delta\langle H_I\rangle = -\Delta S_S$, with $\beta\Delta\langle H_I\rangle = -2\pi\beta\lambda(e^{-\gamma a} - e^{-\gamma b}) \neq 0$. The evolution is autonomous and the interaction-energy term still cannot be dropped. At zero temperature the dynamics and energies are well defined, but the relative-entropy terms of the finite-temperature ledger cannot be used with β^{-1} treated as a finite energy unit.

5.2 An autonomous two-qubit recurrence: correlation without a record

Let a system qubit and an environment qubit evolve under the time-independent, mutually commuting Hamiltonian

$$H = \frac{\omega_0}{2}\sigma_z + \frac{\Omega}{2}\tau_z + g\,\sigma_z\tau_z, \qquad \rho_{SE}(0) = \frac{\mathbb{1} + r\sigma_x}{2}\otimes\tau_E, \quad \tau_E = \frac{e^{-\beta\Omega\tau_z/2}}{Z_E}, \tag{10}$$

with $g, \Omega, \beta > 0$ and $0 < r < 1$. Put $m = \tanh(\beta\Omega/2)$, $t_c = \pi/(4g)$ and $t_r = \pi/(2g)$.

[1] The oscillatory decoherence function, survival-gain table and figure in the July 2026 example were not solutions of its stated pure-dephasing model. Propositions 11 and 12 replace them and restore the interaction-energy term.

Proposition 13 (Exact saturation of the correlation budget)**.** *The coherence of the system is* $\frac{r}{2}e^{-i\omega_0 t}\big(\cos 2gt + im\sin 2gt\big)$*, of modulus* $\frac{r}{2}\sqrt{\cos^2 2gt + m^2\sin^2 2gt}$*. The environment marginal stays* τ_E*, and* $\Delta E_S = \Delta E_E = \Delta\langle H_I\rangle = W = 0$*. On the discharging interval*

$$\beta\mathcal{S}[t_c, t_r] = h\Big(\frac{1+rm}{2}\Big) - h\Big(\frac{1+r}{2}\Big) = I(S{:}E; t_c) > 0,$$

where h *is the binary entropy in nats, so the budget of Corollary 9(iii) is attained with equality. The charging interval* $[0, t_c]$ *has the opposite value and* $\mathcal{S}[0, t_r] = 0$*. The two conditional environment states for the two pointer values coincide, so the environment holds no information about the pointer:* $\chi(\text{pointer}{:}E) = 0$ *at all times.*

Proof. The Hamiltonian is diagonal in the product basis. The off-diagonal element of S acquires the phase $e^{-i\omega_0 t}e^{-2igst}$ when the environment is in the τ_z eigenstate $s = \pm 1$, which has weight $(1 \mp m)/2$; averaging gives the coherence. The diagonal blocks are unchanged, so the energies and the environment marginal are constant and the global entropy is constant; hence $I(S{:}E; t) = S(\rho_S(t)) - S(\rho_S(0))$. At t_r the coupling factor of the evolution operator is $e^{-i(\pi/2)\sigma_z\tau_z} = -i\sigma_z \otimes \tau_z$, so the evolution operator is a product of local unitaries and the correlation vanishes. With the energies constant, $\beta\mathcal{S} = -\Delta S_S$, which gives the displayed values. The environment state conditional on either pointer value is τ_E, because τ_E commutes with τ_z. □

For $(\beta, \Omega, g, r) = (1.2, 1.1, 0.3, 0.7)$, $m = 0.578363$, $t_c = 2.617994$, $t_r = 5.235988$ and $\beta\mathcal{S}[t_c, t_r] = 0.1860846$. The model realizes the charge–discharge cycle of Proposition 10 with an autonomous constant coupling, and it is exact. Its environment remembers in the sense that the correlation returns at a fixed time; it does not remember in the sense of holding a readable record. The reduced dynamics is not CP-divisible, since the coherence revives.

5.3 Records: the collision model

The collision model of the companion papers supplies correlations of a third kind [25, 26]. A pointer qubit with $H_R = \varepsilon P_1$ starts in $\sqrt{p}\,|0\rangle + \sqrt{1-p}\,|1\rangle$; memories $E_1, \ldots, E_n$ with degenerate Hamiltonians start in $|0\rangle$, and the k-th collides once through $P_0 \otimes \mathbb{1} + P_1 \otimes R_k$ with $\langle 0|R_k|0\rangle = c_k$. The collision can be generated by $g_k(t)P_1 \otimes \sigma_y$, which commutes with H_R and vanishes at both ends of the pulse, so $W = 0$ and $\Delta\langle H_{\text{int}}\rangle = 0$. With $r = \prod_k c_k$ and $s_n = H_p(r)$, where $H_p(r) = h\big((1 + \sqrt{1 - 4p(1-p)(1-r^2)})/2\big)$, a fragment F holds $\chi(X{:}E_F) = H_p(r_F)$ nats about the pointer label X [25].

Proposition 14 (Ledger of the collision model)**.** *With the environmental reference* $\mathbb{1}/2^n$*:*

1. *from the coherent initial state,* $\Delta I(S{:}E) = 2s_n$*,* $\Delta D = -s_n$ *and* $\beta\mathcal{S} = -s_n$*;*
2. *from the diagonal initial state* $pP_0 + (1-p)P_1$*, every record quantity is the same,* $\Delta I = s_n$*,* $\Delta D = -s_n$ *and* $\beta\mathcal{S} = 0$*;*
3. *the exact inverse protocol, which needs access to* S *and all memories, gives* $\Delta I = -2s_n$*,* $\Delta D = +s_n$ *and* $\beta\mathcal{S} = +s_n$*, and the full forward–backward cycle has* $\mathcal{S} = 0$*.*

Proof. The global state is pure in case (1), so $I(S{:}E) = 2S(\rho_E) = 2s_n$, and $D(\rho_E \| \mathbb{1}/2^n) = n \ln 2 - s_n$. Populations of S are conserved and $W = 0$, so $\beta\mathcal{S} = -\Delta S_S = -s_n$, in agreement with (7). In case (2) the global entropy is $h(p)$ throughout, $S(\rho_S) = h(p)$ is unchanged and $S(\rho_E) = s_n$, so $I = s_n$. Case (3) reverses case (1). □

For $p = 1/2$, $c_k = 0.8$ and $n = 60$, $s_{60} = 0.693147$ nats, equal to $\ln 2$ to nine digits. Records form along a CP-divisible reduced dynamics, and the diagonal case shows that the formation of records does not by itself cost the system free energy; the memories' initial purity deficit supplies the record budget [26].

5.4 Who can use which correlation

	Lorentz–Drude dephasing	Two-qubit recurrence	Collision records
Correlation built	yes	yes	yes
Returns to S by itself	never	at t_r, fixed by g	no
Pointer information in E	yes, spread over a continuum of modes	none ($\chi = 0$)	yes, in many separate copies
Who can use it	an agent with coherent control of the bath	the autonomous dynamics, at one time	an agent with fragment readout and feedback permissions
Endpoint terms	$\Delta\langle H_I \rangle \neq 0$	zero	zero

Table 1: Three forms of system–environment correlation.

In the accounting identity, consuming one nat of mutual information contributes $+k_B T$ to $\mathcal{S}$, with the other terms retained. The models show that the identity does not say who can consume it. Correlation created by a structured bath can be dissipated without ever returning, although the bath holds information about the pointer spread over its modes. Correlation held by a single environmental degree of freedom with a commensurate spectrum can return at a fixed time, without anyone reading anything. Correlation copied into records can be read by many agents at their own times, and the rest of this paper studies what such an agent must do to turn a record into work.

6 The value of a record, and of a model of one's own readout

6.1 The task

The pointer S of Section 5.3 has a degenerate Hamiltonian during the task ($\varepsilon = 0$) and has been decohered by memories the agent does not read, so that its state is diagonal: $\rho_S = pP_0 + (1-p)P_1$, correlated with the memories through X. The agent reads a fragment

F by a measurement N and stores the outcome in a classical memory M. It then runs a feedback protocol on S and a bath B at inverse temperature β, chosen according to the stored value m, in which the system Hamiltonian returns to zero and the coupling vanishes at both ends. The work extracted is $W_{\text{ext}} = -W$. The task is to extract as much work as possible on average.

Lemma 15 (Extraction from a qubit)**.** *For a qubit with degenerate Hamiltonian in a state ρ, every such protocol extracts on average at most $k_B T\, D(\rho\|\mathbb{1}/2) = k_B T[\ln 2 - S(\rho)]$. If the diagonal of ρ is a distribution P and Q is a distribution with full support, the protocol that quenches the levels to $E_x = -k_B T \ln Q(x)$, lets the system thermalize and returns the levels quasistatically to degeneracy extracts*

$$k_B T\big[\ln 2 - H(P) - D(P\|Q)\big].$$

Proof. For the bound: the joint state of system and bath evolves unitarily from $\rho\otimes\tau_B$; since the system Hamiltonian is zero at both ends and the coupling vanishes there, $W = \Delta E_B$. Subadditivity gives $S(\rho') + S(\rho'_B) \geq S(\rho) + S(\tau_B)$, and $D(\rho'_B\|\tau_B) \geq 0$ gives $\beta\Delta E_B \geq S(\rho'_B) - S(\tau_B) \geq S(\rho) - S(\rho') \geq S(\rho) - \ln 2$. For the protocol: the quench costs $\sum_x P(x)E_x = k_B T[H(P) + D(P\|Q)]$; thermalization at fixed levels costs no work; the quasistatic return costs the free-energy difference $-k_B T \ln 2 - (-k_B T \ln \sum_x Q(x)) = -k_B T \ln 2$. The work done on the system is $k_B T[H(P) + D(P\|Q) - \ln 2]$. Off-diagonal elements do not affect it, since the levels are diagonal and the bath relaxes the system to the Gibbs state of the quenched levels. □

Quasistatic values throughout this section are ideal limiting values. A full-support posterior needs finite quenched gaps, but reversible thermalization and return are still limits of physical protocols. A posterior on the boundary additionally needs a divergent level gap; it is treated by a full-support approximation.

6.2 Memory is necessary, and its value is the accessible information

Theorem 16 (Value of a stored record)**.** *In the task above, let $W_0 = k_B T[\ln 2 - h(p)]$ be the largest average work obtainable without the record. For every measurement N, every processing of its outcome into M and every feedback protocol,*

$$\mathbb{E}[W_{\text{ext}}] - W_0 \leq k_B T\, I(X{:}M) \leq k_B T\, I_{\text{acc}}(X{:}E_F) \leq k_B T\, \chi(X{:}E_F) = k_B T\, H_p(r_F).$$

The first inequality is approached by the quasistatic protocol of Lemma 15 with $Q = P(\cdot \mid m)$ when every posterior $P(\cdot \mid m)$ has full support; otherwise it is the supremum over such protocols. For $p = 1/2$ the Helstrom measurement gives $I(X{:}M) = I_{\text{acc}}(X{:}E_F) = \ln 2 - h\big((1 - \sqrt{1 - r_F^2})/2\big)$.

Proof. Given $M = m$ the state of S is $\sum_x P(x \mid m)P_x$, because S is diagonal and correlated with F only through X. Lemma 15 bounds the extracted work by $k_B T[\ln 2 - H(X \mid M = m)]$, with equality in the matched quasistatic limit. Averaging over m and subtracting W_0 gives $k_B T[h(p) - H(X \mid M)] = k_B T\, I(X{:}M)$. Since M is obtained from a measurement on F

followed by processing, $I(X{:}M) \le I_{\mathrm{acc}}(X{:}E_F)$, and Holevo's bound gives $I_{\mathrm{acc}} \le \chi$ [18]. The value of χ and the equal-prior accessible-information formula are the pure-fragment results of Paper E [25]. For mixed record ports the first inequality follows from the same extraction argument; their actual information replaces $H_p(r_F)$. □

The theorem gives the memory link of the route in the strongest form available for this task. A work gain ΔW is possible only if the agent holds a memory with $I(X{:}M) \ge \beta\Delta W$, and only if it reads a fragment with $H_p(r_F) \ge \beta\Delta W$; the fragment lengths of Paper E then say how many environmental memories it must control. Conversely, the gain $k_B T\, I(X{:}M)$ is the matched quasistatic limit. With $p = 1/2$ and $c_k = 0.8$, one memory gives at most $0.1927\, k_B T$ to an agent with a classical memory although its Holevo information is 0.3251; five memories give $0.5668\, k_B T$ against 0.6385.

The gap between I_{acc} and χ has a thermodynamic reading. An agent that does not measure the fragment but keeps it and acts on S and F jointly can, in the asymptotic limit, use F as quantum side information, for which the relevant quantity is the conditional entropy $S(X \mid F) = H(X) - \chi(X{:}E_F)$ [19]; the Holevo value is then the ceiling. The accessible information is the value of the same record to an agent that first writes it into a classical memory.

6.3 A model of one's own readout is necessary

Theorem 17 (Every protocol assumes a posterior)**.** *For every feedback protocol* Π *on* S *and a bath, of the kind allowed in the task, there is a distribution* Q_Π *with full support such that for every diagonal state with populations* P

$$\mathbb{E}_\Pi[W_{\mathrm{ext}} \mid P] \le k_B T\big[\ln 2 - H(P) - D(P\|Q_\Pi)\big],$$

with equality for all P *when* Π *is the protocol of Lemma 15 designed for* Q_Π*. Consequently, if the agent runs* Π_m *after reading* m*,*

$$\mathbb{E}[W_{\mathrm{ext}}] - W_0 \le k_B T\Big[I(X{:}M) - \sum_m P(m)\, D\big(P(\cdot \mid m) \,\|\, Q_{\Pi_m}\big)\Big],$$

with equality for the matched two-step protocols. The record has negative value whenever the expected divergence exceeds $I(X{:}M)$*, and the value* $k_B T\, I(X{:}M)$ *is attained only if* $Q_{\Pi_m} = P(\cdot \mid m)$ *for every* m *with* $P(m) > 0$*.*

Proof. For a fixed protocol the extracted work is an affine function of the initial system state, $E(\rho) = -\operatorname{tr}\big[(U^\dagger H_{\mathrm{end}} U - H_{\mathrm{start}})(\rho \otimes \tau_B)\big]$, and by Lemma 15 it lies below the convex function $f(\rho) = k_B T\, D(\rho\|\mathbb{1}/2)$. On diagonal states $f(P) = k_B T[\ln 2 - H(P)]$, whose gradient $k_B T \ln P$ (up to a constant) takes every value as P ranges over distributions with full support. Choose Q_Π so that the tangent $T(P) = f(Q_\Pi) + \langle \nabla f(Q_\Pi), P - Q_\Pi\rangle = k_B T[\ln 2 - H(P) - D(P\|Q_\Pi)]$ has the same slope as E. Then $E - T$ is constant, and at $P = Q_\Pi$ it equals $E(Q_\Pi) - f(Q_\Pi) \le 0$. So $E \le T$ everywhere. The quasistatic two-step protocol for Q approaches T by Lemma 15. Averaging over m and subtracting W_0 gives the second display, and the divergence vanishes only when the distributions coincide. □

The extra dissipation $k_B T\, D(P\|Q)$ of a protocol tuned to Q and run on P is a special case of the mismatch cost of Kolchinsky and Wolpert, for a process whose output does not depend on its input [20]; the tangent argument adds that every protocol, tuned or not, has such a Q. The distribution Q_Π is what the protocol assumes about the system, whether or not anything in the agent represents it explicitly. An agent that obtains $P(\cdot \mid m)$ from the prior and its readout channel $P(m \mid x)$, that is, from how its own apparatus responds to the two pointer values, acts on a correct assumption only if its model of that channel is correct where the posterior depends on it.

What the agent needs is therefore not a full description of its apparatus but the part on which the posterior depends: given the prior, the likelihood ratio of its readout. Getting that part wrong has a price that no choice of protocol avoids. With one memory of overlap 0.8 and the Helstrom readout, the true error is 0.2 and the record is worth $0.1927\, k_B T$. An agent that believes its error is 0.01 loses $0.4287\, k_B T$ and ends $0.2359\, k_B T$ below an agent with no memory; one that believes 0.1 keeps $0.1483\, k_B T$; one that believes 0.4 keeps $0.1012\, k_B T$. Overconfidence in one's own readout is more expensive than underconfidence of the same size because the quench pushes the level of the excluded value up by $-k_B T \ln Q$.

6.4 Calibration from references

Let the readout be the binary channel of T-DOME III, $P(y = +1 \mid x) = (1 + gx + b)/2$ with pointer values written $x = \pm 1$, gain g and bias b, $|g| + |b| < 1$, and prior $1/2$ [29]. The agent estimates $\theta = (g, b)$ from n reference reads of each known pointer value.

Proposition 18 (Metric and calibrated loss)**.** *Let $P_\theta(m \mid x)$ be a smooth family and $L(\hat\theta) = \sum_m P_\theta(m) D(P_\theta(\cdot \mid m) \| P_{\hat\theta}(\cdot \mid m))$ the expected divergence of Theorem 17. Then*

$$L(\hat\theta) = \tfrac{1}{2}(\hat\theta - \theta)^\top G\,(\hat\theta - \theta) + O(|\hat\theta - \theta|^3), \qquad G = J_{XM} - J_M,$$

where J_{XM} is the Fisher information of the joint distribution of (x, m) with x drawn from the prior and J_M that of the marginal of m. If $\hat\theta$ is estimated from n reference reads per pointer value with second-moment matrix $\mathbb{E}(\hat\theta - \theta)(\hat\theta - \theta)^\top = (2nJ_{XM})^{-1} + o(1/n)$, third absolute moments $O(n^{-3/2})$, and $\mathbb{E}\big[L(\hat\theta)\mathbf{1}\{|\hat\theta - \theta| > \epsilon\}\big] = o(1/n)$ for some $\epsilon > 0$, then

$$\mathbb{E}\, L = \frac{1}{4n}\Big[\dim\theta - \operatorname{tr}\big(J_M J_{XM}^{-1}\big)\Big] + o(1/n).$$

For the gain–bias channel this is $\mathbb{E}\, L = \dfrac{1}{4n}\Big(1 + \dfrac{g^2}{1 - b^2}\Big) + o(1/n)$. The tail condition is needed: the plug-in frequency estimate meets the other conditions but returns a boundary value with positive probability, at which the model posterior excludes an outcome the true posterior allows, so its expected loss is infinite for every n; add-one-half estimates satisfy it.

Proof. For each m, $D(P_\theta(\cdot \mid m) \| P_{\hat\theta}(\cdot \mid m)) = \tfrac{1}{2}\delta^\top J_{X|m}\delta + O(|\delta|^3)$ with $\delta = \hat\theta - \theta$ and $J_{X|m}$ the Fisher information of the posterior family. The chain rule for Fisher information, $J_{XM} = J_M + \sum_m P(m) J_{X|m}$, gives G. On a fixed interior neighborhood, Taylor's remainder has expectation $O(n^{-3/2})$. Outside it, the assumed loss tail is $o(n^{-1})$, and the third-moment bound makes the quadratic tail $O(n^{-3/2})$. Thus $\mathbb{E}\, L = \tfrac{1}{2}\operatorname{tr}(G\Sigma) + o(n^{-1})$, with Σ the error

second-moment matrix including bias. With $\Sigma = (2nJ_{XM})^{-1}$, $\frac{1}{2}\operatorname{tr}(G\Sigma) = \frac{1}{4n}\operatorname{tr}\big((J_{XM} - J_M)J_{XM}^{-1}\big)$. For the gain–bias channel with prior 1/2, the efficient reference estimator of T-DOME III has covariance $\frac{1}{2n}\begin{pmatrix} 1-g^2-b^2 & -2gb \\ -2gb & 1-g^2-b^2 \end{pmatrix} = (2nJ_{XM})^{-1}$, the marginal of m is Bernoulli with parameter $(1+b)/2$, so $J_M = \operatorname{diag}(0, (1-b^2)^{-1})$, and $\operatorname{tr}\big(J_M J_{XM}^{-1}\big) = (1-g^2-b^2)/(1-b^2)$. □

Lemma 19 (Tail control for add-one-half calibration)**.** *For a fixed interior gain–bias channel, let independent $K_\pm \sim \mathrm{Bin}(n, p_\pm)$, $p_\pm = (1+b\pm g)/2$. Set $\hat{p}_\pm = (K_\pm+1/2)/(n+1)$, $\hat{g} = \hat{p}_+ - \hat{p}_-$ and $\hat{b} = \hat{p}_+ + \hat{p}_- - 1$. These estimates meet all moment and tail hypotheses of Proposition 18.*

Proof. The estimates are a linear transform of two independent frequencies. Their bias is $O(n^{-1})$ and their second-moment matrix is $(2nJ_{XM})^{-1} + O(n^{-2})$; the fourth centred binomial moment is $O(n^2)$, so their third absolute error moment is $O(n^{-3/2})$. Every estimated likelihood is at least $1/[2(n+1)]$. For the fair prior every model posterior probability is therefore at least $1/[4(n+1)]$, giving $0 \le L \le \log[4(n+1)]$. For any fixed sufficiently small neighborhood of the interior true parameter, Hoeffding's inequality bounds its complement by Ce^{-cn}, with $c > 0$. Its contribution to expected loss is at most $C\log[4(n+1)]e^{-cn} = o(n^{-1})$. □

The matrix G is the nonnegative posterior Fisher form on the agent's model of itself, a metric on its identifiable directions, and Proposition 18 gives it a thermodynamic meaning: a calibration error δ costs $\frac{1}{2}k_BT\,\delta^\top G\delta$ of work per use. The subtraction of J_M has a simple content. Unlabelled readings already carry information about the bias, which enters the marginal of m, and the posterior is less sensitive to that direction. For $g = 0.55$ and $b = 0$ the leading term is $1.3025/(4n)$; exact enumeration over the reference outcomes, with estimates $(k+\frac{1}{2})/(n+1)$ for the two response probabilities, gives $\mathbb{E}\,L = 0.02064$, 0.005113 and 0.001273 for $n = 16$, 64 and 256, ratios 1.014, 1.005 and 1.001 to the leading term. The same channel carries a record worth $0.1600\,k_BT$, and an agent that assumes $g = 0.8$ without calibrating loses $0.0666\,k_BT$ per use.

Corollary 20 (When calibration pays)**.** *Suppose each reference read costs w_{ref} and the calibrated model is used U times before it is renewed. To leading order the total cost $2nw_{\mathrm{ref}} + Uk_BT\kappa/(4n)$, with $\kappa = 1 + g^2/(1-b^2)$, is minimized at $n^* = \sqrt{U\kappa k_BT/(8w_{\mathrm{ref}})}$, and the overhead per use is $\sqrt{2\kappa k_BT w_{\mathrm{ref}}/U}$. If the known pointer states must be prepared from the thermal state of the degenerate pointer, $w_{\mathrm{ref}} \ge k_BT\ln 2$ by Landauer's bound. Calibration is worthwhile when this overhead is smaller than the loss of the uncalibrated model.*

Proof. Minimize $2nw + Uk_BT\kappa/(4n)$ over $n > 0$. Preparing a pure qubit state from the maximally mixed state with a bath at temperature T costs at least $k_BT\ln 2$ [12, 21]. □

With $g = 0.55$, $b = 0$ and $w_{\mathrm{ref}} = k_BT\ln 2$, the overhead per use is $0.134\,k_BT$ for $U = 100$, $0.0425\,k_BT$ for $U = 1000$ and $0.00425\,k_BT$ for $U = 10^5$, against a record value of $0.1600\,k_BT$. A model of one's own readout is worth building only if it is used many times before it goes stale. How long it stays fresh depends on how fast the apparatus drifts, which is the subject of T-DOME III.

Proposition 21 (Drift law for self-calibration; proved in Paper III)**.** *Let the scalar parameter follow a Gaussian random walk with increment variance ν^2 per use. Suppose each reference has conditional Fisher information at most J, unlabelled observations provide no information about that parameter, and the loss is globally $\frac{1}{2}k_BTG(\hat{\theta}-\theta)^2$ with constant $G > 0$. References cost w_{ref} each. Every predictable calibration policy has long-run average loss plus reference cost at least $c\, w_{\mathrm{ref}}^{1/3}(k_BTG\nu)^{2/3}J^{-1/3} - \frac{1}{2}k_BTG\nu^2$, where $c = 3 \cdot 4^{-2/3}$. With a Gaussian prior and Gaussian observations of controllable real precision at linear cost, a persistent Kalman estimate at rate $\rho^* = \big(k_BTG\nu/(4w_{\mathrm{ref}}\sqrt{J})\big)^{2/3}$ attains $3\rho^* w_{\mathrm{ref}}$ to leading order [29]. For whole reading counts, the continuous-precision optimum is a relaxation bound.*

The rate follows from the steady state of a Kalman filter: with ρ references per use the posterior variance is $\nu/\sqrt{\rho J}$ to leading order, and optimizing $w_{\mathrm{ref}}\rho + \frac{1}{2}k_BTG\nu/\sqrt{\rho J}$ gives the exponent. Paper III proves the lower bound by telescoping the posterior Cramér–Rao recursion, including predictable choices made from earlier records. It also proves the exact predictive-relative-entropy optimum of Consciousness III under a Gaussian prior and controllable precision. Free information changes the optimum: in this static readout experiment, unlabelled readings inform the directions where $J_M \neq 0$; in an acting body, the entire correlated history may inform additional parameters. Such a history needs its joint likelihood. For the gain–bias channel, the work loss is a local divergence, so the quadratic comparison requires Proposition 18's expansion and tail control. A bounded physical gain also needs its actual transition law; clipping a Gaussian walk changes the filtering problem. Smoother drift and vector parameters remain open.

7 Reading, harvesting and resetting

After a harvest the private memory must be retained, reset or replaced. A pure replacement of dimension d_M carries free energy $k_BT \ln d_M$ relative to its degenerate thermal state. Moving the old record to a spent register postpones reset and consumes storage; it does not restore a finite memory bank. We first price reset with a declared side-information menu.

Proposition 22 (Read–harvest–reset task bound)**.** *Let the agent read F into M, extract work from S with feedback on M, and reset M to a standard state by a protocol that couples M and possibly a second fragment G to a bath, leaves the marginal of G unchanged, and vanishes at the endpoints, all registers having degenerate Hamiltonians. The reset bath is thermal and initially independent of MG. No additional purity-bearing workspace or correlated side information is available to the reset. Exact blanking denotes the vanishing-error limit when a finite full-rank thermal bath cannot realize it. Then the net average work satisfies*

$$\beta\big(\mathbb{E}[W_{\mathrm{ext}}] - W_0 - W_{\mathrm{reset}}\big) \leq I(X{:}M) - S(M \mid E_G) = I(X{:}M) - H(M) + \chi(M{:}E_G).$$

Without G, the right side is $-H(M \mid X) \leq 0$, so this task has no positive net gain over the blind harvest. If the records are perfect ($r_F = r_G = 0$), the bound equals $h(p)$ and is approached by reading with a controlled-NOT, extracting with the matched protocol and resetting M by a controlled-NOT from G; with exactly perfect records it is reached only as a limit.

Proof. Theorem 16 bounds the extraction gain by $k_B T\, I(X{:}M)$. For the reset, the joint state of M, G and the bath evolves unitarily from a state in which the bath is thermal and uncorrelated with MG; with degenerate Hamiltonians and vanishing coupling at the endpoints, $W_{\mathrm{reset}} = \Delta E_B$. As in Lemma 15, $\beta \Delta E_B \geq S(\rho_{MG}) - S(\rho'_{MG})$, and with M pure at the end and the marginal of G unchanged, $S(\rho'_{MG}) = S(\rho_G)$, so $\beta W_{\mathrm{reset}} \geq S(M \mid E_G)$. For classical M, $S(M \mid E_G) = H(M) - \chi(M{:}E_G)$. Without G, $I(X{:}M) - H(M) = -H(M \mid X)$. With perfect records, M is a copy of X, $\chi(M{:}E_G) = H(M) = h(p)$, and the controlled-NOT from G maps M to its standard state at no work. □

For pure conditional fragment states with $p = 1/2$, $c_k = 0.8$ and Helstrom readings of five-memory fragments, $I(X{:}M) = 0.5668$, $H(M) = \ln 2$ and $\chi(M{:}E_G) = 0.5231$, so the net task gain is at most $0.3968\, k_B T$ with a second five-memory copy and at most $-0.1263\, k_B T$ without one; with seven-memory fragments the bounds are 0.5511 and −0.0611. Quantum conditional-entropy erasure has asymptotic realizations with the corresponding collective control and side-information permissions [19]. This does not give an arbitrary finite local eraser those permissions. The following classical record pair supplies a direct realization within the present task. If pure blanks are supplied instead, their entropy capacity can lower reset work and can itself fund work extraction; their consumed free energy must enter the input ledger. The harvest's information ceiling alone does not cap harvest minus a work-producing reset.

7.1 Two mixed ports: feedback and an actual conditional reset

Use the fair diagonal pointer and the controlled-X recording protocol of Paper F [26]. The two independent initial ports are $\eta_\alpha = (\mathbb{1} + \alpha Z)/2$, with $0 < \alpha < 1$ and $\epsilon = (1-\alpha)/2$. Their endpoint Hamiltonians are zero. The recording gates act as

$$(x, n_1, n_2) \longmapsto (x, m = x \oplus n_1, g = x \oplus n_2), \qquad X \sim \mathrm{Bern}(1/2), \quad N_1, N_2 \overset{\mathrm{iid}}{\sim} \mathrm{Bern}(\epsilon).$$

Actual Z readout uses the diagonal port M as the feedback control; no extra free copy of its value is introduced. The other physical port G is retained and is accessible to the reset. Preparation, gates and any controller that reads their values remain part of the apparatus.

Proposition 23 (Mixed-record task realization)**.** *In this protocol, put $t = 2\epsilon(1-\epsilon)$. Matched feedback on M has work gain over the blind harvest $k_B T[\ln 2 - h(\epsilon)]$. With G available as a reversible reset control, the infimum reset work is $k_B T h(t)$, and the supremum net task gain is*

$$k_B T\big[\ln 2 - h(\epsilon) - h(t)\big]. \tag{11}$$

Without G, the infimum reset cost is $k_B T \ln 2$ and the net gain is $-k_B T h(\epsilon)$. The supremum in (11) *is approached as a quasistatic, vanishing-reset-error limit and is positive for sufficiently small ϵ. It does not restore the prepared record source or the thermal bath supply.*

Proof. The actual joint record law is $P(m, g \mid x) = P_\epsilon(m \mid x) P_\epsilon(g \mid x)$. Thus M is fair, $H(X \mid M) = h(\epsilon)$ and $I(X{:}M) = \ln 2 - h(\epsilon)$. Lemma 15 with the two full-support posteriors gives the stated harvest. The matched branch protocol ends with a thermal source independent of the records, and leaves the classical joint law of MG unchanged.

The reversible controlled-NOT from G to M changes its physical content to $Z = M \oplus G = N_1 \oplus N_2$. The pair (Z, G) has law $\mathrm{Bern}(t) \otimes \mathrm{Bern}(1/2)$: for each fixed pair of noises the fair X makes G uniform. Hence $S(M \mid G) = h(t)$ before the reset, and after this gate the noisy logical bit is independent of the retained control.

Choose $t = q_0 > q_1 > \cdots > q_\ell = \zeta > 0$. At stage i, swap the degenerate logical memory with a fresh independent Gibbs qubit whose gap is $\Delta_i = k_B T \ln[(1 - q_i)/q_i]$. This is a driven swap, with endpoint interaction off. It resets the logical memory to error ζ, leaves G unchanged and independent of it, and has

$$\beta W_{\mathrm{reset}} = \beta Q_{\mathrm{bath}} = \sum_{i=1}^{\ell} (q_{i-1} - q_i) \ln \frac{1 - q_i}{q_i} \longrightarrow h(t) - h(\zeta).$$

The limit is the thermal-swap construction of Paper F and Reeb–Wolf [26, 21]. Letting $\zeta \downarrow 0$ reaches the entropy lower bound $h(t)$. The post-parity register is reset to the standard zero value, so the parity gate is not undone. Without G the same construction starts from the fair M and approaches cost $\ln 2$. Combining these reset costs with the matched harvest proves the claims. □

At $\epsilon = 0.05$ ($\alpha = 0.9$), the dimensionless harvest is 0.494632, conditional reset costs 0.313956 and net task gain is 0.180676; without the second record the net value is −0.198515. Both ports are partially mixed. Their total initial purity deficit is $2[\ln 2 - h(\epsilon)]$, paid before acquisition; their one-port record values saturate that budget in Paper F. The recording gates themselves have zero net endpoint work. Harvesting is a separate source–bath operation, not work extracted by the recording gate.

The retained G need not be a perfect copy of X: its correlation with the private memory is the useful reset resource. In the present menu it can fund a positive task balance while its marginal is unchanged. Its correlation with the harvested source, the two prepared ports, the driven gaps, clock and fresh Gibbs qubits have not all returned to their input states. A subsequent use must supply a new compatible source–record state and account for that preparation. More generally, an upkeep cost w can fit within the task only if $k_B T[I(X{:}M) - S(M \mid E_G)] \geq w$, before charging any omitted record supply or apparatus costs. This connects readable records to an action budget without identifying it with a sustainable complete-device cycle.

8 Operational temporal relations

The accounting identity does not derive a temporal order. Two notions of influence between times can be defined after a reduced model is specified, and they differ.

Definition 24 (Kernel memory and intervention influence)**.** *For a reduced model with Nakajima–Zwanzig kernel $\mathcal{K}$ and tolerance $\epsilon > 0$, write $s \rightsquigarrow_{\mathcal{K}} t$ if $\|\mathcal{K}(t, s)\| > \epsilon$ for some $s < t$; this is kernel memory. For a record-forming protocol, write $s \rightsquigarrow_{\mathrm{rec}} t$ if an intervention on the system at time s changes the records written at time t, as in the diagnostic of Paper F [26].*

The total linear response of a state at t to a perturbation at s is not a memory diagnostic: a Markov semigroup propagates a perturbation $\delta\rho$ as $e^{(t-s)\mathcal{L}}\delta\rho \neq 0$. The two relations of Definition 24 come apart in both directions. In the collision model the reduced dynamics is a product of dephasing channels with a time-local generator, so there is no kernel memory beyond a single step, while an intervention after the j-th collision changes every later record and none of the earlier ones (the fresh-fragment intervention result of Paper F). In the two-qubit model the coherence revives, so the reduced dynamics is not a semigroup and its Nakajima–Zwanzig kernel is not local in time, while the environment never holds a record. A correspondence between a memory-influence relation and nested record algebras therefore holds in neither direction in general; in fresh-memory collision models the record relation and the intervention order coincide, and kernel memory is absent.

9 The cost of keeping records

Landauer's principle prices logically irreversible resets, not passive storage, sensing, correlation maintenance or reversible computation. If a memory implementation irreversibly erases N_{erase} unbiased bits into a bath at temperature T, its reset work obeys $W_{\text{reset}} \geq k_B T \ln 2\, N_{\text{erase}}$; reversible storage and measurement can postpone this cost until an irreversible operation is performed [12, 13]. No lower bound proportional to stored mutual information follows without an implementation model, and Proposition 22 shows how strongly the reset cost depends on what the eraser can access.

To describe how much of a history is worth keeping, let $\{X_t\}$ be a stationary process describing the environment as seen by the agent. Its entropy rate is $h_\mu = \lim_{k\to\infty} h_k$ with $h_k = H(X_k \mid X_1, \ldots, X_{k-1})$ and $h_1 = H(X_1)$, and its excess entropy, or predictive information, is

$$\mathbf{E} = I(\overleftarrow{X}; \overrightarrow{X}) = \sum_{k=1}^{\infty} \big(h_k - h_\mu\big) \tag{12}$$

whenever the series converges [22, 24, 23]. For a symmetric two-state Markov chain with flip probability $0 < \varepsilon < 1/2$, $h_1 = \ln 2$ and $h_k = h(\varepsilon)$ for $k \geq 2$, so $\mathbf{E} = \ln 2 - h(\varepsilon)$. Finite excess entropy does not prove that a finite physical memory captures every useful feature, and a positive entropy rate does not bound the predictive information from below by h_μ times a memory depth.

Remark 25 (Raw-archive overload)**.** *Consider a specified implementation: an uncompressed archive receives records at r_{raw} bits per unit time, has capacity M bits and evaluates the entire archive once per control cycle. Without deletion it fills after M/r_{raw}. If it irreversibly resets discarded records at rate r_{erase}, Landauer gives $\dot{W}_{\text{reset}} \geq k_B T \ln 2\, r_{\text{erase}}$. For a long-memory or non-stationary source with divergent $\mathbf{E}$, the predictively relevant history may also grow; that is an assumption about the source. The archive can then exceed a finite storage, throughput or reset budget. This is an engineering overload scenario for one implementation.*

An agent using such an archive can respond by enlarging capacity, sampling less, uncomputing reversibly, keeping sufficient statistics, compressing lossily or accepting stale data. Each choice of what to keep needs a criterion that ranks stored correlations by their value.

Section 6 supplies one such value in work units for a single record; Paper II studies what a bounded agent reads, where a preferred frame arises from the agent's history rather than from the task ensemble. Neither compression, nor Clifford coordinates, nor symmetry breaking, nor a "self" is shown here to be thermodynamically necessary in general.

10 A classical filtering illustration

The following hidden Markov model compares finite observation windows with a Bayes filter in an aliased environment. Its reward is classification accuracy, not the survival functional, so it illustrates the task-level value of retained history rather than testing a theorem.

Proposition 26 (A finite-history classification separation)**.** *Let a fair binary hidden state flip with probability ε per step. Conditional on the hidden path, observations are independent, with $\Pr(O = 1 \mid X = 1) = 1/2 + \delta$ and $\Pr(O = 1 \mid X = 0) = 1/2 - \delta$, where $0 < \delta < 1/2$. A decision using only the last k observations, the known model and independent private randomness has classification risk at least*

$$R_k \geq \tfrac{1}{2}(1 - 2\delta)^k.$$

For odd n, the majority of n observations ending at the decision time has risk

$$R_{\mathrm{maj,n}} \leq (n-1)\varepsilon + \exp\bigl(-2n\delta^2\bigr).$$

For every fixed k, some finite n and positive ε make the second bound smaller than the first.

Proof. Every observation pattern of length k, conditional on either terminal hidden value, has probability at least $(1/2-\delta)^k$: each emission factor has that lower bound, and averaging over hidden paths preserves it. For the fair prior the optimal risk is $\frac{1}{2}\sum_o \min\{P(o \mid X = 0), P(o \mid X = 1)\}$. Summing the lower bound over 2^k patterns gives the first inequality; randomization cannot improve the optimal rule.

With probability at least $1-(n-1)\varepsilon$ there is no hidden flip in the n-observation episode. On that event the observations are independent samples with mean $1/2 + \delta$ or $1/2 - \delta$, so Hoeffding's inequality [31] bounds the majority error by $\exp(-2n\delta^2)$. The complement is charged its full probability. Choose odd n so that the exponential is below half the first bound, then choose $\varepsilon > 0$ small enough to make the remaining term smaller than that half. □

For $\delta = 0.05$, $k = 8$, $n = 1001$ and $\varepsilon = 10^{-4}$, these bounds are $R_{\mathrm{maj,n}} \leq 0.106704 < 0.215234 \leq R_8$. The count can be stored in $\lceil \log_2(n+1) \rceil = 10$ bits, in addition to the episode clock, decision and calculation resources; keeping every raw reading would require n bits instead. The comparison excludes additional correlated side information for the window rule. Acquisition, counter updates and any reset of its workspace still require an implementation budget; the classification bounds do not price them. It establishes a finite task separation, not a fixed physical memory that beats all window lengths, and not a bound on thermodynamic profit.

The hidden state $s_t \in \{0, 1\}$ keeps its value with probability $1 - \varepsilon$, $\varepsilon \in [10^{-3}, 10^{-0.7}]$, so the correlation length is $\ell \sim 1/\varepsilon$. Observations are aliased: $\Pr(o_t = A \mid s_t = 0) = 0.5 + \delta$

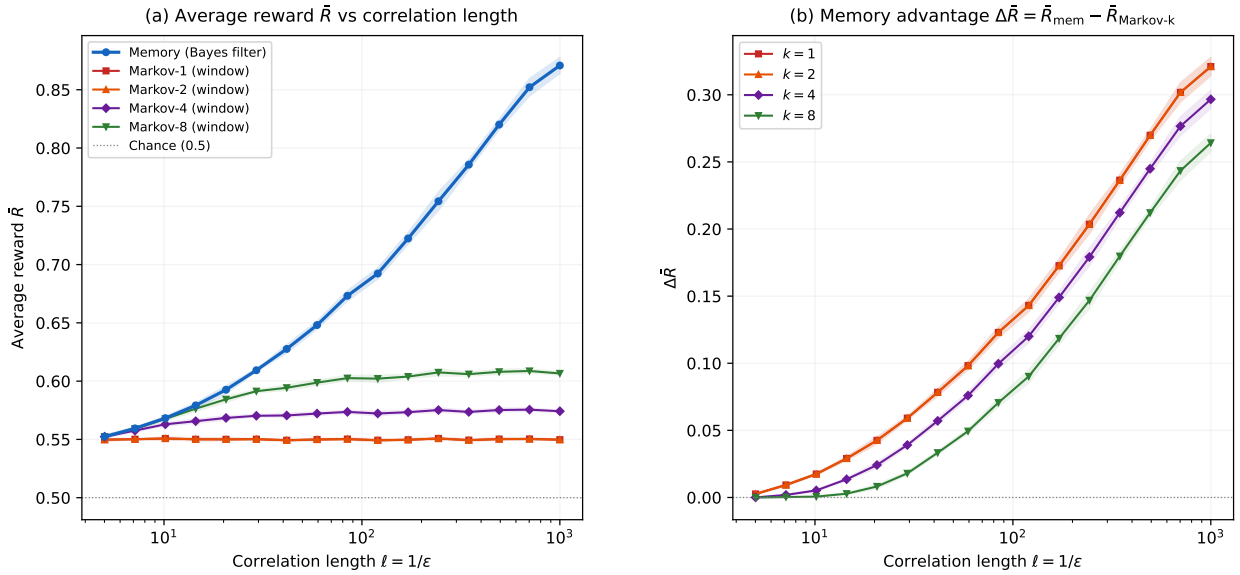


Figure 1: **Finite-window saturation and Bayes-filter advantage.** Historical sweep: $T = 10^5$ total steps, 2000 discarded, 10 seeds. Shading uses 1.96 times the across-seed standard deviation (with NumPy's population convention) divided by $\sqrt{10}$; these are normal-approximation error bands, not rigorous finite-sample confidence intervals. **(a)** Average reward $\bar{R}$ against correlation length $\ell = 1/\varepsilon$. The Bayes filter (blue, bold) rises monotonically; Markov-k agents saturate at k-dependent levels. **(b)** Gap $\Delta\bar{R} = \bar{R}_{\text{mem}} - \bar{R}_{\text{Markov-}k}$, which increases with ℓ and is larger for smaller k.

and $\Pr(o_t = A \mid s_t = 1) = 0.5 - \delta$ with $\delta = 0.05$, so a single observation carries 0.0072 bits about the state. The reward is 1 when the action equals s_t. All agents know the model and compute exact posteriors; Markov-k agents ($k \in \{1, 2, 4, 8\}$) use only the last k observations from a uniform prior, and the memory agent runs the full Bayes filter. The supplied script runs $T = 10^5$ total steps, discards the first 2000, and uses 10 seeds at 16 logarithmically spaced values of ε. The supplementary 16-row summary reports this sweep.

The reward of the Bayes filter increases with the correlation length, while each Markov-k agent saturates at a level ordered by k, all below the filter for $\ell \gtrsim 20$ (Figure 1). The Markov-1 and Markov-2 curves nearly coincide at small ℓ because short windows add little in this aliasing regime. The sampled long-correlation regime shows an advantage over the four specified windows. For the given model and observation history, the full posterior Bayes decision minimizes classification risk. The example does not establish tight saturation constants or superiority across different reading and resource menus. Theorem 16 supplies the conversion to work for a different task, in which the retained statistic is used for feedback on a system whose state it predicts. The supplementary source record gives the script, historical sweep and legacy-data distinction; these classification data do not computationally verify a survival-functional theorem.

11 Discussion

11.1 The memory–self–introspection route

In the task of Section 6, a work gain requires a memory carrying the relevant correlation. A protocol matched to its posterior realizes the value $k_B T\, I(X{:}M)$; every error in that posterior has a computable price. Reference calibration builds the required readout model, and reuse pays its preparation cost through the square-root amortization rule. Paper III proves the stated scalar Gaussian drift law. These are the concrete links in the series' memory–self–introspection route. Proposition 22 adds the reset condition for its allowed menu, and Proposition 23 realizes positive task value with two partially mixed records. A useful second record reduces the reset price; a full apparatus cycle still has to renew the source, recording stock and bath supply. The conditions belong to this task and its allowed protocols. Driven Markovian systems can persist without memory or a self-model.

The results concern function. They show what an agent must hold and do to cash a record, and they do not show that holding a model of one's own readout is, or produces, experience. The relation between such functional self-models and first-person organization is treated in the Consciousness series with its own arguments.

The task fixes the record source, pointer basis and feedback device. Paper II addresses how an agent selects what it reads; Paper III addresses detection, the price of tracking, and repair selected by the history of its body-gated access. The three correlation models here remain distinct: the Lorentz–Drude bath carries pointer information without designated fragments, the two-qubit environment carries none, and collision records have no autonomous recurrence. Paper III combines the collision records, a bounded changing instrument, a field-coupled actuator, harvest and reset on one ledger [29]. It exhibits a positive operating margin after body upkeep under supplied food, record purity, drive preparation, phase reference and timing. The calculation does not make those inputs products of the loop. Paper III also constructs a continuing finite joint filter: a bounded changing instrument and controlled body generate the history that revises their model, with a uniform quantization-risk bound and adaptive reference timing. The updating mechanism is thus explicit under its stated transition law and resource permissions. Its processor, reference supply and drive preparation still need one sustainable apparatus ledger.

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Reference Frames Selected by History: The Capability and Cost of a Bounded Agent's Self-Frame

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Abstract

With limited access, a system must learn how a task appears through its own channels before it can choose useful reads. We study this retained relation and the foreground it selects as a self-frame. For a Gaussian task with an unknown, isotropically distributed coupling and $b < D$ linear reads per step, the posterior mean determines the optimal reading subspace and prediction. Beating a history-independent foreground requires a calculable amount of information about the coupling. We distinguish the sharp information price of subspace alignment from the signed information needed to use that subspace: perfect alignment alone can leave every prediction uninformative. In-frame evidence has a further blind spot. It cannot locate the omitted direction or, when signal and noise strengths are unknown, distinguish world noise from its own misalignment. Exact drift curves show when a frozen model becomes worse than no memory and when a retained frame loses its advantage. Off-frame probes revise the frame at a cost in reads and storage; for eight channels and three reads, the specified policy lowers static risk from 0.86 to 0.26. Weighting predictions by their calculated reliability extends the drift range in which this policy beats the fixed-foreground comparator. These task-specific results supply a formation mechanism within the memory–self–introspection route. A learned protective encoding supplies a separate physical example of history changing the relations used by a body. A quantum reference gives a second, finite resource realization of a self-frame. Its advantage in any bounded decision problem is bounded by its asymmetry. For a single qubit phase token we obtain the exact optimal discrimination probability and its loss under dephasing.

1 Introduction

1.1 The question

A system that persists by using its records cannot use all of them at once. It reads some channels and not others, keeps some correlations and lets others go. Whatever rule makes

that choice embodies a view of where the task lies in the system's own coordinates. If the system's coupling to the world is not known in advance, that view has to be learned, and once learned it decides which evidence the system will see next. This paper asks four questions about such a view. When is it informationally necessary? How can it be formed from the agent's own records? What capability does it buy? What does it cost, including what it prevents the agent from seeing?

The series calls this organized view the ego. The word is used here for a definite object: a retained model of how a task-relevant direction sits in the agent's own channels, together with the foreground of reads that the model selects. We call the pair a self-frame. It is a model of the agent's own coupling, because the direction it describes is the image of a world direction under that coupling.

1.2 Position within the series

This paper is the second of the T-DOME (Thermodynamic Dynamics of Observer-Memory Entanglement) series, the third line of a programme whose other lines are HAFF and Q-RAIF.

HAFF-E supplies the actual record protocol [5]; Q-RAIF supplies physical interfaces and their prices [6]. Here we ask how the retained history organizes the reader itself.

Paper I bounds the extra work available from a record by $k_B T\, I(X{:}M)$ in a matched feedback task, explains when that bound is attained, and includes preparation, updates and old correlations in the complete account [3]. It leaves open how an agent comes to select what it keeps. Paper III studies drift detection, the price of tracking a changing instrument, and history-selected repair of body-gated access [4]. The present paper sits between them. Its self-frame is formed from records, it decides which records are gathered next, and its blind spot is the reason a calibration loop is needed at all.

The main results progress from the posterior capability (Theorem 2) to the information needed to select and use a foreground (Theorem 4 and Proposition 5), then to its blind spot and drift. Section 8 gives an explicit revision policy with exact stationary risk. Section 5 treats supplied quantum references separately; Section 9 connects the formation mechanism to learned protection.

1.3 Related work

Bounded rationality [11], rate–distortion theory [12, 13], rational inattention [14] and the information bottleneck [15] treat the choice of what to keep as an optimization under an information constraint. Budgeted learning studies learners that may observe only a few attributes of each example; Cesa-Bianchi, Shalev-Shwartz and Shamir show, among other results, that a budget on the attributes available at prediction time can make arbitrary accuracy unattainable even when a perfect linear predictor exists [16] . The present setting is close to theirs, with a different emphasis: the unknown is the agent's own coupling, the foreground is a function of the agent's memory, and the questions are the information that the foreground must carry and the evidence it excludes. Reference frames as physical resources are reviewed by Bartlett, Rudolph and Spekkens [17]; Skotiniotis and Gour give the accessible information of frame-alignment protocols an operational role for the G-asymmetry [18] .

Theorem 4 is the classical counterpart for a direction on a sphere: an application of the Gibbs variational principle, which we do not claim as new mathematics. Its use here is to price the self-frame of a bounded agent and to connect that price to the agent's capability and blind spot.

2 The budgeted reader

2.1 Channels, task and coupling

At each step t the agent's D channels register a fresh input $x_t \sim \mathcal{N}(0, I_D)$, independent of everything before. After the agent responds, the consequence

$$y_t = \rho\, u_t \cdot x_t + \xi_t, \qquad \xi_t \sim \mathcal{N}(0, s^2), \tag{1}$$

is revealed, with $\rho > 0$ and $|u_t| = 1$. Unless stated otherwise the agent knows ρ and s; Theorem 12(iii) drops this. The agent is scored by the squared error of its anticipation $\hat{y}_t$ of y_t. More generally the consequence may be a vector $y_t = A_t x_t + \xi_t \in \mathbb{R}^p$ with noise covariance of trace ν; we write $W = A^\top A$.

The direction u_t is expressed in the agent's channel coordinates. If e is the task-relevant direction in world coordinates and $R_t \in O(D)$ is the agent's coupling, which maps its channels to the world, then $u_t = R_t^\top e$. A model of u_t is therefore a model of the agent's own coupling as far as this task is concerned. The records cannot separate a change of R_t from a change of e; only their relation enters. Distinguishing the two requires a reference with a known world direction, the role played by reference reads in Paper III, or a known push through an actuator that shares the agent's sensing coupling; Paper III shows that pushes through an independent coupling do not help [4].

Symmetric ensemble. The initial coupling is Haar distributed, so u_0 is uniform on the sphere S^{D-1}, and more generally the law of A_0 is invariant under $A_0 \mapsto A_0 O$ for $O \in O(D)$. Nothing in the task specification prefers any channel.

2.2 The reading budget

At each step the agent may make at most b linear reads $w_1 \cdot x_t, \ldots, w_b \cdot x_t$ with orthonormal w_j chosen from what it retains. The span of the reads used for the response is its *foreground* P_t, an orthogonal projection of rank $k \le b$. Everything the agent retains from past steps is its memory M_t; it may contain past reads, past outcomes and any summaries of them. The foreground and the response are functions of M_t and of the current reads only.

A full probe reads all D channels. With $b < D$ it cannot be made at every step. The resource protocol charges reads per step, start-up probes and retained numbers or bits; it does not assign a thermodynamic price. Section 8.3 relates it to the work currency of Paper I.

Definition 1 (Self-frame)**.** A self-frame is a retained state from which the agent computes, at each step, an estimate of how the task direction sits in its own channels and the foreground that this estimate selects. Its alignment at step t is $c_t = \mathbb{E}|P_t u_t|^2$ for rank-one tasks.

2.3 Comparators

All comparisons in this paper charge the same b reads per step and the same start-up probes.

- *No memory*: $\hat{y} = 0$.
- *No self-frame*: a foreground fixed independently of the history (for definiteness the first b channels), with in-frame coefficients learned from outcomes.
- *Fixed self-frame*: a foreground formed once from the start-up probes, with in-frame coefficients learned afterwards.
- *Frozen self-model*: frame and coefficients formed once and never revised.
- *Calibratable self-frame*: $b - 1$ coordinate probes plus one orthogonal residual read, which together supply the response along the current frame and revise it.

3 What a foreground is worth

Theorem 2 (Capability of a budgeted reader)**.** *Let M be any memory independent of the current input, let P be a rank-k foreground that is a function of M, and let $\hat{y}$ be any function of (Px, M). For the vector task,*

$$\mathbb{E}|y - \hat{y}|^2 \;\geq\; \nu + \mathbb{E}\,\mathrm{tr}\,W - \mathbb{E}\,\mathrm{tr}\big(P\,\widehat{W}_M P\big) \;\geq\; \nu + \mathbb{E}\,\mathrm{tr}\,W - \mathbb{E}\sum_{i\leq k}\lambda_i(\widehat{W}_M), \tag{2}$$

where $\widehat{W}_M = \mathbb{E}[A \mid M]^\top \mathbb{E}[A \mid M]$ and $\lambda_1 \geq \lambda_2 \geq \cdots$ are its eigenvalues. The first bound is attained by $\hat{y} = \mathbb{E}[A \mid M]Px$; the second by taking P to be a top-k eigenspace of $\widehat{W}_M$. For the rank-one task (1)*, with $m = \mathbb{E}[u \mid M]$,*

$$\mathbb{E}(y - \hat{y})^2 \;\geq\; s^2 + \rho^2\big(1 - \mathbb{E}|Pm|^2\big) \;\geq\; s^2 + \rho^2\big(1 - \mathbb{E}|Pu|^2\big). \tag{3}$$

Proof. Write $x = Px + (I - P)x$. Given M, the input x is independent of (A, M), and for fixed P the two Gaussian components are independent of each other. Hence $\mathbb{E}[(A(I-P)x) \mid Px, M] = \mathbb{E}[A \mid M]\,\mathbb{E}[(I-P)x] = 0$ and $\mathbb{E}[APx \mid Px, M] = \mathbb{E}[A \mid M]Px$. The Bayes response is therefore $\mathbb{E}[A \mid M]Px$, and its risk is $\mathbb{E}|y|^2 - \mathbb{E}|\mathbb{E}[A \mid M]Px|^2 = \nu + \mathbb{E}\,\mathrm{tr}\,W - \mathbb{E}\,\mathrm{tr}(P\widehat{W}_M P)$, using $\mathbb{E}[xx^\top] = I$. No other response has smaller risk. Ky Fan's maximum principle gives $\mathrm{tr}(P\widehat{W}P) \leq \sum_{i\leq k}\lambda_i(\widehat{W})$ for every rank-k projection, with equality on a top-k eigenspace. For the rank-one task $\widehat{W} = \rho^2 mm^\top$, which gives the first inequality in (3). For the second, $\mathbb{E}|Pu|^2 = \mathbb{E}\,\mathbb{E}[|Pu|^2 \mid M] \geq \mathbb{E}|P\,\mathbb{E}[u \mid M]|^2$ by Jensen's inequality, since P is fixed given M. □

Corollary 3 (Three limits)**.** *(a)* No history. *If M is independent of the coupling, then $\mathbb{E}[A \mid M] = 0$ in the symmetric ensemble and every response has risk at least $\nu + \mathbb{E}\,\mathrm{tr}\,W$, the risk of $\hat{y} = 0$.*

(b) Known task. *If M determines A, the best rank-k risk is $\nu + \sum_{i>k} \lambda_i(W)$, attained by the top-$k$ eigenspace of W. Each added dimension strictly helps while $\lambda_{k+1} > 0$, and the optimal subspace is unique when $\lambda_k > \lambda_{k+1}$. If $W = \lambda I$, every rank-k subspace is optimal.*

(c) Frame-free level. *If the foreground is independent of the coupling, then $\mathbb{E}|Pu|^2 = k/D$ and every policy has risk at least $s^2 + \rho^2(1 - k/D)$ in the rank-one task.*

Proof. (a) Invariance gives $\mathbb{E}[A] = \mathbb{E}[A]O$ for all $O \in O(D)$, hence $\mathbb{E}[A] = 0$ for $D \geq 2$; use (2). (b) Here $\widehat{W} = W$. (c) For P independent of the uniform u, $\mathbb{E}|Pu|^2 = \operatorname{tr} P/D$. □

Theorem 2 has the same form as Paper I's theorem that every feedback protocol assumes a posterior [3]: whatever a budgeted policy represents, its performance is bounded by the alignment of the foreground it actually uses. The agent need not store an explicit direction. It does need some retained state from which its foreground is computed, and no response computed from that state can do better than the alignment of its foreground allows.

Part (b) is the known-task case of rank selection, a spectral variational fact. It locates the source of any preferred subspace. When W is anisotropic with a known eigenbasis, the selection is fixed by the task, an explicit asymmetry. When the ensemble is symmetric, as here, no subspace is preferred before records arrive. A preferred foreground can then come only from the agent's history, which is the subject of the next section.

4 A frame is a record of the coupling

For uniform u and a fixed rank-k projection P, the overlap $t = |Pu|^2$ has the Beta law $\beta_{k,D} = \mathrm{Beta}(k/2, (D-k)/2)$ with mean k/D and moment generating function $\mathbb{E}e^{\kappa t} = {}_1F_1(k/2; D/2; \kappa)$. Define

$$\Lambda^*_{D,k}(c) = \sup_{\kappa \geq 0} \Big[\kappa c - \ln {}_1F_1\big(\tfrac{k}{2}; \tfrac{D}{2}; \kappa\big) \Big] \tag{4}$$

for $0 \leq c < 1$. It is the upper half of the Legendre transform of the log moment generating function of $\beta_{k,D}$: convex, zero for $c \leq k/D$ and increasing above.

Theorem 4 (The information a self-frame must carry)**.** *In the symmetric rank-one ensemble, let P be a rank-k foreground that is a function of the agent's memory. Then*

$$I(U; P) \;\geq\; \Lambda^*_{D,k}\big(\mathbb{E}|PU|^2\big). \tag{5}$$

For $k/D \leq c < 1$ the bound is attained: if P is Haar distributed and, given P, U has density proportional to $e^{\kappa|PU|^2}$ with respect to the uniform measure, with $\kappa \geq 0$ chosen so that $\mathbb{E}|PU|^2 = c$, then U is uniform and $I(U; P) = \Lambda^*_{D,k}(c)$. (For $\kappa < 0$ the same construction gives the lower branch of the Legendre transform, which (5) does not use.) Consequently, a policy whose risk is at most $r < s^2 + \rho^2(1 - k/D)$ has a foreground with

$$I(U; P) \;\geq\; \Lambda^*_{D,k}\Big(1 - \frac{r - s^2}{\rho^2}\Big) > 0. \tag{6}$$

Proof. Since U is uniform, $I(U;P) = \mathbb{E}_P\, D(\mathcal{L}(U \mid P)\|\mathrm{Unif})$. Fix a value of P and map $u \mapsto |Pu|^2$. Under the uniform law the image is $\beta_{k,D}$; let Q_P be the image of $\mathcal{L}(U \mid P)$. The data-processing inequality gives $D(\mathcal{L}(U \mid P)\|\mathrm{Unif}) \geq D(Q_P\|\beta_{k,D})$. The Gibbs variational inequality $D(Q\|\beta) \geq \kappa\mathbb{E}_Q t - \ln \mathbb{E}_\beta e^{\kappa t}$ for every κ gives $D(Q_P\|\beta_{k,D}) \geq \Lambda^*_{D,k}(\mathbb{E}_{Q_P} t)$. Averaging over P and using convexity of Λ^* (Jensen) proves (5). In the attaining construction, rotation invariance of the uniform measure and of the Haar law of P makes the marginal of U uniform. Given P, the law of U is the exponential tilt of the uniform law by $\kappa|Pu|^2$; its image is the corresponding tilt of $\beta_{k,D}$, both inequalities are equalities, and the relative entropy is $\kappa c - \ln {}_1F_1(k/2; D/2; \kappa)$ with κ chosen so that the tilted mean is c. For (6), Theorem 2 gives $\mathbb{E}|PU|^2 \geq 1 - (r - s^2)/\rho^2$, and Λ^* is increasing on $[k/D, 1)$. □

Proposition 5 (Information needed to use an aligned frame)**.** *Let U be uniform on S^{D-1}, $D \geq 2$, and let $m_M = \mathbb{E}[U \mid M]$. Set*

$$\Psi_D(a) = \sup_{t \geq 0}\{ta - \log \mathbb{E}e^{tU_1}\}, \qquad 0 \leq a \leq 1,$$

with $\Psi_D(1) = +\infty$; the inequalities use extended relative entropy. For any memory and any allowed rank-k foreground,

$$I(U;M) \geq \mathbb{E}\Psi_D(|m_M|) \geq \frac{D}{2}\mathbb{E}|m_M|^2 \geq \frac{D}{2}\mathbb{E}|Pm_M|^2. \tag{7}$$

Consequently a predictor with risk at most $r < s^2 + \rho^2$ needs $I(U;M) \geq \frac{D}{2}[1 - (r - s^2)/\rho^2]$ nats. This bound concerns the signed conditional mean needed for prediction; it is not asserted to be sharp.

Proof. For each posterior, the relative-entropy variational inequality applied to $tv \cdot U$, with $v = m_M/|m_M|$, gives $D(\mathcal{L}(U \mid M)\|\mathrm{Unif}) \geq \Psi_D(|m_M|)$. At $m_M = 0$ take $t = 0$. The uniform sphere has odd moments zero and

$$\mathbb{E}U_1^{2j} = \frac{(2j-1)!!}{D(D+2)\cdots(D+2j-2)} \leq \frac{(2j-1)!!}{D^j}.$$

Termwise comparison of the moment-generating series gives $\log \mathbb{E}e^{tU_1} \leq t^2/(2D)$. Choosing $t = D|m_M|$ gives the second inequality in Eq. (7); averaging gives the mutual information. Projection contracts the norm. Finally Theorem 2 implies $\mathbb{E}|Pm_M|^2 \geq 1 - (r - s^2)/\rho^2$ for any such predictor. □

The attaining law in Theorem 4 is invariant under $U \mapsto -U$ conditional on P. Its posterior mean is therefore zero: a memory consisting only of that P has no predictive advantage, despite its high alignment. At the limit, knowing the unoriented line $UU^\top$ gives alignment one and still leaves the sign undetermined. Outcome-bearing records, not merely an axis label, supply the signed relation required by the response. The alignment table below is an information bound for choosing a subspace, not an achievable risk–information curve. For a finite memory its entropy bounds both the alignment price and the signed price. The step-capacity argument below likewise applies to either necessary information bound.

D	$c = 0.5$	$c = 0.9$	$c = 0.99$	ratio to $\frac{D}{2}\log_2\frac{1}{1-c}$ at $c = 0.9$
2	0	1.18	2.92	0.354
3	0.21	2.80	6.19	0.561
8	2.46	11.07	22.76	0.833
16	6.45	24.35	49.33	0.916
64	30.49	104.07	208.78	0.979
256	126.50	422.97	846.58	0.995

Table 1: Minimum frame information $\Lambda^*_{D,1}(c)$ in bits for a one-dimensional foreground (Theorem 4). For $D = 2$, $c = 1/2$ is the frame-free alignment.

Corollary 6 (Price of forming a frame)**.** *(a) A memory from which a foreground of alignment c is computed holds at least $\Lambda^*_{D,k}(c)/\ln 2$ bits.*

(b) Each step, whatever reads are made, conveys at most $C_1 = \frac{1}{2}\ln(1 + \rho^2/s^2)$ nats about a static coupling. Forming a foreground of alignment c from scratch therefore takes at least $\Lambda^*_{D,k}(c)/C_1$ outcome-bearing steps.

Proof. (a) $I(U;P) \le I(U;M) \le H(M)$. (b) Let H_{t-1} be the history. The reads at step t are functions of x_t and H_{t-1}, so $I(U;\text{reads}_t, y_t \mid H_{t-1}) \le I(U; y_t \mid x_t, H_{t-1})$. Given (x_t, H_{t-1}) the outcome has variance at most $\rho^2 x_t^\top \Sigma x_t + s^2$, with Σ the posterior covariance of u, and given U as well its entropy is that of $\mathcal{N}(\cdot, s^2)$. The Gaussian maximum-entropy bound and Jensen's inequality give at most $\frac{1}{2}\ln(1 + \rho^2 \operatorname{tr}\Sigma/s^2) \le C_1$, since $\mathbb{E}x_t^\top \Sigma x_t = \operatorname{tr}\Sigma \le 1$. The chain rule sums the steps, and $I(U;P) \le I(U;H_n)$. □

Table 1 lists $\Lambda^*_{D,1}$. For $D = 2$ the transform is $\sup_\kappa[\kappa(2c-1) - \ln I_0(\kappa)]$, the von Mises form on the doubled angle; the two expressions agree to 10^{-15} in the check. A Laplace evaluation of the moment generating function for large D gives $\Lambda^*_{D,1}(c) = \frac{D}{2}\ln\frac{1}{1-c}\,(1 + o(1))$ at fixed c, and the last column shows the approach. A self-frame thus costs about half a natural logarithm of the inverse misalignment per channel, the same per-direction precision cost that appears in finite self-models generally [7].

Remark 7 (Symmetry broken by the record)**.** If the agent's update rule is equivariant, so that rotating all channel coordinates rotates its records, memory and foreground alike, then the marginal law of the foreground over histories is Haar: the ensemble prefers no subspace. Conditional on one history the foreground is concentrated, and $I(U;P)$ measures how far. This is the precise sense in which a reference frame is selected spontaneously here. The symmetry of the task ensemble is intact; each history breaks it, and the frame is the trace that the history leaves. The reading budget makes a selection valuable (Corollary 3(c)); the records decide which frame is selected.

Remark 8 (Group-theoretic and quantum settings)**.** Nothing above uses the full orthogonal group beyond its transitivity on directions. If the channels carry a representation of a smaller group acting transitively on the sphere, for example $SO(3)$ acting on the pure quaternions of $Cl(0,2) \cong \mathbb{H}$, the overlap law and Theorem 4 are unchanged. In quantum settings an agent without a frame is restricted to group-averaged operations, and alignment resources

are quantified by asymmetry measures [17, 18]. The next section gives a quantum task bound and an exact finite reference. The spherical information price is not identified with quantum asymmetry.

5 A quantum reference changes the available reading

The classical foreground is a policy selected by memory. A retained quantum reference changes what a symmetry-restricted policy can measure. This is an operational bridge to a fixed-point algebra, rather than an identification of that algebra with a classical subspace.

Proposition 9 (Bounded task value of a retained reference)**.** *Let a compact group G act on a finite reference system, with twirling $\mathcal{G}$. In a specified task the reference starts in τ, independent of the hidden task label; target states and permissions are unchanged when the reference is replaced by $\mathcal{G}(\tau)$. The same class of physical protocols is allowed in both experiments. If losses lie in $[0, L]$, their optimal Bayes risks satisfy*

$$|R^*(\tau) - R^*(\mathcal{G}(\tau))| \leq \frac{L}{2}\|\tau - \mathcal{G}(\tau)\|_1 \leq L\sqrt{\frac{\mathcal{A}_G(\tau)}{2}}, \tag{8}$$

where $\mathcal{A}_G(\tau) = D(\tau\|\mathcal{G}(\tau))$ uses natural logarithms. For jointly G-invariant measurement permissions, an invariant reference induces only invariant effects on the target.

Proof. For each hidden label and each allowed protocol, the input trace distance is $\|\tau - \mathcal{G}(\tau)\|_1/2$. Contractivity bounds the total variation of its outcome laws by the same number. A loss in $[0, L]$ changes by at most L times that variation. Averaging and taking the infimum over the unchanged protocol class gives the first bound. Quantum Pinsker gives the second. The twirled-state relative entropy equals the entropy increase under twirling [2, Proposition 2 and Theorem 3]; the task bound is its combination with distinguishability. If a joint effect commutes with $U_S(g) \otimes U_R(g)$ and τ is invariant, partial expectation over R gives an effect commuting with $U_S(g)$. □

Proposition 10 (One finite phase reference and its passive lifetime)**.** *Two equiprobable target states are $\rho_\pm = (I \pm X)/2$. The retained reference $\tau_r = (I + rX)/2$, $0 \leq r \leq 1$, is prepared relative to their source phase. Allowed joint effects commute with $Z_S + Z_R$. The exact optimal success probability is*

$$P^*(r) = \frac{1}{2} + \frac{r}{4}. \tag{9}$$

With independent phase-dephasing of the unused reference, $r(t) = r_0 e^{-\kappa t}$. A required advantage $\Delta > 0$ above chance can persist without renewal only until $t \leq \kappa^{-1} \ln[r_0/(4\Delta)]$, for $4\Delta \leq r_0$.

Proof. The optimum invariant measurement is the Helstrom measurement of the two jointly twirled input states. Their difference is

$$\mathcal{G}_{SR}(\rho_+ \otimes \tau_r - \rho_- \otimes \tau_r) = \frac{r}{4}(X \otimes X + Y \otimes Y).$$

Its nonzero eigenvalues are $r/2$ and $-r/2$, in the common-charge space spanned by $|01\rangle, |10\rangle$. The trace norm is r, giving Eq. (9); its eigenprojections are permitted. Dephasing multiplies the reference's transverse Bloch vector by $e^{-\kappa t}$, and substitution proves the lifetime. □

Corollary 11 (One noisy carrier as a record and a frame)**.** *Use $X_S + X_R$ as the conserved charge and $(I \pm Z)/2$ as the equiprobable target states. A retained reference with Bloch vector $(0, y, z)$ gives optimal invariant-measurement success*

$$P^*_{\text{frame}} = \frac{1}{2} + \frac{\sqrt{y^2 + z^2}}{4}.$$

Take the reference to be the initially positive record in HAFF-B's transverse-qubit model [1], with known elapsed time and known generator. If $E^*_{\text{record}}(t)$ is the optimal error for reading its two possible initial signs directly, then

$$P^*_{\text{frame}}(t) = \frac{3}{4} - \frac{1}{2} E^*_{\text{record}}(t). \tag{10}$$

Thus the same physical carrier loses its record distinguishability and its one-use reference advantage on the same slow decay scale, even though its exact fixed algebra is scalar.

Proof. A rotation about the charge axis brings the transverse reference vector to one axis without changing its length. It commutes with the symmetry and maps allowed effects to allowed effects. The calculation of Proposition 10, with the Pauli axes relabelled, therefore gives the first equation. The two noisy record states have opposite Bloch vectors of length $\sqrt{y^2 + z^2}$, so their Helstrom error is $(1 - \sqrt{y^2 + z^2})/2$, proving Eq. (10). □

The baseline reference $r = 0$ permits only diagonal target effects and cannot distinguish ρ_+ from ρ_-. Retaining $r > 0$ makes a phase-sensitive reading possible without a non-invariant joint measurement. It does not create an absolute phase: the entire input may be jointly twirled with no change of the reading probabilities. This is the established relational-reference mechanism [17], here used to connect task capability and passive noise. A fresh symmetric history cannot create $r > 0$ by covariant operations alone; each conditional output of an invariant-outcome covariant instrument is still invariant. The source-relative preparation is therefore a priced input, not a frame created from an uninformative record.

The lifetime concerns an unused token followed by one reading. Repeated measurement can disturb it further and requires a reuse account. If a protocol must keep its advantage in this drifting task, it must obtain new phase information, protect that information or refresh the reference; renaming a stale token a self-model does not change its capability. This is a finite quantum counterpart of the self-frame's calibration problem, not a sufficient condition for experience or a theorem that all tasks need asymmetry. Charge-label tasks, for example, can be solved with invariant effects.

6 The frame shapes its own evidence

Once a foreground is chosen, the agent's next records are read through it. The following theorem says what those records can and cannot reveal.

Theorem 12 (In-frame identifiability and its blind spot)**.** *Fix a rank-k foreground P and consider in-frame records (Px_t, y_t) from a static coupling.*

(i) Their joint law depends on (ρ, s, u) only through $b = \rho Pu \in \operatorname{ran} P$ and $V = \rho^2 + s^2$.

(ii) With ρ and s known, the likelihood is constant on each fibre $\{u : Pu = w\}$, a sphere of dimension $D - k - 1$ in the background. The posterior law of the background direction $(I - P)u/|(I - P)u|$ is not updated by any number of in-frame records.

(iii) With ρ and s unknown, every triple (ρ', s', u') with $\rho' Pu' = b$, $\rho'^2 + s'^2 = V$ and $|b| \le \rho' \le \sqrt{V}$ produces the same law. The risk reduction available by reorienting the frame, $\rho'^2 - |b|^2$, ranges over $[0, V - |b|^2]$ within this family: the unexplained variance may be entirely world noise or entirely the agent's own misalignment.

(iv) No rule using only in-frame records and the frame achieves risk below $V - |b|^2 = s^2 + \rho^2(1 - |Pu|^2)$, and in-frame regression attains it asymptotically.

Proof. (Px, y) is Gaussian with mean zero, $\operatorname{Cov}(Px) = P$, $\operatorname{Cov}(Px, y) = \rho Pu$ and $\operatorname{Var} y = \rho^2 + s^2$, which proves (i). Parts (ii) and (iii) follow, since the likelihood of any sample is a function of (b, V); with ρ, s known, b fixes Pu and nothing else, and the conditional law of the background direction given Pu is therefore unchanged by the data. For (iv), given u the part $\rho(I - P)u \cdot x + \xi_t$ of y_t has variance $V - |b|^2$ and is independent of Px_t and of all earlier records, so no rule based on them removes it; regression of y on Px estimates b consistently and attains $V - |b|^2$. □

Theorem 12 states the cost side of a self-frame. A frame that serves the agent well also decides which evidence it will receive, and that evidence cannot show where the missing part of the task has gone. Part (iii) adds a second limitation: from its own in-frame records the agent cannot know whether anything is missing. A large residual is compatible with a well-aligned frame in a noisy world and with a misaligned frame in a quiet one. Separating the two requires either a known noise level or reads outside the frame. This is the agent-level form of the distinction between a change of the world and a change of the agent's own coupling.

Corollary 13 (The two-channel case: an even residual)**.** *Let $D = 2$, $k = 1$, foreground v and $u = \cos\theta\, v + \sin\theta\, v_\perp$. The in-frame law depends on θ only through $\cos\theta$, so θ and $-\theta$ are indistinguishable, and the in-frame residual variance $s^2 + \rho^2 \sin^2\theta$ is even in θ. The Fisher information of one in-frame record about θ is*

$$F(\theta) = \frac{\rho^2 \sin^2\theta \left(\rho^2 + s^2 + \rho^2 \cos^2\theta\right)}{\left(s^2 + \rho^2 \sin^2\theta\right)^2} = \frac{\rho^2(2\rho^2 + s^2)}{s^4}\,\theta^2 + O(\theta^4). \tag{11}$$

One off-frame read along $v_\perp$ gives $\operatorname{Cov}(v_\perp \cdot x, y) = \rho \sin\theta$, which is odd in θ and resolves the sign.

Proof. The covariance of $(v \cdot x, y)$ is $\Sigma = \begin{pmatrix} 1 & a \\ a & V \end{pmatrix}$ with $a = \rho\cos\theta$. For a Gaussian family with parameter-dependent covariance, $F = \frac{1}{2}\operatorname{tr}(\Sigma^{-1}\Sigma'\Sigma^{-1}\Sigma')$; with $a' = -\rho\sin\theta$ and $\det\Sigma = V - a^2$ this equals $a'^2(a^2 + V)/(V - a^2)^2$, which is (11). □

Paper III takes this even residual law as its detection model, proves that it cannot identify the sign of the mismatch, and shows that a regular even record law with a nonzero quadratic leading perturbation has Fisher information quadratic and relative entropy quartic in the mismatch near calibration, which fixes the time needed to detect drift [4]. Corollary 13 derives the structure from a budgeted self-frame: the residual is even because the frame reads only one of two channels, and the sign is the unobservable fibre of Theorem 12. A single off-frame read is the minimal probe that breaks the symmetry. The derivation fixes the structure (evenness, a two-point fibre, quadratic information), not the Gaussian mean model used in Paper III's computations.

7 Drift: fixed and frozen frames

Let the coupling drift. In continuous time we take u_t to be Brownian motion on S^{D-1} with generator $\sigma\Delta_{S^{D-1}}$. A time-homogeneous discrete Markov step equivariant under $O(D)$, including reflections, has the corresponding scalar structure: $\mathbb{E}[u_{t+1} \mid u_t] = \lambda_1 u_t$, and the degree-two harmonics have multiplier λ_2. Equivariance under $SO(D)$ suffices for $D \geq 3$, but not for a directed drift on the circle.

Theorem 14 (Fixed and frozen frames under drift)**.** *For spherical Brownian motion with generator $\sigma\Delta$,*

$$\mathbb{E}[u_t \cdot v] = e^{-(D-1)\sigma t}\, u_0 \cdot v, \qquad \mathbb{E}[(u_t \cdot v)^2] - \tfrac{1}{D} = e^{-2D\sigma t}\big((u_0 \cdot v)^2 - \tfrac{1}{D}\big). \tag{12}$$

(a) A frozen self-model with response $\hat{y} = g{\cdot}x$, $g = \rho u_0$ fixed, has risk $s^2 + 2\rho^2\big(1 - e^{-(D-1)\sigma t}\big)$. It is worse than having no memory after $t^* = \ln 2/((D-1)\sigma)$, and worse than the frame-free level once $2(1 - e^{-(D-1)\sigma t}) > 1 - b/D$. If g is estimated from n_0 full probes, the risk is $s^2 + 2\rho^2 + \epsilon_0 - 2\rho^2 e^{-(D-1)\sigma t}$ with $\epsilon_0 = (D(\rho^2 + s^2) + \rho^2)/n_0$.

(b) A fixed foreground spanned by u_0 and $b - 1$ orthonormal directions orthogonal to it, with exact in-frame coefficients, has risk $s^2 + \rho^2(1 - b/D)\big(1 - e^{-2D\sigma t}\big)$. It never rises above the frame-free level and approaches it at rate $2D\sigma$.

For an $O(D)$-equivariant discrete step the same formulas hold with $e^{-(D-1)\sigma}$ and $e^{-2D\sigma}$ replaced by λ_1 and λ_2 per step. The discrete risks need not be monotone when these multipliers are negative.

Proof. The functions $u \mapsto u \cdot v$ and $u \mapsto (u \cdot v)^2 - 1/D$ are spherical harmonics of degrees one and two, with Laplace–Beltrami eigenvalues $-(D-1)$ and $-2D$; their expectations decay accordingly. For an $O(D)$-equivariant step the stabilizer of u forces $\mathbb{E}[u' \mid u] = \lambda_1 u$ and $\mathbb{E}[u'u'^{\top} \mid u] = aI + \lambda_2 uu^{\top}$, with $Da + \lambda_2 = 1$. Iteration gives the same first- and second-moment formulas with the corresponding powers of λ_1 and λ_2. (a) The risk is $s^2 + \mathbb{E}|\rho u_t - g|^2 = s^2 + \rho^2 + |g|^2 - 2\rho\, g \cdot \mathbb{E}u_t$. With g the mean of $y_i x_i$ over n_0 probes, $\mathbb{E}[g \mid u_0] = \rho u_0$ and $\mathbb{E}|g - \rho u_0|^2 = \epsilon_0$ by the Isserlis identity $\mathbb{E}[(u \cdot x)^2 x_i^2] = 1 + 2u_i^2$. (b) The risk is $s^2 + \rho^2(1 - \mathbb{E}|Pu_t|^2)$, and (12) gives $\mathbb{E}(u_t \cdot u_0)^2 = \frac{1}{D} + (1 - \frac{1}{D})e^{-2D\sigma t}$ and $\mathbb{E}(u_t \cdot w)^2 = \frac{1}{D}(1 - e^{-2D\sigma t})$ for $w \perp u_0$; summing over the b directions gives the stated curve. □

The two Brownian curves separate two failure modes that a single visibility angle cannot tell apart. A self-model that keeps both its frame and its coefficients becomes confidently wrong: its predictions remain as large as before while their direction decorrelates, and after t^* it does worse than predicting nothing. A frame that keeps learning its in-frame coefficients cannot become worse than the frame-free reader, because in-frame regression shrinks the coefficient on a stale direction, but it loses its whole advantage. Neither can recover the advantage without information from outside the frame, by Theorem 12. Paper III compares these curves with the time needed to detect drift and finds that a residual-variance test detects slow drift long before t^* [4].

Proposition 15 (Directed drift of a planar frame)**.** *Let $D = 2$ and $u_t = (\cos\varphi_t, \sin\varphi_t)$, with $\varphi_{t+1} = \varphi_t + \Delta_t$ modulo 2π, where the increments are independent with one common law. Write $\mu_j = \mathbb{E}e^{ij\Delta_t}$. For a known initial direction, a frozen predictor $g = \rho u_0$ has risk*

$$R_{\text{frozen}}(t) = s^2 + 2\rho^2[1 - \operatorname{Re}(\mu_1^t)].$$

A fixed one-dimensional foreground along u_0, using its exact current regression coefficient, has risk

$$R_{\text{axis}}(t) = s^2 + \frac{\rho^2}{2}[1 - \operatorname{Re}(\mu_2^t)].$$

These laws are $SO(2)$-equivariant. Reflection symmetry makes both multipliers real and recovers the scalar discrete formulas.

Proof. Independence gives $\mathbb{E}e^{ij(\varphi_t - \varphi_0)} = \mu_j^t$. The two risks are respectively $s^2 + 2\rho^2\mathbb{E}[1 - \cos(\varphi_t - \varphi_0)]$ and $s^2 + \rho^2\mathbb{E}\sin^2(\varphi_t - \varphi_0)$. Taking real parts proves both identities. For deterministic increments $\Delta_t = \phi$, they involve $\cos(t\phi)$ and $\cos(2t\phi)$, not $(\cos\phi)^t$ and $(\cos 2\phi)^t$. □

A directed rotation can repeatedly return a stale frame to alignment. The Brownian crossing time is therefore a property of that diffusion, not a universal lifetime inferred from rotational symmetry.

8 A calibratable frame and its price

8.1 The policy and its exact stationary risk

The calibratable agent stores a vector $g_t \in \mathbb{R}^D$, its estimate of $\theta_t = \rho u_t$. Before the input arrives it chooses a set S_t of $b - 1$ channel indices uniformly without replacement, independently of the world. It reads those coordinate directions and, as its last read, the normalized component of g_t orthogonal to their span. If this component vanishes it fills the slot with any unused orthogonal direction. These are exactly b orthonormal reads. From them it computes $g_t \cdot x_t$ and responds $\hat{y}_t = \gamma g_t \cdot x_t$, with $\gamma \in [0, 1]$. No repeated or additional read is required: coordinate values and the orthogonal residual reconstruct the noiseless linear functional algebraically. Each coordinate is probed with probability $p = (b-1)/D$. For each probed channel i it updates $g_{t+1,i} = g_{t,i} + \alpha(y_t x_{t,i} - g_{t,i})$; unprobed coordinates are kept. The frame-free comparator stores coefficients β_t on the fixed channels $S = \{1, \ldots, b\}$, reads them every step, updates them by the same rule with $p = 1$, and responds $\gamma\,\beta_t \cdot x_{t,S}$.

Proposition 16 (Exact stationary risk)**.** *Assume $1 < b < D$, $0 < \alpha < 2$, u_0 uniform, a drift step with $\mathbb{E}[u_{t+1} \mid u_t] = \lambda_1 u_t$ and a uniform invariant marginal, with factor $\lambda_1 = 1 - \kappa \in [0, 1]$, fresh inputs and noise, and probe choices independent of the world (correlations among the probe indicators of different channels do not matter). Let m be the number of tracked coordinates ($m = D$ for the calibratable frame, $m = b$ for the frame-free reader), p the probe probability ($p = 1$ for the frame-free reader), $T = \rho^2 m/D$ and $a = p\alpha(2 - \alpha)$. The tracking error e_t of the tracked coordinates has stationary moments*

$$\begin{aligned} c_\infty &= \mathbb{E}[e \cdot \theta] = -\frac{\kappa T}{1 - \lambda_1(1 - p\alpha)}, \\ A_\infty &= \mathbb{E}|e|^2 = \frac{p\alpha^2 m\big(\rho^2 + s^2 + \rho^2/D\big) + 2\kappa T + 2\kappa(1 - p\alpha)c_\infty}{a}, \end{aligned} \tag{13}$$

and the stationary risk with weight γ is

$$R(\gamma) = s^2 + \rho^2 - 2\gamma\,(T + c_\infty) + \gamma^2\,(T + 2c_\infty + A_\infty). \tag{14}$$

In particular $R(1) = s^2 + (\rho^2 - T) + A_\infty$, and the reliability weight $\gamma^* = (T + c_\infty)/(T + 2c_\infty + A_\infty)$ gives $R(\gamma^*) = s^2 + \rho^2 - (T + c_\infty)^2/(T + 2c_\infty + A_\infty) \le s^2 + \rho^2$; the weight lies in $[0, 1]$ under the stated nonnegative first-harmonic assumption.

Proof. Coordinate i is probed with indicator χ_i, $\mathbb{E}\chi_i = p$. The probe value yx_i has conditional mean θ_i given the past and the current coupling, and conditional variance $\rho^2 + s^2 + \rho^2 u_i^2$ by the Isserlis identity; its average over the uniform marginal of u_t, which holds at every t, is $\rho^2 + s^2 + \rho^2/D$. After the update the error is $e_i' = (1 - \alpha\chi_i)e_i + \alpha\chi_i n_i$ with n_i conditionally centred, so $\mathbb{E}|e'|^2 = (1 - a)\mathbb{E}|e|^2 + p\alpha^2 m(\rho^2 + s^2 + \rho^2/D)$ and $\mathbb{E}[e' \cdot \theta] = (1 - p\alpha)\mathbb{E}[e \cdot \theta]$. The drift step then replaces θ by $\theta + \Delta$ and e' by $e' - \Delta$, where Δ is independent of e' given u_t, $\mathbb{E}[\Delta \mid u_t] = -\kappa\theta$ on the tracked coordinates, $\mathbb{E}|\Delta|^2 = 2\kappa T$ and $\mathbb{E}[\Delta \cdot \theta] = -\kappa T$. Hence

$$\mathbb{E}|e_{t+1}|^2 = \mathbb{E}|e'|^2 + 2\kappa(1 - p\alpha)\mathbb{E}[e_t \cdot \theta_t] + 2\kappa T, \qquad \mathbb{E}[e_{t+1} \cdot \theta_{t+1}] = \lambda_1(1 - p\alpha)\mathbb{E}[e_t \cdot \theta_t] - \kappa T.$$

Both recursions are affine with contraction factors $1 - a$ and $\lambda_1(1 - p\alpha)$ of modulus less than one, so the moments converge to the fixed point (13) from any start. The response $\gamma\, g \cdot x$ has risk $s^2 + \mathbb{E}|\theta - \gamma g|^2$ with g supported on the tracked coordinates; expanding with $g = \theta_m + e$ gives (14). The minimizing weight follows by completing the square, and $R(\gamma^*) \le R(0) = s^2 + \rho^2$. □

Only the one-step correlation $\lambda_1 = \mathbb{E}[u_{t+1} \cdot u_t]$ of the drift enters. For spherical Brownian motion sampled at unit time, $\kappa = 1 - e^{-(D-1)\sigma}$. The weight γ^* is a property of the agent's own tracking error. An agent that knows its drift factor and noise level can compute it from (13); an agent that knows only its residuals cannot, because by Theorem 12(iii) its in-frame residual mixes its tracking error with world noise. The reliability weight is thus a small self-assessment, and its effect is measurable.

8.2 Capability and benefit threshold

Table 2 compares the five policies of Section 2 under one protocol. The exact stationary risk agrees with the simulation to within the Monte Carlo error at every drift level, including

κ	Frozen	No frame	Fixed	Calib.	Calib. (th.)	Calib. γ^*	No frame γ^*	Alignment
0	0.414	0.863	0.340	0.2608	0.2610	0.2607	0.863	0.991
1.40×10^{-5}	0.812	0.872	0.559	0.2856	0.2848	0.2852	0.872	0.967
8.75×10^{-5}	1.934	0.897	0.856	0.3367	0.3361	0.3350	0.897	0.921
3.50×10^{-4}	2.448	0.915	0.920	0.4188	0.4190	0.4116	0.914	0.849
1.40×10^{-3}	2.426	0.938	0.941	0.5743	0.5742	0.5477	0.935	0.718
5.54×10^{-3}	2.464	0.994	0.995	0.8411	0.8396	0.7568	0.985	0.506
1.23×10^{-2}	2.449	1.049	1.049	1.0383	1.0402	0.8969	1.030	0.353
2.15×10^{-2}	2.430	1.095	1.096	1.1740	1.1736	0.9944	1.064	0.238
4.61×10^{-2}	2.423	1.169	1.169	1.2516	1.2518	1.1043	1.116	0.128

Table 2: Time-averaged risk in the closed loop ($D = 8$, $b = 3$, $\rho = 1$, $s = 0.5$, start-up $n_0 = 64$ full probes, 24 000 steps, first 8 000 discarded, 200 trajectories with common random numbers). No memory gives 1.25. "Calib. (th.)" is $R(1)$ from Proposition 16; learning rates are set from the formulas, not tuned on the simulation. Alignment is $\mathbb{E}(u \cdot g/|g|)^2$ of the calibratable frame. Standard errors are at most 0.016 for the frame-free and fixed columns (dominated by the sample of couplings), at most 0.05 for the frozen column, and at most 0.0015 for the calibratable columns; the paired difference between the calibratable and frame-free readers has standard error at most 0.016.

levels where the calibratable frame loses. With a static coupling the calibratable frame reaches alignment 0.991 and risk 0.261, against 0.863 for the frame-free reader and 0.340 for the frame fixed after start-up. The frozen self-model is worse than no memory from $\kappa = 8.75 \times 10^{-5}$ on. The fixed frame becomes indistinguishable from the frame-free reader at moderate drift, as Theorem 14(b) predicts.

Without the reliability weight the calibratable frame loses to the frame-free reader between $\kappa = 1.23 \times 10^{-2}$ and 2.15×10^{-2}: it keeps responding at full strength along a direction it can no longer track. With γ^* for both readers it remains better through $\kappa = 4.61 \times 10^{-2}$. Table 3 gives the crossing drift κ_c from the exact formulas for other dimensions and budgets. Weighting by self-assessed reliability raises κ_c by factors from 2.1 ($D = 4$, $b = 3$) to 13.8 ($D = 32$, $b = 3$). Without weighting, $\kappa_c D^2/(b-1)$ stays between 0.14 and 0.94 over the computed cases: the sampled scaling is $\kappa_c \propto (b-1)/D^2$, or D^{-2} at fixed b, rather than $[(b-1)/D]^2$.

Corollary 17 (Static limit)**.** *With $\kappa = 0$ the stationary risk of the calibratable frame is $s^2 + \alpha D(\rho^2 + s^2 + \rho^2/D)/(2-\alpha)$ at weight 1, which tends to s^2 as $\alpha \to 0$, while the frame-free reader stays at or above $s^2 + \rho^2(1 - b/D)$. Under a reading budget $2 \le b < D$, a calibratable self-frame is therefore strictly better than every frame-free policy for all sufficiently slow drift.*

Proof. Set $\kappa = 0$ in (13). The frame-free bound is Corollary 3(c). Continuity of (13) in κ at fixed α extends the strict inequality to small κ. □

8.3 What the agent pays

The exact curves use noiseless linear reads and real-valued coefficients. Orthonormalizing the probe span is a computation on retained g and chosen indices before the input arrives. It

D	b	κ_c, weight 1	κ_c, weight γ^*
4	2	1.47×10^{-2}	3.82×10^{-2}
4	3	1.73×10^{-2}	3.68×10^{-2}
8	2	7.66×10^{-3}	3.49×10^{-2}
8	3	1.29×10^{-2}	5.78×10^{-2}
8	5	1.85×10^{-2}	7.19×10^{-2}
16	2	2.85×10^{-3}	2.23×10^{-2}
16	3	5.07×10^{-3}	4.00×10^{-2}
16	5	8.45×10^{-3}	6.44×10^{-2}
32	2	9.16×10^{-4}	1.26×10^{-2}
32	3	1.68×10^{-3}	2.31×10^{-2}
32	5	2.93×10^{-3}	3.93×10^{-2}

Table 3: Drift $\kappa_c = 1 - \lambda_1$ at which the calibratable frame and the frame-free reader have equal stationary risk, each with learning rate optimized from Proposition 16 ($\rho = 1$, $s = 0.5$). Below κ_c the calibratable frame is better.

needs arithmetic and working storage, not another observation. These resources are counted separately from the b reads. If the implemented response coefficient differs by a vector with mean squared norm at most ϵ^2, the squared-risk increase is bounded by $2\epsilon\sqrt{R - s^2} + \epsilon^2$, by Cauchy–Schwarz. Thus a strict benefit margin survives sufficiently precise finite arithmetic; precision is an additional paid input. Independent noisy hardware reads require their own covariance and calibration account.

All readers above use b reads per step and the same $n_0 D$ start-up reads. They differ in retained state. The frame-free reader keeps b coefficients. The fixed frame keeps b coefficients and its b directions. The calibratable frame keeps the D numbers of g, and Theorem 4 says that some such growth is unavoidable: alignment 0.991 in eight channels requires at least $\Lambda^*_{8,1}(0.991)$ nats in its foreground, 23.3 bits. In time the calibratable frame spends $b - 1$ of its b reads off its own frame. Paper I prices these resources when they are implemented thermodynamically. In Paper I's matched task, a record adds at most $k_B T$ times its task mutual information. Isothermal erasure of independent unbiased degenerate bits without usable side information costs at least $k_B T \ln 2$ per bit; general updates instead require the conditional-entropy and free-energy ledger (a known bit need not cost that amount). Reference preparation requires the free-energy difference for its specified Hamiltonian and initial state [3]. A pure degenerate one-bit reference has price $k_B T \ln 2$; an arbitrary reference does not have that universal price. The present paper keeps reads, probes and stored numbers as separate entries and does not convert them into a single price, because the conversion depends on the implementation.

9 Connections within the theory

9.1 A frame can change the body's protective relations

Foundation III gives a finite body implementation [10]. A diagnostic record of an unknown noise sign selects and later decodes an odd or even protective code on two body bits holding an unknown logical qubit and its inaccessible old reference. The controller sees the record, never the sign. In that protocol the learned code beats both fixed codes when $p > .51430317693$; at $p = .9$ its ideal loss is .0527213975, with early damage included. The full gate program, paid ports and source/return construction belong to Foundation III. This is a separate implementation, not an identification of Gaussian subspace alignment with quantum protection. The available encoders and diagnostic coupling remain inputs; Paper III studies history-selected maintenance when body access becomes unreliable. The positive subject-formation criterion proposed here is that learned relations organize the same system's future access, protection and revision, rather than merely naming a preferred basis. Paper III's body-gated repair gives this criterion an acting interpretation: the acquired model selects which future access is worth restoring. This is an interpretive connection between the models, not a theorem of general subject formation.

Paper I. Paper I establishes the task value of correlations and the conditions under which their preparation and maintenance pay [3]. Its feedback protocols imply a posterior; a record is not valuable independently of its task and implementation. Theorem 2 is the reading-side counterpart: every budgeted policy implies a frame, and its capability is the alignment of that frame. Theorem 4 prices the frame in information, as Paper I prices the record in work.

Paper III. Corollary 13 derives the even residual model that Paper III assumes, including the sign ambiguity and the quadratic Fisher signal. Paper III's reference reads play the role of the known world direction in Section 2: they separate a change of the agent's coupling from a change of the world, which in-frame records cannot do (Theorem 12(iii)). Its memory-under-drift proposition and Proposition 16 have the same structure, one for a scalar instrument parameter and one for a frame in D channels; the second shows that the cost of tracking a frame grows with the number of channels the frame excludes.

Consciousness III. Theorem 12 is an instance of a self-shaped unobservable fibre: what the agent cannot learn about itself is set by its own current frame and moves when the frame moves. Consciousness III develops the general theory of such fibres for finite self-models, including a floor for self-knowledge under drift [7].

Imprint and Brain Anchor. The reader of *The World as Imprint* learns which of two instrument axes it has and retains the corresponding information [8]. The self-frame here is its continuous, budgeted generalization, with the added feature that the learned model controls which records arrive. Brain Anchor's learned relation weights are another case of a frame formed from records and selecting later use [9].

10 Discussion

10.1 What is established

Within the budgeted reader, a self-frame is informationally necessary for any risk below $s^2+\rho^2(1-b/D)$ (Corollary 3 and Theorem 4); it can be formed from the agent's own records, at a price of at least $\Lambda^*_{D,k}(c)/C_1$ outcome-bearing steps and $\Lambda^*_{D,k}(c)/\ln 2$ bits (Corollary 6); it buys the capability computed in Theorem 2 and Proposition 16; and it has a blind spot that only reads outside the frame can repair (Theorem 12). The necessity is created by the reading budget. With $b = D$ the frame-free level falls to s^2 and selecting a foreground gains nothing, although the agent still needs memory of its coefficients.

For the memory–self–introspection route, this gives the self link a definite content. Task-relevant memory becomes a self-frame here when it selects reads through a learned model of the system's own coupling. The blind spot then gives a reason for acquiring new relations outside that frame. A system may implement this calibration function without representing a human-like self. Each statement concerns this task class. It is not claimed that every persistent system needs a self-frame; a system that can read everything it needs, or whose coupling is known, does not.

The self-frame is a functional object. Nothing here shows that holding one is, or produces, experience. The Consciousness series treats the relation between such functional self-models and first-person organization with its own arguments.

10.2 Drift boundaries and two conjectures

Proposition 18 (A drift factor does not determine the tracking problem)**.** *Let $U_t = S_tV$ with V initially uniform on the sphere and fixed, while $S_t \in \{-1, 1\}$ flips independently with probability $\kappa/2$ each step. This rotation-invariant Markov law has the same $\lambda_1 = 1-\kappa$ as a spherical diffusion with that first harmonic. Given a retained exact axis V, a legal reader of that axis has excess risk at most*

$$R - s^2 \le 4\rho^2 \left(\frac{w\kappa}{2} + e^{-wC_1}\right), \qquad C_1 = \frac{1}{2}\ln(1+\rho^2/s^2), \tag{15}$$

using only the last w outcome-bearing reads. In particular it has $O(\kappa\log(1/\kappa))$ excess risk at fixed signal-to-noise ratio.

Proof. Read $z_j = V \cdot x_j$ and estimate the current sign by the sign of $\sum_{j=t-w}^{t-1} z_j y_j$. Predict $\rho\widehat{S}_t V\cdot x_t$. A flip in the w transitions joining these outcomes to the current step has probability at most $w\kappa/2$. On the no-flip event, relabel the common sign positive. For one sample, $y = \rho z + s\eta$, with z, η independent standard normals. At $a = \rho/s^2$,

$$\mathbb{E}e^{-azy} = \mathbb{E}e^{-a\rho z^2 + a^2 s^2 z^2/2} = (1+\rho^2/s^2)^{-1/2}.$$

Chernoff's inequality bounds a wrong sign by e^{-wC_1}. A sign error costs $4\rho^2$ in squared prediction loss; the current Gaussian input is independent of the sign estimate. Taking $w = \lceil \log(1/\kappa)/C_1 \rceil$ proves the order. □

An exact axis is a supplied resource in this statement. A paid initialization with sufficiently small axis error adds a corresponding prediction error; finite precision is not free. The axis can also be learned without access to S_t: for $i \neq j$, $\mathbb{E}[y_t^2 x_{t,i} x_{t,j}] = 2\rho^2 V_i V_j$, while $\mathbb{E}[y_t^2 x_{t,i}^2] = \rho^2 + s^2 + 2\rho^2 V_i^2$. Cycling through coordinate pairs uses two legal reads per training step. These expectations are unchanged by sign flips and identify $VV^\top$ as training grows. Training time, stored moments and the sign window are resources. This law preserves an axis rather than diffusing $D-1$ independent tangent directions. Thus λ_1 suffices for the moment formula of Proposition 16, but cannot justify a universal diffusion tracking lower bound.

Conjecture 19 (Rate optimality of probe calibration)**.** *For the rank-one task with $2 \leq b < D$ noiseless linear reads per step, let U_t be spherical Brownian motion sampled at unit intervals, with generator $\sigma\Delta$ and $\kappa = 1 - e^{-(D-1)\sigma}$. Policies may retain their paid observed history and randomize, but receive no external coupling truth or additional drift reference. Their long-run average risk is at least $s^2 + c_0\rho^2 \min\{1, D\sqrt{\kappa(\rho^2+s^2)/(\rho^2(b-1))}\}$ for a constant $c_0 > 0$ independent of D, b, κ, in the local diffusion regime. Initialization is charged and does not supply future increments.*

The mechanism is that each off-frame read informs one background direction, and $D-1$ directions drift independently. For a single coordinate observed with probability p per step under a random walk with increment variance q, the posterior Cramér–Rao recursion gives a stationary error of order $\sqrt{q/(pF)}$, with F the Fisher information per observation; summing over $D-1$ coordinates with $p = (b-1)/D$, $q \simeq 2\rho^2\kappa/D$ and $F \simeq (\rho^2+s^2)^{-1}$ gives the stated scaling, which Proposition 16 attains with optimized α for small κ. The strongest objection is that the coordinates are not independent: an adaptive policy can probe the directions in which its uncertainty is largest, and the response read itself carries information about the magnitude of the misalignment (Corollary 13). These could improve the constant; the conjecture would fail if they changed the exponent of κ or the power of D.

Conjecture 20 (Sharp quantum counterpart of the frame price)**.** *For a specified quantum direction-estimation task and a compact group acting transitively on its directions, the minimum retained reference resource needed for a given capability admits a sharp task-dependent variational bound whose classical direction limit is Theorem 4.*

Proposition 9 supplies a general bounded-risk inequality, and Proposition 10 supplies an exact quantum task. What remains is sharpness and the classical limit, not the existence of any risk bound. Equal asymmetry need not give equal capability in different tasks, so the task family cannot be omitted. The noise-selected algebra of Consciousness I is a symmetry-fixed algebra; the present reference enlarges effective measurement permissions through a state, not by claiming that every fixed algebra becomes a larger abstract algebra.

10.3 Open problems

The task class is linear and Gaussian with a rank-one or fixed-rank target. Several competing frames and structured non-Gaussian inputs remain open. Paper III joins a binary self-frame to a world-changing loop: acting cannot identify parameters moved by its record-law symmetry, while explicit experiments in its stationary nondegenerate model identify

persistence, squared gain and flip reliability [4]. Symmetry invariance is therefore a necessary condition, not a general identification theorem. A continuous frame with matrix-valued loop gains, and a thermodynamic implementation of the reading and retention budgets, are next steps.

A Simulation protocol for Table 2

Parameters: $D = 8$, $b = 3$, $\rho = 1$, $s = 0.5$. The coupling starts uniform and moves by the rotation-invariant step $u \mapsto (u + \eta g_\perp)/|u + \eta g_\perp|$ with $g_\perp$ the projection of a standard Gaussian vector orthogonal to u, for $\eta \in \{0, 0.002, 0.005, 0.01, 0.02, 0.04, 0.06, 0.08, 0.12\}$; $\lambda_1 = \mathbb{E}(1 + \eta^2 \chi^2_{D-1})^{-1/2}$ is computed by quadrature. Each agent receives the same $n_0 = 64$ start-up probes, from which g_0 is the mean of $y_i x_i$. The frozen agent responds $g_0 \cdot x$. The frame-free agent reads channels 1–3; the fixed agent reads $g_0/|g_0|$ and two fixed random orthonormal directions orthogonal to it; both learn their three coefficients by the update of Section 8 with $p = 1$. The calibratable agent chooses two coordinate probes uniformly without replacement and reads those axes plus the normalized component of g orthogonal to them. It reconstructs $g \cdot x$ from these three orthonormal reads and updates only the probed coordinates. The recorded noiseless simulation used the algebraically equivalent nonorthogonal functionals; this equivalence need not hold for hardware with independent read noise. Learning rates and weights are set by minimizing the risks of Proposition 16 over $\alpha \in [10^{-4}, 0.999]$ (floor 0.002). Risks are evaluated exactly given the state, $s^2 + |\rho u - c|^2$ with c the agent's effective coefficient vector, and averaged over steps 8 000–23 999 and over 200 trajectories that share the world's random numbers across agents. The code is `td2_loop.py`; the thresholds of Table 3 are computed by `td2_threshold.py`, and the analytic checks of Theorems 2–14 by `td2_checks.py`.

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Keeping a Self-Model Calibrated: Drift Detection, the Price of Keeping Up, and Reliability in a Closed Loop

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Abstract

A learned instrument model helps an observer decide which parts of its own access are worth maintaining. We study this decision together with drift detection and paid calibration. Smooth even record laws delay linear-drift detection to order $\varepsilon^{-4/5}$, compared with $\varepsilon^{-2/3}$ for first-order probes. For scalar Gaussian transitions, no free background information and quadratic loss, the all-policy long-run cost lies between $C - \frac{1}{2}k_BTG\nu^2$ and C, with $C \propto \nu^{2/3}$. The bounds have the same slow-drift leading term; the upper bound uses a Gaussian prior and reads with controllable continuous precision, while integer reads inherit the relaxation lower bound. In an acting body, stationary correlations identify gain and actuator success, and labelled standards supply orientation. Adaptive reference timing has a strict advantage and an exact optimum in a finite-rate example. When the body's working state gates reading and processing, latched repair changes future access and the kernel between acquired records. Retained body history raises task value and, under positive gain and sign drift, can reverse optimal repair at equal sign confidence. Two posterior moments determine terminal access value; continuing enquiry can require a third moment and the joint belief. In a two-observer quadratic task, estimation-optimal and task-optimal authority weights differ. These constructions use one acquired model for task permission, future enquiry and repair under declared physical supplies.

Keywords: self-calibration, self-models, Fisher information, change detection, posterior Cramér–Rao bound, drift tracking, information thermodynamics, feedback control

1 Introduction

1.1 The question

Paper I bounds the usable value of a stored record by $k_BT\,I(X{:}M)$ and associates every feedback protocol with an implicit matching distribution [1]. This distribution need not be

represented inside the agent. When an agent does represent its own readout channel and decodes with a wrong model, it loses part of the record's value and can fall below an agent without memory. Paper II studies a retained self-frame that selects reads under a channel budget; in-frame records cannot locate its missing task direction [2]. Here such a represented model is checked against an external reference or the pointer value selected by the body's working state. The later body constructions show how acquired records form that model and change its next operations.

We ask how quickly drift is detected, what calibration costs, and how the retained model changes action when the body's function determines future access.

1.2 Position within the series

This paper completes the T-DOME (Thermodynamic Dynamics of Observer-Memory Entanglement) series, the third line of a programme whose other lines are HAFF and Q-RAIF.

The three T-DOME papers follow a route that the programme calls memory–self–introspection. An agent keeps records of its world (Paper I), organizes its access to them through a model of itself (Paper II), and checks that model against references while it acts (this paper). The links share their objects. The gain–bias channel whose calibrated loss Paper I computes is the instrument of Section B; the even residual law that Paper II derives from a budgeted frame is the detection model of Section 3; the conditional drift law cited in Paper I is proved in Section 5. Each link is established in a stated model.

1.3 Main results

The analytical chain runs from detection (Section 3) through all-policy Gaussian maintenance costs (Section 5) to tracking (Section 6) and implementation (Appendix F). The joint body model of Section 7 supplies the main acting example. Section 8 then makes working access conditional on the observer's body: actual history changes harvest, permission to act, and paid repair. Sign confidence is not a sufficient repair statistic. Terminal access has a two-moment value, while continued enquiry generally needs the joint belief. The older numerical checks and gain–bias bank models are retained in Appendices A–B; the finite-filter implementation certificates are in Appendix H.

1.4 Relation to internal selection

Paper G of HAFF gives a typed model of internal selection: a program register holds a choice, a fixed executor turns program states into couplings or readouts, records are outcomes of instruments, and a fixed rule rewrites the program from the records [8]. A finite system can select its own readouts and revise that selection. Within this target/program architecture, a selected target operation is not a complete implementation of the larger executor acting on both registers; a stored description or an additional selector is not excluded. The calibration loop uses this architecture. The self-model is the program, decoding and command rules are the executor, and calibration is the revision rule. History changes the program and the operations it selects. The executor and the labelled standard specify how those changes are used.

1.5 Related work

The Fisher information metric was introduced by Rao [13] and characterized by Čencov [14]; Amari developed the natural gradient and information geometry [15, 16]. The Bayesian Cramér–Rao (van Trees) inequality [17] and its recursive form for filtering, the posterior Cramér–Rao bound of Tichavský, Muravchik and Nehorai [18], are the estimation tools of Sections 4–5; the Kalman filter [19] attains the latter in the linear Gaussian case. The trade-off between noise and lag in adaptive estimation, and the persistent-excitation condition, are standard in adaptive control [21]. Thermodynamic length and minimum-dissipation protocols are due to Crooks and Sivak [22, 23], and Fisher information enters entropy production through thermodynamic uncertainty relations and information geometry [24, 25]. Second-order cybernetics, the control of the observing system itself, goes back to von Foerster, building on Ashby's cybernetics [26, 27]. The data-processing bound for hypothesis tests used in Section 3 is standard [28]. We do not claim that the telescoping inequality behind Theorem 11 is new to filtering theory; its use here is to settle the drift law of Paper I for all policies and to give it a thermodynamic reading.

2 Setting

2.1 Inputs from Papers I and II

Paper I supplies the record's matched work value, the posterior mismatch loss and its calibrated expansion [1]. Its gain–bias example has leading loss $k_B T(1 + g^2/(1 - b^2))/(4n)$ after n add-half reference reads per sign. Paper II supplies a retained foreground, an in-frame blind spot and, for its two-channel reader, the even information law [2]

$$F(\theta) = \frac{\rho^2 \sin^2\theta\,(\rho^2 + s^2 + \rho^2\cos^2\theta)}{(s^2 + \rho^2\sin^2\theta)^2}. \tag{1}$$

An off-frame read resolves its sign fibre. The Gaussian drift law cited by Paper I is proved here; its assumptions do not identify either reader with the bounded acting body below.

2.2 Fisher information and the Cramér–Rao bounds

Definition 1 (Fisher information)**.** Let $\{p(x \mid \theta) : \theta \in \Theta \subset \mathbb{R}^d\}$ be a family of densities for which differentiation and integration may be interchanged. Its Fisher information matrix is

$$g_{ij}(\theta) := \mathbb{E}_\theta\big[\partial_i \log p(x \mid \theta)\, \partial_j \log p(x \mid \theta)\big] = -\mathbb{E}_\theta\big[\partial_i\partial_j \log p(x \mid \theta)\big]. \tag{2}$$

Where it is positive definite it is a Riemannian metric on Θ.

By Čencov's theorem, on finite probability simplices the Fisher metric is, up to a constant factor, the unique Riemannian metric invariant under statistically sufficient embeddings [14]. This makes it the canonical local measure of distinguishability; it does not make a Fisher-based detector physically mandatory, and it does not choose the agent's likelihood or objective. The metric is the Hessian of relative entropy,

$$D\big(p_\theta \,\|\, p_{\theta+d\theta}\big) = \tfrac{1}{2}\, g_{ij}(\theta)\, d\theta^i\, d\theta^j + O(|d\theta|^3), \tag{3}$$

and it bounds estimation: for an unbiased estimator from n independent observations, $\mathrm{Cov}(\hat{\theta}) \succeq g(\theta)^{-1}/n$ [13, 28]. For a correlated stream use the Fisher information of its joint likelihood. A decorrelation time alone does not justify replacing the sample count by T/τ_E.

2.3 Standing assumptions

Definition 2 (Standing assumptions)**.** Sections 3–6 assume the following. Section B states its own inputs.

(C1) **Self-model.** The agent's model of itself is a parameter ϑ in a finite-dimensional manifold Θ: a frame of Paper II, the gain and bias of an instrument, or a scalar mismatch coordinate. The family of records $p(e \mid \vartheta)$ that the agent uses to assess it is specified.

(C2) **Drift.** The parameter that best describes the instrument, $\vartheta^*(t)$, moves. Each result states its drift law: linear, spherical Brownian, a Gaussian random walk, or bounded speed.

(C3) **Regularity.** Fisher informations are finite and differentiation under the integral sign is allowed to the order used. Uniform lower bounds on information are assumed separately where they are needed.

3 Detecting drift from one's own records

A self-model goes stale when the instrument it describes changes. The agent's first question is whether it can notice this from the records it already gathers. Paper II shows that records read through a frame depend on the frame's misalignment only through an even function of it. This section works out what evenness implies.

3.1 Even models

Proposition 3 (Sign non-identifiability)**.** *If a record family satisfies $p(e \mid \theta) = p(e \mid -\theta)$ for all θ, then the joint law of any number of conditionally independent records is invariant under $\theta \mapsto -\theta$, and no statistic of the records distinguishes θ from $-\theta$.*

Proof. The joint likelihood is a product of factors that are each invariant under the sign change. □

Paper II derives this structure instead of assuming it: a frame that reads one of two channels sees an in-frame law that depends on the angle θ to the coupling only through $\cos\theta$, and the sign is the unobservable fibre of its blind spot [2]. Evenness has a second consequence, which governs how fast drift can be noticed at all.

Proposition 4 (Information and relative entropy of an even model)**.** *Let $p(e \mid \theta)$ be even in θ and smooth enough near $\theta = 0$ that the expansions below may be integrated term by term (this holds for the Gaussian families of Corollary 5). Write $\ell(\theta) = \log p(e \mid \theta)$ and*

$$\kappa := \mathbb{E}_0\big[\ell''(0)^2\big].$$

Assume $\kappa > 0$ for the sharp detection exponents below. Then the score vanishes at $\theta = 0$, so the Fisher information $\mathcal{I}_1(0)$ of one record is zero, and

$$\mathcal{I}_1(\theta) = \kappa\,\theta^2 + O(\theta^4), \qquad D\big(p_\theta \,\|\, p_0\big) = \frac{\kappa}{8}\,\theta^4 + O(\theta^6). \tag{4}$$

Proof. For each e the function ℓ is even, so $\ell'(0) = \ell'''(0) = 0$ and $\ell'(\theta) = \theta\,\ell''(0) + O(\theta^3)$. Since $p_\theta - p_0 = O(\theta^2)$, $\mathcal{I}_1(\theta) = \mathbb{E}_\theta[\ell'(\theta)^2] = \theta^2\mathbb{E}_0[\ell''(0)^2] + O(\theta^4)$. For the relative entropy write $p_\theta/p_0 = 1+\eta$ with $\eta = \frac{1}{2}\theta^2 H + O(\theta^4)$ and $H = p_0''/p_0 = \ell''(0) + \ell'(0)^2 = \ell''(0)$. Normalization gives $\mathbb{E}_0\eta = 0$, and $(1+\eta)\ln(1+\eta) = \eta + \frac{1}{2}\eta^2 + O(\eta^3)$, so $D(p_\theta\|p_0) = \frac{1}{2}\mathbb{E}_0\eta^2 + O(\theta^6) = \frac{1}{8}\theta^4\mathbb{E}_0 H^2 + O(\theta^6)$. □

Corollary 5 (Three record families)**.** *(a)* Gaussian mean model, $e \mid \theta \sim \mathcal{N}(a\theta^2, \sigma_e^2)$*:* $\mathcal{I}_1(\theta) = 4a^2\theta^2/\sigma_e^2$ *and* $D(p_\theta\|p_0) = a^2\theta^4/(2\sigma_e^2)$ *exactly, so* $\kappa = 4a^2/\sigma_e^2$*.*

(b) Paper II's in-frame record $(v\cdot x, y)$ *with* $D = 2$*:* $\mathcal{I}_1 = F(\theta)$ *of* (1)*, and*

$$D(p_\theta\|p_0) = \frac{\rho^2(1-\cos\theta)}{s^2} - \frac{1}{2}\ln\Big(1 + \frac{\rho^2\sin^2\theta}{s^2}\Big), \qquad \kappa = \frac{\rho^2(2\rho^2+s^2)}{s^4}. \tag{5}$$

(c) An off-frame probe record $(v_\perp\cdot x, y)$*:* $D(p_\theta\|p_0) = -\frac{1}{2}\ln\big(1 - \rho^2\sin^2\theta/(\rho^2+s^2)\big)$*, and its Fisher information*

$$F_\perp(\theta) = \frac{\rho^2\cos^2\theta\,(\rho^2\sin^2\theta + \rho^2 + s^2)}{(\rho^2 + s^2 - \rho^2\sin^2\theta)^2} \tag{6}$$

is positive at calibration, $F_\perp(0) = \rho^2/(\rho^2+s^2)$*.*

Proof. (a) is the normal location family with mean $a\theta^2$. For (b) and (c) the record is a zero-mean Gaussian pair with covariance $\Sigma_\theta = \left(\begin{smallmatrix} 1 & c(\theta) \\ c(\theta) & V \end{smallmatrix}\right)$, $V = \rho^2 + s^2$, where $c = \rho\cos\theta$ in (b) and $c = \rho\sin\theta$ in (c). Then $D(p_\theta\|p_0) = \frac{1}{2}[\mathrm{tr}\big(\Sigma_0^{-1}\Sigma_\theta\big) - 2 - \ln(\det\Sigma_\theta/\det\Sigma_0)]$ gives the two relative entropies, and the Fisher information of a Gaussian family with parameter-dependent covariance, $\frac{1}{2}\,\mathrm{tr}(\Sigma^{-1}\Sigma'\Sigma^{-1}\Sigma') = c'^2(c^2+V)/(V-c^2)^2$, gives (1) and (6). Expanding (5) to fourth order gives $\kappa/8 = \rho^2(2\rho^2+s^2)/(8s^4)$, in agreement with $F(\theta)/\theta^2 \to \rho^2(2\rho^2+s^2)/s^4$. □

Model (a) is a convenient assumption; model (b) is what a budgeted frame actually produces. Proposition 4 shows that the quadratic onset of information does not depend on the choice: evenness gives the expansion; a nonzero κ gives its quadratic onset. The constant κ is the only feature of the model that enters the leading order. For records at rate r under linear drift $\theta(t) = \theta_0 + \varepsilon t$, the accumulated local information $J_T := \sum_{t_m \le T}\mathcal{I}_1(\theta(t_m)) \simeq r\int_0^T \mathcal{I}_1(\theta_0 + \varepsilon t)\,dt$ is

$$J_T = \frac{\kappa r\varepsilon^2 T^3}{3} + O(\varepsilon^4 T^5) \quad (\theta_0 = 0), \qquad J_T = \frac{4a^2 r}{3\sigma_e^2\varepsilon}\big[(\theta_0+\varepsilon T)^3 - \theta_0^3\big] \quad \text{in model (a).} \tag{7}$$

It enters the estimation bound of Section 4. Detection is governed by a different quantity, the accumulated relative entropy.

3.2 How long detection takes

Fix a horizon T and let records arrive at rate r, conditionally independent given the mismatch path $\theta_m = \theta(m/r)$. A test is a function of the records up to time T with the values "calibrated" and "drifted". It has errors (α, β) if it says "drifted" with probability at most α when $\theta_m = 0$ for all m, and "calibrated" with probability at most β under the drifted law; for a random drift path the drifted law is the mixture over paths. Write $d(p\|q) = p \ln \frac{p}{q} + (1-p) \ln \frac{1-p}{1-q}$.

Proposition 6 (Detection time)**.** *Every test with errors (α, β), $\alpha + \beta < 1$, satisfies*

$$\sum_{m \le rT} \mathbb{E}\, D\big(p_{\theta_m} \,\|\, p_0\big) \;\ge\; d(1-\beta \,\|\, \alpha), \tag{8}$$

with the expectation over the drift path if it is random. Consequently, as the drift rate tends to zero:

(a) for an even model with $D(p_\theta\|p_0) \le \kappa\theta^4/8 + C\theta^6$ for all θ and linear drift $\theta(t) = \varepsilon t$, $T \ge \big(40\, d/(\kappa r \varepsilon^4)\big)^{1/5}(1 + o(1))$;

(b) if probe positions are fixed in advance with counting function $N_\perp(t) = prt + o(t)$ for $p > 0$, their record law has $D(p_\theta\|p_0) \le \frac{1}{2}F_1\theta^2 + C\theta^4$ with $F_1 > 0$ its Fisher information at calibration, as off-frame probes do, and the others are as in (a), $T \ge \big(6\, d/(pF_1 r\varepsilon^2)\big)^{1/3}(1 + o(1))$;

(c) for Paper II's reader with $D = 2$ under spherical Brownian drift with generator $\sigma\Delta$, $T \ge \big(2\, d/(\kappa r \sigma^2)\big)^{1/3}(1 + o(1))$,

where $d = d(1-\beta\|\alpha)$.

Proof. Let P_0 and P_1 be the laws of the records up to T under the two hypotheses. The test's decision is a function of the records, so by the data-processing inequality $D(P_1\|P_0)$ is at least the relative entropy between the two laws of the decision, which are Bernoulli with parameters $p_1 \ge 1-\beta$ and $p_0 \le \alpha$; since $d(p_1\|p_0)$ increases in p_1 and decreases in p_0 for $p_1 > p_0$, it is at least $d(1-\beta\|\alpha)$ [28]. For a fixed path the chain rule gives $D(P_1\|P_0) = \sum_m D(p_{\theta_m}\|p_0)$; for a random path P_1 is a mixture and joint convexity gives $D(P_1\|P_0) \le \mathbb{E}\sum_m D(p_{\theta_m}\|p_0)$. This proves (8). For (a), summing the hypothesis over $m \le rT$ gives $\kappa r \varepsilon^4 T^5/40\,(1 + O(\varepsilon^2 T^2))$, and the solution $T \propto \varepsilon^{-4/5}$ has $\varepsilon T \to 0$. For (b), summation by parts gives $\sum_{t_m \le T,\, m \text{ probe}} t_m^2 = prT^3/3 + o(T^3)$. The probe records therefore give $pF_1 r\varepsilon^2 T^3/6\,(1 + o(1) + O(\varepsilon^2 T^2))$ and the in-frame records only $O(\varepsilon^4 T^5)$, which vanishes at $T \propto \varepsilon^{-2/3}$. For (c), the generator $\sigma\Delta$ on the circle makes the unwrapped angle a Brownian motion with variance $2\sigma t$, and the relative entropy (5) is periodic and bounded, so $D(p_\theta\|p_0) \le \kappa\theta^4/8 + C\theta^6$ for all real θ. With $\mathbb{E}\theta_t^4 = 12\sigma^2 t^2$ and $\mathbb{E}\theta_t^6 = 120\sigma^3 t^3$ the sum is $\kappa r \sigma^2 T^3/2\,(1 + O(\sigma T))$, and $\sigma T \to 0$ at $T \propto \sigma^{-2/3}$. The families of Corollary 5 satisfy the hypotheses of (a) and (b): in (a) with $C = 0$, in (b) and (c) because the relative entropies are bounded and smooth. □

The bound holds for any test, whatever statistic it uses. For a known drift it is also of the right order: the log-likelihood ratio has mean equal to the accumulated relative entropy and, for small drift, variance about twice that mean, so the likelihood-ratio test succeeds at a constant multiple of the bound. In the example of Section A it reaches equal errors of 0.05 at about 1.3 times the bound. Evenness therefore costs time. A frame that sees only its own channel needs a time of order $\varepsilon^{-4/5}$ to notice a drift of rate ε; a stream that includes probes outside the frame needs a time of order $\varepsilon^{-2/3}$, which is shorter for slow enough drift at any fixed fraction p of probes.

Corollary 7 (Detection before failure)**.** *For Paper II's reader with $D = 2$ under spherical Brownian drift, a frozen self-model becomes worse than no memory at $t^* = \ln 2/\sigma$. Let the agent compare the mean S_T of the squared in-frame residuals $e_m = y_m - \rho\, v \cdot x_m$ over a window of length T with the threshold that gives type I error α under calibration. For every $\beta > 0$ there is a constant K, depending on α, β, ρ, s and r, such that at $T = K\sigma^{-2/3}$ the type II error is at most β for all small enough σ. Together with Proposition 6(c), the detection time is of order $\sigma^{-2/3}$, and its ratio to t^* tends to zero: slow drift can be detected from in-frame records long before the frozen model fails.*

Proof. Paper II's frozen-model curve with $D = 2$ gives t^*. Given the path, $e_m \sim \mathcal{N}(0, s^2 + 2\rho^2(1 - \cos\theta_m))$ independently, and under calibration nS_T/s^2 is χ^2_n with $n = rT$, so the threshold is $s^2(1 + z_\alpha\sqrt{2/n}\,(1 + o(1)))$. Write $\theta_{uT} = \sqrt{2\sigma T}\,B_u$ with B a standard Brownian motion. Then the conditional mean of $S_T - s^2$ is $2\sigma T\rho^2 Z_n(1 + O(\sigma T))$ with $Z_n \to Z = \int_0^1 B_u^2\,du$, and its conditional fluctuation is $s^2\sqrt{2/n}\,(1 + o(1))$ times a standard normal variable. At $T = K\sigma^{-2/3}$ the ratio of the shift to the null scale is $\sqrt{2}\,\rho^2\sqrt{r}\,K^{3/2}Z/s^2$, independent of σ, so the type II error tends to $P\big(N + \sqrt{2}\rho^2\sqrt{r}K^{3/2}Z/s^2 \le z_\alpha\big)$ with N standard normal and independent of Z. Since $Z > 0$ almost surely, this tends to zero as $K \to \infty$. □

For a general predetermined probe schedule, let χ_m indicate a probe at $t_m = m/r$. Under the linear drift of Proposition 6(b), the leading term of the probe entropy upper bound is

$$\frac{F_1\varepsilon^2}{2}\sum_{m\le rT}\chi_m t_m^2,$$

with remainder $O(\varepsilon^4 rT^5)$ under its global entropy bounds. The prefix-density assumption makes this sum $prT^3/3 + o(T^3)$. If only the horizon count $\sum\chi_m = prT + o(T)$ is specified, placing all probes first gives $p^3rT^3/3$, while placing them last gives $[1 - (1-p)^3]rT^3/3$, to leading order. These are the sharp extremes because the weights t_m^2 increase. A probe fraction alone therefore does not determine the detection constant. The placement is a genuine use of the access budget, even though all three schedules have the same $\varepsilon^{-2/3}$ scaling of the necessary detection time at fixed $p > 0$.

Detection is not correction. An even record can signal that the frame has moved but not in which direction (Proposition 3). One off-frame read or one reference read supplies the sign and a first-order signal at the same time. This is the division of labour in Paper II's calibratable frame and in Section B: most reads serve the task through the frame, and a small budget of reads outside it keeps the frame correct.

4 Estimating the mismatch with a prior self-model

For an agent that retains a model of its instrument, earlier calibration records supply a prior π on the drift parameter $\delta\vartheta$. Its Fisher information measures how firmly that estimate is held. Paper II's foreground uses such a retained model; Paper I's implicit protocol distribution alone does not require one.

Definition 8 (Rigidity of a self-model)**.** The rigidity of a prior self-model π is its Fisher information

$$\mathcal{I}_{\text{self}} := \int \big(\partial_{\delta\vartheta} \log \pi(\delta\vartheta)\big)^2 \, \pi(\delta\vartheta)\, d(\delta\vartheta). \tag{9}$$

Theorem 9 (Van Trees bound with a prior self-model)**.** *Let $\delta\vartheta$ lie in a compact interval $[a, b]$, let the prior π be C^1 with finite rigidity and vanish at a and b, and let the record consist of n_{eff} conditionally independent samples (or a joint experiment with that exact information), each with Fisher information $\mathcal{I}_1(\delta\vartheta)$ under a regular likelihood. Then every estimator $\delta\hat{\vartheta}$ based on the record satisfies*

$$\mathbb{E}\big[(\delta\hat{\vartheta} - \delta\vartheta)^2\big] \;\geq\; \frac{1}{n_{\text{eff}}\, \mathbb{E}_\pi[\mathcal{I}_1(\delta\vartheta)] + \mathcal{I}_{\text{self}}}. \tag{10}$$

Proof. The van Trees inequality [17] bounds the Bayesian mean squared error of any estimator by the inverse of the prior-averaged Fisher information of the record plus the Fisher information of the prior; the boundary conditions remove the boundary term of its integration by parts. Conditional independence makes the record's information $n_{\text{eff}}\mathcal{I}_1$. □

The prior helps when it is well placed and hinders when it is not. In the conjugate Gaussian location model, with constant per-sample precision $\bar{\mathcal{I}}_1$, the posterior mean is the precision-weighted average

$$\hat{\vartheta} = \frac{n_{\text{eff}}\bar{\mathcal{I}}_1\, \hat{\vartheta}_{\text{MLE}} + \mathcal{I}_{\text{self}}\, \vartheta_0}{n_{\text{eff}}\bar{\mathcal{I}}_1 + \mathcal{I}_{\text{self}}} \tag{11}$$

of the maximum-likelihood estimate and the prior centre ϑ_0. A rigid self-model centred on the wrong value pulls every estimate toward it and resists recalibration; a soft one follows the noise. Without a model of the drift one can only say that an optimal rigidity exists when the loss is lower semicontinuous on a compact set of rigidities. Under a random-walk drift the optimum is exact (Corollary 15).

5 The price of keeping up

5.1 The drift law

Let a scalar self-parameter, for instance the gain of the agent's instrument, drift as a Gaussian random walk $\theta_k = \theta_{k-1} + \omega_k$, with independent $\omega_k \sim \mathcal{N}(0, \nu^2)$ per use and θ_0 drawn from a prior with finite Fisher information Q_0 ($Q_0 = 1/P_0$ for a Gaussian prior of variance P_0). In use k the agent makes $\rho_k \geq 0$ reference reads, each costing w and carrying Fisher information at most J about θ_k under the pointwise or Bayesian conditional-information

budget of Lemma 10, together with any unlabelled reads, which carry expected information at most J_0 per use at no extra cost. It then forms an estimate $\hat{\theta}_k$ from all its records so far and incurs a loss. The number ρ_k may be a real reading intensity, and it may depend on the records of earlier uses. Define $\bar{\rho}_k = \mathbb{E}\rho_k$, $\psi(Q) = (Q^{-1} + \nu^2)^{-1}$ and

$$Q_k = \psi(Q_{k-1}) + J_0 + J\bar{\rho}_k, \qquad P_k = Q_k^{-1} \qquad (k \geq 1). \tag{12}$$

Lemma 10 (Predictable design and stopped reading scores)**.** *Fix a use k with θ_k unchanged during its reads. Let $\mathcal{H}_{j-1}$ contain the preceding complete acquired history, selected designs and private coins, including earlier reads of this use, and set $\mathcal{G}_{j-1} = \mathcal{H}_{j-1} \vee \sigma(\theta_0, \dots, \theta_k)$. Assume a parameter-independent, nonanticipating design rule and regular dominated likelihoods. The next reading score s_j has conditional mean zero given $\mathcal{G}_{j-1}$, and put $I_j = \mathbb{E}[s_j^2 \mid \mathcal{G}_{j-1}]$. Either the pointwise bound $I_j \leq J$, or the weaker Bayesian bound $\mathbb{E}[I_j \mid \mathcal{H}_{j-1}] \leq J$ under the actual joint prior and experiment, suffices. For a stopping count ρ_k with $\mathbb{E}\rho_k < \infty$ and $\{\rho_k \geq j\} \in \mathcal{H}_{j-1}$, the Bayesian expected reading information is at most $J\mathbb{E}\rho_k$. A separately declared free record channel contributes its joint conditional-information budget J_0.*

Proof. The joint likelihood factors as $p(\theta_0)\prod_k p(\theta_k \mid \theta_{k-1})\prod_j p(z_{kj} \mid \theta_k, d_{kj})$ times design probabilities. At the realized record the selected designs are fixed under differentiation, so design contributes no parameter score. The scores are martingale differences in the enlarged filtration $\mathcal{G}_j$, not necessarily in $\mathcal{H}_j$. Writing $A_j = \mathbf{1}\{\rho_k \geq j\}$, orthogonality and the tower property give

$$\mathbb{E}S^2_{\rho_k \wedge m} = \sum_{j \leq m} \mathbb{E}[A_j I_j] = \sum_{j \leq m} \mathbb{E}[A_j \mathbb{E}(I_j \mid \mathcal{H}_{j-1})] \leq J\mathbb{E}(\rho_k \wedge m).$$

The same argument bounds the squared L^2 tail by $J\mathbb{E}(\rho_k - m)_+ \to 0$. The transition information blocks of the posterior Cramér–Rao recursion are unchanged by this factorization. The conditional-average alternative proves a Bayesian information bound, not a separate bound at every θ_k. Parameter-dependent design or informative status records need their own scores; an unconditional per-read average alone does not justify stopping on the record. □

Theorem 11 (Drift law)**.** *Let $\ell : (0, \infty) \to [0, \infty)$ be a continuous function such that the expected loss in use k is at least $\ell(P_k)$ for every estimator. Then every reading schedule and every estimator satisfy*

$$\liminf_{K \to \infty} \frac{1}{K} \sum_{k=1}^{K} \big(w\bar{\rho}_k + \mathbb{E}[\mathrm{loss}_k]\big) \;\geq\; \inf_{0 < P \leq P_{\mathrm{free}}} h(P), \qquad h(P) = \frac{w}{J}\Big(\frac{\nu^2}{P(P+\nu^2)} - J_0\Big) + \ell(P), \tag{13}$$

where $P_{\mathrm{free}} = 1/Q_{\mathrm{free}}$ and Q_{free} is the positive fixed point of $Q = \psi(Q) + J_0$ ($P_{\mathrm{free}} = \infty$ if $J_0 = 0$). For the quadratic loss $\mathbb{E}(\hat{\theta}_k - \theta_k)^2$ times a constant c the hypothesis holds with $\ell(P) = cP$. If the reads are Gaussian, the loss of the posterior-optimal decoder is exactly $\ell(P_k)$, and arbitrary nonnegative observation precision is an allowed control, and the initial prior is Gaussian, a constant precision attaining a minimizing P^* attains the bound. When reads must be counted in whole units, the same expression is a lower bound from a continuous-precision relaxation; equality requires an admissible schedule.

Proof. Information bound. For a random walk with increment variance ν^2 the posterior Cramér–Rao recursion of Tichavský, Muravchik and Nehorai [18] has $D_{11} = -D_{12} = -D_{21} = \nu^{-2}$ and $D_{22} = \nu^{-2} + I_k$, where I_k is the expected Fisher information of the records of use k about θ_k, and it reads $\mathcal{J}_k = D_{22} - D_{21}(\mathcal{J}_{k-1} + D_{11})^{-1}D_{12} = \psi(\mathcal{J}_{k-1}) + I_k$, with $\mathbb{E}(\hat{\theta}_k - \theta_k)^2 \geq \mathcal{J}_k^{-1}$. Lemma 10 supplies $I_k \leq J_0 + J\bar{\rho}_k$ for predictable designs, including finite-mean within-use stopping counts. Whole-use Gaussian precision has its declared conditional information, without a discrete stopping argument. Since ψ is increasing, induction gives $\mathcal{J}_k \leq Q_k$, so the mean squared error is at least P_k, and the quadratic loss is at least cP_k.

Telescoping. By (12), $J\bar{\rho}_k = Q_k - \psi(Q_{k-1}) - J_0$. Summing,

$$\sum_{k=1}^{K}\big(Q_k - \psi(Q_{k-1})\big) = \sum_{k=1}^{K-1}\big(Q_k - \psi(Q_k)\big) + Q_K - \psi(Q_0) \;\geq\; \sum_{k=1}^{K-1}\big(Q_k - \psi(Q_k)\big) - \psi(Q_0),$$

and $Q - \psi(Q) = \nu^2/(P(P+\nu^2))$ for $P = 1/Q$. Hence

$$\sum_{k=1}^{K}\big(w\bar{\rho}_k + \mathbb{E}[\mathrm{loss}_k]\big) \;\geq\; \sum_{k=1}^{K-1} h(P_k) + \ell(P_K) - \frac{w}{J}\big(\psi(Q_0) + J_0\big) \;\geq\; \sum_{k=1}^{K-1} h(P_k) - \frac{w}{J}\big(\psi(Q_0) + J_0\big).$$

Each term of the sum is paired with a single P_k: no convexity is needed.

Range. Since $\bar{\rho}_k \geq 0$, $Q_k \geq f(Q_{k-1})$ with $f(Q) = \psi(Q) + J_0$ increasing, so $Q_k \geq f^k(0)$, and $f^k(0)$ increases to Q_{free}. For every $\eta > 0$, $P_k \leq (1+\eta)P_{\text{free}}$ for all large k. Dividing by K, letting $K \to \infty$ and then $\eta \to 0$, which is allowed because h is continuous, gives (13).

Attainment. With a Gaussian prior and Gaussian measurements, the allowed constant precision makes Q_k converge to $1/P^*$. A posterior-optimal decoder with loss exactly $\ell(P_k)$ then has average cost converging to $h(P^*)$. □

Corollary 12 (Paper I's drift law)**.** *Take Paper I's quadratic loss $\frac{1}{2}k_BTG(\hat{\theta} - \theta)^2$ per use and $J_0 = 0$, and let*

$$C = 3 \cdot 4^{-2/3}\, w^{1/3}\, (k_BTG\nu)^{2/3}\, J^{-1/3}. \tag{14}$$

Every schedule and estimator has long-run average reference cost plus loss at least $C - \frac{1}{2}k_BTG\nu^2$ per use. With a Gaussian prior and reads of controllable real precision, the best stationary policy achieves at most C, so the least achievable average L^* in that class satisfies $C - \frac{1}{2}k_BTG\nu^2 \leq L^* \leq C$. The optimal reference rate is $\rho^* = \big(k_BTG\nu/(4w\sqrt{J})\big)^{2/3}(1 + o(1))$ as $\nu \to 0$, and $C = 3w\rho^*$ to leading order. The optimal estimator is a persistent memory: the steady-state Kalman filter updates its estimate with the constant weight $\alpha^* = P^*J\rho^* = \nu\sqrt{\rho^* J}\,(1 + o(1))$, which is the memory of Proposition 43 with that weight.

Proof. With $c = \frac{1}{2}k_BTG$ and $Q = P + \nu^2$, the inequalities $\nu^2/Q^2 \leq \nu^2/(P(P+\nu^2)) \leq \nu^2/P^2$ give $h(P) \geq (w/J)\nu^2/Q^2 + cQ - c\nu^2$ and $h(P) \leq (w/J)\nu^2/P^2 + cP$. Both $\min_{Q>0}[(w/J)\nu^2/Q^2 + cQ]$ and $\min_{P>0}[(w/J)\nu^2/P^2 + cP]$ equal $3 \cdot 2^{-2/3}c^{2/3}(w\nu^2/J)^{1/3} = C$. Theorem 11 gives the lower bound and, with Gaussian reads, the upper bound. At the stationary point $\nu^2/(P^*(P^*+\nu^2)) = J\rho^*$ and $P^* = \nu/\sqrt{\rho^* J}\,(1 + o(1))$; minimizing $w\rho + c\nu/\sqrt{\rho J}$ gives ρ^* and the value $3w\rho^*$. The stationary Kalman gain is the posterior variance times the precision of the use's reads, $P^*J\rho^*$. □

Paper I conjectured this law with an unspecified universal constant, the rate ρ^* and the value $3w\rho^*$ [1]. Corollary 12 proves it, with the constant $3 \cdot 4^{-2/3} \approx 1.191$, for every schedule including time-varying and adaptive ones in its scalar Gaussian-transition model, for any read family whose Fisher information is bounded by J, and for the quadratic loss defined by Paper I's metric G. For Paper I's exact divergence loss the statement holds to leading order in the calibration error; a global version holds with G replaced by any $G' > 0$ such that the loss is at least $\frac{1}{2}k_B T G'(\hat{\theta}-\theta)^2$ on the explored range, under a uniform quadratic separation of posteriors on that range. Injectivity alone does not guarantee a positive quadratic minorant.

Corollary 13 (The relative-entropy floor of Consciousness III for all policies)**.** *In the Gaussian self-model of Consciousness III [3], let the initial prior be Gaussian and the self-state $\mathcal{N}(\theta_t, v)$ drift with increment variance q, with readings $\theta_t + \mathcal{N}(0, r)$ costing w each. Every predictable observation-precision schedule and decoder has long-run average relative-entropy loss plus reading cost at least*

$$\min_{P>0}\Big[\frac{wrq}{P(P+q)} + \tfrac{1}{2}\ln\Big(1+\frac{P}{v}\Big)\Big],$$

For a Gaussian initial prior and controllable real observation precision at linear cost, the constant-rate Kalman schedule attains it. Integer reading counts give the corresponding relaxation bound. The stationary constant-rate optimum computed in Consciousness III is therefore optimal among all policies.

Proof. Let $\mathcal{H}_k$ contain the full readings, chosen precisions, and private randomness through use k. A precision chosen from $\mathcal{H}_{k-1}$ does not contribute an extra parameter-dependent likelihood factor. Gaussian conjugacy therefore gives, path by path,

$$P_k^{-1} = (P_{k-1}+q)^{-1} + \rho_k/r, \qquad \theta_k \mid \mathcal{H}_k \sim \mathcal{N}(m_k, P_k).$$

The posterior predictive law is $\mathcal{N}(m_k, v + P_k)$, with conditional optimal relative-entropy loss $\ell(P_k) = \frac{1}{2}\ln(1 + P_k/v)$. Put $Q_k = P_k^{-1}$ and $\psi(Q) = (Q^{-1}+q)^{-1}$. For every realized observation schedule,

$$\sum_{k=1}^{K}[w\rho_k + \ell(P_k)] = \sum_{k=1}^{K-1}\left[\frac{wrq}{P_k(P_k+q)} + \ell(P_k)\right] + \ell(P_K) + wr[Q_K - \psi(Q_0)].$$

The terminal terms are at least $-wr\psi(Q_0)$. Take expectations and divide by K. A constant allowed precision at the minimizer attains the resulting bound. □

5.2 What the floor means

Remark 14 (Local thermal comparison)**.** At a fixed interior gain g, the Bernoulli reference information and Paper I's local loss metric are $J = G = 1/(1-g^2)$. Substitution into the exact Gaussian comparison yields

$$3 \cdot 4^{-2/3}(\ln 2)^{1/3}\frac{k_B T \nu^{2/3}}{(1-g^2)^{1/3}} - \frac{k_B T \nu^2}{2(1-g^2)}. \tag{15}$$

This expression gives the local quadratic comparison at thermally prepared reference cost $k_B T \ln 2$. A Gaussian random walk leaves the physical gain interval almost surely, so this substitution alone does not prove an infinite-time lower bound for the Bernoulli instrument. A bounded-drift theorem or a uniform finite-window limit is needed for that conclusion. The Gaussian theorem itself retains its all-policy strength. An alternative coordinate is $p = \sin^2 \phi$, whose Bernoulli Fisher information is 4 where $0 < p < 1$. This alone cannot supply a global quadratic KL minorant on the circle: ϕ and $\pi - \phi$ give identical laws while their circle distance can be nonzero. A bounded-instrument extension must use an identifiable quotient or chart, establish its loss separation and treat its actual transition law; the Gaussian information recursion is not coordinate-invariant under that change.

Paper I's amortization rule says that the overhead of a static calibration falls as $U^{-1/2}$ with the number U of uses, and so can be made as small as desired. The exact Gaussian comparison removes that freedom: with no free information its price per use has order $k_B T \nu^{2/3}$. For $g = 0.55$ the local Gaussian comparison value is $0.012\, k_B T$ per use at $\nu = 10^{-3}$, where the calibration error at the optimum is about 0.1 in g and the leading-order statement is accurate to a few per cent; the quadratic-loss value at $\nu = 10^{-2}$ is $0.055\, k_B T$, but there the error is about 0.23 and the divergence loss is about 7% below its quadratic approximation (Section A). A record read through the same channel is worth $0.160\, k_B T$ per use in Paper I's example. For a bounded physical gain, a uniform information bound and a quadratic loss minorant must be paired with the actual transition law; clipping the Gaussian walk changes its information recursion. With the quadratic loss the two would be equal near $\nu = 0.05$, where calibration errors are no longer small, so that crossing is only indicative. A physical persistence calculation must combine the actual record reset and the cost of its admissible calibration policy; Section G.1 supplies such a bounded example.

Corollary 15 (Optimal rigidity under drift)**.** *In the optimal stationary policy of Corollary 12, the information that the self-model carries into each use from earlier uses is* $\mathcal{I}^*_{\text{self}} = 1/(P^* + \nu^2)$*, the information gathered in the use is* $J\rho^*$*, and their ratio is* $(1 - \alpha^*)/\alpha^* = (\nu\sqrt{\rho^* J})^{-1}(1 + o(1))$.

Proof. At stationarity $Q^* = \psi(Q^*) + J\rho^*$ with $\psi(Q^*) = 1/(P^* + \nu^2)$, and the Kalman weight is $\alpha^* = J\rho^*/Q^*$. □

A more rigid self-model would ignore real drift; a softer one would spend references on noise. The trade-off that Section 4 could only assert exists has, under drift, an exact optimum set by the drift speed and the price of a reference.

Remark 16 (What the drift law does not cover)**.** (i) *Free information.* If unlabelled reads inform the parameter ($J_0 > 0$), the floor is lower, and when $\nu/\sqrt{J_0}$ is small enough the agent can track without references at an error of about $\nu/\sqrt{J_0}$; the $\nu^{2/3}$ law then no longer describes the optimum. In Paper I's setting with prior 1/2, unlabelled reads inform the bias and not the gain. In a world that the agent changes, an unlabelled read has mean $gx + b$ with the world state x unknown, so world reads alone do not separate the instrument from the world, which is Paper II's blind spot in another form. Section B uses references for this reason; Section 7 shows that a closed loop supplies free information about some invariant self-parameters in its stationary nondegenerate model, while the symmetry obstructs absolute sign identification.

(ii) *Linear drift.* Under the linear drift of Proposition 43, a memory with a constant weight has mean squared error $\nu^2(1-\alpha)^2/\alpha^2 + \alpha\sigma^2/(2-\alpha)$; optimizing the weight and the number of reads gives a cost proportional to $\nu^{2/5}$, worse than $\nu^{2/3}$. A memory that also estimates the drift rate drives the error to zero under exactly linear drift, so linear drift has no positive floor. (iii) *Vector parameters.* For a self-model with several drifting parameters the same telescoping, applied to traces, gives a lower bound by relaxation; whether it is attained when one reference informs several parameters at once is open.

Conjecture 17 (Smoother drift)**.** *Let the self-parameter be an m-fold integrated random walk, so that its m-th differences are independent $\mathcal{N}(0,\nu^2)$ variables, and let the other inputs be as in Theorem 11 with quadratic loss. Then every policy pays at least $c_m\, w^{(2m-1)/(4m-1)}\, \nu^{2/(4m-1)}$ per use, where $c_m > 0$ depends only on m, J and the loss constant, and the stationary Kalman filter with a constant reference rate attains this order.*

The stationary part is a steady-state computation: for an m-fold integrator observed with noise variance R the stationary position error scales as $\nu^{1/m} R^{(2m-1)/(2m)}$, and optimizing $R = 1/(\rho J)$ gives the exponent $2/(4m-1)$, which is $2/3$, $2/7$ and $2/11$ for $m = 1, 2, 3$; a numerical solution of the Riccati equation reproduces these slopes. The open part is the lower bound for time-varying and adaptive schedules. The telescoping of Theorem 11 does not extend directly, because the information recursion is matrix-valued and the reads inform only the first component. The conjecture would fail if a schedule that alternates bursts of reads with silence could exploit the smoothness to beat the stationary exponent.

6 The calibration loop

6.1 The natural-gradient update

A loop that keeps the self-model current can move ϑ along the steepest descent of a frame loss in the geometry of its own records,

$$\dot{\vartheta} = -\eta\, g(\vartheta)^{-1} \nabla_\vartheta L(\vartheta, t), \tag{16}$$

where $\eta > 0$ is the adaptation rate, g is the Fisher metric of the records the loop uses, and $L(\vartheta, t)$ is the risk of acting at time t with self-model ϑ. For Paper II's rank-one frame v, $L(v,t) = s^2 + \rho^2\big(1 - (v \cdot u_t)^2\big)$, which is smallest for the frame aligned with u_t. The update is invariant under reparametrization of Θ; this matters because no coordinate system on a space of frames or instrument models is preferred. It is a definite algorithm. An implementation must supply the likelihood, evaluate or approximate its score and realize the update law; accumulating an unlabelled second moment of residuals is not in general a Fisher computation.

The metric must be nondegenerate where the loop operates, and records read through the frame fail exactly there.

Proposition 18 (The in-frame metric degenerates at calibration)**.** *For Paper II's two-channel reader with the frame at angle θ to the coupling, the metric of in-frame records*

is $F(\theta)$ of (1), and the natural gradient of $L = s^2 + \rho^2 \sin^2\theta$ in this metric is

$$\frac{\partial_\theta L}{F(\theta)} = \frac{\rho^2 \sin 2\theta}{F(\theta)} = \frac{2\rho^2}{\kappa\,\theta}\big(1 + O(\theta^2)\big),$$

which diverges at $\theta = 0$. If a fraction $p \in (0, 1]$ of the records are off-frame probes, the metric $(1-p)F(\theta) + pF_\perp(\theta)$ is at least $p\rho^2 \cos^2\theta/(\rho^2 + s^2)$, and for $|\theta| < \pi/2$ its absolute natural gradient is bounded by $2(\rho^2 + s^2)|\tan\theta|/p$.

Proof. The first statement follows from $F(\theta) = \kappa\theta^2 + O(\theta^4)$ (Corollary 5). In (6) the factor $\rho^2 \sin^2\theta + \rho^2 + s^2$ is at least the denominator's base $\rho^2 + s^2 - \rho^2 \sin^2\theta$, so $F_\perp(\theta) \geq \rho^2 \cos^2\theta/(\rho^2 + s^2 - \rho^2 \sin^2\theta) \geq \rho^2 \cos^2\theta/(\rho^2 + s^2)$; divide $\rho^2 \sin 2\theta$ by p times this. □

A loop that uses only in-frame records therefore takes ever larger natural-gradient steps as it approaches calibration, and by Proposition 3 its records do not even fix the sign of the step. This is the geometric form of Paper II's blind spot, and the remedy is the same: a fixed fraction of reads outside the frame makes the metric positive at calibration, with value $pF_\perp(0) = p\rho^2/(\rho^2 + s^2)$.

The local strong-convexity tracking theorem and proof are retained in Appendix E.1. Its assumptions require a nondegenerate metric and a smooth geodesic neighborhood; they do not repair the in-frame degeneracy. Persistent excitation is a separate input.

Definition 19 (Persistent excitation)**.** The records are persistently exciting with constant $\alpha_{\mathrm{PE}} > 0$ if for some window T

$$\frac{1}{T}\int_t^{t+T} g(\vartheta(s))\,ds \;\succeq\; \alpha_{\mathrm{PE}}\, I \qquad \text{for all } t. \tag{17}$$

Pointwise regularity and a positive entropy rate of the environment do not by themselves give a uniform lower bound of this kind, and no equality between α_{PE} and m_L is assumed. Proposition 18 is a concrete failure of persistent excitation: in-frame records near calibration have $\alpha_{\mathrm{PE}} \to 0$, and probes restore it.

6.2 The four links of the loop

The series uses the following model architecture. It is a template, not a necessity theorem.

Definition 20 (Four links)**.** (I) **Records.** A world leaves readable, stable records, which an agent can store and use (HAFF; Paper I gives their value).

(II) **Reading and control interface.** Instruments read the world and actuators change it, each with a cost and a legal form (Q-RAIF; Foundation III's legal feedback channels [11]). The programme's running example of an internal algebra, a Clifford representation, can serve as the channel space; nothing here depends on it.

(III) **Self-frame.** A retained model of how the task sits in the agent's own channels, formed from records and selecting later reads (Paper II).

(IV) **Calibration.** References, memory and reliability weighting that keep (III) correct while the instrument drifts, and that decide how much authority over the world each reading carries (this paper).

Each link has an exact result in a stated model, listed in Section 10. Section B joins (II)–(IV) in one finite model in which the agent's actions change what it reads next. Section 8 makes the body's working state a condition of access and identifies a task domain where body history is necessary for a higher harvest. The template is not shown to be necessary for persistence: a system that can read everything it needs, or whose coupling never changes, needs neither (III) nor (IV).

A separate-register implementation can add reset, reference preparation and slow-driving actuation costs without double counting. Appendix F states that conditional bound and its full proof. These implementation charges are not a universal persistence budget.

7 A joint loop and its reference standard

The memory–self–introspection route can be joined around a body that the observer reads and changes. Records consume memory purity; a model of the instrument selects feedback and harvest levels; calibration revises that model. We first give the ledger and its identification boundary, then a finite-block operating regime and a continuing joint filter that uses its body history to revise both its instrument model and its reference decisions. A further construction lets the body determine whether those readings and updates can occur; latched recovery, when physically available, preserves later access. A labelled operation supplies orientation relative to its dispatch map, the calibration counterpart of HAFF Paper G's final cut [8]. Record purity, field preparation and timing are supplied resources in these constructions.

7.1 The model

Definition 21 (Joint model)**.** One cycle is one use; entropies are in nats and energies in units of k_BT.

(i) *World.* A persistent *body* bit $B_t = \pm 1$, flipped by a bath with probability d per cycle ($c = 1 - 2d$), and a *food* pointer $X_t = \pm 1$, fresh each cycle with $P(X_t = +1) = \frac{1}{2}$. Both have degenerate Hamiltonians and are diagonal in the pointer basis.

(ii) *Records.* Each cycle fresh pure memories collide with each carrier through pointer-controlled couplings [5, 6]: the food leaves two disjoint fragments F and G of overlap r, the body a fragment F_B and a copy G_B. Record formation costs a diagonal carrier nothing and consumes memory purity (Paper I, ledger of the collision model).

(iii) *Instrument.* Reading a fragment through the setting that matches its record gives $P(m \mid x) = (1 + \sigma g\, mx)/2$ with $0 \le g \le \sqrt{1 - r^2}$, the Helstrom gain. The *magnitude* g follows a specified bounded transition law; the unbounded Gaussian model of Section 5 is an exact comparison and requires a separate bridge to this physical instrument; the *sign* $\sigma = \pm 1$ may invert with probability η per use; the *setting* is one of D, and a

read with the wrong setting returns a fair coin. The same instrument reads food and body. (This is the gain–bias channel of Section B.2 with $b = 0$, and the setting is a discrete restriction of Paper II's direction family.) For uniform settings, the D choices are coordinate axes $\{e_1, \ldots, e_D\} \subset S^{D-1}$; exact identification removes $\ln D$ nats of setting uncertainty. This is not Paper II's continuous spherical alignment price, and the present model supplies these discrete settings rather than deriving their formation.

(iv) *Actions.* The flip of Q-RAIF Paper B, $U = e^{-i\lambda XQ}$, which with drive occupation N acts as $s\,X\rho X + (1-s)\rho$, $s = 1 - \epsilon_N$ [9], and costs $\theta_\omega(N + a\lambda^2)$ per opportunity, $\theta_\omega = \hbar\bar{\omega}/k_BT$, $a = 1$ if switched on. On the food: flip if $M = -1$, then harvest with fixed levels $E_\pm = -\ln \hat{q}(\pm)$ set from the agent's estimate, by the level protocol of Paper I. On the body: with a gate probability p_{g} known to the agent, flip if the reading is -1.

(v) *Memory and calibration.* The record register M is reset against the copy G, as in the cycle bound of Paper I, or against blank qubits. References are fragments of pointers prepared in a known state and read with the same instrument. A *self-label* is the recorded quench work of the harvest, which equals the energy of the level that the flipped food occupies and so records the food value relative to the agent's own level assignment.

With an ideal actuator, no drift, the matched setting and no body, the food cycle is Paper I's read–harvest–reset cycle; the body alone with a known instrument is the persistence task of Q-RAIF Paper B; the instrument alone is the drifting channel of Section 5. After the flip the food has $P(X' = +1) = q = (1+s\sigma g)/2$, and the harvest returns $\ln 2 - H_2(q) - d(q\|\hat{q})$, with $H_2(x) = -x\ln x - (1-x)\ln(1-x)$, by Paper I's lemma on extraction from a qubit, so $N \to \infty$ recovers the value $I(X{:}M)$ of Paper I.

7.2 The ledger

Every read leaves an outcome that the agent must eventually erase.

Proposition 22 (Reading tax)**.** *Let a classical register R with finite alphabet be updated once per cycle by a deterministic map $(R_t, Y_t) \mapsto R_{t+1}$, after which the outcome register is returned to a standard state, all registers having degenerate Hamiltonians and the update coupling them to a thermal bath with couplings that vanish at the endpoints and no other system correlated with (R_t, Y_t). Then $\beta \sum_{t\le K} W_t \ge \sum_{t\le K} H(Y_t \mid R_t) - \ln|R|$. Under these isolation assumptions, every reference read costs at least $\mathbb{E}\, H_2((1+g)/2)$ per read in the long run, whatever its preparation costs.*

Proof. As in the cycle bound of Paper I, unitarity and $D(\rho'_B\|\tau_B) \ge 0$ give $\beta W_t \ge H(R_t, Y_t) - H(R_{t+1}) = H(Y_t \mid R_t) + H(R_t) - H(R_{t+1})$; the last two terms telescope to at least $-\ln|R|$. Condition also on the sign, known input and register history. Fresh sensing noise gives $H(Y_t \mid R_t) \ge \mathbb{E}H_2((1+g_t)/2)$. A retained correlated outcome copy or newly consumed pure workspace must instead enter the reset ledger. □

The tax distinguishes a free reference input from a free cyclic processor. Under Proposition 22's isolation assumptions, an outcome stream still has a reset cost even when reference preparation is free. In a valid scalar Gaussian comparison this can supply the positive reading price w and preserve the $\nu^{2/3}$ order. For the local Bernoulli comparison, adding preparation and separate outcome erasure changes w from $\ln 2$ to $\ln 2 + H_2((1+g)/2)$; the leading comparison rises by a factor 1.21 at $g = 0.55$. The bounded physical transition and joint likelihood must still be established.

Proposition 23 (Purity available to a reset)**.** *Suppose the reset may couple M, a copy G whose marginal is retained, and n_B initially pure, independent blank qubits to a thermal bath. All registers have degenerate endpoint Hamiltonians. Then*

$$\beta W_{\text{reset}} \geq S(M \mid E_G) - n_B \ln 2.$$

Consequently, for a uniform food pointer,

$$\beta \,\text{net} \leq \chi(M{:}E_G) - H(M \mid X) + \Pi_B \leq \Pi_G + \Pi_B - H(M \mid X), \qquad \Pi_B = n_B \ln 2,$$

provided the record model gives $\chi(X{:}E_G) \leq \Pi_G$ and $M - X - E_G$ is the stated conditional product model. If the allowed reset is additionally required to consume nonnegative work, the first upper bound can also be capped by $I(X{:}M)$.

Proof. The entropy balance for reset gives the first inequality. Subtract it from the harvest bound $\beta W_{\text{harvest}} \leq I(X{:}M)$, and use $S(M \mid E_G) = H(M) - \chi(M{:}E_G)$ and $I(X{:}M) - H(M) = -H(M \mid X)$. Data processing gives $\chi(M{:}E_G) \leq \chi(X{:}E_G) \leq \Pi_G$. Pure blanks can themselves fund work during reset; only the added nonnegative-reset-work condition permits the cap by the harvest alone. □

In the uniform-pointer task, the second copy and pure blanks are resources for reset. Preparing all consumed purity from the agent's own thermal resources removes that contribution to profit. The five-memory copy of Paper I supplies $\chi(M{:}E_G) = 0.5231$ from $5 \ln 2 = 3.466$ nats of initial purity, while a blank qubit supplies $\ln 2$ of entropy capacity. The copy also serves as a readable record.

Theorem 24 (Ledger of the joint model)**.** *Use the food feedback of Definition 21, reset against the declared copies with no additional pure blanks, and account separately for every remaining processor reset and reference preparation. Let $\mathcal{C}_t$ denote the classical information retained and available to a reset, whose marginal the reset preserves. Condition each register's state on that information. With separate baths and no shared erasure accounting, the long-run average net free energy per cycle satisfies*

$$\beta\,\overline{\text{net}} \leq \mathbb{E}\big[\ln 2 - H_2(q_t)\big] - \mathbb{E}\big[S(M_t \mid E_G, \mathcal{C}_t)\big] - \mathbb{E}\big[\ell_t + w\rho_t\big] - \bar{\tau} - \theta_\omega \mathbb{E}\big[N_t + a_t \lambda_t^2\big] - U_{\text{body}}, \quad (18)$$

where $\ell_t = d(q_t \| \hat{q}_t)$ is the harvest mismatch, $w \geq \ln 2$ the reference preparation cost, $\bar{\tau}$ the cost of the remaining outcome and model-register resets, and $U_{\text{body}} \geq \mathbb{E}[S(Y_t \mid E_{G_B}, \mathcal{C}_t)] + \theta_\omega \mathbb{E}[N_B + a\lambda^2]$ the body upkeep. Without retained side information the conditional entropy reduces to $H(M_t) - \chi(M_t{:}E_G)$. Proposition 22 applies to an isolated processor among the remaining resets. To apply Theorem 11, the complete likelihood, transition law and loss

must meet its hypotheses. For constant known s and fixed sign, the local loop-gain coordinate $\vartheta = s\sigma g$ has reference information $1/(s^2(1-g^2))$; a marginal self-label has information $1/(1-\vartheta^2)$. Retaining the food reading changes the label likelihood and requires its full conditional information.

Proof. Apply Paper I's extraction lemma and conditional reset bound to each state conditioned on $\mathcal{C}_t$, then average. Book the other processor resets, reference preparation and Q-RAIF B's drive and switching work once each. The displayed Bernoulli informations follow by differentiating their likelihoods. Local quadratic harvest loss alone does not supply the global loss premise of Theorem 11. Under the separate-register assumptions the actual costs add to the displayed bound. □

Beyond leading order the calibration loss has a closed form. An agent with a belief of variance P about its gain sets its levels to $\mathbb{E}q$, and its expected harvest is $\ln 2 - H_2(\mathbb{E}[q \mid \text{records}])$: the agent earns what its belief allows, and the loss relative to a known gain is the Jensen gap $H_2(\mathbb{E}q) - \mathbb{E}H_2(q)$, which tends to $Ps^2/(2(1-s^2g^2))$ as $P \to 0$ and saturates for broad beliefs. The harvest is increasing in s, for a fixed gain belief. Its Jensen-gap mismatch penalty can also increase with s, so total harvest and mismatch penalty have different monotonicities.

7.3 What acting can reveal

Let Z_t be the hidden world state, θ the hidden parameters (instrument, actuators, world dynamics, record formation, initial law μ_θ), and let the agent's record be $(Y_1, A_1, \ldots, Y_t, A_t)$, with readings $Y_t \sim K^{\mathrm{s}}_\theta(\cdot \mid Z_t)$, actions A_t chosen by an arbitrary adaptive policy with private randomness, and transitions $Z_{t+1} \sim K^{\mathrm{w}}_\theta(\cdot \mid Z_t, A_t)$.

Definition 25 (Loop symmetry)**.** A group $\mathcal{G}$ acting on world states and parameters is a symmetry of the loop if $K^{\mathrm{s}}_{g\theta}(y \mid gz) = K^{\mathrm{s}}_\theta(y \mid z)$, $K^{\mathrm{w}}_{g\theta}(g\,\cdot \mid gz, a) = K^{\mathrm{w}}_\theta(\cdot \mid z, a)$ and $\mu_{g\theta} = g_*\mu_\theta$ for all $g \in \mathcal{G}$.

Theorem 26 (Acting cannot see the symmetry of its loop)**.** *If $\mathcal{G}$ is a loop symmetry, then for every policy the law of the agent's record is the same under $g\theta$ as under θ, and $\mathbb{E}_{g\theta}L(Z_t) = \mathbb{E}_\theta L(gZ_t)$ for every loss L. A parameter function that $\mathcal{G}$ moves has no consistent estimator from the agent's records, however it acts. If $\gamma \in \mathcal{G}$ exchanges a binary loss, $L(\gamma z) = 1 - L(z)$, then $\mathbb{E}_\theta L(Z_t) + \mathbb{E}_{\gamma\theta} L(Z_t) = 1$ for every policy and every t.*

Proof. Write the joint density of hidden variables and records as $\mu_\theta(z_0)\prod_t K^{\mathrm{s}}_\theta(y_t \mid z_t)\,\pi_t(a_t \mid \cdot)\,K^{\mathrm{w}}_\theta(z_{t+1} \mid z_t, a_t)$ and substitute $z \mapsto gz$ in every hidden variable. The policy factors do not involve z, and the others are invariant by Definition 25. Integrating out the hidden variables gives equal record laws, and the same substitution gives the loss identity. □

In the joint model, $\gamma : (B, X, \sigma) \mapsto (-B, -X, -\sigma)$ with everything else fixed is a loop symmetry for any agent that only reads and flips: the instrument depends on σx, the flip commutes with $x \mapsto -x$, the bath and the collisions are symmetric, and the gate coin is independent. The body loss $\mathbf{1}\{B = -1\}$ is exchanged. Every such agent therefore satisfies $P_\theta(B_t = -1) + P_{\gamma\theta}(B_t = -1) = 1$: if it holds its body at loss L it holds it at $1 - L$ in

the world related to it by γ, with identical records. Exact enumeration over twenty random adaptive policies of five steps gives record laws that agree to 1.2×10^{-16} in total variation; one reference read per step separates them by at least 0.30. This is the observation of the Imprint Chain that its calibration statistic depends only on ν^2 [7], extended to every policy and to the task loss. For a linear loop, $x_{t+1} = ax_t + \kappa u_t + f + w_t$, $y_t = gx_t + b + e_t$, the affine maps $x \mapsto \lambda x + \mu$ with $(g, b, \kappa, f) \mapsto (g/\lambda,\ b - g\mu/\lambda,\ \lambda\kappa,\ \lambda f + \mu(1-a))$ form a loop symmetry when the initial and process-noise laws are transformed as well; acting can identify the loop gain $g\kappa$ and never g or κ separately (Kalman likelihoods under random affine maps and dithered adaptive policies agree to 6×10^{-14}).

Proposition 27 (What the closed loop identifies)**.** *In the stationary body loop with constant gain and sign, $0 < g \le 1$, $d \in (0,1) \setminus \{1/2\}$, and the matched setting, gate coin A_t of known probability $p_{\mathrm{g}} \in (0,1)$ and flip success s,*

$$\mathbb{E}[Y_{t+1}Y_t \mid A_t = 0] = cg^2, \qquad \mathbb{E}[Y_{t+2}Y_t \mid A_t = A_{t+1} = 0] = c^2g^2,$$
$$\mathbb{E}[Y_{t+1}Y_t \mid A_t = 1] = cg^2(1-s) + cs\,\mathbb{E}Y_t.$$

Writing the three moments as K_0, K_2, K_1, and $\mu = \mathbb{E}Y_t$, one has

$$c = K_2/K_0, \qquad g^2 = K_0^2/K_2, \qquad s = (K_1 - K_0)/(c\mu - K_0).$$

Stationarity gives $\mu = cp_{\mathrm{g}}sg^2/(1 - c + cp_{\mathrm{g}}s)$, so the denominator is nonzero. Thus c, g^2 and s are determined by the agent's own records, without references, and σ is not. With a wrong setting the first and third moments vanish.

Proof. $B_{t+1} = \xi_t B_t \Phi_t$ with $\mathbb{E}\xi_t = c$ independent and $\Phi_t = -1$ if the flip is switched on and succeeds. For $A_t = 0$, $\mathbb{E}[Y_{t+1}Y_t] = \sigma gc\,\mathbb{E}[B_tY_t] = cg^2$. For $A_t = 1$, $\mathbb{E}[B_t\Phi_tY_t] = \sigma g + 2s\,\mathbb{E}[B_t\mathbf{1}\{Y_t = -1\}]$ and $\mathbb{E}[B_t\mathbf{1}\{Y_t = -1\}] = (\mathbb{E}B_t - \sigma g)/2$; multiply by σgc and use $\sigma g\,\mathbb{E}B_t = \mathbb{E}Y_t$. The lag-two identity iterates the first. □

A frame error is thus visible in the agent's own loop as a loss of loop correlation; locating the right setting still requires trying the others, which is Paper II's blind spot in binary form. What no record reveals is fixed by the following statement. Call an operation *symmetry-breaking* if its action singles out a world state in a way that the symmetry does not respect: a reference prepared in the state $+1$, a reset to $+1$, harvest levels with the higher level on -1, a bath relaxing toward a known state. Each carries a label ζ saying which state it singles out, and that label belongs to the operation's dispatch map in the sense of Paper G.

Proposition 28 (No internal standard)**.** *Let $\mathcal{G}$ act also on the labels of all symmetry-breaking operations by relabelling. If the labelled sensing, transition and initial kernels are jointly covariant, $\mathcal{G}$ remains a loop symmetry. No policy identifies an absolute jointly relabelled orientation. A system can form and retain relational standards.*

Proof. Apply the assumed joint covariance in Theorem 26. Set-theoretic closure of a kernel family alone does not imply covariance. □

The model locates an operational reference standard in a labelled kernel: an external reference, an actuator reset or level assignment, a known environmental asymmetry, or a previously interpreted record. Jointly relabelling every such kernel preserves the record law. A system can nevertheless construct relational standards and carry them between operations. Paper G places the final cut of internal selection at the dispatch map of the top executor [8]; here the corresponding calibration object is the labelled operation whose orientation the loop uses. The symmetry theorem does not derive the physical origin of that operation.

7.4 Three self-parameters, three laws

The magnitude and the setting are invariant under the sign symmetry; the sign is not.

Proposition 29 (Free loop information changes the exponent)**.** *In Theorem 11 let unlabelled reads carry $J_0 > 0$ per use at no extra cost, and let the loss be cP. Put $P_f = \frac{1}{2}\big(-\nu^2 + \sqrt{\nu^4 + 4\nu^2/J_0}\big)$. If $c \le (w/J)J_0(2P_f+\nu^2)/(P_f(P_f+\nu^2))$, which holds for $\nu \le \nu_\times = 2wJ_0^{3/2}/(Jc)\,(1+o(1))$, every policy pays at least $cP_f = c\nu/\sqrt{J_0}\,(1+O(\nu\sqrt{J_0}))$ per use, and with Gaussian reads this is attained with no references.*

Proof. $\phi(P) = \nu^2/(P(P+\nu^2))$ is convex and decreasing, so h of Theorem 11 is convex. P_f solves $\phi(P) = J_0$, and $|\phi'(P_f)| = J_0(2P_f+\nu^2)/(P_f(P_f+\nu^2))$, so under the condition $h' \le 0$ on $(0, P_f]$ and the infimum is $h(P_f) = cP_f$. As $\nu \to 0$, $P_f = \nu/\sqrt{J_0} - \nu^2/2 + O(\nu^3)$. The Kalman filter on the free reads alone has stationary variance P_f. □

The body moments establish static identifiability under Proposition 27's nondegeneracy conditions. A joint information-rate bound for the correlated record stream is a further step: the information of one overlapping product is not that rate. Proposition 29 gives an exact Gaussian realization of the linear-in-ν price when a genuine free observation channel is supplied. For self-labels the marginal Bernoulli information is $1/(1-\vartheta^2)$; retaining the earlier food read changes the conditional likelihood and its information.

Theorem 30 (Sign inversions)**.** *Suppose that sign inversions and body-bath flips are independent Bernoulli events with probabilities $\eta, d \in (0,1)$ per use. The agent knows the initial sign, has no symmetry-breaking operation, and reads only records formed in the current cycle. Put*

$$b_* = \max\{|1-2\eta|, |1-2d|\}.$$

For every read-and-flip policy,

$$\big|P(B_t = -1) - \tfrac{1}{2}\big| \le \tfrac{1}{2}b_*^t.$$

In particular the loss converges to $1/2$. If $\eta \le \min\{d, 1-d\}$, its lower bound is $\frac{1}{2}[1-(1-2\eta)^t]$.

Proof. Let a_j be the exclusive-or of the bath flip and the instrument inversion. Condition on the apparent-body trajectory, the a_j, and all independent reading and actuation randomness. This is more information than the agent receives. Conditional inversion probabilities are

$$q_1 = \frac{\eta(1-d)}{\eta(1-d) + d(1-\eta)}, \qquad q_0 = \frac{\eta d}{\eta d + (1-\eta)(1-d)}.$$

Conditional on that information, the absolute sign bias is $\prod_{j=1}^{t}|1-2q_{a_j}|$. The absolute conditional body bias has the same value, since the apparent body is then fixed. Hence

$$|\mathbb{E}B_t| \le \left(\sum_{a=0}^{1} P(a)|1-2q_a|\right)^t = \big(|1-\eta-d|+|\eta-d|\big)^t = b_*^t.$$

The a_j are independent with a law unaffected by the policy. This proves both sides of the bound. The special case follows from $b_* = 1-2\eta$ under the stated condition. □

The limit $\frac{1}{2}$ is the loss of an agent that does nothing. An inversion is the image under γ of a world kick that the loop corrected, so the records never show it. The food fares worse: after an inversion the flip pushes it the wrong way, the harvest runs at a loss, and an agent that never reads its energy store cannot tell. Reading the store is a self-label, whose standard is the agent's own level assignment. The Bayes-optimal policy, computed by backward induction over all adaptive policies for eight cycles, respects the bound in every case with $\eta \le d$ (for $g = 0.9$, $d = 0.1$, $s = 0.95$, $\eta = 0.05$: 0.307 against 0.285 at $t = 8$). Consciousness III shows that an observer can be confidently and undetectably wrong on a two-point fibre of its protocol [3]; Theorem 30 adds that acting does not change the protocol in the relevant sense, and that inversions make the error grow to chance. The hypothesis of fresh records matters. If a stored record of an earlier body state, made by the environment's fixed couplings, is read again later, the two reads compare σ at the two times, so the world's stable records detect changes of the sign though not the sign. Records of Paper F's kind [6] thus carry an agent's calibration across time, and the standard sits in the past.

Proposition 31 (Reference cost under exogenous and adaptive timing)**.** *Let a reference cost $w > 0$ and reveal the current sign exactly, and let body loss cost $V > 0$ per use. For $0 \le r < 1$, define*

$$F_r(G) = \frac{G}{2} - \frac{1-r^G}{2(1-r)}, \qquad C_r = \inf_{G\in\mathbb{N}} \frac{w + VF_r(G)}{G}.$$

Reference times fixed in advance, or randomized independently of the record, give long-run average cost at least C_{b_*}, where b_* is defined in Theorem 30. For reference times chosen from the records, the bound C_{r_*} holds with

$$r_* = \max_{a=0,1} |1-2q_a|.$$

If $\eta \to 0$ with $d \in (0,1)$ and w, V fixed, these benchmarks satisfy

$$C_{b_*} = \sqrt{2wV\eta}\,(1+o(1)), \qquad C_{r_*} = \sqrt{2wV\eta\,\min\{d/(1-d),(1-d)/d\}}\,(1+o(1)).$$

When a finite gap attains the first infimum, an ideal periodic read-and-control policy attains it; otherwise the no-reference limit gives its value $V/2$. It need not bound record-adaptive schedules.

Proof. With an exogenous gap, the mean loss at age j after a reference is at least $(1-b_*^j)/2$, by the proof of Theorem 30. For an adaptive gap one cannot use this unconditional mean

after conditioning on the absence of another reference. Instead, the same proof gives the conditional bound $(1-r_*^j)/2$ for every apparent-event history. The event that a policy has not yet requested another reference is determined by such a history and its private randomness, so this bound also holds on each adaptive gap. Summing over complete gaps and the terminal incomplete gap gives at least $KC_r - O(w)$ over K uses. Equivalently, the piecewise linear interpolation of F_r is convex, and an exogenous mixture of gap lengths cannot beat the best single length for the corresponding benchmark. Writing $r = 1 - 2a$ gives $F_r(G) = aG(G-1)/2 + O(a^2G^3)$; minimization has $G \sim \sqrt{2w/(Va)}$ and cost $\sqrt{2wVa}(1+o(1))$. For small η, $(1-b_*)/2 = \eta$ and $(1-r_*)/2 = \eta \min\{d/(1-d), (1-d)/d\}(1+O(\eta))$. Ideal reads reveal the apparent state, and ideal flips can hold it aligned with the optimal sign estimate, attaining the exogenous benchmark. □

Remark 32 (Record-adaptive references improve the benchmark)**.** Take $\eta = d = 1/10$, $w = 1/10$ and $V = 1$. Read the apparent body exactly. After accounting for the preceding commanded flip, request a reference precisely when the apparent body has changed sign. Then control the body using the last reference's sign estimate. Such events have probability $p = 2\eta(1-\eta) = 9/50$. Between them the conditional inversion probability is $q_0 = 1/82$. The stationary posterior sign bias is

$$z = p + (1-p)(1-2q_0)z = \frac{9}{10}.$$

The loss is $1/20$, and references cost $9/500$, giving total cost $17/250 = 0.068$. The exogenous-gap benchmark is $1/10$. An exact finite-state computation with sensing gain and flip success both 0.999 gives cost 0.0687253, so the separation persists away from the ideal limit. The bound for adaptive timing is $C_{r_*} = 0.0429971$ here.

Proposition 33 (Exact adaptive value at equal inversion and bath rates)**.** *In Proposition 31, take $\eta = d = 1/10$, $w = 1/10$ and $V = 1$. The minimum long-run average cost, with ideal apparent-body reads and ideal flips, over all adaptive reference policies is*

$$\rho_* = \frac{2089311}{43698622} = 0.0478118280\ldots.$$

Request a reference after an apparent innovation or after five consecutive quiet transitions since the last reference. This policy attains the value, which also bounds policies with less informative sensing or control.

Proof. An apparent innovation has probability $p = 9/50$ and conditional sign bias zero; a quiet transition has probability $z = 41/50$ and multiplies the bias by $u = 40/41$. The oracle therefore has states $k = 1, 2, \ldots$ for k quiet transitions after a reference and a zero-bias state. At state k, continued control has loss $l_k = (1-u^k)/2$; a reference resets the bias and costs w. For threshold $K = 5$, a renewal cycle has

$$D_K = \sum_{j=0}^{K-1} z^j, \qquad C_K = w + \sum_{j=1}^{K-1} z^j l_j, \qquad \rho_* = C_K/D_K.$$

Self-parameter	Sign symmetry	Without any standard	With references or labels
Setting (switches)	invariant	detected by the loop (Proposition 27); found by search	not needed
Magnitude (random walk ν)	invariant	static body moments identify gain; a valid Gaussian free channel gives price $c\nu/\sqrt{J_0}$ (Proposition 29)	$\nu^{2/3}$ (Theorem 11); marginal self-labels avoid pointer preparation; full retained history determines their information
Sign (inversion rate η)	variant	loss $\to \frac{1}{2}$ for fresh-record read-and-flip policies (Theorem 30)	positive square-root lower bound for adaptive timing; exact finite-rate optimum in Proposition 33

Table 1: What acting buys for each self-parameter of the joint model.

To verify global optimality, set $v_k = w$ for $k \geq 5$ and set $v_k = l_k - \rho_* + pw + zv_{k+1}$ backwards for $k = 4, 3, 2, 1$. Exact rational arithmetic gives

$$-\rho_* + pw + zv_1 = 0, \quad v_k < w \ (1 \leq k < 5), \quad l_5 > \rho_*.$$

Thus $v_k = \min\{w, l_k - \rho_* + pw + zv_{k+1}\}$ for every k; monotonicity of l_k proves the stopping inequality for all $k \geq 5$. The zero-bias state also stops, since $1/2 > \rho_*$. These bounded Bellman inequalities telescope over every adaptive, randomized or nonstationary policy, leaving a bounded terminal term. Dividing by the horizon proves the lower bound. Ideal readings reveal the apparent innovations, and ideal flips attain their conditional Bayes body loss, so the threshold policy is an admissible implementation of the oracle optimum. Less informative records are a garbling of this oracle. A less reliable flip can be simulated by an ideal actuator using a private coin, with its storage and reset charged to the policy. Neither restriction improves the oracle value. □

In the exogenous benchmark with $V = 1$, the small-η calculation is within 4% of $\sqrt{2w\eta}$ over the tested $\eta \leq 10^{-4}$ range. A periodic noisy-reference policy gives excess cost within a factor 2–3 of $\sqrt{2wV(1-2L_0)\eta}$, where $L_0 = 0.052$ is its no-inversion loss. Its fitted exponent is 0.41–0.43, consistent with the growing reference-block size. Proposition 33 settles the stated finite-rate ideal example. Proposition 31 supplies the general adaptive lower bound; its constant need not be sharp.

7.5 Operating and continuing regimes

A bounded two-gain block model has a supplied-resource operating margin of $0.02396\, k_B T$ per food use with a squeezed drive, body loss 0.08725 and explicit purity input. Its complete

ledger and comparison are in Appendix G.1. A separate continuing eight-state filter carries instrument and body history through every use. Under the same reference cap, adaptive timing gives body loss .10319, versus .11726 for periodic timing. Its transition/likelihood, two-seed check, 350-bit quantization certificate and resource boundary are retained in Appendix G.2. The block margin cannot be imported into that continuing protocol without its own apparatus ledger.

8 When the body controls access

The preceding loop receives a reading and can process it at every use, even when its body is at the loss state. A body that supports those operations changes the problem. Keep the bounded instrument, record couplings, field drive and body bath, but let $B = +1$ enable the main instrument, reference ports, model logic and food protocol. At $B = -1$ these ports are disconnected. The model register is held, not updated. A disabled reader cannot acquire a reference or ask a running estimator to repair it.

Definition 34 (Access-gated joint loop)**.** At each working use, read the body with likelihood $(1 + y\sigma g_G)/2$, update the model, perform the food task and write a recovery command $\ell \in \{0, 1\}$ to a latch. The working body is not actively flipped; the bath disables it with probability d. During disability the latch and model register remain unchanged. A separate standby port, if present, executes that stored command through the same field drive; it obtains no new instrument reading. The recovery probability per disabled slot is

$$r_\ell^{\mathrm{rec}} = d + \ell s(1 - 2d), \qquad 0 < d < \tfrac{1}{2}.$$

The instrument continues its gain and sign transitions in all wall-clock slots. Disabled-slot fragments and enable flags leave through output streams that the reader cannot later acquire; there is no bank of delayed readings on resumption. The reader obtains no elapsed-time counter on recovery. A resumed use is another working invocation, with its prior obtained from the event kernel below. Harvest work enters the physical ledger but is not read back as an additional self-label unless a protocol explicitly permits that read. The latch is written after all records acquired in its working invocation.

An independent enable flag copies the pointer value of B and controls the standby drive while the main processor is held. In the diagonal body model the repair leaves $+1$ fixed and sends -1 to $+1$ with probability s. A CPTP extension has Kraus operators

$$D_0 = P_+, \qquad D_1 = \sqrt{1 - s}\, P_-, \qquad D_2 = \sqrt{s}\, |+\rangle\langle -|, \qquad \sum_j D_j^\dagger D_j = I.$$

It dephases the body blocks and is nonunital. The enable flag, its preparation or reset, and the body's free-energy change enter the ledger. One explicit resource model supplies a fresh pure flag per wall-clock slot and retains the used flag in a charged output stream, consuming $\ln 2$ of flag purity per slot. Cyclic reuse requires a conditional reset and protection protocol. Table 2 collects these resource declarations. With no standby, the symmetric bath still recovers the body at rate d.

Proposition 35 (The kernel between working uses)**.** *Let* K *be the four-state gain/sign factor of* T_a *and fix a stored command with recovery rate* $r_{\mathrm{rec}} > 0$*. If* L *is the number of wall-clock slots between successive working invocations, then*

$$P(L = 1) = 1 - d, \qquad P(L = k) = d r_{\mathrm{rec}} (1 - r_{\mathrm{rec}})^{k-2} \quad (k \geq 2), \tag{19}$$

$$\mathbb{E}L = 1 + d/r_{\mathrm{rec}}, \qquad \pi_{\mathrm{w}} = \frac{r_{\mathrm{rec}}}{r_{\mathrm{rec}} + d}, \tag{20}$$

$$\mathsf{K}_\ell = (1 - d)\mathsf{K} + d r_{\mathrm{rec}} \mathsf{K}^2 [I - (1 - r_{\mathrm{rec}})\mathsf{K}]^{-1}. \tag{21}$$

π_{w} *is the long-run working fraction. A right eigenmode of* K *of eigenvalue* λ *has event eigenvalue*

$$f_{r_{\mathrm{rec}}}(\lambda) = (1 - d)\lambda + \frac{d r_{\mathrm{rec}} \lambda^2}{1 - (1 - r_{\mathrm{rec}})\lambda}.$$

The event kernel remains valid conditionally when the latch is chosen from the preceding working record.

Proof. The body either remains working for the next slot or enters a geometric recovery wait. This gives (19) and its mean. The stationary flow balance is $\pi_{\mathrm{w}} d = (1 - \pi_{\mathrm{w}}) r_{\mathrm{rec}}$; renewal averaging gives the same fraction. Instrument noise is independent of the body and recovery noise conditional on the selected latch. Sum K^L with the displayed gap probabilities. Since $\|(1 - r_{\mathrm{rec}})\mathsf{K}\|_\infty = 1 - r_{\mathrm{rec}} < 1$, the sum is the positive resolvent in (21). Applying it to an eigenvector gives $f_{r_{\mathrm{rec}}}$. No recursion is executed in disabled slots: the finite controller uses the matrix for its last stored command on the next invocation. □

The working body provides a relative calibration standard. With static sign and gain, its working readings are independent samples of $(1 + y\sigma g)/2$; their mean identifies σg for $g > 0$. With a fixed latch their Fisher information rate is $\pi_{\mathrm{w}}/(1 - g^2)$ in the signed-gain coordinate. Under a history-dependent latch, information is counted on actual acquired uses, as in Consciousness III [3]. The transformation $(B, X, \sigma) \mapsto (-B, -X, -\sigma)$ exchanges enabled and disabled blocks and hence fails to preserve this access kernel. The orientation comes from the declared body–permission coupling. Jointly relabelling that coupling, every operation and the task preserves the absolute-orientation obstruction.

The readout must be informative: $g = 0$ or wrong-setting reads give no instrument information. Continued access is a separate condition. In a hardware variant with an absorbing disabled state and conditional failure hazard at least $h > 0$ on every working use, every policy has

$$P(N \geq n) \leq (1 - h)^{n-1}, \qquad \mathbb{E}N \leq h^{-1},$$

where N is its total number of working readings. Successive conditioning proves the bound, so only finitely many readings occur almost surely. If two static parameters have positive overlapping likelihoods on finite stopping records, their stopped-record laws have positive overlap and their equal-prior Bayes error is strictly positive. Waiting longer cannot identify them. The absorbing variant removes passive recovery as well as standby.

Proposition 36 (Calibration through unobserved disability)**.** *Take known* $0 < g < 1$*, sign inversion probability* $0 < \eta < 1/2$*, one body reading on resumption and no additional current*

directional observation. Conditional on a stored recovery command, every sign estimator has average error at least

$$\min\{e_{\mathrm{rec}}, (1-g)/2\}, \qquad e_{\mathrm{rec}} = \frac{1 - f_{r_{\mathrm{rec}}}(1-2\eta)}{2} > 0.$$

Increasing r_{rec} lowers this oracle floor and increases the working fraction. For small η, $e_{\mathrm{rec}} = \eta/\pi_{\mathrm{w}} + O(\eta^2)$.

Proof. Grant the reader the true sign at its preceding working invocation. Across an unobserved gap L, its sign bias is multiplied by $(1-2\eta)^L$. Averaging gives the prior error e_{rec}. The current body reading is a binary symmetric channel of error $e = (1-g)/2$. For prior error $q \le 1/2$, its two-output Bayes error is

$$\min\{(1-q)(1-e), qe\} + \min\{(1-q)e, q(1-e)\} = \min\{q, e\}.$$

An actual history is less informative than the oracle history. For $0 < \lambda < 1$, differentiation gives

$$\frac{\partial f_{r_{\mathrm{rec}}}(\lambda)}{\partial r_{\mathrm{rec}}} = \frac{d\lambda^2(1-\lambda)}{[1-(1-r_{\mathrm{rec}})\lambda]^2} > 0.$$

The working fraction is also increasing. A paid elapsed-time counter or another current reference changes the available record and therefore changes this bound. □

Filtering at actual working invocations uses the positive four-state event kernels of Proposition 35. The certificate in Appendix H needs 150 ideal model bits and bounds asymptotic total-variation error by 1.247×10^{-7} for the stated switching rates. Its precision is fixed; identifying a continuous parameter to arbitrarily growing precision requires growing memory.

Proposition 37 (A domain where history is necessary for the food task)**.** *Fix the recovery command, a known static $0 < g < 1$, an unknown static sign with equal prior, one body reading per working invocation and no further directional sample before the food action. Within the joint food protocol, choose harvest levels from the body model before the fresh food reading M, then choose whether to flip from M. Without retained history, the maximal harvest is*

$$J_1 = \ln 2 - H_2((1+sg^2)/2).$$

Let $a_g = (1+g)/2$, $b_g = (1-g)/2$, and retain the latest three body readings. For their number k of positive results set

$$p_k = \tfrac{1}{2}\binom{3}{k}(a_g^k b_g^{3-k} + b_g^k a_g^{3-k}), \quad m_k = \frac{a_g^k b_g^{3-k} - b_g^k a_g^{3-k}}{a_g^k b_g^{3-k} + b_g^k a_g^{3-k}}.$$

The achievable harvest

$$J_3 = \sum_{k=0}^{3} p_k[\ln 2 - H_2((1+sg|m_k|)/2)]$$

is strictly larger than J_1. Thus a target harvest in $(J_1, J_3]$ requires history within these permissions. An already learned sign encoded in a program, latch or weight counts as retained history.

Proof. The fair food reading has no marginal sign information. A body-only record of posterior sign mean m allows a food polarization of at most $sg|m|$: for arbitrary flip probabilities depending on M, its magnitude is $sg|m|$ times the difference of two probabilities, whose absolute value is at most one. Choosing the appropriate orientation attains that endpoint, and matched levels attain the stated harvest. Without history $m = gY$. Three-sample Bayes gives p_k, m_k. The function $F(m) = \ln 2 - H_2((1 + sgm)/2)$ is strictly convex, with $F''(m) = (sg)^2/[1 - (sgm)^2] > 0$. Posterior means form a martingale, and two extra informative noisy samples make its conditional value nonconstant. Strict Jensen therefore gives $\mathbb{E}F(m_3) > \mathbb{E}F(m_1)$. A three-bit sliding record acquires no additional samples and remains held during disability. □

For $g = 3/5$ and $s = 49/50$, exact finite body–food–drive path enumeration gives $J_1 = 0.0635939$ and $J_3 = 0.1202672$. If levels may instead depend on the current food reading, the posterior-matched protocol family of Paper I replaces sg by g in both formulas and retains the strict history gap.

Performance is averaged over wall-clock slots. The fixed-latch harvest gain is $\pi_{\mathrm{w}}(J_3 - J_1)$ before the extra memory cost. The drive of Section G.2 changes the working fraction from $1/2$ to about 0.949083 when standby is enabled. Body readings at known $B = +1$ contain fresh instrument noise uncorrelated with the body copy. Table 2 therefore charges their accessible reset information separately, together with holding and processing. The finite-block operating margin uses a different body protocol.

Resource	Declared supply or charge	Availability and accounting
Food and body fragments	$4n \ln 2$ purity nats per wall slot if continuously allocated	Acquired and reset only on working uses; disabled outputs are not banked
Standby enable flag	$\ln 2$ supplied purity nats per wall slot in the fresh-flag implementation	Used outputs leave the reader; cyclic reuse needs a reset/protection ledger
Two drive packets	$2\theta_\omega N$ per wall slot	Unused prepared packets remain charged
Drive switching	$\theta_\omega \lambda^2[\ell(1 - \pi_{\mathrm{w}}) + \pi_{\mathrm{w}}/2]$	Recovery on disabled slots; food on working slots
Body outcomes and model	Conditional erasure, update and reliable-hold costs	No processing during disability; the body copy does not erase fresh sensing noise

Table 2: Resource declarations for access-gated comparisons. Purity is an input, not a proved total energy cost. Preparation from thermal resources and any remaining apparatus work must also be charged.

For a fully known instrument and fixed active pipeline, let v denote net value per working use after its active costs, and put $\kappa = \theta_\omega \lambda^2$. If all base supplies, including fresh flags, are

allocated equally in wall time, the two fixed latches have residual values

$$U_\ell = \pi_{\mathrm{w},\ell} v - (1 - \pi_{\mathrm{w},\ell})\kappa\ell - C_{\mathrm{base}}, \qquad U_1 - U_0 = \frac{(r_1^{\mathrm{rec}} - d)v - 2d\kappa}{2(r_1^{\mathrm{rec}} + d)}.$$

This identifies the priced gain from making future calls possible. Flag costs that depend on the latch must be added, not hidden in a common base. For an uncertain instrument, the future task value and the evidence lost while disabled must also enter the control problem. The known-parameter price threshold motivates a stronger history-dependent hypothesis.

Proposition 38 (Equal sign confidence, opposite paid repairs)**.** *There is a static, clocked version of Definition 34 in which two legally acquired histories at the same working slot have equal sign confidence and opposite optimal repair decisions. Take* $d = 1/20$, $s = 49/50$, $H = 12$, *and a uniform prior on the four static states* $(\sigma, g) \in \{\pm 1\} \times \{3/20, g_+\}$. *At slot* $t = 10$, *compare the history with positive readings at slots* 1 *and* 10 *and disability at slots* $2, \ldots, 9$, *with ten working readings of which eight are positive. Choose* g_+ *so that their sign confidences agree, with*

$$0.959375749239844 < g_+ < 0.959375749239846.$$

Both confidences are $0.876966618\ldots$, *whereas their posterior means of the unsigned gain are* $0.678161237\ldots$ *and* $0.249069192\ldots$.

Each working invocation acquires one body reading and permits the food protocol of Proposition 37, or abstention from that food task. This witness does not acquire or read back the harvest-work self-label of Definition 21: its histories contain body readings, gate outcomes and the priced clock. The harvest work still enters the physical ledger. Food switching costs $\kappa/2$ *per executed task and latched recovery costs* κ *per disabled slot. Every working invocation also pays a declared mandatory processing charge* c_{p}. *A paid clock signals the slot and horizon; base record, flag, drive and clock supplies have the same wall-time charges for both latch choices. In this witness, body-reading erasure and food reset are supplied operations, excluded from* $V(b)$ *below rather than assigned zero universal cost. Proposition 41 establishes the corresponding opposite-repair phenomenon with the terminal reading and reset charges included. At* $\kappa = 2/25$, $c_{\mathrm{p}} = 0$, *the optimal latch differences satisfy*

$$Q_1(b) - Q_0(b) \in (0.00559146, 0.00559147), \qquad Q_1(b') - Q_0(b') = -0.004272.$$

At $\kappa = 9/4000$, $c_{\mathrm{p}} = 3/25$, *their differences satisfy*

$$Q_1(b) - Q_0(b) \in (0.00587980, 0.00587982),$$
$$Q_1(b') - Q_0(b') \in (-0.00432522, -0.00432520).$$

Proof. For a gain g put $a_g = (1 + g)/2$ and $b_g = (1 - g)/2$. The posterior after n readings, k positive, is proportional to $a_g^k b_g^{n-k}$ for sign $+1$ and $b_g^k a_g^{n-k}$ for sign -1. Let $c_{n,k}(x)$ be its positive-sign probability with gains $3/20, x$. The rational function $c_{2,2}(x) - c_{10,8}(x)$ has negative and positive values at the displayed rational endpoints. Its denominator is positive, so continuity gives an exact root. Both histories can be generated by the same legal policy that holds latch zero until slot ten. Their gate paths have probabilities $d^2(1-d)^7$ and $(1-d)^9$,

independent of the static instrument; every required reading has positive likelihood. Gate and clock records therefore leave the count-based posteriors unchanged.

Write $m(b) = \mathbb{E}_b[\sigma g]$ and

$$F(x) = \ln 2 - H_2((1+x)/2), \qquad V(b) = \max\{0, F(s|m(b)|) - \kappa/2\} - c_{\mathrm{p}}, \qquad A(b) = \mathbb{E}_b V(b^Y),$$

where b^Y is the posterior after one further working reading. Only two future slots remain. If slot eleven works, its optimal continuation is common to the two current latches. It writes latch zero: a recovery switched on at slot twelve cannot create a working opportunity within this horizon. If slot eleven is disabled, the stored latch pays $\kappa\ell$ there; recovery makes slot twelve working with probability r_ℓ, otherwise the same charge is paid once more. Consequently

$$Q_1(b) - Q_0(b) = d\big[(r_1 - d)A(b) - \kappa(2 - r_1)\big], \qquad r_1 = d + s(1 - 2d). \tag{22}$$

This is the full Bellman difference for the stated action family.

The numerical intervals follow from rational enclosures, including the gain-root bracket. For $0 \le x < 1$ use

$$F(x) = \sum_{j=1}^{24} \frac{x^{2j}}{2j(2j-1)} + R_{24}(x), \qquad 0 \le R_{24}(x) \le \frac{x^{50}}{50 \cdot 49(1 - x^2)}.$$

Rational interval evaluation gives, for the first price pair, $A(b) \in (0.2236, 0.2237)$ and $A(b') = 0$. For the second it gives $A(b) \in (0.1360, 0.1361)$ and $A(b') \in (-0.0954, -0.0953)$; retaining more digits gives the stated intervals. Substitution in (22) proves the strict decisions. □

The histories distinguish how much a future callable task is worth even when their confidence about its orientation is identical. The processing charge $3/25$ is a stipulated implementation price, not a universal erasure lower bound. The first witness needs no such charge. We now use the close gains and positive drift rates of the continuing instrument, with reading and food-reset charges from its ledger.

Proposition 39 (Two moments determine terminal access value)**.** *Immediately before a final working read let $X = \sigma g_G$ have posterior law b, and put $m = \mathbb{E}_b X$ and $v = \mathbb{E}_b X^2$. Use the body reading $P(Y = y \mid X) = (1 + yX)/2$, charge its conditional entropy, and permit either the posterior-matched food protocol or abstention. For overlap r, flip success s and switching charge $\kappa/2$, the expected working-use value under these separate conditional-reset prices is*

$$\begin{gathered} A(m, v) = -H_2((1 + |m|)/2) + \sum_{y=\pm 1} \frac{1 + ym}{2}\, M(m_y), \qquad m_y = \frac{m + yv}{1 + ym}, \\ M(z) = \max\{0, S_r(0) - S_r(|z|) - H_2((1 + s|z|)/2) - \kappa/2\}. \end{gathered} \tag{23}$$

Here S_r is defined in Section G.1; a term of zero observation probability is taken by continuity. A post-record decision with two wall slots remaining has

$$Q_1(b) - Q_0(b) = d[(r_1 - d)A(m_2, v_2) - \kappa(2 - r_1)], \qquad r_1 = d + s(1 - 2d), \tag{24}$$

where m_2, v_2 are the moments of $b\mathsf{K}^2$.

Proof. Bayes' rule gives $P(Y = y) = (1+ym)/2$ and $\mathbb{E}[X \mid Y = y] = (m+yv)/(1+ym)$. The food harvest is $\ln 2 - H_2((1 + s|m_y|)/2)$, its copy-assisted reset costs $\ln 2 - S_r(0) + S_r(|m_y|)$, and switching costs $\kappa/2$. Their difference, optimized against abstention, is $M(m_y)$. The body-outcome charge is $H_2((1 + |m|)/2)$. These terms give A. For the two-slot difference repeat the branch calculation in (22); the instrument undergoes two hidden transitions before the final reading. □

Two moments suffice for this terminal task, not for the continuing joint filter. The prices are conditional-entropy limits for specified separate resets. Base packets, flags, clock, reliable holding and processor implementation have the common supplied charges of Table 2; the expression is not a total cyclic apparatus profit. Additional action-dependent costs must be included in its value before deciding to repair.

One formed model, different decisions. The body model carried between invocations is a joint belief over (G, σ), formed by the permitted record history. It has three uses within this protocol. The food controller acts only when $M(m_y) > 0$; otherwise it withholds permission to spend the switching and reset resources. The repair controller compares (24) with zero. A controller that can use a later invocation for further evidence must also forecast what today's record will make learnable. These are not three independently supplied assessments of the observer. They use different functions of the same acquired belief. A finite horizon and read quota, or a specified continuing budget, determine which functions are worth retaining and computing.

Proposition 40 (Terminal compression need not close enquiry)**.** *Use the same body likelihood as Proposition 39 and write $t = \mathbb{E}_b X^3$. Conditional on the present working read,*

$$P(Y = y) = \frac{1 + ym}{2}, \qquad m_y = \frac{m + yv}{1 + ym}, \qquad v_y = \frac{v + yt}{1 + ym}. \tag{25}$$

For a static instrument and two available working invocations, consider the body-only retained-history protocol: use the terminal food/abstention rule on each invocation, retain the body record, and do not use harvest-work labels or the food record as an extra calibration record. Its two-invocation value, before common supplied charges, is

$$A_2(b) = -H_2((1 + |m|)/2) + \sum_{y=\pm 1} \frac{1 + ym}{2}[M(m_y) + A(m_y, v_y)]. \tag{26}$$

Thus (m, v) is sufficient for the last invocation but need not determine the value of retaining a record for the next. With the four distinct supported values $\{-g_+, -g_-, g_-, g_+\}$, moments of orders zero through three determine the entire belief; fewer need not. Under positive drift, propagate that belief with the legal event or wall-slot kernel before applying the next likelihood.

Proof. Multiply the prior by $(1 + yX)/(1 + ym)$ and integrate X and X^2. This gives (25). The first invocation pays its outcome entropy and chooses $M(m_y)$; its retained outcome determines the prior of the second, whose value is $A(m_y, v_y)$. Conditioning and averaging prove (26). For the four-state support, moment recovery is an invertible Vandermonde system. To see the failure of two-moment closure without relying on a dimension count, put $a = g_-$,

$c = g_+$ and perturb any strictly positive belief in the stated order by $e(a/c, -1, 1, -a/c)$. Normalization, m and v are unchanged, whereas t changes by $2ea(a^2 - c^2) \neq 0$. For example, with $a = 0.90\sqrt{1 - 0.8^{14}}$, $c = 0.98\sqrt{1 - 0.8^{14}}$, $s = 0.979992$ and $\kappa = 0.00225388$, start from $b = (.02, .08, .10, .80)$ and take $e = .02$. Both beliefs have $(m, v) \simeq (.765001391, .892279800)$, but $\sum_y P(y)A(m_y, v_y)$ is respectively .181752936 and .180874313. The continuous entropy expressions in (23) give this strict difference. These positive-belief examples establish failure of universal compression; they are not claimed as the two histories reached from the uniform prior in Proposition 41. □

The invocation quota in this proposition is real: one body reading per invocation, no extra calibrated input and no read during a blackout. Record retention has holding, processing and eventual reset costs; the gain in (26) pays for it only if it exceeds those prices. An optional additional enquiry must be compared with foregoing that invocation or using it for the current task, including its drive, switching and record charges. The formulas are conditional task values, not a free probe or a claim of recurrent optimality. In a finite-horizon control problem the legal belief, remaining quota and available clock information suffice for the usual Bellman recursion. If the clock is absent, the disabled duration must instead be marginalized with (21); it cannot be recovered from unobserved slots. The fixed-support filter and its precision certificate implement this common model; they do not supply processor work or a cyclic reset for free.

Proposition 41 (Paid-repair decisions under positive drift)**.** *Set* $r = 0.8^7$, $h = \sqrt{1 - r^2}$, $g_- = 0.90h$, $g_+ = 0.98h$, $\delta = 3 \times 10^{-4}$, $d = 1/20$, $s = 0.979992$ *and* $\kappa = 0.00225388$. *Start from the uniform four-state prior. Take a priced clock, horizon* $H = 12$*, and a working decision at slot ten. Use the terminal protocol of Proposition 39, with no harvest-work read-back or further processing surcharge. At* $\eta = 2 \times 10^{-4}$*, a lower-confidence history optimally repairs while a higher-confidence history declines. There is also a strictly positive* η_* *in*

$$.00019999976122194292 < \eta_* < .00019999976122194293$$

at which two reachable histories have exactly equal sign confidence and opposite optimal repairs. The histories and rounded values are shown in Table 3.

Inversion rate	Slot history	Sign confidence	$Q_1 - Q_0$
0.0002	$- \times - - - - - + + ++$	.969070314	+.002191561
0.0002	$+ + - + - - - - - --$	.999973536	−.000178194
η_*	$- + + + \times \times \times + - +$	.999514604	+.000256112
η_*	$+ - - \times - + - - + -$	.999514604	−.000845366

Table 3: Working body readings are + or −; × is a disabled slot. All decisions use the same two-slot horizon, prices and action family. In the last pair confidence is exactly equal at a root in the stated interval, not merely equal after rounding.

Proof. Hold latch zero through the preceding slots. Each displayed gate path then has positive probability, independently of the instrument. All reading likelihoods and instrument

transitions are positive. Starting from the uniform prior, propagate with K between slots and multiply by $(1 + y\sigma g_G)/2$ on working readings. Thus these are reachable beliefs, not assigned posteriors.

The confidence difference of the last two histories is a continuous function of η. Outward interval evaluation gives opposite signs at the two stated rational endpoints, proving existence of an exact root. Over that whole interval the two differences in (24) lie respectively in $(.00025611, .00025612)$ and $(-.00084537, -.00084536)$. At $\eta = .0002$ the first pair lies in $(.00219156, .00219157)$ and $(-.00017820, -.00017819)$. The entropy, square-root and Bayes operations in (23) enclose these values directly; every posterior denominator is strictly positive. □

The enclosures evaluate the displayed Bayes and entropy equations directly, rather than using sampled trajectories to decide the four histories.

The first pair needs no tuned parameter and rules out an increasing confidence-threshold repair rule at the continuing instrument's own rates. The second rules out sign confidence as a sufficient decision statistic in a nearby positive-rate model. It does not assert a unique root. The restoring reader assigns more weight to the higher gain; the declining reader expects less value from a working invocation. Confidence in orientation and the value of accessible tasks are different properties of the same acquired history.

The finite horizon is material. A continuing observer must also price evidence gained on later invocations, lost evidence during disability, and the registers that carry its model. Recurrent optimal maintenance without an elapsed-time counter remains a stronger target. The two moments above explain a terminal decision without adding history as an independent reward.

9 Consequences for the other links

Paper II. Let the agent add a known push p_t, independent of the input, through an actuator with gain k, so that $y_t = \rho\, u \cdot x_t + k\rho\, u_A \cdot p_t + \xi_t$, with one in-frame read per step. If sensing and acting share one coupling ($u_A = u$), the regressions of y on the push and on the in-frame read, with the residual variance, determine u including its background, ρ, k and s; the blind spot of Paper II disappears with a single read per step when $u \cdot v \neq 0$, $k\rho \neq 0$ and sign conventions for k, ρ are fixed, and the remaining symmetry between actuator gain and input scale is fixed by the known input law. If the actuator coupling is independent of the sensing coupling, pushes determine $k\rho u_A$ and nothing about the background of u. In a simulation with $D = 6$ the shared coupling recovers ρ and k within 2% and the direction of u within 0.016; with an independent coupling the direction error is 1.3. A shared sensing/action frame supplies this constructive route. At $u \perp v$ the scale parameters remain confounded; varying the read direction repairs the degeneracy.

HAFF Paper C. The viability bound of an agent depends on it only through the product of its sensing and control influences [4], which is invariant under the sign symmetry, while viability itself is not: an inverted agent has the same influences and fails with probability $\frac{1}{2}(1 + (1 - 2\nu)(1 - 2\mu)(1 - 2q))$. Positive influences can be certified from the agent's own records; viability cannot be certified without a standard.

Several observers. Applying Theorem 26 with the sign symmetry acting on two agents at once, their relative sign is invariant and detectable from agreement, and a common inversion is not. The common error that agreement can never expose, which Foundation II measures [10], is the direction shared by the observers' standards.

10 Discussion

10.1 The memory–self–introspection route

The shared channel, loss metric and frame residual were specified in Section 1.2. The body adds a further use of the learned model: deciding which future access is worth restoring. Proposition 41 proves that sign confidence can select the wrong repair even at positive drift rates; Propositions 39–40 identify the extra retained information needed for terminal and prospective tasks. The body–permission coupling carries a relational standard. Wiring, transition laws and resource supplies remain specified inputs, and joint relabelling preserves the absolute-orientation obstruction of Theorem 26. A cyclic processor with holding and preparation costs, and a continuous self-frame with matrix-valued loop gains, remain stronger targets.

10.2 What is established and what is open

Within the stated models the results are proved, and several are sharp. The detection bound holds for every test; it is reached within a factor of about 1.3 when the drift path is known, and its order is reached by a simple residual-variance test under Brownian drift. The scalar quadratic drift law holds for every schedule and estimator; Gaussian prior/readings and controllable real precision attain its optimum, so the price of keeping up is known exactly to leading order, with its constant. The task-risk formula of Section B is exact for every joint law of the reports.

The main open problems are these. The drift law is proved for a scalar parameter and a Gaussian random walk; vector parameters, non-Gaussian increments (for which the same proof gives a weaker bound with the increment variance replaced by the inverse Fisher information of the increment law) and smoother drift (Conjecture 17) remain. Paper II conjectures that probe calibration of a D-channel frame is rate optimal for spherical Brownian/local isotropic diffusion, not for every rotation-invariant drift law [2]; the telescoping of Theorem 11 applied per background direction is a possible route, but the directions of a frame on a sphere are not independent random walks. No rule is given that learns the task risk of Proposition 45 online, and the gain proxy of Section B is not shown to be the best measure of reliability. Fisher information is the natural local quantity for these questions; nothing here shows that a physical system must compute it.

The positive-rate repair witness establishes a finite-horizon mechanism, not an optimal recurrent policy. Its extension must combine continuing maintenance, a finite processor and the cost of acquiring its decision information. A physical cycle must supply holding, processing, reference preparation and heat transport together. BRAIN's component-maintenance model [12] treats the corresponding question when learned relations span several ports; it has

Linear drift ε	10^{-2}	3×10^{-3}	10^{-3}
In-frame records, exact (asymptotic)	54 (49)	137 (129)	324 (312)
Probe records, exact (asymptotic)	58 (58)	130 (130)	271 (271)
Brownian drift σ	10^{-2}	10^{-3}	10^{-4}
In-frame records, exact (asymptotic)	17 (11)	68 (53)	282 (245)
Residual-variance test reaches errors 0.05	84	356	1565
Frozen self-model fails at $t^* = \ln 2/\sigma$	69	693	6931

Table 4: Detection times in steps for Paper II's reader ($\rho = 1$, $s = 0.5$, one record per step, errors 0.05). The rows labelled "exact (asymptotic)" are necessary times: "exact" sums the exact relative entropies, with the unwrapped angle normal of variance $2\sigma t$ under Brownian drift, and "asymptotic" is the formula of Proposition 6. The test row is the smallest horizon at which the residual-variance test of Corollary 7, with the exact χ^2 threshold, reaches both errors 0.05 (Monte Carlo).

a different repair clock and cannot inherit the single-body event kernel without an explicit mapping.

10.3 Function and experience

Reflexivity in this paper means second-order control: a system that monitors and adjusts its own monitoring. The results say what such control can detect, what it costs and what it buys in a loop. The access gate also makes the reliability of future self-knowledge depend on physical maintenance. Its working and disabled states are operational permissions, not claims about sleep, death or experience. They do not show that a calibrated self-model is, or produces, experience. The Consciousness series treats the relation between functional self-models and first-person organization with its own arguments [3].

A Worked example

A.1 Detecting drift in Paper II's two-channel reader

Take Paper II's reader with $D = 2$, $\rho = 1$, $s = 0.5$ and one record per step ($r = 1$). Then $\kappa = 36$ for in-frame records and $F_\perp(0) = 0.8$ for probes. For errors $\alpha = \beta = 0.05$, $d(0.95\|0.05) = 2.650$. Table 4 gives the smallest horizon at which the exact relative entropies of Corollary 5, summed over the records, reach d. By Proposition 6 no test with these errors can succeed earlier.

For known linear drift, the likelihood-ratio test reaches errors 0.05 at about 1.3 times the necessary horizon (4000 repetitions per point). The path-free residual-variance test needs roughly five times the Brownian lower bound; it precedes frozen-model failure at $\sigma = 10^{-3}, 10^{-4}$, but not at 10^{-2}. Its $T\sigma^{2/3}$ approaches the predicted constant 3.28. Near calibration the in-frame natural gradient is $0.0556/\theta$, whereas a probe fraction p supplies metric $0.8p$ at calibration.

ν per use	Floor (k_BT)	C (k_BT)	ρ^* (references per use)	α^*
10^{-2}	0.05511	0.05515	0.0265	1.95×10^{-3}
3×10^{-3}	0.02471	0.02471	0.0119	3.9×10^{-4}
10^{-3}	0.01188	0.01188	0.0057	9.1×10^{-5}
10^{-4}	0.002560	0.002560	0.0012	4.2×10^{-6}

Table 5: The local Gaussian comparison for a drifting gain ($g = 0.55$, $b = 0$, $w = k_BT \ln 2$). "Floor" is the Gaussian quadratic comparison $\min_P h(P)$ of Theorem 11 for the quadratic loss; C is the leading term of Corollary 12. For Paper I's divergence loss the entries are leading-order values, accurate for $\nu \lesssim 10^{-3}$. A record read through the same channel is worth $0.160\, k_BT$ (Paper I).

A.2 The local Gaussian comparison for a drifting gain

Let the gain of the gain–bias channel drift with $g \approx 0.55$, $b = 0$, so $J = G = 1.4337$, and let references cost $w = k_BT \ln 2$. Table 5 gives the local Gaussian comparison of Remark 14, its leading term C, the optimal reference rate and the optimal memory weight.

At $w = J = c = 1$ and $\nu = 10^{-2}$, average-cost value iteration on a variance grid agrees with the Gaussian floor to relative error 1.3×10^{-5}; 4000 Kalman trajectories agree within Monte Carlo error. Integer binomial reads provide a distinct comparison: at $w = 10^{-4} k_BT$, $\nu = 3 \times 10^{-3}$, the best tested grid policy costs 1.10 times the relaxed floor. For Corollary 13, $q = 10^{-4}$, $r = v = 1$, $w = 10^{-3}$ gives optimum 0.0054878. The source map identifies the full scans and fixed-seed outputs.

B A drifting instrument in a closed loop

Two finite bank models couple the estimate $(\hat{g}, \hat{b})$ to an action that changes the next world reading. Their instruments, baths and results remain separate. They specify their own assumptions and do not use (C1)–(C3).

B.1 The world and the observation map

The bank has M binary variables $z_j \in \{\pm 1\}$ with mean $x = M^{-1} \sum_j z_j$. In each epoch every variable is updated independently: it keeps its value with probability a, is replaced by a variable biased toward the agent's command u with probability c, and is replaced by a variable biased toward the environmental input d with probability r, where $a + c + r = 1$ and the biased variable is $+1$ with probability $(1 + v)/2$ for the bias $v = u$ or $v = d$. For $|u|, |d| \le 1$ this is a convex combination of the identity and two resets, hence a legal channel (Foundation III, section on legal finite feedback [11]). The probability of $+1$ is therefore $(1 + az_j + cu + rd)/2$, and

$$x_{k+1} = a\, x_k + c\, u_k + r\, d_k + \zeta_k, \qquad \mathbb{E}[\zeta_k \mid \text{past}] = 0, \quad \mathbb{E}[\zeta_k^2 \mid \text{past}] \le M^{-1}. \tag{27}$$

The count of $+1$ variables is updated exactly from two binomial draws. The simulations hold that count for efficiency and never give it to the agent. We use $(a, c, r) = (0.65, 0.25, 0.10)$.

An instrument with gain g and bias b, $|g| + |b| \le 1$, reads a uniformly random variable and returns $y = \pm 1$ with probability $(1 + gz_j + b)/2$. The index is not returned. A reference read presents an input known to be $+1$ or -1 and returns $+1$ with probability $(1 \pm g + b)/2$. Gain and bias are fixed within an epoch and drift between epochs. From n_s world reads with mean $\bar{y}$ and the current estimates the agent forms

$$\hat{x} = Q\left(\frac{\bar{y} - \hat{b}}{\max(\hat{g},\, 0.30)}\right), \tag{28}$$

where Q clips to $[-1, 1]$ and rounds to a grid of 1025 points, and it acts with $u = -h\hat{x}$, $h = 0.8$. The total number of reads per agent and epoch is 512. World loss is $x^2/4$, averaged over trajectories and over epochs after a burn-in of 200.

B.2 What reference reads measure

Proposition 42 (Reference reads)**.** *Let $p \sim \mathrm{Bin}(n, \pi_+)$ and $m \sim \mathrm{Bin}(n, \pi_-)$ be independent, with $\pi_\pm = (1 \pm g + b)/2$ and $|g| + |b| < 1$. Then $(\hat{g}, \hat{b}) = \big((p - m)/n,\ (p + m)/n - 1\big)$ is unbiased with covariance*

$$\Sigma_{\mathrm{ref}} = \frac{1}{2n}\begin{pmatrix} 1 - g^2 - b^2 & -2gb \\ -2gb & 1 - g^2 - b^2 \end{pmatrix},$$

which is the inverse of the Fisher information matrix of the $2n$ reads about (g, b). The estimator is therefore efficient.

Proof. $\mathrm{Var}\, p = n(1-(g+b)^2)/4$ and $\mathrm{Var}\, m = n(1-(b-g)^2)/4$, which gives the covariance by linearity. The pair $(p/n, m/n)$ is the maximum-likelihood estimator of the mean parameters (π_+, π_-) of a product of binomials and attains the Cramér–Rao bound $\mathrm{diag}(\pi_\pm(1 - \pi_\pm))/n$. The map to (g, b) is linear and invertible, and efficiency is preserved under it. □

The frequency estimator above is unbiased and efficient for mean-square error, but its boundary values make the plug-in expected divergence infinite at every finite sample size. To use Paper I's calibrated-work expansion, replace each estimated response probability k/n by $(k + 1/2)/(n + 1)$, or use another regularization satisfying its tail condition. The leading covariance is unchanged, and the $1/n$ work loss is then an asymptotic statement for that regularized estimator.

For that regularized estimator, Paper I computes the leading calibrated loss: acting on the estimated readout model costs $k_B T(1 + g^2/(1 - b^2))/(4n)$ per use to leading order, $0.00254\, k_B T$ for $n = 128$ at $g = 0.55$, $b = 0$. A reference read costs a world read, so n_r is a budget. The estimates are also kept in memory. A persistent memory with weight α on the new reference estimate, $\hat{\theta}_k = Q[(1 - \alpha)\hat{\theta}_{k-1} + \alpha\tilde{\theta}_k]$ for $\theta \in \{g, b\}$, trades noise against lag.

Proposition 43 (Memory under drift)**.** *Let $0 < \alpha \le 1$, $\beta = 1 - \alpha$ and $e_k = \hat{\theta}_k - \theta_k$. Suppose the reference estimate has the form $\tilde{\theta}_k = \theta_k + \xi_k$, where θ_k is determined by the information available before the reference reads of epoch k (a deterministic drift qualifies, and the true instrument state may enter without giving the observer access to it) and ξ_k has*

conditional mean zero given that information and variance at most σ^2, that $\theta_k \in [-1,1]$ and $|\theta_k - \theta_{k-1}| \le \nu$, and that Q clips to $[-1,1]$ and rounds to a grid of spacing Δ. Then

$$\|e_k\|_{L^2} \le \sqrt{\big(\beta(\|e_{k-1}\|_{L^2}+\nu)\big)^2 + \alpha^2\sigma^2} + \frac{\Delta}{2}, \qquad \|e_k\|_{L^2} \le X_\sharp + \beta^k\big(\|e_0\|_{L^2} - X_\sharp\big)_+,$$

with $X_\sharp = \dfrac{\Delta/2 + \beta^2\nu + \sqrt{\beta^2(\Delta/2+\nu)^2 + (1-\beta^2)\alpha^2\sigma^2}}{1-\beta^2}$. *If $0 < \alpha < 1$, the noise variance is exactly σ^2, and the drift is the idealised linear drift $\theta_k = \theta_0 + \nu k$ without rounding or clipping, the steady-state error has mean $-\nu(1-\alpha)/\alpha$ and variance $\alpha\sigma^2/(2-\alpha)$. Hence the persistent memory has smaller mean-square error than a memoryless estimate (weight $\alpha = 1$, error variance σ^2) exactly when*

$$\nu^2 < \frac{2\alpha^2\sigma^2}{(1-\alpha)(2-\alpha)}.$$

Proof. Put $y_k = \beta\hat\theta_{k-1} + \alpha\tilde\theta_k$, so that $y_k - \theta_k = \beta e_{k-1} - \beta(\theta_k - \theta_{k-1}) + \alpha\xi_k$. The noise ξ_k is orthogonal in L^2 to the first two terms, which are determined before it, so $\|y_k - \theta_k\|^2 \le \beta^2(\|e_{k-1}\| + \nu)^2 + \alpha^2\sigma^2$. Clipping to an interval that contains θ_k does not increase the distance to θ_k, and rounding adds at most $\Delta/2$; this is the first inequality. Write $\varphi(X)$ for its right-hand side, which is increasing with $\varphi'(X) = \beta^2(X+\nu)/\sqrt{\beta^2(X+\nu)^2 + \alpha^2\sigma^2} \le \beta$. The larger root $X_\sharp$ of $(X - \Delta/2)^2 = \beta^2(X+\nu)^2 + \alpha^2\sigma^2$ is the displayed expression, and it satisfies $X_\sharp \ge \Delta/2$, so $\varphi(X_\sharp) = X_\sharp$ and $\varphi(X_\sharp + t) \le X_\sharp + \beta t$ for $t \ge 0$. The second inequality follows by induction. For linear drift without rounding or clipping the recursion is linear, $e_k = \beta e_{k-1} - \beta\nu + \alpha\xi_k$, which gives the stated mean and variance. The mean-square comparison is $\alpha\sigma^2/(2-\alpha) + \nu^2(1-\alpha)^2/\alpha^2 < \sigma^2$. □

For Model A's $\alpha = 0.25$ and $n_r = 128$ reads per sign, the unrounded linear-drift comparison gives $\nu^* \approx 0.0161$ per epoch. It places the crossover at the scale of the observed four-to-eight-fold drift bracket; it omits the loop, gain floor and periodic drift. The Gaussian-transition optimum of Theorem 11 instead uses the stationary Kalman gain and real precision. A constant-weight estimator under linear drift has the different $\nu^{2/5}$ price of Remark 16.

The world is read through the same estimates. The next statement bounds the error of the decoded state in terms of the errors of the estimates.

Proposition 44 (Error of the decoded state)**.** *Let $g \ge g_{\min} > 0$ and let $\bar y$ be the mean of n_s readings of the bank, which are independent given the state x. Let $\hat x = Q_x\big[\mathrm{clip}\big((\bar y - \hat b)/\max(\hat g, g_{\min})\big)\big]$, where Q_x rounds to a grid of spacing Δ_x that contains ± 1. Then*

$$\|\hat x - x\|_{L^2} \le \min\left(2,\ \frac{n_s^{-1/2} + \|\hat g - g\|_{L^2} + \|\hat b - b\|_{L^2}}{g_{\min}} + \frac{\Delta_x}{2}\right).$$

Proof. Write $\bar y = gx + b + \nu$ with $\mathbb{E}[\nu \mid x] = 0$ and $\mathbb{E}[\nu^2 \mid x] \le n_s^{-1}$, and $\hat g' = \max(\hat g, g_{\min}) \ge g_{\min}$. Then $(\bar y - \hat b)/\hat g' - x = [(g - \hat g')x + (b - \hat b) + \nu]/\hat g'$. Since $g \ge g_{\min}$ one has $|g - \hat g'| \le |g - \hat g|$, and $|x| \le 1$, so the numerator has L^2 norm at most $\|g - \hat g\| + \|b - \hat b\| + n_s^{-1/2}$, and the division by $\hat g' \ge g_{\min}$ gives the bound before clipping. Clipping to $[-1,1] \ni x$ does not increase the distance to x, and rounding adds at most $\Delta_x/2$. Both $\hat x$ and x lie in $[-1,1]$, so the error is at most 2. □

Model A has $\nu_g \le 0.005357$, $\nu_b \le 0.006585$ and $g \ge 0.30$. With $\alpha = 0.25$, $n_r = 128$, $\sigma^2 = 1/(2n_r)$ and grid spacing $2/1024$, Proposition 43 gives limiting L^2 errors at most 0.0351 and 0.0376. Proposition 44, with $n_s = 256$, then bounds decoded-state error by 0.4518, compared with observed epoch-averaged error 0.135. These are expectation bounds for the stated drift, not simultaneous guarantees or bounds for the zero-gain fault.

B.3 One observer: the loop closes on the world

Model A. A single agent, a drifting instrument and a bath $d_k = 0.60 + 0.25\sin(2\pi k/177)$. Normal drift is

$$g(k) = 0.55 + 0.17\sin(2\pi k/420) + 0.03\sin(2\pi k/67),$$
$$b(k) = 0.14\sin(2\pi k/300) + 0.025\sin(2\pi k/43),$$

and a drift factor s replaces k by sk. The static instrument has $g = 0.55$, $b = 0$ and the same bath. Three policies share the interface and the budget of 512 reads. *Frozen*: 512 world reads, calibrated once from 2048 reads per sign at the start and never again. *Memoryless*: 256 world reads and 128 reads per sign each epoch, using only that epoch's references ($\alpha = 1$). *Persistent*: the same reads, $\alpha = 0.25$. Each policy pays the 4096 start-up reads. The agent does not know the drift functions.

Table 6 retains the static, normal-drift and fast-drift comparisons. A separate implementation and larger normal-drift run agreed within 1.7 combined standard errors; their full scans are identified in the source map.

Instrument	Frozen	Memoryless	Persistent
static instrument	3.426±0.036	3.589±0.009	3.520±0.008
drift ×1	5.299±0.043	3.652±0.010	3.591±0.010
drift ×4	4.642±0.040	3.654±0.010	3.622±0.009
drift ×8	4.543±0.042	3.643±0.008	3.802±0.008

Table 6: World loss $x^2/4$ ($\times 10^{-3}$, ± one standard error over 160 independent trajectories, 1200 epochs, burn-in 200).

At normal drift, persistent calibration lowers loss by 1.7% relative to memoryless calibration and 32.2% relative to the frozen policy. Freezing is best for the static instrument. The persistent-minus-memoryless difference changes from -0.032 ± 0.013 at four-fold drift to $+0.159 \pm 0.012$ at eight-fold drift ($\times 10^{-3}$). Memory therefore helps only within a drift range at this weight and read allocation.

B.4 Two observers: reliability and authority

Model B. Two agents read the same bank through different instruments (the instruments and the public event of Section 5 of Foundation II [10], here with $d_k = 0.6\,c_k$), each with the calibration of Section B.2 and the same budget of 512 reads. Each agent forms $\hat{x}_i$ by (28) and issues $u_i = -h\hat{x}_i$. The world receives the joint command $u = wu_1 + (1-w)u_2$, so $|u| \le h$ whenever $w \in [0, 1]$. The agents may also move their bias estimates toward the partner's

report, either with equal weight or weighted by $\mathrm{clip}(\hat{g}_j/0.4, 0, 1)^2$, which requires exchanging $\hat{g}$ as well as $\hat{x}$ (22 bits per sender and epoch instead of 11, so 44 for the pair instead of 22). The fixed rule is $w = \frac{1}{2}$. The convex rule is $w = s_1/(s_1 + s_2)$ with $s_i = \mathrm{clip}(\hat{g}_i/0.4, 0, 1)^2$, and $w = \frac{1}{2}$ if both vanish. In the convex rule the more reliable agent receives more of the authority, and the total never exceeds the cap h.

One-step task risk. Let the reports be $\hat{\boldsymbol{x}} = x\mathbf{1} + \epsilon$ with errors ϵ of any joint law, and let the weights $w \geq 0$, $\mathbf{1}^\top w = 1$, be determined by the information $\mathcal{F}$ available when they are chosen, which may include the reference reads of the epoch. Write $q = ch$, $z = (a-q)x + rd$, $R = \mathbb{E}[\epsilon\epsilon^\top \mid \mathcal{F}]$ (an uncentred second moment) and $b = \mathbb{E}[z\epsilon \mid \mathcal{F}]$. Weights are compared with the conditional laws of the reports and of the world held fixed.

Proposition 45 (Task risk of a weight vector)**.** *Suppose $|u| \leq 1$ and $|d| \leq 1$, so that all branch probabilities of the bank step lie in $[0, 1]$, and the action is the linear rule $u = -h\, w^\top \hat{\boldsymbol{x}}$. Then*

$$\mathbb{E}\big[\tfrac{1}{4}x'^2 \,\big|\, \mathcal{F}\big] = K_{\mathcal{F}} + \frac{1 - M^{-1}}{4}\left(q^2\, w^\top R w - 2q\, b^\top w\right),$$

where $K_{\mathcal{F}} = \frac{1}{4}\big[(1 - M^{-1})\mathbb{E}[z^2 \mid \mathcal{F}] + M^{-1}\mathbb{E}[1 - a^2 + a^2x^2 \mid \mathcal{F}]\big]$ does not depend on w.

Proof. Given x, u and d the variables move independently, so $\mathbb{E}[x' \mid x, u, d] = ax + v$ and $\mathrm{Var}(x' \mid x, u, d) = M^{-1}(1 - a^2 - v^2 - 2axv)$ with $v = cu + rd$. Hence $\mathbb{E}[x'^2 \mid x, u, d] = (1 - M^{-1})(ax + v)^2 + M^{-1}(1 - a^2 + a^2x^2)$. With $u = -h(x + w^\top\epsilon)$ one has $ax + v = z - q\, w^\top\epsilon$. Taking the conditional expectation over ϵ gives the formula. □

The proposition needs no assumption that the errors are unbiased, centred, or independent between the two agents. Quantisation and the gain floor can sit inside ϵ. If the action is saturated or non-linear the formula no longer applies, and for $q = 0$ the weights do not change the one-step risk. When the errors are unbiased given x, d and $\mathcal{F}$, $b = 0$ and R is the error covariance. The risk is a convex quadratic in w. For two agents with $w = (t, 1 - t)$, $q > 0$ and $D = R_{11} + R_{22} - 2R_{12} > 0$, its minimiser over $t \in [0, 1]$ is

$$t_* = \mathrm{clip}\left(\frac{R_{22} - R_{12} + (b_1 - b_2)/q}{D},\, 0,\, 1\right). \tag{29}$$

If $R = \mathrm{diag}(v_1, v_2) + \tau^2\mathbf{1}\mathbf{1}^\top$ and $b = 0$, the best combined error is $\tau^2 + v_1v_2/(v_1 + v_2)$: weighting removes the private part of the error and cannot touch the common part.

Proposition 46 (Estimation-optimal and task-optimal weights differ)**.** *Take $M = 20$, $x = 0.2$ (twelve variables at $+1$), $(a, c, r) = (0.65, 0.25, 0.10)$, $h = 0.8$, $d = -0.7$. Let ϵ_1 be 0 or 0.2 and ϵ_2 be -0.1 or 0.1, each with probability $\frac{1}{2}$ and independent. The weight that minimises the joint estimation error is $t = \frac{1}{3}$; the weight that minimises the next-step world loss is $t = \frac{2}{3}$.*

Proof. The estimation error is $t^2\, 0.02 + (1 - t)^2\, 0.01$, minimised at $t = \frac{1}{3}$. Here $q = 0.2$, $z = 0.02$, $R = \mathrm{diag}(0.02, 0.01)$ and $b = (0.002, 0)$, so (29) gives $t_* = \frac{2}{3}$. In every branch $|u| \leq 0.32$, so the step is legal. The values in Table 7 come from enumerating all four error branches and both binomial draws exactly, and they agree with Proposition 45. □

Allocation rule	t	Joint estimation error	Next-step world loss
Smallest joint estimation error	0.3333	0.006667	0.007525
Inverse centred variance	0.5000	0.007500	0.007501
Smallest next-step world loss	0.6667	0.010000	0.007493

Table 7: The example of Proposition 46, by exact enumeration.

Agent 1's mean error 0.1 offsets $z = 0.02$, making more authority useful despite worse estimation. Task-optimal allocation needs the effects of action as well as sensor reliability. The observable gain proxy below is not the covariance R; no online learner for the true-error quantities R, b is supplied.

The linear term $-2q\, b^\top w$ also marks where Paper I's metric applies. Paper I prices a calibration error by $\frac{1}{2}k_B T\, \delta^\top G \delta$ when the downstream protocol is matched to the posterior the agent assumes, and then a calibrated self-model is a stationary point of the loss. With a fixed linear action rule it need not be one, and a biased report can lower the world loss, as in Proposition 46.

Four cells. We cross the two interventions that can be applied separately: the update rule (equal or weighted alignment) and the authority rule (fixed or convex). One agent's instrument is damaged for 100 epochs at a random start: a bias fault raises its bias by a nominal 0.30 before the legal clip, a gain fault sets its gain to 0.05 or 0 ("zero gain": the current reading carries no information about x). Table 8 gives the increase in world loss during the window relative to a matched run without the fault. The commands in all runs are $u_i = -h\hat{x}_i$ with $\hat{x}_i$ on the 1025-point grid, so $|u_i| \le h$ exactly; the largest $|u|$ applied over 105 runs was 0.8000 and no epoch exceeded the cap. Every cell is charged the same message budget, 22 bits per sender and epoch ($\hat{x}$ and $\hat{g}$) and 44 for the pair, although the unweighted, fixed cell does not use $\hat{g}$.

Policy	Bias fault	Gain 0.05	Gain 0
Unweighted, fixed $w = \frac{1}{2}$	+0.01±0.01	+2.05±0.02	+2.53±0.02
Unweighted, convex authority	+0.03±0.01	+1.11±0.02	+1.30±0.02
Weighted, fixed $w = \frac{1}{2}$	+0.01±0.01	+1.32±0.02	+1.58±0.02
Weighted, convex authority	+0.01±0.01	+0.34±0.02	+0.31±0.02
References only, no coupling	+0.00±0.01	+1.46±0.02	+1.80±0.02

Table 8: Increase of world loss in the fault window ($\times 10^{-3}$, mean of three seed blocks of 800 trajectories ± combined standard error; the different policies use different random draws, so the comparison between rows is not a paired test).

Weighted alignment limits a failed report's influence on the other estimate; convex authority limits its influence on the world while preserving the command cap. Together they reduce the zero-gain fault increase by 87.8% relative to equal alignment and fixed authority. The interaction, -0.042 ± 0.035 ($\times 10^{-3}$), is unresolved. Fault-free losses range from 3.156 to 3.167 ($\times 10^{-3}$), without a claim of equality.

B.5 What the two models give

Both models are finite banks with a declared instrument family, a declared bath and a declared cost vector (reads, start-up references, bits of memory and message). Model A shows a self-model built from reference reads improving the agent's effect on its world, with a memory whose useful length is limited by the drift. Model B shows that reliability can be used in two separate places, in what an agent believes about its partner and in how much of the world it is allowed to move, and that the second place accounts for the larger part of the reduction when a sensor fails. The two models use different instruments and baths and are not one experiment.

C Simulation protocols for Section B

Model A. Use the parameters and policies of Section B.3 and start with $n_+ = \lfloor 0.6M \rfloor$, $M = 4096$. Each epoch reads the pre-action bank and references, updates the estimates, computes (28), then applies (27). The agent receives only read counts and its stored estimates. It retains 22 estimate bits and an 11-bit command index, pays 4096 startup references, 256 references per adaptive epoch and 4096 bank-step applications. Counters, index generation and arithmetic workspace are additional resources; no heat or work is inferred from these counts. Run lengths and errors are specified in Table 6.

Model B. The same bank and step, with the public event c_k (switching with probability 0.01 per epoch) and $d_k = 0.6\, c_k$. The instruments are

$$\begin{aligned} g_i(k) &= 0.55 + 0.17\sin(2\pi k/420 + \phi_i) + 0.03\sin(2\pi k/67 + \phi_i), \\ b_i(k) &= 0.10\sin(2\pi k/250) + 0.03\sin(2\pi k/53) \\ &\quad + 0.10\sin(2\pi k/300 + \psi_i) + 0.025\sin(2\pi k/43 + \psi_i), \end{aligned}$$

with $\phi = (0, 1.7)$ and $\psi = (0.4, 2.9)$, clipped to $|b| \le 0.98 - |g|$.

Each agent makes 512 reads per epoch, of which $2n_r$ with $n_r = 16$ are reference reads, and starts from 2048 reference reads per sign. Order within an epoch: both agents read the world and references on the same pre-action state; each updates its own calibration and forms $\hat{x}_i$; one synchronous exchange; the commands $u_i = -0.8\,\hat{x}_i$ and the joint command are formed from the values before the exchange; the world takes one joint step; the alignment update for the next epoch uses the exchanged values. Equal alignment: $\hat{b}_i \leftarrow Q(\hat{b}_i + 0.25\, e_i)$ with $e_i = \bar{y}_i - (\hat{g}_i \hat{x}_j + \hat{b}_i)$. Weighted alignment multiplies the step by $\mathrm{clip}(\hat{g}_j/0.4, 0, 1)^2$. The fault window lasts 100 epochs from a start drawn uniformly from $[800, 2400)$ for each trajectory; faulted and no-fault runs share it. Runs of 3000 epochs, burn-in 200, 800 trajectories per run, three seed blocks. The world and each agent draw from separate random streams addressed by seed, epoch and event, so a fault in one instrument does not shift the draws of the world or of the other agent; because the agents act on the world, faulted and no-fault worlds differ afterwards.

Reproducibility. The supplemental source map lists checks by manuscript table, proposition or equation label, with named scripts, archived outputs and hashes, including the

two Model A implementations and Model B. Checks cover detection, drift prices, gauge symmetry, sign inversion, body moments, finite-block margins, continuing filtering and repair. Their individual assumptions and seeds are retained with the scripts; one experiment's output is not a certificate for a different model.

D Thermodynamic implementation inputs

D.1 Thermodynamic length

Let $\lambda(t)$, $t \in [0,\tau]$, be a protocol in control-parameter space. For conjugate forces $F_i = -\partial_i H$, define the friction tensor by

$$\zeta_{ij}(\lambda) = \beta \int_0^\infty \langle \delta F_j(0)\delta F_i(t)\rangle_\lambda \, dt.$$

The thermodynamic length is $\mathcal{L} = \int_0^\tau \sqrt{\zeta_{ij}\dot\lambda^i\dot\lambda^j}\, dt$ [22]. Within this slow-driving linear-response approximation, $W_{\mathrm{ex}} \geq \mathcal{L}^2/\tau$. Constant thermodynamic speed saturates the bound along a fixed path; minimizing also over paths gives a shortest geodesic. Where the normalized correlation integrals τ_{ij} are defined, $\zeta_{ij} = k_B T\, \tau_{ij} g_{ij}$ entry by entry. A common scalar relaxation time gives $\zeta = k_B T \tau_{\mathrm{relax}} g$ only when all the relevant entries have that same time [23].

E Local tracking of a moving optimum

E.1 Tracking a moving optimum

Let $\vartheta^*(t)$ be the minimizer of $L(\cdot,t)$, differentiable in t, and let $\|\dot\vartheta^*\|_g$ be its speed in the metric g. As a monitoring function take the squared geodesic distance

$$V(t) := d_g\big(\vartheta(t),\vartheta^*(t)\big)^2. \tag{30}$$

Theorem 47 (Loop tracking bound)**.** *Suppose the trajectory of* (16) *stays in a region where $d_g(\cdot,\vartheta^*(t))^2$ is smooth and $L(\cdot,t)$ is geodesically m_L-strongly convex for some $m_L > 0$. Then*

$$\frac{dV}{dt} \;\leq\; -2\eta m_L\, V + 2\sqrt{V}\,\|\dot\vartheta^*\|_g. \tag{31}$$

Consequently: (a) V decreases whenever $\sqrt{V} > \|\dot\vartheta^*\|_g/(\eta m_L)$; (b) if $\|\dot\vartheta^*\|_g \leq \Lambda$, then

$$\limsup_{t\to\infty} V(t) \;\leq\; \frac{\Lambda^2}{(\eta m_L)^2}; \tag{32}$$

(c) if ϑ^* is constant, $V(t) \leq V(0)\, e^{-2\eta m_L t}$.

Proof. The natural gradient $g^{-1}\nabla L$ is the Riemannian gradient $\operatorname{grad} L$. At ϑ the gradient of $d_g(\cdot,\vartheta^*)^2$ is $-2\exp_\vartheta^{-1}\vartheta^*$. Strong convexity along the geodesic from ϑ to ϑ^* gives $L(\vartheta^*) \geq L(\vartheta) + \langle \operatorname{grad} L(\vartheta), \exp_\vartheta^{-1}\vartheta^*\rangle + \frac{1}{2}m_L V$, and since ϑ^* is the minimizer, $L(\vartheta) \geq L(\vartheta^*) + \frac{1}{2}m_L V$. Adding the two, $\langle \operatorname{grad} L(\vartheta), -\exp_\vartheta^{-1}\vartheta^*\rangle \geq m_L V$. The contribution of the update to dV/dt is $\langle -2\exp_\vartheta^{-1}\vartheta^*, -\eta \operatorname{grad} L\rangle \leq -2\eta m_L V$. The contribution of the moving optimum is $\langle -2\exp_{\vartheta^*}^{-1}\vartheta, \dot\vartheta^*\rangle \leq 2\sqrt{V}\|\dot\vartheta^*\|_g$ by Cauchy–Schwarz. This gives (31) and (a). For (b), $W = \sqrt{V}$ obeys $\dot W \leq -\eta m_L W + \Lambda$ wherever $V > 0$, so $\limsup W \leq \Lambda/(\eta m_L)$. For (c) set $\dot\vartheta^* = 0$. □

For the two-channel frame the loss $s^2 + \rho^2\sin^2\theta$ has $L'(0) = 0$ and $L''(0) = 2\rho^2$, so its Riemannian Hessian at calibration is $2\rho^2/g(0)$. When probes make $g(0) > 0$ (Proposition 18), continuity gives strong convexity in a neighbourhood of calibration with m_L close to $2\rho^2/g(0)$, and the theorem applies there. Without probes $g(0) = 0$ and it does not apply.

Under drift the loop cannot converge to zero mismatch; it holds the mismatch within the neighbourhood (32), which shrinks with ηm_L and grows with the drift speed. Raising η is not free: a physical implementation has to relate the adaptation rate to its energy budget (Section F), and in a stochastic loop a faster update also follows the noise, which is the trade-off Section 5 prices exactly. A second constant governs how well the metric itself is conditioned.

F Thermodynamic cost

F.1 Cost components

A physical calibration loop senses, computes, actuates and prepares references. What each costs depends on how it is realized.

Sensing and computing. Measurement, copying and logic can in principle be reversible, so reading a stream carries no positive cost by itself. If sensor and work registers irreversibly erase Shannon information at rates R_s and R_c bits per unit time, Landauer's bound gives $\dot W_{\text{sense}} \geq k_B T \ln 2\, R_s$ and $\dot W_{\text{compute}} \geq k_B T \ln 2\, R_c$.

References. A reference read requires an input in a known state. Preparing a known pointer state from the thermal state of a degenerate pointer costs at least $k_B T \ln 2$ per reference (Paper I). At R_{ref} references per unit time this is $k_B T \ln 2\, R_{\text{ref}}$, unless references are recycled.

Actuation. Moving the self-model from ϑ to a new estimate $\hat\vartheta^*$ is a finite-time transformation of whatever physical parameter stores it. In linear response, a protocol of duration τ_{recalib} and thermodynamic length $\mathcal{L}(\vartheta,\hat\vartheta^*)$ dissipates at least $\mathcal{L}^2/\tau_{\text{recalib}}$, a rate of at least $\mathcal{L}^2/\tau_{\text{recalib}}^2$ if repeated every τ_{recalib} [23].

F.2 A conditional implementation bound

Proposition 48 (Conditional implementation cost)**.** *Suppose that (i) sensing, computing, reference preparation and actuation use separate physical degrees of freedom with no shared erasure accounting; (ii) sensor and work registers erase Shannon information at rates R_s and R_c, and R_{ref} references per unit time are prepared from thermal states; and (iii) actuation*

follows a linear-response protocol of duration τ_{recalib} *and thermodynamic length* $\mathcal{L}$*, repeated every* τ_{recalib}*. Then*

$$\dot{W}_{\mathrm{loop}} \;\geq\; k_B T \ln 2 \left(R_s + R_c + R_{\mathrm{ref}}\right) + \frac{\mathcal{L}^2}{\tau_{\mathrm{recalib}}^2}. \tag{33}$$

Proof. Landauer's bound applies to the two declared erasure streams and to each reference preparation; the Sivak–Crooks bound divided by the protocol duration gives the actuation rate. Separability allows the terms to be added without double counting. Dropping separability, or using reversible sensing and computation without resets in the window, removes the corresponding terms. □

The proposition and the drift law count different things. Remark 14 counts references and the loss of acting on an imperfect self-model, in the exact Gaussian comparison. Proposition 48 adds the costs of the registers and actuators of one implementation. Together they bound what a physically realized calibration loop pays per use. Papers I and II supply no universal maintenance or processing cost that could be added to these without a concrete implementation, so no general persistence budget is claimed.

G Finite-block margins and continuing filtering

G.1 A resource-conditioned operating margin

A lower bound on drive occupation cannot by itself supply an achievable gate. We instead choose the displaced squeezed Gaussian family of Q-RAIF B and use its exact characteristic function for both error and energy. If its mean occupation is N, set

$$V_Q = \frac{1}{2N+1}, \quad \lambda^2 = \frac{\pi^2(2N+1)}{16N(N+1)}, \quad \alpha^2 = \frac{\pi^2}{16\lambda^2}, \quad s = \frac{1+\exp(-2\lambda^2 V_Q)}{2}.$$

Then $\alpha^2 + (V_Q + V_Q^{-1} - 2)/4 = N$. For the coherent comparison use $V_Q = 1$, $\lambda^2 = \pi^2/(16N)$ and the same formula for s. Each opportunity receives a fresh independent scheduled packet, costing $\theta_\omega N$ whether or not it is switched on. Actual switching adds $\theta_\omega \lambda^2$. Incoming field energy and switching work thus refer to the same state.

The gain is physical and bounded. In blocks of $L = 4096$ food uses, draw independently one of $g_- = 0.90h$ and $g_+ = 0.98h$, with equal prior probabilities, where $r = 0.8^n$ and $h = \sqrt{1-r^2}$. Keep that gain constant through the block, with matched setting and known positive sign. At the start take $m = 32$ references of the input $+1$. The count K has likelihood

$$p_g(k) = \binom{m}{k}[(1+g)/2]^k[(1-g)/2]^{m-k}; \qquad \bar{g}_k = \frac{g_- p_{g_-}(k) + g_+ p_{g_+}(k)}{p_{g_-}(k) + p_{g_+}(k)}.$$

The reference pointer preparation costs $m \ln 2$. We conservatively erase its outcomes separately for another $m \ln 2$, and erase the $(m+1)$-state count register for $\ln(m+1)$ per block. This finite-memory filter incurs no Gaussian approximation or truncation.

Write

$$S_r(x) = H_2\left(\frac{1+\sqrt{r^2+(1-r^2)x^2}}{2}\right).$$

The posterior-optimal food levels give income $\ln 2 - H_2((1+s\bar{g}_k)/2)$. The food reset against the untouched quantum copy costs $\ln 2 - S_r(0) + S_r(\bar{g}_k)$; the switching probability is exactly $1/2$. Its net, averaged over the exact law of K, is

$$A = \mathbb{E}_K[S_r(0) - S_r(\bar{g}_K) - H_2((1+s\bar{g}_K)/2)] - \theta_\omega(N+\lambda^2/2) - \frac{2m\ln 2 + \ln(m+1)}{L}.$$

The reset is the reversible conditional-entropy limit for the stated classical–quantum state, retaining the copy marginal. Its attainability uses quasistatic processing across independent replicas of the complete block protocol, conditioned on the retained count. An implementation with reset overhead below a positive margin has the same sign of net work.

Economy B adds one body read and drive opportunity per food use. The body starts each block uniform and is thermalized by a work-free swap with a fresh equilibrium bit at the block boundary. With bath flip probability $d = 0.05$ and feedback on every negative read,

$$x_{g,t+1} = 0.9[(1-s)x_{g,t} + sg], \qquad x_{g,0} = 0.$$

Given $K = k$, define $x_k = \mathbb{E}[x_{g,t} \mid k]$, $y_k = \mathbb{E}[g x_{g,t} \mid k]$, and $p_\pm = (1 \pm y_k)/2$. The conditional body-copy biases are $b_\pm = (x_k \pm \bar{g}_k)/(1 \pm y_k)$. Body erasure therefore costs

$$H_2(p_+) - S_r(x_k) + \sum_\pm p_\pm S_r(b_\pm),$$

and its drive costs $\theta_\omega[N + p_-\lambda^2]$. Subtract their exact block average from A to get B. This policy retains only K, ignores body readings as calibration data, and requires no unproved body-stream information rate.

At $n = 7$ and $\theta_\omega = 0.01$, the squeezed construction leaves $0.02396\,k_BT$ per food use after body upkeep, with body loss 0.08725; the coherent comparison leaves $-0.09847\,k_BT$. A sufficiently small irreversible overhead preserves the positive squeezed margin. This is a maintained body supplied by a record and drive stream, with a gain that changes between blocks. It is not an infinite Gaussian-gain process.

Record formation consumes $2n\ln 2$ of fresh memory purity per food use in A and $4n\ln 2$ in B, plus $mn\ln 2/L$ for reference fragments. Those memories are supplied by the environment and their purity is explicitly an input, not work produced by the cycle. Replenishing them from the agent's own thermal resources debits that input and eliminates the apparent profit. The drive additionally presupposes its preparation, phase reference and scheduled port; uncomputed apparatus overhead can be tolerated only within the displayed positive margin. The example identifies a nonempty region of positive operating margin under these resource declarations. It does not derive the resources or the reference standard from the loop.

G.2 A continuing joint filter and adaptive references

The block calculation redraws a prior and resets the body at known times. A continuing observer instead carries its model through the changes that its own history reveals. Keep the

n	θ_ω	Drive	N	A: margin	B: margin	Body loss	B: purity
3	0.01	coherent	6.220	-0.1607	-0.5798	0.1402	8.334
3	0.01	squeezed	4.220	-0.0910	-0.4899	0.1377	8.334
5	0.01	coherent	7.343	0.0533	-0.2522	0.1037	13.890
5	0.01	squeezed	4.758	0.1393	-0.1402	0.1013	13.890
7	0.01	coherent	7.865	0.1552	-0.0985	0.0896	19.446
7	0.01	squeezed	5.019	0.2492	0.0240	0.0873	19.446
5	0.05	coherent	2.522	-0.1104	-0.4709	0.1097	13.890
5	0.05	squeezed	2.454	0.0036	-0.3540	0.1032	13.890
7	0.05	coherent	2.708	-0.0196	-0.3317	0.0955	19.446
7	0.05	squeezed	2.587	0.1069	-0.1999	0.0891	19.446
10	0.10	coherent	1.587	-0.0810	-0.3884	0.0945	27.780
10	0.10	squeezed	1.919	0.0410	-0.3007	0.0840	27.780

Table 9: Achievable conditional-entropy limit in units of $k_B T$. Margins are harvest minus the listed reset, reference and drive costs after food and record resources have been supplied. Each row optimizes food-only drive within its family, then uses that same state for food and body. The B: purity column is external record-memory purity per food use, $(4n+mn/L)\ln 2$, in addition to the listed operating costs; A uses $(2n + mn/L)\ln 2$. Body loss is averaged after each step over the full block. Calibration costs 0.011684 per food use.

$n = 7$ fragments and the same single-pass squeezed drive, $N = 5.01898$, $s = 0.979992$ and $\lambda^2 = 0.225388$. Let the gain switch between $g_- = 0.90h$ and $g_+ = 0.98h$ with probability $\delta = 3 \times 10^{-4}$ per use, and the sign invert with probability $\eta = 2 \times 10^{-4}$. The body bath has $d = 0.05$. These are bounded transitions valid for arbitrarily long operation. The body, instrument and stored model continue through every reference-budget boundary.

The hidden state $Z_t = (G_t, \sigma_t, B_t)$ has eight values. Each use gives one body reading with likelihood $(1 + y\sigma g B)/2$. A requested reference gives one known-input reading with likelihood $(1 + z\sigma g)/2$. Its request can depend on the body reading and earlier records. The observer then selects a body flip by the sign of its posterior body mean, and food levels by its posterior signed gain. It receives two fresh scheduled drive packets per use, one for each task, and a maximum of $m = 32$ reference ports per $L = 4096$ uses. The ports, their timing and their input labels are supplied equally to the compared policies; an unused port creates no additional income.

For body command $a \in \{0, 1\}$, let T_a be the eight-state transition matrix. Its gain and sign factors are the symmetric switching kernels of rates δ and η. The body flips with probability d for $a = 0$, and with probability $d + s(1 - 2d)$ for $a = 1$. All entries of T_a are positive. Given the selected actions and realized records, the filter predicts with T_a and multiplies by the two likelihoods, omitting the second when no reference is requested. The request rule contributes no extra parameter-dependent likelihood factor. The fair food reading alone supplies no instrument information and is used only for its task.

Positive transition and observation kernels give a uniform finite-register certificate for this filter (Proposition 49, Appendix H). It controls quantization error even when actions and reference requests are chosen from the rounded posterior.

This filter forms its next prior from the preceding posterior and the actual transition

law. It does not request a newly assigned gain prior or reprepare the body. For the stated parameters, $a_0 = 3\times10^{-9}$, $D = 39.14572$ and $\tau = 0.999999993681$. A log grid with $\Delta = 2^{-44}$ uses 350 model bits; its asymptotic total-variation bound is 2.25×10^{-6} and its one-step excess-risk bound is 4.50×10^{-6}. These conservative uniform bounds concern quantization of a correct filter. They do not certify a wrong transition family or the arithmetic implementation. The 350-bit state is an abstract finite controller; its workspace, update throughput and erasure protocol remain to be implemented.

For a resource-matched comparison, the periodic policy uses one reference every 128 steps. The adaptive policy requests a reference whenever the posterior probability of the less likely sign exceeds 0.04, subject to the same 32-port cap. Both update the joint model from every body reading and use the same body-control rule. Starting with the uniform eight-state prior, 128 independent trajectories of 65536 steps, with the first 8192 discarded, give

	Body loss	References/use	Food mismatch
Periodic	0.11726 ± 0.00083	0.0078125	0.06726 ± 0.00048
Adaptive	0.10319 ± 0.00066	0.005704 ± 0.000020	0.05226 ± 0.00014.

Errors are one standard error across independent trajectories. Food mismatch is the posterior Jensen gap $H_2((1+s|\mathbb{E}[\sigma g \mid \mathcal{H}_t]|)/2)-\mathbb{E}[H_2((1+sg)/2) \mid \mathcal{H}_t]$ in nats. The adaptive gain mean has mean-square error 0.000353, compared with the stationary prior variance 0.001529. A separate seed and shorter horizon give body losses 0.11730 and 0.10344. Requesting too early is costly under the cap: threshold 0.01 exhausts references prematurely and gives body loss 0.19567 in that shorter run. Adaptive timing needs an allocation rule as well as a confidence statistic.

Every policy is allocated the same food and record stream, drive state and 32-port reference budget. The external record purity is at most $(4n + mn/L)\ln 2 = 19.44603$ nats per food use, and reference preparation plus conservative separate erasure is charged at the same supplied-cap rate $2m\ln 2/L$. The posterior is updated from the body's own records and the policy acts on that body, closing the inference–control loop. This establishes continuing model revision and an operational benefit at finite memory. It does not close the apparatus resource loop: conditional-copy reset, model-register processing, field preparation, phase reference and the clock still need one attainable cyclic implementation. The block margin of Table 9 cannot be imported into this continuing protocol without that ledger.

H Finite-filter certificates

H.1 Continuing eight-state filter

Proposition 49 (Finite continuing filter)**.** *Let $a_0 = \min_{a,i,j}(T_a)_{ij} > 0$, and let every observation likelihood lie in $[l_-, l_+]$ with $l_- > 0$. Suppose at most two observations precede each filter update. Set*

$$\epsilon_0 = a_0(l_-/l_+)^2, \quad A = -\ln\epsilon_0, \quad D = \max_{a,i,j,k,l} \ln \frac{(T_a)_{ik}(T_a)_{jl}}{(T_a)_{il}(T_a)_{jk}}, \qquad \tau = \tanh(D/4) < 1.$$

After the first prediction, every exact update has probability at least ϵ_0 in each state. Round each of the seven log ratios $\ln(p_i/p_8)$ to a grid of spacing Δ, then normalize their exponentials.

This uses at most

$$7\left\lceil \log_2\left(\frac{2A}{\Delta}+3\right)\right\rceil$$

bits for the model. For the exact posterior p_t^* conditioned on the same realized history and the rounded filter $\tilde{p}_t$,

$$d_H(\tilde{p}_t,p_t^*)\le \tau^t d_H(\tilde{p}_0,p_0^*)+\frac{\Delta(1-\tau^t)}{1-\tau},\qquad \|\tilde{p}_t-p_t^*\|_{\mathrm{TV}}\le \tanh\bigl(d_H(\tilde{p}_t,p_t^*)/4\bigr).$$

Here $d_H(p,q)=\max_i \ln(p_i/q_i)-\min_i \ln(p_i/q_i)$. A one-step action minimizing a loss in $[0,1]$ under $\tilde{p}_t$ has excess conditional risk at most $2\|\tilde{p}_t-p_t^*\|_{\mathrm{TV}}$. With a finite reference counter and clock, the eight-state plant and rounded controller form a finite Markov chain and possess an invariant distribution under the supplied streams.

Proof. Prediction gives each probability at least a_0. The product of at most two positive likelihoods reduces that bound by at most $(l_-/l_+)^2$. Hence the seven log ratios lie in $[-A,A]$, and rounding increases their range by at most $\Delta/2$ at each end. In the Hilbert metric, positive diagonal likelihood multiplication and normalization are isometries, and multiplication by T_a contracts by at most τ, the Birkhoff coefficient. This is the standard projective stability mechanism for finite hidden Markov filters [20]. Rounding changes the oscillation of the log ratios by at most Δ. The triangle inequality gives $e_t\le \tau e_{t-1}+\Delta$, which iterates to the displayed bound even when actions and reference requests depend on earlier rounded states. To obtain the total-variation constant, let $m\le p_i/q_i\le M$, with $M/m=e^e$. Normalization and the chord bound for $(x-1)_+$ give

$$\|p-q\|_{\mathrm{TV}}\le \frac{(M-1)(1-m)}{M-m}\le \tanh(e/4),$$

where maximizing over m at fixed M/m gives $m=e^{-e/2}$; distance zero is immediate. Comparing a minimizing action under each distribution costs at most twice that distance for a bounded loss. The model grid, budget counter, clock and eight-state plant are finite, so their joint transition has an invariant probability. This last statement need not give a unique invariant distribution or an optimal reference policy, and it concerns the declared plant and controller rather than the whole resource-supplying apparatus. □

H.2 Filter on working invocations

The hard access gate has zero likelihoods on excluded body states, so Proposition 49's eight-state certificate cannot simply be inherited. Filtering at actual working invocations avoids this issue. Each of the two event matrices is positive. For the present switching rates,

$$\min_{\ell,i,j}(\mathsf{K}_\ell)_{ij}\ge a=(1-d)\delta\eta=\frac{57}{10^9},\qquad g_{\max}<0.98.$$

One body likelihood multiplication gives a posterior lower bound $a/99$, hence a log-ratio range less than 22. Event cross-ratios are at most a^{-2}, so their common projective contraction coefficient is at most $(1-a)/(1+a)$. Round the three log ratios on a grid $\Delta=2^{-44}$. The

proof of Proposition 49, now applied once per working invocation, gives 150 ideal model bits and

$$\limsup \|\tilde{p}_n - p_n^*\|_{\mathrm{TV}} \le \frac{\Delta(1+a)}{8a} < 1.247 \times 10^{-7}.$$

For a one-step loss in $[0,1]$, regret is at most twice the total-variation distance and hence less than 2.494×10^{-7}. The latch, startup state, arithmetic, workspace and protection are separate from these model bits. In wall time the plant, held finite register and latch form a finite Markov chain; they have an invariant probability under the supplied streams. The claim neither processes new data during disability nor certifies a cyclic processor.

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